

Notes on Quantum Mechanics

Augustin-Ignace des Rameaux Leung Mouk-yin |



Richard P. Feynman

UBI ERAS QUANDO PONEBAM FUNDAMENTA TERRÆ ; INDICA MIHI, SI HABES INTELLIGENTIAM. QUIS POSUIT MENSURAS EJUS, SI NOSTI ; VEL QUIS TETENDIT SUPER EAM LINEAM ; SUPER QUO BASES ILLIUS SOLIDATÆ SUNT ; AUT QUIS DEMISIT LAPIDEM ANGULAREM EJUS, CUM ME LAUDARENT SIMUL ASTRA MATUTINA, ET JUBILARENT OMNES FILII DEI ; QUIS CONCLUSIT OSTIIS MARE, QUANDO ERUMPEBAT QUASI DE VULVA PROCEDENS;CUM PONEREM NUBEM VESTIMENTUM EJUS, ET CALIGINE ILLUD QUASI PANNIS INFANTIÆ OBVOLVEREM ; CIRCUMDEDI ILLUD TERMINIS MEIS, ET POSUI VECTEM ET OSTIA, ET DIXI: USQUE HUC VENIES, ET NON PROCEDES AMPLIUS, ET HIC CONFRINGES TUMENTES FLUCTUS TUOS. NUMQUID POST ORTUM TUUM PRÆCEPISTI DILUCULO, ET OSTENDISTI AURORÆ LOCUM SUUM ; ET TENUISTI CONCUTIENS EXTREMA TERRÆ, ET EXCUSSISTI IMPIOS EX EA ; RESTITUETUR UT LUTUM SIGNACULUM, ET STABIT SICUT VESTIMENTUM: AUFERETUR AB IMPIIS LUX SUA, ET BRACHIUM EXCELSUM CONFRINGETUR. NUMQUID INGRESSUS ES PROFUNDA MARIS, ET IN NOVISSIMIS ABYSSI DEAMBULASTI ; NUMQUID APERTÆ SUNT TIBI PORTÆ MORTIS, ET OSTIA TENEBROSA VIDISTI ; NUMQUID CONSIDERASTI LATITUDINEM TERRÆ ; INDICA MIHI, SI NOSTI, OMNIA!

Contents

Chapter 1 Invitation: From Old Quantum Theory to Quantum States	1
1.1 The First Crisis: Radiation and the Exchange of Energy	2
1.1.1 Black-Body Radiation: the Classical Laws	2
1.1.2 The Rayleigh–Jeans Law and the Ultraviolet Catastrophe	3
1.1.3 Planck’s Law and the Energy Quantum	5
1.1.4 Einstein’s Light Quantum and the Photoelectric Effect	7
1.1.5 The Specific Heat of Solids	8
1.1.6 Einstein’s A and B Coefficients	9
1.2 The Second Crisis: the Structure of the Atom and its Spectra	10
1.2.1 The Electron and the Nuclear Atom	10
1.2.2 The Regularities of Spectra	11
1.2.3 Bohr’s Model of the Hydrogen Atom	12
1.2.4 The Franck–Hertz Experiment	14
1.2.5 The Bohr–Sommerfeld Theory	15
1.2.6 The Correspondence Principle	16
1.2.7 The Failures of the Old Quantum Theory	17
1.3 The Third Crisis: Waves, Particles, and Statistics	18
1.3.1 The Compton Effect	18
1.3.2 De Broglie’s Matter Waves	19
1.3.3 Bose, Einstein, and the Statistics of Quanta	20
1.3.4 The Exclusion Principle and Electron Spin	21
1.4 The Birth of Modern Quantum Mechanics	22
1.4.1 Heisenberg and the Abandonment of Orbits	22
1.4.2 Schrodinger’s Wave Mechanics	24
1.4.3 Born’s Probability Interpretation	25
1.4.4 The Equivalence of the Two Pictures	26
1.4.5 Dirac, Jordan, and von Neumann: the General Formalism	27
1.4.6 Why a New Kinematics Was Needed	28
1.5 Two-State Experiments and the Language of Amplitudes	29
1.5.1 The Interference of Single Quanta	30
1.5.2 Polarization as a Two-State System	32
1.5.3 Sequential Polarizers and the Failure of Boolean Logic	33
1.5.4 Stern–Gerlach Experiments and Incompatible Questions	34
1.5.5 The Operational Meaning of State	35
Chapter 2 From Experimental Facts to States of the Hilbert Space	37
2.1 Preparations, Alternatives, and Frequencies	38

2.2	The Two Composition Principles for Amplitudes	39
2.3	The Born Rule: from Amplitudes to Frequencies	41
2.4	The Linear Space of States	44
2.5	Rays: Which Vector Is the State?	46
2.6	The Inner Product as a Transition Amplitude	48
2.7	Orthogonality as Experimental Distinguishability	51
2.8	Bases as Complete Sets of Alternatives	52
2.9	Hilbert Space, Finite and Infinite	55
2.10	The State-Space Axioms: Summary and Open Debts	58
	Chapter 2 Exercise	59
Chapter 3 Measurement, Operators, and Spectra		62
3.1	From Selection to Measurement: Reproducible Filters	63
3.2	Projectors: the Operators of Yes-or-No Experiments	66
3.3	Projection-Valued Measures: the Bookkeeping of Complete Questions	69
3.4	Observables: Numerical Labels and Expectation Values	72
3.5	Why Self-Adjoint: Real Outcomes and the Spectral Theorem	74
3.6	Degeneracy and Compatible Observables	78
3.7	Commutators and Incompatible Experiments	81
3.8	Uncertainty Relations from Noncommutativity	83
3.9	The Continuous Spectrum: an Honest Preview	86
3.10	The Measurement Axioms: Summary and Open Debts	88
	Chapter 3 Exercise	89
Chapter 4 Evolution Operators, Symmetries, and Conservation Laws		92
4.1	Between Interrogations: the Evolution Postulate	93
4.2	Stationary States and the Spectral Resolution of $U(t)$	96
4.3	Spin in a Magnetic Field: Precession Solved Exactly	98
4.4	Time-Dependent Hamiltonians: the Time-Ordered Exponential	100
4.5	Symmetry Transformations: Active, Passive, and Wigner's Theorem	102
4.6	Continuous Symmetries as Lie Groups	104
4.7	The Galilean Group and its Algebra	106
4.7.1	Definition and Parametrization of the Galilei Group	106
4.7.2	The Lie Algebra: Complete Derivation	107
4.7.3	Projective Representations and Bargmann's Theorem	109
4.7.4	Unitary Irreducible Representations	111
4.7.5	Physical Implications of Galilean Invariance	113
4.7.6	Examples	115
4.8	The Special Status of Time	116
4.9	Conservation Laws, Good Quantum Numbers, and Degeneracy	118
4.10	Discrete Symmetries: Parity and Time Reversal	120

4.11 The Evolution Postulate and the Symmetry Principles: Summary and Open Debts . . .	123
Chapter 4 Exercise	125
Chapter 5 Operator Methods, Pictures and Representations	128
5.1 One Number, Two Readings: the Schrödinger and Heisenberg Pictures	129
5.2 Splitting the Motion: the Interaction Picture	131
5.3 The Position Representation: Wave Functions and the Form of Momentum	132
5.4 The Momentum Representation and the Canonical Commutator, Built at Last	134
5.5 Ehrenfest's Theorem and the Classical Limit	136
5.6 The Harmonic Oscillator by Ladder Operators	138
5.7 Coherent States: the Oscillator's Classical Limit	141
5.8 The Forced Oscillator: an Exact Solution in the Interaction Picture	143
5.9 The Propagator: the Free Particle and the Spreading Packet	145
5.10 Operator Methods, Pictures, and Representations: Summary and Open Debts	148
Chapter 5 Exercise	150
Chapter 6 The Theory of Rotations and Angular Momenta	153
6.1 The Rotation Group, Euler Angles, and Finite Rotations	154
6.2 Angular Momentum Algebra: the Spectrum of J^2 and J_z	159
6.3 Spin 1/2 and Pauli Matrices	163
6.4 Orbital Angular Momentum: Spherical Harmonics, Central Potential, Parity	168
6.5 Schwinger's Oscillator Model of Angular Momentum	173
6.6 Angular Momentum Coupling: CG Coefficients, Racah Form, Recurrences, and 3-j Symbols	177
6.7 Irreducible Tensor Operators and the Wigner–Eckart Theorem	184
6.8 The Hydrogen Atom: Fruit of Rotational Symmetry	191
6.9 Summary and Open Debts	196
Chapter 6 Exercise	199
Chapter 7 Composite Systems and Entanglement	204
7.1 Tensor Products of Hilbert Spaces	205
7.2 The Density Operator	208
7.3 Entanglement: Separability and the Schmidt Decomposition	212
7.4 The Einstein–Podolsky–Rosen Argument	216
7.5 Bell Inequalities and the Aspect Experiments	219
7.6 The Measurement Problem Revisited	224
7.7 Summary and Open Debts	227
Chapter 7 Exercise	229
Chapter 8 Perturbative Methods and the Theory of Scattering	233
8.1 Time-Independent Perturbation: the Non-Degenerate Case	234

8.2	Degenerate Perturbation Theory: Zeeman, Fine Structure, and Stark	238
8.3	The Variational Method: the Ground State of Helium	242
8.4	Time-Dependent Perturbation: Transition Amplitudes and Fermi's Golden Rule . . .	244
8.5	Formal Scattering Theory: States, S-Matrix, and Cross-Sections	250
8.5.1	The Formalism: Møller Operators and the S-Matrix	250
8.5.2	Stationary Scattering Theory as a Special Case	254
8.5.3	Time-Dependent Scattering Theory: the Dynamical Foundation	258
8.5.4	Symmetry of Scattered States	260
8.6	The Born Approximation	263
8.7	Perturbative Methods and the Theory of Scattering: Summary and Open Debts . . .	265
	Chapter 8 Exercise	267

Chapter 9 The Path Integral Formulation, as According to Feynman 270

9.1	The Propagator Reviewed: Kernel, Composition, and the Short-Time Kernel	271
9.2	The Feynman Construction: Time-Slicing, the Path Sum, and the Free Particle Verified	274
9.3	The Classical Limit: Stationary Phase, the Role of Action, and Hamilton-Jacobi Repaid	278
9.4	The Harmonic Oscillator: Exact Evaluation by Gaussian Integral	283
9.5	The Aharonov-Bohm Effect: Path Topology and a Measurable Phase	288
9.5.1	Setup: Solenoid, Flux, and Multiply Connected Space	288
9.5.2	Path Integral with a Magnetic Field	289
9.5.3	Two-Path Interference and the AB Phase Shift	290
9.5.4	Gauge Invariance	291
9.5.5	Topological Character: Sum over Winding Numbers	291
9.5.6	Experimental Status	292
9.5.7	Example: Two-Slit Interference Pattern	292
9.5.8	Advantage and Limitation	292
9.6	Semiclassical Approximation and WKB: Bohr-Sommerfeld Repaid	293
9.6.1	Motivation from the Path Integral	293
9.6.2	The WKB Ansatz and the $\hbar$ -Expansion	294
9.6.3	The WKB Wave Function	295
9.6.4	Breakdown at Turning Points and Connection Formulas	295
9.6.5	Bohr-Sommerfeld Quantization: The Central Repayment	296
9.6.6	Verification: The Harmonic Oscillator	297
9.6.7	Tunneling: The Gamow Formula	298
9.6.8	The Langer Correction	300
9.6.9	De Broglie Packet Dispersion Revisited	300
9.6.10	Advantage and Limitation of WKB	300
9.6.11	Example: Double-Well Tunneling Splitting	301
9.7	The Spin Path Integral	301
9.7.1	Motivation: Why a Path Integral for Spin?	302

9.7.2	Spin Coherent States	302
9.7.3	The Spin Path Integral Construction	303
9.7.4	Example: Spin-1/2 in a Magnetic Field	304
9.8	The Coherent State Path Integral for Bosons	305
9.8.1	Holomorphic (Bargmann) Coherent States	305
9.8.2	The Bosonic Coherent-State Path Integral Construction	306
9.8.3	Example: The Harmonic Oscillator Revisited	307
9.8.4	Generating Functional and Wick's Theorem	307
9.8.5	From Mechanical Coherent States to the Functional Integral over Fields . . .	308
9.8.6	Five Principles	309
9.8.7	Ledger Balanced	310
9.8.8	Experiment–Formalism Dictionary	312
9.8.9	Exit Question	312
9.9	Exercises for Chapter 9	313
	Chapter 9 Exercise	313
Chapter 10 The Second Quantization, and the Way towards Statistical Mechanics		319
10.1	Identical Particles: the Symmetrization Postulate and Fock Space	320
10.2	Creation and Annihilation Operators: the Algebraic Engine	324
10.3	Observables in the Second-Quantized Formalism	327
10.4	The Ideal Bose Gas, and Bose–Einstein Condensation	330
10.5	The Ideal Fermi Gas: Degeneracy Pressure and the Fermi Surface	334
10.6	The Density Operator and the Quantum Partition Function	338
10.7	The Second Quantization: Summary, the Repaid Debt Ledger, and Open Doors . . .	341
	Chapter 10 Exercise	344
Chapter A Mathematical Supplement		348
A.1	Hilbert Space: Completeness, Separability, and Domains	348
A.2	The Dirac Delta as a Distribution: the Gelfand Triplet	349
A.3	The Spectral Theorem for Unbounded Self-Adjoint Operators	350
A.4	Stone's Theorem	351
A.5	The Trotter Product Formula	352
A.6	Fresnel Integrals and Conditional Convergence	352
Appendix B Group Theory Supplement		354
B.1	Lie Groups and Lie Algebras	354
B.2	Representations	356
B.3	SU(2) Representation Theory	358
B.4	Wigner D-Matrices	360
B.5	The Clebsch–Gordan Series	363
B.6	Racah Algebra: 6- j and 9- j Symbols	365

B.7	Casimir Operators	366
B.8	Covering Groups	367
B.9	Irreducible Tensor Operators and the Wigner–Eckart Theorem	367
B.10	$SO(4)$ and the Kepler Problem	368
B.11	General Semisimple Lie Algebras	370
B.12	The Lorentz and Poincaré Groups	372
B.13	Racah Algebra in Depth	374
B.14	Realizations of $SU(2)$ and $SU(n)$	376
Appendix C Gaussian Units and Constants		378
C.1	The Gaussian System	378
C.2	Conversion Table: Gaussian to SI	378
C.3	Fundamental Constants	379
Appendix D Probability Theory Essentials		381
D.1	The Ensemble Interpretation	381
D.2	Conditional Probability	382
D.3	Expectation, Variance, and Moments	382
D.4	The Law of Large Numbers	383
D.5	Statistical Independence	383
D.6	Density Operators as Quantum Probability Measures	383

Chapter 1 Invitation: From Old Quantum Theory to Quantum States



Note This chapter is introductory in character. It surveys the historical and experimental route from the crises of classical physics to the modern notion of the quantum state, and it presupposes nothing beyond classical mechanics, electromagnetism, and elementary thermodynamics. Subsections marked with a star contain optional derivations; they may be skipped on a first reading without loss of continuity.

The development of Quantum Mechanics as it is can be roughly divided into two phases: that of the **Old Quantum Theory** and that of the **Modern Quantum Mechanics**, of which, the latter in particular will be discussed in this note. Nevertheless, the author considers it helpful for the reader to review, at least briefly, the historical development of both. This is not merely for practical orientation, although many readers may be content with a purely formal treatment of quantum mechanics. Rather, the historical path also helps reveal the conceptual difficulties that gave rise to the modern theory.

The author further maintains that such a study contributes not only to a better understanding of the physical nature of quantum mechanics, but also, for the believing reader, to a renewed wonder at the providence of the Creator, who brought all things into being *ex nihilo* and whose creative command is poetically recalled in the words *Fiat lux*. For the sake of brevity, however, many details of the historical “plot” will be omitted. The main line of development will be retained insofar as it prepares the way for the modern formulation.

At the end of the nineteenth century, the basic architecture of physics appeared remarkably secure. Newtonian mechanics supplied the language of trajectories and forces; Maxwell’s equations unified electricity, magnetism, and light; thermodynamics and statistical mechanics connected heat with molecular motion. The great confidence of the period did not come from naivete. It came from genuine success. Planetary motion, mechanical engineering, optics, electrical technology, and the kinetic theory of gases all suggested that nature was, at least in principle, intelligible by continuous fields and deterministic motion.

The anomalies that led to quantum mechanics were therefore especially disturbing. They were not merely small numerical disagreements. Black-body radiation, atomic spectra, and the photoelectric effect touched the foundations of energy exchange, radiation, and matter itself. The history of early quantum theory is the history of attempts to preserve as much of classical physics as possible while introducing just enough discontinuity to match experiment. Its greatness lies in this restraint; its limitation lies in the same place.

The narrative of this chapter is organized not by the calendar alone but by the crises that forced each new concept into existence. The first crisis concerned *radiation and the exchange of energy*: the cavity spectrum, the photoelectric effect, and the specific heats of solids compelled the introduction of the energy quantum. The second crisis concerned *the structure of the atom*: the discovery of the electron, Rutherford’s nucleus, and the regularities of spectroscopy demanded a mechanics of stationary

states, and received a provisional answer in the Bohr–Sommerfeld theory. The third crisis concerned *waves, particles, and statistics*: Compton scattering, de Broglie’s matter waves, Bose’s counting, and the discovery of spin destroyed any hope that the quantum could be accommodated inside a classical ontology of particles and fields. Out of these three crises came the synthesis of 1925–1927: matrix mechanics, wave mechanics, the probability interpretation, and finally the representation-independent formalism of Dirac, Jordan, and von Neumann. The chapter closes by returning to the simplest experiments of all — several of them older than the theory itself — in which the new grammar of states and amplitudes can be read directly, preparing the systematic development of the following chapters.

1.1 The First Crisis: Radiation and the Exchange of Energy

Of the three crises, the crisis of radiation was chronologically the first and conceptually the deepest, because it arose inside the two most successful theories of the century. Cavity radiation is an electromagnetic problem; its equilibrium is a thermodynamic problem; its spectrum is a statistical problem. That a contradiction should appear precisely at the triple point of Maxwell’s electromagnetism, thermodynamics, and Boltzmann’s statistics was the first indication that the difficulty was not one of detail but of principle.

1.1.1 Black-Body Radiation: the Classical Laws

Thermal radiation is electromagnetic radiation emitted from the surface of an object due to the object’s internal energy, which, traditionally, is treated by studying the radiation by the moving particles. If an object is heated sufficiently, it starts to emit light at the red end of the visible spectrum, as it becomes red hot. Heating it further causes the color to change from red to yellow, white, and blue, as it emits light at increasingly shorter wavelengths (higher frequencies).

The systematic study began with Kirchhoff. In 1859–1860 he proved, from thermodynamic equilibrium alone, that the radiation inside a cavity whose walls are held at temperature T has a spectral density $u(\nu, T)$ that depends only on the frequency and the temperature, and not on the material, shape, or constitution of the walls. A body that absorbs every incident frequency — a *black body* — emits precisely this universal spectrum. The black-body problem was thereby posed with uncommon clarity: determine the universal function $u(\nu, T)$.

By the late nineteenth century, thermal radiation had also been fairly well characterized experimentally, and two phenomenological laws were firmly established. Stefan’s empirical law of 1879, derived thermodynamically by Boltzmann in 1884 from the radiation pressure $p = u/3$, gives the total power emitted per unit area:

$$J = \sigma T^4. \quad (1.1)$$

Wien’s displacement law of 1893 gives the relation between the temperature and the wavelength at which the radiation is strongest:

$$\lambda_m T = b. \quad (1.2)$$

In 1896 Wien proposed moreover an explicit spectral formula,

$$\frac{du_\nu}{d\nu} = a\nu^3 \exp\left(-\frac{\alpha\nu}{T}\right), \quad (1.3)$$

which agreed beautifully with the measurements then available at high frequencies, and which for several years was believed to be the universal law. The decisive experiments of Rubens and Kurlbaum in 1900, performed at long wavelengths in the far infrared by means of residual rays, showed systematically increasing deviations from Wien's formula: at low frequency the spectral density appeared to grow linearly with T , as no exponential law can. The experimental situation in 1900 was therefore double-faced: one formula (Wien's) was exact at high frequency and wrong at low frequency, while the low-frequency behavior still awaited any theoretical expression.

Sketch of Wien's Thermodynamic Argument

Wien's displacement law is remarkable in that it was obtained without any model of the radiating matter, and for this reason it survived every later revolution. The argument is thermodynamic. Consider radiation enclosed in a perfectly reflecting sphere whose radius R is slowly varied. A plane wave reflected from a receding wall suffers a small Doppler shift; summing the shifts over the successive reflections of an isotropic radiation field during an adiabatic compression gives

$$\frac{\delta\nu}{\nu} = -\frac{\delta R}{R} \implies \nu V^{1/3} = \text{const}, \quad (1.4)$$

so that every frequency is diluted in proportion as the cavity expands. On the other hand, the first law applied to the radiation field, with the electromagnetic pressure $p = u/3$ and no heat exchange, gives

$$d(uV) + p dV = 0 \implies u \propto V^{-4/3}, \quad (1.5)$$

and comparison with the Stefan–Boltzmann law $u \propto T^4$ shows that $T \propto V^{-1/3}$ along an adiabat. Hence the ratio ν/T is preserved by the compression, and a spectrum that was in equilibrium remains in equilibrium only if the spectral density has the scaling form

$$\frac{du_\nu}{d\nu} = \nu^3 \varphi\left(\frac{\nu}{T}\right), \quad (1.6)$$

with a single unknown function φ of one variable. Equivalently, in wavelength language, $du_\lambda/d\lambda = \lambda^{-5}\psi(\lambda T)$. The position of the maximum then satisfies $\lambda_m T = \text{const}$, which is the displacement law. The whole content of the black-body problem is thus reduced to the determination of one function of one variable — and thermodynamics is silent about what that function is. It is worth observing that Planck's constant first entered physics precisely as a parameter in this unknown function.

1.1.2 The Rayleigh–Jeans Law and the Ultraviolet Catastrophe

The definitive classical statement of the spectral problem was given by Rayleigh in 1900 and completed by Jeans in 1905. It views the thermal radiation in a cavity as an infinite number of stationary electromagnetic modes, each of which is mechanically a harmonic oscillator and therefore contributes an average energy $k_B T$ according to the equipartition theorem of Boltzmann ($\frac{1}{2}k_B T$ of kinetic and $\frac{1}{2}k_B T$ of potential energy). Since each mode supports two independent polarizations, the mean energy

per cavity mode is $2k_B T$, and the mean energy density per unit frequency is

$$\frac{du_\nu}{d\nu} = 2k_B T \frac{dn}{d\nu}, \quad (1.7)$$

where n is the density of oscillation modes. To calculate n , let us consider a rectangular resonant cavity with dimensions (a, b, c) . The stable stationary modes of the electromagnetic wave require the following:

$$\frac{a}{\lambda/2} = m, \quad \frac{b}{\lambda/2} = n, \quad \frac{c}{\lambda/2} = q. \quad (1.8)$$

Therefore, the wavevector $\mathbf{k}$ can be written as:

$$\mathbf{k} = \pi \left(\frac{m}{a}, \frac{n}{b}, \frac{q}{c} \right). \quad (1.9)$$

If $a, b, c \gg \lambda$, then the number of modes in the thin layer $|\mathbf{k}| \in (k - \frac{dk}{2}, k + \frac{dk}{2})$ is

$$dN = \frac{1}{8} \frac{4\pi k^2 dk}{\left(\frac{\pi}{a}\right)\left(\frac{\pi}{b}\right)\left(\frac{\pi}{c}\right)} = \frac{abc k^2 dk}{2\pi^2}, \quad (1.10)$$

where the factor $\frac{1}{8}$ appears because, for stationary waves, negative m 's, n 's and q 's constitute a repetitive numbering. The number density is therefore

$$dn = \frac{dN}{abc} = \frac{k^2 dk}{2\pi^2}. \quad (1.11)$$

Substituting $k = 2\pi\nu/c$ into the mode density,

$$\frac{dn}{d\nu} = \frac{dn}{dk} \frac{dk}{d\nu} = \frac{k^2}{2\pi^2} \cdot \frac{2\pi}{c} = \frac{(2\pi\nu/c)^2}{\pi c} = \frac{4\pi\nu^2}{c^3}, \quad (1.12)$$

and multiplying by the mean energy per mode $2k_B T$ yields the Rayleigh–Jeans law:

$$\frac{du_\nu}{d\nu} = 2k_B T \cdot \frac{4\pi\nu^2}{c^3} = \frac{8\pi k_B T \nu^2}{c^3}. \quad (1.13)$$

This is the Rayleigh–Jeans law. It agrees with the experimental results well at large wavelengths (or, equivalently, low frequencies) — indeed it is precisely the linear-in- T behavior that Rubens and Kurlbaum had observed — but it disagrees catastrophically at short wavelengths. Since the mode density grows like ν^2 and every mode carries the same mean energy $k_B T$, the total predicted energy density diverges:

$$u = \int_0^\infty \frac{8\pi k_B T \nu^2}{c^3} d\nu = \infty. \quad (1.14)$$

Classical physics thus predicted that a body at any temperature should radiate energy away at an infinite rate into the high-frequency modes. This result, which is clearly wrong, is known as the **ultraviolet catastrophe**.¹

The ultraviolet catastrophe is often presented as a dramatic embarrassment of classical physics, and rightly so. Yet it is worth observing why the classical calculation was so persuasive. It used Maxwell's theory, normal modes in a cavity, and the equipartition theorem of statistical mechanics. These were not speculative ingredients. They were among the most successful concepts available. The catastrophe therefore did not arise from a careless approximation, but from applying trusted principles beyond the domain in which nature permits them to remain valid.

¹The name is due to Ehrenfest (1911), who used it in a famous analysis of which features of the radiation problem genuinely require the quantum hypothesis.

1.1.3 Planck's Law and the Energy Quantum

The first formula that described the full spectrum of thermal radiation was announced by Max Planck on 19 October 1900, before the Berlin meeting of the German Physical Society. Guided by the newest measurements of Rubens and Kurlbaum, Planck proposed an interpolation between the two experimentally secure limits: Wien's exponential law at high frequency, and the linear Rayleigh behavior at low frequency. His interpolation,

$$\frac{du_\nu}{d\nu} = \frac{8\pi h\nu^3}{c^3} \frac{1}{\exp(\frac{h\nu}{k_B T}) - 1}, \quad (1.15)$$

reproduces Wien's law when $h\nu \gg k_B T$ and the linear law $8\pi k_B T \nu^2 / c^3$ when $h\nu \ll k_B T$. Rubens compared the formula with his measurements within days and found agreement everywhere. A formula guessed to fit the data had become exact; it remained to explain why.

Planck supplied the explanation on 14 December 1900, in what he later described as “an act of desperation.”² He returned to Boltzmann's combinatorial method, but with a decisive technical change. To distribute a total energy $E = P\epsilon$ among N material oscillators of frequency ν , the energy had to be divided into P indivisible *elements* ϵ ; only then is the number of complexions,

$$W = \frac{(N + P - 1)!}{N! (P - 1)!}, \quad (1.16)$$

finite, and only then does the entropy $S = k_B \log W$ have a maximum that determines the equilibrium distribution. With Stirling's approximation $\log n! \simeq n \log n - n$ for large n , and neglecting unity against N and P ,

$$S = k_B [(N + P) \log(N + P) - N \log N - P \log P]. \quad (1.17)$$

Thermal equilibrium at fixed oscillator number N and total energy $E = P\epsilon$ requires $\partial S / \partial E = 1/T$. Since $dE = \epsilon dP$,

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{1}{\epsilon} \frac{\partial S}{\partial P} = \frac{k_B}{\epsilon} \log \frac{N + P}{P}, \quad (1.18)$$

hence $(N + P)/P = \exp(\epsilon/k_B T)$, and the mean energy per oscillator is

$$\bar{E} = \frac{P\epsilon}{N} = \frac{\epsilon}{\exp(\epsilon/k_B T) - 1}. \quad (1.19)$$

Finally, consistency with Wien's displacement law — which, being thermodynamic, Planck trusted absolutely — requires the energy element to be proportional to the frequency. The argument is cited without proof:³

$$\epsilon = h\nu, \quad (1.20)$$

with $h = 6.55 \times 10^{-27}$ erg s, a new universal constant, called by Planck the *quantum of action*

²In a letter to R. W. Wood of 1931, Planck recalled the derivation as “an act of desperation,” undertaken because “a theoretical interpretation had to be found at any price, no matter how high.” The same letter makes clear how reluctant a revolutionary he was.

³Wien's scaling law requires the oscillator entropy to be a function of $\bar{E}/\nu$ alone; the entropy computed from the combinatorics above has this property if and only if $\epsilon \propto \nu$. The full derivation, which shows that the entropy $S = k_B [(\bar{E}/\epsilon + 1) \log(\bar{E}/\epsilon + 1) - (\bar{E}/\epsilon) \log(\bar{E}/\epsilon)]$ becomes a function of $\bar{E}/\nu$ only when $\epsilon \propto \nu$, is reproduced in standard texts on the old quantum theory. In this sense the relation $E = h\nu$ was not an additional guess but a consequence of demanding agreement with the one piece of radiation thermodynamics that was model-independent.

(Wirkungsquantum). The energy element is called *quantum* by Planck, which means in Latin *portion*. From the same fit Planck extracted $k_B \simeq 1.35 \times 10^{-16}$ erg/K, and therewith the best values then available for Avogadro's number and the elementary charge.

Derivation of Planck's Law

Precisely speaking, if one has a discrete admitted emission energy $\epsilon = h\nu$, so that the oscillator levels are $E_n = n\epsilon$, $n = 0, 1, 2, \dots$, then the partition function is geometric,

$$Z = \sum_{n=0}^{\infty} e^{-n\beta\epsilon} = \frac{1}{1 - e^{-\beta\epsilon}}, \quad \beta = \frac{1}{k_B T}, \quad (1.21)$$

and the average energy, by means of Boltzmannian statistical mechanics, is

$$\bar{E}_\nu = -\frac{d}{d\beta} \log Z = -\frac{d}{d\beta} \log \frac{1}{1 - e^{-\beta\epsilon}} = \frac{d}{d\beta} \log(1 - e^{-\beta\epsilon}) = \frac{\epsilon e^{-\beta\epsilon}}{1 - e^{-\beta\epsilon}} = \frac{\epsilon}{\exp(\beta\epsilon) - 1}. \quad (1.22)$$

(The modern quantum oscillator has levels $(n + \frac{1}{2})\epsilon$; the additive zero-point energy does not radiate and is irrelevant to the spectrum.)⁴ Substituting $\epsilon = h\nu$ into the mean energy and multiplying by the mode density $8\pi\nu^2/c^3$ derived in the Rayleigh–Jeans argument above gives the spectral energy density

$$\frac{du_\nu}{d\nu} = \frac{8\pi\nu^2}{c^3} \cdot \frac{h\nu}{\exp(h\nu/k_B T) - 1} = \frac{8\pi h\nu^3}{c^3} \frac{1}{\exp(\frac{h\nu}{k_B T}) - 1}, \quad (1.23)$$

which is precisely Planck's Law.⁵

In other words, the energy emitted by an oscillator was quantized. According to Planck, the quantum of energy E of a light quantum is proportional to its frequency ν .

Planck's law was the first quantum theory in physics, and Planck won the 1918 Nobel Prize in Physics “in recognition of the services he rendered to the advancement of Physics by his discovery of energy quanta”. At the time, however, Planck's view was that quantization was purely a heuristic mathematical construct, rather than (as is now believed) a fundamental change in our understanding of the world.

The advantage of Planck's proposal was immediate and quantitative: it gave the correct black-body spectrum across all frequencies, reducing to the Rayleigh–Jeans law at low frequency and avoiding the ultraviolet divergence at high frequency. Its limitation was equally important. Planck had quantized the energy exchange of oscillators, but he had not yet supplied a new mechanics of microscopic systems. The oscillator was still, in language and imagination, a classical oscillator with a strange rule attached. The quantum had entered physics as a successful restriction, not yet as a coherent kinematics.

⁴This is different from the modern perspective, which, similar to Rayleigh and Jeans' formulation, views the photons as the “harmonic oscillators”. Planck's quanta were properties of the material emitters; the quantization of the radiation field itself belongs to a later stage of the theory.

⁵Planck's original derivation passed through a detailed-balance argument between the emission and absorption of his oscillators; a fully quantum version of such a balance argument was given later by Einstein, and is reproduced in Section 1.1.6.

1.1.4 Einstein's Light Quantum and the Photoelectric Effect

Planck's hypothesis concerned the exchange of energy between matter and radiation. Einstein's analysis of 1905 — his only paper he himself called “revolutionary” — went further: the radiation field itself behaves as if its energy were carried in localized quanta. His argument was thermodynamic, not spectroscopic. In the regime where Wien's exponential law is valid, the entropy of monochromatic radiation of energy E enclosed in a volume V depends on the volume as

$$S - S_0 = k_B \frac{E}{h\nu} \log \frac{V}{V_0}, \quad (1.24)$$

which is exactly the volume dependence of the entropy of $n = E/h\nu$ mutually independent particles. Monochromatic radiation of low density therefore behaves, in Einstein's words, as if it “consisted of mutually independent energy quanta of magnitude $h\nu$.”

The heuristic value of the light quantum was displayed at once in the photoelectric effect. If light of frequency ν falls upon a metal surface, electrons are emitted only when the frequency exceeds a threshold. Increasing the intensity below threshold does not liberate electrons; increasing the frequency above threshold increases the kinetic energy of the emitted electrons. Lenard had established these facts experimentally in 1902, and they sit awkwardly with any wave theory, in which the energy delivered to an electron should accumulate with intensity and time. The central relation of Einstein's theory is

$$K_{\max} = h\nu - \phi, \quad (1.25)$$

where ϕ is the work function of the metal. The frequency fixes the size of the elementary energy transfer, while the intensity mainly affects the number of emitted electrons. Millikan's painstaking measurements of 1914–1916 — undertaken, by his own account, in the expectation of disproving Einstein's equation — instead confirmed it in every detail and yielded a value of h agreeing with Planck's radiation value to better than one percent. Einstein received the Nobel Prize for 1921 (awarded in 1922) “for his discovery of the law of the photoelectric effect” — significantly, not for the light quantum itself, which the committee still considered too speculative.

This was difficult to reconcile with the classical wave theory of light. In the classical picture, increasing the intensity should increase the energy delivered to the electron, and sufficiently intense light should eventually eject electrons even at low frequency. Experiment says otherwise. Thus quantization could no longer be regarded merely as a peculiar property of material oscillators in Planck's model. Light itself had acquired a particle-like aspect.

The lesson is not simply that light consists of small corpuscles in the old mechanical sense. Interference and diffraction remain wave phenomena. Rather, the photoelectric effect reveals an incompatibility inside the classical picture: the same radiation field displays wave-like propagation and particle-like exchange of energy. The old theory could name both aspects, but it could not yet describe them within a single kinematical framework.

Historically, this step was controversial precisely because the wave theory of light was so well confirmed. Einstein's proposal explained the data with striking simplicity, but it seemed to endanger the continuity of the electromagnetic field. Later experiments, especially Millikan's careful measure-

ments of the photoelectric effect and Compton's scattering of X-rays by electrons, made the light quantum increasingly difficult to dismiss. The photon concept had a great advantage: it linked energy transfer to frequency in a way that experiments directly displayed. Its limitation was that it sharpened the wave-particle problem without solving it. A particle picture explains localized exchange; a wave picture explains interference. Quantum theory would have to keep both facts without reducing either to the other in the old classical sense.

1.1.5 The Specific Heat of Solids

The quantum hypothesis might have remained a peculiarity of radiation had Einstein not shown, in 1907, that it also governs the thermal motion of matter. The classical law of Dulong and Petit (1819) assigns to every solid a molar heat capacity of about $3R$, corresponding by equipartition to $3N$ independent oscillators of three dimensions. Yet it had long been known that some substances, notably diamond, fall far short of this value, and that heat capacities generally decrease at low temperature — a decrease no classical argument can produce, since equipartition knows no temperature.

Einstein's model treated the N atoms of a solid as $3N$ independent harmonic oscillators, all of one frequency ν_E , quantized according to Planck's rule. With the mean oscillator energy $\bar{E}_\nu = h\nu_E/(\exp(h\nu_E/k_B T) - 1)$ from the preceding sections, the total internal energy is $U = 3N\bar{E}_\nu$. The heat capacity $C_V = \partial U/\partial T$ follows by differentiation. Setting $\Theta_E = h\nu_E/k_B$,

$$C_V = \frac{\partial}{\partial T} \left[3N \frac{h\nu_E}{\exp(h\nu_E/k_B T) - 1} \right] = 3Nk_B \frac{\partial}{\partial T} \left[\frac{\Theta_E}{\exp(\Theta_E/T) - 1} \right] \\ = 3Nk_B \Theta_E \left[-\frac{\exp(\Theta_E/T)}{(\exp(\Theta_E/T) - 1)^2} \right] \left(-\frac{\Theta_E}{T^2} \right) = 3Nk_B \left(\frac{\Theta_E}{T} \right)^2 \frac{e^{\Theta_E/T}}{(e^{\Theta_E/T} - 1)^2}, \quad (1.26)$$

with $\Theta_E = h\nu_E/k_B$. For $T \gg \Theta_E$ this reduces to the Dulong–Petit value $3Nk_B$; for $T \ll \Theta_E$ it falls exponentially toward zero, because an oscillator whose quantum exceeds $k_B T$ cannot be thermally excited at all. The quantum of action thus sets the scale at which degrees of freedom freeze out. (The observed low-temperature law is $C_V \propto T^3$, not exponential; Debye repaired this in 1912 by replacing the single frequency with the elastic spectrum of the solid, quantized by the same rule.)

The importance of this success can hardly be overstated. Nernst had just formulated his heat theorem (the future third law of thermodynamics), which required heat capacities to vanish at absolute zero, and the measurements of Nernst and Lindemann in 1910–1911 confirmed Einstein's curve in detail. Quantization was no longer a property of the exotic radiation field: it was visible in a block of copper cooled in liquid hydrogen. It was primarily this calorimetric evidence that persuaded Nernst to convene the first Solvay Congress of 1911 under the title “The Theory of Radiation and the Quanta,” at which the quantum hypothesis was for the first time treated as a central problem of physics rather than a Berlin eccentricity.

The advantage of the specific-heat theory was that it carried the quantum from radiation into ordinary matter and made it accessible to the thermometer as well as the spectroscope. Its limitation was the same as Planck's, now more keenly felt: the oscillators were still classical objects obeying classical mechanics in every respect except one unexplained restriction on their energies. Physics in 1911 possessed a quantum rule without a quantum mechanics.

1.1.6 Einstein's A and B Coefficients

In 1916–1917 Einstein returned to the radiation problem and gave what remains the most transparent derivation of Planck's law, and in doing so uncovered two features of the quantum world that no later theory has abandoned. Consider two stationary states of an atom, of energies $E_m < E_n$ and degeneracies g_m, g_n , in equilibrium with radiation of spectral density $\rho(\nu)$ at the Bohr frequency $h\nu = E_n - E_m$. Einstein postulated three elementary processes: *absorption*, occurring at a rate $B_m^n \rho(\nu)$ per atom in the lower state; *stimulated emission*, at a rate $B_n^m \rho(\nu)$ per atom in the upper state; and *spontaneous emission*, at a rate A_n^m independent of the radiation field. Thermal equilibrium requires

$$N_n (A_n^m + B_n^m \rho) = N_m B_m^n \rho, \quad \frac{N_n}{N_m} = \frac{g_n}{g_m} e^{-h\nu/k_B T}, \quad (1.27)$$

the second relation being Boltzmann's distribution. Substituting the Boltzmann factor for N_n/N_m into the detailed-balance equation,

$$\frac{g_n}{g_m} e^{-h\nu/k_B T} (A_n^m + B_n^m \rho) = B_m^n \rho, \quad (1.28)$$

and solving for the spectral density,

$$\rho(\nu) = \frac{(g_n/g_m) e^{-h\nu/k_B T} A_n^m}{B_m^n - (g_n/g_m) e^{-h\nu/k_B T} B_n^m} = \frac{g_n A_n^m}{g_m B_m^n e^{h\nu/k_B T} - g_n B_n^m}. \quad (1.29)$$

Two demands now fix everything. As $T \rightarrow \infty$ the exponential $e^{h\nu/k_B T} \rightarrow 1$, so the denominator tends to $g_m B_m^n - g_n B_n^m$. For the density to diverge in this limit (the classical Rayleigh–Jeans regime has no finite bound), the denominator must vanish:

$$g_m B_m^n = g_n B_n^m. \quad (1.30)$$

With this relation, $\rho(\nu) = (A_n^m/B_n^m)/(e^{h\nu/k_B T} - 1)$. Consistency with the classical Rayleigh–Jeans limit at high temperature fixes the ratio. Expanding $e^{h\nu/k_B T} \simeq 1 + h\nu/k_B T$,

$$\rho(\nu) \simeq \frac{A_n^m}{B_n^m} \frac{k_B T}{h\nu}, \quad (1.31)$$

which must equal the Rayleigh–Jeans spectral density $8\pi\nu^2 k_B T/c^3$. Hence

$$\frac{A_n^m}{B_n^m} = \frac{8\pi h\nu^3}{c^3}. \quad (1.32)$$

Substitution yields Planck's law,

$$\rho(\nu) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}, \quad (1.33)$$

now derived not from a model of oscillators but from the statistical equilibrium of the elementary processes themselves.

In the same investigation Einstein showed, by analyzing the momentum fluctuations of a molecule in the radiation field, that every elementary process transfers a momentum $h\nu/c$ in a definite direction: emission and absorption are not spherical waves but directed events. And because the spontaneous rate A_n^m governs only the probability of an emission, not its instant or its direction, the theory, in Einstein's own words, “leaves the duration and direction of the elementary processes to chance.” Statistical causality had entered fundamental physics not as an admission of ignorance but as a deduction from

experiment.

The advantage of the A - and B -coefficient theory was threefold: it derived Planck's law from first principles of detailed balance; it established the momentum of the light quantum years before Compton's experiment displayed it; and it introduced the stimulated emission on which every laser would later depend. Its limitation was that it presupposed precisely what the old quantum theory could not explain — the existence of discrete stationary states with definite transition probabilities — and it replaced the missing dynamics with the word “chance.” What that chance meant, and whether it could ever be reduced to an underlying determinism, became one of the defining questions of the new mechanics.

1.2 The Second Crisis: the Structure of the Atom and its Spectra

While the radiation problem disturbed the foundations of electromagnetism and thermodynamics, a second and independent crisis was maturing in the physics of matter. The discovery of the electron had shown that atoms have parts; Rutherford's scattering experiments showed that the parts are arranged with almost all the mass concentrated in a minute nucleus; and the whole structure, by every law of classical electrodynamics, ought to have collapsed in a fraction of a microsecond while radiating a continuous smear of light. Instead, atoms are stable, identical, and radiate at sharply discrete frequencies whose regularities were known with spectroscopic precision. The reconciliation of these facts was the work of Bohr, and its generalization the work of Sommerfeld; their common failure prepared the way for everything that follows.

1.2.1 The Electron and the Nuclear Atom

The first constituent of the atom was isolated by J. J. Thomson in 1897. By deflecting cathode rays in crossed electric and magnetic fields he showed that they consist of particles of a definite charge-to-mass ratio e/m , nearly two thousand times larger than that of the hydrogen ion, and identical whatever the cathode material or the residual gas. The electron was universal: every atom contains such particles, and chemistry itself became, in principle, a problem of their arrangement. Thomson's own “plum pudding” model distributed the electrons within a uniformly positively charged sphere; it accounted qualitatively for atomic size but made no sharp experimental predictions.

The model was destroyed by experiment. In 1909 Geiger and Marsden, working in Rutherford's laboratory, observed that alpha particles scattered from a thin gold foil were occasionally deflected through angles greater than 90° — about one particle in eight thousand. Rutherford's later recollection is deservedly famous: “It was quite the most incredible event that has ever happened to me in my life. It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you.” A diffuse positive sphere can bend a fast, massive alpha particle only by a fraction of a degree at each encounter; large single deflections require the entire positive charge and nearly the entire mass to be concentrated in a volume smaller than 10^{-14} m. In 1911 Rutherford published the

nuclear atom, with the hyperbolic-orbit cross section

$$\frac{d\sigma}{d\Omega} = \left(\frac{Ze^2}{4E} \right)^2 \frac{1}{\sin^4(\theta/2)}, \quad (1.34)$$

which Geiger and Marsden verified point by point.

The nuclear atom was mechanically triumphant and electrodynamically impossible. An electron circling a nucleus is an accelerated charge, and by Larmor's formula it radiates at the rate $P = 2e^2a^2/3c^3$. For a circular orbit of radius r , Newton's second law with the Coulomb force gives $mv^2/r = e^2/r^2$, whence $v^2 = e^2/mr$ and the centripetal acceleration is $a = v^2/r = e^2/mr^2$. The radiated power is therefore

$$P = \frac{2e^2}{3c^3} \left(\frac{e^2}{mr^2} \right)^2 = \frac{2e^6}{3m^2c^3r^4}. \quad (1.35)$$

The orbital energy is $E = -e^2/2r$, so $dE/dt = (e^2/2r^2) \dot{r}$. Equating the energy loss to the radiated power, $dE/dt = -P$,

$$\frac{e^2}{2r^2} \dot{r} = -\frac{2e^6}{3m^2c^3r^4} \implies \dot{r} = -\frac{4e^4}{3m^2c^3r^2}. \quad (1.36)$$

Separating variables and integrating from an initial radius a_0 down to zero,

$$\tau = \int_0^{a_0} \frac{3m^2c^3r^2}{4e^4} dr = \frac{3m^2c^3}{4e^4} \cdot \frac{a_0^3}{3} = \frac{m^2c^3a_0^3}{4e^4} = \frac{a_0^3}{4r_e^2c} \approx 1.6 \times 10^{-11} \text{ s}, \quad (1.37)$$

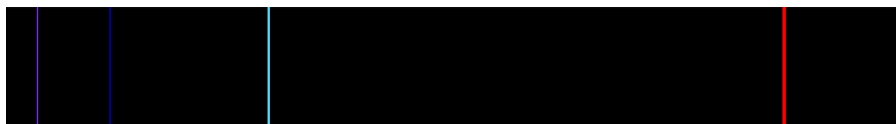
where $r_e = e^2/mc^2$ is the classical electron radius. The atom should shrink to a point in a hundredth of a microsecond, radiating a continuous spectrum as the orbit spirals inward. Nor does classical physics contain any parameter from which the observed size of atoms, $\sim 10^{-10}$ m, could be constructed: the only classical length associated with the electron is $r_e \sim 10^{-15}$ m. The stability, the size, and the spectrum of every atom in the universe were three unexplained gifts.

The advantage of Rutherford's model was that it reduced chemistry to a definite mechanical picture: a nucleus of charge $+Ze$ surrounded by Z electrons. Its limitation was that the picture, taken literally, refuted itself within 10^{-11} seconds. The second crisis was therefore a crisis not of missing data but of a contradiction: the atom that experiment demanded could not exist according to the theory that had discovered it.

1.2.2 The Regularities of Spectra

One of the most mysterious puzzles of contemporary physics is the structure of atoms and their emission spectra. When a gas is heated, it gives off light only at discrete frequencies. For example, the visible light given off by hydrogen consists of four different colors. The quantum hypothesis is apparently related to these phenomena, yet a thorough scrutiny of the hydrogen atom should be conducted before the mysteries of this radiation should be revealed. By the end of the nineteenth century, a sim-

Figure 1.1: Visible Light Emission of the Hydrogen Atom. Adrignola Merikanto (under CC0)



ple rule known as Balmer’s formula (1885) showed how the frequencies of the different lines related to each other, though without explaining why this was, or making any prediction about the intensities. The formula also predicted some additional spectral lines in ultraviolet and infrared light that had not been observed at the time. These lines were later observed experimentally, raising confidence in the value of the formula. Balmer’s formula states

$$\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{n^2} \right). \quad (1.38)$$

In the present era, the R for the atoms are called the Rydberg constant, which has the dimension $[L]^{-1}$; for hydrogen $R_H \simeq 1.097 \times 10^7 \text{ m}^{-1}$. Rydberg showed in 1890 that the spectra of other elements could be organized by similar differences of *terms*, and the Lyman, Paschen, Brackett, and Pfund series of hydrogen all take Balmer’s form with the integer 4 replaced by 1, 9, 16, 25, . . .

The decisive phenomenological principle was stated by Ritz in 1908. The **combination principle** asserts that every spectral line can be written as a difference of two spectral terms,

$$\nu_{mn} = T_m - T_n, \quad (1.39)$$

so that the frequency of any line is determined by a pair of atomic levels, and new lines can be predicted as combinations of known ones. Spectroscopy thereby possessed, years before any theory, an exact algebra of its own: the observable quantities were not orbits or motions but pairs of states. Twenty years later this algebra of pairs would reappear as the multiplication rule of matrices (Section 1.4.1).

The advantage of the term phenomenology was its completeness: a few dozen constants summarized thousands of lines, and the combination principle converted a chaos of frequencies into a table of levels. Its limitation was that the table cried out for a dynamical interpretation that classical physics could not supply. Why should an atom possess a discrete set of preferred energies at all?

1.2.3 Bohr’s Model of the Hydrogen Atom

In 1913, H. M. Hansen asked Niels Bohr about Balmer’s formula. Bohr recalled that “As soon as I saw Balmer’s formula, the whole thing was immediately clear to me.” In *On the Constitution of Atoms and Molecules*, he proposed a new model of the atom that included quantized electron orbits. This result is derived through the postulation of four hypotheses, listed as follows:

1. The quantization of angular momentum, i.e., (orbit) angular momentum takes on a discrete set of values: $L = mvr = n\hbar$. This implies the orbit of electrons is also discrete, instead of a continuum.
2. The correspondence principle. The properties of the hydrogen atom should reduce to the classical results as $n \rightarrow \infty$. As for a general system, this should hold as well insofar as all quantum numbers go to infinity.
3. When an atom makes a transition from one stationary state to another, it emits or absorbs radiation whose frequency is given by the frequency condition $\omega = \frac{\Delta E}{\hbar}$.
4. Apart from the angular momentum, any other generalized coordinate and momentum variables satisfy the condition (Bohr–Sommerfeld condition):

$$\oint_{\text{over the phase curve}} p dq = n_q \hbar. \quad (1.40)$$

Bohr's calculation uses only circular orbits. Let us use not only circular orbits but also ellipsic ones so as to show the quantization of both angular and radial variables. The application of Hamilton-Jacobi equations is natural.

The action can be written

$$S = -Et + L\theta + \oint p_r dr. \quad (1.41)$$

With $\partial S/\partial t = -E$, $\partial S/\partial \theta = L$, and $\partial S/\partial r = p_r$, the Hamilton-Jacobi equation becomes

$$-E + \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} - \frac{Ze^2}{r} = 0, \quad (1.42)$$

and solving for p_r gives

$$p_r = \sqrt{2mE - \frac{L^2}{r^2} + \frac{2mZe^2}{r}}. \quad (1.43)$$

In the case of a constrained solution, the energy E is typically negative. Therefore the equation above can be clarified as:

$$p_r = \sqrt{-2m|E| - \frac{L^2}{r^2} + \frac{2mZe^2}{r}}. \quad (1.44)$$

The radial action integral is evaluated by the substitution $u = 1/r$, which transforms the integrand into a form amenable to contour integration; the evaluation is cited without proof.⁶ The result is

$$\oint p_r dr = 2 \int_{r_{\min}}^{r_{\max}} \sqrt{-2m|E| - \frac{L^2}{r^2} + \frac{2mZe^2}{r}} dr = 2\pi \left(Ze^2 \sqrt{\frac{m}{2|E|}} - L \right) = n_r h. \quad (1.45)$$

Along with the quantization of angular momentum $L = n_\theta \hbar$ (from the angular action integral $\oint p_\theta d\theta = 2\pi L = n_\theta \hbar$), the radial condition gives

$$2\pi \left(Ze^2 \sqrt{\frac{m}{2|E|}} - n_\theta \hbar \right) = n_r h = 2\pi n_r \hbar, \quad (1.46)$$

hence $Ze^2 \sqrt{m/2|E|} = (n_r + n_\theta) \hbar =: n \hbar$. Squaring and rearranging,

$$\frac{Z^2 e^4 m}{2|E|} = n^2 \hbar^2, \quad |E| = \frac{Z^2 e^4 m}{2n^2 \hbar^2}, \quad (1.47)$$

so that

$$E = -|E| = -\frac{Z^2 e^4 m}{2(n_r + n_\theta)^2 \hbar^2} = -\frac{Z^2 e^4 m}{2n^2 \hbar^2}. \quad (1.48)$$

Therefore the energy level solely depends on the sum $n_r + n_\theta$, which we denote by n , and call by **principal quantum number**. A direct consequence of this result is the precise value of Rydberg's constant. By letting $n = 2$ and n , and thus by the emission frequency postulate one has

$$\frac{hc}{\lambda} = \frac{Z^2 e^4 m}{2\hbar^2} \left(\frac{1}{4} - \frac{1}{n^2} \right). \quad (1.49)$$

and so

$$\frac{1}{\lambda} = \frac{Z^2 e^4 m}{4\pi \hbar^3 c} \left(\frac{1}{4} - \frac{1}{n^2} \right), \quad R = \frac{Z^2 e^4 m}{4\pi \hbar^3 c}. \quad (1.50)$$

⁶Under $u = 1/r$, $dr = -du/u^2$, and the integral $\oint \sqrt{-A + B/r - C/r^2} dr$ with $A = 2m|E|$, $B = 2mZe^2$, $C = L^2$ becomes $2 \int_{u_{\min}}^{u_{\max}} \sqrt{-A + Bu - Cu^2} du/u^2$. Completing the square and evaluating by residues yields $2\pi(B/2\sqrt{A} - \sqrt{C})$. The contour integration is a standard exercise in advanced classical mechanics; the steps are reproduced in textbooks on the old quantum theory (e.g., Born, *The Mechanics of the Atom*, 1927).

435 The result matches the experimental value very accurately.

436 The confirmations came quickly. The formula describes not only hydrogen but every one-electron
437 ion; in particular the Pickering series, observed in stellar spectra in 1896 and long misattributed to
438 hydrogen, follows at once for $Z = 2$ and was thereby explained as ionized helium, with the small
439 finite-nuclear-mass correction furnishing one of the most precise agreements in all of spectroscopy.
440 In 1913–1914 Moseley measured the characteristic X-ray frequencies of the elements and found

$$441 \quad \sqrt{\nu} \propto (Z - 1), \quad (1.51)$$

442 exactly as Bohr's model with $n = 2 \rightarrow 1$ predicts; Moseley's law established the atomic number Z ,
443 rather than the atomic weight, as the true ordering principle of the periodic table, and predicted the
444 places of elements not yet discovered.

445 The advantage of Bohr's theory was not merely that it reproduced a known formula. It connected
446 the Rydberg constant with the electron mass, charge, Planck's constant, and the Coulomb force. It
447 also explained atomic stability in a provisional way: the electron did not spiral continuously into
448 the nucleus because only certain stationary orbits were permitted. In this respect the theory was a
449 remarkable bridge between classical mechanics and the quantum hypothesis.

450 But the bridge was visibly unfinished. The theory used classical orbits while forbidding most
451 of them. It asserted stationary states without deriving why these states do not radiate. It assigned
452 transition frequencies from energy differences, but did not describe the transition as a dynamical pro-
453 cess. It succeeded brilliantly for hydrogen-like atoms, yet its extension to more complicated atoms
454 required increasingly artificial rules. The theory therefore revealed both the power and the poverty of
455 quantizing classical motion by hand.

456 1.2.4 The Franck–Hertz Experiment

457 Bohr's stationary states had been inferred from spectroscopy; the Franck–Hertz experiment of
458 1914 exhibited them in a collision. Franck and Hertz accelerated electrons through mercury vapor
459 and measured the current reaching a collector. As the accelerating voltage is raised, the current rises
460 until 4.9 V, where it drops sharply; it rises again until 9.8 V, drops again, and so on periodically.
461 Electrons with less than 4.9 eV of kinetic energy cannot excite the mercury atom and pass through
462 it elastically; electrons with more lose precisely 4.9 eV in a single inelastic collision, and the vapor
463 simultaneously emits its ultraviolet resonance line at 253.7 nm, whose photon energy is

$$464 \quad h\nu = \frac{hc}{\lambda} = 4.9 \text{ eV}. \quad (1.52)$$

465 An atom accepts energy from an electron beam only in the same discrete portions in which it emits
466 light. (Franck and Hertz initially interpreted the 4.9 V threshold as an ionization potential; Bohr's read-
467 ing of it as an excitation between stationary states was the historically decisive one.) The experiment
468 earned Franck and Hertz the 1925 Nobel Prize.

469 The advantage of the Franck–Hertz experiment was its directness: stationary states were no longer
470 an inference from line spectra but a fact of collision physics, measurable with a voltmeter. Its limitation
471 was that it deepened the same mystery it confirmed. The atom evidently possessed discrete internal

states of definite energy, yet nothing in the theory explained either their stability or the rule by which one state changed into another.

1.2.5 The Bohr–Sommerfeld Theory

Bohr’s rules had been stated for the hydrogen atom. Their generalization to arbitrary separable mechanical systems was given independently by W. Wilson in 1915 and by A. Sommerfeld in 1915–1916: for every independent periodic degree of freedom of a multiply periodic system, the action integral over one cycle is an integer multiple of Planck’s constant,

$$\oint p_i dq_i = n_i h \quad (i = 1, 2, \dots, f). \quad (1.53)$$

The hydrogen calculation of the previous section is precisely this machinery applied to the Kepler problem, and the two integers n_r, n_θ are its first fruits. Sommerfeld’s systematic development, codified in his *Atombau und Spektrallinien*, made three great additions.

First, elliptical orbits. With two quantum numbers the electron possesses a family of ellipses for each principal quantum number $n = n_r + n_\theta$, and the energy’s dependence on the sum alone — the degeneracy — is a peculiarity of the pure $1/r$ potential, rooted in the conservation of the Runge–Lenz vector.

Second, the relativistic fine structure. The velocity of the inner electron is a fraction $Z\alpha$ of c , where

$$\alpha = \frac{e^2}{\hbar c} \simeq \frac{1}{137.04} \quad (1.54)$$

is Sommerfeld’s fine-structure constant, and the relativistic mass variation lifts the degeneracy:

$$E_{n,k} = -\frac{RhcZ^2}{n^2} \left[1 + \frac{\alpha^2 Z^2}{n} \left(\frac{1}{k} - \frac{3}{4n} \right) + \dots \right], \quad k = n_\theta = 1, 2, \dots, n. \quad (1.55)$$

Paschen’s precision measurements of the ionized-helium fine structure in 1916 agreed with this formula within experimental error, and the agreement was celebrated as the crown of the old quantum theory. Retrospectively the triumph was partly accidental: Sommerfeld’s classical-relativistic formula coincides, for reasons no one at the time could have suspected, with the exact result of Dirac’s relativistic electron theory, in which spin replaces the orbital quantum number in just the right way.

Third, space quantization. A third integer, m , fixes the projection of the angular momentum on an external axis, so that the orientation of an orbit in space is itself quantized. The direct experimental display of this quantization came with the Stern–Gerlach experiment of 1922, whose full significance, however, belongs to a later section (Section 1.5). Finally, Ehrenfest’s *adiabatic principle* (1916) supplied the theory with its internal anchor: the quantities that remain invariant under slow variation of the external parameters are precisely the action integrals, and hence precisely the quantities that may be quantized. The old quantum theory thereby knew, in a precise sense, which of its rules were consistent.

The advantage of the Bohr–Sommerfeld theory was generality: it provided a uniform machinery — classical mechanics plus action quantization plus correspondence arguments — applicable to atoms, X-ray levels, the Stark and Zeeman effects, and molecular bands, and for one-electron systems it was

quantitatively superb. Its limitation was that the machinery was a hybrid sewn onto classical dynamics rather than derived from anything deeper, and outside the separable, multiply periodic problems the seams split open at once.

1.2.6 The Correspondence Principle

The old quantum theory was not merely a list of numerical tricks. It was guided by a powerful requirement, the **correspondence principle**: quantum results must reproduce classical physics in the domain where classical physics is already known to work. For atomic systems this usually means the limit of large quantum numbers. In Bohr's language, the motion of an electron in a highly excited orbit should resemble the motion predicted by classical mechanics, and the radiation associated with transitions between nearby highly excited states should approach the classical orbital frequency.

The principle may be expressed schematically as follows. If n is large and the transition is between neighboring levels, then

$$\nu_{n+1,n} = \frac{E_{n+1} - E_n}{h} \quad (1.56)$$

should approach the characteristic frequency of the corresponding classical motion. The content of this statement is not that quantum theory is classically intuitive. Rather, it says that classical mechanics is a limiting description of a deeper theory. The old theory therefore had two obligations at once: it had to preserve classical mechanics where classical mechanics was successful, and it had to violate classical expectations where experiment required it.

Bohr and his school pressed the principle beyond frequencies into the territory the theory could not otherwise reach. Classically, the intensity and polarization of radiation are governed by the Fourier amplitudes of the orbital motion; by correspondence, the quantum transition probabilities between highly excited states should be governed by the same amplitudes, and selection rules should follow from the presence or absence of the relevant Fourier component. In Kramers's hands this program produced, in 1924–1925, a formula for the dispersion of light by atoms written entirely in terms of transitions between pairs of states — the Kramers–Heisenberg formula — in which the unobservable orbit had quietly been replaced by observable transition quantities. The bridge to the new mechanics was being built from the far side.

This principle remained essential in the transition to modern quantum mechanics. It helped determine which mathematical quantities in the new theory should correspond to position, momentum, energy, and angular momentum. But by itself it could not supply the new kinematics. It could say how the unknown theory must behave in a limit; it could not say what a quantum state is.

This is why the correspondence principle should be treated with care. Its advantage is methodological: it prevents quantum theory from becoming arbitrary by demanding contact with the classical world. It also preserves the genuine truth contained in classical mechanics. Its limitation is that it points backward as much as forward. It tells the new theory how to resemble the old theory when quantum numbers are large, but it does not explain what replaces an orbit, a trajectory, or a simultaneous assignment of position and momentum when quantum effects are essential.

1.2.7 The Failures of the Old Quantum Theory

The success of the Bohr–Sommerfeld theory should not be underestimated. It gave an astonishing account of the hydrogen spectrum and taught physicists that atomic stability and spectral discreteness were not accidental details to be repaired inside classical mechanics. Yet precisely because the theory was half classical and half quantum, its triumphs also revealed its limitations.

First, it retained the language of definite orbits while denying that all classical orbits were physically possible. The electron was still imagined as moving through a trajectory in space, but the permitted trajectories were selected by extra quantization rules. These rules were imposed from outside the classical equations; they were not consequences of the classical dynamics itself.

Second, the theory had no satisfactory account of quantum transitions. It could assign energies to stationary states and use the frequency condition

$$h\nu = E_i - E_f \quad (1.57)$$

to predict emitted or absorbed radiation. But it did not explain what occurs during the transition, why a transition takes place, or how to compute its probability from first principles. Einstein's coefficients had shown that the transitions are genuinely chancy; the old theory could register the chance but not derive it.

Third, the old theory gave only a partial account of spectral intensities and selection rules. It could often locate spectral lines, but it could not generally explain why some lines are bright, some are weak, and some are absent. The actual data of spectroscopy are not only frequencies; they include probabilities, intensities, degeneracies, and rules of allowed transformation.

Fourth, the old theory did not generalize cleanly. Hydrogen-like atoms were its great triumph, but already helium, the next atom, defeated it: van Vleck's careful Bohr–Sommerfeld calculation of 1922 gave an ionization potential in plain disagreement with experiment, and no arrangement of quantized classical orbits could be made to work. The anomalous Zeeman effect was worse still, forcing Landé to introduce half-integer quantum numbers that made no sense within the whole-number machinery. Multi-electron atoms, molecular spectra, and magnetic effects revealed the poverty of the orbit picture. The later discovery of spin sharpened this difficulty: a two-valued degree of freedom appears in experiments without being reducible to a little classical rotation of a charged body.

Finally, wave-particle duality remained unresolved. Planck and Einstein had shown that radiation exchanges energy in quanta; de Broglie suggested that matter has wave-like character; interference phenomena continued to demand wave behavior. The old theory possessed pieces of the truth, but not the grammar in which those pieces could be spoken together.

The old quantum theory thus ended not with a refutation but with an accumulation of structural failures, all pointing the same way: the unobservable scaffolding of orbits had to go, and the observable algebra of transitions, already visible in Ritz's combination principle and in the Kramers–Heisenberg formula, had to become the foundation. This is precisely the point from which the modern theory departed.

1.3 The Third Crisis: Waves, Particles, and Statistics

By 1923 the quantum of energy was an established experimental fact, but its nature was more obscure than ever. Light interfered like a wave and recoiled like a particle; electrons were detected as particles and yet, it would soon appear, diffracted like waves; and the statistics by which quanta were counted turned out to be the statistics of no classical objects at all. The third crisis was therefore ontological: it concerned what kind of thing a quantum is. Its resolution required abandoning not merely this or that classical equation, but the classical division of the world into particles and waves altogether.

1.3.1 The Compton Effect

The single most persuasive experiment for the light quantum was performed by A. H. Compton in 1923. Compton scattered monochromatic X-rays from graphite and measured the wavelength of the scattered radiation as a function of angle. He found, beside an unshifted line, a shifted line whose wavelength increased with the scattering angle by exactly the amount predicted if the X-ray quantum collides with a single quasifree electron as one relativistic particle with another.

The kinematics is elementary. Let the incident and scattered photons have energies $h\nu$, $h\nu'$ and momenta $h\nu/c$, $h\nu'/c$, and let the electron, initially at rest, recoil with momentum $\mathbf{p}$. Conservation of energy and momentum reads

$$h\nu + mc^2 = h\nu' + \sqrt{(pc)^2 + m^2c^4}, \quad \mathbf{p} = \frac{h\nu}{c} \hat{\mathbf{n}} - \frac{h\nu'}{c} \hat{\mathbf{n}}', \quad (1.58)$$

where θ is the angle between $\hat{\mathbf{n}}$ and $\hat{\mathbf{n}}'$. Eliminating the electron variables,

$$p^2 = \frac{h^2}{c^2} (\nu^2 + \nu'^2 - 2\nu\nu' \cos \theta), \quad [h(\nu - \nu') + mc^2]^2 = p^2c^2 + m^2c^4, \quad (1.59)$$

Expanding the square on the left of the second equation,

$$h^2(\nu - \nu')^2 + 2mc^2h(\nu - \nu') + m^2c^4 = p^2c^2 + m^2c^4, \quad (1.60)$$

cancelling m^2c^4 and substituting $p^2c^2 = h^2(\nu^2 + \nu'^2 - 2\nu\nu' \cos \theta)$ from the first equation,

$$h^2(\nu^2 - 2\nu\nu' + \nu'^2) + 2mc^2h(\nu - \nu') = h^2(\nu^2 + \nu'^2) - 2h^2\nu\nu' \cos \theta. \quad (1.61)$$

Cancelling $h^2(\nu^2 + \nu'^2)$ from both sides,

$$-2h^2\nu\nu' + 2mc^2h(\nu - \nu') = -2h^2\nu\nu' \cos \theta, \quad (1.62)$$

which simplifies to

$$mc^2h(\nu - \nu') = h^2\nu\nu'(1 - \cos \theta). \quad (1.63)$$

To convert from frequency to wavelength, substitute $\nu = c/\lambda$, $\nu' = c/\lambda'$, so that $\nu - \nu' = c(\lambda' - \lambda)/(\lambda\lambda')$ and $\nu\nu' = c^2/(\lambda\lambda')$. Then

$$mc^2h \cdot \frac{c(\lambda' - \lambda)}{\lambda\lambda'} = h^2 \cdot \frac{c^2}{\lambda\lambda'} (1 - \cos \theta), \quad (1.64)$$

and cancelling the common factor $c^2h/(\lambda\lambda')$ yields the Compton formula

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{mc} (1 - \cos \theta), \quad \frac{h}{m_e c} = 2.43 \times 10^{-12} \text{ m}. \quad (1.65)$$

The recoil electrons themselves were subsequently photographed in a cloud chamber by Compton and Simon, and their energies and directions were found to satisfy the same two-body kinematics. No wave theory of scattering can produce either the shift or the recoil.

The crisis can be measured by the remedies that serious physicists were willing to contemplate. In 1924 Bohr, Kramers, and Slater proposed to save the wave picture by sacrificing the exact conservation of energy and momentum in individual elementary processes, retaining them only statistically. Within a year the coincidence experiment of Bothe and Geiger had shown that in individual Compton events the scattered quantum and the recoil electron appear together, as strict conservation demands. Even the law of energy conservation had been put on trial, and the light quantum had survived it.

The advantage of the Compton effect was its finality: after 1923 the light quantum could no longer be described as a manner of speaking, for it possessed not only energy $h\nu$ but momentum $h\nu/c$, and collided accordingly. Its limitation was that it made the wave-particle contradiction completely unavoidable. The same X-rays whose particle collisions Compton measured had been produced and monochromatized by Bragg diffraction — a wave phenomenon — in the very same apparatus.

1.3.2 De Broglie's Matter Waves

The suggestion of de Broglie gave the old quantum theory one of its most suggestive interpretations. If light, formerly understood as a wave, also behaves as a particle in energy exchange, perhaps matter, formerly understood as particles, also possesses a wave-like character. De Broglie's argument of 1924 was guided by relativity: a particle of rest mass m_0 carries in its rest frame an internal oscillation of frequency m_0c^2/h , and the demand that this oscillation remain in phase with a wave of frequency E/h in every frame — the “harmony of phases” — fixes the wave's dispersion uniquely. The proposed relation

$$\lambda = \frac{h}{p} \quad (1.66)$$

turns Bohr's angular momentum rule into a standing-wave condition. For a circular orbit,

$$2\pi r = n\lambda \quad (1.67)$$

is equivalent, via $\lambda = h/p = h/mv$, to

$$2\pi r = n \frac{h}{mv} \implies mvr = n \frac{h}{2\pi} = n\hbar. \quad (1.68)$$

What had looked like an arbitrary restriction on orbits could now be read as the condition that the electron wave fit consistently around the orbit.

This was an immense conceptual advantage. Quantization no longer appeared merely as an imposed arithmetic rule; it resembled the discreteness of normal modes in acoustics or optics. But the limitation remained: if the electron is a wave, what physical quantity does the wave represent? It cannot be simply an ordinary material density, for electrons are detected as localized events. Nor can it be an ordinary classical probability distribution, because probabilities do not interfere with one another. The matter-wave idea therefore brought the theory closer to the modern language of amplitudes, but it did not by itself provide the interpretation of those amplitudes.

The experimental verdict was delivered by diffraction. Elsasser pointed out in 1925 that a crystal

grating should diffract slow electrons if de Broglie was right. In 1927 Davisson and Germer, studying the scattering of electrons from a nickel single crystal, found a pronounced reflection maximum at precisely the angle and energy predicted by $\lambda = h/p$ with the crystal's known spacing; independently, G. P. Thomson passed electrons through thin metallic foils and obtained the ring patterns characteristic of wave diffraction by a polycrystal.⁷ De Broglie received the 1929 Nobel Prize. Matter waves were no longer a hypothesis but a measurement.

1.3.3 Bose, Einstein, and the Statistics of Quanta

In June 1924 Einstein received from Dacca a short paper by an unknown lecturer, S. N. Bose, with a request for help in publication. Bose had derived Planck's law without any appeal to electromagnetism at all. Treat the radiation in a cavity as a gas of light quanta; divide the one-particle phase space into cells of volume h^3 , so that the frequency interval $(\nu, \nu + d\nu)$ contains

$$A_s = \frac{8\pi\nu_s^2 V}{c^3} d\nu_s \quad (1.69)$$

cells (the factor two being the two polarizations); and count, as Boltzmann taught, the number of ways of realizing each distribution of quanta among cells. Bose's decisive step was to count the cells as distinguishable but the quanta as indistinguishable: what matters is only how many quanta occupy each cell, never which ones. The number of complexions is then

$$W = \prod_s \frac{(A_s + N_s - 1)!}{N_s! (A_s - 1)!}, \quad (1.70)$$

and maximizing the entropy $k_B \log W$ at fixed total energy yields the equilibrium distribution. Applying Stirling's approximation $\log n! \simeq n \log n - n$ to each factor, and ignoring unity against large A_s, N_s ,

$$\log W \simeq \sum_s [(A_s + N_s) \log(A_s + N_s) - N_s \log N_s - A_s \log A_s]. \quad (1.71)$$

Maximizing $S = k_B \log W$ with respect to the occupation numbers N_s , subject to the constraint $\sum_s N_s h\nu_s = E$ (fixed total energy), introduces a single Lagrange multiplier β :

$$\frac{\partial}{\partial N_s} (\log W - \beta \sum_r N_r h\nu_r) = 0 \implies \log \frac{A_s + N_s}{N_s} - \beta h\nu_s = 0, \quad (1.72)$$

hence $(A_s + N_s)/N_s = e^{\beta h\nu_s}$, and the mean occupation per cell is

$$\frac{\bar{N}_s}{A_s} = \frac{1}{e^{\beta h\nu_s} - 1}. \quad (1.73)$$

The thermodynamic definition of temperature, $\partial S / \partial E = 1/T$, identifies $\beta = 1/k_B T$; the identification is cited without proof from equilibrium thermodynamics.⁸ The result is the Planck mean

⁷History here permitted itself a small irony: J. J. Thomson received the Nobel Prize for showing that the electron is a particle, and his son G. P. Thomson received it (with Davisson, 1937) for showing that the electron is a wave.

⁸In a rigorous treatment, $\beta = 1/k_B T$ follows from the definition $1/T = \partial S / \partial E$. The absence of a number-conservation Lagrange multiplier (chemical potential) reflects the fact that photons are freely created and absorbed at the cavity walls, so the photon number is not independently conserved.

occupation,

$$\frac{\bar{N}_s}{A_s} = \frac{1}{e^{h\nu_s/k_B T} - 1}, \quad (1.74)$$

which is Planck’s law. Einstein recognized the derivation’s importance at once, translated the paper himself, and saw it into print.

Einstein then did something Bose had not attempted: he applied the same counting to an ideal gas of material particles. The result was a new statistics — **Bose–Einstein statistics** — differing from Boltzmann’s at low temperature, where the gas exhibits a phase transition of an entirely new kind, the **Bose–Einstein condensation**, in which a finite fraction of the particles accumulates in the single lowest state.⁹ Two years later Fermi and Dirac completed the picture with the statistics obeyed by particles subject to Pauli’s exclusion principle. That integer-spin particles obey Bose–Einstein statistics and half-integer-spin particles Fermi–Dirac statistics — the spin-statistics theorem — was proved by Pauli in 1940 within relativistic quantum field theory.

The advantage of Bose’s counting was that it made the particle picture of light self-consistent for the first time: Planck’s law followed from quanta alone, without waves, provided the quanta were counted as indistinguishable. Its limitation was profound and pregnant: indistinguishability is not a property of any classical objects. Classical particles can always be labeled by their trajectories. The new statistics was an early, quantitative sign that a quantum is neither a classical particle nor a classical wave, but something for which classical language has no ready category — a sign that the later formalism would honor by representing states as vectors in a space where permutations act trivially (see the chapter on second quantization).

1.3.4 The Exclusion Principle and Electron Spin

The last discoveries of the old period were forced by the spectroscopy of many-electron atoms. Pauli’s analysis of the alkali spectra and the anomalous Zeeman effect convinced him, in 1925, that the electron requires a fourth quantum number capable of only two values — in his phrase, a “classically non-describable two-valuedness” — and that no two electrons in an atom may agree in all four quantum numbers. This **exclusion principle** explained at one stroke the closing of the electron shells and hence the structure of the periodic table, which the orbit theory had never been able to derive.

Later the same year Uhlenbeck and Goudsmit identified the two-valued degree of freedom with an intrinsic angular momentum: the electron carries a spin of magnitude $\hbar/2$ and a magnetic moment twice as large as classical electromagnetism would assign to a rotating charge.¹⁰ The factor of two — the anomalous gyromagnetic ratio — was precisely what the anomalous Zeeman patterns required, and Thomas’s relativistic precession of 1926 repaired the one remaining discrepancy in the fine structure. The Stern–Gerlach experiment of 1922, which had sent a beam of silver atoms through an inhomoge-

⁹The condensation was observed directly only in 1995, in dilute gases of alkali atoms at microkelvin temperatures, seventy years after its prediction.

¹⁰The idea had occurred earlier to Kronig, who discussed it with Pauli and was so thoroughly discouraged by him that he never published; Uhlenbeck and Goudsmit’s manuscript was nearly withdrawn for the same reason, and survived because Ehrenfest had already sent it to the journal.

neous magnetic field and observed two discrete spots instead of a continuous smear, could now be read as the direct display of the two spin orientations of the valence electron; its modern analysis occupies Section 1.5.

The advantage of the spin hypothesis was enormous: it closed the periodic table, explained the Zeeman patterns, and gave the two-valuedness a definite mechanical content. Its limitation was equally significant for everything that follows. Spin- $\frac{1}{2}$ is a physical observable taking exactly two values, with no continuous classical analogue; no model of a rotating body reproduces its properties, and the attempt to picture it classically fails quantitatively by the factor of two. Here, inside the atom itself, experiment had produced the simplest possible quantum system: two alternatives, and nothing in between.

1.4 The Birth of Modern Quantum Mechanics

By the middle of the 1920s, the old quantum theory had become both indispensable and unsatisfactory. Its successes showed that quantization was not a passing curiosity; its failures showed that quantization could not remain a set of external restrictions imposed on classical motion. The decisive change came when physicists stopped asking how to choose the allowed classical orbits and began asking what should replace the classical description of motion itself.

Two formulations appeared almost simultaneously. Heisenberg's matrix mechanics began from observable transition data and abandoned electron orbits. Schrodinger's wave mechanics began from de Broglie's matter waves and recast bound states as eigenvalue problems. At first these theories looked very different, almost opposed in spirit: one algebraic and discontinuous, the other analytic and wave-like. Their eventual equivalence revealed one of the deepest lessons of quantum mechanics: the representation is not the physical essence. A quantum state can be represented by a column of coefficients, a wave function, or another set of coordinates adapted to the experiment.

1.4.1 Heisenberg and the Abandonment of Orbits

Heisenberg's 1925 work began from a radical act of discipline. If electron orbits inside atoms are not observable, then perhaps the theory should not use them as fundamental objects. Spectroscopy gives transition frequencies and intensities. These quantities naturally carry two labels: an initial stationary state and a final stationary state. Instead of a classical quantity $x(t)$ along an orbit, one considers an array of transition quantities

$$x_{mn}, \tag{1.75}$$

where m and n label stationary states.

The time dependence associated with a transition is governed by Bohr's frequency condition:

$$x_{mn}(t) = x_{mn} \exp(2\pi i \nu_{mn} t), \quad h\nu_{mn} = E_m - E_n. \tag{1.76}$$

The multiplication rule for such quantities is not arbitrary. It is inherited from the way Fourier components combine in classical mechanics, but with a decisive change of labels. For a classical periodic

motion one may write

$$x(t) = \sum_{\alpha} x_{\alpha} \exp(2\pi i \nu_{\alpha} t), \quad y(t) = \sum_{\beta} y_{\beta} \exp(2\pi i \nu_{\beta} t). \quad (1.77)$$

The product $x(t)y(t)$ has components obtained by convolution, because frequencies add:

$$(xy)_{\gamma} = \sum_{\alpha+\beta=\gamma} x_{\alpha} y_{\beta}. \quad (1.78)$$

In spectroscopy, however, the relevant frequency is not attached to a single orbit harmonic but to a pair of energy levels. The Ritz combination principle gives

$$\nu_{mk} + \nu_{kn} = \frac{E_m - E_k}{h} + \frac{E_k - E_n}{h} = \nu_{mn}. \quad (1.79)$$

Thus the analogue of adding Fourier frequencies is composing transitions through an intermediate state. If a transition from m to n can pass through intermediate states k , then the product of two quantities has components

$$(AB)_{mn} = \sum_k A_{mk} B_{kn}. \quad (1.80)$$

Born recognized this rule as matrix multiplication. The noncommutativity is now easy to understand: AB sums over paths in which B acts first and A acts second, while BA sums over the reversed order. Since the intermediate amplitudes need not be the same, the crucial consequence is that, in general,

$$AB \neq BA. \quad (1.81)$$

Thus noncommutativity entered quantum mechanics not as an ornament of later formalism, but as a structural replacement for classical kinematics.

Born and Jordan, and soon Dirac, developed this into a mechanics of noncommuting quantities. The canonical variables satisfy

$$QP - PQ = i\hbar I. \quad (1.82)$$

Historically, this relation was also guided by the correspondence principle. In classical Hamiltonian mechanics, the Poisson bracket of position and momentum is

$$\{q, p\} = 1. \quad (1.83)$$

Dirac observed that the quantum commutator plays the role of the Poisson bracket multiplied by $i\hbar$:

$$\{A, B\} \longrightarrow \frac{1}{i\hbar}[A, B], \quad [A, B] = AB - BA. \quad (1.84)$$

Applied to q and p , this rule gives $[Q, P] = i\hbar I$. This is not a proof in the Euclidean sense; it is a correspondence deduction, tying the new algebra to the old mechanics in the limit where actions are large compared with $\hbar$. The relation has no analogue in classical multiplication, where position and momentum are ordinary functions on phase space and therefore commute. It says that the order of operations matters. The old orbit in phase space is replaced by an algebra of possible transformations and measurements.

The new mechanics proved itself almost immediately. In January 1926 Pauli solved the hydrogen atom purely within matrix mechanics, exploiting the Runge–Lenz vector as a matrix constant of motion, and recovered the Balmer spectrum without a single wave or differential equation. The austere algebra could do everything the orbit picture could do, and it could do things the orbit picture could

not.

The advantage of matrix mechanics was its faithfulness to spectroscopic data. It dealt directly with transition frequencies, intensities, and selection rules, precisely where the old theory had been weakest. It also naturally described discrete spectra. Its limitation was its initial opacity. It did not offer an intuitive picture of an electron moving in space, nor did it immediately explain why a familiar-looking differential equation should emerge for particles in potentials. It was a powerful algebraic grammar, but at first it seemed physically austere.

1.4.2 Schrodinger's Wave Mechanics

Schrodinger's route was quite different. Inspired by de Broglie's matter waves and by Hamilton's optical analogy, he sought an equation whose stationary solutions would give the allowed states of an electron. Hamilton had already observed in the 1830s that the equations of mechanics in action form are formally identical to the equations of geometrical optics in eikonal form: a system of trajectories resembles a system of rays. If matter possesses a wave nature, then classical mechanics should stand to some deeper wave mechanics as geometrical optics stands to wave optics — valid when the wavelength is negligible, and powerless to explain diffraction. Schrodinger's program was to write down the wave equation of which classical mechanics is the short-wavelength limit. In the nonrelativistic case, the central equation became

$$i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi. \quad (1.85)$$

One quick deduction of its form begins from a free matter wave

$$\psi(x, t) = \exp \left[\frac{i}{\hbar} (p \cdot x - Et) \right]. \quad (1.86)$$

For this wave,

$$-i\hbar \nabla \psi = p\psi, \quad i\hbar \frac{\partial \psi}{\partial t} = E\psi. \quad (1.87)$$

The classical nonrelativistic energy relation is

$$E = \frac{p^2}{2m} + V(x). \quad (1.88)$$

Replacing E and p by the differential operations that reproduce their eigenvalues on plane waves,

$$E \mapsto i\hbar \frac{\partial}{\partial t}, \quad p \mapsto -i\hbar \nabla, \quad (1.89)$$

gives Schrodinger's equation. Again, this is not a derivation from classical mechanics alone. It is a controlled guess: de Broglie's wave relations supply the phase, the classical energy relation supplies the dispersion, and the correspondence principle tells us how to translate energy and momentum into operations on amplitudes.

For states of definite energy, one writes

$$\psi(x, t) = \phi(x) \exp \left(-\frac{iEt}{\hbar} \right), \quad (1.90)$$

and obtains the time-independent equation

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + V\right)\phi = E\phi. \quad (1.91)$$

The allowed energies are therefore eigenvalues of a differential operator.

This was mathematically and physically illuminating. The discreteness of bound-state energies no longer had to be imposed through Bohr-Sommerfeld integrals; it could arise from boundary conditions on wave functions. A bound electron resembles a standing wave, and the hydrogen atom can be treated without literal electron orbits. In a celebrated series of four communications of 1926, *Quantisierung als Eigenwertproblem*, Schrodinger solved the hydrogen atom, the harmonic oscillator, and the rigid rotor, developed perturbation theory, and computed the Stark effect. In this sense wave mechanics preserved something intuitive from classical physics: differential equations, spatial functions, and analogies with optics.

Yet the same intuitive strength concealed a danger. Schrodinger initially hoped that the wave might be a real charge distribution. But localized detections, scattering events, and the individuality of particles resisted this interpretation. A wave spread through space does not look like a point-like detection on a screen. Thus wave mechanics had a clear mathematical form before it had a clear physical interpretation.

1.4.3 Born's Probability Interpretation

Born's interpretation supplied the missing bridge. In his scattering papers of 1926, Born argued that the wave function of a scattered particle should not be read as a material density but as a probability amplitude: the wave determines not where the particle is, but where it is likely to be found.¹¹ For a normalized wave function, the probability of finding the particle in a spatial region Ω is

$$\int_{\mathbb{R}^3} |\psi(x)|^2 d^3x = 1, \quad P(\Omega) = \int_{\Omega} |\psi(x)|^2 d^3x. \quad (1.92)$$

Thus ψ itself is not a directly observable quantity. Its modulus squared gives probabilities, while its phase can influence later probabilities through interference.

The compatibility of this interpretation with Schrodinger's equation can be checked directly. Let

$$\rho(x, t) = |\psi(x, t)|^2. \quad (1.93)$$

Using Schrodinger's equation and its complex conjugate, one obtains the continuity equation

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= \psi^* \frac{\partial \psi}{\partial t} + \frac{\partial \psi^*}{\partial t} \psi = \psi^* \left(\frac{i\hbar}{2m} \nabla^2 \psi - \frac{i}{\hbar} V \psi \right) + \left(-\frac{i\hbar}{2m} \nabla^2 \psi^* + \frac{i}{\hbar} V \psi^* \right) \psi \\ &= \frac{i\hbar}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = -\nabla \cdot \left[\frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*) \right] =: -\nabla \cdot \mathbf{j}, \end{aligned} \quad (1.94)$$

where the third equality uses $\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* = \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*)$, giving the probability current

$$\mathbf{j} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*). \quad (1.95)$$

¹¹The interpretation appeared first in a footnote of Born's preliminary communication, in which the probability of a scattering direction is taken proportional to the squared modulus of the corresponding wave amplitude. Born had to wait until 1954 for the Nobel Prize awarded for this step.

Integrating over all space, and assuming the wave function vanishes sufficiently fast at infinity, gives

$$\frac{d}{dt} \int_{\mathbb{R}^3} |\psi(x, t)|^2 d^3x = 0. \quad (1.96)$$

Thus the squared norm is conserved by the dynamics. This is a major advantage of the wave equation as a probability-amplitude equation: it preserves the total probability without an additional postulate.

This interpretation transformed the meaning of the wave equation. The wave function is not a classical wave of matter in the same sense that a water wave is a wave of water. It is a complex amplitude encoding the statistical predictions of a preparation. The advantage of Born's rule is that it explains why wave mechanics can produce localized experimental outcomes while retaining interference. Its limitation is conceptual: it makes probability fundamental in a way foreign to classical mechanics. The theory predicts distributions of outcomes, not hidden trajectories of the old kind. The "chance" that Einstein had been forced to admit in his theory of radiation had become irreducible.

1.4.4 The Equivalence of the Two Pictures

The relation between matrix mechanics and wave mechanics was clarified by Schrodinger, Dirac, and others. A wave function can be expanded in a basis of stationary states:

$$\psi(x) = \sum_n c_n \phi_n(x). \quad (1.97)$$

If the basis is orthonormal, then

$$\int \phi_m^*(x) \phi_n(x) d^3x = \delta_{mn}, \quad c_n = \int \phi_n^*(x) \psi(x) d^3x. \quad (1.98)$$

In this representation, the state is described by the coefficients c_n . Operators acting on wave functions acquire matrix elements

$$A_{mn} = \int \phi_m^*(x) A \phi_n(x) d^3x. \quad (1.99)$$

Indeed, if

$$(A\psi)(x) = \sum_m d_m \phi_m(x), \quad (1.100)$$

then

$$d_m = \sum_n A_{mn} c_n. \quad (1.101)$$

The differential operator has become a matrix acting on the coefficient column. In the same way, Schrodinger's equation becomes

$$i\hbar \dot{c}_m = \sum_n H_{mn} c_n. \quad (1.102)$$

Thus the matrix and wave pictures are not rival descriptions of different worlds. They are different representations of the same abstract structure.

This equivalence is historically decisive for the present notes. It suggests that the physical state should not be identified too quickly with any one representation, whether an orbit, a matrix column, or a wave function in space. What is invariant is the network of transition amplitudes, probabilities, and transformations. The language of Hilbert space is the mathematical distillation of this insight: states

are vectors up to phase, measurable alternatives correspond to decompositions into basis vectors, and physical transformations act linearly on amplitudes.

The modern theory therefore did not arise by simply choosing a more complicated equation of motion. It arose by replacing the classical kinematics of points in phase space with a kinematics of states, amplitudes, and noncommuting transformations. It remained to give this one theory its final, representation-independent form.

1.4.5 Dirac, Jordan, and von Neumann: the General Formalism

The synthesis was completed in a remarkably short time. Dirac and Jordan, independently in 1926–1927, formulated the **transformation theory**: the physical content of the theory lies in the transition amplitudes between the eigenstates of arbitrary observables, and the wave function and the matrix column are merely two coordinate systems for one and the same abstract vector. To handle observables with continuous spectra, Dirac introduced the δ function; his *Principles of Quantum Mechanics* of 1930 presented the whole theory in the language of kets, bras, and operators that this note also uses.¹²

Von Neumann then gave the theory its rigorous mathematical foundation. In papers of 1927 and definitively in his *Mathematische Grundlagen der Quantenmechanik* of 1932, he identified the arena of quantum states as the abstract Hilbert space, observables as self-adjoint operators upon it, measurable spectra as their spectra in the sense of functional analysis, and statistical predictions as the trace rule $\langle A \rangle = \text{tr}(\rho A)$. In the same years Heisenberg’s uncertainty relation (1927) exhibited the quantitative limit on the joint determination of canonically conjugate quantities, and Bohr’s doctrine of complementarity, announced at Como in September 1927, gave the conceptual frame in which mutually exclusive experimental arrangements complete rather than contradict one another; both themes are developed systematically in the chapter on measurement. Wigner and Weyl brought the theory of group representations into quantum mechanics, preparing the modern understanding of symmetry that a later chapter will develop. The fifth Solvay Congress of 1927, devoted to “Electrons and Photons,” is conventionally taken as the closing scene of the founding period and the opening of the Einstein–Bohr debate on its meaning.

The advantage of the Dirac–Jordan–von Neumann formulation was its completeness and its generality: one theory, representation-independent, mathematically precise, embracing the matrix and wave pictures as special cases and finite-dimensional spin systems and infinite-dimensional scattering systems alike. Its limitation was of a different order from all the limitations recorded above. The formalism was finished before its interpretation was settled; what a measurement is, and what the state vector knows, remained — and in part remains — an open question. But the question could now be stated precisely, and these notes will take it up with the same precision.

¹²Dirac’s bra-ket notation itself was introduced in a paper of 1939; the 1930 book still wrote the inner product in the older form (ψ, ϕ) .

1.4.6 Why a New Kinematics Was Needed

Looking back over the three crises, one can locate the exact point at which classical physics broke. It was not a force law that failed, nor any particular equation of motion. What failed was the concept of *state* itself, and with it the concept of a measurable quantity. A failure of this kind is called **kinematical**, in distinction from a dynamical one: kinematics fixes what kind of object a state is and what questions may be asked of it, while dynamics merely says how the state evolves in time. The kinematics of classical mechanics is quickly stated. The state of a particle is a point of phase space — a simultaneous assignment of position and momentum; an observable is a function on phase space; and the result of a measurement is, in principle, nothing but the value of that function, possessed by the system before and after the measurement alike.

Against this scheme, each crisis of the old quantum theory contributed one precise objection, and the modern theory may be read as the minimal kinematics answering all of them at once.

First, the radiation crisis and the specific heats showed that the energies available to a bounded microscopic system may form a discrete set. The state of such a system is therefore not an arbitrary point of a continuum: it is one of a discrete list of experimentally distinguishable alternatives — the stationary states of Bohr, exhibited directly in the collisions of Franck and Hertz.

Second, the combination principle of Ritz showed that the observable data of atomic physics are attached not to single states but to *pairs* of states: a spectral line lives on a transition. Physical quantities are therefore not numbers possessed by a state, but relations between two states — the form in which Heisenberg received them when he built matrix mechanics.

Third, the interference of light and of matter showed that alternatives do not combine as probabilities do. Two open paths can produce fewer counts at a detector than one. The quantities that add must therefore be capable of cancelling: they are complex *amplitudes*, and the probabilities are their squared moduli. Born's rule, read backward from experiment, is the statement that nature composes alternatives by amplitudes and interrogates them by probabilities.

Fourth, the commutator $[Q, P] = i\hbar I$ showed that the order of physical operations matters. If two observables multiplied in different orders give different results, then no state can assign sharp pre-existing values to both, and the classical picture of the state as a complete catalogue of properties is untenable in principle rather than merely in practice.

Fifth, the equivalence of the matrix and wave mechanics showed that the state itself is none of its representations: not the orbit, not the wave function, not the column of coefficients, but that which is common to all of them — an abstract object to which every experimental basis assigns its own list of amplitudes.

A kinematics meeting these five requirements must therefore provide: for every experiment a set of mutually exclusive alternatives; complex amplitudes for the transitions between preparations and outcomes; probabilities as squared moduli of amplitudes; a noncommutative composition of operations; and a notion of state independent of the basis in which it is described. This is precisely the Hilbert-space kinematics of Dirac and von Neumann, whose systematic construction occupies the next chapter. Before that construction, it is worth seeing the five requirements at work in the simplest

possible arena, where each of them can be traced back to a definite experiment.

A Synoptic Chronology

For convenience, the principal events of the chapter are collected below.

Table 1.1: A synoptic chronology of the developments described in this chapter.

Year	Event
1801–03	Young: the interference of light
1809	Malus: polarization by reflection; the cosine-squared law
1859–60	Kirchhoff: universality of the black-body spectrum
1879–84	Stefan’s law; Boltzmann’s thermodynamic derivation
1885–90	Balmer’s formula; Rydberg’s term analysis
1891–92	Zehnder and Mach: the interferometer
1893–96	Wien’s displacement law and exponential spectrum
1897	Thomson: the electron
1900	Rubens–Kurlbaum; Planck’s formula and the quantum of action
1905	Einstein: the light quantum; Jeans completes Rayleigh’s law
1907	Einstein: the specific heat of solids
1908	Ritz’s combination principle
1909	Taylor: interference at feeble light
1909–11	Geiger–Marsden scattering; Rutherford’s nucleus; first Solvay Congress
1913–14	Bohr’s atom; Moseley’s law; Franck–Hertz experiment
1915–16	Wilson–Sommerfeld quantization; relativistic fine structure
1916–17	Einstein’s A and B coefficients
1922	Stern–Gerlach experiment
1923	Compton effect
1924	Bose’s counting; de Broglie’s thesis; Bohr–Kramers–Slater theory
1925	Bothe–Geiger refutes BKS; Pauli exclusion; electron spin; Heisenberg’s mechanics; Born–Jordan and Dirac complete it
1926	Schrodinger’s four papers; Born’s probability rule; equivalence; Fermi–Dirac statistics
1927	Davisson–Germer and G. P. Thomson; uncertainty; complementarity; Dirac–Jordan transformation theory; fifth Solvay Congress
1930–32	Dirac’s <i>Principles</i> ; von Neumann’s <i>Foundations</i>

1.5 Two-State Experiments and the Language of Amplitudes

The kinematics sketched above was distilled by its founders from the whole of atomic physics, but its content is already fully visible in experiments so simple that the apparatus can return only one of two answers: this detector or that one, transmitted or absorbed, upper spot or lower spot. A system interrogated by such an experiment is called, relative to that experiment, a **two-state system**. Two-state systems are not a modern invention. Three families of them have already crossed this narrative: interference between two optical paths, known since Young’s double-slit experiments and sharpened into

the interferometers of Zehnder and Mach in the 1890s; polarization, whose cosine-squared law Malus discovered in 1809; and the two spots of Stern and Gerlach's silver beam of 1922. After 1927 these old devices were revisited with single quanta, one at a time, and each was found to exhibit precisely the requirements of the preceding section: alternatives that combine by amplitudes, probabilities that depend on the basis of the question, measurements that prepare rather than merely reveal.

The notation in which these experiments are now described is Dirac's, devised for the transformation theory. An experimentally distinguishable alternative is written as a ket, as in $|u\rangle$ or $|+z\rangle$; the superposition of alternatives is written as a linear combination of kets; the amplitude for a system prepared in the alternative $|\psi\rangle$ to be found in the alternative $|\phi\rangle$ is written $\langle\phi|\psi\rangle$; and the probability of that transition is $|\langle\phi|\psi\rangle|^2$. These conventions merely transcribe Born's rule and the superposition of amplitudes into a compact script. The experiments of this section show why no simpler script is sufficient. The advantage of beginning with two-state systems is that every essential novelty appears without the analytic machinery of wave equations on continuous space; the limitation is equally clear, for real particles move in space, carry momentum, and may possess infinitely many alternatives. The two-state examples show the grammar in its simplest sentence, not the whole language.

1.5.1 The Interference of Single Quanta

Interference was, historically, the fortress of the wave theory of light: the fringes of Young's double-slit experiments of 1801–1803 were its founding evidence. The fortress was tested at its weakest point by G. I. Taylor in 1909. Photographing the shadow of a needle in light so feeble that, on the quantum hypothesis, only one light quantum at a time could be present in the apparatus, he found after exposures lasting up to three months that the interference fringes were entirely unchanged. The energy arrives in indivisible portions; the pattern is unaffected. Dirac later compressed the paradox into a sentence: each photon “interferes only with itself.” Interference is therefore not an interaction among many quanta, but a property of the alternatives open to each one separately.

The cleanest device in which to see this is the Mach–Zehnder interferometer, introduced by Zehnder and Mach in 1891–1892 in the classical tradition of interferometry. In its usual optical form, a source emits light toward a first beam splitter. Part of the beam is transmitted and part is reflected, so that two spatially separated arms are produced. Mirrors redirect the two arms toward a second beam splitter, where they are recombined and sent toward two detectors.

The important feature is that the two alternatives are spatially distinct between the two beam splitters, but the final detection probabilities depend on recombining them. If one blocks either arm, the remaining beam reaches the detectors without interference. If both arms are open, the two possibilities must be considered together. Let $|u\rangle$ and $|l\rangle$ denote the upper and lower paths. A first beam splitter prepares a superposition of paths. In one common convention,

$$|u\rangle \mapsto \frac{1}{\sqrt{2}} (|u\rangle + |l\rangle). \quad (1.103)$$

If a phase shifter is placed in the lower arm, the state becomes

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|u\rangle + e^{i\varphi} |l\rangle). \quad (1.104)$$

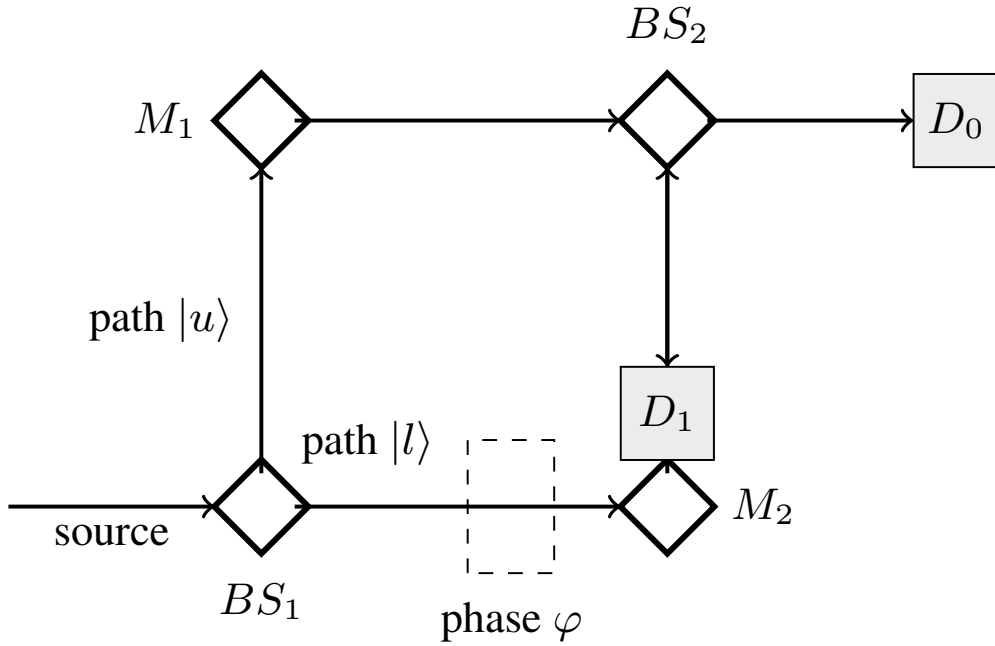


Figure 1.2: A schematic Mach–Zehnder interferometer. The first beam splitter creates two path alternatives, a phase shifter changes the relative phase, and the second beam splitter recombines the alternatives before detection.

The phase φ does not by itself represent a classical probability. The probabilities of finding the particle in the two arms remain equal before the second beam splitter. Nevertheless, after the paths are recombined, this relative phase changes the final detection probabilities. This is the experimental heart of the device: a phase difference accumulated in one arm, though invisible as an immediate path probability, becomes visible as a redistribution of counts between the two detectors. A second, identical beam splitter recombines the paths. In the convention where each beam splitter acts as $|u\rangle \mapsto (|0\rangle + |1\rangle)/\sqrt{2}$ and $|l\rangle \mapsto (|0\rangle - |1\rangle)/\sqrt{2}$ on the output-mode basis $\{|0\rangle, |1\rangle\}$, the state after recombination is

$$|\psi_{\text{out}}\rangle = \frac{1}{2} \left[(1 + e^{i\varphi}) |0\rangle + (1 - e^{i\varphi}) |1\rangle \right], \quad (1.105)$$

and the detection probabilities are

$$P_0 = |1 + e^{i\varphi}|^2/4 = (2 + 2 \cos \varphi)/4 = \cos^2 \frac{\varphi}{2}, \quad P_1 = |1 - e^{i\varphi}|^2/4 = (2 - 2 \cos \varphi)/4 = \sin^2 \frac{\varphi}{2}. \quad (1.106)$$

The detector statistics depend on a phase difference between alternatives.

This is the cleanest reason ordinary probability is insufficient. If the particle had merely chosen the upper or lower arm with probability $\frac{1}{2}$, the final probability would be an ordinary weighted sum. Changing a phase in one arm would not continuously move counts from one detector to the other. But the experiment requires amplitudes associated with alternatives to be added before the squared modulus is taken. Interference is therefore not a decorative wave effect; it is a rule for composing alternatives.

The same apparatus also clarifies the role of which-path information. If a detector is placed in one arm, or if the two arms are marked in principle by some distinguishable physical trace, the

interference disappears. The alternatives then behave more like exclusive classical alternatives. Thus the experiment does not say simply that a particle “really went both ways” in an ordinary spatial sense. Rather, it says that when the experimental arrangement does not distinguish the alternatives, the theory must combine their amplitudes before assigning probabilities to final outcomes.

The interferometer is advantageous because it separates two ideas that are often confused: probability and phase. A relative phase can change future probabilities even when it does not change the immediate probabilities for the two paths. The limitation is that the path labels $|u\rangle$ and $|l\rangle$ may tempt one to imagine that the particle really follows one path while the other path is merely a calculational fiction. The interference pattern is precisely the warning against that simplification: the alternatives must be treated together until the experimental arrangement makes them distinguishable.

1.5.2 Polarization as a Two-State System

Polarization is the oldest two-state phenomenon in physics. In 1809 Malus, examining light from a window reflected in a pane of glass through a crystal of Iceland spar, discovered polarization by reflection, together with the law that bears his name: an analyzer transmits the fraction $\cos^2 \theta$ of an incident polarized beam, θ being the angle between the orientations of polarizer and analyzer. Classical wave optics explains the law by the projection of the field vector. For single quanta the situation is sharper: a photon meeting a polarizer has exactly two outcomes, transmitted or absorbed, and Malus’ law survives only as a statement of probabilities.

Let $|H\rangle$ and $|V\rangle$ denote horizontal and vertical polarization. These are perfectly distinguishable alternatives for an apparatus that measures polarization in the horizontal-vertical basis. A photon can nevertheless be prepared in a diagonal polarization state

$$|D\rangle = \frac{1}{\sqrt{2}} (|H\rangle + |V\rangle), \quad (1.107)$$

or in the anti-diagonal state

$$|A\rangle = \frac{1}{\sqrt{2}} (|H\rangle - |V\rangle). \quad (1.108)$$

If a photon in state $|D\rangle$ is measured by a horizontal-vertical polarizing apparatus, the two outcomes occur with equal probability:

$$P(H|D) = |\langle H|D\rangle|^2 = \frac{1}{2}, \quad P(V|D) = |\langle V|D\rangle|^2 = \frac{1}{2}. \quad (1.109)$$

But it would be misleading to say that the photon secretly had either horizontal or vertical polarization before the measurement. The same state $|D\rangle$ is a definite state for a diagonal polarizer. In the diagonal basis it is not a mixture of alternatives; it is one of the basis states.

This is already a departure from the classical notion of state. The question “what polarization does the photon have?” is incomplete until one specifies the experimental basis in which the question is asked. Different bases correspond to different decompositions of the same state.

Nothing of this is special to light. Relative to any experiment with two distinguishable outcomes, which we may call $|0\rangle$ and $|1\rangle$ without loss of generality, a general pure state of any quantum system

has the form

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (1.110)$$

The complex numbers α and β are not probabilities themselves. They are **probability amplitudes**: their squared moduli give the probabilities of the two outcomes in this basis, and their relative phase governs the outcomes of other measurements, as the interferometer has already shown.

For a polarizer oriented at an angle θ from the horizontal axis, the transmitted polarization state may be written

$$|\theta\rangle = \cos \theta |H\rangle + \sin \theta |V\rangle. \quad (1.111)$$

If the incoming photon is horizontally polarized, the transmission probability is

$$P(\theta|H) = |\langle\theta|H\rangle|^2 = \cos^2 \theta, \quad (1.112)$$

which is Malus' law. Thus an elementary optical experiment already suggests the rule: probabilities are squared inner products of state vectors.

The strength of the polarization example is that it makes basis dependence visible with ordinary optical equipment. It also shows why quantum theory cannot be merely a theory of ignorance about a hidden classical alternative. A diagonal polarization is not half horizontal and half vertical in the same sense that a box contains either a red or a blue ball with unknown color. It is a definite preparation that becomes probabilistic only relative to another measurement basis. The limitation of the example is that polarization is often described by classical electromagnetic waves as well; one must be careful not to confuse the classical wave description of a beam with the single-quantum state description of a photon.

1.5.3 Sequential Polarizers and the Failure of Boolean Logic

Consider now three polarizers. If a horizontally polarized beam first meets a vertical polarizer, no light passes in the ideal case:

$$\langle V|H\rangle = 0. \quad (1.113)$$

Classically, one might expect that inserting an additional filter between them can only reduce the transmitted intensity. But if a diagonal polarizer is placed between the horizontal and vertical polarizers, then some light is transmitted. In the ideal single-photon description, the probability is

$$P(H \rightarrow D \rightarrow V) = |\langle D|H\rangle|^2 |\langle V|D\rangle|^2 = \frac{1}{4}. \quad (1.114)$$

The middle polarizer changes the state preparation for the final measurement. It is not merely a passive question about a pre-existing property.

This experiment gives an early warning that the logic of quantum propositions is not simply the Boolean logic of subsets of a classical phase space. The propositions “horizontal,” “vertical,” and “diagonal” are tied to experimental operations, and these operations need not commute. The formal theory will express this noncommutativity through operators, but the phenomenon itself is already visible in elementary polarization experiments.

This is a decisive advantage of the sequential-polarizer experiment. It shows that measurements

are not merely windows through which one passively reads pre-existing properties. They can prepare new states. Its limitation, however, is that it still does not display all the drama of quantum mechanics, since absorption by a polarizer may be imagined too mechanically. The Stern–Gerlach experiment below makes the same logical structure harder to dismiss as a peculiarity of light.

1.5.4 Stern–Gerlach Experiments and Incompatible Questions

The Stern–Gerlach experiment, already encountered above as the first direct display of spatial quantization (Section 1.3.4), provides a two-state system not built from light polarization or spatial paths, but from intrinsic angular momentum. The original experiment, performed by Otto Stern and Walther Gerlach in 1922, sent a beam of neutral silver atoms through a strongly inhomogeneous magnetic field. Classically, if the atoms carried tiny magnetic moments pointing in arbitrary directions, the magnetic force should have produced a continuous smear on the detection screen. Instead, the beam split into two separated spots.

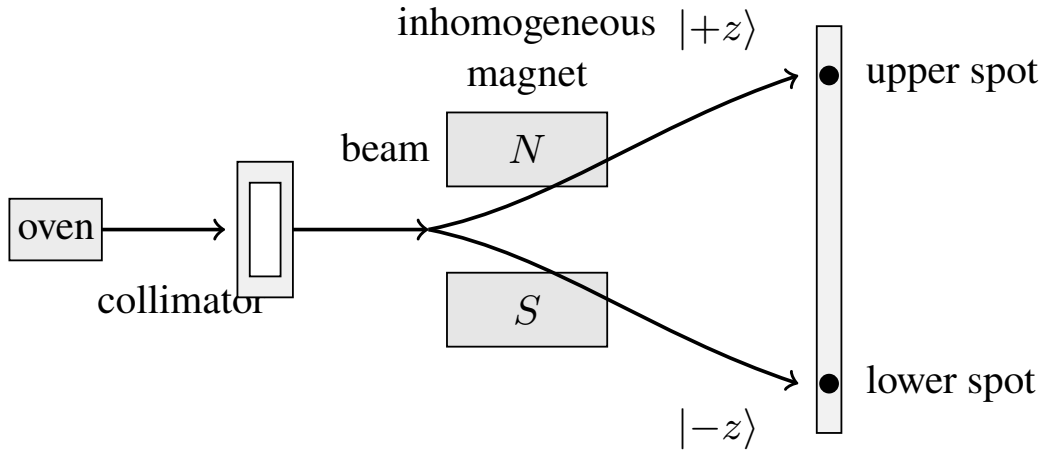


Figure 1.3: A schematic Stern–Gerlach experiment. A collimated atomic beam passes through an inhomogeneous magnetic field and separates into two discrete beams rather than a continuous spread.

The historical experiment was first interpreted before the modern concept of electron spin had fully crystallized, but in retrospect it displays one of the simplest quantum facts: a component of angular momentum can have discrete outcomes. For a spin- $\frac{1}{2}$ system measured along the z -axis, denote the two possible outcomes by

$$|+z\rangle, \quad |-z\rangle. \quad (1.115)$$

A particle prepared in $|+z\rangle$ and immediately measured again along z gives the same result with certainty. The first apparatus can therefore be used as a preparation device.

The most revealing form of the experiment is sequential. Send an unpolarized beam through a z -oriented Stern–Gerlach apparatus and block the lower output, keeping only the $|+z\rangle$ beam. If this selected beam is sent through a second z -oriented apparatus, it does not split again into two equal parts; it goes entirely into the $+z$ output. This shows that the first apparatus has prepared a definite state relative to the same later question.

Now measure the same prepared state along the x -axis by rotating the magnetic field gradient. The relevant basis is

$$|+x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle + |-z\rangle), \quad |-x\rangle = \frac{1}{\sqrt{2}}(|+z\rangle - |-z\rangle). \quad (1.116)$$

Thus a particle prepared as $|+z\rangle$ gives the two x -outcomes with equal probability. More strikingly, if one selects the $|+x\rangle$ beam and then measures z again, the original certainty of the z -outcome is lost. The intermediate x -measurement has not merely revealed a hidden x -property; it has prepared a new state relative to the later z -measurement.

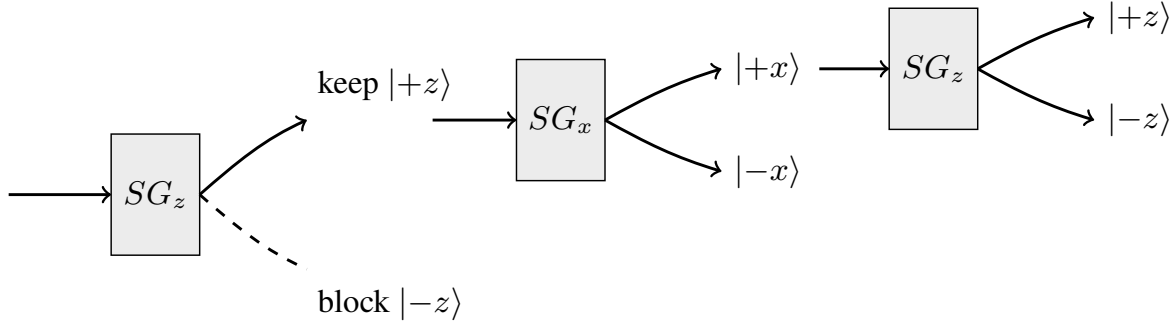


Figure 1.4: Sequential Stern–Gerlach measurements. A z -measurement can prepare $|+z\rangle$, but an intervening x -measurement destroys the certainty of a later z -measurement.

The moral is central. The state $|+z\rangle$ is not a catalogue assigning definite values to both z -spin and x -spin. Instead, it is a rule for assigning amplitudes to possible outcomes in whatever basis an experiment chooses. The incompatibility of different measurement directions is not ignorance in the ordinary sense, but a structural feature of the theory.

The Stern–Gerlach experiment has a special historical and conceptual role because spin is not a disguised position variable. The two outcomes are discrete, reproducible, and associated with spatial direction, yet they cannot be explained as the revelation of a tiny classical arrow with pre-existing components along all axes. The advantage of the spin- $\frac{1}{2}$ system is therefore its purity: it displays the finite-dimensional Hilbert-space structure without relying on wave motion through two visible arms. Its limitation is that a full account of spin ultimately requires the representation theory of rotations, which will only appear after the general theory of symmetries has been developed.

1.5.5 The Operational Meaning of State

These three families of experiments — paths, polarizations, and spin components — suggest a more careful definition. A physical state should be understood operationally as an equivalence class of preparation procedures: two preparations define the same state if they give the same probabilities for every possible later experiment. This definition does not require us to imagine an unseen list of all properties possessed by the system. It asks only what statistical predictions are reproducibly associated with a preparation.

In the two-state examples, such a preparation is represented by a ray in a complex vector space.

1139 Multiplying a vector by an overall nonzero complex number does not change the physical state, since
1140 normalized probabilities are unchanged. In particular,

$$1141 \quad |\psi\rangle \quad \text{and} \quad e^{i\chi} |\psi\rangle \quad (1.117)$$

1142 represent the same pure state. Relative phases, however, are physically meaningful, as the Mach–
1143 Zehnder interferometer shows.

1144 We are thus led to the following preliminary rules:

- 1145 1. mutually distinguishable alternatives are represented by basis vectors;
- 1146 2. a general pure state is a linear superposition of these alternatives;
- 1147 3. complex amplitudes, not probabilities, are the quantities that combine linearly;
- 1148 4. probabilities are obtained from squared moduli of amplitudes;
- 1149 5. changing the experimental question corresponds to changing the basis.

1150 These rules are not yet the full Hilbert-space formalism, but they contain its seed. The next chapter
1151 will develop this seed systematically, replacing the two-dimensional examples by the general language
1152 of vectors, inner products, linear operators, and measurement.

Chapter 2 From Experimental Facts to States of the Hilbert Space

 **Note** *This chapter is constructive in character. Taking only the experimental logic of Chapter 1 as input, it assembles the axioms of the quantum state space step by step: the two composition principles for amplitudes, the Born rule connecting amplitudes with frequencies, the linear structure of superpositions, rays as the physical states, the inner product as a transition amplitude, orthogonality as experimental distinguishability, orthonormal bases as complete sets of alternatives, and finally the Hilbert space — finite- and infinite-dimensional — as the natural arena of quantum states. It presupposes nothing beyond Chapter 1 and elementary linear algebra. Starred material may be omitted on a first reading without loss of continuity.*

Chapter 1 ended in debt. Its history of the three crises distilled five kinematical requirements that any successor of classical mechanics must meet (Section 1.4.6): discrete alternatives in place of a continuum of states; observable data attached to pairs of states rather than to single states; complex amplitudes composed before probabilities; an order-sensitive composition of physical operations; and a notion of state belonging to none of its representations. Its closing study of two-state experiments (Section 1.5) then showed all five requirements at work in the simplest laboratory situations — the Mach–Zehnder interferometer, the polarizer, the Stern–Gerlach magnet — and condensed them into a preliminary grammar of kets, amplitudes, and squared moduli. That grammar was read off the experiments; it was not yet a theory. What the Dirac–Jordan–von Neumann synthesis achieved abstractly (Section 1.4.5), this chapter achieves from below, clause by clause.

The route of the repayment is deliberate. We first fix the operational vocabulary — what an experiment is, what its raw data are, and what the word *state* can legitimately mean — and find that the data are frequencies, and that frequencies by themselves obey no simple law of combination. We then go beneath the frequencies to the quantities that do combine simply: the two composition principles for amplitudes, and the Born rule that returns from amplitudes to frequencies. Only after that does geometry enter. The addition of amplitudes forces the set of states to be a linear space; the Born rule forces the physical state to be a ray rather than a vector and elevates the inner product to the primitive physical object; orthogonality emerges as perfect distinguishability, and an orthonormal basis as a complete set of alternatives. The chapter closes with the Hilbert space as the finished arena, including an honest preview of the infinite-dimensional case, and with the axioms collected in one place.

One debt will be left open on purpose. The fourth requirement of Chapter 1 — that the order of physical operations matters — is motivated at every step of what follows, for the experiments of this chapter are chains of stages whose order cannot be permuted; but its formal repayment requires the theory of linear maps on the state space, which belongs to the next chapter. The same honesty applies to the continuum: it is previewed, not perfected. What this chapter promises is narrower and, for that reason, more solid: by its end, the statement *the state is a ray of a Hilbert space* will have been earned experimentally, clause by clause.

2.1 Preparations, Alternatives, and Frequencies

Every experiment considered in this book has the same anatomy. A **preparation** produces a system in a reproducible condition: an oven and a collimator for a silver beam, a laser and a polarizer for a photon, an electron gun and an accelerating voltage. The system then passes through a **process** — an interferometer, a crystal, a magnet, or perhaps nothing at all — and ends at a **registration**: a counter clicks, a spot appears on a screen, a grain of emulsion blackens. The raw data of physics are clicks and spots. Everything else in the theory is an organization of clicks and spots.

The vocabulary of that organization is fixed once and for all. An **alternative** is a possible outcome of a given apparatus: this detector rather than that one, transmission rather than absorption, the upper spot rather than the lower. Alternatives are **mutually exclusive** when a single run of the experiment can realize at most one of them; transmission and absorption at the same polarizer are mutually exclusive, and so are the two spots of a Stern–Gerlach magnet. A set of alternatives is **complete** for an apparatus when it is exhaustive: every run lands somewhere in the set. Completeness has an immediate quantitative shadow. If, out of N runs prepared in the same way, the counter for the n -th alternative fires N_n times, the relative frequencies N_n/N stabilize, as N grows, at numbers P_n which — precisely because the set is exhaustive — satisfy

$$\sum_n P_n = 1. \quad (2.1)$$

These frequencies are the whole of the data. Counters register frequencies; no counter has ever registered an amplitude.

From such data alone the word *state* receives its first precise meaning. Chapter 1 defined a state operationally as an equivalence class of preparation procedures (Section 1.5.5); we now write the definition in the language of frequencies. Call two preparation procedures Π_1 and Π_2 equivalent, and say that they prepare the same state, if they yield the same frequencies for every later experiment:

$$\Pi_1 \sim \Pi_2 \iff P(E, n | \Pi_1) = P(E, n | \Pi_2) \quad \text{for every experiment } E \text{ and every alternative } n. \quad (2.2)$$

The definition is uncompromisingly operational. A state is nothing but a rule for predicting frequencies; two preparations that predict identically are the same state, whatever they may look like on the bench. A polarizer followed by a long free flight prepares the same photon state as the polarizer alone, because no later experiment can tell the two procedures apart.

One further empirical clause belongs to the vocabulary of this section, and the whole of quantum kinematics is hidden inside it. Suppose an outcome can be reached through several mutually exclusive alternatives. If the arrangement makes the alternatives *distinguishable in principle* — if it deposits anywhere in the world a record of which alternative was realized, whether or not any observer ever reads that record — then the frequencies add classically: the frequency of the outcome is the sum of the frequencies of the alternatives. If the arrangement leaves the alternatives unmarked, they do not. We record this dichotomy here as a fact of nature; the rule that governs it is the content of Section 2.2.

The cleanest proof of the fact is one of the quietest experiments of its decade. In an atom interferometer, a collimated beam of atoms is split — for instance by diffraction from a standing wave

of light — into two spatially separated paths for the atomic center of mass, and the paths are recombined downstream; the atom counts at the outputs exhibit interference fringes exquisitely sensitive to the relative phase of the two path amplitudes. Now tag the paths. Before recombination, a microwave pulse leaves the atom’s internal hyperfine state correlated with the path it took, so that an atom arriving through the upper path carries a different internal label from an atom arriving through the lower. The tag exerts no classical force worth the name on the center of mass; the trajectories, the separations, and the timings are unchanged. Yet the fringes vanish: the visibility collapses the moment the paths become distinguishable in principle, because the internal state of the very atom inside the apparatus could now be inspected to reveal the path. If the tag is withdrawn — if the internal labels of the two paths are made identical again before recombination — the fringes return.¹

The moral is twofold. First, an alternative is not a hidden choice made by the particle; it is a feature of the whole arrangement, preparation and process and registration together. The atom did not secretly take one path and then fail to remember it; the unmarked arrangement simply does not contain two alternatives, and the marked arrangement does. Second, the operative notion is distinguishability *in principle*: what matters is whether the arrangement contains the which-way information, not whether anyone looks at it. This clause will return in every section of the chapter.

The advantage of the vocabulary fixed here is that every term is cashed out in procedures: preparation, alternative, frequency, state. A theory built on such words cannot drift far from the bench, because its primitives are bench words. Its limitation is that the vocabulary describes the data but supplies no law for them. The marking experiment shows that frequencies obey no universal sum rule, for they add when alternatives are marked and refuse to add when they are not. Frequencies are the data, but they are not the quantities that combine simply; to find those we must go one level beneath the counters, and that is the work of the next section.

2.2 The Two Composition Principles for Amplitudes

The experiments of Chapter 1 — the Mach–Zehnder interferometer, the chain of polarizers, the sequence of Stern–Gerlach magnets (Section 1.5) — showed that nature keeps two sets of books. Counters accumulate frequencies, but the frequencies can be predicted only through intermediate quantities that add where frequencies do not, and that multiply where frequencies merely chain. We now distill those intermediate quantities into the first axiom of the kinematics.

Axiom 1 (amplitudes). To each transition from an alternative of the preparation to an alternative of the registration, nature assigns a complex number, the **amplitude** of the transition. An amplitude therefore always lives on a *pair* of states, a beginning and an end. In this form the axiom honors the third requirement of Chapter 1 — complex amplitudes are composed before probabilities — and, because the primitive quantity attaches to a pair, also the second (Section 1.4.6). The two rules by

¹The experiment was performed with rubidium atoms by S. Dürr, T. Nonn, and G. Rempe, *Nature* **395**, 33 (1998). The which-way information is stored in the atom’s own internal states, and the fringe visibility collapses without any measurable mechanical disturbance of the paths; the restoration of the fringes upon withdrawal of the tag is the conditional content of the marking clause.

which amplitudes are composed are the foundational pair of this chapter.

The addition principle. If an outcome can be reached through several mutually indistinguishable alternatives, the amplitude for the outcome is the sum of the partial amplitudes:

$$A_{\text{tot}} = A_1 + A_2 \quad \implies \quad P = |A_1 + A_2|^2 \neq |A_1|^2 + |A_2|^2 = P_1 + P_2. \quad (2.3)$$

The experimental proof is the Mach–Zehnder interferometer of Chapter 1 (Figure 1.2). Each detector is reached through two alternatives, the upper path and the lower, and the two contributions can cancel at a dark port even though neither contribution vanishes by itself: closing either arm brightens that detector. Probabilities, being non-negative, cannot do this. Whatever adds at the output of an interferometer must be capable of cancellation, and only the amplitudes add. The inequality above is therefore not a defect of the theory but its first success: the fringe pattern *is* the difference between $|A_1 + A_2|^2$ and $|A_1|^2 + |A_2|^2$.

The multiplication principle. If a process is a chain of successive stages with definite intermediate alternatives, the amplitude of the chain is the product of the stage amplitudes:

$$A(\psi \rightarrow \phi \rightarrow \chi) = \langle \chi | \phi \rangle \langle \phi | \psi \rangle. \quad (2.4)$$

Here the amplitude for the transition $\psi \rightarrow \phi$ has been written in the bra-ket notation inherited from Chapter 1 (Section 1.5), whose formal status will be fixed in Section 2.6; for the moment it is a compact script for the quantity of Axiom 1. The experimental proof is the chain of filters of Chapter 1. A horizontally polarized photon meets a diagonal polarizer and then a vertical one, and the amplitude of the chain factorizes as $\langle V | D \rangle \langle D | H \rangle$; combined with the rule of Section 2.3 this will reproduce the measured law $P(H \rightarrow D \rightarrow V) = 1/4$ (Section 1.5). The same factorization governs the sequential Stern–Gerlach arrangement: a z -preparation, an x -stage, a final z -question.

Both principles can be exhibited on a single optical bench with no spatial two-path interferometer at all. Prepare a single photon in horizontal polarization and let it traverse a sequence of oriented wave plates — a half-wave plate, say, followed by a quarter-wave plate — and finally an analyzer. Each plate is cut from birefringent crystal, and its fast and slow axes define, for that plate, two perpendicular polarization alternatives of its own. A photon entering a plate is assigned an amplitude for each of the plate’s two axis-alternatives; inside the plate the two alternatives accumulate a relative phase set by the plate’s thickness and birefringence; at the exit face they recombine. The amplitude to leave the plate with a given polarization is thus a sum of two contributions, one through the fast alternative and one through the slow — the addition principle, exercised inside a single element. A horizontally polarized photon can in this way leave a quarter-wave plate with a nonzero amplitude to be registered as right-circularly polarized: the conversion $H \rightarrow R$ resolves into the plate’s two axis-alternatives, each gathering its own phase through the crystal,

$$A(H \rightarrow R) = e^{i\delta_f} \langle R | f \rangle \langle f | H \rangle + e^{i\delta_s} \langle R | s \rangle \langle s | H \rangle, \quad (2.5)$$

and each summand is already a product of two stage amplitudes, one into and one out of an axis-alternative. The chaining from plate to plate then exercises the second principle in full: the amplitude to traverse the whole cascade — say the conversion $H \rightarrow D \rightarrow R$, one plate turning horizontal into diagonal and the next turning diagonal into right-circular — is the product of the per-plate amplitudes, so that rotating any one plate — changing the relative phase it imposes — redistributes the counts at

the final analyzer. The measured conversion of horizontally polarized single photons into circularly polarized ones, and the phase-dependent flicker of the analyzer counts as the plates are turned, are the two composition principles read directly off one bench. Note that the two alternatives here live on the polarization of one and the same photon; no beam is ever split in space.

Why must the arena of the amplitude be the complex numbers? The interferometer excludes anything smaller. Real non-negative quantities cannot cancel: two open paths can give fewer counts at a detector than one, so the quantities that add must not be probabilities or intensities. A single signed real quantity could cancel, but it could not carry what the experiment shows to be physical. As the phase shifter in one arm of the interferometer is dialed, the detector counts vary continuously and periodically; the amplitude of an alternative must therefore store a continuously adjustable *phase* as well as a magnitude. The minimal number system holding a magnitude and a phase is the complex number,

$$A = |A| e^{i\varphi}, \quad (2.6)$$

where φ is, operationally, the quantity a phase-shifter dial reads.²

One feature of the composition law must be registered honestly before we go on. Chaining stage X and then stage Y is a different experiment from chaining Y and then X : the sequential Stern–Gerlach arrangement of Chapter 1 (Figure 1.4) makes this vivid, for a z -selection followed by an x -interrogation does not prepare what an x -selection followed by a z -interrogation prepares. Nothing in the two principles is violated by this. In either order the amplitudes still add over indistinguishable alternatives and multiply over successive stages; what is order-sensitive is the *processes* themselves. The formal machinery for that order-sensitivity — linear maps on the state space — will be built in the chapters on measurement and on dynamics. Here we locate, without yet repaying, the fourth requirement of Chapter 1 (Section 1.4.6): the experiments insist that order matters, and the theory must be built so that it can.

The advantage of the two principles is their economy. The entire combinatorics of quantum kinematics — every interference pattern, every cascade of filters, every correlation of counters — is generated by one addition and one multiplication of complex numbers, and the rest of this chapter is, in a sense, only an unfolding of their consequences. Their limitation is equally plain: counters never click in complex numbers. The principles say how amplitudes combine, but no rule given so far turns an amplitude into a frequency. Supplying that rule is the business of the next section.

2.3 The Born Rule: from Amplitudes to Frequencies

The bridge from amplitude to frequency was first crossed in a footnote. In the scattering papers of 1926, Born proposed that the wave of wave mechanics determines not where a particle is but how likely it is to be found, taking the probability proportional to the squared modulus of the amplitude

²Still larger number systems are not excluded by this argument alone. Quaternionic and other generalizations are constrained only by later consistency requirements — among them the demand that amplitudes for composite systems behave sensibly — and are not pursued in this book.

(Section 1.4.3).³ Chapter 1 told that story; the task of this section is to state the rule as an axiom of the kinematics and to display its empirical content.

Axiom 2 (the Born rule). If a system is prepared in the state ψ and registered against the alternative ϕ , the probability of the transition is the squared modulus of its amplitude:

$$P(\psi \rightarrow \phi) = |\langle \phi | \psi \rangle|^2. \quad (2.7)$$

The transition probability is $|\langle \phi | \psi \rangle|^2$: with this single clause the amplitude calculus of Section 2.2 is connected to the counters of Section 2.1.

The rule is a statement about ensembles. Given a preparation that can be repeated, it predicts the long-run frequency with which the alternative ϕ appears among identically prepared systems; about a single run it claims nothing beyond what frequencies assert — a point the example below makes concrete.⁴

The Born rule is exactly the partner the two composition principles need. Add first, then square: two indistinguishable alternatives with amplitudes A_1 and A_2 give

$$P = |A_1 + A_2|^2 = |A_1|^2 + |A_2|^2 + 2 \operatorname{Re}(A_1^* A_2), \quad (2.8)$$

and the last term — the interference term, carrying the relative phase — is the fringe pattern itself. When the alternatives are marked, their indistinguishability fails, the cross term vanishes, and the classical sum rule of Section 2.1 is recovered. Multiply first, then square: a chain of successive stages gives $P(\psi \rightarrow \phi \rightarrow \chi) = |\langle \chi | \phi \rangle \langle \phi | \psi \rangle|^2 = |\langle \chi | \phi \rangle|^2 |\langle \phi | \psi \rangle|^2$, which for the polarizer chain of Chapter 1 reads $P(H \rightarrow D \rightarrow V) = |\langle D | H \rangle|^2 |\langle V | D \rangle|^2 = 1/4$ (Section 1.5) — the transmission frequencies of successive filters multiply, as they must.

Finally, the frequencies over a complete set of alternatives sum to one (Section 2.1), so the amplitudes from any preparation ψ to the alternatives $\{e_n\}$ of a complete set must satisfy

$$\sum_n |\langle e_n | \psi \rangle|^2 = 1. \quad (2.9)$$

This condition, stated here at the level of frequencies, will become the unit-norm condition on state vectors once the inner product is formalized in Section 2.6.

The empirical content of the axiom is displayed with unmatched directness in the electron biprism experiments of the Tonomura group. In the biprism, a thin charged filament bends the electron trajectories on either side toward one another, so that the amplitude of a single electron arrives at the detector through two indistinguishable alternatives; the source is run so feebly that the apparatus contains, with overwhelming probability, at most one electron at a time. The detector records the arrivals one by one, and the film shows what the Born rule means. The first dots fall at apparently random positions; no regularity governs any single event, and none is predicted for any of them. As the dots accumulate — hundreds, then thousands, then hundreds of thousands of single electrons — the pattern converges with increasing sharpness to the computed $|A_1 + A_2|^2$ of the biprism geometry. The film *is* the frequency bridge: the probability belongs to the ensemble, while each single electron, in

³M. Born, *Z. Phys.* **37**, 863 (1926); **38**, 803 (1926). The historical setting of the proposal is recounted in Section 1.4.3; here we record only its axiomatic content.

⁴The elements of probability theory assumed in this book — ensembles, conditional frequencies, statistical independence — are collected in Appendix D.

Dirac’s phrase, interferes only with itself — the same lesson, now for matter, that Taylor’s feeble-light photographs taught for light in Chapter 1 (Section 1.5).⁵

A Plausibility Argument for the Born Rule

Why the squared modulus, and not the first power, or the fourth, or some other function of the amplitude? A rigorous answer exists but lies beyond this chapter; what can be given here is a plausibility argument, heuristic in character, showing that the two composition principles themselves leave little room for anything else. Suppose, then, that the probability of a transition is a function of its amplitude alone, $P = F(A)$. This is already a physical hypothesis: it asserts that the amplitude exhausts what the transition knows.

The multiplication principle constrains F severely. In a staged experiment the amplitudes of successive stages multiply, while the observed frequencies of successive filters multiply as well: the fraction of systems passing both of two independent stages is the product of the fractions passing each. Consistency therefore demands $F(A_1 A_2) = F(A_1) F(A_2)$, and the associativity of stage composition — three stages may be bracketed in either order, $(A_1 A_2) A_3 = A_1 (A_2 A_3)$ — demands that this rule hold uniformly for chains of any length. Together with $F(1) = 1$ for the trivial stage that changes nothing, and the boundedness $0 \leq F \leq 1$, the only continuous solutions are the powers of the modulus, $F(A) = |A|^\alpha$ for some $\alpha \geq 0$: no dependence on the phase can serve, since $F(e^{i\varphi}) F(e^{-i\varphi}) = F(1) = 1$ with both factors in $[0, 1]$ forces $F \equiv 1$ on every phase.

The addition principle then fixes the exponent. Indistinguishable alternatives may be permuted, regrouped, or subdivided without changing the physical arrangement; more demanding, the total probability must be one no matter which complete set of alternatives resolves the state. A two-alternative instance shows what this costs: a state with $|a|^\alpha + |b|^\alpha = 1$ at $a = b$ carries total probability $2^{\alpha/2-1}$ against the rotated set $\{(|e_1\rangle \pm |e_2\rangle)/\sqrt{2}\}$, which equals one only for $\alpha = 2$. The same exponent also keeps the bookkeeping coherent: $|A_1 + A_2|^2$ decomposes into $|A_1|^2 + |A_2|^2$ plus an interference term carried entirely by the relative phase, so a single rule accounts both for the fringes of indistinguishable alternatives and for the classical sum rule of marked ones. One arrives, heuristically, at $P = |A|^2$, up to the overall normalization convention fixed by the completeness condition above.

Two warnings belong with this argument. First, it is offered as orientation, not as derivation. Rigorous uniqueness theorems for the probability rule — Gleason’s theorem and its later refinements — do exist, but they presuppose the measurement formalism of the next chapter and are only signposted here.⁶ Second, the argument deliberately does *not* assume the freedom to multiply a state by an overall phase. That freedom is a *consequence* of the Born rule, proved in Section 2.5; to assume it here would be circular.

The advantage of the Born rule is that with a single axiom the amplitude calculus becomes predictive: compose amplitudes by the two principles, square one modulus, and the frequencies of every

⁵A. Tonomura, J. Endo, T. Matsuda, T. Kawasaki, and H. Ezawa, *Am. J. Phys.* **57**, 117 (1989).

⁶A. M. Gleason, *J. Math. Mech.* **6**, 885 (1957). Gleason’s result characterizes the probability measures on the closed subspaces of a Hilbert space; even its statement requires the language of projections, which belongs to the next chapter.

experiment in this chapter are computed. Its limitation is written into the Tonomura film: the rule speaks only of ensembles, is silent about the individual event, and says nothing about what a measurement *is*; those questions require the operator formalism of the next chapter. For the present chapter one consequence matters most. Since only squared moduli are observed, the addition of amplitudes must be a well-defined operation on states themselves — and that requirement, taken seriously, forces the state space to be a linear space. To that structure we now turn.

2.4 The Linear Space of States

The composition principles of Section 2.2 add amplitudes over indistinguishable alternatives; so far, addition has been an operation on complex numbers attached to processes. The Born rule of Section 2.3 forces a stronger reading. Consider two preparable states $|\psi_1\rangle$ and $|\psi_2\rangle$ and an arrangement that feeds each run with one or the other while leaving the alternative “which one” unresolved in principle — no record, anywhere, of which channel was realized. What such an arrangement prepares is not a statistical blend of the two states but a new state,

$$|\psi\rangle = \alpha |\psi_1\rangle + \beta |\psi_2\rangle, \quad (2.10)$$

whose amplitude for any later test alternative is, by the addition principle, the sum of the partial amplitudes. The two-state families of Chapter 1 (Section 1.5) are the standing evidence: a diagonal photon is prepared by an arrangement that does not resolve horizontal against vertical, and its state is written $(|H\rangle + |V\rangle)/\sqrt{2}$; the interferometer prepares $(|u\rangle + e^{i\varphi}|l\rangle)/\sqrt{2}$ precisely when the arrangement does not resolve path against path. Superposition is an operation on states, and the set of all states preparable relative to a given system closes under it.

That set, denoted $\mathcal{H}$, therefore carries the structure of a vector space over the complex numbers. Its axioms are the familiar ones; what matters here is that each of them is an operational statement.

- *Closure under superposition.* For any two preparable states and any complex weights, $\alpha |\psi_1\rangle + \beta |\psi_2\rangle$ is again a preparable state: coherent re-blending never leads out of the arena.
- *Scalar multiplication.* The vector $c |\psi\rangle$ has every amplitude of $|\psi\rangle$ re-weighted by the common factor c ; multiplying a state vector is the coherent re-weighting of all its amplitudes at once. Writing $c = re^{i\chi}$, the modulus r rescales every amplitude together, and the phase $e^{i\chi}$ rotates every amplitude together — an operation whose physical status is the subject of Section 2.5.
- *Associativity and commutativity of addition.* An arrangement that leaves three alternatives unresolved does not care how the alternatives are grouped or ordered: $(|\psi_1\rangle + |\psi_2\rangle) + |\psi_3\rangle = |\psi_1\rangle + (|\psi_2\rangle + |\psi_3\rangle)$, and the sum is independent of the order of its terms.
- *Distributivity.* Coherent re-weighting distributes over superposition, $c(|\psi_1\rangle + |\psi_2\rangle) = c|\psi_1\rangle + c|\psi_2\rangle$: re-weighting the blend is the same as blending the re-weighted parts.
- *The zero vector.* There is a vector 0 with $|\psi\rangle + 0 = |\psi\rangle$; it assigns amplitude zero to every test alternative and represents the vacuous alternative — see the note below.
- *Additive inverses and the unit scalar.* The vector $-|\psi\rangle$ carries every amplitude of $|\psi\rangle$ with reversed sign — a shift of π against the rest of any superposition — and $1|\psi\rangle = |\psi\rangle$.



Note The zero vector is not a physical state. A state is a rule assigning frequencies to the outcomes of experiments on an existing ensemble (Section 2.1); the zero vector assigns amplitude zero to every alternative and therefore describes no ensemble at all. It enters the formalism as the additive identity of $\mathcal{H}$, and the equality $|\psi\rangle + (-|\psi\rangle) = 0$ records the cancellation of two alternatives, not an exotic preparation. Much later we will meet a state called the vacuum; it will be an honest nonzero vector, for the vacuum is a preparation with definite, testable frequencies.

With the vector space in place, Dirac's ket is formalized: $|\psi\rangle$ simply names a vector of $\mathcal{H}$, exactly the usage that Chapter 1 practiced informally (Section 1.5). To each vector $|\phi\rangle$ there will correspond a dual object $\langle\phi|$; its full definition waits for the inner product of Section 2.6, and until then the bracket $\langle\phi|\psi\rangle$ remains what it was in Chapter 1 — a notation for the transition amplitude.

A set of vectors is *linearly independent* when no member is a superposition of the others; the *span* of a set is the collection of all superpositions of its members; the *dimension* of $\mathcal{H}$ is the number of vectors in any maximal independent set. At this stage these are algebraic notions and nothing more. Their operational reading — the dimension is the maximal number of mutually exclusive alternatives that a single experiment on the system can return — will be earned in Sections 2.7 and 2.8.

Already at the algebraic level, however, the dimension can be made tangible. Take photon polarization, and choose the alternatives of a horizontal–vertical analyzer as the two basic kets. Every polarization state is then a column of two complex amplitudes, and the four states of Chapter 1's bench, augmented by the two circular ones, read

$$\begin{aligned} |H\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, & |V\rangle &= \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\ |D\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, & |A\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \\ |R\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, & |L\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \end{aligned} \quad (2.11)$$

Horizontal and vertical alone span the space, since $|D\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$ is a superposition of them; conversely $|H\rangle = (|D\rangle + |A\rangle)/\sqrt{2}$ and $|V\rangle = (|D\rangle - |A\rangle)/\sqrt{2}$ show the diagonal pair to span it equally well. Any two of $|H\rangle, |V\rangle, |D\rangle, |A\rangle$ are linearly independent — the determinant of each such pair is nonzero — while any three vectors of the display are necessarily dependent: two complex weights exhaust the space, and no third alternative is independent of an independent pair. The circular states $|R\rangle$ and $|L\rangle$ are no exception; for instance $|R\rangle = ((1+i)|D\rangle + (1-i)|A\rangle)/2$, as the columns verify at a glance. Their columns carry the factor i whose experimental necessity is established in Section 2.6, and they add no third direction. Polarization is a two-dimensional state space, and we can see that it is one before dimension has acquired any operational meaning.

What Chapter 1 read off three separate devices is now a theorem of the structure. Relative to any experiment with exactly two mutually exclusive alternatives, call the corresponding kets $|0\rangle$ and $|1\rangle$; they span the state space of the system relative to that experiment, and every preparable state has the

form

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (2.12)$$

where the constraint on the weights is the normalization of Section 2.3: the two alternatives are complete, so their squared amplitudes must sum to one. The general two-state form is thus recovered from the vector-space structure plus the Born rule, not re-assumed device by device.

The advantage of the linear structure is that it makes the theory computable: every consequence of superposition follows from the two composition principles by pure algebra, and the whole kinematics of states is purchased at the price of a single operation — the addition of amplitudes. Its limitation is subtler, and Chapter 1 already recorded it for the wave function: linearity makes the state space dangerously easy to reify. The vector $\alpha |\psi_1\rangle + \beta |\psi_2\rangle$ is a bookkeeping object for amplitudes, not a material wave propagating in some auxiliary space; it is not made of ψ_1 -stuff and ψ_2 -stuff, any more than the diagonal photon of Chapter 1 was half horizontal and half vertical. The states $|\psi_1\rangle$ and $|\psi_2\rangle$ are alternatives the arrangement leaves unresolved, not ingredients the system contains. A vector space, moreover, still knows nothing of length or angle; and since the Born rule tells us that an overall factor drops out of every frequency, we must ask which vectors are *the same state* before we ask about angles between different ones.

2.5 Rays: Which Vector Is the State?

The linear space of Section 2.4 supplies the candidates; we now say precisely which candidate is the state. The criterion is operational and was fixed in Section 2.1: two preparations define the same state if and only if they yield the same frequencies for every later experiment. Apply it to two vectors that differ by an overall phase. For every test alternative ϕ , the Born rule of Section 2.3 gives

$$|\langle\phi| e^{ix}\psi\rangle|^2 = |e^{ix}\langle\phi|\psi\rangle|^2 = |\langle\phi|\psi\rangle|^2. \quad (2.13)$$

Every outcome frequency of every conceivable experiment is unchanged, so no experiment can distinguish the two preparations: by the operational definition, $|\psi\rangle$ and $e^{ix}|\psi\rangle$ are the same state. Chapter 1 stated this as a remark alongside its operational definition of state (Section 1.5.5); with the Born rule in hand it has become a theorem. The converse holds as well: if two vectors deliver identical frequencies for every test alternative, they differ at most by such a phase (the proof is the saturation clause of Cauchy–Schwarz, Section 2.6). Testing against a complete set of alternatives fixes the modulus of every amplitude, and testing against superposed alternatives then fixes every relative phase except one overall angle, which no frequency can reach.

The physical state is therefore not a vector but a **ray** — the one-dimensional subspace

$$\{c|\psi\rangle : c \in \mathbb{C}, c \neq 0\}. \quad (2.14)$$

To compute, one picks a representative of the ray, and the standing convention is the normalized one,

$$\langle\psi|\psi\rangle = 1. \quad (2.15)$$

This is the completeness condition of Section 2.3, the sum of $|\langle e_n|\psi\rangle|^2$ over a complete set of alternatives, read as a statement about the vector itself; its full inner-product meaning is sealed in Section 2.6.

The convention costs nothing: the ray theorem has already removed the freedom of a complex factor, normalization spends the modulus of that factor, and the only freedom left in the representative is the phase $e^{i\chi}$ — exactly the freedom that the theorem declares meaningless.

The states themselves, freed of representatives, are the points of the complex projective space

$$\text{states of the system} \longleftrightarrow \mathbb{CP}^{n-1} = (\mathbb{C}^n \setminus \{0\})/\sim, \quad |\psi\rangle \sim e^{i\chi} |\psi\rangle, \quad (2.16)$$

where $n = \dim \mathcal{H}$. Nothing in what follows requires the machinery of projective geometry; the display says only that the space of states has one point per ray and carries no privileged representative. For a two-state system this space is a sphere, on which the two alternatives of any two-outcome experiment appear as antipodal points — a picture whose precise metric meaning must wait for the inner product of Section 2.6 and for the rotation theory of a later chapter.

That the overall phase is meaningless does not make every phase meaningless. In

$$|\psi(\chi)\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\chi} |1\rangle), \quad (2.17)$$

multiplying the whole vector by $e^{i\alpha}$ changes nothing, by the ray theorem. But χ is the phase of the $|1\rangle$ -component *relative* to the $|0\rangle$ -component, and it is physical: test the state against the alternative $|0'\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ — a member of a different basis, an experiment that resolves the superposition — and the Born rule gives

$$P(0' | \psi(\chi)) = \left| \frac{1 + e^{i\chi}}{2} \right|^2 = \frac{1}{4} |1 + e^{i\chi}|^2 = \frac{1}{4} |e^{i\chi/2}(e^{-i\chi/2} + e^{i\chi/2})|^2 = \frac{1}{4} |2 \cos \frac{\chi}{2}|^2 = \cos^2 \frac{\chi}{2}, \quad (2.18)$$

so the counts move continuously with χ from certainty to exclusion. This is precisely the fringe law of Chapter 1's interferometer (Figure 1.2), where χ was the setting of a phase shifter in one arm: the dial redistributes counts between the detectors without changing any immediate path probability. The phase of a state vector by itself is unobservable; the relative phase between the components of a superposition is among the most directly measurable quantities in physics.

One experiment exhibits both faces of the theorem in a single arrangement. In a perfect-crystal neutron interferometer, a neutron beam is divided into two spatially separated, coherent arms and recombined before detection — a Mach–Zehnder built of neutrons. In one arm the spin of the neutron, a spin- $\frac{1}{2}$ degree of freedom, is subjected to a magnetic field arranged to turn it through one full revolution. The experimental fact is this: a full 2π turn does not restore the spin state but multiplies its amplitude by -1 , and only a 4π turn restores it.⁷ Considered on either arm alone, the factor -1 is an overall phase and is invisible: every experiment performed solely on that arm gives unchanged frequencies, as the ray theorem demands. But inside the interferometer the same factor is a *relative* phase of π between the two arms, and the fringe law of this section applies: with the field switched off and a phase φ scanned in one arm, the counts follow $P_0(\varphi) \propto \cos^2(\varphi/2)$; with the 2π rotator inserted

⁷H. Rauch, A. Zeilinger, G. Badurek, A. Wilfing, W. Bauspiess, and U. Bonse, *Physics Letters* 54A (1975), 425; S. A. Werner, R. Colella, A. W. Overhauser, and C. F. Eagen, *Physical Review Letters* 35 (1975), 1053. The derivation of the sign from the theory of rotations is a debt owed by the chapter on angular momentum; none of its machinery is needed to state the fact.

they follow

$$P_0^{(2\pi)}(\varphi) \propto \cos^2 \frac{\varphi + \pi}{2} = \sin^2 \frac{\varphi}{2}, \quad (2.19)$$

the complementary pattern — the observed fringes are shifted by exactly half a period, and a second full turn, 4π altogether, restores the original one. The unobservability of the global phase and the observability of the relative phase are thus read off in one and the same experiment.

A closing remark belongs here, advanced or not at all. The ray theorem says that relative phases within a superposition are observable, and everything so far supports it — within the alternatives of one experiment. But there are known pairs of states between which no relative phase is ever observable: no experiment has exhibited interference between a state of integer and a state of half-integer total angular momentum, or between states of different electric charge. Such pairs are said to be separated by a *superselection rule*.⁸ The correct statement of the kinematics is then sector by sector: within each sector every ray can be superposed with every other, and the full space of states is a family of projective spaces rather than a single one. We record the clause and proceed; nothing in the present chapter depends on it.

The advantage of the ray picture is its economy: one mathematical point for one operational state, no redundancy, and the single remaining degree of freedom of a representative — its overall phase — certified meaningless by experiment itself. Its limitation is that it decides sameness but not difference: the ray picture tells us which vectors are one state, but not yet how far apart two different states lie, and the superselection clause warns that even the reach of superposition is an empirical, sector-by-sector fact rather than a theorem. To speak of rays, norms, and rotated bases quantitatively, we need the one sesquilinear object that the Born rule keeps invoking — the inner product, read physically as the transition amplitude itself.

2.6 The Inner Product as a Transition Amplitude

Since Chapter 1 we have written $\langle \phi | \psi \rangle$ for the amplitude of the transition $\psi \rightarrow \phi$, and Sections 2.2–2.5 have spent the notation freely. It is time to say exactly what object it is. The definition is physical: $\langle \phi | \psi \rangle$ is the amplitude that a system prepared in the state $|\psi\rangle$ is registered in the state $|\phi\rangle$. Nothing more primitive is available: Chapter 1’s second kinematical requirement (Section 1.4.6) was that the observable data live on *pairs* of states, and the inner product is that pair-datum in axiomatic form. Every geometric notion of the state space — length, angle, orthogonality — will be derived from it, and each of its properties will turn out to be an experimental statement.

We take the properties one at a time; together they constitute the third axiom of the chapter. The first is *conjugate symmetry*,

$$\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*. \quad (2.20)$$

Its experimental warrant is the reciprocity of transition frequencies: $|\langle \phi | \psi \rangle|^2 = |\langle \psi | \phi \rangle|^2$, the fact that a pair of polarizers transmits the same fraction of an incident beam whichever way the bench

⁸The systematic statement is due to G. C. Wick, A. S. Wightman, and E. P. Wigner (1952); the angular-momentum example finds its explanation in the rotation theory of a later chapter, and the charge example in the quantum theory of fields.

is run, and that the sequential Stern–Gerlach arrangements of Chapter 1 return the same fractions in reverse order. Frequencies force only the moduli to agree; declaring the amplitudes themselves to be conjugates is the weakest strengthening compatible with the addition principle, and no experiment has ever required more.

The second property is *linearity in the second argument*,

$$\langle \phi | \alpha \psi_1 + \beta \psi_2 \rangle = \alpha \langle \phi | \psi_1 \rangle + \beta \langle \phi | \psi_2 \rangle. \quad (2.21)$$

This is not a new assumption but the addition principle of Section 2.2 restated in the language of state vectors: the amplitude to reach ϕ through an unresolved superposition is the sum of the partial amplitudes. Combined with conjugate symmetry it yields conjugate-linearity in the first argument, $\langle \alpha \phi_1 + \beta \phi_2 | \psi \rangle = \alpha^* \langle \phi_1 | \psi \rangle + \beta^* \langle \phi_2 | \psi \rangle$. A form linear in one argument and conjugate-linear in the other is called *sesquilinear*; the asymmetry between the two arguments is a bookkeeping convention, and physics books and mathematics books make it oppositely.

The third property is *positivity*,

$$\langle \psi | \psi \rangle \geq 0, \quad \langle \psi | \psi \rangle = 0 \iff |\psi\rangle = 0. \quad (2.22)$$

Its experimental reading: the modulus squared of $\langle \psi | \psi \rangle$ is the probability that a ψ -prepared system passes a ψ -test — the total probability of finding the system in an alternative compatible with its own preparation. A preparation that never passes its own test prepares nothing, and nothing is the zero vector of Section 2.4.

With positivity comes the norm of a state vector,

$$\|\psi\| = \sqrt{\langle \psi | \psi \rangle}, \quad (2.23)$$

and the convention $\|\psi\| = 1$ now seals both the normalization of Section 2.3 and the choice of representatives of Section 2.5: for a normalized state, the completeness condition $\sum_n |\langle e_n | \psi \rangle|^2 = 1$ over a complete set of alternatives is precisely the statement $\langle \psi | \psi \rangle = 1$ expanded over that set.

One theorem is admitted at this point, because it answers an experimental question. The *Cauchy–Schwarz inequality* states that

$$|\langle \phi | \psi \rangle|^2 \leq \langle \phi | \phi \rangle \langle \psi | \psi \rangle, \quad (2.24)$$

with equality if and only if $|\phi\rangle$ and $|\psi\rangle$ lie on the same ray.⁹ For normalized states it reads $|\langle \phi | \psi \rangle|^2 \leq 1$: a transition probability never exceeds one. This closes a consistency question left open since Section 2.3: the Born rule interprets $|\langle \phi | \psi \rangle|^2$ as a frequency for *every* pair of states, and it is the inequality that guarantees the quantity to lie in $[0, 1]$ state-space-wide, not merely for the pairs checked on particular benches. The question “can a preparation be found in a test alternative more often than always?” is answered by an inequality, not by an axiom; and the saturation clause is the ray theorem’s converse in quantitative dress — overlap one means sameness of state.

⁹The proof is displayed in the note below.

 **Note** The proof of (2.24): apply positivity (2.22) to the vector $|\psi\rangle - \lambda|\phi\rangle$ with $\lambda = \langle\phi|\psi\rangle / \langle\phi|\phi\rangle$:

$$\begin{aligned} 0 &\leq \|\psi - \lambda\phi\|^2 = \langle\psi - \lambda\phi|\psi - \lambda\phi\rangle \\ &= \langle\psi|\psi\rangle - \lambda\langle\phi|\psi\rangle^* - \lambda^*\langle\phi|\psi\rangle + |\lambda|^2\langle\phi|\phi\rangle \\ &= \langle\psi|\psi\rangle - \frac{|\langle\phi|\psi\rangle|^2}{\langle\phi|\phi\rangle}. \end{aligned} \quad (2.25)$$

Rearranging gives (2.24). Equality holds iff $\|\psi - \lambda\phi\| = 0$, i.e. $|\psi\rangle = \lambda|\phi\rangle$, which is precisely the statement that the two vectors lie on the same ray.

The dual object announced in Section 2.4 can now be defined. To each vector $|\phi\rangle$ corresponds the **bra** $\langle\phi|$, the linear map $\psi \mapsto \langle\phi|\psi\rangle$ that eats a vector and returns the amplitude of the transition into ϕ . The composite symbol $\langle\phi|\psi\rangle$ — bra meeting ket — is hereby fixed as the notation of this book. A bra is a bookkeeping device for amplitudes; nothing about observables is asserted by it.

There remains the promissory note of Section 2.2: why the amplitudes must be *complex*. The bench that closes the question is the polarization bench one last time, now completed by circular analyzers. A right-circular analyzer prepares the state $|R\rangle$: single photons that pass it exhibit three facts. First, any linear analyzer, at whatever orientation, transmits them with probability $\frac{1}{2}$; in particular $|\langle H|R\rangle|^2 = |\langle V|R\rangle|^2 = |\langle D|R\rangle|^2 = |\langle A|R\rangle|^2 = \frac{1}{2}$. Second, an R -prepared photon passes a second right-circular analyzer with certainty. Third, it never passes a left-circular analyzer: $\langle L|R\rangle = 0$, where $|L\rangle$ is the state prepared by the left-circular device. Run the bench in the opposite order and the same fractions return, instance by instance — reciprocity again.

Now suppose, for contradiction, that amplitudes were real. The first fact, restricted to the horizontal-vertical pair, forces $|R\rangle$ to be a real equal-weight superposition of $|H\rangle$ and $|V\rangle$; enumerating the signs, the only such vectors are $(\pm|H\rangle \pm |V\rangle)/\sqrt{2}$, which are $|D\rangle$ and $|A\rangle$ up to overall sign. But then a diagonal analyzer would transmit the R -photons with probability 1 or 0 — never $\frac{1}{2}$, as the same first fact demands of the diagonal pair. A real state space therefore cannot place two circular polarizations on equal footing with all linear ones while keeping them mutually exclusive. Complex amplitudes accommodate all three facts at once:

$$|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle), \quad |L\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle), \quad (2.26)$$

for then

$$\langle R|L\rangle = \frac{1}{2}(\langle H| - i\langle V|)(|H\rangle - i|V\rangle) = \frac{1}{2}(1 + (-i)(-i)) = 0, \quad |\langle D|R\rangle|^2 = \left|\frac{1+i}{2}\right|^2 = \frac{1}{2}, \quad (2.27)$$

and the reversed order gives $\langle R|D\rangle = (1-i)/2 = \langle D|R\rangle^*$: the same bench, run both ways, exhibits orthogonality, reciprocity, and conjugate symmetry in one arrangement. No real bilinear form can make two equal-weight superpositions of the same pair orthogonal while both remain equal-weight with respect to the diagonal pair; the i in $|R\rangle$ is not a decoration but the piece of the kinematics forced by the existence of circular polarization. The debt of Section 2.2 is discharged.

The advantage of the inner-product axiom is that its every clause is an experimental report — reciprocity, the addition principle, the reliability of preparations — so that geometry is made physical at a stroke: length is reliability, angle is overlap, and the kinematics of state pairs is complete. Its

limitation is that it prices single transitions only. The inner product says what it costs a preparation to pass a test; it does not yet say what an observable is, what values it can return, or how it acts on a state. Those questions belong to the theory of operators and measurement, and they begin in the next chapter. What the inner product does immediately is measure overlap; and its extreme cases decide the most. Overlap one is sameness of state. Overlap zero has a name in the laboratory — perfect distinguishability — and it deserves a section of its own.

2.7 Orthogonality as Experimental Distinguishability

Two states are called *orthogonal* when the transition between them never happens:

$$\langle \phi | \psi \rangle = 0 \iff P(\phi | \psi) = |\langle \phi | \psi \rangle|^2 = 0. \quad (2.28)$$

Chapter 1 supplied the anchor cases (Section 1.5): crossed polarizers, $\langle V | H \rangle = 0$, an H -prepared photon never passing a V -analyzer; and the two beams of a Stern–Gerlach magnet, a $|+z\rangle$ -prepared atom never appearing in the $|-z\rangle$ output of a second z -oriented apparatus. The definition makes the laboratory statement precise in both directions. If $\langle \phi | \psi \rangle = 0$, then any experiment whose complete set of alternatives contains ϕ gives zero counts in its ϕ -channel on a ψ -prepared ensemble, while it returns ϕ with certainty on a ϕ -prepared one: a single ideal run of that experiment decides with certainty which of the two preparations is at hand. Conversely, if some complete set of alternatives containing ϕ never fires its ϕ -outcome on ψ -preparations, then the Born rule of Section 2.3 forces $\langle \phi | \psi \rangle = 0$. Orthogonality and perfect single-shot distinguishability are one and the same property, the first named by the algebra, the second by the bench.

Sequential Stern–Gerlach filtering (Figure 1.4) exhibits the two ends of the spectrum in one family of arrangements. A beam selected as $|+z\rangle$ and sent into an apparatus oriented to select $|-z\rangle$ finds that channel certainly dark: $|\langle -z | +z \rangle|^2 = 0$, certain exclusion. Rotate the second magnet to the x -axis and both of its channels fire equally often: since $|+x\rangle = (|+z\rangle + |-z\rangle)/\sqrt{2}$, the Born rule gives $|\langle +x | +z \rangle|^2 = \frac{1}{2}$, and at this orientation no experiment can do better than even odds — irreducible confusion. Between the two extremes, the overlap varies continuously, and with it the degree to which the two preparations can be told apart.

That is the general situation. When $0 < |\langle \phi | \psi \rangle|^2 < 1$, no experiment can decide with certainty in a single shot which of the two preparations is at hand: any test that sometimes accepts ϕ will also accept a ψ -prepared system with frequency $|\langle \phi | \psi \rangle|^2$. Distinguishability is therefore not a binary verdict but a spectrum, and the squared overlap is its operational measure — the frequency with which a ψ -ensemble masquerades as ϕ in the ϕ -test itself. Which experiment minimizes the confusion between an arbitrary pair, and how repeated runs drive the error down, are questions of measurement theory proper and are deferred to the next chapter and its literature.¹⁰

Families of mutually orthogonal normalized states,

$$\langle \psi_m | \psi_n \rangle = \delta_{mn}, \quad m, n = 1, \dots, k, \quad (2.29)$$

¹⁰The general problem of optimal discrimination between non-orthogonal states requires the measurement formalism of the next chapter; what is asserted here is only the negative theorem, which follows from the Born rule alone.

are families of mutually exclusive alternatives: a single ideal experiment can have all of them as its outcomes, since no member of the family ever masquerades as another. Their number is bounded by the dimension of the state space, $k \leq \dim \mathcal{H}$, and a maximal such family has exactly $\dim \mathcal{H}$ members.¹¹ The algebraic dimension of Section 2.4 is hereby tied to counting: it is the size of the largest set of perfectly distinguishable preparations.

Non-orthogonal preparations can be refined into exclusive ones. Given two normalized states with $\langle \psi_1 | \psi_2 \rangle \neq 0$, form

$$|\psi'_2\rangle = \frac{|\psi_2\rangle - |\psi_1\rangle \langle \psi_1 | \psi_2 \rangle}{\sqrt{1 - |\langle \psi_1 | \psi_2 \rangle|^2}}, \quad \langle \psi_1 | \psi'_2 \rangle = 0. \quad (2.30)$$

Read physically, the step subtracts from the second preparation its $|\psi_1\rangle$ -component — the piece responsible for every ψ_1 -test it passes — and normalizes what remains; the remainder never triggers a ψ_1 -test, as the linearity of the inner product confirms in one line: $\langle \psi_1 | \psi'_2 \rangle$ is proportional to $\langle \psi_1 | \psi_2 \rangle - \langle \psi_1 | \psi_1 \rangle \langle \psi_1 | \psi_2 \rangle = 0$. The pair $|\psi_1\rangle, |\psi'_2\rangle$ can then serve as the two alternatives of a single two-outcome experiment. Iterated on any list of independent preparations, each stage subtracting the components along all directions already carved out, the procedure produces a mutually exclusive family: the Gram–Schmidt orthogonalization of algebra is filter refinement written as a formula, carving a preparation into components no two of which can be confused.

The advantage of the orthogonality concept is that it is decidable by frequencies alone: zero against nonzero overlap is the sharpest statement the theory makes about a pair of states, and it requires no hypotheses about hidden labels the systems might carry. Its limitation lies in the clause that gives it strength — “distinguishable” means distinguishable by some allowed experiment *in principle*, not by the apparatus standing on the bench today. This is the same clause that powered the marking experiment of Section 2.1, where tagging the alternatives in principle, with no force acting on the system, was enough to destroy the interference. A maximal set of perfectly distinguishable alternatives is exactly what an ideal experiment can resolve at once — and that is precisely what a basis is.

2.8 Bases as Complete Sets of Alternatives

Section 2.7 identified mutually orthogonal sets of vectors with the mutually exclusive alternatives of an experiment, and it called a maximal such set — one to which no further perfectly distinguishable alternative can be added — the sharpest question the system allows. Such a maximal orthonormal set is an **orthonormal basis** of $\mathcal{H}$. Explicitly, the alternatives $|e_1\rangle, |e_2\rangle, \dots$ resolved by an ideal apparatus obey

$$\langle e_m | e_n \rangle = \delta_{mn}, \quad (2.31)$$

and the only vector orthogonal to all of them is the zero vector. There is one basis per ideal experimental question: the z - and x -oriented Stern–Gerlach magnets of Chapter 1 (Section 1.5) define two different bases of the same two-dimensional state space, as do the H/V and D/A settings of a polar-

¹¹The proof is linear algebra: an orthonormal family is linearly independent (apply $\langle \psi_m |$ to a vanishing combination), whence the bound; a maximal independent family spans the space, whence equality. The spanning half of the statement is where bases enter, in Section 2.8; details are collected in Appendix A.

ization analyzer. Completeness is here an experimental statement before it is a mathematical one: the alternatives exhaust the possible outcomes, so that in every run exactly one of them is realized, and the count of alternatives in the most refined question is the count promised, in advance, by requirement (i) of Chapter 1 (Section 1.4.6).

The first consequence is the **expansion theorem**, the payoff of the whole development of Sections 2.2–2.7: every state expands in every basis, with the transition amplitudes as coefficients,

$$|\psi\rangle = \sum_n c_n |e_n\rangle = \sum_n |e_n\rangle \langle e_n|\psi\rangle, \quad c_n = \langle e_n|\psi\rangle. \quad (2.32)$$

Indeed, with this choice of c_n the algebraic proof is one line:

$$\langle e_m|\psi - \sum_n c_n |e_n\rangle\rangle = \langle e_m|\psi\rangle - \sum_n c_n \underbrace{\langle e_m|e_n\rangle}_{\delta_{mn}} = c_m - c_m = 0.$$

The difference $|\psi\rangle - \sum_n c_n |e_n\rangle$ is therefore orthogonal to every $|e_m\rangle$, hence vanishes by maximality and the positivity of the norm. The physical reading is what matters: *the amplitude to go anywhere is the sum over a complete set of intermediate alternatives of the product of the stage amplitudes*. Prepared as ψ and resolved by the basis apparatus, the system reaches the alternative $|e_n\rangle$ with amplitude $\langle e_n|\psi\rangle$ — the multiplication principle of Section 2.2; the unprejudiced superposition of all channels then reconstitutes the state — the addition principle. If instead the intermediate channel is registered, marked in the sense of Section 2.1, the alternatives become distinguishable in principle, and it is the probabilities, not the amplitudes, that add.

Bracketing the expansion with an arbitrary registration alternative $\langle\phi|$ fuses the two composition principles of Section 2.2 into a single statement:

$$\langle\phi|\psi\rangle = \sum_n \langle\phi|e_n\rangle \langle e_n|\psi\rangle. \quad (2.33)$$

This is the **general composition law**, the keystone equation of the chapter. Every rule met so far is a special case of it: $|\phi\rangle = |e_m\rangle$ recovers the expansion coefficients; $|\phi\rangle = |\psi\rangle$ returns the normalization of Section 2.3; and when the intermediate alternatives are marked rather than coherent, the same insertion degrades into the classical sum of products of probabilities that Section 2.3 exhibited for the sequential-filter experiments of Chapter 1. The law holds for *any* complete set of intermediate alternatives: the value $\langle\phi|\psi\rangle$ on the left does not depend on which basis was inserted on the right, and this basis-independence of the sum is itself an experimental statement — the amplitude between a preparation and a registration is a property of the pair, not of the question one chooses to interpose.

The expansion theorem has an abbreviation used so constantly in the literature that it must be displayed here, together with an explicit caution. If the symbol $|e_n\rangle \langle e_n|$ is agreed to act on any ket $|\psi\rangle$ by yielding the vector $|e_n\rangle \langle e_n|\psi\rangle$, then the expansion (2.32) says that $\sum_n |e_n\rangle \langle e_n|$ returns every ket unchanged, and one writes

$$\sum_n |e_n\rangle \langle e_n| = I. \quad (2.34)$$

In this chapter, (2.34) is a bookkeeping shorthand and nothing more: a one-line compression of the expansion theorem. No notion of operator has been introduced, and the right-hand side is not yet “the identity operator on $\mathcal{H}$ ”; that reading, and the algebra of the individual symbols $|e_n\rangle \langle e_n|$, belongs to

the operator theory of the next chapter.

The coefficients of the expansion carry the probabilities of the basis alternatives, $P_n = |c_n|^2$, so the exhaustiveness of the question appears as **Parseval's identity** — the normalization condition of Section 2.3, now derived rather than postulated. Writing $|\phi\rangle = \sum_n b_n |e_n\rangle$, the composition law also delivers the inner product in coordinates:

$$\langle\phi|\psi\rangle = \sum_n b_n^* c_n, \quad \langle\psi|\psi\rangle = \sum_n |c_n|^2 = 1. \quad (2.35)$$

The alternatives are exhaustive: the system is certainly found in one of them.

One unifying paragraph discharges a debt incurred in Section 2.4, where the dimension of $\mathcal{H}$ was defined algebraically and its operational meaning was promised here. Three counts coincide: the number of vectors in any orthonormal basis (the algebraic dimension of Section 2.4); the maximal number of mutually exclusive alternatives (Section 2.7); and the number of answers of the most refined ideal experiment the system supports. The dimension of a state space is therefore not an algebraic ornament but an experimental count. For the spin- $\frac{1}{2}$ system and for photon polarization the count is two; for the energy alternatives of a bound atom it is countably infinite — a possibility whose consistency Section 2.9 now has to redeem.

A change of basis is a change of the experimental question. Let $\{|e_n\rangle\}$ and $\{|e'_m\rangle\}$ be two bases, two ideal apparatuses interrogating the same system. The same state carries two columns of coefficients, related by inserting the composition law:

$$c'_m = \langle e'_m|\psi\rangle = \sum_n \langle e'_m|e_n\rangle \langle e_n|\psi\rangle = \sum_n U_{mn} c_n, \quad U_{mn} = \langle e'_m|e_n\rangle. \quad (2.36)$$

The entries of the **overlap matrix** U are the transition amplitudes between the alternatives of the two questions; nothing in its definition refers to operators, and none is needed. Inserting the primed basis into $\langle e_m|e_n\rangle = \delta_{mn}$ gives

$$\sum_k U_{km}^* U_{kn} = \sum_k \langle e_m|e'_k\rangle \langle e'_k|e_n\rangle = \langle e_m|e_n\rangle = \delta_{mn}. \quad (2.37)$$

A matrix with this property is called *unitary*. The unitarity of the overlap matrix and the conservation of probability are one and the same fact: both experimental questions are exhaustive. The vocabulary of “unitary operators” belongs to the next chapter; (2.37) is, strictly, a relation between two coordinate systems for the same physics.

The content of these displays is already familiar in the two-state laboratory of Chapter 1. Let the second question be a Stern–Gerlach apparatus rotated to the axis $\mathbf{n}$ with polar angles (θ, φ) relative to z . Its two alternatives, expressed in the z -basis, are

$$|+\mathbf{n}\rangle = \cos \frac{\theta}{2} |+\mathbf{z}\rangle + e^{i\varphi} \sin \frac{\theta}{2} |-\mathbf{z}\rangle, \quad |-\mathbf{n}\rangle = -e^{-i\varphi} \sin \frac{\theta}{2} |+\mathbf{z}\rangle + \cos \frac{\theta}{2} |-\mathbf{z}\rangle, \quad (2.38)$$

an orthonormal pair, as direct substitution in (2.31) confirms. A beam prepared in $|+\mathbf{z}\rangle$ and interrogated along $\mathbf{n}$ emerges in the $+\mathbf{n}$ channel with frequency

$$|\langle +\mathbf{z} | +\mathbf{n} \rangle|^2 = \cos^2 \frac{\theta}{2}, \quad (2.39)$$

the spin analogue of Malus' cosine-squared law (Section 1.5). An honesty clause, mirroring that of Section 2.5: the half-angle law (2.39) is recorded here purely as an experimental fact of rotated

Stern–Gerlach apparatuses. Its *derivation* — the half-angle $\theta/2$ is the kinematics of a spinor under rotation, rather than a property of ordinary vectors in space — is a debt owed to the rotation theory of the chapter on angular momentum, where the two-valuedness of the spin- $\frac{1}{2}$ state acquires its group-theoretic explanation. No $SU(2)$ machinery is spent in this chapter. The special case $\theta = \pi/2$, $\varphi = 0$ is Chapter 1’s x -basis, $|\pm x\rangle = (|+z\rangle \pm |-z\rangle)/\sqrt{2}$ (Section 1.5); its overlap matrix is concretely

$$U^{(z \rightarrow x)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (2.40)$$

with rows indexed by the x -alternatives and columns by the z -alternatives. The unitarity relation (2.37) follows by direct multiplication:

$$UU^\dagger = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I.$$

The example plants a seed that the chapters on dynamics and on angular momentum will harvest: rotations of the apparatus act on states through overlap matrices, and the family of all such matrices carries the structure of the rotation group.

This is the place to repay requirement (v) of Chapter 1 (Section 1.4.6). The state is the abstract vector $|\psi\rangle$; every basis assigns it a coordinate column; no column is the state. The physics lives in the overlaps $\langle\phi|\psi\rangle$, basis-independent amplitudes that the composition law (2.33) computes in any basis with the same value. What the equivalence proofs of 1926 established historically (Section 1.4.4) — that matrix mechanics and wave mechanics are one theory written in two coordinate systems — appears here as a theorem of the structure rather than a surprise.

The advantage of the basis calculus is freedom. One computes in the basis matched to the apparatus on the bench, translates to any other by an overlap matrix, and reduces every question about alternatives to arithmetic with amplitudes; the state space becomes fully navigable. Its limitation is the seduction of coordinates. A column of coefficients is a representation, not the state — the very reification against which Section 2.4 warned — and the completeness shorthand (2.34), for all its convenience, invites an operator reading that this chapter has deliberately not yet earned.

Everything so far is exact for finite sets of alternatives. But the physics of Chapter 1 refuses finiteness: spectra come with infinitely many lines, and a particle can be found at infinitely many positions. What survives of the basis calculus when the number of alternatives is unlimited?

2.9 Hilbert Space, Finite and Infinite

Two facts of Chapter 1 overflow the finite algebra developed so far: the line spectra of the atoms enumerate countably infinitely many energy alternatives, and the diffraction experiments register particles at continuously many positions. Both extensions are made by a single analytic step, and that step is physics, not mathematical taste. A preparation procedure is never exact; it can only be refined, stage by stage, and a sequence of preparations that settles down — whose states come arbitrarily close to one another — must settle down *to a state*. A theory in which such a sequence of physical states had no limit state would be incomplete in the operational sense: it would declare an arbitrarily well-

approximated preparation unphysical. The requirement is made precise in one display. A sequence $|\psi_1\rangle, |\psi_2\rangle, \dots$ is called a **Cauchy sequence** if its members eventually cluster, and the space is required to contain their limit:

$$\|\psi_n - \psi_m\| \longrightarrow 0 \quad (n, m \rightarrow \infty) \quad \implies \quad \exists |\psi\rangle \in \mathcal{H} : \|\psi_n - \psi\| \longrightarrow 0. \quad (2.41)$$

The left-hand side says that the preparations become indistinguishable from one another at any prescribed tolerance; the right-hand side says that the theory recognizes a limiting preparation.

An inner-product space obeying (2.41) is called **complete**, and a complete inner-product space is a **Hilbert space**. This book adds one further assumption, **separability**: the space admits a countable orthonormal basis, so that the alternatives of any ideal experiment can be enumerated.¹² Every state space of this book is therefore a separable complex Hilbert space. No axiom of Sections 2.1–2.8 is touched by the extension; only the bookkeeping of infinity is new.

The countable case comes first. With a basis $|e_1\rangle, |e_2\rangle, \dots$ indexed by the natural numbers, the expansion theorem and the composition law of Section 2.8 survive verbatim, the sums now being series that converge in norm:

$$|\psi\rangle = \sum_{n=1}^{\infty} c_n |e_n\rangle, \quad \sum_{n=1}^{\infty} |c_n|^2 < \infty, \quad \langle\phi|\psi\rangle = \sum_{n=1}^{\infty} b_n^* c_n. \quad (2.42)$$

The space of coefficient columns with $\sum_n |c_n|^2 < \infty$ is called ℓ^2 ; Parseval's identity is unchanged, and with it the Born probabilities over the complete set of alternatives still sum to one. This is the arena of every bound system in this book: the energy alternatives of the harmonic oscillator and of the particle in a box are countably infinite, and their concrete theory arrives in the chapter on operator methods and representations.

The continuous case must be handled honestly. A particle in the diffraction experiments of Chapter 1 can be registered at any of continuously many positions; *position is a label* for such registration alternatives — nothing more is claimed here, and no position observable has been defined. The label is introduced as a limit of discrete alternatives, by the mechanism exhibited in the example below; the shorthand itself is quickly stated. One writes

$$\psi(x) = \langle x|\psi\rangle \quad (2.43)$$

for the amplitude of the state $|\psi\rangle$ to be registered at the position x — the **wave function** of the state — and one extends the orthonormality condition (2.31) to the continuum by writing

$$\langle x|x'\rangle = \delta(x - x'). \quad (2.44)$$

 **Note** The ket $|x\rangle$ is not a vector of $\mathcal{H}$: its formal “norm” $\langle x|x\rangle = \delta(0)$ diverges, and no normalization convention repairs the defect. The symbol $\delta(x - x')$ is a distribution, not a function, and (2.44) is a bookkeeping shorthand — a powerful calculus whose honest justification, the rigged Hilbert space, is deferred to Appendix A.¹³ The genuine vectors of the continuum theory are the square-integrable wave

¹²The rigorous development — completeness, separability, and the theorem that every separable infinite-dimensional Hilbert space is isomorphic to ℓ^2 — is deferred to Appendix A.

¹³The physical exploitation of the position representation — wave packets, propagators, and the recovery of Chapter 1's matter waves (Section 1.3.2) — begins in the chapter on operator methods and representations.

functions,

$$\int |\psi(x)|^2 d^3x < \infty, \quad (2.45)$$

which form the space $L^2(\mathbb{R}^3)$; every statement made with the shorthand can be re-expressed, at the price of some care, among such vectors alone.

The mechanism by which discrete alternatives fuse into a continuum is visible in a single diffraction experiment. Let N narrow slits at positions $x_n = nd$ be equally illuminated, so that each slit is an alternative $|n\rangle$ with $\langle n|\psi\rangle = 1/\sqrt{N}$, and let a distant screen register arrivals at the point x . By the composition law, the arrival amplitude is the sum over the slit alternatives,

$$A_N(x) \propto \frac{1}{\sqrt{N}} \sum_{n=1}^N e^{ik_n x}, \quad k_n = \frac{nk d}{L}, \quad (2.46)$$

in the Fraunhofer regime, with $k = 2\pi/\lambda$ and L the screen distance: the multiplication principle supplies the stage amplitudes $\langle x|n\rangle \propto e^{ik_n x}$ and the addition principle sums them. Now let the slits multiply and close up, $N \rightarrow \infty$ and $d \rightarrow 0$ with the aperture $a = Nd$ held fixed. The sum becomes an integral over a continuous aperture coordinate x' ,

$$\begin{aligned} A(x) &\propto \int_{-a/2}^{a/2} e^{ikx'x/L} dx' \\ &= \frac{L}{ikx} e^{ikx'x/L} \Big|_{-a/2}^{a/2} \\ &= \frac{L}{ikx} (e^{ikax/2L} - e^{-ikax/2L}) \\ &= \frac{2L}{kx} \sin \frac{kax}{2L}, \end{aligned} \quad (2.47)$$

the familiar diffraction envelope of a single aperture. The discrete slit alternatives have fused into continuously many position alternatives labeled by x' ; the sum over n has become an integral over x' . If the composition law itself is to survive the fusion, with $\sum_n$ replaced by $\int dx'$, one must be able to write

$$\langle x|\psi\rangle = \int \langle x|x'\rangle \langle x'|\psi\rangle dx' \quad (2.48)$$

for every state, and this is possible only if $\langle x|x'\rangle$ vanishes for $x \neq x'$ while its integral over x' is one — precisely the two demands that define the symbol $\delta(x - x')$. The δ -normalization (2.44) is therefore not an additional axiom but the price of extending the keystone law (2.33) to a continuum of alternatives; the note above states exactly what that price buys.

It is worth listing what the extension to infinite alternatives does and does not change. Unchanged: the Born rule (with probability densities replacing probabilities over continuous labels), rays as physical states, superposition, orthogonality as perfect distinguishability, and the existence of countable orthonormal bases. Changed: only the level of analytic care — convergence of series, the status of shorthand kets such as $|x\rangle$ — and not a single axiom.

The advantage of the completed arena is its minimality together with its reach. The separable complex Hilbert space is the smallest structure meeting all five kinematical requirements of Chapter 1 (Section 1.4.6) — discrete alternatives, data attached to pairs of states, amplitudes composed before

probabilities, order-sensitive processes (previewed in Section 2.2 and still owed a formal theory), and representation-independence — and it houses a two-state system and a particle in space under one roof. Its limitation is structural: the Hilbert space says what states *are*, not yet what observables *do*. A position label is an alternative, not a measurement; and the magnets and polarizers of Chapter 1 act on states in ways this kinematics has not described.

The arena is finished. It remains to collect the axioms, to take stock of which of Chapter 1’s debts this chapter has paid, and to name the question that this kinematics cannot answer.

2.10 The State-Space Axioms: Summary and Open Debts

The construction of this chapter is complete, and its content compresses into five statements. Each is an axiom in the formal sense and an experimental report in the operational one; the warrant and the sections that earned it are attached to each.

(S1) States. The states of a quantum system are the rays of a separable complex Hilbert space $\mathcal{H}$ (Sections 2.4, 2.5, 2.9). *Warrant:* superpositions of alternatives are preparable, and an overall phase is invisible to every experiment.

(S2) Amplitudes. To each pair of states is attached a complex transition amplitude, the inner product $\langle\phi|\psi\rangle$ (Section 2.6). *Warrant:* the observable data live on preparation–registration pairs — requirement (ii) of Chapter 1 (Section 1.4.6).

(S3) Born rule. The frequency of the transition $\psi \rightarrow \phi$ on an ensemble of identically prepared systems is $P(\psi \rightarrow \phi) = |\langle\phi|\psi\rangle|^2$ (Section 2.3). *Warrant:* the interference and counting experiments of Chapter 1, read through Born’s interpretation of 1926 (Section 1.4.3).

(S4) Composition. Amplitudes add over indistinguishable alternatives and multiply over successive stages; over any complete set of alternatives they obey the general composition law $\langle\phi|\psi\rangle = \sum_n \langle\phi|e_n\rangle \langle e_n|\psi\rangle$ (Sections 2.2, 2.8). *Warrant:* the interferometers and sequential filters of Chapter 1 (Section 1.5).

(S5) Ideal experiments. An ideal experiment is a complete set of mutually exclusive alternatives, represented by an orthonormal basis of $\mathcal{H}$ (Sections 2.1, 2.7, 2.8). *Warrant:* maximal sets of perfectly distinguishable states have exactly $\dim \mathcal{H}$ members.

The chapter’s integration of experiment and formalism is collected, in lieu of a new example, in the two-column dictionary of Table 2.1: every laboratory notion of Chapter 1 has acquired a precise formal object, and no formal object has been admitted without a laboratory warrant.

The mainline duty of this book is a ledger of debts, and the ledger must now be stated plainly. *Repaid:* requirements (i), (ii), (iii), and (v) of Chapter 1 (Section 1.4.6) — discrete alternatives (S5), data on pairs of states (S2), amplitudes composed before probabilities (S3, S4), and a representation-independent notion of state (Section 2.8) — while the two-state grammar of Chapter 1 (Section 1.5) has served throughout as running evidence, cited and never re-derived. *Deferred, and here declared openly:* requirement (iv), the order-sensitivity of physical operations, has been *motivated* only — a stage X followed by a stage Y is experimentally a different arrangement from Y followed by X (Section 2.2) — but its formal repayment requires the theory of linear maps on $\mathcal{H}$, and is owed by the

Table 2.1: The experiment–formalism dictionary assembled in this chapter.

Experimental notion	Formal object
A single alternative of an apparatus	a basis vector
Indistinguishable alternatives	a sum of amplitudes
Successive processes	a product of amplitudes
A registered frequency	the squared modulus of an amplitude
Perfect distinguishability	orthogonality
A complete experimental question	an orthonormal basis
A change of the experimental question	an overlap matrix
A physical state	a ray of $\mathcal{H}$

operator theory of the next chapter and the transformation theory of the chapter on dynamics. The continuum shorthand of Section 2.9 owes its rigor to Appendix A and its physical development to the chapter on operator methods and representations. And the filtering of Section 2.7 remained, deliberately, at Chapter 1’s operational level of selection-as-preparation; the theory of measurement proper belongs to the next chapter. The kinematics is complete; the dynamics and the theory of observation are not yet born.

The state is a ray. But the experiments that forced this kinematics upon us — polarizers in sequence, magnets in sequence — act on states, and their order matters. How do observables act on states? That is the theory of operators and measurement, and it is the next chapter.

Chapter 2 Exercise

1. (★) (Section 2.1) Two preparation procedures are applied to a beam of single photons. Procedure A passes each photon through a linear polarizer oriented at 45° to the horizontal. Procedure B splits each photon coherently into H and V components of equal amplitude and recombines them with zero relative phase. By enumerating the outcome frequencies of both preparations in the H/V , D/A , and R/L bases, show that A and B define the same operational state. Procedure C is procedure B with an additional quarter-wave relative phase $\pi/2$ inserted in one arm: exhibit a basis in which C is operationally distinct from A, and identify the state C prepares.
2. (★) (Section 2.2) A symmetric three-way splitter divides the amplitude equally among three paths, $1/\sqrt{3}$ into each, and path n accumulates an adjustable phase φ_n ($n = 1, 2, 3$). The paths are recombined at a second symmetric element, each of whose output ports sums the three path amplitudes with weight $1/\sqrt{3}$ and with relative phases $\{0, 0, 0\}$ (port C) and $\{0, 2\pi/3, 4\pi/3\}$ (port D). Assemble the total amplitudes at C and D from the two composition principles — products along each stage, sums over the paths — and compute the two detector probabilities P_C and P_D as functions of $\varphi_1, \varphi_2, \varphi_3$. Now tag each path with an orthogonal internal marker state, and show that every cross term $2\cos(\varphi_m - \varphi_n + \dots)$ disappears from both ports.
3. (★★) (Section 2.3) Relative to an orthonormal basis $\{|e_1\rangle, |e_2\rangle, |e_3\rangle\}$, a preparation ψ has the unnormalized amplitudes $(2, i, -1)$, and a second preparation ϕ has the unnormalized amplitudes $(1, 1, i)$. (a) Normalize both states. (b) Compute the probabilities of the three outcomes

for each preparation, and verify that they are exhaustive. (c) Compute $\langle\phi|\psi\rangle$ and $\langle\psi|\phi\rangle$, and verify the reciprocity law $P(\psi \rightarrow \phi) = P(\phi \rightarrow \psi)$.

4. (**) (Section 2.4) Take the six polarization states

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \quad |A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}, \quad |R\rangle = \frac{|H\rangle + i|V\rangle}{\sqrt{2}}, \quad |L\rangle = \frac{|H\rangle - i|V\rangle}{\sqrt{2}}, \quad (2.49)$$

together with $|H\rangle$ and $|V\rangle$, written as columns in the H/V basis. (a) Show that the six states span $\mathbb{C}^2$. (b) Determine which pairs of distinct states among the six are linearly independent, and state what your answer implies for the number of mutually exclusive alternatives of a photon's polarization. (c) Express $|A\rangle$ in the R/L basis, and verify that the coefficients are the transition amplitudes demanded by the expansion theorem.

5. (**) (Section 2.5) In a neutron interferometer the state before recombination is $(|u\rangle + e^{i\varphi}|l\rangle)/\sqrt{2}$, and the detector at the bright port registers $P_0 = \cos^2(\varphi/2)$, the Mach–Zehnder law of Chapter 1 (Section 1.5, Figure 1.2). A spin rotator inserted in the lower arm performs a 2π rotation of the neutron's spin, multiplying that arm's amplitude by -1 — the experimental fact asserted in Section 2.5, whose derivation is owed to the chapter on angular momentum. (a) Compute the modified fringe law $P_0(\varphi)$ in the presence of the rotator, and show that the interference pattern is shifted by half a period. (b) Show that the same factor -1 applied to *both* arms leaves all detector statistics unchanged, and state precisely which phase — global or relative — each part of this experiment probes.

6. (**) (Section 2.6) Let $|\psi\rangle$ and $|\phi\rangle$ be normalized. (a) Using only positivity and conjugate symmetry, applied to the vector $|\chi\rangle = |\psi\rangle - \langle\phi|\psi\rangle|\phi\rangle$, derive the probability bound

$$|\langle\phi|\psi\rangle|^2 \leq 1, \quad (2.50)$$

and show that equality holds if and only if $|\psi\rangle$ and $|\phi\rangle$ belong to the same ray. (b) Verify the bound and its saturation clause on the explicit spin- $\frac{1}{2}$ pairs $(|+z\rangle, |+\mathbf{x}\rangle)$ and $(|+z\rangle, \cos(\theta/2)|+z\rangle + \sin(\theta/2)|-z\rangle)$.

7. (***) (Sections 2.7 and 2.8) A Stern–Gerlach apparatus is rotated by an angle θ about the y -axis, so that its axis $\mathbf{n}$ lies in the x - z plane (azimuth $\varphi = 0$). Take the rotated basis as given — by Section 2.8's example,

$$|+\mathbf{n}\rangle = \cos\frac{\theta}{2}|+z\rangle + \sin\frac{\theta}{2}|-z\rangle, \quad |-\mathbf{n}\rangle = -\sin\frac{\theta}{2}|+z\rangle + \cos\frac{\theta}{2}|-z\rangle, \quad (2.51)$$

whose construction is a debt owed to the rotation theory of the chapter on angular momentum; the present exercise is verification only. (a) Expand $|+\mathbf{x}\rangle = (|+z\rangle + |-z\rangle)/\sqrt{2}$ in the $\{|+\mathbf{n}\rangle, |-\mathbf{n}\rangle\}$ basis. (b) Compute the four overlaps $\langle\pm\mathbf{n}|\pm z\rangle$, i.e. all entries of the overlap matrix U with rows indexed by the $\mathbf{n}$ -alternatives and columns by the z -alternatives. (c) Verify explicitly that U obeys $\sum_k U_{km}^* U_{kn} = \delta_{mn}$.

8. (***) (Sections 2.8 and 2.9) (a) Let $\{|e_n\rangle\}$ be an orthonormal set. Prove that

$$\sum_n |\langle e_n|\psi\rangle|^2 = 1 \quad \text{for every normalized } |\psi\rangle \quad (2.52)$$

if and only if the set is complete, i.e. no nonzero vector is orthogonal to all of them. Describe the failure mode of a truncated basis by computing the left-hand side for the set $\{|+z\rangle\}$

alone when the system is prepared in $|+x\rangle$. (b) Chapter 1 introduced de Broglie's matter wave
 (Section 1.3.2), in which a particle of definite momentum is represented in position space by
 $\psi_\lambda(x) = A e^{2\pi i x/\lambda}$. (This chapter introduces no momentum label $|p\rangle$; the de Broglie wave func-
 tion is the whole input.) Compute the formal "norm" $\int_{-\infty}^{+\infty} |\psi_\lambda(x)|^2 dx$ and show that it diverges.
 Then regularize by confining the wave to a box $[-L/2, L/2]$ with periodic boundary conditions,
 normalize for finite L , and discuss the limit $L \rightarrow \infty$. Explain why no vector of $\mathcal{H} = L^2(\mathbb{R})$
 represents this alternative, and reconcile your conclusion with the δ -normalization shorthand of
 Section 2.9.

1995

Chapter 3 Measurement, Operators, and Spectra



Note *This chapter is inductive in character. Taking the state-space axioms of Chapter 2 as input, it constructs the theory of measurement and observables from the experimental logic of filters and counters: an ideal measurement as a reproducible filter, projectors as the operators of yes-or-no experiments, projection-valued measures as the bookkeeping of complete experimental questions, observables as self-adjoint operators, the spectral theorem proved in finite dimension and honestly deferred in the infinite one, degenerate spectra and compatible observables, commutators as the arithmetic of incompatibility, and the uncertainty relations as its quantitative shadow. The continuous spectrum is previewed and referred to Appendix A; the chapter closes with the measurement axioms collected in one place. It presupposes nothing beyond Chapter 2 and elementary linear algebra.*

The preceding chapter closed its ledger with an explicit question. With the state-space axioms in hand — states are rays of a separable complex Hilbert space, amplitudes are inner products, frequencies are squared moduli of amplitudes, and an ideal experiment is a complete set of mutually exclusive alternatives (Section 2.10) — it asked: “The state is a ray. But the experiments that forced this kinematics upon us — polarizers in sequence, magnets in sequence — act on states, and their order matters. How do observables act on states?” This chapter is the answer, and it is an answer of the same inductive kind as the question: no formal object is admitted here that has not first been earned from an experimental arrangement.

The answer itself is not new, and this chapter does not pretend otherwise. The general formalism was found in one stroke by the founders of the theory: von Neumann’s treatise of 1932 placed observables as self-adjoint operators on a Hilbert space and cast their statistical predictions into a trace rule; Heisenberg’s relation of 1927 set a quantitative limit on the joint determination of certain pairs of quantities; and Bohr’s doctrine of complementarity, announced at Como in September of that year, named the conceptual frame in which mutually exclusive experimental arrangements complete rather than contradict one another (Section 1.4.5). Chapter 1 recorded those achievements and announced that they would be developed systematically in the chapter on measurement. This is that chapter. But where the synthesis was found whole, it is here re-earned piece by piece, so that every clause of the formalism is tied to the laboratory fact that forces it; the slow route — filters, counters, and the arithmetic of repeatable questions — is the route this book has chosen.

The route is deliberately narrow at the start. An ideal measurement is first characterized operationally, as a reproducible filter: a device that sorts an incoming ensemble into a complete set of mutually exclusive outputs, whose selected outputs pass the same test again with certainty (Section 3.1). Repeatability is the single new empirical clause of the chapter, and it does more work than any axiom in this book: together with the Born rule and the state-space geometry already established, it *forces* the conditional state after a sharp measurement outcome — a theorem, not a postulate. The filter’s action on states is the first operator of the book, the projector (Section 3.2); a complete experimental question, with its outcomes grouped, ignored, or refined, is a projection-valued measure (Section 3.3); attaching instrument readings to the outcomes compresses the whole measure into one operator, the demand

that all readings be real forces that operator to be self-adjoint (Section 3.5), and the spectral theorem then reverses the arrow: every self-adjoint operator encodes an ideal experiment. Degenerate spectra lead to compatible observables and to the commutator as the measure of incompatibility (Sections 3.6 and 3.7), and the commutator, through the Cauchy–Schwarz theorem admitted in Chapter 2, yields the uncertainty relations (Section 3.8). One clause is honestly left open. The theory built here speaks only of interrogations: a preparation, a question, an answer. What law governs the state between two interrogations no axiom of this chapter contains; that is the next chapter’s opening question, and it is posed, with the ledger balanced, at this chapter’s end (Section 3.10).

Two old debts guide the construction. The fourth kinematical requirement of Chapter 1 — that the order of physical operations matters (Section 1.4.6) — and the matrix multiplication rule of 1925, with its insistence that $AB \neq BA$ (Section 1.4.1), receive their kinematical repayment here: successive experiments compose as operator products, and noncommutation is the arithmetic of filter order. The open debts of Chapter 2 are repaid in kind: the theory of measurement proper, deferred at its close; the operator reading of the completeness shorthand (2.34); the uniqueness theorem for the probability rule signposted in Section 2.3; and the uncertainty relations, built on the Cauchy–Schwarz inequality (2.24). What remains open is declared, as always, at the exit.

3.1 From Selection to Measurement: Reproducible Filters

The state space of the preceding chapter was built out of experiments, yet the experiments themselves were left deliberately schematic. An ideal experiment was identified with a complete set of mutually exclusive alternatives and represented by an orthonormal basis of $\mathcal{H}$ (Sections 2.8 and 2.10); a filter — a polarizer passing one component and absorbing the rest, a magnet whose unwanted beam is blocked — served as selection-as-preparation (Section 2.7), an operational device for making states. What was not given is what this section must now supply: a precise statement of what an ideal measurement *is*. The frequencies recorded against a basis are the raw data of the kinematics, but the act of interrogation that produces them has so far been only a silent change of basis. The theory of measurement begins by listening to the device.

A filter does two things at once, and measurement theory is the recognition that the two are one. Faced with an incoming ensemble, it *interrogates*: the counters at its output ports accumulate frequencies, and those frequencies are the probabilities of the alternatives, read off through the Born rule (2.7). And it *prepares*: an output port, once selected, delivers a system in a definite state, exactly as the selections of the preceding chapter prepared $|+z\rangle$ beams and H -polarized photons. Interrogation and preparation are two readings of a single act.

The cleanest realization is an optical-bench device that will serve as this section’s running example. A **polarizing beam splitter** sorts single photons by polarization without destroying either component: the horizontally polarized component is transmitted, the vertically polarized component is reflected, and both outputs leave the device as usable beams. Sent one photon at a time, in the feeble-light discipline of Chapter 1’s two-state experiments (Section 1.5), an unpolarized or diagonally polarized input registers as single clicks at two counters, never both at once. The device is a

filter with its two faces made visible: its port counts interrogate the incoming state in the H/V basis, and each of its output beams is a prepared state, H at the transmitted port and V at the reflected one.

One further property of such a device is an experimental fact, and it is the single new empirical clause of this chapter. Take either output of the beam splitter and interrogate it with a second, identical analyzer: the transmitted H beam passes the second test as H with certainty, and the reflected V beam as V . The same fact was met on the magnetic bench of Chapter 1: a beam selected as $|+z\rangle$, re-interrogated along z , returns $+z$ with certainty (Figure 1.4). A filter's answer, once given, is *reproducible*: an immediate re-asking of the same question returns the same answer. This is not a theorem about states; it is a property of good instruments, and it is what distinguishes a measurement from a one-shot scattering.

The postulate of this section can now be stated. It comes as two axioms with a theorem between them, and the honesty of the chapter requires that the three not be confused.

Axiom M1 (ideal measurement). An ideal measurement of a system is a complete set of mutually exclusive alternatives $\{|e_n\rangle\}$ — an orthonormal basis of $\mathcal{H}$ in the sense of Section 2.8 — realized by a **reproducible filter**: an apparatus with one output port per alternative, whose selected output, immediately re-interrogated by the same question, returns the same alternative with certainty. In terms of the outcome probabilities, repeatability is the empirical identity

$$P(n \text{ then } n) = p_n \cdot 1 = p_n, \quad (3.1)$$

the probability of n at the first interrogation times *one* at the second.

Axiom M2 (outcome probabilities). For a system prepared in the state ψ , the probability of outcome n is the Born rule of the preceding chapter (2.7),

$$p_n = |\langle e_n | \psi \rangle|^2, \quad \sum_n p_n = 1, \quad (3.2)$$

the frequencies of the port counts on an ensemble of identically prepared systems; the sum is one because the alternatives are exhaustive.

Between the two axioms sits a fact that is proved, not postulated — the first theorem of measurement theory.

Theorem (rank-1 update). Conditional on the outcome n of an ideal measurement performed on the state ψ , the state of the selected output is the ray of $|e_n\rangle$.

The proof uses nothing beyond repeatability and the state-space geometry of the preceding chapter. A selected output is a preparation, so the axioms of Chapter 2 assign it a ray; let $|\chi\rangle$ be a normalized representative of the state conditional on outcome n . Repeatability (M1) demands that an immediate re-interrogation with the same question return n with certainty, $|\langle e_n | \chi \rangle|^2 = 1$. By the saturation clause of the Cauchy–Schwarz inequality (2.24) — equality holds if and only if the two states belong to the same ray — this forces

$$|\chi\rangle = e^{i\alpha} |e_n\rangle, \quad \text{or, in unnormalized bookkeeping form,} \quad |\psi\rangle \longmapsto |e_n\rangle \langle e_n | \psi \rangle, \quad (3.3)$$

where the overall phase is physically irrelevant by the ray theorem of Section 2.5. The second form of (3.3) keeps the record of the incoming state: the squared norm of the right-hand side is $|\langle e_n | \psi \rangle|^2 = p_n$, so the unnormalized output carries the outcome's probability as its length. The conditional ray is forced

mathematics: given that re-asking returns the same answer with certainty, no state other than the ray of $|e_n\rangle$ is available. The symbol $|e_n\rangle \langle e_n|\psi\rangle$ is used here operationally — extract the $|e_n\rangle$ -component of $|\psi\rangle$ — and receives its precise operator status in Section 3.2.

The bookkeeping of honesty must be kept carefully, for it is the brand of this chapter. The experimental input is repeatability and nothing more: that a $|+z\rangle$ -selected beam, sent through a second z -magnet, returns $+z$ with certainty is a fact of the bench (Figure 1.4), and its optical twin is the re-test of the beam-splitter outputs described above. Everything else in the update rule is deduced: the Born rule prices the outcomes (M2), and the saturation clause of (2.24) then forces the conditional ray. For a *sharp* question — one whose alternatives are rays, with no two outcomes sharing an answer — the state after a measurement is therefore proved, not postulated. An axiom appears only where repeatability stops forcing the answer. When several orthogonal alternatives carry the same instrument reading, repeatability confines the conditional state to a subspace but does not fix it within the subspace; the update rule for that *degenerate* case is the one genuinely new measurement axiom of the chapter, and it is stated at the point of first consumption, in Section 3.6 (Axiom M3).

The beam splitter exhibits every clause of the postulate on one bench. A photon prepared diagonal, $|D\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$ in the notation of Section 2.4, exits the two ports at fifty-fifty — the port counts realize $p_H = p_V = \frac{1}{2}$ by (M2) — and each output thereafter behaves as H or as V respectively: re-tested, the transmitted beam never fires the V counter, and the reflected beam never fires the H counter. The update theorem is read off directly: selection at a port is conditioning on an outcome, and the selected beam's subsequent statistics are those of the corresponding basis ray. The example also shows what the word *ideal* idealizes away. An absorptive Polaroid sheet is a filter with only one port: it transmits H and destroys V , so that no re-testable V output exists. Its transmitted beam is the same prepared state, but the rejected alternative leaves no beam at all, and with it no possibility of confirming repeatability for both outcomes; the beam splitter, not the sheet, is the complete instrument.

The same theorem quantifies what an intermediate measurement disturbs. Chapter 1's decisive arrangement — a z -preparation, an x -interrogation, a final z -question (Figure 1.4) — showed that the certainty of the first z -answer is lost when the x -question is interposed. The mechanism is now arithmetic. After the x -stage returns an outcome, say $+x$, the update theorem leaves the system in the ray of $|+x\rangle = (|+z\rangle + |-z\rangle)/\sqrt{2}$, and the final z -question prices that state by (M2): $|\langle +z | +x \rangle|^2 = \frac{1}{2}$; the $-x$ branch contributes the same. The final counts are $+z$ and $-z$ in equal shares, whatever the first stage prepared. The intermediate measurement has not revealed a pre-existing x -value while leaving the z -value intact; it has re-prepared the state. This is the marking clause of Section 2.1 at work: the x -outcomes are registered alternatives, distinguishable in principle, and between marked alternatives it is probabilities, not amplitudes, that combine — the coherence that would have carried the z -certainty through the x -stage is gone.

One variant of the staged arrangement must be named now, though its full bookkeeping belongs to a later chapter. Suppose the outcomes of an intermediate measurement are recorded — the alternatives remain marked — but no port is selected: every branch is allowed to proceed, and later counters see the whole ensemble. This is a **non-selective measurement**. Its description in the language of this chapter

is outcome by outcome: the ensemble is the union of its branches, each branch a pure conditional state weighted by its outcome probability, and later frequencies are the probability-weighted sums of the branches' frequencies. A description of the whole ensemble by a single statistical object exists — the density operator — and is built in the chapter on composite systems; it is not needed here, because the branch-by-branch account is complete.

Note Terminology. *Throughout this book, measurement means the ideal case of this section: a complete set of alternatives realized by a reproducible, minimally disturbing filter — the idealization of von Neumann and Lüders. Real instruments approximate it to varying degrees; the absorptive sheet above is already an approximation with one port missing. Generalized measurement schemes, in which the outcomes need not correspond to orthogonal alternatives, exist and have their uses, but they lie outside the main line of this book.*¹

The advantage of defining measurement by reproducible filtering is that the postulate is phenomenological through and through: every clause — completeness of the alternatives, the Born pricing of outcomes, the conditional ray — is either a bench fact or a deduction from bench facts, and the theory of measurement is thereby anchored in the same operational soil as the state space itself. Its limitation is that the idealization is strong. Repeatability is a property of selected instruments in selected regimes, not a universal law of devices, and the minimally disturbing clause will need its own axiom the moment outcomes become degenerate (Section 3.6). What the definition does immediately is turn the filter into a mathematical object. The rule *keep the $|e_n\rangle$ -component of a state and discard the rest* is the first instance of a map on $\mathcal{H}$ — an operator — and the next section makes it precise and finds its algebra.

3.2 Projectors: the Operators of Yes-or-No Experiments

The update rule of Section 3.1 contains the first operator of this book, still disguised as a book-keeping device. Conditional on outcome n , the filter of an ideal measurement maps every incoming state to the vector $|e_n\rangle \langle e_n|\psi\rangle$: it keeps the component of $|\psi\rangle$ along the selected alternative and discards the rest. Give this map a name,

$$P_n |\psi\rangle := |e_n\rangle \langle e_n|\psi\rangle \quad \text{for every } |\psi\rangle, \quad (3.4)$$

and the symbol $|e_n\rangle \langle e_n|$ of the completeness shorthand (2.34) acquires at last its promised operator reading. The map is linear: it acts on a superposition term by term, because the amplitude $\langle e_n|\psi\rangle$ is linear in $|\psi\rangle$ by the inner-product axioms of Section 2.6. Linearity is not an extra assumption about filters; it is inherited from the superposition structure of the state space itself.

With the first instance in hand, the general concept costs nothing. An **operator** on $\mathcal{H}$ is a linear map from $\mathcal{H}$ to itself: a rule A taking every vector $|\psi\rangle$ to a vector $A|\psi\rangle$, with $A(\alpha|\psi_1\rangle + \beta|\psi_2\rangle) = \alpha A|\psi_1\rangle + \beta A|\psi_2\rangle$. An operator is fixed by its action on any basis, and its action is recorded by its

¹The appropriate formalism is that of positive operator-valued measures; the projection-valued measures of Section 3.3 are the orthogonal case. The mainline route of this book reaches the general case only through the composite-system theory of a later chapter.

matrix elements in that basis,

$$A_{mn} := \langle e_m | A | e_n \rangle, \quad (3.5)$$

for the expansion theorem (2.32) reassembles the action on a general vector from them, $A|\psi\rangle = \sum_{m,n} |e_m\rangle A_{mn} \langle e_n|\psi\rangle$. In the basis of its own alternatives the filter (3.4) has the simplest matrix of all: a single nonzero entry, a 1 in the n -th diagonal place, the arithmetic shadow of a device that passes one alternative and annihilates the rest.

Operators compose, and the composition is the arithmetic of staged experiments. If a process B is followed by a process A , the action of the chain is $A(B|\psi\rangle) =: (AB)|\psi\rangle$, and inserting a basis shows the matrix of the chain to be the product of the matrices,

$$(AB)_{mn} = \sum_k A_{mk} B_{kn}, \quad (3.6)$$

the row-times-column rule of 1925 recorded in Chapter 1 (Section 1.4.1). The repayment promised there is now delivered in both directions. Physically: two successive stages of an experiment compose as a product of operators, because each stage acts on the output of the one before. Formally: the product is order-sensitive, $AB \neq BA$ in general, which is precisely the fourth kinematical requirement of Chapter 1 (Section 1.4.6) — *the order of physical operations matters* — receiving its formal repayment. Two qualifications keep the ledger honest. What is repaid here is the kinematical share of the requirement, the composition of given operations; the transformation theory — how operations are generated, and what becomes of a state while no interrogation is in progress — is the dynamical share, and it belongs to the next chapter. And the machinery that quantifies the difference $AB - BA$, the commutator, is built in Sections 3.6 and 3.7.

One operation on operators is needed at once, first for the classification of this section and then for every expectation value of the chapter. The **adjoint** of an operator A is the operator $A^\dagger$ defined by

$$\langle \phi | A^\dagger | \psi \rangle := \langle \psi | A | \phi \rangle^* \quad \text{for every pair } |\phi\rangle, |\psi\rangle. \quad (3.7)$$

In any basis the definition reads $(A^\dagger)_{mn} = A_{nm}^*$: the matrix of the adjoint is the conjugate transpose, which both proves that the adjoint exists and fixes it uniquely. (The domain questions that this definition raises in infinite dimension are deferred to Appendix A; the operators of this chapter act on finite-dimensional spaces, or are read formally.) The product rule follows from two applications of (3.7). In any basis, $(AB)^\dagger_{mn} = (AB)_{nm}^* = (\sum_k A_{nk} B_{km})^* = \sum_k A_{nk}^* B_{km}^* = \sum_k (A^\dagger)_{kn} (B^\dagger)_{mk} = (B^\dagger A^\dagger)_{mn}$, so $(AB)^\dagger = B^\dagger A^\dagger$: the adjoint of a chain is the chain of the adjoints, reversed.

The filter operators (3.4) obey an algebra of their own, and every clause of it is an experimental report. For any $|\psi\rangle$, the first follows from the definition and $\langle e_n | e_n \rangle = 1$: $P_n^2 |\psi\rangle = P_n(|e_n\rangle \langle e_n | \psi \rangle) = |e_n\rangle \langle e_n | e_n \rangle \langle e_n | \psi \rangle = |e_n\rangle \langle e_n | \psi \rangle = P_n |\psi\rangle$. The second uses orthonormality $\langle e_m | e_n \rangle = \delta_{mn}$: $P_m P_n |\psi\rangle = |e_m\rangle \langle e_m | e_n \rangle \langle e_n | \psi \rangle = \delta_{mn} |e_m\rangle \langle e_n | \psi \rangle$, vanishing for $m \neq n$. Thus

$$P_n^2 = P_n, \quad P_m P_n = 0 \quad (m \neq n). \quad (3.8)$$

The first clause, **idempotence**, is repeatability in algebraic dress: a state that has passed the n -th filter is unchanged by passing it again — Axiom M1 applied to the output $|e_n\rangle$, since $P_n |e_n\rangle = |e_n\rangle$. The second clause, **mutual exclusivity**, is the orthogonality of distinct alternatives (2.28): a system certainly in the n -th output is certainly absent from the m -th, and the m -th filter annihilates what the n -

th passed. Together the two clauses say that the filters of one experiment carve $\mathcal{H}$ into non-overlapping rays, each preserved by its own filter and destroyed by every other.

The third clause is a debt discharged. Summing the filter operators over all outcomes of the experiment gives

$$\sum_n P_n = I, \quad (3.9)$$

the **resolution of identity**: the identity operator, which leaves every state unchanged, is the sum of an experiment's filters. This is precisely the completeness shorthand (2.34) of the preceding chapter, displayed there with an explicit caution — a bookkeeping compression of the expansion theorem, awaiting an operator reading. The reading is now given: the symbols $|e_n\rangle\langle e_n|$ are operators, their sum is the identity operator on $\mathcal{H}$, and inserting (3.9) into any amplitude reproduces the general composition law (2.33). The alternatives of a complete question exhaust the space of states; that is what the decomposition of the identity says.

With the adjoint in hand, the operations on states met so far fall into two great families, classified by how A and $A^\dagger$ are related. The **filters** of this section are self-adjoint idempotents: $P_n^\dagger = P_n$, proved by $\langle\phi|P_n^\dagger|\psi\rangle = \langle\psi|P_n|\phi\rangle^* = \langle\psi|e_n\rangle^*\langle e_n|\phi\rangle^* = \langle e_n|\psi\rangle\langle\phi|e_n\rangle = \langle\phi|P_n|\psi\rangle$ via (3.7) and (3.4); and $P_n^2 = P_n$ from (3.8). They are irreversible — the components they discard are gone, and no operator undoes a filter; information has been traded for an answer. The **relabelings** are their opposite in every respect. An operator U with

$$U^\dagger U = I \quad (3.10)$$

is called **unitary**; it preserves every inner product, $\langle U\phi|U\psi\rangle = \langle\phi|\psi\rangle$, and is therefore invertible, its inverse being $U^\dagger$. Unitary operators are already on the page in disguise: the overlap matrix of two bases obeyed exactly this law (2.37), and the operator $U = \sum_n |e'_n\rangle\langle e_n|$, carrying the orthonormal basis $\{|e_n\rangle\}$ to the basis $\{|e'_n\rangle\}$ of (2.36), is its operator reading. A relabeling discards nothing; it is the same physics described against a different complete question. These two families — self-adjoint filters and unitary relabelings — are the first instances of the two great operator classes of quantum mechanics, and much of the rest of this book is their theory. One flag is planted here for the next chapter: relabelings so far are changes of the question, applied at an instant. Continuous families of relabelings, and the question of what carries a state *between* interrogations, belong to the theory of evolution.

The algebra of this section is exercised, and the two-state dictionary extended, on the polarization bench of Section 3.1. In the H/V basis the filters of the two complete polarization questions are explicit 2×2 matrices,

$$P_H = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad P_V = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad P_D = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_A = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad (3.11)$$

built from the columns of Section 2.4 by the rule (3.4): the entries of P_D are $\langle e_m|D\rangle\langle D|e_n\rangle$ with $|D\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$, and likewise $|A\rangle = (|H\rangle - |V\rangle)/\sqrt{2}$. Direct multiplication verifies the

whole projector algebra in arithmetic.

$$P_H + P_V = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = I, \quad P_D + P_A = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = I, \quad (3.12)$$

$$P_D^2 = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^2 = \frac{1}{4} \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = P_D, \quad P_D P_A = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0. \quad (3.13)$$

The sequential-filter law of Chapter 1 then becomes a product of matrices. An H -polarized photon meeting a D -filter and then a V -filter emerges in the state

$$P_V P_D P_H |H\rangle = \frac{1}{2} |V\rangle, \quad \|P_V P_D P_H |H\rangle\|^2 = \frac{1}{4}, \quad (3.14)$$

the measured $1/4$ of the three-polarizer experiment (Section 1.5), recovered from operator arithmetic exactly as the multiplication principle (2.4) recovered it from amplitudes. And the order-sensitivity of filters is now visible in numbers:

$$P_D P_H = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} \neq P_H P_D = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad (3.15)$$

filter order matters, in arithmetic at last. One identification is recorded here for later consumption. The two benches of this chapter — polarization and spin — realize one and the same two-state overlap structure, and the identification $H \leftrightarrow | +z \rangle$, $V \leftrightarrow | -z \rangle$, $D \leftrightarrow | +x \rangle$, $A \leftrightarrow | -x \rangle$ makes the arithmetic of either bench available to the other; it is what will allow $P_D - P_A$ to be read as the x -axis spin operator in Section 3.7.

A single filter answers a yes-or-no question: is the system in the alternative $|e_n\rangle$, or not? A complete experiment asks all of its yes-or-no questions at once, and the family comes with a bookkeeping of its own — outcomes grouped and ignored, probabilities assigned consistently to every grouping. That bookkeeping is the projection-valued measure, the subject of the next section.

3.3 Projection-Valued Measures: the Bookkeeping of Complete Questions

The projector of Section 3.2 is the operator of a single yes-or-no question. A complete experiment asks its whole family of such questions at once, and the family comes with a bookkeeping of its own: outcomes can be grouped into sets, sets can be coarse-grained, and probabilities must be assigned consistently to all of them. The bookkeeping device is the **projection-valued measure**. Given an ideal experiment with alternatives $\{|e_n\rangle\}$ and projectors $P_n = |e_n\rangle\langle e_n|$, assign to every set Ω of outcomes the projector

$$P_\Omega = \sum_{n \in \Omega} P_n, \quad (3.16)$$

the sum of the projectors of the outcomes in the set, projecting onto their span. The assignment obeys the three clauses of a measure,

$$P_{\emptyset} = 0, \quad P_{\text{all}} = I, \quad P_{\Omega_1 \cup \Omega_2} = P_{\Omega_1} + P_{\Omega_2} \text{ for disjoint } \Omega_1, \Omega_2, \quad (3.17)$$

the last of them by the exclusivity clause of (3.8), which keeps the sum of two projectors a projector exactly when their ranges are orthogonal. In finite dimension the whole assignment is equivalently a complete family $\{P_n\}$ with $\sum_n P_n = I$ and $P_m P_n = \delta_{mn} P_n$ — the projector algebra of (3.8) and (3.9), now read as the axioms of an experimental question rather than the properties of a single filter.

The first business of the bookkeeping is **coarse-graining**. If the distinction among the outcomes $n \in \Omega$ is ignored — their counters wired to a single register — the fine-grained experiment is replaced by a coarser one whose single outcome Ω carries the projector (3.16): rank-1 projectors fuse into a projector of higher rank, onto the subspace spanned by the fused alternatives. Coarse-graining is a physical operation on the description, not a notational one: it is the experiment in which some resolutions are deliberately unrecorded. Its extreme case fuses every outcome: the **trivial measurement** $\{I\}$, the experiment with one port, asked of every system and answered yes always — “did anything happen?” It is the unique projection-valued measure with a single projector, and it will reappear below as the formal opposite of an informative measurement.

The Born rule prices every set of outcomes at once. For a system prepared in the state ψ , the probability of the outcome set Ω is

$$p(\Omega) = \langle \psi | P_{\Omega} | \psi \rangle, \quad (3.18)$$

which for a single outcome is (M2) of Section 3.1 and for a general set is the sum of its outcomes’ probabilities — additivity over disjoint sets being exactly the third clause of the measure. In the language of probability theory, (3.18) is a probability measure on the projectors of *one experiment*. The experimental warrant for that additivity is the marked-alternatives sum rule of Section 2.1: the outcomes of a realized measurement are registered alternatives, distinguishable in principle, and their frequencies add classically.

This is the exact sense in which classical logic survives inside quantum mechanics, and its boundary is equally exact. *Within* one projection-valued measure, the logic of outcomes is Boolean: the alternatives are mutually exclusive and exhaustive, their propositions combine by union and intersection of outcome sets, and the ordinary rules of probability apply without exception. *Across* two different measures — the z -question and the x -question of a Stern–Gerlach bench — the propositions of one share no common refinement with the propositions of the other: no third experiment has outcomes that are simultaneously answers to both, and the conjunction “spin up along z and spin up along x ” has no operational meaning. Chapter 1 exhibited this on the bench as the failure of Boolean logic among sequential polarizers and sequential magnets (Section 1.5); the exact formal statement of the failure is now on the page: two projection-valued measures whose projectors do not all commute define propositions belonging to no single Boolean algebra. How much two families of questions fail to fit together is measured by the commutator, the subject of Sections 3.6 and 3.7.

The converse question — must *every* consistent pricing of an experiment’s projectors come from a state? — has a celebrated answer, and it is signposted here rather than proved. Suppose a rule assigns

to every projector on $\mathcal{H}$ a number between 0 and 1, additively over orthogonal families: the prices of mutually orthogonal projectors summing to the identity add to 1. On any space of dimension three or greater, every such rule is necessarily of the form

$$p(P) = \text{tr}(\rho P) \quad (3.19)$$

for a unique positive operator ρ of unit trace,² where tr denotes the **trace**, the sum of the diagonal matrix elements in any orthonormal basis — a sum whose value is independent of the basis chosen. This is Gleason’s theorem of 1957, the uniqueness result signposted in the preceding chapter’s discussion of the Born rule (Section 2.3), now statable because the language of projections it requires is on the page: the geometry of the state space leaves no freedom in the probability rule beyond the choice of state itself. Its proof is long and belongs to the mathematical literature; its content is the deepest warrant this book has for the universality of the Born rule.



Note *Marking consistency.* The probability bookkeeping of this section applies to staged experiments in which the intermediate outcomes leave a which-way record: probabilities, not amplitudes, propagate from stage to stage, per the marking clause of Section 2.1. It must not be confused with the coherent insertion of a basis into an amplitude, (2.33), where no record exists and amplitudes combine. Coarse-graining a projection-valued measure — ignoring which outcome occurred — belongs to the marked case; recombining the alternatives so that no record survives belongs to the coherent case. The difference is physical, and the example below keeps the two apart at every stage.

The canonical projection-valued measure of this book is the sequential Stern–Gerlach bench of Chapter 1 (Figure 1.4), and it exhibits every clause of the bookkeeping. The z -apparatus is the two-element measure $\{P_{+z}, P_{-z}\}$, with $P_{\pm z} = |\pm z\rangle\langle\pm z|$. Blocking the $-z$ port and keeping the $+z$ beam is conditioning on an outcome: the selected ensemble is described by the updated state of Section 3.1, and all later frequencies are computed from it. Letting both separated beams pass unresolved — no port blocked, the deflection record intact — is a different physical arrangement: a non-selective z -measurement in the language of Section 3.1. It is not the trivial measure. The two beams are marked alternatives — the which-way record exists in the world, whether or not anyone reads it — so later questions see added probabilities, and the counts confirm the addition: an ensemble passed unresolved through the z -stage and then interrogated along x splits fifty-fifty, exactly the sum of the two branches’ separate statistics, with no interference between them.

The trivial measure $\{I\}$ is reserved for the genuinely record-free arrangement: the two beams coherently recombined, the separation undone before any which-way trace survives — or no magnet present at all. Then there is one outcome, answered with certainty, and the state proceeds unchanged. The distinction is quantitative, and the full z - x - z chain makes it sharp. Compute the final z -counts of the three-stage arrangement in two independent ways. By the update theorem of Section 3.1: the first z -stage returns $+z$ with certainty; the x -stage returns $\pm x$ with probabilities $\frac{1}{2}$ each, leaving the conditional states $|\pm x\rangle$; the final z -stage prices each branch at $|\langle +z | \pm x \rangle|^2 = \frac{1}{2}$, so the recapture probability is $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$. By the projector products of Section 3.2: the probability of a history

²A. M. Gleason, *J. Math. Mech.* **6**, 885 (1957). The operator ρ is the *density operator*, whose theory is built in the chapter on composite systems; it is named here once, as the theorem requires it, and used no further in this chapter. The pure states of this book are the rank-1 case, $\rho = |\psi\rangle\langle\psi|$, for which $\text{tr}(\rho P) = \langle\psi|P|\psi\rangle$ — the Born rule (3.18) recovered.

is the squared norm of the corresponding operator chain applied to the initial state,

$$\|P_{+z}P_{+x}|+z\rangle\|^2 = |\langle +z|+x\rangle|^2|\langle +x|+z\rangle|^2 = \frac{1}{4}, \quad (3.20)$$

each of the four histories carrying probability $1/4$, and the two histories ending at $+z$ summing to $1/2$. The two computations agree, as they must — the update theorem and the projector chain are one rule in two notations — and both rely on the marking clause at the intermediate stage: the x -outcomes are recorded alternatives, and it is their probabilities that propagate. Formally, ignoring the x -outcome is the non-selective x -measurement; its branch projectors sum to the identity as operators, $P_{+x} + P_{-x} = I$, but the physical operation is not the identity, for the branches are marked. “The x -question was asked and its answer ignored” and “the x -question was never asked” are different experiments with different counts — one-half recapture in the first, perfect recapture in the second — and the difference between them is precisely the bookkeeping this section exists to keep.

A projection-valued measure knows *which* alternatives occur and with what probabilities, but a physical quantity carries more than a list of alternatives: its outcomes are *numbers* — deflections, dial readings, fringe positions. Attaching numerical readings to the outcomes of a measure turns statistics into expectation values, and compresses the whole measure into a single operator. That is the next section’s construction.

3.4 Observables: Numerical Labels and Expectation Values

A projection-valued measure, in the sense of Section 3.3, knows *which* alternatives an experiment resolves and with what frequencies they occur. A physical quantity carries one more piece of structure: every outcome comes with a *number* — a deflection on a scale, a dial reading, a fringe position. Attaching numbers to outcomes turns the bookkeeping of complete questions into statistics.

An **observable** is an ideal experiment whose outcomes are labeled by real numbers $a_1, a_2, \dots$, one instrument reading per alternative. We take the non-degenerate case first: the readings are all distinct, and the experiment’s projectors are the rank-1 filters P_n of Sections 3.1–3.2, one ray per reading. The degenerate case, in which a single reading labels a whole subspace of alternatives, is taken up in Section 3.6.

Repeat the experiment on an ensemble of identically prepared systems and record the numbers. The frequencies $p_n = \langle \psi | P_n | \psi \rangle = |\langle e_n | \psi \rangle|^2$ are the Born frequencies of the experiment (Sections 2.3 and 3.3, (3.18)), and the long-run mean of the recorded readings is their weighted average,

$$\langle a \rangle_\psi = \sum_n a_n p_n. \quad (3.21)$$

This is an ensemble statement in the sense of Section 2.3: $\langle a \rangle_\psi$ is a property of the preparation, not of any single run, and it is measured by repeating the experiment, not by refining one. The scatter of the recorded numbers around the mean is measured by the **variance**,

$$\sigma_a^2 = \sum_n (a_n - \langle a \rangle_\psi)^2 p_n \geq 0, \quad (3.22)$$

a sum of nonnegative terms. A preparation is called *sharp* for this experiment when $\sigma_a^2 = 0$: then every term vanishes separately, and since the readings are distinct, a single outcome n_0 must carry all

the probability, $|\langle e_{n_0} | \psi \rangle|^2 = 1$. By the saturation clause of the Cauchy–Schwarz inequality (2.24) — equality holds if and only if the two states share a ray — the state is then the ray of $|e_{n_0}\rangle$ itself. The variance vanishes if and only if the state lies in a single outcome’s ray: the sharp preparations of an experiment are exactly its own alternatives, and every reading returned on such a preparation is the same number a_{n_0} .

Frequencies and readings together determine the statistics completely, and the whole of it compresses into a single operator. Assemble readings and projectors into

$$A := \sum_n a_n P_n. \quad (3.23)$$

Bracketing with an arbitrary state and using $\langle \psi | P_n | \psi \rangle = p_n$ gives

$$\langle A \rangle_\psi := \langle \psi | A | \psi \rangle = \sum_n a_n |\langle e_n | \psi \rangle|^2 = \sum_n a_n p_n = \langle a \rangle_\psi, \quad (3.24)$$

the ensemble mean (3.21) read off as one matrix element. The notation on the left is fixed once and for all: $\langle A \rangle_\psi$ always means $\langle \psi | A | \psi \rangle$. In the experiment’s own basis $\{|e_n\rangle\}$ the operator is diagonal, with the readings down the diagonal — the matrix of an observable is its table of outcomes.

The projector algebra of Section 3.2 ($P_m P_n = \delta_{mn} P_n$, (3.8)) turns powers of A into powers of the readings: $A^2 = \sum_{m,n} a_m a_n P_m P_n = \sum_n a_n^2 P_n$, and by induction $A^k = \sum_n a_n^k P_n$ for every power, so that every polynomial obeys $p(A) = \sum_n p(a_n) P_n$. The same formula serves as the definition of an arbitrary function of an observable,

$$f(A) := \sum_n f(a_n) P_n, \quad (3.25)$$

for any f defined on the readings: apply the function to each dial mark and keep the projectors. In particular the variance takes the operator form

$$(\Delta A)^2 := \langle A^2 \rangle_\psi - \langle A \rangle_\psi^2, \quad (3.26)$$

The equality with σ_a^2 above is the expansion

$$\sigma_a^2 = \sum_n (a_n - \langle a \rangle_\psi)^2 p_n = \sum_n (a_n^2 - 2a_n \langle a \rangle_\psi + \langle a \rangle_\psi^2) p_n = \langle A^2 \rangle_\psi - 2\langle A \rangle_\psi^2 + \langle A \rangle_\psi^2 = \langle A^2 \rangle_\psi - \langle A \rangle_\psi^2 = (\Delta A)^2, \quad (3.27)$$

using $\sum_n p_n = 1$ and $\sum_n a_n p_n = \langle A \rangle_\psi$.

Two honesty clauses attend the compression of an experiment into an operator. The first concerns convention. The assignment of numbers to outcomes is an instrument-maker’s choice — zeros, units, and scales are conventions: re-zeroing the scale shifts every reading, $a_n \mapsto a_n + c$, and with it the mean, while the projectors, the frequencies, and every physical prediction remain untouched. The PVM is the physics; the numerical labels are its bookkeeping. The second clause runs in the opposite direction and will be load-bearing in Section 3.5: although one mean fixes nothing, the mean in *every* state fixes the operator uniquely. The workhorse is the **polarization identity**, displayed here at its first use: for any operator A and any pair of vectors,

$$\langle \phi | A | \psi \rangle = \frac{1}{4} \left[\langle \psi + \phi | A | \psi + \phi \rangle - \langle \psi - \phi | A | \psi - \phi \rangle + i \langle \psi + i\phi | A | \psi + i\phi \rangle - i \langle \psi - i\phi | A | \psi - i\phi \rangle \right], \quad (3.28)$$

as expanding the right-hand side by linearity confirms:

$$\langle \psi \pm \phi | A | \psi \pm \phi \rangle = \langle \psi | A | \psi \rangle + \langle \phi | A | \phi \rangle \pm \langle \psi | A | \phi \rangle \pm \langle \phi | A | \psi \rangle, \quad (3.29)$$

$$\langle \psi \pm i\phi | A | \psi \pm i\phi \rangle = \langle \psi | A | \psi \rangle + \langle \phi | A | \phi \rangle \mp i \langle \psi | A | \phi \rangle \pm i \langle \phi | A | \psi \rangle. \quad (3.30)$$

The four-diagonal combination in (3.28) then telescopes: the $\langle \psi | A | \psi \rangle$ and $\langle \phi | A | \phi \rangle$ terms cancel, the $\pm i \langle \phi | A | \psi \rangle$ terms cancel, and the surviving $\langle \phi | A | \psi \rangle$ contributions add to $4 \langle \phi | A | \psi \rangle$. Every matrix element of A is therefore a fixed linear combination of diagonal values $\langle \chi | A | \chi \rangle$; and the diagonal values on arbitrary vectors are fixed by the means on normalized states, since rescaling $\chi \mapsto c\chi$ rescales $\langle \chi | A | \chi \rangle$ by $|c|^2$. If two operators have the same mean in every state, their difference has vanishing diagonal values everywhere, hence vanishing matrix elements by (3.28): the operators coincide. The operative constraint is the quantifier — *every* state, not some states, and not an average over states. Section 3.5 will spend precisely this display.

The spin component along an axis $\mathbf{n}$ makes the encoding concrete. Take the Stern–Gerlach apparatus of Chapter 1 (Section 1.5) rotated to the axis $\mathbf{n}$, with the two output ports carrying the readings ± 1 in units of $\hbar/2$. The observable is the difference of the two port projectors,

$$\sigma_{\mathbf{n}} := P_{+\mathbf{n}} - P_{-\mathbf{n}}, \quad S_{\mathbf{n}} = \frac{\hbar}{2} \sigma_{\mathbf{n}}, \quad (3.31)$$

where $|\pm \mathbf{n}\rangle$ are the rotated alternatives of Chapter 2, (2.38). Send in a beam prepared in $|+z\rangle$, and let θ be the polar angle of $\mathbf{n}$ relative to z . The half-angle law (2.39) — recorded as an experimental fact of rotated magnets, its derivation owed to the chapter on angular momentum — gives the port frequencies $p_+ = \cos^2(\theta/2)$ and $p_- = \sin^2(\theta/2)$, and with readings ± 1 the mean is

$$\langle +z | \sigma_{\mathbf{n}} | +z \rangle = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = \cos \theta, \quad \Delta \sigma_{\mathbf{n}} = \sin \theta. \quad (3.32)$$

The spread follows from $\sigma_{\mathbf{n}}^2 = P_{+\mathbf{n}} + P_{-\mathbf{n}} = I$ by the projector algebra, so $(\Delta \sigma_{\mathbf{n}})^2 = 1 - \cos^2 \theta$ by (3.26). The mean traces the classical projection of a unit arrow along $\mathbf{n}$ onto the z -axis — yet no classical arrow exists: a single atom returns only ± 1 , and it is the ensemble of counts that imitates the projection, the moral of Chapter 1’s two-state experiments (Section 1.5). The spread tells the complementary story: along the preparation axis ($\theta = 0$) every atom returns $+1$ and the variance vanishes, the sharpness proved above; perpendicular to it ($\theta = \pi/2$) the ports fire equally, the mean vanishes, and the spread is maximal. The statistics of a turned magnet, computed end-to-end from Chapter 2’s data.

The construction of this section ran from experiment to operator: the PVM came first, and $A = \sum_n a_n P_n$ was assembled from it. The arrow must now be reversed. Not every operator can be an observable’s encoding — which ones are, and why exactly the self-adjoint ones?

3.5 Why Self-Adjoint: Real Outcomes and the Spectral Theorem

Section 3.4 assembled an operator from an experiment: alternatives first, then readings, then $A = \sum_n a_n P_n$. This section reverses the arrow and asks which operators can arise this way. The answer comes in two theorems, and both are proved. The first shows that the reality of instrument readings forces a definite symmetry on the operator; the second — the spectral theorem — shows

that every operator with that symmetry encodes an ideal experiment. Between them they close the observable concept.

Every instrument reading is a real number, so the mean (3.24) of an observable is real in every state. This single experimental fact already fixes the symmetry:

$$\langle \psi | A | \psi \rangle \in \mathbb{R} \text{ for every } |\psi\rangle \iff A = A^\dagger, \quad (3.33)$$

with the adjoint as defined in (3.7). The proof is the polarization identity at work. If $A = A^\dagger$, then $\langle \psi | A | \psi \rangle^* = \langle \psi | A^\dagger | \psi \rangle = \langle \psi | A | \psi \rangle$, and the mean is real. Conversely, suppose the mean is real in every state. Then for every vector $|\chi\rangle$,

$$\langle \chi | A^\dagger | \chi \rangle = \langle \chi | A | \chi \rangle^* = \langle \chi | A | \chi \rangle, \quad (3.34)$$

so A and $A^\dagger$ have identical diagonal values in every state; the polarization identity (3.28) then gives $\langle \phi | A^\dagger | \psi \rangle = \langle \phi | A | \psi \rangle$ for every pair $|\phi\rangle, |\psi\rangle$, which is to say $A^\dagger = A$. The operative constraint is again the quantifier: reality in *every* state — not in some states, and not on average over states. An operator whose mean happens to be real on a subset of preparations need not be self-adjoint, and encodes no observable of the present kind.

An operator with $A = A^\dagger$ is called **self-adjoint**, or **Hermitian**. In finite dimension the two words coincide, and this book uses them interchangeably; where the distinction acquires content — the continuous spectra previewed in Section 3.9 — only “self-adjoint” survives, and the present section says so once, here.

The sharp preparations of Section 3.4 — the states of zero variance, one per outcome ray — appear on the operator side as the solutions of the **eigenvalue equation**

$$A |a_n\rangle = a_n |a_n\rangle, \quad (3.35)$$

where the number a_n is an *eigenvalue* and $|a_n\rangle \neq 0$ an *eigenvector* of A . Two properties follow from self-adjointness in one line each. First, eigenvalues are real: bracketing (3.35) with $\langle a_n|$ gives

$$a_n \langle a_n | a_n \rangle = \langle a_n | A | a_n \rangle \in \mathbb{R} \implies a_n \in \mathbb{R}. \quad (3.36)$$

Second, eigenvectors of distinct eigenvalues are orthogonal: if $A |a_m\rangle = a_m |a_m\rangle$ and $A |a_n\rangle = a_n |a_n\rangle$, then

$$a_n \langle a_m | a_n \rangle = \langle a_m | A | a_n \rangle = \langle A^\dagger a_m | a_n \rangle = a_m \langle a_m | a_n \rangle \implies (a_m - a_n) \langle a_m | a_n \rangle = 0, \quad (3.37)$$

using the reality of a_m in the third step, so $a_m \neq a_n$ forces $\langle a_m | a_n \rangle = 0$. Eigen-readings are real and distinct readings are mutually exclusive — the two facts an instrument requires of its dial.

The converse is the structural theorem of the chapter.

Spectral theorem (finite-dimensional). *Every self-adjoint operator A on a finite-dimensional Hilbert space $\mathcal{H}$ possesses an orthonormal basis of eigenvectors. Grouping the basis vectors by eigenvalue, A takes the spectral form $A = \sum_\nu a_\nu P_\nu$, with a_ν its distinct real eigenvalues and P_ν the projector onto the a_ν -eigenspace.*

Proof. By induction on $d = \dim \mathcal{H}$. For $d = 1$ every nonzero vector is an eigenvector and there is nothing to prove. Let $d \geq 2$. The characteristic polynomial $\chi_A(\lambda) = \det(\lambda I - A)$ is a polynomial of

degree d , and by the fundamental theorem of algebra it has at least one root a_1 .³ Then $\det(a_1 I - A) = 0$, so $a_1 I - A$ annihilates some nonzero vector; normalized, that vector is an eigenvector $|a_1\rangle$ with eigenvalue a_1 , and a_1 is real by the reality argument above. Let W be the orthogonal complement of $|a_1\rangle$. The subspace W is invariant under A : for $|w\rangle \in W$,

$$\langle a_1 | A | w \rangle = \langle A^\dagger a_1 | w \rangle = \langle A a_1 | w \rangle = a_1 \langle a_1 | w \rangle = 0, \quad (3.38)$$

so $A | w \rangle \in W$. The restriction of A to W is self-adjoint on the $(d - 1)$ -dimensional space W , so by the induction hypothesis W has an orthonormal basis of eigenvectors of A ; adjoining $|a_1\rangle$ yields an orthonormal eigenbasis of $\mathcal{H}$. Collecting the basis vectors carrying the same eigenvalue a_ν , their span is the a_ν -eigenspace with projector P_ν , and the expansion of A in its own eigenbasis reads $A = \sum_\nu a_\nu P_\nu$. $\square$

The physical reading of the theorem is the closure of the observable concept: the theory has *no orphan operators*. Every self-adjoint operator A is some ideal experiment's encoding — its eigenbasis supplies the complete set of alternatives, its eigenvalues the readings, and Section 3.4's construction recovers the full statistics. Combined with (3.33) the circle is closed: an ideal experiment with real labels yields a self-adjoint operator, and every self-adjoint operator yields an ideal experiment. The vocabulary is fixed accordingly: Hermitian, self-adjoint, and observable-encoding name one class of operators.

Two honesty pointers bound the theorem's reach. First, the proof just given is finite-dimensional. In infinite dimension the spectral theorem survives, but at a price — spectra become continuous, operators unbounded, and domain questions inseparable from statements; the honest development belongs to Appendix A, and Section 3.9 previews the physics in one labeled preview, no more. Second, the induction above is silent about multiplicity: when a root of χ_A repeats, the eigenspace P_ν has rank greater than one, a single outcome no longer selects a ray, and the same quantity can be resolved further by other questions. That is degeneracy, and it is the next section's subject.

The two-state system exhibits the whole theorem at a glance, because there every self-adjoint operator can be diagonalized explicitly. A Hermitian 2×2 matrix has the form $\begin{pmatrix} \alpha & \beta \\ \beta^* & \gamma \end{pmatrix}$ with α, γ real; writing $\alpha = a_0 + a_z$, $\gamma = a_0 - a_z$, and $\beta = a_x - i a_y$ parameterizes it as

$$A = a_0 I + a_x \sigma_x + a_y \sigma_y + a_z \sigma_z, \quad a_0, a_x, a_y, a_z \in \mathbb{R}, \quad (3.39)$$

in terms of three operators already earned. The first is $\sigma_z = P_{+z} - P_{-z}$, diagonal with entries ± 1 . The second is $\sigma_x = P_{+x} - P_{-x}$, whose matrix is the arithmetic of Section 3.2's polarization example, read through the identification $D \leftrightarrow | + x \rangle$, $A \leftrightarrow | - x \rangle$ recorded there. The third is constructed here: the y -magnet, whose alternatives are

$$|\pm y\rangle := \frac{|+z\rangle \pm i |-z\rangle}{\sqrt{2}}, \quad (3.40)$$

the rotated states of (2.38) at polar angle $\theta = \pi/2$, azimuth $\varphi = \pi/2$ — taken, like all of (2.38), as experimental input — and $\sigma_y := P_{+y} - P_{-y}$. Complex amplitudes are not optional in this construction: no real equal-weight superposition of $|+z\rangle$ and $|-z\rangle$ remains equal-weight with respect to the x -

³This is the one analytic input of the proof; the fundamental theorem of algebra is proved in the algebra texts and is not derived here. Everything else is finite-dimensional linear algebra.

alternatives as well — the only real candidates are $|\pm x\rangle$ themselves, which the x -question prices 1 and 0 — the necessity-of- i computation of Section 2.6 once again. As matrices in the z -basis,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (3.41)$$

the middle one from the spectral sum $|+y\rangle\langle+y| - |-y\rangle\langle-y|$.

Diagonalizing (3.39) is now explicit. Write $A = a_0 I + \mathbf{a} \cdot \boldsymbol{\sigma}$ with $\mathbf{a} = (a_x, a_y, a_z)$ and $\mathbf{a} \cdot \boldsymbol{\sigma} := a_x \sigma_x + a_y \sigma_y + a_z \sigma_z$. Expanding the square,

$$\begin{aligned} (\mathbf{a} \cdot \boldsymbol{\sigma})^2 &= (a_x \sigma_x + a_y \sigma_y + a_z \sigma_z)^2 \\ &= a_x^2 \sigma_x^2 + a_y^2 \sigma_y^2 + a_z^2 \sigma_z^2 + a_x a_y (\sigma_x \sigma_y + \sigma_y \sigma_x) + a_y a_z (\sigma_y \sigma_z + \sigma_z \sigma_y) + a_z a_x (\sigma_z \sigma_x + \sigma_x \sigma_z). \end{aligned} \quad (3.42)$$

Each $\sigma_i^2 = I$ from the projector algebra ($\sigma_i = P_{+i} - P_{-i}$, and $P_{+i}P_{-i} = 0$ with $P_{+i} + P_{-i} = I$ gives the square I). The cross terms cancel in pairs: multiplying the 2×2 matrices displayed above verifies $\sigma_x \sigma_y + \sigma_y \sigma_x = 0$, $\sigma_y \sigma_z + \sigma_z \sigma_y = 0$, $\sigma_z \sigma_x + \sigma_x \sigma_z = 0$. Hence $(\mathbf{a} \cdot \boldsymbol{\sigma})^2 = (a_x^2 + a_y^2 + a_z^2)I = |\mathbf{a}|^2 I$. An eigenvalue λ of $\mathbf{a} \cdot \boldsymbol{\sigma}$ then obeys $\lambda^2 = |\mathbf{a}|^2$; and since $\mathbf{a} \cdot \boldsymbol{\sigma}$ is traceless and, for $\mathbf{a} \neq 0$, not a multiple of the identity, both signs occur. In the z -basis,

$$\mathbf{a} \cdot \boldsymbol{\sigma} = \begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix} = |\mathbf{a}| \begin{pmatrix} \cos \theta & e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & -\cos \theta \end{pmatrix}, \quad (3.43)$$

where (θ, φ) are the polar angles of the unit vector $\mathbf{n} := \mathbf{a}/|\mathbf{a}|$. Multiplying this matrix onto the column of $|+\mathbf{n}\rangle$ from (2.38):

$$\begin{aligned} (\mathbf{a} \cdot \boldsymbol{\sigma}) |+\mathbf{n}\rangle &= |\mathbf{a}| \begin{pmatrix} \cos \theta & e^{-i\varphi} \sin \theta \\ e^{i\varphi} \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\varphi} \sin \frac{\theta}{2} \end{pmatrix} = |\mathbf{a}| \begin{pmatrix} \cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2} \\ e^{i\varphi} (\sin \theta \cos \frac{\theta}{2} - \cos \theta \sin \frac{\theta}{2}) \end{pmatrix} \\ &= |\mathbf{a}| \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\varphi} \sin \frac{\theta}{2} \end{pmatrix} = +|\mathbf{a}| |+\mathbf{n}\rangle, \end{aligned} \quad (3.44)$$

using $\cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2} = \cos \frac{\theta}{2}$ and $\sin \theta \cos \frac{\theta}{2} - \cos \theta \sin \frac{\theta}{2} = \sin \frac{\theta}{2}$. The same multiplication for $|-\mathbf{n}\rangle$ yields $-|\mathbf{a}| |-\mathbf{n}\rangle$. The general two-state observable therefore has eigenvalues $a_0 \pm |\mathbf{a}|$ and spectral decomposition

$$A = (a_0 + |\mathbf{a}|)P_{+\mathbf{n}} + (a_0 - |\mathbf{a}|)P_{-\mathbf{n}}. \quad (3.45)$$

The spectral data — an axis $\mathbf{n}$ and two readings $a_0 \pm |\mathbf{a}|$ — are the operator: every Hermitian operator on $\mathbb{C}^2$ is a Stern–Gerlach magnet with a zero-shifted dial, and “Hermitian operators encode spectral data” is here made literal. The honesty clause of Chapter 2 stands unchanged: the half-angle form (2.38) of the eigenstates remains an experimental input, and its derivation is a debt owed to the rotation theory of the chapter on angular momentum. Nothing in this example is rotation theory; it is 2×2 arithmetic read through the spectral theorem.

The advantage of the self-adjointness theorems is closure. The observable concept, built inductively from filters and counters, now exhausts its subject: real readings force $A = A^\dagger$, the spectral theorem guarantees that every such A is an ideal experiment, and no further structure — no hidden label, no privileged basis — is either needed or permitted. Its limitation is twofold. The proof is finite-dimensional, and the continuum case must be re-earned with the analytic care deferred to Appendix A.

And the theorem treats one observable at a time: it certifies each question separately and says nothing about when two questions can be asked together. When eigenvalues repeat, a single question no longer exhausts the alternatives, and resolving the degeneracy needs several questions at once — questions that must fit together. Whether they do is the subject of the next section.

3.6 Degeneracy and Compatible Observables

The spectral theorem of Section 3.5 grouped repeated eigenvalues into a single clause of the sum: when the characteristic polynomial has a repeated root a_ν , its eigenspace $\mathcal{H}_\nu$ has dimension $d_\nu > 1$, and the observable reads

$$A = \sum_{\nu} a_{\nu} P_{\nu}, \quad \text{rank } P_{\nu} = d_{\nu} \geq 1, \quad (3.46)$$

with P_ν the projector onto $\mathcal{H}_\nu$. Such an eigenvalue is called **degenerate**. A degenerate outcome does not select a ray; it selects a subspace. The instrument's dial cannot tell the vectors of $\mathcal{H}_\nu$ apart, although its mutually orthogonal vectors are perfectly distinguishable in the sense of Chapter 2 (orthogonal states, (2.28)): a finer experiment exists in principle, and the present one has simply not resolved it. Two questions then force themselves. What is the state after a degenerate outcome? And when can a second observable resolve what the first leaves unresolved? The first is answered by a new axiom — the measurement postulate's one genuinely new clause. The second is the theory of compatible observables.

Section 3.1 proved the update rule for sharp questions: conditional on a non-degenerate outcome, the state is the outcome's ray ((3.3), proved from repeatability, the Born rule, and the saturation clause of (2.24)). For a degenerate outcome repeatability still constrains, but no longer determines, the conditional state, and the gap is closed by stipulation. We record it as the third measurement axiom.

Axiom M3 (the Lüders update). *Conditional on the outcome ν of an ideal measurement, whose projector P_ν has any rank, the state updates as*

$$|\psi\rangle \mapsto P_\nu |\psi\rangle \quad (\text{unnormalized}); \quad |\psi_\nu\rangle = \frac{P_\nu |\psi\rangle}{\sqrt{p_\nu}}, \quad p_\nu = \langle\psi|P_\nu|\psi\rangle, \quad (3.47)$$

with p_ν the probability of the outcome. The probability is itself a computation, not an assumption: p_ν is the summed Born probability of the subspace's alternatives, $p_\nu = \sum_{n \in \nu} p_n = \langle\psi| \sum_{n \in \nu} P_n |\psi\rangle = \langle\psi|P_\nu|\psi\rangle$ by (3.18) and the coarse-graining of Section 3.3. For a rank-1 projector, $P_\nu |\psi\rangle = |e_\nu\rangle\langle e_\nu|\psi\rangle$, and (3.47) reproduces the proved update of Section 3.1; the axiom's new content is entirely in the higher-rank case.

 **Note** [Honesty bookkeeping: what is proved and what is axiomatic here] Repeatability alone forces the conditional state to lie within the ν -subspace, by the same certainty-plus-saturation logic as Section 3.1: an immediate re-interrogation with the question $\{P_\nu, I - P_\nu\}$ must return ν with certainty, so the conditional state $|\chi\rangle$ obeys $\langle\chi|I - P_\nu|\chi\rangle = 0$, and the positivity of the norm forces $(I - P_\nu)|\chi\rangle = 0$, i.e. $|\chi\rangle \in \mathcal{H}_\nu$. But repeatability does not fix the state within $\mathcal{H}_\nu$: any vector of the subspace re-passes the ν -test with certainty. The Lüders rule adds the genuinely new clause that the ideal filter preserves the incoming state's relative amplitudes inside the subspace — it resolves ν without re-preparing within

ν . This is the minimally-disturbing idealization, and it is named as one: real instruments may do more, and an outcome may re-prepare the subspace state entirely. Section 3.1's bookkeeping of selective and non-selective measurements consumed only the rank-1 theorem; the compatibility definition below, and every later degenerate-outcome update in this book, consume the present axiom.

Two observables A and B are called **compatible** when a B -measurement after an A -measurement never disturbs A 's outcomes: for every input state, every A -outcome ν , and every B -outcome μ that may follow, an immediate re-test of A returns ν with certainty. This is repeatability across questions, read off sequential filters; for degenerate A , the update rule (3.47) is exactly what “after” and “re-test” mean — the state goes from $P_\nu^A |\psi\rangle$ to $P_\mu^B P_\nu^A |\psi\rangle$, and the re-test interrogates the latter. Compatibility is decidable by the algebra of the two operators alone, through the **commutator**, defined here at its first printing:

$$[A, B] := AB - BA. \quad (3.48)$$

It enters as the compatibility diagnostic; its physics — the arithmetic of filter order — is developed in Section 3.7.

Compatibility theorem (finite-dimensional). For self-adjoint A and B on a finite-dimensional $\mathcal{H}$, the following are equivalent:

$$A \text{ and } B \text{ compatible} \iff \text{common orthonormal eigenbasis} \iff [A, B] = 0. \quad (3.49)$$

Proof. Write $A = \sum_\nu a_\nu P_\nu^A$ and $B = \sum_\mu b_\mu P_\mu^B$ in spectral form, with eigenspaces $\mathcal{H}_\nu^A$ and $\mathcal{H}_\mu^B$.

(i) *Common eigenbasis $\Rightarrow$ non-disturbance.* Let $\{|k\rangle\}$ be a common orthonormal eigenbasis, $A|k\rangle = \alpha_k|k\rangle$, $B|k\rangle = \beta_k|k\rangle$. After an A -outcome ν the state lies in $\mathcal{H}_\nu^A$, the span of those $|k\rangle$ with $\alpha_k = a_\nu$. Each B -projector P_μ^B is a sum of rank-1 projectors $|k\rangle\langle k|$ selected by $\beta_k = b_\mu$, and each $|k\rangle\langle k|$ maps every common eigenvector either to itself or to zero; hence P_μ^B maps $\mathcal{H}_\nu^A$ into itself. The post- B state $P_\mu^B P_\nu^A |\psi\rangle$ therefore remains in $\mathcal{H}_\nu^A$, and an immediate re-test of A returns ν with certainty, for every input state and every μ .

(ii) *Non-disturbance $\Rightarrow$ common eigenbasis.* Suppose compatibility holds, and take any $|\psi\rangle \in \mathcal{H}_\nu^A$ — an A -preparation returning ν with certainty. After any B -outcome μ , non-disturbance demands that an A re-test still return ν with certainty, so by the certainty argument of the note above, $P_\mu^B |\psi\rangle \in \mathcal{H}_\nu^A$. Every B -projector thus maps every A -eigenspace into itself. A two-line lemma turns this invariance into commutation of the projectors: since $P_\mu^B P_\nu^A |\psi\rangle$ already lies in $\mathcal{H}_\nu^A$ for every $|\psi\rangle$, one has $P_\nu^A P_\mu^B P_\nu^A = P_\mu^B P_\nu^A$; taking the adjoint of this equality gives $P_\nu^A P_\mu^B P_\nu^A = P_\nu^A P_\mu^B$, and comparing the two,

$$[P_\nu^A, P_\mu^B] = 0 \quad \text{for all } \nu, \mu. \quad (3.50)$$

Now $B = \sum_\mu b_\mu P_\mu^B$ leaves each $\mathcal{H}_\nu^A$ invariant, and its restriction to $\mathcal{H}_\nu^A$ is self-adjoint there; by the spectral theorem of Section 3.5, $\mathcal{H}_\nu^A$ carries an orthonormal basis of eigenvectors of B , each of which is automatically an eigenvector of A with eigenvalue a_ν . Assembling these bases over all ν yields a common orthonormal eigenbasis of $\mathcal{H}$.

(iii) *The commutator clause.* If the spectral projectors commute, then

$$[A, B] = \sum_{\nu, \mu} a_\nu b_\mu [P_\nu^A, P_\mu^B] = 0, \quad (3.51)$$

since each operator is the spectral sum of its projectors. Conversely, if $[A, B] = 0$, then for $|\psi\rangle \in \mathcal{H}_\nu^A$,

$$A(B|\psi\rangle) = BA|\psi\rangle = a_\nu B|\psi\rangle, \quad (3.52)$$

so $B|\psi\rangle \in \mathcal{H}_\nu^A$: the same invariance as in step (ii), and the same spectral-theorem argument diagonalizes B within each A -eigenspace, producing a common eigenbasis. The three clauses of (3.49) are therefore equivalent. $\square$

The theorem's laboratory content is that compatible observables refine one another. Measuring A and then B is one compound experiment whose outcomes are the pairs (a_ν, b_μ) ; neither stage disturbs the other, and the joint question is sharper than either question alone. The endpoint of such refinement is a set $\{A, B, \dots\}$ of mutually commuting observables whose common eigenspaces are all one-dimensional — a **complete set of commuting observables** (CSCO). The joint outcome then labels a ray uniquely:

$$A|a_\nu, b_\mu, \dots\rangle = a_\nu |a_\nu, b_\mu, \dots\rangle, \quad B|a_\nu, b_\mu, \dots\rangle = b_\mu |a_\nu, b_\mu, \dots\rangle, \quad \dots \quad (3.53)$$

and the readings $(a_\nu, b_\mu, \dots)$ are a complete experimental address of the state: the most refined preparation the system allows. This completes Chapter 2's dimension-as-count story (Section 2.8): the number of joint outcomes of a CSCO — the size of the most refined question — is the dimension of $\mathcal{H}$, now counted by resolving power rather than by orthogonality alone.

 **Note** [*Product-form caution*] *No tensor products appear in this chapter. Every example in this section lives in a single degree of freedom, or in a direct sum of internal sectors in the sense of Chapter 2's superselection remark (Section 2.5): sectors added, never factors multiplied. Composite systems, entanglement, and the tensor product belong to the chapter on composite systems and entanglement; nothing here depends on them.*

The CSCO idea is exhibited by an atom carrying two long-lived internal spin- $\frac{1}{2}$ doublets — a ground doublet and a metastable doublet whose magnetic responses differ, i.e. whose Landé factors are unequal. The internal state space is the four-dimensional *direct sum* $\mathcal{H} = \mathcal{H}_g \oplus \mathcal{H}_m$ of the two doublet spaces — explicitly a sum, not a product, in the sense of the note above. Denote the S_z -eigenstates within the doublets by $|g, \pm\rangle$ and $|m, \pm\rangle$, with eigenvalues $\pm\hbar/2$. A first inhomogeneous magnetic field separates the two manifolds: it realizes the “which-doublet” observable M , each of whose eigenvalues is twofold degenerate. A second Stern–Gerlach stage, applied within each emerging beam, resolves S_z , whose eigenvalues are likewise twofold degenerate — one alternative per manifold. In the basis $(|g, +\rangle, |g, -\rangle, |m, +\rangle, |m, -\rangle)$ both observables are diagonal,

$$M = \text{diag}(m_g, m_g, m_m, m_m), \quad S_z = \frac{\hbar}{2} \text{diag}(1, -1, 1, -1), \quad (3.54)$$

and because both are diagonal in the same basis their commutator vanishes: $MS_z = S_zM = \text{diag}(m_g\frac{\hbar}{2}, -m_g\frac{\hbar}{2}, m_m\frac{\hbar}{2}, -m_m\frac{\hbar}{2})$ so $[M, S_z] = 0$. Neither question alone is complete: the M -outcome g leaves the two-dimensional eigenspace spanned by $|g, \pm\rangle$, and the S_z -outcome $+\hbar/2$ leaves the span of $|g, +\rangle$ and $|m, +\rangle$. Jointly they resolve four rays: the common eigenspaces of M and S_z are the one-dimensional spans of

$|g, +\rangle, |g, -\rangle, |m, +\rangle, |m, -\rangle$, so $\{M, S_z\}$ is a CSCO, and a joint reading such as $(m_g, +\hbar/2)$ fixes the state uniquely.

The same physics appears in a single suitably chosen gradient. Because the two doublets respond to the field with nonzero Landé factors of different magnitudes, the four states $|g, \pm\rangle, |m, \pm\rangle$ carry four distinct magnetic responses, and one inhomogeneous field separates the beam into four spots — the common refinement of both questions at once, possible precisely because the questions commute. The sequential reading is the compatibility theorem on the bench: an M -filter selecting the ground manifold, followed by an S_z -filter selecting $+$, followed by an M re-test returns g with certainty, for the S_z -stage acts within the M -eigenspace it inherits. The example is schematic, and is meant to be: the two doublets and their differing magnetic response are stipulated at the level of an idealized atom, with no species identified and no atomic-structure theory spent. The staged M -filter is likewise an idealized selector: no single magnetic gradient separates the two manifolds with S_z unresolved, for the force on each state is proportional to $g m_j$ and splits every doublet symmetrically; a real realization must address the manifolds selectively, for instance by optical shelving.

Compatibility is the exception, not the rule. The generic pair of questions does not fit together, and the commutator (3.48) — introduced above as a diagnostic — measures exactly how much. That measurement is the next section's work.

3.7 Commutators and Incompatible Experiments

The compatibility theorem of Section 3.6 pronounces a verdict — a pair of observables either admits a common eigenbasis or it does not — but a verdict is not yet a measure. The experiments of the historical chapter showed that the failure has a magnitude: a diagonal polarizer inserted between horizontal and vertical passes half the photons where crossed polarizers passed none, and an x -magnet inserted between two z -magnets restores counts that the z -pair alone had extinguished (Section 1.5). The quantity that carries this magnitude is already on the page. It is the commutator, defined in Section 3.6 where the compatibility theorem first needed it ((3.48)); this section elevates it from a diagnostic to the quantitative measure of incompatibility, and discharges with it one of the oldest unpaid debts of the book's kinematics.

The algebra of the commutator is read off its definition (3.48) by expansion:

$$[A, B] = AB - BA = -(BA - AB) = -[B, A], \quad (3.55)$$

$$[A, \alpha B + \beta C] = A(\alpha B + \beta C) - (\alpha B + \beta C)A = \alpha AB + \beta AC - \alpha BA - \beta CA = \alpha[A, B] + \beta[A, C] \quad (3.56)$$

$$[A, BC] = A(BC) - (BC)A = ABC - BCA = ABC - BAC + BAC - BCA = (AB - BA)C + B(A - CA) \quad (3.57)$$

To summarize:

$$[A, B] = -[B, A], \quad [A, \alpha B + \beta C] = \alpha[A, B] + \beta[A, C], \quad [A, BC] = [A, B]C + B[A, C]. \quad (3.58)$$

One structural fact will be spent in the next section. For self-adjoint A and B the commutator is

2714 anti-self-adjoint,

$$2715 \quad [A, B]^\dagger = (AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = BA - AB = -[A, B] \quad (A = A^\dagger, B = B^\dagger), \quad (3.59)$$

2716 so that $i[A, B]$ is again self-adjoint: the commutator of two observables is not an observable, but i
 2717 times it is, and its expectation value in every state is real. This is the fact that Section 3.8 turns into
 2718 the uncertainty relations.

2719 With the commutator, the fourth kinematical requirement of the historical chapter (Section 1.4.6)
 2720 — *the order of physical operations matters* — receives its kinematical repayment. Successive mea-
 2721 surements compose as operator products (Section 3.2): “the A -question, then the B -question” acts on
 2722 a state as the product BA , and the same questions in the opposite order as AB . The difference of the
 2723 two experiment orders is therefore the single operator $AB - BA = [A, B]$: noncommutation is the
 2724 arithmetic of filter order. What the sequential polarizers and the sequential Stern–Gerlach magnets of
 2725 the historical chapter exhibited on the bench (Section 1.5) is now a computation — where the com-
 2726 mutator vanishes the two orders agree and the questions refine one another (Section 3.6), and where
 2727 it does not, no preparation can serve both orders alike. The repayment is deliberately partial. The
 2728 kinematical share of the requirement — operations compose noncommutatively, and operator multi-
 2729 plication *is* that composition — is hereby closed; the dynamical share, the order-sensitive composition
 2730 of transformations and of time evolution, remains the evolution chapter’s debt, declared open here as
 2731 it was at the close of the states chapter (Section 2.10).

2732 The historically first instance of the algebra predates the formalism itself. The matrix mechanics
 2733 of the historical chapter was built on the rule that observable data multiply noncommutatively, and
 2734 on the canonical relation deduced from the correspondence with classical mechanics (Section 1.4.1),
 2735 which in the language of this chapter reads

$$2736 \quad [Q, P] = i\hbar I. \quad (3.60)$$

2737 The relation is *named* here, not built: the physical theory of the position and momentum operators —
 2738 their definitions, their domains, and the sense in which no finite-dimensional pair can obey the relation
 2739 — belongs to the preview of Section 3.9 and to the chapter on operator methods and representations.
 2740 What this section keeps is the identification itself: the canonical commutator is the historically first
 2741 measured incompatibility, and Section 3.8 will read its uncertainty relation off it directly.

2742 The two-state system exhibits the whole structure in miniature. Section 3.2 computed the polarization-
 2743 filter algebra and recorded the identification $H \leftrightarrow |+\rangle$, $V \leftrightarrow |-\rangle$, $D \leftrightarrow |+\rangle$, $A \leftrightarrow |-\rangle$; with it,
 2744 the matrix of $\sigma_x = P_{+x} - P_{-x}$ is already on the page, $\sigma_z = P_{+z} - P_{-z}$ is the observable of Section 3.4
 2745 ((3.31)), and σ_y is the operator constructed in Section 3.5 with eigenstates $(|+\rangle \pm i|-\rangle)/\sqrt{2}$. Direct
 2746 multiplication then gives

$$2747 \quad [\sigma_z, \sigma_x] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = 2i\sigma_y. \quad (3.61)$$

2748 The physical content of (3.61) is the difference of two benches. The product $P_{+x}P_{+z}$ is “ z -filter, then
 2749 x -filter”; the product $P_{+z}P_{+x}$ is the same questions in the opposite order; and their difference acting
 2750 on a state *is* the commutator $[P_{+x}, P_{+z}]$ at work — the same order-sensitivity as (3.61), in arithmetic.
 2751 The z - x - z recapture probability of the historical chapter (Figure 1.4) is the norm of such a chain:

$P_{+z}P_{+x}|+z\rangle = |+z\rangle\langle +z|+x\rangle\langle +x|+z\rangle = \frac{1}{2}|+z\rangle$, whose squared norm $1/4$ is the probability of the history $(+z \rightarrow +x \rightarrow +z)$; the $-x$ branch contributes another $1/4$, and the recapture probability $1/2$ is recovered from operator arithmetic alone, with no fresh appeal to the bench. The two remaining products of the cycle, by the same multiplication,

$$[\sigma_x, \sigma_y] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = 2i\sigma_z, \quad (3.62)$$

and $[\sigma_y, \sigma_z] = 2i\sigma_x$ by the identical computation with the cyclic permutation $(x, y, z) \mapsto (y, z, x)$; the three relations together are the complete incompatibility structure of the two-state system, every two-state commutator being a linear combination of them. Their deeper meaning, as the algebra of the rotation group, is a debt of the chapter on angular momentum; no rotation theory is used or claimed here.

One conceptual remark is permitted before the quantitative work begins. That mutually exclusive experimental arrangements — this magnet orientation or that one, never both at once — can each be required for a complete account of the same system is the doctrine Bohr announced at Como under the name of complementarity, in the historical setting recorded by the historical chapter (Section 1.4.5). Noncommutativity is its precise kinematical form: the arrangements exclude one another because their operators do not commute, and the commutator measures how strongly. This book develops no philosophy beyond that pointer, for the physics of the doctrine is quantitative: noncommutativity casts a shadow on the statistics, a shadow in which no state can be sharp for both questions at once. That shadow is the next section.

3.8 Uncertainty Relations from Noncommutativity

Section 3.7 closed with a qualitative claim: where $[A, B] \neq 0$, no preparation can serve both questions alike. This section makes the claim quantitative. The tool is the variance of Section 3.4 ((3.26)), restated for the occasion in operator form: with $\langle A \rangle_\psi := \langle \psi | A | \psi \rangle$ ((3.24)), the spread of the A -histogram on an ensemble prepared in $|\psi\rangle$ is

$$(\Delta A)^2 = \langle (A - \langle A \rangle_\psi)^2 \rangle_\psi = \|(A - \langle A \rangle_\psi) |\psi\rangle\|^2, \quad (3.63)$$

the squared norm of a vector — and it is this norm form that admits the Cauchy–Schwarz inequality admitted in the states chapter ((2.24)).

Theorem (Robertson). *For self-adjoint A, B and any state $|\psi\rangle$,*

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle_\psi|. \quad (3.64)$$

The proof is three lines. Write $A' := A - \langle A \rangle_\psi I$ and $B' := B - \langle B \rangle_\psi I$; both are self-adjoint, and $(\Delta A)^2 = \|A' |\psi\rangle\|^2$, $(\Delta B)^2 = \|B' |\psi\rangle\|^2$. The Cauchy–Schwarz inequality ((2.24)), applied to the two vectors $A' |\psi\rangle$ and $B' |\psi\rangle$, gives

$$(\Delta A)^2 (\Delta B)^2 = \|A' |\psi\rangle\|^2 \|B' |\psi\rangle\|^2 \geq |\langle \psi | A' B' | \psi \rangle|^2. \quad (3.65)$$

The expectation on the right decomposes into its Hermitian and anti-Hermitian parts,

$$\langle A' B' \rangle_\psi = \frac{1}{2} \langle A' B' + B' A' \rangle_\psi + \frac{1}{2} \langle [A, B] \rangle_\psi, \quad (3.66)$$

where $[A', B'] = [A, B]$ has been used; the first term is real and the second, by the anti-self-adjointness of the commutator (Section 3.7), is purely imaginary. The modulus therefore splits,

$$|\langle A'B' \rangle_\psi|^2 = \frac{1}{4} \langle A'B' + B'A' \rangle_\psi^2 + \frac{1}{4} |\langle [A, B] \rangle_\psi|^2 \geq \frac{1}{4} |\langle [A, B] \rangle_\psi|^2, \quad (3.67)$$

and the square root of the two displays is (3.64).⁴ The discarded term is information, not waste: retaining it sharpens (3.64) to $(\Delta A)^2(\Delta B)^2 \geq \frac{1}{4} \langle A'B' + B'A' \rangle_\psi^2 + \frac{1}{4} |\langle [A, B] \rangle_\psi|^2$, in which the covariance of the two observables cooperates with their commutator.⁵ In the examples of this chapter the covariance term vanishes, and (3.64) is the operative bound.

The operational reading of (3.64) is exact and austere. The quantities ΔA and ΔB are ensemble spreads — the widths of the two histograms accumulated when the A -question and the B -question are put, in separate runs, to members of an identically prepared ensemble, in the ensemble discipline of Section 3.1. The inequality is a statement about those two histograms, state by state: whenever $\langle [A, B] \rangle_\psi \neq 0$, no preparation in the state $|\psi\rangle$ makes both distributions narrow. It is not a statement about the errors of single measurements, nor about one apparatus disturbing another, nor about any individual event; such stories may be told as heuristics, and the theorem needs none of them.

The two-state system gives the first instance. From the cyclic commutators of Section 3.7 ((3.61) and its cycle),

$$\Delta\sigma_x \Delta\sigma_z \geq |\langle \sigma_y \rangle_\psi|. \quad (3.68)$$

In the state $|+z\rangle$ the right side vanishes, and the inequality permits exactly what Section 3.4 computed ((3.32)): the z -question is sharp, $\Delta\sigma_z = 0$, while the x -question is maximally spread, $\Delta\sigma_x = 1$ — one component certain, the other maximally uncertain, and the bound silent because the state has no y -inclination. In a state tilted toward the y -axis the right side is nonzero, and both perpendicular spreads must pay for it. The lesson is general: the bound is state-dependent, and where the mean commutator vanishes it simply falls silent.

For the canonical pair of Section 3.7, $[Q, P] = i\hbar I$, the mean commutator is state-independent, and (3.64) reads

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (3.69)$$

This is stated as the formal consequence of the canonical commutator. The rigorous meaning of Δx and Δp — for the unbounded position and momentum operators, with their domains and their non-normalizable eigenstates — is deferred to Appendix A and to the chapter on operator methods and representations. The physics of (3.69), however, can be exhibited now, quantitatively, on a bench the book has already built.

The relation entered physics in 1927, announced together with the doctrine of complementarity in the synthesis whose setting the historical chapter records (Section 1.4.5); its historical heuristic companion was Heisenberg's microscope thought experiment, in which the position of an electron is

⁴The displayed algebra is finite-dimensional, or formal: for unbounded operators the vectors $A'|\psi\rangle$ and $B'|\psi\rangle$ must exist, and the products AB and BA must share a domain on which the decomposition runs. The rigorous statement is deferred to Appendix A.

⁵The strengthening is Schrödinger's: E. Schrödinger, *Sitzungsber. Preuss. Akad. Wiss. Phys.-Math. Kl.* **19**, 296 (1930); the relation (3.64) itself was proved in this generality by H. P. Robertson, *Phys. Rev.* **34**, 163 (1929).

resolved by scattered light and the recoil of the scattering disturbs the momentum.⁶ That argument is named here, not used: it needs scattering kinematics and a model of instrumental disturbance, neither of which this chapter has built — and the theorem above needs neither. The chapter’s quantitative example is not scattering but diffraction, where the uncertainty physics is read off a single measured envelope.

A single slit of width a realizes the physics directly. The slit is a position filter: it transmits only amplitudes localized in its interval, the continuum’s version of a projector — the interval projectors made precise in Section 3.9. A particle that has passed the slit therefore carries a transverse position prepared to a width $\Delta x \sim a$. Its far-field amplitude on a screen at distance L is the aperture integral of the states chapter ((2.47)),

$$A(x) \propto \int_{-a/2}^{a/2} e^{ikx'x/L} dx' = \frac{2L}{kx} \sin \frac{kax}{2L}, \quad (3.70)$$

whose intensity envelope first vanishes at $kax/2L = \pi$, that is, at the transverse wave number $k_x = kx/L = 2\pi/a$. Reading the momentum scale off the envelope with de Broglie’s relation $p = h/\lambda$ (Section 1.3.2), the transverse momentum distribution reaches its first zero at $p_x = \hbar k_x = h/a$: the slit has prepared a momentum spread $\Delta p_x \sim h/a$. The product of the two widths is

$$\Delta x \Delta p_x \sim a \cdot \frac{h}{a} = h, \quad (3.71)$$

the uncertainty physics of (3.69) read off the diffraction envelope, with the aperture width cancelling out of the product: narrowing the slit buys its position definition with a proportionally broader momentum envelope.

The honesty clause of this example is mandatory, and it concerns the meaning of the word *width*. The envelope is proportional to $\text{sinc}^2(kax/2L)$, and that distribution has *no finite second moment*: the integral $\int x^2 \text{sinc}^2(ax) dx$ diverges, so the standard deviation of the arrival positions does not exist, and the Δp_x of the estimate is not a Robertson standard deviation. It is a width-to-first-minimum — an operational width, whose numerical constant depends on the convention adopted — exactly as $\Delta x \sim a$ is the width of the transmitting interval and not a standard deviation. The example therefore exhibits the physics of the uncertainty relation without pretending to saturate (3.69); the precise saturation story, and the preparations that achieve it, belong to the chapter on operator methods and representations.

The advantage of the Robertson relation is its generality together with its austerity: one algebraic line — noncommutativity, filtered through the admitted Cauchy–Schwarz theorem — quantifies the incompatibility of every pair of questions, for every state, with no model of any instrument. Its limitations are written into its form. The bound is state-dependent and falls silent exactly where the mean commutator vanishes, as the spin instance showed; it constrains ensemble spreads and says nothing about single events or single devices; and for the pair on which it matters most, position and momentum, its rigorous content owes Appendix A an analysis of unbounded operators and their domains. The relation is the exact quantitative shadow of noncommutativity — and a shadow, by its nature, is not the object.

The canonical pair’s alternatives are continuous, and the example has just spent them: the slit

⁶W. Heisenberg, *Z. Phys.* **43**, 172 (1927).

was called a projector onto an interval and the momentum a de Broglie scale, both honestly but neither formally. What, exactly, becomes of the measurement theory when the alternatives form a continuum? The answer is a preview — labeled, bounded, and tied to Appendix A — and it is the next section.

3.9 The Continuous Spectrum: an Honest Preview

Everything proved so far lives where the alternatives can be counted. The diffraction physics of Section 3.8 has already spent a continuum: the slit prepared a position, the screen registered one, and both were spoken of honestly but informally. This section is the chapter's one labeled preview of the continuous case. It states what the measurement theory becomes when the alternatives form a continuum; it uses only the shorthand the states chapter has already licensed; and it assigns its two honest debts in advance — rigor to Appendix A, physics to the chapter on operator methods and representations. Nothing later in the book will silently reuse more than is exhibited here.

The mechanism is already built; it only needs to be re-read as measurement. In the states chapter, N illuminated slits fused into a continuous aperture as $N \rightarrow \infty$ with the aperture held fixed ((2.46), (2.47)): each slit was one alternative, and the fusion replaced the sum over alternatives by an integral. Measurement-theoretically, each slit was a rank-1 alternative with its projector, and the fused continuum therefore assigns a projector not to a point but to an *interval*:

$$P_I = \int_I |x\rangle \langle x| \, dx, \quad (3.72)$$

written with the states chapter's δ -normalization shorthand ((2.44)). The family of all such P_I is a projection-valued measure on the line, in exactly the sense of Section 3.3 with sums replaced by integrals: $P_\emptyset = 0$; $P_{\mathbb{R}} = I$, formally the continuum composition law ((2.48)); and $P_{I \cup J} = P_I + P_J$ for disjoint intervals, since integrals add over disjoint domains.

Note *The warning of the states chapter stands verbatim ((2.44) and the note attached to it there): the ket $|x\rangle$ is not a vector of $\mathcal{H}$ — its formal norm diverges, and no convention repairs the defect — and (3.72) is a calculus whose honest justification, the rigged Hilbert space, is Appendix A's debt. Every statement made with the shorthand can be re-expressed, at the price of some care, among the genuine vectors of $L^2(\mathbb{R})$ ((2.45)).*

The position measurement is now a computation. For a state with wave function $\psi(x) = \langle x|\psi\rangle$ ((2.43)), the probability that the position registration falls in the interval I is the expectation of its projector,

$$p(I) = \langle \psi | P_I | \psi \rangle = \int_I |\psi(x)|^2 \, dx, \quad (3.73)$$

which is Born's 1926 rule (Section 1.4.3) — announced in the historical chapter as an interpretation of the wave function — recovered here as the expectation of the interval projectors: the continuum's instance of the Born rule in projection-valued-measure language ((3.18)). The mean position is the expectation of the same measure,

$$\langle x \rangle_\psi = \int_{-\infty}^{+\infty} x |\psi(x)|^2 \, dx, \quad (3.74)$$

and the whole statistic compresses, formally, into the symbol $(X\psi)(x) = x\psi(x)$, the *position operator*. The name carries its status on its face: X is unbounded, acts only on a domain within $L^2(\mathbb{R})$, and possesses no eigenvectors in $\mathcal{H}$ at all.⁷ In this chapter, accordingly, X remains exactly what the shorthand licenses: a labeled compression of the interval projection-valued measure, not an operator in the developed sense of Sections 3.2–3.5.

Momentum is named, not built. De Broglie’s relation (Section 1.3.2) singles out the plane waves $e^{ipx/\hbar}$ as the candidates for momentum-sharp states; but such a wave has divergent norm — an exercise of the states chapter computes it directly (Section 2.9) — so the momentum-sharp candidates are no more vectors of $\mathcal{H}$ than $|x\rangle$ is. The momentum operator, and with it the canonical commutator’s proper home, is the representations chapter’s debt; the relation $[Q, P] = i\hbar I$ itself was named, not built, in Section 3.7.

It is worth listing, as the states chapter did for its own continuum, what the fusion of alternatives preserves and what it breaks. *Surviving*, with sums systematically replaced by integrals: the projection-valued-measure language of Section 3.3 — the projectors P_Ω , coarse-graining, the Born rule ((3.18)); the spectral decomposition of Section 3.4, whose continuum form underlies (3.72); expectation values and variances, as above. *Broken*: the eigenkets — $|x\rangle \notin \mathcal{H}$, and the eigenvalue equation $X|x\rangle = x|x\rangle$ ((3.35)) has no normalizable solutions; and with the eigenkets goes the finite-dimensional spectral theorem itself, whose honest infinite-dimensional successor — projection-valued measures on $\mathbb{R}$, unbounded self-adjoint operators, their domains — is Appendix A’s debt, as Section 3.5 declared.

The diffraction screen of Section 3.8 is the continuum’s first instrument, and it is already built. Each bin I of the screen’s surface is an interval alternative, and the counts accumulated in it realize P_I : run by run, the histogram of arrivals converges to the distribution

$$|\psi_{\text{screen}}(x)|^2 \propto \text{sinc}^2\left(\frac{kax}{2L}\right), \quad (3.75)$$

the squared aperture amplitude of the states chapter ((2.47)) — the chapter’s flagship example recycled as the continuous measure’s first measurement. The cumulative counts up to the position x evaluate the same measure on a half-line,

$$F(x) = \int_{-\infty}^x |\psi_{\text{screen}}(x')|^2 dx', \quad (3.76)$$

and the family $\{F(x)\}$ is the cumulative distribution of that measure — its expectation on the preparation — read out directly, bin by bin, on the bench.

The measurement theory is complete, the continuum honestly included: what the finite theory proved, the preview states, and what the preview cannot prove, it owes by name. It remains to collect the axioms in one place, to balance the ledger of debts, and to name the question that measurement theory itself cannot answer.

⁷These are precisely the features — unboundedness, domains, continuous spectra — that the infinite-dimensional spectral theorem of Appendix A is built to handle; the finite-dimensional theorem of Section 3.5 proves nothing about X .

3.10 The Measurement Axioms: Summary and Open Debts

The construction of this chapter is complete, and its content compresses into five statements. As with the states chapter, each is an axiom in the formal sense and an experimental report in the operational one — with one exception, honestly labeled: the higher-rank update of Section 3.6 is the chapter’s single genuinely new axiom, and it says so in its place.

(M1) Ideal measurement. An ideal measurement is a complete set of mutually exclusive alternatives realized as a reproducible filter: an immediate re-interrogation of a selected output returns the same outcome with certainty (Section 3.1). *Warrant:* the re-tested beams of the sequential experiments — a $|+z\rangle$ -selected beam re-asked the z -question answers $+z$ with certainty (Figure 1.4).

(M2) Outcome probabilities. The outcome n of an ideal measurement on the state $|\psi\rangle$ occurs with probability $p_n = \langle\psi|P_n|\psi\rangle$, the expectation of the outcome’s projector (Sections 3.1–3.3; (3.18)). *Warrant:* the Born rule of the states chapter ((2.7)), of which this is the restatement in projector language.

(M3) State update. Conditional on a non-degenerate outcome n , the state is the ray of $|e_n\rangle$ — a *theorem*, proved from repeatability, the Born rule, and the saturation clause of the Cauchy–Schwarz inequality ((3.3), Section 3.1). Conditional on a degenerate outcome ν with projector P_ν , the state updates as $|\psi\rangle \mapsto P_\nu |\psi\rangle$, renormalized ((3.47), Section 3.6): the Lüders rule, the measurement postulate’s one genuinely new axiom, adopted exactly where repeatability stops forcing the answer. *Warrant:* repeatability forces the conditional state into the outcome’s subspace; the preservation of the incoming relative amplitudes within it is the minimally-disturbing idealization, named as one.

(M4) Observables. An observable is a projection-valued measure whose outcomes carry real numerical labels — equivalently, a self-adjoint operator $A = \sum_n a_n P_n$ ((3.23)); in finite dimension every self-adjoint operator is the spectral encoding of an ideal experiment, and the instrument readings are its eigenvalues (Sections 3.4–3.5). *Warrant:* instrument readings are real numbers, and reality in every state forces $A = A^\dagger$ ((3.33)); the spectral theorem then leaves no orphan operators.

(M5) Compatibility. Two observables are compatible — each question undisturbed by the other — if and only if they possess a common orthonormal eigenbasis, if and only if their commutator vanishes ((3.49), Section 3.6); the quantitative shadow of incompatibility is the Robertson bound $\Delta A \Delta B \geq \frac{1}{2} |\langle[A, B]\rangle_\psi|$ ((3.64), Section 3.8). *Warrant:* the staged-filter experiments, and the Cauchy–Schwarz theorem admitted in the states chapter.

The chapter’s integration of experiment and formalism is collected, in lieu of a new example, in Table 3.1, which adds the operator rows to the experiment–formalism dictionary of the states chapter (Table 2.1).

The mainline duty is a ledger of debts, and it must now be stated plainly. *Repaid:* the four measurement debts of the states chapter — the operator reading of the completeness shorthand ((2.34)), discharged as the resolution of identity ((3.9), Section 3.2); the measurement postulate proper, dis-

Table 3.1: The experiment–formalism dictionary, continued: the operator rows of this chapter, extending Table 2.1.

Experimental notion	Formal object
A reproducible filter	a projector
A complete experimental question	a projection-valued measure
A numerical instrument	a self-adjoint operator
An instrument reading	an eigenvalue
A sharp preparation	an eigenstate
Joint refinability of two questions	a vanishing commutator
Irreducible joint spread	the Robertson bound

charged as the filter definition and the rank-1 update theorem of Section 3.1 together with the Lüders axiom of Section 3.6; the Gleason signpost, stated in projector language in Section 3.3; and the uncertainty relations, proved from the admitted Cauchy–Schwarz theorem in Section 3.8. Repaid also are two older debts of the historical chapter: requirement (iv), the order-sensitivity of physical operations (Section 1.4.6), discharged kinematically in Sections 3.2 and 3.7 — successive processes compose as operator products, and noncommutation is the arithmetic of filter order — and the matrix-mechanics debt (Section 1.4.1), discharged as the commutator theory of Section 3.7. *Planted, and declared openly:* unitary relabelings, and the exit question below, to the chapter on evolution; the canonical pair’s physical theory — position and momentum as operators, their domains, the commutator’s proper home — to the chapter on operator methods and representations, with the preview of Section 3.9; the rotational meaning of the two-state commutator algebra, to the chapter on angular momentum; density operators, Gleason’s ρ , the bookkeeping of non-selective measurement, and the measurement problem, to the chapter on composite systems; spectral rigor for the continuum, to Appendix A. The dynamical share of requirement (iv) remains, as declared in Section 3.7, the evolution chapter’s.

An observable is a self-adjoint operator, and a measurement is a projector’s work. But the theory so far speaks only of interrogations: a preparation, then a question, then an answer. Between two measurements the counters are silent — and the silence is an experimental fact with content, for interrogating identically prepared ensembles after different waiting times does not answer as “nothing happened” would predict: the next experiment’s counts depend on the delay. Yet no axiom given so far contains the waiting time. What law governs the state between measurements? That is the theory of evolution, and it is the next chapter.

Chapter 3 Exercise

- (*) (Section 3.1) A Stern–Gerlach z – x – z chain (Figure 1.4) is fed with atoms prepared in $|+z\rangle$. Apply the rank-1 update theorem of Section 3.1 stage by stage: after the x -stage, and again after the final z -stage, write the conditional state for each recorded outcome. Enumerate the four outcome histories ($\pm x$ then $\pm z$), compute the probability of each as a product of stage probabilities, and verify that the four joint probabilities sum to one. Show that the final counts

are $+z$ and $-z$ in equal shares $1/2$, whichever x -outcome was recorded — the x -selection has re-prepared the state.

2. (★) (Section 3.2) Using the $\mathbb{C}^2$ columns of the states chapter, write the polarization projectors $P_D = |D\rangle\langle D|$ and $P_A = |A\rangle\langle A|$ as explicit 2×2 matrices in the H/V basis, where $|D\rangle = (|H\rangle + |V\rangle)/\sqrt{2}$ and $|A\rangle = (|H\rangle - |V\rangle)/\sqrt{2}$. Verify idempotence $P_D^2 = P_D$, mutual exclusivity $P_D P_A = 0$, and completeness $P_D + P_A = I$ by direct multiplication. Compute $\|P_V P_D P_H |H\rangle\|^2$ and compare with the value $1/4$ measured for the polarizer chain of the historical chapter (Section 1.5).
3. (★★) (Section 3.3) A spin-1 Stern–Gerlach apparatus has three outcomes, with projectors $P_+ = \text{diag}(1, 0, 0)$, $P_0 = \text{diag}(0, 1, 0)$, $P_- = \text{diag}(0, 0, 1)$ in the basis $\{|+1\rangle, |0\rangle, |-1\rangle\}$. (a) Verify the projection-valued-measure axioms of Section 3.3 for the family $\{P_+, P_0, P_-\}$. (b) Coarse-grain to the two-outcome measure $\{P_+ + P_-, P_0\}$, and state in one sentence which experimental question it asks. (c) For the state $(1, 1, 1)^T/\sqrt{3}$, compute all outcome probabilities in both the fine and the coarse measure, and verify consistency: the coarse outcome “ $\pm$ ” must carry probability $2/3$, the sum of its two fine constituents, and the total probability must remain one throughout.
4. (★★) (Section 3.4) Take the half-angle law as given ((2.39), experimental input). A spin- $\frac{1}{2}$ system is prepared in the state

$$|\psi\rangle = \cos \frac{\theta'}{2} |+\rangle + e^{i\varphi'} \sin \frac{\theta'}{2} |-\rangle, \quad (3.77)$$

and interrogated along the axis $\mathbf{n}$ with polar angles (θ, φ) , using the observable $\sigma_{\mathbf{n}} = P_{+\mathbf{n}} - P_{-\mathbf{n}}$ ((3.31)) and the rotated basis $|\pm\mathbf{n}\rangle$ of the states chapter ((2.38)). (a) Compute $\langle\sigma_{\mathbf{n}}\rangle_{\psi}$ and $\Delta\sigma_{\mathbf{n}}$ as functions of (θ, φ) and (θ', φ') . (b) Show that $\Delta\sigma_{\mathbf{n}} = 0$ exactly when $\mathbf{n}$ coincides with the preparation axis (θ', φ') , and interpret: along which axis is the preparation sharp, and along which axis is the spread maximal?

5. (★★) (Section 3.5) Let $A = (\sigma_x + \sigma_z)/\sqrt{2}$, with σ_x, σ_z the 2×2 matrices of Sections 3.2 and 3.4. (a) Diagonalize A explicitly: find its eigenvalues and its eigenbasis, expressed in the z -basis. (b) The spectral theorem of Section 3.5 promises that A encodes a Stern–Gerlach axis: identify the axis $\mathbf{n}$, and verify by direct multiplication that $A = (+1) P_{+\mathbf{n}} + (-1) P_{-\mathbf{n}}$ with $P_{\pm\mathbf{n}} = |\pm\mathbf{n}\rangle\langle\pm\mathbf{n}|$. (c) Generalize: for $A = \mathbf{a} \cdot \boldsymbol{\sigma} := a_x \sigma_x + a_y \sigma_y + a_z \sigma_z$ with real coefficients, show that the eigenvalues are $\pm|\mathbf{a}|$.
6. (★★★) (Section 3.6) In the two-doublet atom of Section 3.6’s example, the state space is the four-dimensional *direct sum* of a ground doublet and a metastable doublet, with basis $\{|g, +\rangle, |g, -\rangle, |m, +\rangle, |m, -\rangle\}$, ordered by manifold and by z -component within each manifold. (a) Write the which-manifold observable M (eigenvalue $+1$ on the ground doublet, -1 on the metastable doublet) and the z -component S_z (eigenvalues $\pm\hbar/2$ within each doublet) as diagonal 4×4 matrices, and show by direct multiplication that $[M, S_z] = 0$. (b) List the four common eigenrays and their joint labels. (c) Exhibit an observable B' — a diagonal matrix of your choice — that commutes with M yet fails to resolve the within-manifold degeneracy. (d) State precisely, in terms of the dimensions of the common eigenspaces, why $\{M, S_z\}$ is a complete set of commuting observables

and $\{M, B'\}$ is not.

7. (★ ★ ★) (Sections 3.7 and 3.8) Let $|+\mathbf{n}_0\rangle$ be the spin state sharp along the unit vector $\mathbf{n}_0 = (n_x, n_y, n_z)$, for which $\langle\sigma_i\rangle = n_i$ and $\Delta\sigma_i = \sqrt{1 - n_i^2}$. (a) Using the cyclic commutators of Section 3.7 ((3.61) and its cycle), compute $\langle[\sigma_i, \sigma_j]\rangle$ for all three pairs. (b) Verify each Robertson inequality ((3.64)) explicitly; the identity $(1 - n_x^2)(1 - n_z^2) = n_y^2 + n_x^2 n_z^2$, and its permutations, may be useful. (c) Identify the states and the pairs for which the inequality is saturated *trivially*, with vanishing right side, and state what that means physically: when is the bound silent, and when is it informative?
8. (★ ★ ★) (Sections 3.8 and 3.9) The far-field amplitude behind a slit of width a is the aperture integral ((2.47)),

$$A(x) \propto \frac{2L}{kx} \sin \frac{kax}{2L}, \quad (3.78)$$

with $k = 2\pi/\lambda$ and L the screen distance. (a) Locate the first minimum x_1 of the intensity envelope. (b) Convert x_1 to a transverse momentum via de Broglie's relation $p = h/\lambda$ (Section 1.3.2); taking the momentum to first minimum as the operational spread Δp_x , show that $a \Delta p_x = h$, with the numerical constant displayed explicitly. (c) Show that the second moment $\int x^2 |A(x)|^2 dx$ of the sinc² envelope diverges, and explain in one paragraph why the Robertson standard deviation of Section 3.8 is therefore not the operative width in this estimate — and why the estimate is nonetheless physically meaningful.

Chapter 4 Evolution Operators, Symmetries, and Conservation Laws



Note *This chapter is deductive in character. Taking the state-space axioms of Chapter 2 and the measurement axioms of Chapter 3 as established input, it adds exactly one new axiom — the evolution postulate — and derives its consequences: the Schrödinger equation as the postulate’s theorem-form, the spectral resolution of the evolution operator, two exact two-state dynamics (spin precession and magnetic resonance), the time-ordered exponential for driven systems, symmetry transformations and Wigner’s theorem, continuous symmetries treated as Lie groups, the Galilean group with its algebra derived and its central charge exhibited, the special status of time, conservation laws with good quantum numbers and selection rules, and the discrete symmetries of space and time. Two deep theorems (Stone’s and Wigner’s) are stated without proof and cited with pointers to the literature; everything provable with the machinery at hand is proved. It presupposes nothing beyond Chapters 2 and 3.*

The preceding chapter closed with a question it had proved its own axioms cannot answer (Section 3.10):

An observable is a self-adjoint operator, and a measurement is a projector’s work. But the theory so far speaks only of interrogations: a preparation, then a question, then an answer. Between two measurements the counters are silent — and the silence is an experimental fact with content, for interrogating identically prepared ensembles after different waiting times does not answer as “nothing happened” would predict: the next experiment’s counts depend on the delay. Yet no axiom given so far contains the waiting time. What law governs the state between measurements? That is the theory of evolution, and it is the next chapter.

This chapter is that theory.

The register of the answer differs from that of the first three chapters, and the difference is the book’s, not merely the chapter’s. Chapters 1–3 were inductive: from the crises of the old quantum theory and from the two-state experiments they distilled the state-space axioms (S1)–(S5) and the measurement axioms (M1)–(M5), admitting no formal object that had not first been earned on a bench. With those axioms in hand the book enters its second phase, in which the axioms are input and the theorems are output. Exactly one new axiom is added — the evolution postulate of Section 4.1 — and everything else in the chapter is either proved from it and its predecessors, or cited honestly with the proof pointed to. Deduction does not mean detachment from the bench: by the charter of this phase, every section still lands on a measurable quantity — precession and resonance frequencies, survival and beat laws, selection rules and degeneracies, and one phase between two orders of moving a system that no counter has yet exhibited.

The question is as old as the theory, and the shape of its answer was glimpsed at the theory’s birth. Heisenberg’s transition quantities of 1925 oscillated at the Bohr frequencies by construction,

because the observed spectral lines demanded it (Section 1.4.1); Schrödinger’s wave equation of 1926, a controlled guess in Chapter 1’s accounting, supplied a law of motion in one representation; and the distinction drawn there between kinematics and dynamics — kinematics fixes what kind of object a state is and what questions may be asked of it, while dynamics says how the state evolves in time (Section 1.4.6) — is exactly the border this chapter crosses. Wigner and Weyl brought the theory of group representations into quantum mechanics within a few years, preparing the modern understanding of symmetry that Chapter 1 deferred to a later chapter (Section 1.4.5). This is that chapter as well: the law of waiting and the theory of symmetry turn out to be one subject, because evolution is the model continuous family of transformations.

The route is deductive, but the landings are experimental. Section 4.1 states the evolution postulate and proves its infinitesimal form; Section 4.2 resolves the evolution operator into the spectrum of its generator and identifies what the eigenvalues physically are; Sections 4.3 and 4.4 solve the two-state dynamics exactly, first for a static field and then for a driven one, repaying the oldest two-state debt of the book, the Stern–Gerlach spin of Chapter 1. The chapter then turns to the transformations whose theory evolution completes: symmetry transformations, classified by Wigner’s theorem (Section 4.5); continuous symmetry families, recognized as Lie groups (Section 4.6); the Galilean group as the central worked example, with mass emerging as a central, superselected charge (Section 4.7); and the special status of time (Section 4.8). The harvest follows: conservation laws, good quantum numbers, degeneracy, and selection rules (Section 4.9); parity and time reversal (Section 4.10); and at the exit, the ledger balanced and the next chapters’ questions named (Section 4.11).

4.1 Between Interrogations: the Evolution Postulate

The question is now posed with all its machinery in place. A system is prepared in the state $|\psi(0)\rangle$ at time $t = 0$; it is then left alone; at time t it is interrogated. The counters of that interrogation depend on the delay — that is the experimental content of the preceding chapter’s exit question — so the preparation and the delay together fix a state $|\psi(t)\rangle$, and waiting defines a map $|\psi(0)\rangle \mapsto |\psi(t)\rangle$. What can be said about that map a priori, before any particular system is studied? Four clauses, each a bench report with its operative constraint named, fix its form completely; and the form they fix is this chapter’s single new axiom.

(i) *Linearity*. Between the preparation and the interrogation no record exists anywhere in the world of which alternative the system took — that is what “left alone” means. By the marking clause of Chapter 2 (Section 2.1), alternatives that leave no record combine by amplitudes, and amplitudes add: a superposition of preparations must evolve into the corresponding superposition of evolved states. The map of waiting is therefore a linear map on $\mathcal{H}$. The operative constraint is the absence of a record; where a record exists, the experiment is an interrogation and the law of this section does not apply to it.

(ii) *Unitarity*. At any delay, any complete question may be asked, and its outcome probabilities must sum to one: for every projection-valued measure, $\sum_n \langle \psi(t) | P_n | \psi(t) \rangle = \langle \psi(t) | \psi(t) \rangle$ by the Born rule (3.18) and the resolution of identity (3.9). The map of waiting therefore preserves the norm

of every state. A linear map that preserves the norm of every vector preserves every inner product — inner products are fixed linear combinations of norms, by the polarization identity (3.28) — and is therefore unitary (3.10). One loophole must be closed honestly, and it is closed inside the axiom itself. Probability sums alone would also permit an *antilinear* norm-preserving map, of which complex conjugation in a fixed basis is the prototype. The composition law of clause (iv) below excludes the loophole: it gives $U(t) = U(t/2) U(t/2)$, so every member of the family is a square, and the square of *either* norm-preserving kind is linear — conjugating the scalars twice restores them. Hence every $U(t)$ is linear, hence unitary. The antilinear possibility is genuine physics, not a pedant’s scruple: it returns exactly once in this book, where the reversal of motion forces it (Section 4.10), and its classification theorem is Wigner’s (Section 4.5).

(iii) *Continuity*. The delayed counts vary continuously with the delay: no frequency has ever been observed to jump as a waiting time is dialed through any value. In the theory’s language, $U(t) |\psi\rangle$ approaches $|\psi\rangle$ as $t \rightarrow 0$, for every prepared state — strong continuity.

(iv) *Composition*. Waiting s and then waiting t is one way of waiting $s + t$, so $U(t + s) = U(t) U(s)$; and the law of waiting does not depend on when the preparation happened — the homogeneity of time — so one and the same family $U(t)$ serves at every epoch, with $U(0) = I$. It is this clause that makes the family a *one-parameter group*: composition is addition of the parameter. The vocabulary is Section 4.6’s; the fact is the bench’s.

None of these clauses is a theorem, and this section does not pretend otherwise: each is a plausibility argument whose operative constraint has been named, and together they fix the form of the law. That form we now postulate.

Axiom D1 (evolution between interrogations). *Between a preparation and the next interrogation, the state of an isolated system is carried by a family of linear operators $U(t)$, one for each waiting time:*

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad U(t + s) = U(t) U(s), \quad U(0) = I, \quad U(t)^\dagger U(t) = I, \quad (4.1)$$

with the family strongly continuous in t .

The axiom’s whole content is infinitesimal: a family obeying the composition law is determined by its behavior at $t = 0$, and the object sitting there is the theory’s first generator. In finite dimension the statement is elementary. A continuous unitary family is differentiable; differentiating $U(t)^\dagger U(t) = I$ at $t = 0$ gives $U'(0)^\dagger = -U'(0)$, so the operator $i\hbar U'(0)$ equals its adjoint. In the general case the differentiability itself is deep, and its theorem is cited, not proved.

Theorem (Stone, stated without proof). *Every strongly continuous one-parameter unitary family (4.1) has a unique self-adjoint generator H :*

$$U(t) = e^{-iHt/\hbar}, \quad H := i\hbar \left. \frac{dU}{dt} \right|_{t=0}, \quad (4.2)$$

where the exponential is the function $e^{-ixt/\hbar}$ of the observable H in the sense of Chapter 3’s function-of-an-observable rule (3.25).¹ The insertion of $\hbar$ is, at this stage, dimensional only: $U'(0)$ is a rate,

¹M. H. Stone, *Ann. of Math.* **33**, 643 (1932). For unbounded H the exponential is defined through the spectral resolution, and the domain questions it raises belong to Appendix A; the spectral reading of the display is made explicit in Section 4.2.

with units of inverse time, and multiplication by the quantum of action gives the generator the *units* of energy. Whether the generator *is* the energy observable is a physical question, not settled by a choice of units; it is earned in Section 4.2.

Theorem (Schrödinger equation; the theorem-form of D1). *Every state evolving by (D1) obeys*

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle, \quad (4.3)$$

with H the generator (4.2). The proof is one line of calculus on the family law:

$$\left. \frac{dU}{dt} \right|_t = \lim_{\varepsilon \rightarrow 0} \frac{U(t + \varepsilon) - U(t)}{\varepsilon} = \lim_{\varepsilon \rightarrow 0} \frac{U(\varepsilon) - I}{\varepsilon} U(t) = U'(0) U(t) = -\frac{i}{\hbar} H U(t),$$

and applying both sides to $|\psi(0)\rangle$ gives (4.3). The equation is therefore honestly labeled: it is not a new axiom but the infinitesimal register of (D1), and the two say one thing. In finite dimension the loop closes elementarily, without Stone's theorem: the equation $i\hbar dU/dt = HU$ with $U(0) = I$ has the unique solution $U(t) = e^{-iHt/\hbar}$, as differentiation of the exponential series shows at once. The generator H is called the **Hamiltonian** of the system. The name is so far only a name; its physical reading as the energy is earned by the Bohr-frequency identification of Section 4.2 and sealed by the time-translation analysis of Sections 4.7 and 4.8.

The equation has stood in this book twice before, and both appearances are now discharged. Chapter 1 recorded Schrödinger's wave equation of 1926 as a controlled guess, and its equivalence section recorded the component law $i\hbar \dot{c}_m = \sum_n H_{mn} c_n$ of matrix mechanics (Section 1.4.4). What (4.3) adds is deductive status: the law of motion is proved from (D1), abstract and representation-free — no longer a guess, and no longer tied to any representation. Its wave-mechanical specialization remains the next chapter's.

One operational remark belongs to this section, stated once and not developed. Throughout this chapter, t is the experimenter's waiting time, read from the laboratory clock: a parameter of the description, not an observable of the system — no counter registers it. The structural consequences of that status (there is no time operator, and the Hamiltonian's role is dual) are analyzed in Section 4.8.



Note *Two laws, two jurisdictions. Between a preparation and the next interrogation, the state obeys (D1) alone: the evolution is continuous, deterministic, and reversible. At an interrogation, the measurement axioms (M1)–(M5) take over: the outcomes are random with Born-rule frequencies, and the conditional state is the Lüders update (3.47) — discontinuous and irreversible. The joint protocol of the theory is: evolve by (4.3) between interrogations; update by (3.47) at one. Which physical event constitutes an interrogation — when nature applies the second clause rather than the first — the formalism does not say; that is the measurement problem, and it belongs to the chapter on composite systems.*

The theorem's first consequence is a survival law, and it is measurable. Ask the simplest delayed question of all: is the system still in the state in which it was prepared? The survival amplitude is expanded from the exponential series of (4.2):

$$\begin{aligned} \langle \psi(0) | \psi(t) \rangle &= \langle \psi(0) | e^{-iHt/\hbar} | \psi(0) \rangle = \langle \psi(0) | \left(I - \frac{i}{\hbar} Ht - \frac{1}{2\hbar^2} H^2 t^2 + O(t^3) \right) | \psi(0) \rangle \\ &= 1 - \frac{i}{\hbar} \langle H \rangle_\psi t - \frac{1}{2\hbar^2} \langle H^2 \rangle_\psi t^2 + O(t^3). \end{aligned}$$

For the squared modulus, set $a := \frac{t}{\hbar} \langle H \rangle_\psi$ and $b := \frac{t^2}{2\hbar^2} \langle H^2 \rangle_\psi$, so that $\langle \psi(0) | \psi(t) \rangle = 1 - ia - b + O(t^3)$.

Then

$$\begin{aligned} P(t) &:= |\langle \psi(0) | \psi(t) \rangle|^2 = |1 - ia - b|^2 + O(t^4) = (1 - b)^2 + a^2 + O(t^4) \\ &= 1 - 2b + a^2 + O(t^4) = 1 - \frac{t^2}{\hbar^2} \langle H^2 \rangle_\psi + \frac{t^2}{\hbar^2} \langle H \rangle_\psi^2 + O(t^4) = 1 - \frac{t^2}{\hbar^2} (\langle H^2 \rangle_\psi - \langle H \rangle_\psi^2) \end{aligned}$$

Collecting the variance via (3.26):

$$\langle \psi(0) | \psi(t) \rangle = 1 - \frac{i}{\hbar} \langle H \rangle_\psi t - \frac{t^2}{2\hbar^2} \langle H^2 \rangle_\psi + O(t^3), \quad P(t) = 1 - \frac{(\Delta H)^2}{\hbar^2} t^2 + O(t^4), \quad (4.4)$$

with $(\Delta H)^2 := \langle H^2 \rangle_\psi - \langle H \rangle_\psi^2$ the generator's variance (3.26) in the prepared state. Two readings of the display are experimental. First, no survival curve ever begins linearly: every decay starts flat, with a curvature set by the spread of the preparation over the generator's eigenvalues. Second, the flatness is exploitable: re-interrogating the survival question n times at intervals t/n returns the survival probability $P(t/n)^n \simeq (1 - (\Delta H)^2 t^2 / \hbar^2 n^2)^n \rightarrow 1$ as $n \rightarrow \infty$ — watched sufficiently often, the state ceases to change. This is the *quantum Zeno effect*, named once and not developed further.²

The advantage of (D1) is its economy: a single axiom carries the entire dynamics of the theory, and every clause of it is a bench report — linearity from the absence of records between interrogations, unitarity from the freedom to ask any complete question at any delay, continuity from the smooth drift of the counts, composition from the homogeneity of time. Its limitation is equally sharp: the postulate says *that* a self-adjoint generator exists, never *which* generator a given system has. The Hamiltonian is physical input, identified case by case; the two-state instances of Sections 4.3 and 4.4 are phenomenological identifications, measured rather than derived, and Section 4.7 will show how far symmetry alone can carry the generator of a free system.

The generator is in hand, but its eigenvalues are so far anonymous numbers: the theorem provides the operator, not their physical reading. What those eigenvalues *are* — and how the whole evolution decomposes into them — is the spectral resolution of $U(t)$, the subject of the next section.

4.2 Stationary States and the Spectral Resolution of $U(t)$

The evolution postulate hands us a self-adjoint generator, and Chapter 3 hands us the language to read it: by the spectral theorem (3.35), H resolves into eigenvalues and projectors, $H = \sum_n E_n P_n$ (3.23), with the degenerate form (3.46) when levels repeat. This section solves the evolution postulate in that language — and in the solution discovers what the eigenvalues physically are.

The exponential of (4.2) is the function $e^{-ixt/\hbar}$ of the observable H , in the sense of Chapter 3's function-of-an-observable rule (3.25):

$$U(t) = e^{-iHt/\hbar} := \sum_n e^{-iE_n t/\hbar} P_n. \quad (4.5)$$

That this family satisfies every clause of (D1) is a computation with the projector algebra (3.8): orthogonality $P_m P_n = 0$ ($m \neq n$) collapses the double sum in $U(t)U(s)$ to $\sum_n e^{-iE_n(t+s)/\hbar} P_n = U(t+s)$; idempotence and the resolution of identity (3.9) give $U(0) = \sum_n P_n = I$; and the reality of the E_n

²B. Misra and E. C. G. Sudarshan, *J. Math. Phys.* **18**, 756 (1977).

gives $U(t)^\dagger U(t) = \sum_n P_n = I$. Term-by-term differentiation gives $i\hbar dU/dt = \sum_n E_n e^{-iE_n t/\hbar} P_n = H U(t)$, the Schrödinger equation (4.3). In finite dimension nothing further is needed; the convergence questions of the infinite case are Appendix A's.³

An eigenray of the generator evolves by a phase and nothing more, and a general preparation decomposes into such eigenrays:

$$|e_n(t)\rangle = e^{-iE_n t/\hbar} |e_n\rangle, \quad |\psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |e_n\rangle, \quad c_n = \langle e_n | \psi(0) \rangle. \quad (4.6)$$

By the ray theorem (Section 2.5), a state changed only by a phase is the same state: a preparation sharp in the generator is physically unchanged by waiting, and every interrogation answers at time t exactly as it answered at time 0. Such eigenrays are called *stationary states*. (They are eigenrays of H ; no wave functions are involved or needed.) All the time dependence of a general state lives in the *relative* phases $e^{-i(E_m - E_n)t/\hbar}$ of its stationary components.

Insert the expansion into the expectation of any fixed operator A :

$$\langle A \rangle(t) = \langle \psi(t) | A | \psi(t) \rangle = \sum_{m,n} c_m^* c_n e^{i(E_m - E_n)t/\hbar} \langle e_m | A | e_n \rangle. \quad (4.7)$$

Every time dependence of every expectation value is a sum of oscillations at the frequencies $\omega_{mn} := (E_m - E_n)/\hbar$ — the *Bohr frequencies* of the generator. This is exactly the time dependence Heisenberg imposed on his transition quantities in 1925, because the observed spectral lines come at such differences (Section 1.4.1); here it is not imposed but derived, the dynamical repayment of that debt. And the empirical warrant Heisenberg had — the combination principle, that spectral lines are differences of terms (Section 1.4.1) — is now turned into an identification, stated with its register made plain: **the eigenvalues of the generator are the energy levels of the system, and the generator H is the energy observable**. The identification is physical, not definitional; its structural sealing — time translation exhibited as the evolution family's place among the spacetime symmetries — is the work of Sections 4.7 and 4.8. Stationary states are now named honestly: they are the energy-sharp preparations.

One clause of (4.7) deserves isolating at once: an operator with no matrix elements between distinct eigenrays, $\langle e_m | A | e_n \rangle = 0$ for $m \neq n$, has a constant expectation in every state. Which operators these are, and why, is the conservation theorem of Section 4.9.

The two-state case computes the display end to end, and it is the book's first exact dynamics. The general two-state generator is Chapter 3's (3.39): $H = a_0 I + \mathbf{a} \cdot \boldsymbol{\sigma}$, with eigenvalues $E_\pm = a_0 \pm |\mathbf{a}|$ and eigenrays the spin alternatives along $\pm \hat{\mathbf{a}}$. Take $\hat{\mathbf{a}} = \hat{\mathbf{z}}$ and prepare the state $|+x\rangle = (|+z\rangle + |-z\rangle)/\sqrt{2}$ of Chapter 1's two-state grammar (Section 1.5). The state decomposes into the two stationary rays,

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left(e^{-i(a_0 + |\mathbf{a}|)t/\hbar} |+z\rangle + e^{-i(a_0 - |\mathbf{a}|)t/\hbar} |-z\rangle \right),$$

and every expectation with off-diagonal content oscillates at the single Bohr frequency $\omega = (E_+ - E_-)/\hbar = 2|\mathbf{a}|/\hbar$:

$$\langle \sigma_x \rangle(t) = \langle \psi(t) | \sigma_x | \psi(t) \rangle = \frac{1}{2} \left(e^{i\omega t} \langle +z | \sigma_x | -z \rangle + e^{-i\omega t} \langle -z | \sigma_x | +z \rangle \right) = \frac{1}{2} (e^{i\omega t} + e^{-i\omega t}) = \cos \omega t,$$

³For an unbounded H the sum is replaced by an integral over the spectrum, and the differentiability statements hold on the generator's domain; see Appendix A.

using $\sigma_x |\pm z\rangle = |\mp z\rangle$ from the Pauli algebra of Chapter 3 (3.61); a complete oscillation of the ensemble's answer between the $+x$ and $-x$ alternatives. Two measurables of this kind carry names. The intensity modulation of a radiating ensemble whose atoms are coherently excited into a superposition of two nearby levels is the *quantum beat*, read off at the Bohr frequency of the pair. And Ramsey's separated-field method — two short interrogations bracketing a free wait — reads the accumulated relative phase directly, the working principle of atomic frequency standards. The overall rate $a_0/\hbar$ appears in no count: a common phase is unobservable (Section 2.5), and only eigenvalue *differences* are measured — the combination principle again, now a theorem-clause of the evolution law.

The beat was abstract: a_0 and α were numbers without an apparatus. The oldest two-state system of this book — the spin of the Stern–Gerlach benches — realizes it literally when a magnetic field does the driving. That realization, solved exactly, is next.

4.3 Spin in a Magnetic Field: Precession Solved Exactly

The Stern–Gerlach benches of Chapter 1 (Section 1.5, Figure 1.3) gave this book its first two-state system, and Pauli's two-valuedness gave the electron its spin (Section 1.3.4). The debt recorded then comes due now in its dynamical share: what does a spin *do* between interrogations? In a static magnetic field the answer is exact, and it is this chapter's flagship computation.

A spin carries a magnetic moment $\boldsymbol{\mu} = \gamma \mathbf{S}$, with $\mathbf{S} = (\hbar/2) \boldsymbol{\sigma}$ the spin operator of Chapter 3 (3.31) and γ the particle's gyromagnetic ratio — a measured constant of the particle, known from the Stern–Gerlach deflection itself and read with precision by the resonance of Section 4.4. The coupling to a field $\mathbf{B} = B\mathbf{n}$ is $H = -\boldsymbol{\mu} \cdot \mathbf{B}$:

$$H = -\gamma \mathbf{B} \cdot \mathbf{S} = -\frac{\gamma \hbar B}{2} (\mathbf{n} \cdot \boldsymbol{\sigma}) =: \frac{\hbar \omega}{2} (\mathbf{n} \cdot \boldsymbol{\sigma}), \quad \omega := -\gamma B.$$

The identification is phenomenological in the sense of Section 4.1's limitation: the form of the generator is measured, not derived. The Larmor frequency ω carries the sign of $-\gamma$ (the electron's γ is negative, its ω positive); the sign fixes only the sense of the precession, and $|\omega|$ is what the bench measures.

The evolution operator follows in closed form. The Pauli algebra recorded with (3.39) gives $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$, so the exponential series splits into its even and odd powers, the two subseries being those of the cosine and the sine:

$$U(t) = e^{-i\omega t (\mathbf{n} \cdot \boldsymbol{\sigma})/2} = \cos \frac{\omega t}{2} I - i \sin \frac{\omega t}{2} (\mathbf{n} \cdot \boldsymbol{\sigma}). \quad (4.8)$$

Choose the field along z and expand the state as $a|+z\rangle + b|-z\rangle$, with a, b the complex coefficients at $t = 0$. By (4.8) with $\mathbf{n} = \hat{\mathbf{z}}$,

$$U(t) = \cos \frac{\omega t}{2} I - i \sin \frac{\omega t}{2} \sigma_z = \begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{+i\omega t/2} \end{pmatrix},$$

so the two components evolve as $a(t) = a e^{-i\omega t/2}$ and $b(t) = b e^{+i\omega t/2}$. Now form the combination

that extracts the off-diagonal content:

$$\langle \sigma_x \rangle(t) + i \langle \sigma_y \rangle(t) = \langle \psi(t) | (\sigma_x + i \sigma_y) | \psi(t) \rangle = \begin{pmatrix} a(t)^* & b(t)^* \end{pmatrix} \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a(t) \\ b(t) \end{pmatrix} = 2 a(t)^* b(t).$$

Inserting the evolved coefficients,

$$\langle \sigma_x \rangle(t) + i \langle \sigma_y \rangle(t) = 2 (a^* e^{+i\omega t/2}) (b e^{+i\omega t/2}) = 2 a^* b e^{+i\omega t} = (\langle \sigma_x \rangle(0) + i \langle \sigma_y \rangle(0)) e^{+i\omega t},$$

and the conjugate combination gives the minus sign. Meanwhile $\langle \sigma_z \rangle(t) = |a(t)|^2 - |b(t)|^2 = |a|^2 - |b|^2 = \langle \sigma_z \rangle(0)$, since the phases cancel in each squared modulus. For the $+x$ recapture, with $|\psi(0)\rangle = |+\rangle = (|+z\rangle + |-z\rangle)/\sqrt{2}$ we have $a = b = 1/\sqrt{2}$, so $a(t) = e^{-i\omega t/2}/\sqrt{2}$, $b(t) = e^{+i\omega t/2}/\sqrt{2}$, and

$$P_{+x \rightarrow +x}(t) = |\langle +x | \psi(t) \rangle|^2 = \left| \frac{1}{\sqrt{2}} (a(t) + b(t)) \right|^2 = \frac{1}{4} |e^{-i\omega t/2} + e^{+i\omega t/2}|^2 = \cos^2 \frac{\omega t}{2}.$$

Collecting the results:

$$\langle \sigma_x \rangle(t) \pm i \langle \sigma_y \rangle(t) = e^{\pm i\omega t} (\langle \sigma_x \rangle(0) \pm i \langle \sigma_y \rangle(0)), \quad \langle \sigma_z \rangle(t) = \langle \sigma_z \rangle(0); \quad P_{+x \rightarrow +x}(t) = \cos^2 \frac{\omega t}{2}. \quad (4.9)$$

The expectation vector rotates about the field axis at the rate ω with its z -component frozen — Larmor precession. The third clause is the exit question of Chapter 3 (Section 3.10) turned into a number: a $|+\rangle$ beam, left to wait in the field and then re-interrogated along x , answers $+x$ with a frequency $\cos^2(\omega t/2)$ that oscillates between one and zero, although nothing was recorded in between.

 **Note** The motion of the expectation vector has an evident geometric reading: the arrow $(\langle \sigma_x \rangle, \langle \sigma_y \rangle, \langle \sigma_z \rangle)$ rotates about the field axis. The reading is correct, but this section has used no rotation theory: every display above is 2×2 matrix arithmetic. The identification of that arithmetic with the geometry of rotations is a theorem of Sections 4.6 and 4.7, not an assumption of this one.

One full period of the precession returns the state with a sign:

$$U(2\pi/\omega) = \cos \pi I - i \sin \pi (\mathbf{n} \cdot \boldsymbol{\sigma}) = -I. \quad (4.10)$$

By the ray theorem (Section 2.5) the sign changes no count: after a full Larmor period every interrogation answers as it did before. Yet the -1 is not nothing. It is exactly the sign Chapter 2 recorded as an experimental fact of neutron interferometry (Section 2.5): a rotation through 4π , not 2π , is required to return a spin- $\frac{1}{2}$ state to itself. And the precession operator $e^{-i\theta(\mathbf{n} \cdot \boldsymbol{\sigma})/2}$ has precisely the form of a rotation representative acting on the two-state system — angle θ about the axis $\mathbf{n}$, with the characteristic half-angle. That Larmor precession is a rotation of the spin state is named here; it is earned where the rotation generators are derived (Section 4.7), and its full interpretation remains the angular-momentum chapter's declared debt.

The measurable landing of the computation is the precession frequency itself. Magnetic resonance — Rabi's molecular-beam method⁴ — reads ω against a known field and hence measures $|\gamma|$; the magnetic moments of the electron, the proton, and the nuclei are known to their many digits by this route.

A static field precesses the spin; an oscillating field can flip it. Flips driven on demand need the evolution operator for a time-dependent Hamiltonian — the one place the postulate (D1) requires generalizing. That generalization, and the exact solution of the resonance problem it buys, are next.

⁴I. I. Rabi, *Phys. Rev.* **51**, 652 (1937). The method's exact theory is the driven problem of Section 4.4.

4.4 Time-Dependent Hamiltonians: the Time-Ordered Exponential

The postulate (D1) governs an isolated system, whose law of waiting carries no clock of its own. A driven system — a spin in a field that the experimenter sweeps, an atom in a switched pulse — has its dynamics prescribed from outside: the Hamiltonian itself depends on time, $H = H(t)$. The generalization of the law of evolution is immediate to state, and its solution form is this section's first business.

The generalized evolution law. *For a driven system the postulate (D1) is generalized in the only way the drive allows: the evolution is a two-parameter family of unitary operators $U(t, t_0)$ obeying*

$$i\hbar \frac{d}{dt} U(t, t_0) = H(t) U(t, t_0), \quad U(t_0, t_0) = I, \quad (4.11)$$

with $|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle$ between interrogations. The composition law survives, but only in its two-time form:

$$U(t_2, t_0) = U(t_2, t_1) U(t_1, t_0), \quad U(t, t_0)^{-1} = U(t_0, t) = U(t, t_0)^\dagger.$$

What is given up is the homogeneity of time: $U(t, t_0)$ depends on both times, not on their difference, because the law itself differs from epoch to epoch. With homogeneity goes energy conservation — a driven system exchanges energy with its drive; the precise statement is Section 4.9's.

When the Hamiltonian commutes with itself at different times, $[H(t), H(t')] = 0$, the solution of (4.11) is the ordinary exponential of the integrated generator, $U(t, t_0) = \exp(-\frac{i}{\hbar} \int_{t_0}^t H(t') dt')$, as differentiation shows at once (Exercise 4 of this chapter). In general it is not: $H(t_1)$ and $H(t_2)$ need not commute, and the order in which the infinitesimal evolutions are applied matters — the order-sensitivity of Chapter 1's fourth requirement now *inside* the dynamics itself. The solution keeps the order explicit. The differential equation with its initial condition is one integral equation,

$$U(t, t_0) = I - \frac{i}{\hbar} \int_{t_0}^t H(t') U(t', t_0) dt',$$

and iterating it into itself — replacing $U(t', t_0)$ under the integral by the whole right-hand side, and repeating — generates a series of nested integrals in which the times are ordered $t_1 \geq t_2 \geq \dots \geq t_n$ by construction. That series *defines* the time-ordered exponential:

$$\begin{aligned} U(t, t_0) &= T \exp\left(-\frac{i}{\hbar} \int_{t_0}^t H(t') dt'\right) \\ &:= I + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n H(t_1) H(t_2) \cdots H(t_n). \end{aligned} \quad (4.12)$$

The symbol T (time-ordering) is an instruction: in every operator product, the factors stand with later times to the left. The nested-integral series on the right — the *Dyson series* — is the definition's content; its existence and convergence for bounded $H(t)$ are matters for Appendix A.⁵

One pointer, and no more, belongs here. Truncated after its first nontrivial term, the series (4.12)

⁵On a finite interval with $\|H(t)\|$ bounded, the n -th term is bounded by $(\sup \|H\| (t - t_0)/\hbar)^n / n!$ and the series converges in norm; the unbounded cases are Appendix A's.

is time-dependent perturbation theory, and its first-order amplitude is the seed of Fermi's golden rule — the later chapter's repayment of Chapter 1's debt of the Einstein coefficients (Section 1.1.6). No truncation is made here; this chapter's driven problem is solved exactly.

The driven two-state system is that problem, and it carries the oldest two-state hardware of the book to its precision form. Take the static field of Section 4.3 along z and add a transverse field rotating circularly in the x - y plane, $\mathbf{B}_1(t) = B_1(\cos \omega t, \sin \omega t, 0)$; with the same moment-coupling as before, $\omega_0 := -\gamma B_0$ and $\omega_1 := -\gamma B_1$, the Hamiltonian is

$$H(t) = \frac{\hbar\omega_0}{2} \sigma_z + \frac{\hbar\omega_1}{2} (\cos \omega t \sigma_x + \sin \omega t \sigma_y).$$

The time dependence is removed by a relabeling: describe the state against a question that has itself been rotated through the angle ωt about z ,

$$|\tilde{\psi}(t)\rangle := e^{i\omega t \sigma_z/2} |\psi(t)\rangle.$$

Two conjugation identities do the work. From $e^{i\theta\sigma_z/2} = \cos(\theta/2)I + i\sin(\theta/2)\sigma_z$ and the Pauli algebra (3.61):

$$\begin{aligned} e^{i\theta\sigma_z/2} \sigma_x e^{-i\theta\sigma_z/2} &= \left(\cos \frac{\theta}{2} I + i\sin \frac{\theta}{2} \sigma_z\right) \sigma_x \left(\cos \frac{\theta}{2} I - i\sin \frac{\theta}{2} \sigma_z\right) \\ &= \cos^2 \frac{\theta}{2} \sigma_x - i\cos \frac{\theta}{2} \sin \frac{\theta}{2} \sigma_x \sigma_z + i\cos \frac{\theta}{2} \sin \frac{\theta}{2} \sigma_z \sigma_x + \sin^2 \frac{\theta}{2} \sigma_z \sigma_x \sigma_z. \end{aligned}$$

Using $\sigma_x \sigma_z = -i\sigma_y$, $\sigma_z \sigma_x = i\sigma_y$, and $\sigma_z \sigma_x \sigma_z = \sigma_z(i\sigma_y) = i\sigma_z \sigma_y = i(-i\sigma_x) = -\sigma_x$ (from $\sigma_z \sigma_y = -i\sigma_x$), the four terms become

$$\cos^2 \frac{\theta}{2} \sigma_x - \cos \frac{\theta}{2} \sin \frac{\theta}{2} \sigma_y - \cos \frac{\theta}{2} \sin \frac{\theta}{2} \sigma_y - \sin^2 \frac{\theta}{2} \sigma_x = (\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) \sigma_x - 2 \cos \frac{\theta}{2} \sin \frac{\theta}{2} \sigma_y = \cos \theta \sigma_x - \sin \theta \sigma_y.$$

The second identity follows by the same arithmetic with σ_y in place of σ_x ; the result is recorded together:

$$\begin{aligned} e^{i\theta\sigma_z/2} \sigma_x e^{-i\theta\sigma_z/2} &= \cos \theta \sigma_x - \sin \theta \sigma_y, \\ e^{i\theta\sigma_z/2} \sigma_y e^{-i\theta\sigma_z/2} &= \sin \theta \sigma_x + \cos \theta \sigma_y, \end{aligned}$$

with $\theta = \omega t$, hence

$$e^{i\omega t \sigma_z/2} (\cos \omega t \sigma_x + \sin \omega t \sigma_y) e^{-i\omega t \sigma_z/2} = \cos \omega t (\cos \omega t \sigma_x - \sin \omega t \sigma_y) + \sin \omega t (\sin \omega t \sigma_x + \cos \omega t \sigma_y) = \cos 2\omega t \sigma_x - \sin 2\omega t \sigma_y,$$

and $e^{i\omega t \sigma_z/2} \sigma_z e^{-i\omega t \sigma_z/2} = \sigma_z$ trivially since σ_z commutes with itself.

Substituting $|\psi(t)\rangle = e^{-i\omega t \sigma_z/2} |\tilde{\psi}(t)\rangle$ into the Schrödinger equation (4.3):

$$i\hbar \frac{d}{dt} (e^{-i\omega t \sigma_z/2} |\tilde{\psi}(t)\rangle) = H(t) e^{-i\omega t \sigma_z/2} |\tilde{\psi}(t)\rangle.$$

The left side carries two contributions — the derivative of the exponential and the derivative of $|\tilde{\psi}(t)\rangle$:

$$i\hbar \left(-\frac{i\omega}{2} \sigma_z\right) e^{-i\omega t \sigma_z/2} |\tilde{\psi}(t)\rangle + i\hbar e^{-i\omega t \sigma_z/2} \frac{d}{dt} |\tilde{\psi}(t)\rangle = H(t) e^{-i\omega t \sigma_z/2} |\tilde{\psi}(t)\rangle.$$

Multiplying from the left by $e^{i\omega t \sigma_z/2}$ and rearranging:

$$i\hbar \frac{d}{dt} |\tilde{\psi}(t)\rangle = e^{i\omega t \sigma_z/2} H(t) e^{-i\omega t \sigma_z/2} |\tilde{\psi}(t)\rangle - \frac{\hbar\omega}{2} \sigma_z |\tilde{\psi}(t)\rangle.$$

The conjugated Hamiltonian expands with the identities above. The static z -term $(\hbar\omega_0/2)\sigma_z$ commutes with $e^{i\omega t \sigma_z/2}$ and is unchanged; the drive term $(\hbar\omega_1/2)(\cos \omega t \sigma_x + \sin \omega t \sigma_y)$ conjugates to $(\hbar\omega_1/2) \sigma_x$ by the line following (3.61). Defining $\delta := \omega - \omega_0$:

$$i\hbar \frac{d}{dt} |\tilde{\psi}(t)\rangle = H_{\text{eff}} |\tilde{\psi}(t)\rangle, \quad H_{\text{eff}} = \frac{\hbar\omega_1}{2} \sigma_x - \frac{\hbar\delta}{2} \sigma_z.$$

The problem is reduced to the static one of Section 4.3: an effective field with Larmor rate $\sqrt{\omega_1^2 + \delta^2}$. Set $\Omega := \sqrt{\omega_1^2 + \delta^2}$ and write $H_{\text{eff}} = \frac{\hbar\Omega}{2} (\hat{\mathbf{n}}_{\text{eff}} \cdot \boldsymbol{\sigma})$ with $\hat{\mathbf{n}}_{\text{eff}} = (\omega_1/\Omega, 0, -\delta/\Omega)$. The evolution in the rotating frame is, by (4.8),

$$\tilde{U}(t) = e^{-iH_{\text{eff}}t/\hbar} = \cos \frac{\Omega t}{2} I - i \sin \frac{\Omega t}{2} (\hat{\mathbf{n}}_{\text{eff}} \cdot \boldsymbol{\sigma}) = \begin{pmatrix} \cos \frac{\Omega t}{2} + i \frac{\delta}{\Omega} \sin \frac{\Omega t}{2} & -i \frac{\omega_1}{\Omega} \sin \frac{\Omega t}{2} \\ -i \frac{\omega_1}{\Omega} \sin \frac{\Omega t}{2} & \cos \frac{\Omega t}{2} - i \frac{\delta}{\Omega} \sin \frac{\Omega t}{2} \end{pmatrix}.$$

For the initial state $|+z\rangle$, the rotating-frame state is $|\tilde{\psi}(t)\rangle = \tilde{U}(t) |+z\rangle$, whose lower component reads $-i(\omega_1/\Omega) \sin(\Omega t/2)$. The physical state is $|\psi(t)\rangle = e^{-i\omega t\sigma_z/2} |\tilde{\psi}(t)\rangle$, so the $|-z\rangle$ component acquires only the phase $e^{+i\omega t/2}$ from the relabeling. Squaring,

$$P_{+\rightarrow-}(t) = |\langle -z|\psi(t)\rangle|^2 = |e^{+i\omega t/2} \left[-i \frac{\omega_1}{\Omega} \sin \frac{\Omega t}{2} \right]|^2 = \frac{\omega_1^2}{\Omega^2} \sin^2 \frac{\Omega t}{2}.$$

The detuning δ measures how far the drive sits from resonance:

$$P_{+\rightarrow-}(t) = \frac{\omega_1^2}{\omega_1^2 + \delta^2} \sin^2 \left(\frac{1}{2} \sqrt{\omega_1^2 + \delta^2} t \right), \quad \delta = \omega - \omega_0. \quad (4.13)$$

The relabeling contributed only a phase to the $|-z\rangle$ amplitude, invisible in the probability. This is the *Rabi formula*, exact. At exact resonance ($\delta = 0$) the spin flips with certainty after a time π/ω_1 , and oscillates between the two alternatives at the rate ω_1 ; off resonance the transfer is incomplete, with peak height $\omega_1^2/(\omega_1^2 + \delta^2)$. Swept in ω , the flip probability traces a resonance curve peaked at $\omega = \omega_0$ with a width set by ω_1 : the curve reads $|\omega_0| = |\gamma|B_0$ directly, and with it the particle's magnetic moment. This is the working principle of the precision moment measurements named in Section 4.3.

Two honesty clauses attach to the example. First, the laboratory drive is usually a field oscillating along a fixed line, $B_1 \cos \omega t$; such a field decomposes into two counter-rotating circular components, and the solution above is exact for the circular drive. The standard treatment of a linear drive keeps the component rotating near the spin's precession sense and drops the counter-rotating one — the *rotating-wave approximation*, named here as what it is. Second, the drive has been treated as a prescribed classical function of time; its own dynamics and back-action belong to a fuller theory.

 **Note** The substitution $|\psi(t)\rangle \mapsto e^{i\omega t\sigma_z/2} |\psi(t)\rangle$ used above is Chapter 3's instantaneous relabeling (Section 3.2) applied at each instant: the same state, described against a question rotated through the angle ωt about z . It is not a new law of evolution, and it is not a picture of dynamics — the operator-carried reading of evolution is the next chapter's subject, flagged at this chapter's exit (Section 4.11).

Evolution — static or driven — is one distinguished continuous family of relabelings. But relabelings, and their active cousins, form a far larger class: the transformations that leave the physics of a system invariant. Classifying what quantum mechanics permits there is the theory of symmetry, and its foundation theorem is Wigner's — the next section.

4.5 Symmetry Transformations: Active, Passive, and Wigner's Theorem

The evolution operator of Section 4.1 carries the state through time while preserving every transition probability. Probability preservation is a far broader notion, and it deserves a name of its own.

An invertible map T that assigns to every preparation a transformed preparation is called a **symmetry transformation** when it preserves every Born-rule overlap:

$$|\langle T\phi | T\psi \rangle| = |\langle \phi | \psi \rangle| \quad (4.14)$$

for every pair of states. The definition is stated on rays, not on vectors, because the Born rule (2.7) is the whole empirical content of the theory: two transformations that agree on every ray are the same transformation, and the arena is the projective space of Section 2.5, whose invariance (2.13) is built into the definition itself.

Two readings of such a map must be distinguished, for they share one operator but not one meaning. In the *passive* reading the system stands still and the question is moved: the same preparation is described against a rotated apparatus, and the two descriptions are related by the overlap matrix of the two bases ((2.36), (2.37)) — the relabelings of the measurement chapter, read here as the passive case of symmetry. In the *active* reading the question stands still and the system is carried to a new physical situation: the beam is translated, the preparation is rotated, the laboratory waits. The Stern–Gerlach bench of Chapter 1 exhibits the unity (Figure 1.3): rotating the magnet and rotating the beam's preparation by the same angle are one transformation, and the measured overlaps are identical in the two readings.

Wigner's theorem states that the definition (4.14) exhausts its own solutions.

Theorem (Wigner, stated without proof).⁶ *Every symmetry transformation (4.14) is induced by an operator U on $\mathcal{H}$, unique up to an overall phase, and U is either **unitary** (3.10) or **antiunitary**:*

$$\langle U\phi | U\psi \rangle = \langle \phi | \psi \rangle \quad (\text{unitary}), \quad \text{or} \quad \langle U\phi | U\psi \rangle = \langle \psi | \phi \rangle \quad (\text{antiunitary}). \quad (4.15)$$

The prototype of the second branch is complex conjugation in a fixed basis, denoted K_0 :

$$\langle K_0\phi | K_0\psi \rangle = \langle \psi | \phi \rangle \quad (4.16)$$

The dichotomy is exercised at once on the two-state system. Let K_0 conjugate the components of a vector in the z -basis: it is antilinear by construction, $K_0(\alpha|\psi\rangle) = \alpha^*K_0|\psi\rangle$; direct 2×2 arithmetic verifies $\langle K_0\phi | K_0\psi \rangle = \langle \psi | \phi \rangle$ and $K_0^2 = I$; and every two-state probability is preserved. Complex conjugation is an antiunitary symmetry of $\mathbb{CP}^1$, in arithmetic. Its geometric content on the two-state system is read off in Exercise 5; its physical identification — which operation of reversal it implements — is deferred to the rotation theory of the angular-momentum chapter.

Two structural remarks carry the chapter forward. First, the theorem's conclusion is an operator *up to a phase*; consequently a *group* of symmetry transformations is represented on states only up to phase, with a composition law of the form $U(g)U(g') = e^{i\xi(g,g')}U(gg')$. That phase freedom, trivial for a single transformation, is load-bearing for families: it is what the next section names a ray representative, and what the Galilean section spends on the algebra's center. Second, symmetry transformations compose, and the composition is order-sensitive: they form a group. The fourth kinematical requirement of Chapter 1 (Section 1.4.6) receives here its dynamical repayment — the kinematical share, the order-sensitive product of given operators (Section 3.7), is completed by the transformation law itself and by the composition of the evolution family of Section 4.1.

⁶E. Wigner, *Gruppentheorie und ihre Anwendung auf die Quantenmechanik der Atomspektren* (1931); a modern proof is V. Bargmann, *J. Math. Phys.* **5**, 862 (1964). The historical setting was recalled in Section 1.4.5.

The advantage of Wigner’s theorem is its exhaustiveness: every invariance of the theory is an operator, and the two cases are the whole story. Its limitation is existential: which symmetries a given system *has* is physics, not theorem — that is the work of the following sections — and the antiunitary branch carries exactly one physical occupant, which we meet at the chapter’s end. Wigner says every symmetry is an operator, up to a phase. The symmetries that come in continuous families say much more: their composition law, differentiated once, hands each family an observable, and differentiated twice, an algebra. The chapter’s own $U(t)$ has been such a family since Section 4.1.

4.6 Continuous Symmetries as Lie Groups

The families of transformations met so far share one law. Waiting twice is waiting once: $U(t_1)U(t_2) = U(t_1+t_2)$, the composition clause of the evolution postulate. Translating twice in one direction is translating once: $T(a_1)T(a_2) = T(a_1 + a_2)$. Rotating twice about one axis is rotating once. A continuous family of transformations whose parameter composes by addition,

$$U(s_1)U(s_2) = U(s_1 + s_2), \quad (4.17)$$

is called a **one-parameter subgroup**, and a family of transformations whose labels compose smoothly is called a **Lie group**. Smoothness is used here in the operational sense of the evolution postulate: small changes of the parameter give near-identical statistics on every preparation.⁷

Stone’s theorem, stated for the evolution family in Section 4.1, applies to every one-parameter subgroup of unitaries: there exists a self-adjoint operator G with

$$U(s) = e^{-isG/\hbar} = I - \frac{i}{\hbar} sG + O(s^2), \quad (4.18)$$

the **exponential map**, and G is called the family’s **generator**. Since the generator is self-adjoint, it is an observable of the theory of the measurement chapter ((3.33), (3.35)): every continuous symmetry family carries its own measurable quantity, and no operator in this chapter is an orphan. The three families of nonrelativistic physics complete the correspondence:

$$T(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{P}/\hbar}, \quad U(t) = e^{-itH/\hbar}, \quad R_{\mathbf{n}}(\theta) = e^{-i\theta J_{\mathbf{n}}/\hbar}, \quad (4.19)$$

translations generated by the momentum $\mathbf{P}$, time translations by the energy H (the identification of Section 4.2, whose structural analysis is Section 4.8’s), rotations by the angular momentum $J_{\mathbf{n}}$ about the axis $\mathbf{n}$. Momentum’s name is earned as energy’s was: by correspondence with the classical generator of translations, by de Broglie’s relation (Section 1.3.2), and by the diffraction bench of the measurement chapter (Section 3.8), which is the momentum meter of this book; the position-representation form of $\mathbf{P}$ is the representations chapter’s debt, not this one’s.

For a family of several parameters, composition of different subgroups need not commute, and the failure is measurable. Let $U_{\epsilon} = e^{-i\epsilon G_1/\hbar}$ and $V_{\epsilon} = e^{-i\epsilon G_2/\hbar}$ be two one-parameter subgroups of one group, and form the group commutator $U_{\epsilon}V_{\epsilon}U_{\epsilon}^{-1}V_{\epsilon}^{-1}$. Set $A = -iG_1/\hbar$, $B = -iG_2/\hbar$. Expanding

⁷The systematic theory of Lie groups and Lie algebras is collected in Appendix B; this section earns the four notions it needs — one-parameter subgroups, the exponential map, generators, structure constants — from families already on the page.

each factor to second order by (4.18):

$$\begin{aligned} e^{\varepsilon A} &= I + \varepsilon A + \frac{1}{2}\varepsilon^2 A^2 + O(\varepsilon^3), & e^{\varepsilon B} &= I + \varepsilon B + \frac{1}{2}\varepsilon^2 B^2 + O(\varepsilon^3), \\ e^{-\varepsilon A} &= I - \varepsilon A + \frac{1}{2}\varepsilon^2 A^2 + O(\varepsilon^3), & e^{-\varepsilon B} &= I - \varepsilon B + \frac{1}{2}\varepsilon^2 B^2 + O(\varepsilon^3). \end{aligned}$$

Multiply the first pair and the second pair, keeping terms through order ε^2 :

$$\begin{aligned} e^{\varepsilon A} e^{\varepsilon B} &= I + \varepsilon(A + B) + \varepsilon^2\left(\frac{1}{2}A^2 + AB + \frac{1}{2}B^2\right) + O(\varepsilon^3), \\ e^{-\varepsilon A} e^{-\varepsilon B} &= I - \varepsilon(A + B) + \varepsilon^2\left(\frac{1}{2}A^2 + AB + \frac{1}{2}B^2\right) + O(\varepsilon^3). \end{aligned}$$

Now multiply the two products, again retaining terms through ε^2 :

$$\begin{aligned} e^{\varepsilon A} e^{\varepsilon B} e^{-\varepsilon A} e^{-\varepsilon B} &= [I + \varepsilon(A + B) + \varepsilon^2 S] [I - \varepsilon(A + B) + \varepsilon^2 S] + O(\varepsilon^3) \quad (S := \frac{1}{2}A^2 + AB + \frac{1}{2}B^2) \\ &= I + \cancel{\varepsilon(A + B)} - \cancel{\varepsilon(A + B)} + \varepsilon^2(2S - (A + B)^2) + O(\varepsilon^3). \end{aligned}$$

Since $2S = A^2 + 2AB + B^2$ and $(A + B)^2 = A^2 + AB + BA + B^2$, their difference is $2S - (A + B)^2 = AB - BA = [A, B]$. Hence

$$e^{\varepsilon A} e^{\varepsilon B} e^{-\varepsilon A} e^{-\varepsilon B} = I + \varepsilon^2 [A, B] + O(\varepsilon^3), \quad (4.20)$$

with $A = -iG_1/\hbar$, $B = -iG_2/\hbar$. The operator commutator of the generators is the infinitesimal shadow of the group's noncommutation, and it is in principle measurable by composing the transformations in both orders on the bench. Since the group commutator is again an element of the group, the brackets of the generators must close among themselves:

$$[G_a, G_b] = i\hbar \sum_c f_{abc} G_c, \quad (4.21)$$

where the numbers f_{abc} are the group's **structure constants** — properties of its composition law, not new physics.

The bench decides one case and the representatives compute another. Translations of the Stern–Gerlach bench in two perpendicular directions, performed in either order, return identical statistics for every preparation: the translation family commutes, $[P_x, P_y] = 0$, and path-independence of translation is the vanishing commutator. Rotations of the two-state system do not commute, and the instance has been on the page since the measurement chapter: $[\sigma_x, \sigma_y] = 2i\sigma_z$ (3.61), the bracket of the two-state rotation family closing on a third generator, with structure constant $f_{xyz} = 2$ in σ -units read off a 2×2 multiplication. One group's constants are decided on the bench; another's are computed from its representatives; the full spacetime family of the next section is treated by exactly this pattern.

One freedom remains to be named before the walk through that family. By Wigner's up-to-a-phase conclusion (Section 4.5), a group's action on *states* is a family of unitaries obeying

$$U(g) U(g') = e^{i\xi(g,g')} U(gg'), \quad (4.22)$$

a **ray representative** of the group. The phase ξ is freedom inherited from the ray theorem (2.13): each representative is defined only up to its own phase, and the composition law inherits the ambiguity. When ξ can be absorbed by rephasing every representative, the representative is true; when it cannot, the generators' brackets are allowed to close on a multiple of the identity — a *central charge*. That possibility is the door, and the Galilean group walks through it.

A continuous symmetry family, in this language, is no longer an unstructured bag of operators: its law is the composition rule, its observable content is the generator list, and its noncommutation

is compressed into a finite table of structure constants. Which families nature uses is physics, not language. One group above all organizes nonrelativistic physics, the transformations of space and time themselves: translations, rotations, boosts, and waiting. Its algebra, derived from the representatives, carries a surprise in its center.

4.7 The Galilean Group and its Algebra

The group is the Galilean group, the complete spacetime symmetry of nonrelativistic physics. Its ten one-parameter subgroups — space translations $T(\mathbf{a})$ (three), time translations $U(t)$ (one, the chapter's own), rotations $R_n(\theta)$ (three, named since Section 4.6), and the **boosts** $B(\mathbf{v})$ (three) — encode every continuous transformation that preserves the structure of nonrelativistic spacetime. A boost carries the system to a copy moving with velocity $\mathbf{v}$ relative to the original — or, in the passive reading of Section 4.5, describes the same system from a uniformly moving frame. The empirical warrant is the Galilean relativity of the bench: experiments performed in a uniformly moving laboratory return the same statistics. The group is the classical inheritance of the theory, and its range of validity is this book's domain. This section gives the Galilean group the systematic treatment it deserves: definition and parametrization, complete derivation of the Lie algebra, the theory of its projective representations (Bargmann's theorem), construction of the unitary irreducible representations, the physical implications of Galilean invariance, and worked examples.

4.7.1 Definition and Parametrization of the Galilei Group

The group manifold. A Galilean transformation is a change of inertial frame specified by a rotation $R \in \text{SO}(3)$, a boost velocity $\mathbf{v} \in \mathbb{R}^3$, a spatial translation $\mathbf{a} \in \mathbb{R}^3$, and a temporal translation $b \in \mathbb{R}$. The transformation acts on spacetime coordinates as

$$\mathbf{r} \mapsto R\mathbf{r} + \mathbf{v}t + \mathbf{a}, \quad t \mapsto t + b. \quad (4.23)$$

The group element is the quadruple $g = (R, \mathbf{v}, \mathbf{a}, b)$. The composition law, read off two successive applications of (4.23), is

$$(R_2, \mathbf{v}_2, \mathbf{a}_2, b_2) (R_1, \mathbf{v}_1, \mathbf{a}_1, b_1) = (R_2 R_1, R_2 \mathbf{v}_1 + \mathbf{v}_2, R_2 \mathbf{a}_1 + \mathbf{v}_2 b_1 + \mathbf{a}_2, b_1 + b_2). \quad (4.24)$$

The identity is $e = (I, \mathbf{0}, \mathbf{0}, 0)$ and the inverse is

$$(R, \mathbf{v}, \mathbf{a}, b)^{-1} = (R^{-1}, -R^{-1}\mathbf{v}, -R^{-1}(\mathbf{a} - \mathbf{v}b), -b). \quad (4.25)$$

The group manifold is the ten-dimensional set

$$G = \text{SO}(3) \ltimes (\mathbb{R}^3 \times \mathbb{R}^3) \ltimes \mathbb{R}^1, \quad (4.26)$$

where the semidirect products record that rotations act on boosts and translations (both transform as vectors under $\text{SO}(3)$), and boosts act on spatial translations (the boost velocity shifts the translation vector). The time translation $\mathbb{R}^1$ is an invariant subgroup. The group is non-compact (translations and boosts are unbounded), ten-dimensional, and non-semisimple (it contains the abelian ideal of translations). The connected component of the identity is all elements with $R \in \text{SO}(3)$; the full

Galilei group also contains the discrete reflections, which are treated separately in Section 4.10.

The ten one-parameter subgroups. Each of the ten parameters generates a continuous family. In the quantum theory they are represented by unitary operators (up to a phase, as the next subsection will detail) on the state space:

$$T(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{P}/\hbar}, \quad U(t) = e^{-iHt/\hbar}, \quad (4.27)$$

$$R_{\mathbf{n}}(\theta) = e^{-i\theta J_{\mathbf{n}}/\hbar}, \quad B(\mathbf{v}) = e^{-i\mathbf{v}\cdot\mathbf{K}/\hbar}, \quad (4.28)$$

where $\mathbf{K} = (K_x, K_y, K_z)$ are the boost generators. The operators $\mathbf{P}$, H , $\mathbf{J}$, and $\mathbf{K}$ are self-adjoint by the spectral theorem of Section 3.5; their commutation relations are the subject of the next subsection.

Conjugation relations. The geometric meaning of each subgroup is captured by how it conjugates the others. These conjugation relations are the group-level input from which the Lie algebra will be derived. Translation composition is abelian: $T(\mathbf{a}_1)T(\mathbf{a}_2) = T(\mathbf{a}_1 + \mathbf{a}_2)$. A rotation conjugates a translation into the rotated translation,

$$R T(\mathbf{a}) R^{-1} = T(R\mathbf{a}), \quad (4.29)$$

which is not an axiom but a derivation from what rotation does to a displacement — the classical input, read off the structure of space itself. Likewise, a rotation conjugates a boost:

$$R B(\mathbf{v}) R^{-1} = B(R\mathbf{v}), \quad (4.30)$$

because boosting in a rotated direction is the rotation of a boost. A boost followed by a translation (and its inverse) shifts the position by $\mathbf{v}t$, so

$$B(\mathbf{v}) T(\mathbf{a}) B(\mathbf{v})^{-1} = T(\mathbf{a} + \mathbf{v}t), \quad (4.31)$$

where t is the parameter of the time translation that the boost and translation flank. These conjugation relations are the group-theoretic statements from which every bracket below will be derived. They are **stated/cited without proof** as the classical geometry of displacements, rotations, and frame changes.

Lie algebra defined. The Galilei algebra $\mathfrak{g}$ is the tangent space at the identity, spanned by the ten generators. Its bracket $[\cdot, \cdot]$ is the matrix (operator) commutator, discovered by the infinitesimal conjugation of one subgroup by another, exactly as the Lie bracket of Section 4.6 was derived from the group commutator. The remainder of this section is the systematic derivation of every bracket.

4.7.2 The Lie Algebra: Complete Derivation

The Lie algebra of the Galilei group, derived from the unitary ray representatives by the construction of Section 4.6, is

$$\begin{aligned} [P_i, P_j] &= 0, & [J_i, J_j] &= i\hbar \epsilon_{ijk} J_k, & [J_i, P_j] &= i\hbar \epsilon_{ijk} P_k, & [J_i, K_j] &= i\hbar \epsilon_{ijk} K_k, \\ [K_i, K_j] &= 0, & [P_i, H] &= 0, & [J_i, H] &= 0, & [K_i, H] &= -i\hbar P_i, \end{aligned} \quad (4.32)$$

together with the center (4.33). Each entry is derived or cited below, bracket by bracket.

(i) Translations commute: $[P_i, P_j] = 0$. This is the bench fact of Section 4.6: translating a system first in the x -direction and then in the y -direction returns the same final state as translating in the opposite

order. The physical assumption is the flatness of space — parallel transport is path-independent, and the translation subgroup is abelian. **Stated/cited without proof.**

(ii) The rotation algebra: $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$. This is the rotation family's own composition law, derived in the angular-momentum chapter (Chapter 6) from the SU(2) algebra and stated here as a structural fact about the spacetime group. Its two-state shadow $[\sigma_x, \sigma_y] = 2i\sigma_z$ has been on the page since the measurement chapter ((3.61)). **Stated/cited without proof** — the full spectral harvest belongs to the angular-momentum chapter.

(iii) Momentum is a vector under rotations: $[J_i, P_j] = i\hbar\epsilon_{ijk}P_k$. This is **proved** from the conjugation display (4.29). Set $R = e^{-i\theta J_n/\hbar}$ with infinitesimal θ and $T(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{P}/\hbar}$:

$$\begin{aligned} RT(\mathbf{a})R^{-1} &= \left(I - \frac{i\theta}{\hbar}J_n + O(\theta^2)\right) \left(I - \frac{i}{\hbar}\mathbf{a}\cdot\mathbf{P} + O(a^2)\right) \left(I + \frac{i\theta}{\hbar}J_n + O(\theta^2)\right) \\ &= I - \frac{i}{\hbar}\mathbf{a}\cdot\mathbf{P} - \frac{\theta}{\hbar^2}(J_n(\mathbf{a}\cdot\mathbf{P}) - (\mathbf{a}\cdot\mathbf{P})J_n) + O(\theta^2, a^2) \\ &= I - \frac{i}{\hbar}\mathbf{a}\cdot\mathbf{P} - \frac{\theta}{\hbar^2}a_j[J_n, P_j] + O(\theta^2, a^2). \end{aligned}$$

The right side of (4.29) is $T(R\mathbf{a}) = I - \frac{i}{\hbar}(R\mathbf{a})_k P_k + O(a^2)$. For an infinitesimal rotation by angle θ about $\hat{n}$, $(R\mathbf{a})_k = a_k + \theta\epsilon_{knj}a_j + O(\theta^2)$. Hence

$$T(R\mathbf{a}) = I - \frac{i}{\hbar}(a_k + \theta\epsilon_{knj}a_j)P_k + O(\theta^2, a^2) = I - \frac{i}{\hbar}\mathbf{a}\cdot\mathbf{P} - \frac{i\theta}{\hbar}\epsilon_{knj}a_jP_k + O(\theta^2, a^2).$$

Equating the $O(\theta)$ terms of the two sides, $-\frac{1}{\hbar^2}a_j[J_n, P_j] = -\frac{i}{\hbar}\epsilon_{knj}a_jP_k$. Since this holds for all $\mathbf{a}$, $[J_n, P_j] = i\hbar\epsilon_{knj}P_k = i\hbar\epsilon_{njk}P_k$. The vector character of $\mathbf{P}$ is a theorem of the group structure.

(iv) Boost generators are vectors under rotations: $[J_i, K_j] = i\hbar\epsilon_{ijk}K_k$. **Proved** by the identical conjugation argument applied to $B(\mathbf{v}) = e^{-i\mathbf{v}\cdot\mathbf{K}/\hbar}$ under rotations, with the classical input that a boosted frame rotated is a rotation of the boosted frame: $RB(\mathbf{v})R^{-1} = B(R\mathbf{v})$ ((4.30)). The expansion is identical to the momentum case with $\mathbf{K}$ replacing $\mathbf{P}$, and yields the bracket above. The vector character of $\mathbf{K}$ enforces that it transforms under rotations exactly like $\mathbf{P}$ — the generators form a vector multiplet of the rotation subalgebra.

(v) Boosts commute: $[K_i, K_j] = 0$. **Stated/cited without proof:** the composition of two Galilean boosts is a single boost by the vector sum of the velocities — a kinematic fact of Galilean relativity. This is the fundamental difference between the Galilei group and the Lorentz group, where boosts do not commute. The vanishing bracket is the infinitesimal statement of the law of addition of velocities in Galilean relativity.

(vi) Space-translation invariance: $[P_i, H] = 0$. **Proved** from the homogeneity of space. Translating a prepared system and then waiting, and waiting and then translating, must return the same final state: $T(\mathbf{a})U(t) = U(t)T(\mathbf{a})$ for all $\mathbf{a}, t$. Expanding each side to first order in $\mathbf{a}$ and t via $T(\mathbf{a}) = I - i(\mathbf{a}\cdot\mathbf{P})/\hbar + O(a^2)$ and $U(t) = I - iHt/\hbar + O(t^2)$:

$$(I - i\mathbf{a}\cdot\mathbf{P}/\hbar)(I - iHt/\hbar) = (I - iHt/\hbar)(I - i\mathbf{a}\cdot\mathbf{P}/\hbar) + O(a^2, t^2),$$

from which the $O(at)$ terms yield $(\mathbf{a}\cdot\mathbf{P})H = H(\mathbf{a}\cdot\mathbf{P})$, i.e. $[P_i, H] = 0$. The physical assumption is the invariance of the law of waiting under spatial translation — the homogeneity of time was already encoded in Axiom (D1), and the homogeneity of space is the additional, independent clause.

(vii) Rotation invariance: $[J_i, H] = 0$. **Proved** by the identical argument from $RU(t) = U(t)R$ for

all rotations and all t , with the physical assumption that the law of waiting is isotropic. No direction in space is dynamically privileged unless the Hamiltonian explicitly breaks rotational symmetry.

(viii) Boost–Hamiltonian bracket: $[K_i, H] = -i\hbar P_i$. **Derived and cited.** This bracket encodes the Galilean transformation of energy — in a boosted frame the energy of a particle acquires a contribution from its momentum: $E' = E + \mathbf{v} \cdot \mathbf{P} + M\mathbf{v}^2/2$. Differentiating with respect to the boost parameter yields $[K_i, H] = -i\hbar P_i$, with the sign convention fixed by the convention (4.39). The bracket is *exact* and is the infinitesimal content of the Galilean boost law. It does not assume a particular form of H ; any theory with Galilean invariance must satisfy this bracket.

The Jacobi identities. The consistency of the algebra is verified by checking the Jacobi identities for all 45 triples of generators. The only non-trivial check involves K_i , K_j , and P_k :

$$[K_i, [K_j, P_k]] + [K_j, [P_k, K_i]] + [P_k, [K_i, K_j]] = [K_i, -i\hbar\delta_{jk}M] + [K_j, i\hbar\delta_{ik}M] + [P_k, 0] = 0,$$

since M is central. Every other triple either vanishes identically or reproduces the rotation algebra's Jacobi identity, which holds by the structure of $\mathfrak{so}(3)$. The algebra is consistent. The tabulation of all 45 independent brackets is collected in Table 4.1.

Table 4.1: Complete bracket table for the Galilei Lie algebra. Rows and columns index the ten generators. Entries give the commutator $[X, Y]$. Blank entries are zero.

	J_i	P_i	K_i	H	M
J_j	$i\hbar\epsilon_{ijk}J_k$	$i\hbar\epsilon_{ijk}P_k$	$i\hbar\epsilon_{ijk}K_k$	0	0
P_j	$-i\hbar\epsilon_{ijk}P_k$	0	$+i\hbar\delta_{ij}M$	0	0
K_j	$-i\hbar\epsilon_{ijk}K_k$	$-i\hbar\delta_{ij}M$	0	$+i\hbar P_i$	0
H	0	0	$-i\hbar P_i$	0	0
M	0	0	0	0	0

4.7.3 Projective Representations and Bargmann's Theorem

The center. The translation–boost bracket,

$$[P_i, K_j] = +i\hbar\delta_{ij}M, \quad [M, \text{every generator}] = 0. \quad (4.33)$$

That a center is possible at all is the ray freedom made algebraic. By Wigner's theorem (Section 4.5), a symmetry group acts on states through operators defined only up to a phase:

$$U(g_1)U(g_2) = e^{i\xi(g_1, g_2)}U(g_1g_2). \quad (4.34)$$

The composition of two infinitesimal representatives produces, at order ε^2 , a commutator that closes on the identity operator — precisely because the phase ξ may be non-trivial. The bracket $[P_i, K_j] = i\hbar\delta_{ij}M$ is that commutator, and M is the central charge that records the obstruction to the ray representative being a true (vector) representation.

Bargmann's theorem. The second cohomology group of the Galilei Lie algebra with real coefficients is one-dimensional:⁸

$$H^2(\mathfrak{g}, \mathbb{R}) \cong \mathbb{R}, \quad (4.35)$$

⁸V. Bargmann, *Ann. Math.* **59**, 1 (1954). The theorem classifies all possible central extensions of the Galilei algebra by $\mathbb{R}$;

generated by the mass cocycle. Consequently, every projective unitary representation of the Galilei group is equivalent to a true unitary representation of the centrally extended group, with the mass parameter M as the single central charge. The theorem both justifies the focus on the extended algebra (4.32)–(4.33) and classifies every possible Galilean-invariant quantum theory: all such theories are parametrized by a single real number, the mass.

The 2-cocycle, explicitly. The phase ξ in the ray-representative law $U(g_1)U(g_2) = e^{i\xi(g_1, g_2)}U(g_1g_2)$ is the Galilean **2-cocycle**. For $g_i = (R_i, \mathbf{v}_i, \mathbf{a}_i, b_i)$ its translation–boost part is

$$\xi(g_1, g_2) = \frac{M}{2\hbar}(\mathbf{a}_1 \cdot \mathbf{v}_2 - \mathbf{a}_2 \cdot \mathbf{v}_1), \quad (4.36)$$

supplemented by terms bilinear in b_i from the time translation (all of the same cohomology class). It is bilinear and antisymmetric under $g_1 \leftrightarrow g_2$ in the $(\mathbf{a}, \mathbf{v})$ sector, and it satisfies the **cocycle condition** $\xi(g_1, g_2) + \xi(g_1g_2, g_3) = \xi(g_2, g_3) + \xi(g_1, g_2g_3)$ — the associativity of the projective product. Rephasing each representative, $U(g) \mapsto e^{i\varphi(g)}U(g)$, shifts the cocycle by a **coboundary** $\xi \mapsto \xi + \delta\varphi$ with $(\delta\varphi)(g_1, g_2) = \varphi(g_1) + \varphi(g_2) - \varphi(g_1g_2)$; coboundaries are the trivial cocycles, removable by rephasing. The cocycle condition is $\delta\xi = 0$; the physical content is the cohomology class $[\xi] \in H^2(\mathfrak{g}, \mathbb{R})$, invariant under rephasing. Bargmann’s theorem is the computation $H^2(\mathfrak{g}, \mathbb{R}) \cong \mathbb{R}$, generated by $[\xi]$ with coefficient M : the cocycle is not a coboundary (no rephasing removes it), and every other cocycle is either a coboundary or proportional to ξ . The integrated phase (4.37), $T(\mathbf{a})B(\mathbf{v}) = e^{-iM\mathbf{a} \cdot \mathbf{v}/\hbar}B(\mathbf{v})T(\mathbf{a})$, is this cocycle evaluated on a pure translation and a pure boost — the position-representation shadow of the central charge M .

The integrated phase. The central bracket integrates to a phase that captures the noncommutation of translations and boosts in a single display. The commutator $[-i\mathbf{a} \cdot \mathbf{P}/\hbar, -i\mathbf{v} \cdot \mathbf{K}/\hbar]$ is central — it commutes with both generators, so all higher BCH commutators vanish. Set $A = -i\mathbf{a} \cdot \mathbf{P}/\hbar$, $B = -i\mathbf{v} \cdot \mathbf{K}/\hbar$:

$$[A, B] = \left(-\frac{i}{\hbar}\right)^2 a_i v_j [P_i, K_j] = -\frac{1}{\hbar^2} a_i v_j (i\hbar \delta_{ij} M) = -\frac{i}{\hbar} M \mathbf{a} \cdot \mathbf{v},$$

which commutes with A and with B . For such a central commutator the Baker–Campbell–Hausdorff series truncates: $e^A e^B = e^{A+B+[A, B]/2}$.⁹ Hence

$$e^A e^B = e^{-iM\mathbf{a} \cdot \mathbf{v}/2\hbar} e^{-i(\mathbf{a} \cdot \mathbf{P} + \mathbf{v} \cdot \mathbf{K})/\hbar}, \quad e^B e^A = e^{+iM\mathbf{a} \cdot \mathbf{v}/2\hbar} e^{-i(\mathbf{a} \cdot \mathbf{P} + \mathbf{v} \cdot \mathbf{K})/\hbar}.$$

Eliminating the common factor $e^{-i(\mathbf{a} \cdot \mathbf{P} + \mathbf{v} \cdot \mathbf{K})/\hbar}$,

$$T(\mathbf{a}) B(\mathbf{v}) = e^{-iM\mathbf{a} \cdot \mathbf{v}/\hbar} B(\mathbf{v}) T(\mathbf{a}). \quad (4.37)$$

Translate-then-boost differs from boost-then-translate by a phase proportional to the mass.

Physical reading and Bargmann’s superselection rule. The phase (4.37) has two profound physical consequences, both identified by Bargmann.

First, M is the mass. The boost action (4.39) identifies M as the inertial parameter: $B(\mathbf{v})^\dagger \mathbf{P} B(\mathbf{v}) = \mathbf{P} + M\mathbf{v}$. In the classical limit, the Galilean boost shifts the momentum by the mass times the veloc-

the conclusion is that the only non-trivial extension is the one parameter M , and that all other apparent phases in (4.34) can be absorbed into a redefinition of the generators.

⁹The identity $e^A e^B = e^B e^A e^{[A, B]}$ when $[A, B]$ is central is verified in Exercise 7(a) by series expansion.

ity, exactly as Newton's second law demands. The mass is the one Galilean invariant carried by every elementary system — it is the central charge of the spacetime algebra.

Second, mass is superselected. In a superposition of two masses m_1 and m_2 , the phase (4.37) would differ for the two components. The difference is a *relative* phase, and by the reasoning of Section 2.5, a relative phase between components of a superposition is in principle observable — it would manifest as interference under a sequence of translations and boosts. Yet no experiment has ever exhibited interference between states of different mass. The consistent reading is that such superpositions do not occur: mass is a *superselected* charge, and the state space decomposes into orthogonal sectors labelled by the mass,

$$\mathcal{H} = \bigoplus_{m \geq 0} \mathcal{H}_m, \quad (4.38)$$

with no physical observable connecting different sectors. The superselection clause of Chapter 2 (Section 2.5) here receives its first derived instance: the central extension of the Galilei algebra *enforces* the splitting. The two other instances the states chapter named — integer versus half-integer angular momentum and electric charge — are reaffirmed as the angular-momentum chapter's and field theory's; the Galilean share is this chapter's.

Action on momentum and the boost–momentum shift. The conjugation relation (4.39),

$$B(\mathbf{v})^\dagger \mathbf{P} B(\mathbf{v}) = \mathbf{P} + M\mathbf{v}, \quad (4.39)$$

is derived directly from the algebra. Expanding for small $\mathbf{v}$:

$$(I + i\mathbf{v} \cdot \mathbf{K}/\hbar) \mathbf{P} (I - i\mathbf{v} \cdot \mathbf{K}/\hbar) = \mathbf{P} + \frac{i}{\hbar} v_i [K_i, \mathbf{P}] = \mathbf{P} + \frac{i}{\hbar} v_i (-i\hbar \delta_{ij} M \hat{e}_j) = \mathbf{P} + M\mathbf{v},$$

using $[K_i, P_j] = -i\hbar \delta_{ij} M$ (the adjoint of (4.33), keeping the sign consistent: $[P_i, K_j] = +i\hbar \delta_{ij} M$ implies $[K_i, P_j] = -i\hbar \delta_{ij} M$ by antisymmetry of the commutator). The infinitesimal form integrates to (4.39).

Advantage. The projective-representation analysis compresses the entire phenomenology of non-relativistic quantum kinematics into a single algebraic fact: the Galilei algebra admits exactly one non-trivial central extension, whose coefficient is the mass. Every quantum system carries a definite mass; no superposition of different masses exists. The analysis is exact and algebraic: no assumption about dynamics beyond Galilean symmetry enters.

4.7.4 Unitary Irreducible Representations

With the algebra (4.32)–(4.33) in hand, the unitary irreducible representations (unirreps) of the centrally extended Galilei group are constructed by the method of induced representations. The construction yields the concrete Hilbert space of a free particle and the explicit action of every generator upon it.

The little-group method. The Galilei group acts on the momentum (energy-momentum) space. The orbit of a reference four-momentum $(E, \mathbf{p})$ under the group is the mass shell:

$$E = \frac{\mathbf{p}^2}{2m}, \quad \mathbf{p} \in \mathbb{R}^3, \quad (4.40)$$

for a fixed mass $m > 0$. (The massless case $m = 0$ admits no physical nonrelativistic interpretation and is stated for completeness only.) The **little group** (stabilizer) of a reference momentum on the mass shell is the subgroup leaving that momentum invariant. Choose the reference momentum $\mathbf{p}_0 = \mathbf{0}$ (a particle at rest). The transformations that preserve $\mathbf{p} = \mathbf{0}$ are rotations and time translations; the little group is $SU(2) \times \mathbb{R}$. Its unitary irreducible representations are labelled by the spin $s = 0, 1/2, 1, \dots$, each of dimension $2s + 1$. The full unirrep of the Galilei group is induced from the little-group representation.

The massive spin- s representation. The representation space is

$$\mathcal{H}_{m,s} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^{2s+1}, \quad (4.41)$$

the space of square-integrable momentum-space wavefunctions $\psi_\sigma(\mathbf{p})$, $\sigma = -s, \dots, s$, with the inner product

$$\langle \psi | \phi \rangle = \sum_{\sigma=-s}^s \int d^3p \, \psi_\sigma^*(\mathbf{p}) \phi_\sigma(\mathbf{p}). \quad (4.42)$$

The generators act as multiplicative and differential operators on these wavefunctions:

$$(\mathbf{P}\psi)_\sigma(\mathbf{p}) = \mathbf{p} \psi_\sigma(\mathbf{p}), \quad (4.43)$$

$$(H\psi)_\sigma(\mathbf{p}) = \frac{\mathbf{p}^2}{2m} \psi_\sigma(\mathbf{p}), \quad (4.44)$$

$$(\mathbf{J}\psi)_\sigma(\mathbf{p}) = (-i\hbar \mathbf{p} \times \nabla_{\mathbf{p}} \psi)_\sigma + \sum_{\sigma'} (\mathbf{S}^{(s)})_{\sigma\sigma'} \psi_{\sigma'}(\mathbf{p}), \quad (4.45)$$

$$(\mathbf{K}\psi)_\sigma(\mathbf{p}) = -i\hbar m \nabla_{\mathbf{p}} \psi_\sigma(\mathbf{p}), \quad (4.46)$$

where $\mathbf{S}^{(s)}$ are the spin- s matrices (the generators of the $(2s+1)$ -dimensional representation of $\mathfrak{su}(2)$).

The mass operator acts as multiplication by m .

Verification of the algebra. The commutation relations (4.32)–(4.33) are verified by direct computation on the representation space. *Translations commute:* $[P_i, P_j] = 0$ trivially since both act as multiplication. *Rotation–momentum:* $[J_i, P_j] = i\hbar \epsilon_{ijk} P_k$ follows from the cross-product term in $\mathbf{J}$ acting on the linear function $\mathbf{p}$ (the spin term commutes with $\mathbf{P}$). *Boost–momentum:* $[K_i, P_j] = -i\hbar m \partial_{p_i} p_j = -i\hbar m \delta_{ij}$, which, together with antisymmetry, yields $[P_i, K_j] = +i\hbar \delta_{ij} M$, correctly. *Boost–Hamiltonian:* $[K_i, H] = -i\hbar m \partial_{p_i} (\mathbf{p}^2/2m) = -i\hbar p_i$ and $[H, P_i] = 0$, giving $[K_i, H] = -i\hbar P_i$, correctly. *Rotation algebra:* $[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$ follows from the commutation of the orbital $(-i\hbar \mathbf{p} \times \nabla_{\mathbf{p}})$ and spin ($\mathbf{S}$) parts, each of which separately satisfies the $\mathfrak{so}(3)$ algebra. Every bracket of (4.32) is satisfied.

Mackey’s induced-representation theorem. The construction above is an instance of Mackey’s general theory of irreducible representations of semidirect products $G = N \rtimes H$ with N abelian. The dual group $\hat{N}$ (here energy–momentum space) carries an action of H ; Mackey’s theorem states that the irreps of G are in bijection with (i) an orbit $\mathcal{O}$ of H in $\hat{N}$ and (ii) an irrep σ of the **little group** (stabilizer) H_{p_0} of a reference point $p_0 \in \mathcal{O}$: the irrep is the representation *induced* from σ on H_{p_0} , lifted to G through the orbit. For the Galilei group, $\hat{N}$ is $(E, \mathbf{p})$ -space, the orbits are the mass shells $E = \mathbf{p}^2/2m$ (labelled by the mass m), and the little group of the rest point $(E_{\text{int}}, \mathbf{0})$ is $SO(3) \times \mathbb{R}$ (rotations and

time translations), whose irreps are the spin- s multiplets. Mackey's theorem then gives the unirreps (m, s) and *only* these — the construction of this subsection is not one method among several but the classification, and the labels (m, s) exhaust the Galilean elementary systems. The Inönü–Wigner contraction of the Poincaré group (below) carries the Mackey classification of the relativistic case into this nonrelativistic one.

The massless case. For $m = 0$, the mass shell degenerates and the representation theory changes. The little group enlarges; the physical representations are infinite-dimensional (continuous spin) and have no known elementary realizations in nonrelativistic physics. The case is named for completeness and is not developed.

Advantage. The explicit unirrep construction grounds the abstract algebra in concrete operators on a concrete Hilbert space. It supplies the operators whose commutation relations were derived algebraically in the preceding subsections; it confirms that the algebra is not empty — for every mass $m > 0$ and spin s , a representation exists; and it makes contact with the momentum representation of Chapter 5, in which $\mathbf{P}$ is diagonal and $H = \mathbf{P}^2/2m$ is the free Hamiltonian. What the unirrep achieves is the proof that the Galilean algebra *requires* the Hamiltonian to take this form, up to internal degrees of freedom.

4.7.5 Physical Implications of Galilean Invariance

Galilean invariance constrains the form of every physical law within its domain. This subsection collects the constraints and their measurable consequences.

Constraint on the Hamiltonian. The three brackets $[P_i, H] = 0$, $[J_i, H] = 0$, and $[K_i, H] = -i\hbar P_i$ together fix the general form of a Galilean-invariant Hamiltonian. For a single particle, the algebra forces

$$H = \frac{\mathbf{P}^2}{2M} + H_{\text{int}}, \quad [P_i, H_{\text{int}}] = 0, \quad [J_i, H_{\text{int}}] = 0, \quad [K_i, H_{\text{int}}] = 0. \quad (4.47)$$

The internal Hamiltonian H_{int} must be a Galilean scalar: it commutes with every generator of the spacetime algebra. For a particle without internal structure, $H_{\text{int}} = \text{const}$, and the free-particle form (??) is recovered. The proof is the computation of (??), restated:

$$[K_i, H - \frac{\mathbf{P}^2}{2M}] = -i\hbar P_i - \frac{1}{2M}((-i\hbar\delta_{ij}M)P_j + P_j(-i\hbar\delta_{ij}M)) = 0.$$

The same quantity commutes with $\mathbf{P}$ and with $\mathbf{J}$, so it is a central element and, for an irreducible representation, proportional to the identity — exactly E_{int} in (??).

Constraint on interactions. For two particles, the most general Galilean-invariant interaction potential is

$$V = V(|\mathbf{r}_1 - \mathbf{r}_2|, |\mathbf{p}_1 - \mathbf{p}_2|^2, (\mathbf{r}_1 - \mathbf{r}_2) \cdot (\mathbf{p}_1 - \mathbf{p}_2), \mathbf{S}_1, \mathbf{S}_2, \dots), \quad (4.48)$$

a function of the relative coordinate and relative momentum, rotationally invariant, and possibly carrying spin dependence. The center-of-mass motion separates exactly:

$$H = \frac{\mathbf{P}_{\text{CM}}^2}{2M_{\text{tot}}} + H_{\text{rel}}, \quad (4.49)$$

with $M_{\text{tot}} = m_1 + m_2$, $\mathbf{P}_{\text{CM}} = \mathbf{p}_1 + \mathbf{p}_2$, and $H_{\text{rel}} = \mathbf{p}_{\text{rel}}^2/2\mu + V$, where $\mu = m_1 m_2 / (m_1 + m_2)$ is the reduced mass. The separation is a theorem of Galilean covariance, not an approximation.

Behavior of wavefunctions under boosts. In the Schrödinger representation, a Galilean boost acts on a wavefunction as

$$\psi'(\mathbf{r}, t) = \exp\left[\frac{im}{\hbar}(\mathbf{v} \cdot \mathbf{r} - \tfrac{1}{2}\mathbf{v}^2 t)\right] \psi(\mathbf{r} - \mathbf{v}t, t). \quad (4.50)$$

The phase factor is the **Galilean phase**. It arises because the Schrödinger equation is not invariant under boosts — the wavefunction acquires a spacetime-dependent phase. Yet every probability density $|\psi'|^2 = |\psi|^2$ and every expectation value of a Galilean-covariant observable transforms correctly. The phase is the position-representation face of the boost-translation phase (4.37): it ensures that the superposition principle operates coherently in every uniformly moving frame.

Galilean covariance of the Schrödinger equation. Under the boost (4.50), the Schrödinger equation transforms as

$$i\hbar \frac{\partial \psi'}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V'\right) \psi', \quad (4.51)$$

with the boosted potential $V'(\mathbf{r}, t) = V(\mathbf{r} - \mathbf{v}t)$. The equation is *covariant*: if ψ satisfies the Schrödinger equation in the original frame, ψ' satisfies it in the boosted frame with the appropriately transformed potential. The covariance ensures that the laws of physics have the same form in all inertial frames — the operational content of Galilean relativity.

Superselection of mass as a theorem. Within each mass sector $\mathcal{H}_m$, every observable commutes with the mass operator. An operator that connects different mass sectors would necessarily fail to commute with the boost generator $\mathbf{K}$ (because the boost–translation phase differs across sectors) and would therefore not be Galilean-covariant. Since all physical observables are Galilean-covariant, mass superselection follows: the mass is a fixed label of every physical preparation, and no coherent superposition of different-mass states exists. This is the operational content of Bargmann’s analysis.

The Inönü–Wigner contraction from the Poincaré group. The Galilei group is the $c \rightarrow \infty$ *contraction* of the Poincaré group. Rescale the Lorentz boost generators $\tilde{\mathbf{K}} = \mathbf{K}/c$ (so that a finite boost is $e^{-i\mathbf{v} \cdot \tilde{\mathbf{K}}/\hbar}$ with $\mathbf{v} = \beta c$); the Poincaré bracket $[\tilde{K}_i, \tilde{K}_j] = -(i\hbar/c^2)\epsilon_{ijk}J_k$ tends to 0 as $c \rightarrow \infty$, recovering the Galilean $[K_i, K_j] = 0$ — the law of Galilean addition of velocities. The remaining brackets are unchanged in the limit, and the relativistic energy–momentum relation $E = \sqrt{\mathbf{p}^2 c^2 + m^2 c^4}$ expands to $mc^2 + \mathbf{p}^2/2m + \dots$, the rest energy (an additive constant, absorbed into the internal E_{int}) plus the nonrelativistic kinetic energy. The contraction is the structural reason the Galilean analysis of this section is exact within its domain, and the reason the Galilean mass superselection has no Poincaré analogue: in the relativistic theory mass is already a Casimir ($P_\mu P^\mu = m^2 c^2$), not a central extension. The first correction away from the contraction — the $O(1/c^2)$ term $[K_i, K_j] \neq 0$ — is the Thomas precession of the historical chapter (Section 1.3.4); the full relativistic structure (the Lorentz group and its $\text{SL}(2, \mathbb{C})$ cover) is developed in Appendix B.8 and is this book’s relativistic boundary.

Discrete Galilean symmetries. The connected Galilei group analysed above comprises only the continuously connected transformations (rotations, translations, boosts, waiting). The *full* Galilei group adjoins two discrete spacetime symmetries: space inversion $P : \mathbf{r} \mapsto -\mathbf{r}, t \mapsto t$, and time reversal

$T : \mathbf{r} \mapsto \mathbf{r}, t \mapsto -t$. Their action on the generators, read off the transformation of the corresponding spacetime displacement, is

	$\mathbf{J}$	$\mathbf{P}$	$\mathbf{K}$	H	M	
P	$+\mathbf{J}$	$-\mathbf{P}$	$-\mathbf{K}$	$+H$	$+M$	(4.52)
T	$-\mathbf{J}$	$-\mathbf{P}$	$+\mathbf{K}$	$+H$	$+M$	

(P is unitary, T antiunitary). Angular momentum $\mathbf{J} = \mathbf{r} \times \mathbf{p}$ is an *axial* vector (P -even, T -odd, since T flips both $\mathbf{r}$ and $\mathbf{p}$); linear momentum $\mathbf{p}$ is *polar* (P -odd, T -odd); the boost generator $\mathbf{K} = m\mathbf{r} - \mathbf{p}t$ is P -odd, T -even. The mass M and the Hamiltonian H are invariant under both. These discrete actions extend the Galilean algebra consistently and are the spacetime-group avatars of the parity and time-reversal symmetries developed physically in Section 4.10: P -invariance yields Laporte's rule (4.63), T -invariance yields Kramers degeneracy (4.66). The full Galilei group is the semidirect product of the connected group with $\mathbb{Z}_2^P \times \mathbb{Z}_2^T$; its projective representations are classified by extending Bargmann's analysis to the discrete generators, a refinement that changes nothing essential at the level of the continuous algebra.

4.7.6 Examples

The Galilean structure is exercised on four concrete systems, each illuminating a different facet of the algebra.

(a) The free particle. For a single particle of mass m and spin s , the unirrep (4.41) is the complete quantum description. The Hamiltonian is $H = \mathbf{P}^2/2m$, with spectrum $\sigma(H) = [0, \infty)$ and generalized eigenstates $|\mathbf{p}, \sigma\rangle$ labelled by momentum and spin projection. The evolution operator $e^{-iHt/\hbar}$ propagates these states as plane waves — the de Broglie wave of Chapter 1 ((1.3.2)) is the momentum eigenstate in the position representation, and its Galilean covariance is the theorem (4.50).

(b) Two-body problem: center-of-mass decomposition. Two particles of masses m_1, m_2 and spins s_1, s_2 are described by the tensor product $\mathcal{H}_{m_1, s_1} \otimes \mathcal{H}_{m_2, s_2}$. Galilean covariance forces the total generators to be the sums of the single-particle generators:

$$\mathbf{P}_{\text{tot}} = \mathbf{P}_1 + \mathbf{P}_2, \quad \mathbf{J}_{\text{tot}} = \mathbf{J}_1 + \mathbf{J}_2, \quad \mathbf{K}_{\text{tot}} = \mathbf{K}_1 + \mathbf{K}_2, \quad M_{\text{tot}} = m_1 + m_2. \quad (4.53)$$

The total Hamiltonian separates into center-of-mass and relative parts (4.49). The separation is exact and is the quantum analogue of the classical reduction of the two-body problem to a one-body problem.

(c) The harmonic oscillator. For a particle of mass m in a harmonic potential $V(\mathbf{r}) = \frac{1}{2}m\omega^2|\mathbf{r}|^2$, the Hamiltonian $H = \mathbf{P}^2/2m + V$ is not invariant under boosts: $[K_i, V] \neq 0$, reflecting that a boosted oscillator sees a displaced and moving potential. The bracket $[K_i, H] = -i\hbar P_i$ is nevertheless satisfied: the boost generates the correct transformation of the full Hamiltonian. The oscillator's spectrum $E_n = \hbar\omega(n_x + n_y + n_z + 3/2)$ is rotationally invariant ($[J_i, H] = 0$) and translationally non-invariant ($[P_i, H] \neq 0$), correctly.

(d) A particle in an external potential. An external potential $V(\mathbf{r})$ breaks Galilean invariance partially. If V is constant, the theory is fully Galilean-invariant (Galilean relativity is trivial for a free

particle). If $V = V(|\mathbf{r}|)$ is central, rotational invariance survives: $[J_i, H] = 0$. If V is periodic, translation invariance is lost but a discrete subgroup survives. The Galilean algebra provides the framework for identifying which symmetries any given potential respects.

The internal two-state systems of this book. Spin and polarization are *internal*: the Galilean boost representative factors, acting on the motional state and as the identity on the internal $\mathbb{C}^2$ — a boost does not turn a spin. A boosted qubit is the same qubit described from a moving frame: pure relabeling, no precession. The non-trivial content is the motional phase: $T(\mathbf{a})B(\mathbf{v})|\psi\rangle$ and $B(\mathbf{v})T(\mathbf{a})|\psi\rangle$ differ by $e^{-iM\mathbf{a}\cdot\mathbf{v}/\hbar}$ — unobservable on a single mass sector by the ray theorem, would-be observable across a mass superposition. The phase's interferometric realization, species-dependent fringe shifts, is named and not developed; the point is the algebra's center and what it forbids.

Advantage of the Galilean analysis. The whole spacetime kinematics of nonrelativistic quantum mechanics fits in one algebra, derived from ten generators and one central charge. The mass emerges as the central charge; the superselection of mass is exhibited as the reading the center forces; the free dynamics' form is constrained by symmetry alone; and every interaction is limited to the few shapes compatible with the algebra. The analysis is exact and algebraic: no assumption beyond Galilean symmetry enters.

Limitation. The group is the *nonrelativistic* kinematics.¹⁰ One generator stands apart from the others already within the domain. $\mathbf{P}$, $\mathbf{J}$, $\mathbf{K}$ are symmetry charges whose families may or may not leave the dynamics invariant; H is the dynamics itself. The asymmetry is not accidental — it is what time *is* in this theory, and it deserves a section of its own.

4.8 The Special Status of Time

The Galilean census of Section 4.7 placed ten generators side by side, and one of them does not belong to the company it keeps. Translations, rotations, and boosts are operations the experimenter may perform or refrain from performing: their generators $\mathbf{P}$, $\mathbf{J}$, $\mathbf{K}$ are symmetry charges, and whether the dynamics respects them is a question of fact, settled case by case by the bracket $[G, H]$ of the next section. Waiting is not such an operation. The time-translation subgroup is not a candidate symmetry of the evolution law; by Axiom (D1) ((4.1)) it *is* the evolution law. This section draws the structural consequences of that asymmetry, and so repays with its full analysis the operational remark of Section 4.1: that t is the reading of the laboratory clock, the experimenter's waiting time, never an observable of the system.

The dual role of H . In the Galilei algebra (4.32), H appears as the generator of the time-translation subgroup, one row of brackets among many. But its place in the theory is double in a way no other generator shares: H generates time translations, and the family it generates *is* the dynamics. To ask whether the evolution is invariant under time translation is to ask whether $[H, H] = 0$, which is

¹⁰In the relativistic upgrade the boosts cease to commute, $[K_i, K_j] \sim -i\epsilon_{ijk}J_k/c^2$, spin mixes with boosts, and the Thomas precession of the historical chapter is the first symptom. This book does not develop the Lorentz group.

a tautology; time translation asks no invariance question because it is the law whose invariance is elsewhere being asked about. The contrast with the pure symmetry charges is threefold. *Parameter versus coordinate*: t labels the waiting between a preparation and an interrogation; in this book's nonrelativistic theory it is not a reading drawn from the system, and no state is sharp in it. *Generator of the law versus generators of transformations*: P, J, K generate physical operations whose invariance status is an open, physical question; H 's family is the law itself. *Conservation by tautology versus conservation by symmetry*: momentum conservation, where it holds, reports a translation invariance of H ; energy conservation reports nothing beyond the homogeneity of time already postulated with (D1) — a tautological shadow that the conservation theorem of Section 4.9 recovers and elevates into the general principle.

The book's own two-state arena makes the dual role literal, and then exhibits, in arithmetic, the obstruction that bars time from the ledger of observables. On the spin bench of Section 4.3, the single operator $H = (\hbar\omega/2)\sigma_z$ is at once the energy observable — eigenvalues $\pm\hbar\omega/2$, whose difference is the Bohr frequency identified in Section 4.2 — and the generator of waiting, whose exponential precesses every transverse polarization ((4.8)). The same matrix does both jobs; that is the dual role made concrete. Against it, a charge such as σ_x is a question: $[\sigma_x, H] \neq 0$, and the answer is read off the motion of $\langle\sigma_x\rangle$ computed in Section 4.3.

No time operator. If t were an observable of the system, there would be a self-adjoint operator T reading it, canonically conjugate to the energy in the sense of the canonical pair named in the measurement chapter ((3.60), cited by name):

$$[T, H] = i\hbar I \implies \text{tr}[T, H] = 0 \neq i\hbar \dim \mathcal{H}. \quad (4.54)$$

In finite dimension the obstruction is the displayed line of arithmetic and nothing more: the trace of a commutator vanishes, and no dimension makes zero equal to $i\hbar \dim \mathcal{H}$. The two-state instance is immediate — for $H = (\hbar\omega/2)\sigma_z$, or for any 2×2 matrix whatever, taking traces of both sides of $[T, H] = i\hbar I$ pits 0 against $2i\hbar$. The general obstruction is spectral rather than arithmetic, and it is named here, not developed.¹¹



Note [Two registers, kept apart] The obstruction (4.54) is exact arithmetic on the finite arena of this book's models; Pauli's argument is a statement about the spectra of realistic, infinite-dimensional systems. They are different proofs of one conclusion, and this book keeps them in different registers: the displayed line is a theorem, the footnote a literature pointer whose rigor belongs to Appendix A. What neither register permits is the conclusion's evasion: there is no time operator, in any dimension.

The time–energy uncertainty relation. The frequently quoted $\Delta E \Delta t \gtrsim \hbar/2$ is not a Robertson relation between two spreads of a state, and the machinery of Section 3.8 cannot make it one: the Robertson bound (3.64) prices the ensemble widths of two observables, and there is no time width of a state, t being a parameter — the formal status of $\Delta x \Delta p$ ((3.69)) has no temporal analog. The relation's precise content is kinetic: the energy width of a preparation sets the rate of the state's own

¹¹W. Pauli, *Die allgemeinen Prinzipien der Wellenmechanik*, in *Handbuch der Physik*, vol. 24/1 (Springer, 1933): a T with $[T, H] = i\hbar I$ would generate rigid translations of the energy, making H unitarily equivalent to $H + \epsilon I$ for every real ϵ and thereby spreading its spectrum over the whole real line — against the boundedness below of every stable system's energy. The spectral and domain questions on which the argument turns for unbounded H are deferred to Appendix A.

change. For any observable A , the timescale $\tau_A := \Delta A / |\mathrm{d}\langle A \rangle_\psi / \mathrm{d}t|$ obeys $\Delta E \tau_A \geq \hbar/2$, with ΔE the energy spread of the state.¹² A preparation of narrow energy spread changes slowly; the decay widths of unstable states — the natural widths of the spectral lines of the historical chapter's spectroscopy — are the same content read from the other side, and their rate theory belongs to the chapter on perturbation methods. What the relation does not say is that a measurement of time carries an uncertainty; there is no such measurement, and no such reading is endorsed here.

Symmetries of the question rearrange the description; symmetries of the dynamics — generators that commute with H , in the language now earned — freeze quantities. That freezing, computed for every Galilean charge at once, is the conservation law, and it is the next section.

4.9 Conservation Laws, Good Quantum Numbers, and Degeneracy

Section 4.8 separated the generators that ask a question — may this family be performed without changing the dynamics? — from the one generator that is the dynamics. This section answers the question, once for every generator, and then reads the answer against the Galilean census of Section 4.7. The instrument is the equation of motion of expectation values.

The expectation-value equation of motion. Let A be any fixed operator and $|\psi(t)\rangle$ any state evolving under Axiom (D1). Differentiating the sandwich $\langle \psi(t) | A | \psi(t) \rangle$,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \langle \psi | A | \psi \rangle &= \left(\frac{\mathrm{d}}{\mathrm{d}t} \langle \psi | \right) A | \psi \rangle + \langle \psi | A \left(\frac{\mathrm{d}}{\mathrm{d}t} | \psi \rangle \right) \\ &= \left(\frac{i}{\hbar} \langle \psi | H \right) A | \psi \rangle + \langle \psi | A \left(-\frac{i}{\hbar} H | \psi \rangle \right) \\ &= \frac{i}{\hbar} \langle \psi | (HA - AH) | \psi \rangle = \frac{i}{\hbar} \langle \psi | [H, A] | \psi \rangle, \end{aligned} \quad (4.55)$$

hence

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle A \rangle_\psi = \frac{i}{\hbar} \langle [H, A] \rangle_\psi \quad (A \text{ fixed}). \quad (4.56)$$

The rate of change of every ensemble mean is priced by one commutator. The statement concerns expectation values — a fixed operator between evolving states. Arithmetically the same display admits a second reading, in which $U^\dagger(t) A U(t)$ is taken to evolve in the state's stead; that reading is the next chapter's, and the ambiguity it opens is registered at this chapter's exit (Section 4.11) — nothing here depends on it. One clause is named for completeness: an operator carrying its own explicit time dependence, $A = A(t)$, contributes the additional term $\langle \partial A / \partial t \rangle_\psi$; the instruments of this chapter are fixed, and the clause will not be used.

Theorem (conservation). For a self-adjoint generator G ,

$$[G, H] = 0 \iff \frac{\mathrm{d}}{\mathrm{d}t} \langle G \rangle_\psi = 0 \text{ in every state } |\psi\rangle. \quad (4.57)$$

The forward direction is (4.56) read off: a vanishing commutator makes the right side zero. For

¹²L. Mandelstam and I. Tamm, *J. Phys. (USSR)* **9**, 249 (1945). The argument applies the Robertson relation (3.64) to the pair (H, A) and converts the commutator into a rate through the expectation-value equation of motion (4.56); the full derivation, with its honesty clause, is this chapter's Exercise 9.

the converse, suppose the mean is constant in every state; then $\langle [H, G] \rangle_\psi = 0$ in every state. The operator $i[H, G]$ is self-adjoint by the anti-self-adjointness of the commutator ((3.59)), and a self-adjoint operator whose diagonal values vanish in *every* state vanishes as an operator, by the polarization identity of the measurement chapter ((3.28)) — the operative constraint is the quantifier: conservation on some subfamily of preparations would force nothing. The theorem's content is in fact distributional, stronger than the constancy of a mean. If $[G, H] = 0$, the compatibility theorem ((3.49)) hands G and H a common eigenbasis, in which the evolution operator (4.5) is diagonal; every amplitude against a G -eigenray therefore changes only by a phase,

$$|\langle g|\psi(t)\rangle|^2 = |\langle g|\psi(0)\rangle|^2, \quad (4.58)$$

and the whole statistics of G — every moment, every outcome probability — is conserved, not merely its average.

The Galilean reading. Read against the algebra (4.32), the theorem is a census of the conserved quantities of nonrelativistic physics, generator by generator. Translation invariance of the law of waiting, $[P_i, H] = 0$, is momentum conservation — the free law of Section 4.7, with the free generator (??) its paradigm. Rotation invariance, $[J_i, H] = 0$, is angular momentum conservation — the symmetry of central problems, whose spectroscopic harvest, the hydrogen atom behind the historical chapter's Balmer terms, is the angular-momentum chapter's to reap. Time independence is energy conservation, the tautological case of Section 4.8: the homogeneity of time is its mother.¹³ The boost stands apart: its bracket $[K_i, H] = -i\hbar P_i$ never vanishes, so boosts furnish no conserved charge; their symmetry content survives only for the free system, where a boost carries motions to motions, and interactions destroy even that.

Good quantum numbers. The eigenvalues of a conserved G are permanent labels: a preparation addressed by g keeps that address under every waiting. A complete set of commuting observables containing H ((3.53)) therefore addresses every state by quantities that either set its phase rate — the energy — or never change at all. Such addresses are the **good quantum numbers**; the permanence of the label is the conservation theorem, and its precision as an address is the measurement chapter's completeness.

Theorem (degeneracy). *If $[G, H] = 0$, every eigenspace of H is invariant under G ; if G does not act as a scalar within an eigenspace, that level is degenerate.* The proof is one line: for $H|E\rangle = E|E\rangle$,

$$H(G|E\rangle) = GH|E\rangle = E(G|E\rangle), \quad (4.59)$$

so G maps each level into itself; within a non-degenerate level every operator is a scalar, hence a non-scalar restriction forces the level to have dimension greater than one. A corollary is worth naming: two conserved generators *whose restrictions to a level fail to commute* — equivalently, whose commutator does not annihilate the level — force that level to be degenerate, since on a one-dimensional space the restrictions would be scalars and scalars commute. The Galilean instance is immediate: a rotation-invariant H inherits the rotation family's noncommutation (4.32) wherever the generators

¹³The general principle — to each continuous symmetry of the dynamics, one conserved quantity — is Noether's theorem of 1918, proved there for classical action principles. In this book's quantum setting its content is precisely the conservation theorem proved above; the classical variational theory is named, not developed.

fail to commute within a level; the classification of the multiplets so forced is the angular-momentum chapter's harvest, flagged here and not built.

Theorem (selection rule). *If $[A, G] = 0$, then $\langle g'|A|g\rangle = 0$ whenever $g' \neq g$ are distinct eigenvalues of G :*

$$(g - g') \langle g'|A|g\rangle = \langle g'|[A, G]|g\rangle = 0 \implies \langle g'|A|g\rangle = 0 \quad (g' \neq g). \quad (4.60)$$

One line, and it is the engine of spectroscopy: an operator that respects a conserved charge connects no states that differ in it. The parity instance is proved in Section 4.10; the angular-momentum rules belong to the angular-momentum chapter; the rates of the transitions the rule allows belong to the chapter on perturbation methods.

The theorem pair is exhibited on hardware the book already owns. The two-doublet atom of the measurement chapter (Section 3.6) carries a which-manifold observable — renamed here in prose, to keep it clear of the Galilean central charge — that commutes with the internal Hamiltonian. The manifold label is therefore a good quantum number: no internal evolution moves population between the doublets, and the metastable doublet's long life *is* a selection rule — decay awaits a coupling that fails to commute with the manifold observable, the coupling to the radiation field whose theory belongs to later chapters and is named here as the mechanism, not developed. The degeneracy theorem is read off the same bench: in zero field the internal Hamiltonian is spin-independent, so every spin component is conserved, and since the components fail to commute within each level's spin space, each manifold's level is at least twofold — the doublets themselves. Metastable lifetimes and forbidden lines, measured in every spectroscopic laboratory, are symmetry bookkeeping.

The conservation machinery is continuous-symmetry machinery, keyed to the Lie families of Sections 4.6 and 4.7. Its discrete cousins — the inversion of space and the reversal of motion — belong to the same spacetime group in its disconnected pieces, and they are no less measurable. They close the chapter's construction, next.

4.10 Discrete Symmetries: Parity and Time Reversal

The spacetime group also has disconnected pieces — transformations no parameter dials continuously to zero — and two of them belong to the main line of physics: the inversion of space and the reversal of motion. They are no less measurable than their continuous cousins; each carries a selection rule, and one of them is this book's single consumer of Wigner's antiunitary branch (Section 4.5).

Parity. Space inversion is a symmetry transformation in the sense of (4.14): a ray map preserving every transition probability, and therefore, by Wigner's theorem, an operator on $\mathcal{H}$ from one of the two branches. It is realized unitarily — nothing in the inversion of space compels the conjugation of i , and the compulsion that does arise for the reversal of motion is the second half of this section, where the contrast will do honest work. Two inversions restore the original situation, so the square of the inversion operator is the identity up to a phase, which the freedom of (2.13) absorbs; the parity operator can therefore be chosen with

$$\Pi^\dagger = \Pi, \quad \Pi^2 = I, \quad \text{eigenvalues } \pm 1, \quad (4.61)$$

both self-adjoint and unitary, an observable in its own right. Its eigenstates are the even states ($\Pi|\psi\rangle = +|\psi\rangle$) and the odd states ($\Pi|\psi\rangle = -|\psi\rangle$). For the interactions of this book's main line, space inversion leaves the dynamics invariant, $[\Pi, H] = 0$, and parity is a good quantum number in the sense of Section 4.9.¹⁴¹⁵

Operators are classified by the same conjugation: A is even or odd according as

$$\Pi A \Pi = +A \quad \text{or} \quad \Pi A \Pi = -A. \quad (4.62)$$

Physical instances are named, not built: the position operator and the electric dipole are odd — their operator theory belongs to the chapters on operator methods and on perturbation theory.

Theorem (the parity selection rule). *For odd A , the matrix elements within either parity sector vanish:*

$$\Pi A \Pi = -A \implies \langle \text{even} | A | \text{even} \rangle = \langle \text{odd} | A | \text{odd} \rangle = 0. \quad (4.63)$$

The proof is two insertions of $\Pi^2 = I$. For even $|e\rangle, |e'\rangle$: $\langle e | A | e' \rangle = \langle e | \Pi (\Pi A \Pi) \Pi | e' \rangle = (+1)(-1)(+1) \langle e | A | e' \rangle$, hence zero; for odd $|o\rangle, |o'\rangle$ the same display comes out with $(-1)(-1)(-1) = -1$, hence zero again. This is Laporte's rule: dipole transitions connect states of opposite parity only, and the spectra of parity-invariant systems divide into the allowed and the forbidden. The rates of the allowed transitions belong to the chapter on perturbation methods.

The rule is read off arithmetic on a two-sector model. Let the state space be the direct sum of an even and an odd sector — the direct-sum license of Section 3.6 — with $\Pi = \text{diag}(+I, -I)$ by blocks. Then $[\Pi, H] = 0$ holds exactly when H is block-diagonal: parity conservation forbids the dynamics itself from mixing the sectors. Even operators are block-diagonal, odd operators off-block-diagonal; a dipole-like operator has only the off-diagonal blocks, and the selection rule (4.63) is the matrix shape read aloud: within a sector every such element is a zero block, across sectors it may live. A spectrum with lines only between the sectors is the arithmetic shadow of Laporte's rule — allowed and forbidden lines as two shapes of matrix.

Time reversal. The reversal of motion is defined operationally: the ray map that exchanges every motion with its reverse, carrying an evolving state to the state that evolves as the film run backward — sending the family $U(t)$ to $U(-t)$ — while leaving a time-reversal-invariant Hamiltonian unchanged. The implementing operator Θ must therefore satisfy, for all t ,

$$\Theta e^{-iHt/\hbar} \Theta^{-1} = e^{+iHt/\hbar} \implies \text{unitary } \Theta : \Theta H \Theta^{-1} = -H; \quad \text{antiunitary } \Theta : \Theta H \Theta^{-1} = H, \quad (4.64)$$

Differentiating both sides of $\Theta e^{-iHt/\hbar} \Theta^{-1} = e^{+iHt/\hbar}$ at $t = 0$:

$$\Theta \left(-\frac{i}{\hbar} H \right) \Theta^{-1} = \frac{d}{dt} e^{+iHt/\hbar} \Big|_{t=0} = \frac{i}{\hbar} H.$$

Multiplying through by $i\hbar$ yields $\Theta H \Theta^{-1} = -\Theta i \Theta^{-1} H$. Now invoke Wigner's dichotomy ((4.15)). If Θ is unitary, $\Theta i \Theta^{-1} = i$, so $\Theta H \Theta^{-1} = -iH/i = -H$. If Θ is antiunitary, $\Theta i \Theta^{-1} = -i$ (by (4.16)),

¹⁴The rule of parity entered spectroscopy before the concept: O. Laporte distilled from the iron arc in 1924 the empirical law that spectral lines fall into two classes connected only across, never within; E. Wigner explained the law in 1927 by introducing the parity quantum number — the first concrete fruit of the symmetry theory of Section 4.5.

¹⁵Parity is not a universal symmetry: the weak interactions violate it (T. D. Lee and C. N. Yang, 1956; C. S. Wu and collaborators, 1957). That physics lies outside this book's main line, and this footnote is its only mention.

so $\Theta H \Theta^{-1} = -(-i)H/i = H$.

The unitary branch $\Theta H \Theta^{-1} = -H$ forces the spectrum of H to be unitarily equivalent to its reflection through zero; realistic systems, whose energies are bounded below and not above, admit no such equivalence, and the branch is rejected. The antiunitary branch $\Theta H \Theta^{-1} = H$ requires exactly time-reversal invariance of the Hamiltonian, and nothing pathological. The choice is forced once for all systems: one law of motion-reversal must cover the spin bench and the atom alike.

 **Note** [*The honesty of the finite arena*] On this book's two-state toys the spectral objection to the unitary branch is vacuous: the spectrum of $H = (\hbar\omega/2) \sigma_z$ is symmetric under reflection through zero, and unitary candidates exist (σ_x conjugates H to $-H$). The antiunitarity of time reversal is therefore not proved by the toys; it is forced by the demand that one law cover every system, the realistic ones included, and seconded by the kinematical argument below. The toys illustrate the conclusion; the universe forces it.

Kinematics independently seconds the same branch. A motion reversed keeps its positions and reverses its momenta: $\Theta X_i \Theta^{-1} = X_i$, $\Theta P_i \Theta^{-1} = -P_i$. Conjugating the canonical commutator named in the measurement chapter ((3.60), cited by name — its construction belongs to the chapter on operator methods) then yields $[X_i, -P_j] = -i\hbar \delta_{ij} I$ on the operator side and $\Theta (i\hbar \delta_{ij} I) \Theta^{-1}$ on the scalar side, which agree only if $\Theta i \Theta^{-1} = -i$. Note also the consonance with Section 4.8: Θ acts on the sign of the *parameter*, not on any coordinate of the system — there is no time operator for it to conjugate.

The antiunitary algebra of Section 4.5 now earns its keep. The square Θ^2 is a symmetry realized unitarily — the square of either Wigner branch is unitary — and, differing from the identity by nothing measurable, it acts as a phase; two cases exhaust the possibilities,

$$\Theta^2 = \pm I, \quad (4.65)$$

stated here without proof. Which case a system realizes is rotation theory: for half-integer spin, $\Theta^2 = -I$ — the time-reversal twin of the spinor sign (4.10), whose reason belongs to the angular-momentum chapter and is flagged as its debt, with this chapter's rotation algebra (4.32) on the page but its spectrum still unbuild.

Kramers degeneracy. If $\Theta^2 = -I$, every energy eigenstate of a time-reversal-invariant Hamiltonian is at least twofold degenerate. The argument is two lines from the antiunitary rule (4.16): applied with $\Theta\psi$ in the first slot,

$$\Theta^2 = -I \implies \langle \psi | \Theta \psi \rangle = \langle \Theta^2 \psi | \Theta \psi \rangle = -\langle \psi | \Theta \psi \rangle \implies \langle \psi | \Theta \psi \rangle = 0, \quad (4.66)$$

so $\Theta |\psi\rangle$ is orthogonal to $|\psi\rangle$ — not a phase times it — while $\Theta H \Theta^{-1} = H$ makes it an eigenstate with the same eigenvalue. The theorem's content is physical: no time-reversal-invariant perturbation can split a Kramers pair. Electric fields are even under motion reversal and magnetic fields odd — their sources are charges at rest and charges in motion, the classical input named — so electrostatic environments of arbitrary complexity leave the doublet intact.¹⁶

The pair is on the bench. For the spin-1/2 doublet $\{|+z\rangle, |-z\rangle\}$, time reversal exchanges the

¹⁶H. A. Kramers, 1930. The degeneracy is exact: its proof needs no knowledge of the Hamiltonian's details, only its invariance under the reversal of motion.

partners up to phase, $\Theta | +z \rangle \propto | -z \rangle$ — the precise phases are rotation theory and belong to the angular-momentum chapter. In zero field the doublet is a Kramers pair, degenerate in every time-reversal-invariant environment; the resonance bench of Section 4.4 registers the fact as an absence — at zero field there is no line to split — and a resonance line appears only when the T -odd field is switched on. The Stern–Gerlach magnet’s splitting (Figure 1.3) is consistent with the theorem only because the field is odd under motion reversal — its currents run backward when the film does — and the theory of that splitting, the Zeeman effect, is named as a preview of the angular-momentum chapter.

The advantage of the discrete symmetries is their economy: they are the spacetime group’s disconnected pieces, and parity is the simplest selection-rule engine in physics — two sectors, one sign, and a spectroscopy of allowed and forbidden lines. Time reversal is the subtlest symmetry of the chapter: the only antiunitary one, a transformation whose square, not itself, carries the deepest physics. The limitation of this section’s account is that the spin content of that square, $\Theta^2 = -I$ for half-integer spin, is declared and not derived: its proof is the representation theory of rotations, owed to the angular-momentum chapter together with the rest of the rotation spectrum.

The chapter’s construction is complete: one new axiom, two cited theorems, one derived algebra, and a column of proved ones. It remains to collect the principles, balance the ledger, and name what the evolution operator itself has left ambiguous.

4.11 The Evolution Postulate and the Symmetry Principles: Summary and Open Debts

The chapter’s content compresses into one new axiom and five symmetry principles. The book-keeping of honesty is restated with them: postulated, (D1) alone; cited without proof, Stone’s theorem and Wigner’s theorem; everything else — the Schrödinger equation, the spectral resolution of $U(t)$, the Lie bracket, the Galilei algebra and its central extension, the boost–translation phase, the conservation, degeneracy, and selection theorems, the parity rule, the Kramers argument — proved.

- (D1) Evolution.** Between interrogations the state is carried by a continuous one-parameter unitary family, $|\psi(t)\rangle = U(t) |\psi(0)\rangle$ with $U(t+s) = U(t)U(s)$ and $U(0) = I$ ((4.1), Section 4.1); the Schrödinger equation is its proved theorem-form with H the self-adjoint generator ((4.3)), and the time-ordered exponential its generalization to time-dependent laws ((4.12), Section 4.4).
- (S1) Symmetry.** A symmetry transformation is a probability-preserving ray map ((4.14)), realized by a unitary or antiunitary operator unique up to a phase — Wigner’s theorem, cited (Section 4.5).
- (S2) Continuous symmetries.** A continuous symmetry family is a Lie group: its one-parameter subgroups are exponentials of self-adjoint generators ((4.18)); the generators’ brackets are the infinitesimal shadow of the family’s composition and close with the structure constants ((4.21)); on states the group acts by ray representatives ((4.22)) (Section 4.6).
- (S3) Spacetime symmetry.** The spacetime symmetry of nonrelativistic states is the Galilean group, whose algebra, derived from the ray representatives, carries the mass as a central charge ((4.32),

(4.33)); the boost–translation phase (4.37) then forces mass superselection as the consistent reading, given the empirical absence of interference across mass sectors (Section 4.7).

(S4) Time. Time is the evolution parameter, not an observable: H ’s role is dual, and no time operator exists ((4.54), Section 4.8).

(S5) Conservation. A generator is conserved exactly when it commutes with H ((4.57)), with good quantum numbers, symmetry-forced degeneracy, and selection rules as its fruits ((4.60), Section 4.9); parity and time reversal are the discrete instances (Section 4.10).

The chapter’s integration of experiment and formalism is collected, in lieu of a new example, in Table 4.2, which adds the transformation rows to the experiment–formalism dictionary of the states and measurement chapters (Tables 2.1 and 3.1).

Table 4.2: The experiment–formalism dictionary, third installment: the transformation rows of this chapter, extending Tables 2.1 and 3.1.

Experimental notion	Formal object
Waiting between interrogations	a one-parameter unitary family
The law of waiting	the self-adjoint generator H
A symmetry	a unitary or antiunitary operator, up to a phase
A continuous symmetry	a one-parameter subgroup, exponential of a generator
The composition law of a symmetry family	the structure constants
The spacetime symmetry of nonrelativistic states	the Galilean group
Mass	the central charge of the Galilei algebra (superselected)
Time	the evolution parameter — not an observable
A conserved quantity	a generator with $[G, H] = 0$
A forbidden transition	a selection rule
Space inversion	the parity operator Π
Reversal of motion	the antiunitary time reversal Θ

The mainline duty is the ledger, and it must be stated plainly. *Repaid here:* the measurement chapter’s exit question — the law of the state between measurements — as Axiom (D1) and its theorem-form (Section 4.1); the two-families flag of the measurement chapter, as the model continuous family of relabelings and their classification (Sections 4.1, 4.5, 4.6); the overlap-matrix seed of the states chapter, as the passive case of symmetry (Section 4.5); the dynamical share of the historical chapter’s fourth requirement (Section 1.4.6) — order-sensitive composition of transformations — as the family law, the group law, and the bracket; the Stern–Gerlach debt, this chapter’s share, as spin precession and magnetic resonance (Sections 4.3 and 4.4); the Bohr-frequency time dependence of 1925 (Section 1.4.1), derived as (4.7); the historical chapter’s controlled guess of 1926, its abstract share, as the theorem (4.3); the Wigner–Weyl pointer of the historical chapter (Section 1.4.5), as the symmetry theory of Sections 4.5–4.7; and the states chapter’s superselection clause, its Galilean share (Section 2.5): the central extension derived ((4.33)), its mechanism identified in the phase freedom of ray representatives, and the superselection reading named with its empirical warrant, the absence of observed interference across mass sectors — while the two instances the states chapter named remain the angular-momentum chapter’s and field theory’s. *Planted, and declared openly:* the pictures of evolution and the operator reading of the equation of motion, position and momentum as operators

with the canonical commutator built, wave functions and propagators, to the next chapter; the rotation spectrum — ladder operators, eigenvalues and multiplets, the half-angle law, the spinor sign's interpretation, $\Theta^2 = -1$ for half-integer spin, and the angular-momentum share of superselection — to the chapter on angular momentum, whose inheritance is now precise: the algebra is derived here ((4.32)), the spectrum is owed there; the truncation of the time-ordered series, transition rates, and the dipole physics behind Laporte's rule, to the chapter on perturbation methods; the proofs of Stone and Wigner and the domains of unbounded generators, to Appendix A; group-theory systematics, to Appendix B; parity violation, outside the main line, one footnote spent in Section 4.10; the relativistic contrast, one footnote spent in Section 4.7.

One number, two readings. A fixed observable's expectation in an evolving state, $\langle \psi(t) | A | \psi(t) \rangle = \langle \psi(0) | U^\dagger(t) A U(t) | \psi(0) \rangle$, can be computed in two ways: carry the evolution on the state — this chapter's choice — or carry it on the operator, $U^\dagger(t) A U(t)$, with the state standing still. Every count of every experiment is indifferent to the choice; yet the second reading re-reads this chapter's every equation, and in it the commutator $[H, A]$, not the state, carries the motion. Which of the two is the dynamics — and must a physical law have a unique answer to that question? That is the theory of pictures, and it is the next chapter. The rotation family, whose algebra this chapter has derived among the Galilean brackets, must then be built in its own right: its spectra, its representations, and the half-angle law it has owed us since the states chapter — the chapter after next.

Chapter 4 Exercise

1. (★) (Section 4.1) Family-law drill. For $H = (\hbar\omega/2) \sigma_z$, exponentiate $U(t) = e^{-iHt/\hbar}$ directly via $\sigma_z^2 = I$, obtaining $U(t) = \cos(\omega t/2) I - i \sin(\omega t/2) \sigma_z$. Verify by explicit multiplication: $U^\dagger(t) U(t) = I$; $U(0) = I$; $U(t+s) = U(t) U(s)$; $U(-t) = U^\dagger(t)$. Verify the Schrödinger equation (4.3) by differentiating $U(t) | \psi(0) \rangle$ for an arbitrary initial state.
2. (★) (Section 4.2) Stationary phases and the Bohr beat. Let $H |E_{1,2}\rangle = E_{1,2} |E_{1,2}\rangle$ with $E_2 > E_1$, and prepare $|\psi(0)\rangle = (|E_1\rangle + |E_2\rangle)/\sqrt{2}$. (a) Write $|\psi(t)\rangle$ using (4.6). (b) Show that the survival probability is $\cos^2(\omega t/2)$ with $\hbar\omega = E_2 - E_1$. (c) Show that $\langle A \rangle(t)$ is constant for every energy-diagonal A , and oscillates at the Bohr frequency ω for $A = |E_1\rangle \langle E_2| + |E_2\rangle \langle E_1|$, confirming the expansion (4.7). (d) State in one sentence why only eigenvalue *differences* of H are observable (the ray theorem, Section 2.5).
3. (★★) (Section 4.3) Larmor precession. Take $H = (\hbar\omega/2) \sigma_z$ with the initial state $|+x\rangle$. (a) Compute $|\psi(t)\rangle$ from (4.8). (b) Compute $\langle \sigma_x \rangle(t)$, $\langle \sigma_y \rangle(t)$, $\langle \sigma_z \rangle(t)$ and verify the precession (4.9). (c) Evaluate the $+x$ -recapture probability at $t = \pi/\omega$ and at $t = 2\pi/\omega$. (d) Show that $|\psi(2\pi/\omega)\rangle = -|+x\rangle$, and state, invoking the ray theorem (Section 2.5), why the sign changes no count of any experiment — the 4π return of (4.10), whose rotational interpretation belongs to the chapter on angular momentum.
4. (★★) (Section 4.4) Time ordering exercised. (a) For $H(t) = f(t) H_0$ with a fixed H_0 , show that the time-ordered exponential (4.12) reduces to

$$U(t, 0) = \exp\left(-\frac{i}{\hbar} H_0 \int_0^t f(t') dt'\right), \quad (4.67)$$

and verify the result by differentiation against (4.11). (b) For the piecewise-constant two-state drive — $H_1 = (\hbar\omega_1/2) \sigma_x$ for $0 < t < \tau$, then $H_2 = (\hbar\omega_2/2) \sigma_z$ for $\tau < t < 2\tau$ — write $U(2\tau, 0)$ as an ordered product of the two segment evolutions. For the input $|+z\rangle$, compute the $+x$ -recapture probability $|\langle +x|U(2\tau, 0)|+z\rangle|^2$ in *both* orderings — drive 1 then drive 2, and drive 2 then drive 1 — and show that they differ:

$$\frac{1}{2} (1 + \sin \omega_1 \tau \sin \omega_2 \tau) \quad \text{versus} \quad \frac{1}{2}. \quad (4.68)$$

Ordering inside the dynamics matters. (c) Show that the $+z$ -survival probability does *not* differ between the orderings — both give $\cos^2(\omega_1 \tau/2)$ — and explain in one sentence why the σ_z segment is invisible to that particular question: *which* question one asks decides whether ordering shows.

5. (★★) (Section 4.5) The Wigner dichotomy in dimension two. Let K_0 be complex conjugation in the z -basis, $K_0(\alpha|+z\rangle + \beta|-z\rangle) = \alpha^*|+z\rangle + \beta^*|-z\rangle$. (a) Verify antilinearity, the antiunitary rule $\langle K_0\phi|K_0\psi\rangle = \langle\psi|\phi\rangle$ ((4.16)), and $K_0^2 = I$. (b) Verify probability preservation on the pair $|+z\rangle, |+x\rangle$: $|\langle K_0(+z)|K_0(+x)\rangle|^2 = |\langle +z|+x\rangle|^2$. (c) Using the spin- n basis of the states chapter ((2.38), experimental input), show that K_0 maps the ray of $|+n\rangle$ with polar angles (θ, φ) to the ray with $(\theta, -\varphi)$.

6. (★★) (Section 4.6) From the group commutator to the algebra bracket. (a) Prove

$$e^{\varepsilon A} e^{\varepsilon B} e^{-\varepsilon A} e^{-\varepsilon B} = I + \varepsilon^2 [A, B] + O(\varepsilon^3) \quad (4.69)$$

by series expansion through second order (the commutator algebra (3.58) may be cited) — the content of (4.20). (b) Interpret in one sentence: the group commutator measures the bracket of the generators. (c) Compute both sides at order ε^2 for $A = -i\sigma_x/2$, $B = -i\sigma_y/2$, and identify the structure constant of the two-state rotation family (in σ -units, from (3.61)); state in one sentence what vanishes instead for two perpendicular spatial translations.

7. (★★) (Section 4.7) The Galilean center. Take $[P_i, K_j] = +i\hbar \delta_{ij} M$ with M central ((4.33)). (a) Prove the central Baker–Campbell–Hausdorff identity

$$e^A e^B = e^B e^A e^{[A, B]} \quad \text{when } [A, B] \text{ commutes with } A \text{ and with } B, \quad (4.70)$$

by series expansion or by the differential method (either, stated). (b) Apply it to $T(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{P}/\hbar}$ and $B(\mathbf{v}) = e^{-i\mathbf{v}\cdot\mathbf{K}/\hbar}$ to derive the boost–translation phase (4.37),

$$T(\mathbf{a}) B(\mathbf{v}) = e^{-iM\mathbf{a}\cdot\mathbf{v}/\hbar} B(\mathbf{v}) T(\mathbf{a}). \quad (4.71)$$

Watch the signs: they work out *with* the stated package. (c) Explain in one paragraph why the phase is unobservable within a single mass sector (the ray theorem, Section 2.5) yet forbids coherent superpositions of different masses — Bargmann’s superselection reading of Section 4.7, connecting to the superselection clause of the states chapter (Section 2.5).

8. (★★) (Section 4.9) Conservation computed. Let $H = (\hbar\omega/2) \sigma_z$. (a) Verify $[\sigma_z, H] = 0$ and show, via the equation of motion (4.56), that $\langle\sigma_z\rangle$ is constant for *any* initial state. (b) Compute

$$\frac{d}{dt}\langle\sigma_x\rangle = -\omega\langle\sigma_y\rangle, \quad \frac{d}{dt}\langle\sigma_y\rangle = \omega\langle\sigma_x\rangle \quad (4.72)$$

from (4.56) and (3.61), and check the result against the precession of Exercise 3. (c) Exhibit one state with $d\langle\sigma_x\rangle/dt \neq 0$ — a non-commuting generator is not conserved, in numbers.

9. (★ ★ ★) (Section 4.8, with the Robertson relation (3.64)) The status of time, honestly handled. 4265
 (a) *The obstruction*: show that no 2×2 matrix T can satisfy $[T, H] = i\hbar I$ (take traces of both 4266
 sides); state in one sentence what extra input the general argument needs (a bounded-below 4267
 energy spectrum — the Pauli footnote of Section 4.8). (b) *The relation that does exist*: apply 4268
 the Robertson relation to the pair (H, A) and use the equation of motion (4.56) to derive 4269

$$\Delta E \Delta A \geq \frac{\hbar}{2} \left| \frac{d\langle A \rangle}{dt} \right|; \quad (4.73) \quad 4270$$

- define the timescale $\tau_A := \Delta A / |d\langle A \rangle/dt|$ and conclude $\Delta E \tau_A \geq \hbar/2$. Compute τ_{σ_x} for 4271
 the precessing state of Exercise 3. (c) *Honesty clause*: explain in one paragraph why (b) is 4272
not a Robertson relation for a “time operator” — t is a parameter, not an observable, and τ_A 4273
 is a timescale defined by the state and the observable jointly (the Mandelstam–Tamm reading, 4274
 footnote of Section 4.8). 4275
10. (★ ★ ★) (Section 4.10) Discrete symmetries in arithmetic. (a) *Parity*: with $\Pi = \sigma_z$ on $\mathbb{C}^2$, 4276
 classify $\sigma_x, \sigma_y, \sigma_z, I$ by operator parity ((4.61)); verify $\langle \pm z | A | \pm z \rangle = 0$ for every odd A 4277
 (the selection rule (4.63)) and exhibit a nonzero even–odd matrix element. (b) *Kramers*: given 4278
 antiunitary Θ with $\Theta^2 = -I$, prove $\langle \psi | \Theta \psi \rangle = 0$ in two lines from the antiunitary rule (4.16) 4279
 ((4.66)); then verify the abstract argument on the doublet $\{|+z\rangle, |-z\rangle\}$ with $\Theta = i\sigma_y K_0$ — 4280
 check $\Theta^2 = -I$ and $\Theta |+z\rangle \propto |-z\rangle$. (The rotation reading of $i\sigma_y K_0$ belongs to the chapter on 4281
 angular momentum: one honesty sentence.) 4282

Chapter 5 Operator Methods, Pictures and Representations



Note This chapter is deductive in character and adds no new axiom. Taking the state-space axioms of Chapter 2, the measurement axioms of Chapter 3, and the evolution postulate of Chapter 4 as established input, it derives: the three pictures of evolution and their equivalence; the position and momentum representations, with the momentum operator's form derived from the translation generator and the canonical commutator computed at last; Ehrenfest's theorem and the classical limit of narrow states; the harmonic oscillator solved by ladder operators, zero-point energy included; coherent states; the forced oscillator solved exactly in the interaction picture; and the free-particle propagator with the spreading of the wave packet. Displays built on the continuum labels $|x\rangle$ and $|p\rangle$ are formal in the licensed-shorthand sense of the states and measurement chapters ((2.44), Section 3.9); the rigor behind them is Appendix A's standing debt, flagged at first use and not argued here. Two theorems are cited without proof — Stone's theorem, already the evolution chapter's, and the Wintner–Wielandt theorem, named at its point of use — and certain convergence facts of the continuum displays are footnoted rather than argued; everything provable with the machinery at hand is proved.

The preceding chapter closed with a question it deliberately did not answer (Section 4.11):

One number, two readings. A fixed observable's expectation in an evolving state, $\langle\psi(t)|A|\psi(t)\rangle = \langle\psi(0)|U^\dagger(t)AU(t)|\psi(0)\rangle$, can be computed in two ways: carry the evolution on the state — this chapter's choice — or carry it on the operator, $U^\dagger(t)AU(t)$, with the state standing still. Every count of every experiment is indifferent to the choice; yet the second reading re-reads this chapter's every equation, and in it the commutator $[H, A]$, not the state, carries the motion. Which of the two is the dynamics — and must a physical law have a unique answer to that question? That is the theory of pictures, and it is the next chapter.

...

This chapter is that theory. Its first two sections answer the question — Sections 5.1 and 5.2 build the three pictures, Schrödinger's, Heisenberg's, and Dirac's interaction picture, and prove their equivalence — and the honest answer is the chapter's thesis: the dynamics is what every picture agrees on. Pictures are to time dependence what representations are to state content: choices of description, with the physics in the intersection.

The question is as old as the theory's two births. Heisenberg's mechanics of 1925 carried the time on its transition quantities, which oscillated at the observed Bohr frequencies by construction (Section 1.4.1); Schrödinger's wave mechanics of 1926 carried it on the state, in the historical chapter's account; and the equivalence demonstrated that same year (Section 1.4.4) taught the lesson this chapter spends: matrix and wave mechanics were two coordinate systems for one structure, and no experimental count ever preferred one. The states chapter's fifth requirement — the state is none of its representations (Section 1.4.6) — is the kinematical half of that lesson; the pictures question is its dynamical half.

A matching debt waits in the kinematics itself. The measurement chapter’s continuum preview named the position operator, the momentum operator, and the canonical commutator $[Q, P] = i\hbar I$ without building any of the three ((3.60), Section 3.9): the position and momentum labels of the diffraction bench have had no operators behind them. Sections 5.3 and 5.4 build them — X as multiplication, P from the translation generator of the evolution chapter ((4.19)), the commutator computed from the representations — and repay with them the historical chapter’s matter-wave debt (Section 1.3.2): de Broglie’s wave appears as a theorem of the representation theory, not as a premise.

With the canonical pair in hand, the book’s two oldest promissory notes fall due. Planck’s oscillator quantization of 1900, the historical chapter’s account of the black-body law, is repaid in Section 5.6, where the discreteness postulated for material resonators is derived and corrected by the zero-point energy; de Broglie’s wave packet is repaid in Section 5.9, where the free evolution of a localized state is computed exactly. Between the two repayments the machinery is exercised: the classical limit made a theorem about narrow states (Section 5.5) and about large excitation (Section 5.7), and a driven oscillator solved exactly in the interaction picture (Section 5.8). The landings remain the bench’s, by the charter of this phase of the book: vibrational spectra and their selection rule, resonant energy uptake against static-drive oscillation, and time-of-flight momentum distributions. The summary of Section 5.10 then balances the ledger and hands the operator triad — generators, commutators, ladders — to the rotation family.

5.1 One Number, Two Readings: the Schrödinger and Heisenberg Pictures

The question quoted at this chapter’s opening is answered here for the systems to which the evolution postulate applies directly: an isolated system, its Hamiltonian time-independent, its evolution the one-parameter family of (4.1). The register in which the evolution chapter worked — the state carried by the family, $|\psi(t)\rangle = U(t)|\psi(0)\rangle$, and every observable a fixed operator — is called the **Schrödinger picture**. The name honors the reading of 1926; the alternative reading is the older of the two. Heisenberg’s mechanics of 1925 carried the time on the observables, its transition quantities oscillating at the Bohr frequencies because the spectral lines demanded it (Section 1.4.1); the states-carry-time reading arrived with wave mechanics the following year. The 1926 demonstration that the two mechanics are one (Section 1.4.4) taught that two coordinate systems can house a single structure — and this section is that lesson’s dynamical face, proved now from the evolution postulate rather than inferred from data.

Form, from any fixed operator A of the Schrödinger picture, the one-parameter family

$$A_H(t) := U^\dagger(t) A U(t), \quad (5.1)$$

with the states frozen at their prepared values, $|\psi_H\rangle := |\psi(0)\rangle$. This is the **Heisenberg picture**: the evolution is carried on the observables, and the state stands still. One operator deserves immediate remark. The generator commutes with its own exponential, $[H, U(t)] = 0$ by (4.2) — the exponential is a function of H in the sense of (3.25) — so $H_H = H$: the Hamiltonian is picture-free, one operator

with the same form in both registers.

Theorem (the Heisenberg equation of motion). *Every Heisenberg-picture operator (5.1) obeys*

$$\frac{dA_H}{dt} = \frac{i}{\hbar} [H, A_H] + \left(\frac{\partial A}{\partial t} \right)_H. \quad (5.2)$$

The proof is two lines from (4.3) and its adjoint. Applied to an arbitrary prepared state, the theorem-form of the evolution postulate gives $dU/dt = -(i/\hbar) HU$ and $dU^\dagger/dt = +(i/\hbar) U^\dagger H$; differentiating the definition,

$$\frac{dA_H}{dt} = \frac{i}{\hbar} U^\dagger H A U - \frac{i}{\hbar} U^\dagger A H U = \frac{i}{\hbar} (H A_H - A_H H),$$

using $[H, U] = 0$. The last term of (5.2) is the explicit-dependence clause: an operator carrying its own time dependence, $A = A(t)$, contributes $(\partial A/\partial t)_H := U^\dagger (\partial A/\partial t) U$; the instruments of this chapter are fixed operators, and the clause will not be used — the same clause was named and dismissed at (4.56). The sign is the evolution chapter's: taking the expectation of (5.2) in the frozen state reproduces the expectation-value equation (4.56) exactly.

Theorem (picture equivalence). *For any two states and any operator,*

$$\langle \phi(t) | A | \psi(t) \rangle = \langle \phi(0) | U^\dagger(t) A U(t) | \psi(0) \rangle = \langle \phi(0) | A_H(t) | \psi(0) \rangle. \quad (5.3)$$

Every matrix element is one number with two readings; hence every transition probability by the Born rule (2.7), every expectation value (3.24), every count of every experiment is picture-independent. Sandwiched between stationary states ((4.6)), the Heisenberg operator's matrix elements read

$$\langle e_m | A_H(t) | e_n \rangle = e^{i(E_m - E_n)t/\hbar} \langle e_m | A | e_n \rangle,$$

the Bohr-frequency expansion (4.7) recovered in the operator register: Heisenberg's imposed time dependence of 1925 is now the definition's shadow, derived where it was postulated.

The equivalence is exhibited on the book's exact two-state dynamics. For the static-field spin of Section 4.3, $H = (\hbar\omega/2) \sigma_z$ with the evolution operator (4.8), the transverse components carried by the operator are

$$\sigma_{x,H}(t) = \cos \omega t \sigma_x - \sin \omega t \sigma_y, \quad \sigma_{y,H}(t) = \sin \omega t \sigma_x + \cos \omega t \sigma_y, \quad \sigma_{z,H}(t) = \sigma_z,$$

stated here and verified in Exercise 1 by two routes: direct conjugation with (4.8) and reduction by the Pauli algebra (3.61), and integration of the Heisenberg equation of motion (5.2).¹ The count check is the point: in the state prepared along $+x$,

$$\langle +x | \sigma_{x,H}(t) | +x \rangle = \cos \omega t,$$

the same cosine the evolution chapter extracted from the evolving state ((4.9)) — the same count from both bookkeepings, in arithmetic.

The equivalence theorem answers the exit question in its own terms. No experiment decides between the pictures, because the pictures differ only in where the evolution is written down; the invariant content — the matrix elements between states — is the dynamics. “Which of the two is the dynamics?” is therefore a question about the description, not about the physics, and a physical law need not have a unique answer to it: the two readings are one law. Pictures are to time dependence

¹The explicit Pauli-algebra reduction and the integration of the coupled differential equations are supplied in Exercise 1; the verification is deferred to the problem set.

what representations are to state content — the states chapter’s fifth requirement, that the state is none of its representations (Section 1.4.6), has its dynamical twin: the dynamics is none of its pictures.

The advantage of the Heisenberg picture is structural: it puts the motion where classical mechanics puts it, on the observables, so that the classical-limit theorem of Section 5.5 is natural in this register, and conserved quantities are literally time-independent operators — the conservation theorem (4.57) re-read: $[G, H] = 0$ if and only if G_H is constant. Its limitation is practical. For a time-dependent law $H(t)$ the operator equation is rarely simpler than the state’s, for the conjugated Hamiltonian $U^\dagger H(t) U$ is no longer $H(t)$ itself, and the equation must be built from the time-ordered family of Section 4.4. The picture that earns its keep on driven problems is the next section’s.

The split of the motion between state and operator has been all-or-nothing here: one register carries it wholly on the state, the other wholly on the observables. A driven problem wants a partial split — the solved part of the law carried one way, the unsolved drive the other. That partial split is Dirac’s picture, and it is the working register of the rest of physics.

5.2 Splitting the Motion: the Interaction Picture

A driven system splits its law into a solved part and a remainder: $H(t) = H_0 + V(t)$, with H_0 time-independent and its evolution family $U_0(t) := e^{-iH_0 t/\hbar}$ known. The reference time is $t = 0$ throughout; at a general reference time t_0 the same definitions stand with $U_0(t, t_0) := e^{-iH_0(t-t_0)/\hbar}$. The **interaction picture** — Dirac’s, from the transformation theory of 1926² — carries the solved part of the motion on the operators and leaves only the remainder to the state:

$$|\psi_I(t)\rangle := U_0^\dagger(t) |\psi(t)\rangle, \quad A_I(t) := U_0^\dagger(t) A U_0(t). \quad (5.4)$$

The split interpolates between the two pictures of Section 5.1, and both reductions check: with $V = 0$ the state stands still and the operators carry all of H_0 ’s motion — the Heisenberg register of the solved system; with $H_0 = 0$ the operators stand still and the state carries everything — the Schrödinger register.

Theorem (the interaction equation of motion). *The interaction state obeys*

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle, \quad V_I(t) := U_0^\dagger(t) V(t) U_0(t). \quad (5.5)$$

The proof is two lines from the generalized evolution law (4.11). Since U_0 is a function of H_0 ((3.25)), one has $dU_0^\dagger/dt = (i/\hbar) U_0^\dagger H_0$; differentiating the definition,

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = -U_0^\dagger H_0 |\psi(t)\rangle + U_0^\dagger (H_0 + V(t)) |\psi(t)\rangle = U_0^\dagger V(t) U_0(t) U_0^\dagger(t) |\psi(t)\rangle,$$

which is (5.5). The drive, and only the drive, moves the interaction state: switch V off and $|\psi_I\rangle$ stands still.

The interaction state is carried between interrogations by a two-parameter family $U_I(t, t_0)$, which obeys the evolution chapter’s law (4.11) with generator $V_I(t)$; its solution is the time-ordered exponential of V_I — the Dyson series (4.12) re-read in the interaction register — and the full evolution factors as $U(t, t_0) = U_0(t, t_0) U_I(t, t_0)$. Every count is likewise interaction-picture-independent:

²P. A. M. Dirac, *Proc. R. Soc. Lond. A* **112**, 661 (1926).

$\langle \phi(t) | A | \psi(t) \rangle = \langle \phi_I(t) | A_I(t) | \psi_I(t) \rangle$ by the unitarity of $U_0(t)$, inserting $U_0 U_0^\dagger = I$ twice. With Section 5.1 this completes the equivalence of the three pictures: three bookkeepings of one invariant content.

One pointer, contractual, and no more. Truncated after its first nontrivial term, the interaction-picture series is time-dependent perturbation theory — the later chapter’s theory, and its repayment route of the historical chapter’s debt of the Einstein coefficients (Section 1.1.6). Nothing is truncated in this chapter: Section 5.8 will sum the series exactly for a drive whose commutator is central, and the exact solution will stand as the benchmark for the later approximations.

The interaction picture also discharges a flag the evolution chapter planted. Its rotating relabeling of the resonance problem was flagged there, in a note, as the measurement chapter’s instantaneous relabeling applied continuously — “not a picture of dynamics” (Section 4.4). The tool was on the page; the concept — the bookkeeping split of the motion — was not. It is on the page now. Take the circularly driven spin of that section and choose the solved part at the *drive* frequency, $H_0 = (\hbar\omega/2) \sigma_z$. The choice is legitimate: H_0 need only be solved, and the static z -field is solved in Section 4.3. The remainder’s interaction form is

$$V_I(t) = U_0^\dagger(t) (H(t) - H_0) U_0(t) = \frac{\hbar\omega_1}{2} \sigma_x - \frac{\hbar\delta}{2} \sigma_z, \quad \delta = \omega - \omega_0,$$

static at every detuning — resonance, $\delta = 0$, merely makes it purely transverse. This is the effective generator H_{eff} of Section 4.4: the relabeled state of that section is the interaction state of this one, and its static-generator equation is (5.5) of this system. The Rabi oscillation (4.13) is thereby re-read, not redone: the interaction state rotates under a static generator at the rate $\sqrt{\omega_1^2 + \delta^2}$, fully transferred between the alternatives on resonance, partially off it — one computation, both bookkeepings, and no new solution claimed.

The pictures are three bookkeepings of one motion — abstract, representation-free, complete. But the measurement chapter left a matching kinematical debt: the position and momentum labels of its continuum preview (Section 3.9) still have no operators behind them. Building those operators is the next two sections’ work, and it begins where the evolution chapter left its own promissory note: the translation generator of (4.19).

5.3 The Position Representation: Wave Functions and the Form of Momentum

The continuum shorthand of the states and measurement chapters is licensed machinery, recalled by citation: the wave function $\psi(x) = \langle x | \psi \rangle$ ((2.43)), the δ -normalization $\langle x | x' \rangle = \delta(x - x')$ ((2.44)), the genuine vectors the square-integrable functions of $L^2(\mathbb{R})$ ((2.45)). Its operator content is one display, the continuum resolution of identity:

$$\int_{-\infty}^{+\infty} dx |x\rangle \langle x| = I. \quad (5.6)$$

This is the operator reading of the continuum composition law (2.48) and the continuum sibling of the measurement chapter’s resolution (3.9): applied to a ket it returns the wave function, and sandwiched

between $\langle x|$ and $|x'\rangle$ it returns the δ -shorthand itself.



Note *The warning of the states and measurement chapters stands verbatim a second time ((2.44) and the note attached to it there, recycled at Section 3.9): the ket $|x\rangle$ is not a vector of $\mathcal{H}$ — its formal norm diverges, and no convention repairs the defect — and the continuum displays of this chapter are a calculus whose honest justification, the rigged Hilbert space, is Appendix A's debt. Every statement made with the shorthand can be re-expressed, at the price of some care, among the genuine vectors of $L^2(\mathbb{R})$ ((2.45)).*

The interval projectors of the measurement chapter ((3.72)) priced the position statistics without supplying an operator. The operator is now built:

$$(X\psi)(x) := x\psi(x), \quad \text{that is,} \quad \langle x|X|\psi\rangle = x\langle x|\psi\rangle. \quad (5.7)$$

The position observable is multiplication by the label. Its expectation is the first moment of the position Born rule, $\langle x\rangle_\psi = \int x|\psi(x)|^2 dx$ ((3.74), itself the expectation of the interval measure (3.73)); and X is self-adjoint on its domain — multiplication by a real function — the domain being the wave functions with $x\psi$ still square-integrable. Unboundedness and domains are the preview's honesty, standing as stated in Section 3.9; their rigorous theory is Appendix A's, pointed to once here.

The evolution chapter built translations as operators, $T(a) = e^{-iaP/\hbar}$ with P the self-adjoint translation generator ((4.19), (4.18)), acting on states by translating the position label (Section 4.7) — and it left the position-representation form of that generator as this chapter's debt. The action is a shift of the argument: actively translating a preparation by a carries a wave function peaked at x_0 to one peaked at $x_0 + a$, so

$$(T(a)\psi)(x) = \psi(x - a). \quad (5.8)$$

Theorem (the position-representation form of momentum). *On wave functions in its domain, the translation generator acts as*

$$(P\psi)(x) = -i\hbar \frac{d\psi}{dx}. \quad (5.9)$$

The proof is a differentiation of the family at $a = 0$. On one side, the exponential map (4.18) gives $T(a) = I - (ia/\hbar)P + O(a^2)$, so $(T(a)\psi)(x) = \psi(x) - (ia/\hbar)(P\psi)(x) + O(a^2)$; on the other, (5.8) gives $\psi(x - a) = \psi(x) - a d\psi/dx + O(a^2)$. Matching the terms linear in a yields (5.9).

The derivation route is contractual, and it is stated plainly. The input is the Galilean generator correspondence — translations are exponentials of the momentum, earned in the evolution chapter from the translation symmetry of the bench ((4.19), Section 4.6); the output is the position-representation form of P . De Broglie's wave is nowhere an input: it is the next section's theorem, not this section's premise. Schrödinger's correspondence guess of 1926, recorded in the historical chapter's account of wave mechanics, stands as this result's anticipation — history, not derivation.

One remark on status. The operator $-i\hbar d/dx$ is the derivative's self-adjoint face on the line's arena: integration by parts on square-integrable functions leaves no boundary term, and the domain questions this one-sentence gloss hides are Appendix A's, as flagged in the note above. With the generator's form derived, the evolution chapter's census is completed in this representation: P generates translations of the position label exactly as H generates waiting, and the diffraction bench's momentum

measurement (Section 3.8) now has a built observable behind it.

The abstract parity operator of the evolution chapter ((4.61)) acts on wave functions as $(\Pi\psi)(x) = \psi(-x)$: the even and odd functions are its eigenspaces of eigenvalue ± 1 . The position operator is odd under it, $\Pi X \Pi = -X$ — multiplication by x changes sign when the argument is reflected — as is every odd power of X . Laporte's rule ((4.63)) is thereby an integral computation. For any even state,

$$\langle \psi_{\text{even}} | X | \psi_{\text{even}} \rangle = \int_{-\infty}^{+\infty} x |\psi_{\text{even}}(x)|^2 dx = 0,$$

the integrand odd on a symmetric line. The abstract selection theorem is made concrete, and its next cashing is already planted: the oscillator eigenfunctions of Section 5.6 alternate in parity.

X and P are now both operators, but the position label is only half the continuum: the plane waves $e^{ipx/\hbar}$ — momentum's candidates since the measurement chapter named them (Section 3.9) — are still unbuilt. Building them is building the momentum representation, and with it the canonical commutator's proper home.

5.4 The Momentum Representation and the Canonical Commutator, Built at Last

Momentum, built in the last section as an operator, has its formal eigenvalue equation $P |p\rangle = p |p\rangle$ — formal in the sense of the note of Section 5.3: $|p\rangle$ is no more a vector of $\mathcal{H}$ than $|x\rangle$ is. Read in the position representation, with $\langle x|p\rangle$ the eigenfunction, (5.9) makes it an elementary differential equation,

$$-i\hbar \frac{d}{dx} \langle x|p\rangle = p \langle x|p\rangle, \quad \text{whence} \quad \langle x|p\rangle = C e^{ipx/\hbar}.$$

The constant is fixed by the shorthand's own discipline. The continuum composition law (2.48) gives $\langle p|p'\rangle = |C|^2 \int e^{i(p'-p)x/\hbar} dx = 2\pi\hbar |C|^2 \delta(p-p')$, and imposing the δ -normalization $\langle p|p'\rangle = \delta(p-p')$ of (2.44), with the conventional phase, yields

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}. \quad (5.10)$$

The repayment is stated plainly. De Broglie's relation of 1924 assigned to every momentum a wave (Section 1.3.2); the measurement chapter named the plane waves the momentum-sharp candidates and built nothing further (Section 3.9). Here the wave is a theorem of the representation theory — the momentum eigenfunction, non-normalizable exactly as the measurement chapter warned, so that momentum-sharp preparations are idealizations exactly as position-sharp ones are. The hypothesis of 1924 is repaid, not assumed.

Every state now has its second face. The **momentum representative** of $|\psi\rangle$ is its amplitude against the momentum eigenkets, and (5.10) makes the transform explicit:

$$\begin{aligned} \varphi(p) &:= \langle p|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{-ipx/\hbar} \psi(x) dx, & \psi(x) &= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} e^{ipx/\hbar} \varphi(p) dp, \\ (P\varphi)(p) &= p \varphi(p), & (X\varphi)(p) &= +i\hbar \frac{d\varphi}{dp}. \end{aligned} \quad (5.11)$$

The first clause is the Fourier transform; the second is the eigenvalue equation in its own representation; the third follows by differentiating under the integral, since $x e^{-ipx/\hbar} = i\hbar (\partial/\partial p) e^{-ipx/\hbar}$ — the signs checked against the transform. The two representations are dual faces of one arena, and Parseval's identity (2.35) re-reads as the norm preservation of the transform: $\int |\psi(x)|^2 dx = \int |\varphi(p)|^2 dp$.

The machinery's first instrument is already on the bench. The single slit of the states and measurement chapters ((2.47); the estimate of Section 3.8) prepares a box wave function of width a , normalized on its interval; its momentum amplitude is the box's Fourier transform by (5.11),

$$\varphi(p) = \sqrt{\frac{a}{2\pi\hbar}} \frac{\sin(pa/2\hbar)}{pa/2\hbar} \propto \text{sinc}\left(\frac{pa}{2\hbar}\right).$$

The aperture amplitude of the states chapter is thereby recognized as a Fourier transform — the exponent of the aperture integral (2.47) is $ip_x x'/\hbar$ with $p_x = \hbar k_x/L$, the arrival coordinate reading the transverse momentum — and the screen's sinc^2 histogram behind the slit (Section 3.9) is $|\varphi(p_x)|^2$ mapped to position: the diffraction bench is the book's first momentum meter. The width product is re-estimated with the honesty clause intact: the sinc^2 distribution has no finite second moment — the box's cusps are the culprit — so no Robertson variance is quoted for it; the width-to-first-zero estimate gives $\Delta x \Delta p_x \sim a \cdot (h/a) = h$, the measurement chapter's estimate (Section 3.8) recovered, not improved upon.

One convention before the peak. The symbols Q and X name one observable: Q is the canonical position operator in abstract statements, X its position-representation face; the notation reports which register is speaking.

Theorem (the canonical commutator, built). *On the common domain of Q and P ,*

$$[Q, P] = i\hbar I. \quad (5.12)$$

The proof is one direct computation in the position representation. For any ψ in the common domain, (5.7) and (5.9) give $(QP\psi)(x) = -i\hbar x d\psi/dx$ and $(PQ\psi)(x) = -i\hbar \frac{d}{dx}(x\psi(x)) = -i\hbar \psi(x) - i\hbar x d\psi/dx$; the difference is $i\hbar \psi(x)$.

The register is stated contractually. The measurement chapter *named* the 1925 relation without building it ((3.60)), in a finite arena that could not house it; the same relation is now *computed* from the representations just built — its proper home, discharging the debt the measurement chapter declared (Section 3.9). Downstream, the relation as physics is cited from here.



Note *The pair's unboundedness comes in two registers, kept apart. Finite-dimensional, displayed and exact: in any finite dimension the trace of a commutator vanishes, so*

$$\text{tr}[Q, P] = 0 \neq i\hbar \dim \mathcal{H}, \quad (5.13)$$

and no matrices realize the canonical pair — the one-line arithmetic of the evolution chapter's no-time-operator display (4.54), recycled. This alone explains why the measurement chapter's finite arena could only name the relation. General, cited and not proved: the pair admits no realization by bounded operators on any Hilbert space — the Wintner–Wielandt theorem,³ cited here as a fact, with Appendix A pointed to at this point of use. The infinite-dimensional arena with unbounded oper-

³A. Wintner, *Phys. Rev.* **71**, 738 (1947); H. Wielandt, *Math. Ann.* **121**, 21 (1949). The rigorous statements, with the domain theory they presuppose, are Appendix A's.

ators is thereby forced, not chosen. The displayed trace argument carries the finite-dimensional claim only; the bounded claim never rests on the display.

The pair also has a bounded register, exact and free of domain caveats. With the exponentials $U(a) := e^{-iaP/\hbar}$ and $V(b) := e^{-ibQ/\hbar}$, the translation action (5.8) and multiplication by a phase give, computed in either order,

$$U(a) V(b) = e^{iab/\hbar} V(b) U(a), \quad (5.14)$$

the *Weyl relation*: exponentiating tames what the commutator leaves wild. The display is read off the representations in one line, $e^{-ib(x-a)/\hbar} = e^{iab/\hbar} e^{-ibx/\hbar}$; its rigorous content, like the pair's, is Appendix A's.

With $[Q, P]$ computed, the canonical uncertainty relation (3.69) is Robertson's corollary (3.64) applied to built operators: $\Delta x \Delta p \geq \hbar/2$, with the variance (3.26) of observables that now exist — its rigor register, unbounded operators and their domains, unchanged and still Appendix A's. Which states saturate the bound? The question is planted here and answered three times in what follows: the oscillator's ground state (Section 5.6), every coherent state (Section 5.7), and the Gaussian packet at its release (Section 5.9).

The advantage of the representation theory is that the continuum is now machinery: position and momentum are operators, their commutator is computed, de Broglie's wave is a theorem, and the Fourier transform moves every problem between its two natural descriptions — the whole kinematics of wave mechanics deduced from the symmetry input of the evolution chapter. Its limitation is honesty's price: the eigenkets are not vectors, the operators are unbounded and defined only on domains, and the rigorous justification — the rigged Hilbert space and the infinite-dimensional spectral theorem — is delegated to Appendix A, not discharged here.

With $[Q, P] = i\hbar I$ built, the abstract dynamics of the evolution chapter can be specialized to mechanical systems: $H = P^2/2m + V(Q)$, the free generator (??) plus a potential. Its equation of motion for the means is one commutator away — and it reads like Newton.

5.5 Ehrenfest's Theorem and the Classical Limit

The canonical commutator (5.12) of Section 5.4 is built, and the abstract dynamics of the evolution chapter can now be specialized to mechanics: one particle on the line, with Hamiltonian $H = P^2/2m + V(Q)$ — the Galilei-compatible free generator (??) of the evolution chapter, plus a potential. Its equation of motion for the means is one commutator away, and it reads like Newton's. This section computes it, and then prices the difference between it and Newton's equation itself: the price is the honest content of the classical limit.

The commutator rules. One identity does all the work. From (5.12) and the derivation property of the commutator ((3.58)), induction on the monomials gives $[Q, P^{n+1}] = [Q, P] P^n + P [Q, P^n] = i\hbar (n+1) P^n$ and, by the same token, $[P, Q^n] = -i\hbar n Q^{n-1}$; linearity extends the rules to polynomials, and the series of an analytic function extends them further (the series acts on the common domain of the unbounded pair — the domain questions are Appendix A's, as flagged since Section 5.3). The

result is the pair of rules

$$[Q, F(P)] = i\hbar F'(P), \quad [P, G(Q)] = -i\hbar G'(Q), \quad (5.15)$$

with the instances $[Q, P^2] = 2i\hbar P$ and $[P, Q^n] = -i\hbar n Q^{n-1}$ exhibited for use below (Exercise 4 of this chapter rehearses the induction).

Ehrenfest's theorem. The Heisenberg equation of motion (5.2) is applied to the fixed operators Q and P , so the explicit-dependence term is absent. Conjugation by $U(t)$ preserves commutators — $[A_H, B_H] = U^\dagger[A, B]U$ — and $[H, U] = 0$ (the Hamiltonian is time-independent) gives $H_H = H$; therefore $[H, Q_H] = U^\dagger[H, Q]U$ and $[H, P_H] = U^\dagger[H, P]U$. With $H = P^2/2m + V(Q)$, the commutators are

$$[H, Q] = \left[\frac{P^2}{2m} + V(Q), Q \right] = \frac{1}{2m} [P^2, Q] = \frac{1}{2m} (P[P, Q] + [P, Q]P) = -\frac{i\hbar}{m} P,$$

$$[H, P] = \left[\frac{P^2}{2m} + V(Q), P \right] = [V(Q), P] = i\hbar V'(Q),$$

using (5.15) in the last equality of each line. Inserting these into (5.2):

$$\frac{dQ_H}{dt} = \frac{i}{\hbar} U^\dagger \left(-\frac{i\hbar}{m} P \right) U = \frac{P_H}{m}, \quad \frac{dP_H}{dt} = \frac{i}{\hbar} U^\dagger (i\hbar V'(Q)) U = -V'(Q_H). \quad (5.16)$$

This is Newton's *form*, held by the operators themselves. Taking means in the frozen state — which, by picture-invariance (5.3), is the mean in the evolved state — gives the theorem's working form,⁴

$$\frac{d\langle Q \rangle}{dt} = \frac{\langle P \rangle}{m}, \quad \frac{d\langle P \rangle}{dt} = -\langle V'(Q) \rangle, \quad \text{i.e.} \quad m \frac{d^2\langle Q \rangle}{dt^2} = -\langle V'(Q) \rangle. \quad (5.17)$$

The same two equations follow from the expectation-value equation of motion of the evolution chapter ((4.56)) applied to Q and P — one arithmetic, two bookkeepings, as Section 5.1 taught.

The correction, priced. The mean does *not* in general follow Newton's equation, for $\langle V'(Q) \rangle \neq V'(\langle Q \rangle)$. The failure is computable. Expand $V'(Q)$ in a Taylor series about the mean position $\langle Q \rangle$:

$$V'(Q) = V'(\langle Q \rangle) + V''(\langle Q \rangle) (Q - \langle Q \rangle) + \frac{1}{2} V'''(\langle Q \rangle) (Q - \langle Q \rangle)^2 + \dots$$

Taking the expectation value term by term, the linear term vanishes because $\langle Q - \langle Q \rangle \rangle = 0$, and the quadratic term is $\frac{1}{2} V'''(\langle Q \rangle) (\Delta x)^2$ by the definition of the variance (3.26); higher-order terms involve higher central moments. The result is

$$\langle V'(Q) \rangle = V'(\langle Q \rangle) + \frac{1}{2} V'''(\langle Q \rangle) (\Delta x)^2 + \dots, \quad (5.18)$$

with $(\Delta x)^2 := \langle (Q - \langle Q \rangle)^2 \rangle$ the position spread. Newton's equation for the mean is therefore exact in the limit of *narrow states*: states whose width is small on the scale over which the force varies. This is the classical limit as a theorem — the historical chapter's correspondence principle repaid in arithmetic, not invoked as a plea: classical mechanics is the theory of means of narrow packets, and the quantum correction is priced per state by $V'''(\Delta x)^2$. No $\hbar \rightarrow 0$ argument enters anywhere in this chapter — the semiclassical machinery belongs to the path-integral chapter, named here once and not used. And the register is one of two: narrow states here, large excitation in Section 5.7; the two are distinct theorems and are not to be conflated.

Two forces make the point exact, and both are cashed later in this chapter. The free particle

⁴P. Ehrenfest, *Z. Phys.* **45**, 455 (1927). Ehrenfest's question was precisely the classical one: under what conditions does the mean of a wave-mechanical description follow the classical trajectory?

has $V' = 0$: the means move inertially in every state — the plant of Section 5.9. The harmonic oscillator, $V = \frac{1}{2}m\omega^2 Q^2$, has V' linear: then $\langle V'(Q) \rangle = V'(\langle Q \rangle)$ exactly, in any state however broad, and the mean obeys $m d^2 \langle Q \rangle / dt^2 = -m\omega^2 \langle Q \rangle$ — every mean, however wild the state, executes the classical oscillation (the riding means of Sections 5.6 and 5.7 are promised here). The contrast with an anharmonic force isolates what the classical limit needs. For the quartic potential $V = \lambda Q^4$, the force is cubic, $V''' = 24\lambda Q$, and (5.18) reads

$$\langle V'(Q) \rangle = 4\lambda \langle Q \rangle^3 + 12\lambda \langle Q \rangle (\Delta x)^2 + \cdots, \quad (5.19)$$

the correction $12\lambda \langle Q \rangle (\Delta x)^2 = \frac{1}{2} V'''(\langle Q \rangle) (\Delta x)^2$ being an anharmonicity-driven departure of the mean from the naive classical trajectory, priced per state (the Taylor series of a cubic closes: the omitted remainder is the third central moment, vanishing for symmetric packets; Exercise 4 of this chapter recomputes both instances). Linearity of the force, or narrowness of the state: the classical description of a mean requires one of the two, and (5.18) says exactly how much of either.

The oscillator has now appeared twice: as the system whose means are exactly classical, and — since the historical chapter's black-body account — as the theory's oldest debt. Both call for its exact solution, and the canonical commutator just built is the whole input. The solution is algebraic, and it is next.

5.6 The Harmonic Oscillator by Ladder Operators

The quadratic potential now receives its exact solution, and the theory repays its oldest debt in the same stroke. The system is the one-dimensional oscillator,

$$H = \frac{P^2}{2m} + \frac{1}{2} m\omega^2 Q^2, \quad (5.20)$$

and the whole solution — spectrum, eigenstates, selection rule — follows from the canonical commutator (5.12) alone. Classically, the Hamiltonian factorizes: $p^2/2m + m\omega^2 q^2/2 = (m\omega/2)(q - ip/m\omega)(q + ip/m\omega)$. Quantum mechanically the two factors fail to commute, and the failure is exactly the physics: it will prove to be the zero-point energy.

The ladder operators. Form the dimensionless pair

$$\begin{aligned} a &:= \sqrt{\frac{m\omega}{2\hbar}} \left(Q + \frac{iP}{m\omega} \right), & a^\dagger &:= \sqrt{\frac{m\omega}{2\hbar}} \left(Q - \frac{iP}{m\omega} \right), \\ Q &= \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger), & P &= -i\sqrt{\frac{m\omega\hbar}{2}} (a - a^\dagger), \end{aligned} \quad (5.21)$$

the last two inverting the first two. Neither is self-adjoint: a is not an observable, and nothing spectral will be claimed for it. Their commutator follows from the definitions and (5.12):

$$[a, a^\dagger] = \frac{m\omega}{2\hbar} \left[Q + \frac{iP}{m\omega}, Q - \frac{iP}{m\omega} \right] = \frac{m\omega}{2\hbar} \left(-\frac{i}{m\omega} [Q, P] + \frac{i}{m\omega} [P, Q] \right) = \frac{m\omega}{2\hbar} \cdot \frac{2\hbar}{m\omega} = I.$$

Substituting the inverse forms of (5.21) into $H = P^2/2m + \frac{1}{2}m\omega^2 Q^2$,

$$\begin{aligned} \frac{P^2}{2m} &= -\frac{\hbar\omega}{4} (a - a^\dagger)^2 = -\frac{\hbar\omega}{4} (a^2 - aa^\dagger - a^\dagger a + (a^\dagger)^2), \\ \frac{1}{2}m\omega^2 Q^2 &= \frac{\hbar\omega}{4} (a + a^\dagger)^2 = \frac{\hbar\omega}{4} (a^2 + aa^\dagger + a^\dagger a + (a^\dagger)^2), \end{aligned}$$

and adding the two lines, the a^2 and $(a^\dagger)^2$ terms cancel while $aa^\dagger + a^\dagger a = 2a^\dagger a + I$ from the commutator just proved; therefore

$$[a, a^\dagger] = I, \quad N := a^\dagger a, \quad H = \hbar\omega \left(N + \frac{1}{2} \right), \quad (5.22)$$

the $1/2$ being the price of noncommutation, computed, not postulated.

Theorem (the oscillator spectrum). *The eigenvalues of N are $n = 0, 1, 2, \dots$, and the energies are $E_n = \hbar\omega (n + \frac{1}{2})$. The proof is the chapter's peak of economy. From $[a, a^\dagger] = I$ and the derivation property of the commutator ((3.58)),*

$$[N, a] = [a^\dagger a, a] = a^\dagger [a, a] + [a^\dagger, a] a = 0 + (-I)a = -a,$$

$$[N, a^\dagger] = [a^\dagger a, a^\dagger] = a^\dagger [a, a^\dagger] + [a^\dagger, a^\dagger] a = a^\dagger I + 0 = +a^\dagger.$$

These are the number-ladder commutators,

$$[N, a] = -a, \quad [N, a^\dagger] = +a^\dagger. \quad (5.23)$$

First, N is positive: $\langle \psi | N | \psi \rangle = \| a | \psi \rangle \|^2 \geq 0$ for every state, so no eigenvalue is negative. Second, if $N | \nu \rangle = \nu | \nu \rangle$ on a normalized state, then $a | \nu \rangle$ — when nonzero — is an eigenstate with eigenvalue $\nu - 1$, and its squared norm is $\langle \nu | a^\dagger a | \nu \rangle = \nu$. Descent by repeated application of a would eventually produce a negative eigenvalue, contradicting positivity, unless it terminates; termination forces ν to be a nonnegative integer n and forces the existence of a state $|0\rangle$ with $a |0\rangle = 0$. Ascent by $a^\dagger$ is unobstructed. Hence

$$N | n \rangle = n | n \rangle, \quad n = 0, 1, 2, \dots; \quad E_n = \hbar\omega \left(n + \frac{1}{2} \right). \quad (5.24)$$

The Planck repayment. The discreteness that Planck postulated in 1900 for the material resonators of his black-body theory — the historical chapter's account — is here *derived*, and corrected by the zero-point energy $\hbar\omega/2$ he did not have; Einstein's 1907 quantum solid, the historical chapter's specific-heat formula, is the physical anchor that makes the repayment physics rather than form: those resonators and that solid are exactly this section's material systems, and the material share of the Planck debt is repaid in full. What this algebra does not touch is the radiation field: the field-mode, light-quantum share of the debt — Einstein's photon of 1905 — belongs to the field-quantization chapter, and no sentence here claims it. Nor does the zero-point shift disturb Planck's spectrum: only level differences enter the exchange frequencies ((4.7)), and a uniform shift of every level is invisible to them.

The ladder actions. With phases fixed by the standard convention, the norm computation above gives

$$a | n \rangle = \sqrt{n} | n - 1 \rangle, \quad a^\dagger | n \rangle = \sqrt{n + 1} | n + 1 \rangle, \quad (5.25)$$

from $\langle n | aa^\dagger | n \rangle = n + 1$ and its adjoint; every eigenstate is built from the ground state, $|n\rangle = (a^\dagger)^n / \sqrt{n!} |0\rangle$ (Exercise 5 of this chapter constructs the first few). The operator N counts excitations of one mode; the vocabulary of creation and annihilation is earned by these actions — its many-mode, particle-number reading is the field-quantization chapter's, planted here in one sentence and used no further.

The ground state: the uncertainty floor. The condition $a |0\rangle = 0$, read in the position representation of Section 5.3 with P as the derivative (5.9), is an ordinary differential equation solved

at sight (the licensed shorthand of that section; the Gaussian it yields is an honest square-integrable function):

$$\left(x + \frac{\hbar}{m\omega} \frac{d}{dx}\right) \psi_0(x) = 0, \quad \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}. \quad (5.26)$$

The uncertainty product in $|0\rangle$ follows from the ladder algebra without any integration. From $Q = \sqrt{\hbar/2m\omega} (a + a^\dagger)$ and $P = -i\sqrt{m\omega\hbar/2} (a - a^\dagger)$, one computes

$$\langle 0|Q|0\rangle = 0, \quad \langle 0|Q^2|0\rangle = \frac{\hbar}{2m\omega} \langle 0|(a + a^\dagger)^2|0\rangle = \frac{\hbar}{2m\omega} \langle 0|aa^\dagger|0\rangle = \frac{\hbar}{2m\omega} \langle 0|(N + I)|0\rangle = \frac{\hbar}{2m\omega},$$

$$\langle 0|P|0\rangle = 0, \quad \langle 0|P^2|0\rangle = \frac{m\omega\hbar}{2} \langle 0|(a - a^\dagger)^2|0\rangle = \frac{m\omega\hbar}{2} \langle 0|aa^\dagger|0\rangle = \frac{m\omega\hbar}{2},$$

since $a|0\rangle = 0$ removes every term in which a acts first. Hence $(\Delta Q)^2 = \hbar/2m\omega$, $(\Delta P)^2 = m\omega\hbar/2$, and

$$\Delta Q \Delta P = \frac{\hbar}{2} \text{ in } |0\rangle,$$

normalized, unique up to a phase. The statistics of the excited states are as direct (Exercise 6 of this chapter):

$$(\Delta Q)^2 = \frac{\hbar}{m\omega} \left(n + \frac{1}{2}\right), \quad (\Delta P)^2 = m\omega\hbar \left(n + \frac{1}{2}\right), \quad \Delta Q \Delta P = \left(n + \frac{1}{2}\right)\hbar \geq \frac{\hbar}{2}, \quad (5.27)$$

with equality only for $n = 0$: the ground state saturates the canonical uncertainty relation (3.69) — the measurement chapter's saturation question, which states reach the floor, answered a first time.⁵ The zero-point energy is that floor read energetically: confinement forbids rest, and even at zero temperature the oscillator holds $\langle H \rangle = \hbar\omega/2$, half kinetic, half potential.

The eigenfunctions. Applying $(a^\dagger)^n$ to the Gaussian generates the excited wave functions: Hermite polynomials in $\sqrt{m\omega/\hbar}x$, times the same Gaussian.⁶ Their defining symmetry is read off the algebra: Q and P are both odd under the parity operator of the evolution chapter ((4.61)), so $a^\dagger$ is odd, the Gaussian ground state is even, and the eigenfunctions alternate, $\Pi\psi_n = (-1)^n \psi_n$ — the position-representation parity calculus of Section 5.3's example, cashed.

The dipole selection rule. The position operator connects only neighboring levels. Reading Q off (5.21) and applying (5.25),

$$\langle n'|Q|n\rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(\sqrt{n+1} \delta_{n',n+1} + \sqrt{n} \delta_{n',n-1} \right), \quad (5.28)$$

so vibrational absorption and emission obey $\Delta n = \pm 1$ — an instance of the evolution chapter's selection-rule engine ((4.60)), and consistent with its Laporte rule ((4.63)): Q is odd, and the allowed neighbors carry opposite parity, exactly as the alternation demands. The rates of the transitions the rule allows belong to the chapter on perturbation methods; the rule itself is exact here. One consequence is registered for the next section: $\langle n|Q|n\rangle = 0$ — the number states have no mean motion at all.

The rule is measured, routinely, in molecular spectroscopy. A diatomic molecule's relative coordinate near its equilibrium separation is an oscillator — the harmonic register of a Morse-type potential, named and not developed — and the model predicts an equally spaced ladder with the $\Delta n = \pm 1$

⁵E. H. Kennard, *Z. Phys.* **44**, 326 (1927): the Gaussian ground state was the first state shown to attain the bound.

⁶Hermite's polynomials H_n , defined by the generating function $e^{2\xi s - s^2} = \sum_n H_n(\xi) s^n/n!$; no properties beyond parity are needed in this chapter.

rule: one dominant vibrational line per neighboring pair, at the ladder spacing ω . The hydrogen chloride fundamental band near $3.46 \mu\text{m}$ is the standard instance, infrared-active exactly because the molecular dipole rides on Q .⁷ On isotopic substitution — H^{35}Cl against H^{37}Cl — the line splits into an isotopic doublet, spaced by the reduced-mass scaling $\omega \propto \mu^{-1/2}$: the ladder spacing measured as a frequency, with the zero-point motion registered as the band's lower edge, no transition beginning below E_0 .

The advantage of the ladder solution is its completeness from one commutator: spectrum, eigenstates, and selection rule fall out algebraically, with no differential equation solved — the wave-mechanical twin of this solution, Schrödinger's 1926 oscillator of the historical chapter's account, is the same physics in the position representation. Its limitation is its specialty: the algebra is keyed to the quadratic potential. The quartic oscillator of Section 5.5 admits no ladders, and its treatment waits on the approximation methods of a later chapter.

The number states are the oscillator's stationary bookkeepings: sharp in energy, and wildly non-classical in their means, since $\langle Q \rangle = 0$ at all times in every $|n\rangle$. Yet Section 5.5 proved that for this force the mean obeys Newton's equation exactly — so states must exist whose means ride the classical orbit. They do: they are the eigenstates of a itself, and the ladder machinery just built constructs them in one line.

5.7 Coherent States: the Oscillator's Classical Limit

The number states answered the spectral question, and their means stand still: $\langle n|Q|n\rangle = 0$ at all times. The classical question is sharper: which states of the oscillator move the way a classical oscillator moves? Section 5.5 proved that for the quadratic force the mean of *any* state obeys Newton's equation exactly; the states that fill out the classical register are the eigenstates of the annihilation operator itself, and the ladder algebra of Section 5.6 constructs them in one line. The register is stated as the construction proceeds: a is not self-adjoint, its eigenstates form no spectral basis, and what the family is instead — an overcomplete set of genuine, normalizable states — is part of the answer.

The displacement operator. For any complex number α , form

$$D(\alpha) := e^{\alpha a^\dagger - \alpha^* a}, \quad D^\dagger(\alpha) a D(\alpha) = a + \alpha I. \quad (5.29)$$

The operator is unitary, its exponent being anti-self-adjoint (the adjoint arithmetic of (3.59)). The shift law is one line: the commutator $[a, \alpha a^\dagger - \alpha^* a] = \alpha I$ is a c -number, so the Hadamard series for the conjugation closes at its first term — the central-bracket arithmetic of Exercise 7 of the evolution chapter (Section 4.7).

The coherent state. Define $|\alpha\rangle := D(\alpha)|0\rangle$: the ground state, displaced. It is an eigenstate of a , since $a|\alpha\rangle = D(\alpha)(D^\dagger(\alpha)aD(\alpha))|0\rangle = D(\alpha)(a + \alpha)|0\rangle = \alpha|\alpha\rangle$ by $a|0\rangle = 0$. Its content in number states follows from disentangling the displacement operator. Set $A := \alpha a^\dagger$ and

⁷The vibrational spectroscopy of diatomic molecules is among the most quantitative fields of experimental physics; the single dominant band and its isotopic splitting are the harmonic model's first two entries, the anharmonic corrections being the Morse register's business.

$B := -\alpha^* a$; then $[A, B] = -|\alpha|^2 [a^\dagger, a] = |\alpha|^2 I$, which commutes with both A and B . The central Baker–Campbell–Hausdorff identity of Exercise 7 of the evolution chapter (Section 4.7) — when $[A, [A, B]] = [B, [A, B]] = 0$, one has $e^{A+B} = e^{-[A,B]/2} e^A e^B$ — gives

$$D(\alpha) = e^{\alpha a^\dagger - \alpha^* a} = e^{-|\alpha|^2/2} e^{\alpha a^\dagger} e^{-\alpha^* a}.$$

Now $e^{-\alpha^* a} |0\rangle = |0\rangle$ because every term beyond the first in its series contains $a |0\rangle = 0$. Expanding the remaining exponential and using $|n\rangle = (a^\dagger)^n / \sqrt{n!} |0\rangle$ of (5.25):

$$|\alpha\rangle := D(\alpha) |0\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \quad a |\alpha\rangle = \alpha |\alpha\rangle, \quad P(n) = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}. \quad (5.30)$$

The number content is a Poisson distribution: $\langle N \rangle = |\alpha|^2$ with spread $\Delta N = |\alpha|$, so the relative number fluctuation is $1/|\alpha|$. Two honesty clauses. First, the series converges in norm, $\sum_n |\alpha|^{2n}/n! = e^{|\alpha|^2} < \infty$: coherent states are genuine vectors of the Hilbert space, not shorthand objects like $|x\rangle$ and $|p\rangle$. Second, the eigenstate property carries no spectral-theorem reading: a is not an observable, and the family is not an eigenbasis but something richer, as the end of the section shows.

Minimum uncertainty at every excitation. The displacement shifts means, not fluctuations. Conjugating Q and P of (5.21) by $D(\alpha)$, using the shift law $D^\dagger(\alpha) a D(\alpha) = a + \alpha I$ of (5.29) and its adjoint $D^\dagger(\alpha) a^\dagger D(\alpha) = a^\dagger + \alpha^* I$, gives

$$\begin{aligned} D^\dagger(\alpha) Q D(\alpha) &= \sqrt{\frac{\hbar}{2m\omega}} (D^\dagger a D + D^\dagger a^\dagger D) = Q + \sqrt{\frac{\hbar}{2m\omega}} (\alpha + \alpha^*) I, \\ D^\dagger(\alpha) P D(\alpha) &= -i\sqrt{\frac{m\omega\hbar}{2}} (D^\dagger a D - D^\dagger a^\dagger D) = P - i\sqrt{\frac{m\omega\hbar}{2}} (\alpha - \alpha^*) I, \end{aligned}$$

so each operator is shifted by a c -number; the variances, which subtract the mean, are unchanged from the ground state's. Hence the spreads of $|\alpha\rangle$ are the ground state's,

$$\Delta Q = \sqrt{\frac{\hbar}{2m\omega}}, \quad \Delta P = \sqrt{\frac{m\omega\hbar}{2}}, \quad \Delta Q \Delta P = \frac{\hbar}{2} \quad \text{for every } \alpha, \quad (5.31)$$

— saturation of the canonical relation (3.69) at every excitation, the measurement chapter's saturation question answered a second time (the computation is Exercise 7 of this chapter). The ground state of Section 5.6 is the member $\alpha = 0$.

The classical orbit. Under the oscillator's evolution each $|n\rangle$ gains its stationary phase ((4.6)), with E_n of (5.24), and the family closes on itself:

$$|\alpha(t)\rangle = e^{-i\omega t/2} |\alpha e^{-i\omega t}\rangle, \quad \langle Q \rangle(t) = \sqrt{\frac{2\hbar}{m\omega}} |\alpha| \cos(\omega t - \arg \alpha). \quad (5.32)$$

The prefactor phase is the zero-point phase, common to every state of the family and unobservable by the ray theorem (Section 2.5); what remains is a coherent state whose amplitude rotates uniformly in the complex plane. The means ride the classical ellipse — $\langle P \rangle(t) = m d\langle Q \rangle/dt$, as (5.17) demands — and Section 5.5's promise is cashed in full: the exact Ehrenfest motion of the quadratic potential, embodied by one family of states. And the fluctuations do not grow with the excitation: ΔQ is fixed while the orbit's amplitude scales as $|\alpha|$, so the relative noise $\Delta Q / \langle Q \rangle_{\max}$ falls as $1/|\alpha|$. At large $|\alpha|$ the state's mean is indistinguishable from a classical oscillator's motion, with quantum noise one part in $|\alpha|$: *the classical limit as large excitation*, a theorem — the chapter's second classical register, distinct from Section 5.5's narrow states.

Overcompleteness. Distinct coherent states are never orthogonal. From the expansion (5.30),

$$\langle \alpha | \beta \rangle = e^{-(|\alpha|^2 + |\beta|^2)/2 + \alpha^* \beta}, \quad |\langle \alpha | \beta \rangle|^2 = e^{-|\alpha - \beta|^2}, \quad (5.33)$$

an overlap that never vanishes, however far apart α and β move. The family is overcomplete: it resolves the identity,

$$\frac{1}{\pi} \int d^2 \alpha |\alpha\rangle \langle \alpha| = I, \quad (5.34)$$

stated here; the verification on the number basis is the starred part of Exercise 7 of this chapter — a basis richer than a basis, and the section's answer to what replaces a spectral basis for a non-self-adjoint operator.

The physical reading is stated with its boundary. These states are the oscillator's classical register embodied, not invoked: a means-carrying, minimum-noise family, manufactured from the ground state by displacement. Laser light is the celebrated approximate realization — the construction is Schrödinger's, the field reading Glauber's⁸ — but the modes of the radiation field are the field-quantization chapter's system, not this one's: the register here is one mechanical oscillator, and no field claim is made.

Coherent states answer which states behave classically. A sharper question follows: how does one *make* one? Drive the oscillator. A prescribed classical current couples to $a + a^\dagger$, its interaction-picture commutator is central — and the exact evolution is precisely a displacement operator. The construction of a coherent state from the vacuum by a classical drive is the interaction picture's first exact triumph, and it is next (Section 5.8).

5.8 The Forced Oscillator: an Exact Solution in the Interaction Picture

Nature's own answer to the question Section 5.7 has just posed — how to *make* a coherent state — is the oldest trick of spectroscopy: drive the oscillator. Couple it to a prescribed external force, let the force act for a while, and ask what state the oscillator is left in. The interaction picture of Section 5.2 was built for exactly this problem, and here it collects its one contracted payment of the chapter: an exact solution, with nothing truncated and nothing left over.

The model. Take the oscillator of Section 5.6 and add a linear coupling to a prescribed classical drive $f(t)$, a real function with the units of a frequency:

$$H(t) = \hbar\omega \left(N + \frac{1}{2} \right) - \hbar f(t) (a + a^\dagger) =: H_0 + V(t).$$

The split is the interaction picture's own: $H_0 = \hbar\omega(N + \frac{1}{2})$ is the solved part, with its spectrum (5.24), and $V(t) = -\hbar f(t)(a + a^\dagger)$ is the drive — proportional to the position operator by (5.21), a force in the everyday sense, pushing the oscillator's coordinate. The honesty clause of the driven spin stands here unchanged.

⁸E. Schrödinger, *Naturwissenschaften* **14**, 664 (1926): the non-spreading oscillator packets; R. J. Glauber, *Phys. Rev.* **131**, 2766 (1963): the coherent states of the radiation field.

Note The drive $f(t)$ is prescribed, not dynamical: its own equation of motion, and the oscillator's back-action on its source, are not modeled. This is the same idealization the evolution chapter made for its driven two-state system (Section 4.4); the theory of the drive's own quanta belongs to the field chapter.

The interaction register. In the interaction picture of (5.4) the operators carry the free motion and the state carries only the drive's. The annihilation operator's free motion is one line of arithmetic from $[N, a] = -a$: differentiating $a_I(t) = U_0^\dagger(t) a U_0(t)$ gives $da_I/dt = (i/\hbar) U_0^\dagger [H_0, a] U_0 = -i\omega a_I$, hence $a_I(t) = a e^{-i\omega t}$, and the drive reads

$$V_I(t) = -\hbar f(t) (a e^{-i\omega t} + a^\dagger e^{i\omega t}). \quad (5.35)$$

The interaction state $|\psi_I(t)\rangle$ obeys the interaction equation of motion (5.5) with this generator, and its evolution is the time-ordered exponential of V_I — the evolution chapter's Dyson series (4.12), which this section now sums exactly.

The exact solution. The drive's commutator at two times is a c -number. From (5.35) and $[a, a^\dagger] = I$ of (5.22),

$$[V_I(t), V_I(t')] = \hbar^2 f(t) f(t') (e^{-i\omega(t-t')} - e^{+i\omega(t-t')}) I = -2i\hbar^2 f(t) f(t') \sin[\omega(t-t')] I, \quad (5.36)$$

central: it commutes with every operator of the problem. Centrality is the whole mechanism of what follows.

Note The Magnus expansion — the representation of the time-ordered exponential as $U_I(t, 0) = e^{\Omega(t)}$ with $\Omega(t) = \sum_{k=1}^{\infty} \Omega_k(t)$, where each Ω_k is a k -fold nested commutator of V_I at different times — is **named, not built** in this chapter; its explicit form is a theorem of the theory of linear differential equations.⁹ Here only its first two terms are required and their forms are stated.

The first term is the integrated generator,

$$\Omega_1(t) = -\frac{i}{\hbar} \int_0^t V_I(t') dt' = i \int_0^t f(t') (a e^{-i\omega t'} + a^\dagger e^{i\omega t'}) dt' = a \cdot i \int_0^t f(t') e^{-i\omega t'} dt' + a^\dagger \cdot i \int_0^t f(t') e^{i\omega t'} dt'.$$

With $\beta(t) := i \int_0^t f(t') e^{i\omega t'} dt'$ one has $\beta(t)^* = -i \int_0^t f(t') e^{-i\omega t'} dt'$, so the first integral is $-\beta(t)^*$ and the second is $\beta(t)$; therefore

$$\Omega_1(t) = \beta(t) a^\dagger - \beta(t)^* a, \quad \beta(t) = i \int_0^t f(t') e^{i\omega t'} dt'.$$

The second term contains the two-time commutator,

$$\Omega_2(t) = \frac{1}{2} \left(-\frac{i}{\hbar} \right)^2 \int_0^t dt_1 \int_0^{t_1} dt_2 [V_I(t_1), V_I(t_2)] = i \Phi(t) I,$$

with $\Phi(t)$ a real phase — the double integral of the drive's two-time product against $\sin[\omega(t_1 - t_2)]$, its explicit form Exercise 8's. The exponent Ω_1 is precisely the exponent of the displacement operator (5.29), and Ω_2 is a phase times the identity. Summing the two terms,

$$U_I(t, 0) = e^{i\Phi(t)} D(\beta(t)), \quad \beta(t) = i \int_0^t f(t') e^{i\omega t'} dt'. \quad (5.37)$$

This is the content of the section: the time-ordered series (4.12) summed exactly — the Magnus

⁹W. Magnus, *Commun. Pure Appl. Math.* **7**, 649 (1954). The expansion's first two terms, which are the only terms needed here, are displayed below; the proof that the series closes at $k = 2$ when $[V_I(t_1), V_I(t_2)]$ is central is supplied in the text.

expansion closing at its second term by centrality, the central Baker–Campbell–Hausdorff arithmetic of Exercise 7 of the evolution chapter (Section 4.7) doing the work. A ground-state start ends exactly in a coherent state:

$$U_I(t, 0) |0\rangle = e^{i\Phi(t)} D(\beta(t)) |0\rangle = e^{i\Phi(t)} |\beta(t)\rangle,$$

by (5.30) — Section 5.7’s family, manufactured from the vacuum by a classical current. The phase $\Phi(t)$ is global and changes no count (the ray theorem, Section 2.5); the displacement $\beta(t)$ is the whole physical answer.

Energy uptake. The excitation the drive has purchased is the coherent state’s Poisson mean of (5.30), $\langle N \rangle(t) = |\beta(t)|^2$; in energy units the oscillator has absorbed

$$\Delta E(t) = \hbar\omega \langle N \rangle(t) = \hbar\omega |\beta(t)|^2, \quad \text{static drive: } |\beta(t)|^2 = \left(\frac{2f_0}{\omega}\right)^2 \sin^2 \frac{\omega t}{2}. \quad (5.38)$$

The classical correspondence is exact here: a classical oscillator driven by the same force law gains precisely $\hbar\omega |\beta(t)|^2$ — the quantum mean is the classical energy, and for $|\beta| \gg 1$ the coherent state embodies it in Section 5.7’s large-excitation register.

The resonance structure. The displacement $\beta(t) = i \int_0^t f(t') e^{i\omega t'} dt'$ reads one Fourier component of the drive: the component at the oscillator’s own frequency. Two drives make the contrast measurable. A *static* force, $f(t) = f_0$, gives $\beta(t) = (f_0/\omega)(e^{i\omega t} - 1)$, the bounded oscillation of (5.38): the oscillator borrows energy from a constant force and hands it back, once per period, the excitation never exceeding $(2f_0/\omega)^2$. A *resonant* drive, $f(t) = f_1 \cos \omega t$, feeds the oscillator at its own frequency:

$$\beta(t) = \frac{if_1}{2} \left(t + \frac{e^{2i\omega t} - 1}{2i\omega} \right), \quad |\beta(t)| \simeq \frac{f_1 t}{2} \quad (t \gg \omega^{-1}),$$

linear growth of the displacement, hence quadratic growth of the energy, $\langle N \rangle \simeq f_1^2 t^2 / 4$, the oscillating second term falling ever further behind the secular first. The oscillator absorbs energy only from the drive’s resonant Fourier component — the resonance criterion of every classical driven oscillator, recovered here as exact quantum arithmetic, with the uptake (5.38) as its measurable.

One honesty paragraph, and it is the boundary of the chapter. Nothing in the derivation was truncated: the time-ordered series was summed to all orders, and it summed because the drive’s commutator is central. The same series, truncated, is the time-dependent perturbation theory of a later chapter — and the exact solution (5.37) is the benchmark against which that chapter’s approximations will be checked.

The oscillator complex is complete: spectrum, classical states, and exact driven dynamics. One debt of the historical chapter remains outstanding — de Broglie’s packet itself: a free particle, localized, released, and watched as it moves. Its exact evolution needs the evolution operator read in the position representation: the propagator.

5.9 The Propagator: the Free Particle and the Spreading Packet

The debt just named is de Broglie’s wave packet itself (Section 1.3.2): a free particle, prepared localized, and released. The evolution operator of the evolution chapter is in hand; what the question

needs is that operator read in the position representation of Sections 5.3 and 5.4. Read so, evolution is an integral kernel, and the kernel is this section's object.

The propagator. Insert the continuum resolution of identity (5.6) into the evolution law $|\psi(t)\rangle = U(t) |\psi(0)\rangle$ and take the position component (2.43) of both sides:

$$\psi(x, t) = \int dx' K(x, t; x', 0) \psi(x', 0), \quad K(x, t; x', 0) := \langle x | U(t) | x' \rangle. \quad (5.39)$$

The kernel K is the transition amplitude to register the particle at x at time t , given that it was prepared at x' at time 0; the states chapter's continuum composition law (2.48) is its mother, and the position Born rule (3.73) gives its first reading: $|K(x, t; x', 0)|^2$ is the probability density of the arrival, per unit length. As $t \rightarrow 0$ the evolution becomes the identity and $K(x, t; x', 0) \rightarrow \delta(x - x')$ in the shorthand sense of (2.44). Like every display built on $|x\rangle$, the kernel lives in the licensed-shorthand register; the rigor questions are Appendix A's.

The composition property. The family law of the evolution operator, $U(t) = U(t - t') U(t')$ from (4.1), read between position labels with a resolution of identity inserted, is a composition property of the kernel:

$$K(x, t; x', 0) = \int dx'' K(x, t; x'', t') K(x'', t'; x', 0). \quad (5.40)$$

Waiting decomposes over the intermediate position: the amplitude to go from x' to x is the amplitude to go by way of x'' , summed over all intermediate positions — the composition principles of the states chapter, exercised on a continuous family of alternatives. This is an operator fact, nothing more: the one-parameter family law re-read in the position representation. It is also a seed. Decomposing the wait into ever more, ever shorter steps, with a resolution of identity inserted at every step, is the germ from which the path-integral chapter will grow its construction.¹⁰ No slicing, no Lagrangian: here the composition property is exactly what (5.40) says, and it is earned already.

The free kernel. For the free particle the generator is the evolution chapter's $H = \mathbf{P}^2/2M + E_{\text{int}}$ (??), specialized to one structureless particle of mass m on the line, $H = P^2/2m$ (the constant internal energy contributes only an overall phase, dropped by the ray theorem of Section 2.5). Momentum commutes with H , so it is conserved (4.57), and the free evolution is trivial in the momentum representation of Section 5.4: each momentum component merely accumulates its phase,

$$\varphi(p, t) = e^{-ip^2 t/2m\hbar} \varphi(p, 0).$$

The kernel follows by the Fourier machinery of (5.11) and the momentum eigenfunctions (5.10): inserting the momentum resolution into (5.39),

$$K_0(x, t; x', 0) = \frac{1}{2\pi\hbar} \int dp e^{ip(x-x')/\hbar} e^{-ip^2 t/2m\hbar} = \frac{1}{2\pi\hbar} \int dp \exp\left[-\frac{it}{2m\hbar} p^2 + \frac{i(x-x')}{\hbar} p\right]. \quad (5.41)$$

Completing the square in the exponent,

$$-\frac{it}{2m\hbar} p^2 + \frac{i(x-x')}{\hbar} p = -\frac{it}{2m\hbar} \left[p - \frac{m(x-x')}{t}\right]^2 + \frac{im(x-x')^2}{2\hbar t},$$

and shifting the integration variable $p \mapsto p + m(x - x')/t$, the Gaussian integration (with the conver-

¹⁰R. P. Feynman, *Rev. Mod. Phys.* **20**, 367 (1948). The construction itself — the short-time kernels and their composition, the classical action as the phase — belongs to that later chapter; none of it is used here.

gence factor of the footnote) yields $\sqrt{2\pi m\hbar/it}$; assembling the prefactors,

$$K_0(x, t; x', 0) = \sqrt{\frac{m}{2\pi i\hbar t}} \exp\left(\frac{im(x - x')^2}{2\hbar t}\right). \quad (5.42)$$

The Gaussian integral is Fresnel's; its conditional convergence is handled honestly with a convergence factor, removed after the integration.¹¹ With the free kernel, the evolution chapter's abstract free generator is cashed at last: (??), named in the symmetry algebra, now has its position-representation face. One sentence belongs to a later chapter: the kernel's phase is the classical action of the straight path, $S = m(x - x')^2/2t$ — named here, developed where the action principle is.

The Gaussian packet. Prepare the minimum-uncertainty packet of the uncertainty floor, a Gaussian of width σ carrying momentum p_0 :

$$\psi(x, 0) = \frac{1}{(2\pi\sigma^2)^{1/4}} e^{-x^2/4\sigma^2} e^{ip_0x/\hbar}, \quad \Delta x = \sigma, \quad \Delta p = \frac{\hbar}{2\sigma},$$

saturating (3.69) at $t = 0$ — the third and last cashing of the saturation story, after the oscillator's ground state (5.26) and the coherent states of Section 5.7. The evolved wave function is the convolution

$$\psi(x, t) = \int_{-\infty}^{+\infty} dx' \sqrt{\frac{m}{2\pi i\hbar t}} \exp\left(\frac{im(x - x')^2}{2\hbar t}\right) \frac{1}{(2\pi\sigma^2)^{1/4}} e^{-x'^2/4\sigma^2} e^{ip_0x'/\hbar}.$$

Both factors are Gaussians; completing the square in x' — the exponent is quadratic, the cross term from the kernel's phase combining with the initial packet's Gaussian — and performing the Gaussian integration gives

$$\psi(x, t) = \frac{1}{(2\pi\sigma(t)^2)^{1/4}} \exp\left(-\frac{(x - p_0t/m)^2}{4\sigma(t)^2}\right) \exp\left(\frac{i}{\hbar} \left[p_0x - \frac{p_0^2t}{2m} + \frac{\hbar^2t(x - p_0t/m)^2}{8m\sigma^2\sigma(t)^2} \right]\right),$$

whence the probability density

$$|\psi(x, t)|^2 = \frac{1}{\sqrt{2\pi\sigma(t)^2}} \exp\left(-\frac{(x - p_0t/m)^2}{2\sigma(t)^2}\right), \quad \sigma(t)^2 = \sigma^2 \left(1 + \left(\frac{\hbar t}{2m\sigma^2}\right)^2\right).$$

The center rides the classical trajectory, $\langle x \rangle(t) = p_0t/m$ — Ehrenfest's inertial law (5.17), cashed — and the width obeys the spreading law

$$\Delta x(t)^2 = \sigma^2 \left(1 + \left(\frac{\hbar t}{2m\sigma^2}\right)^2\right) = \sigma^2 + \left(\frac{\Delta p}{m}\right)^2 t^2. \quad (5.43)$$

The packet spreads *ballistically*, at the velocity uncertainty $\Delta p/m$: the preparation's momentum spread is a spread of velocities, and the position spread it purchases grows linearly in time. The dispersion's origin is the kinetic law $E = p^2/2m$ itself, which gives each momentum component its own phase rate $p^2/2m\hbar$ in $\varphi(p, t)$; de Broglie's relations, derived as representation theory in Section 5.4, are here doing dynamics.

The honest verdict on de Broglie's hope must now be stated. A free packet cannot hold together. Spreading is unavoidable, and it is worse for the better-localized preparation: $\Delta p \propto 1/\sigma$, so the narrower the packet at release, the faster it disperses. The matter-wave packet is not a particle surrogate — what holds the electron together in matter is exactly the binding this section lacks, the oscillator's restoring force of Section 5.6 and the atom's of the next chapter. One further clause of honesty: the spreading packet ceases to saturate. Momentum is conserved, so Δp stands at $\hbar/2\sigma$ forever, while

¹¹With the factor $e^{-\varepsilon p^2}$ the integral is the elementary Gaussian $\int e^{-\alpha p^2 + \gamma p} dp = \sqrt{\pi/\alpha} e^{\gamma^2/4\alpha}$ at $\alpha = \varepsilon + it/2m\hbar$; the limit $\varepsilon \rightarrow 0^+$ exists and gives (5.42). The distributional footing of such oscillatory integrals is Appendix A's.

Δx grows; by (5.43) the product $\Delta x \Delta p$ exceeds $\hbar/2$ for every $t > 0$, the difference priced by a position–momentum correlation the free evolution manufactures (Exercise 9).

The far field: time of flight. At late times the kernel’s phase varies slowly across the initial packet’s support, $e^{imx'^2/2\hbar t} \simeq 1$, and the convolution of (5.39) collapses to a Fourier transform:

$$\psi(x, t) \simeq \sqrt{\frac{m}{it}} e^{imx^2/2\hbar t} \varphi\left(\frac{mx}{t}\right) \quad (t \text{ large}). \quad (5.44)$$

The long-time shape of the packet is a magnified image of its momentum distribution, $|\psi(x, t)|^2 \simeq (m/t) |\varphi(mx/t)|^2$: position at late times is a momentum measurement, with the calibration $p = mx/t$. This is the working principle of time-of-flight spectroscopy: release the preparation — a cold-atom cloud, a molecular beam — let it fly, and read the arrival histogram at a distant detector as $|\varphi(p)|^2$. It is the momentum meter’s second instrument, after the diffraction bench of Section 5.4: the bench maps momentum to angle at a screen, time of flight maps it to arrival position; both read the same $|\varphi(p)|^2$, and the far-field form (5.44) is the Fraunhofer kinship of the states chapter’s aperture limit (2.47) — the far field shows the Fourier transform.

The advantage of the propagator account is its completeness: the free evolution is fully explicit, kernel and packet in closed form, and de Broglie’s picture of 1924 is made exact — the wave repaid in Section 5.4, the packet repaid here, spreading and all. Its limitation is the method’s reach. The computation worked because momentum space made the free evolution trivial and Fourier analysis carried the result back; a potential re-entangles the two representations, and no such closed form exists for K in general. The systematic treatment of that entanglement — building the kernel for short times and composing by (5.40) — is the path-integral chapter’s construction, named here as a pointer and developed there.

The debts are cashed: the pictures question answered, the representations built, Planck and de Broglie repaid, the classical limit theoremized. It remains to collect the principles, balance the ledger, and hand the operator triad — generators, commutators, ladders — to the family whose spectrum still owes us everything since the states chapter: the rotations.

5.10 Operator Methods, Pictures, and Representations: Summary and Open Debts

The chapter’s work is done, and its content compresses into five principles. The bookkeeping of honesty is restated with them: postulated, nothing — this chapter added no axiom to the theory; everything above was proved from the state-space axioms, the measurement axioms, and the evolution postulate, working inside the continuum shorthand licensed by the states and measurement chapters; cited without proof, Stone’s theorem (already the evolution chapter’s), the Wintner–Wielandt theorem named at its point of use, and the elementary convergence facts footnoted where they occur. The rigor ledger — non-normalizable kets, unbounded operators and their domains, the spectral theorem of the infinite-dimensional arena — stands where it has stood since the states chapter: to Appendix A.

(P1) Pictures. The dynamics of an isolated system admits three equivalent bookkeepings — Schrödinger,

Heisenberg, Dirac interaction — and every matrix element, hence every count of every experiment, is picture-independent ((5.3), Sections 5.1 and 5.2). The dynamics is what every picture agrees on.

(P2) Representations. The continuum's labels are operators: position is multiplication ((5.7)); momentum is $-i\hbar d/dx$, derived from the translation generator ((5.9)); their commutator is computed at last, $[Q, P] = i\hbar I$ ((5.12)), necessarily unbounded and tamed in the bounded register by the Weyl relation ((5.14)) (Sections 5.3 and 5.4).

(P3) The classical limit. Classical mechanics is recovered as two theorems, not a correspondence plea: the theory of means of narrow states, priced by the force's curvature ((5.17), (5.18)), and the theory of large excitation, its fluctuations one part in $|\alpha|$ ((5.30), (5.32)) (Sections 5.5 and 5.7).

(P4) The oscillator algebra. One commutator gives the whole physics of the quadratic potential: the ladder pair and the excitation counter, the spectrum $E_n = \hbar\omega(n + \frac{1}{2})$ with its zero-point floor ((5.22), (5.24), (5.26)), the $\Delta n = \pm 1$ vibrational rule ((5.28)), and the displacement family with its exact driven dynamics ((5.29), (5.30), (5.37)) (Sections 5.6–5.8).

(P5) The propagator. Evolution read in the position representation is a kernel, $K(x, t; x', 0) = \langle x|U(t)|x'\rangle$ ((5.39)), composing over intermediate positions ((5.40)); the free kernel is in closed form ((5.42)), and a released packet spreads ballistically ((5.43)), its late-time shape a momentum image ((5.44)) (Section 5.9).

The chapter's integration of experiment and formalism is collected, in lieu of a new example, in Table 5.1, which adds the method rows to the experiment–formalism dictionary of the states, measurement, and evolution chapters (Tables 2.1, 3.1, and 4.2).

Table 5.1: The experiment–formalism dictionary, fourth installment: the method rows of this chapter, extending Tables 2.1, 3.1, and 4.2.

Experimental notion	Formal object
The question of where the motion lives	a picture — indifferent bookkeeping
A position label	the multiplication operator
A momentum	the translation generator $-i\hbar d/dx$
A canonical pair	$[Q, P] = i\hbar I$ (unbounded, Weyl-tamed)
A quantum of vibration	$a^\dagger$ on $ 0\rangle$
A classical orbit	a coherent state $ \alpha\rangle$
A classical drive	a displacement $D(\beta)$
Waiting between two positions	the propagator kernel
A released packet	ballistic spreading

The mainline duty is the ledger, and it must be stated plainly. *Repaid here:* the evolution chapter's exit question — the pictures of evolution — answered as the three bookkeepings and their equivalence (Sections 5.1, 5.2), with the rotating relabeling of the resonance problem identified as an interaction-picture transformation; the continuum shorthand's physics — wave functions, the position operator, the momentum operator's form, and the canonical commutator's proper home (Sections 5.3, 5.4), with the saturation story of (3.69) cashed three times (Sections 5.6, 5.7, 5.9); the propagator and wave-packet

debt of the states chapter (Section 5.9); the historical chapter's Planck debt, its material-oscillator share — the discreteness the 1900 resonators postulated, derived with the zero-point correction (Section 5.6), the field share remaining the field chapter's; the de Broglie debt, the wave as theorem (Sections 5.4, 1.3.2) and the packet's exact spreading law (Section 5.9); the correspondence principle, theoremized in its two registers (Sections 5.5, 5.7); the 1926 equivalence, its dynamical face (Sections 5.1, 1.4.4); and the evolution chapter's two named-not-built promissory notes, the position-representation form of P (Section 5.3) and the abstract free Hamiltonian (Section 5.9). *Planted, and declared openly:* the ladder template for the rotation spectrum, the position representation for orbital wave functions, and the central-potential problem, to the angular-momentum chapter; the interaction picture's truncation register, the exact driven solution (5.37) as its benchmark, and the dipole matrix elements (5.28) whose rates are its business, to the chapter on perturbation methods; the composition property of the propagator (5.40), the action phase of the free kernel (5.42), and the spreading packet as the dispersion its semiclassical theory must reproduce, to the path-integral chapter; the excitation counter as particle counter, the ladder pair as the creation–annihilation prototype, and coherent states as the classical-field register, to the field chapter; and the rigor ledger — the rigged Hilbert space, δ as distribution, unbounded self-adjointness and domains — to Appendix A.

The operator triad of this chapter — a generator, its commutators, and the ladders they permit — has been exercised on the line and on one mode. The rotation family's algebra is on the page since the evolution chapter, derived among the Galilean brackets; its spectrum is not: the eigenvalues, the ladders, the multiplets, and the half-angle law owed since the states chapter all remain unbuilt. The same three tools that built the oscillator's spectrum must now build the rotation family's. That is the theory of angular momentum, and it is the next chapter.

Chapter 5 Exercise

1. (★) (Section 5.1) Pictures drill. Take $H = (\hbar\omega/2) \sigma_z$. (a) Compute $\sigma_{x,H}(t) = U^\dagger(t) \sigma_x U(t)$ by direct conjugation with (4.8), obtaining $\sigma_{x,H}(t) = \cos \omega t \sigma_x - \sin \omega t \sigma_y$, and the companion $\sigma_{y,H}(t) = \cos \omega t \sigma_y + \sin \omega t \sigma_x$. (b) Verify the Heisenberg equation of motion (5.2) on this operator, $d\sigma_{x,H}/dt = (i/\hbar)[H, \sigma_{x,H}] = -\omega \sigma_{y,H}$, using (3.61). (c) Verify picture-invariance (5.3) in arithmetic: $\langle +x | \sigma_{x,H}(t) | +x \rangle = \cos \omega t$, reproducing the state-side precession (4.9).
2. (★) (Section 5.3) Representation drill. (a) From $(T(a)\psi)(x) = \psi(x - a)$ ((5.8)) and $T(a) = e^{-iaP/\hbar}$, derive $(P\psi)(x) = -i\hbar d\psi/dx$ by differentiating the family at $a = 0$. (b) Compute $[Q, P]\psi$ for a general ψ in the common domain, recovering (5.12). (c) Verify that $\langle x|p \rangle = (2\pi\hbar)^{-1/2} e^{ipx/\hbar}$ ((5.10)) solves the formal eigenvalue equation of P with eigenvalue p , and that $\int dp \langle x|p \rangle \langle p|x' \rangle = \delta(x - x')$ — the δ -shorthand discipline of (2.44).
3. (★★) (Section 5.4) The slit, honestly. Take the box wave function $\psi(x) = 1/\sqrt{a}$ on $[-a/2, a/2]$, zero outside. (a) Compute the momentum amplitude $\varphi(p) = \sqrt{a/2\pi\hbar} \text{sinc}(pa/2\hbar)$, with $\text{sinc } u := \sin u/u$, from (5.11), and $|\varphi(p)|^2$. (b) Show that the second moment $\langle p^2 \rangle$ diverges: $|\varphi(p)|^2$ decays only as $1/p^2$ — the sinc^2 honesty clause of the measurement chapter's slit estimate (Section 3.8), whose width was always the width to the first zero. (c) Use the width-to-first-zero estimate to recover $\Delta x \Delta p \sim \hbar$. (d) State in one sentence why the divergence does

not violate the Robertson relation (3.64): the relation prices states in the common domain of the two operators, and the box's cusps are the culprit.

4. (**) (Section 5.5) Ehrenfest computed. (a) From $[Q, P] = i\hbar I$ prove $[Q, P^2] = 2i\hbar P$ and $[P, Q^n] = -i\hbar nQ^{n-1}$ by induction (the rules (5.15)). (b) Derive the two Ehrenfest equations (5.17) for $H = P^2/2m + V(Q)$ from the Heisenberg equation of motion (5.2). (c) Show that for $V = \frac{1}{2}m\omega^2 Q^2$ the mean obeys $m\langle\ddot{Q}\rangle = -m\omega^2\langle Q\rangle$ in any state. (d) For $V = \lambda Q^4$ compute the leading correction of (5.18), $\frac{1}{2}V'''(\langle Q\rangle)(\Delta x)^2 = 12\lambda\langle Q\rangle(\Delta x)^2$, and state in one sentence when the classical description of the mean fails.

5. (**) (Section 5.6) Ladder arithmetic. (a) From the definitions (5.21), using only (5.12), prove $[a, a^\dagger] = I$, $H = \hbar\omega(N + \frac{1}{2})$, $[N, a] = -a$, $[N, a^\dagger] = +a^\dagger$ ((5.22)). (b) Hence show that $a^\dagger|n\rangle$ has N -eigenvalue $n + 1$ and squared norm $n + 1$ — the ladder actions (5.25). (c) Construct $|1\rangle$ and $|2\rangle$ from $|0\rangle$. (d) Derive the ground-state Gaussian (5.26) from $a|0\rangle = 0$ read in the position representation — an ordinary differential equation in a single derivative — including its normalization.

6. (**) (Section 5.6) Oscillator statistics. With $Q = \sqrt{\hbar/2m\omega}(a + a^\dagger)$ and $P = -i\sqrt{m\omega\hbar/2}(a - a^\dagger)$ from (5.21): (a) compute $\langle Q\rangle$, $\langle P\rangle$, $(\Delta Q)^2 = (\hbar/m\omega)(n + \frac{1}{2})$, $(\Delta P)^2 = (m\omega\hbar)(n + \frac{1}{2})$ in $|n\rangle$; (b) conclude $\Delta Q \Delta P = (n + \frac{1}{2})\hbar \geq \hbar/2$ with equality only for $n = 0$ — the Robertson bound (3.64) saturated by the ground state alone; (c) compute $\langle n'|Q|n\rangle$ and verify the $\Delta n = \pm 1$ rule (5.28).

7. (****) (Section 5.7) Coherent-state BCH. (a) Prove $D^\dagger(\alpha) a D(\alpha) = a + \alpha I$ ((5.29)) by the central Baker–Campbell–Hausdorff identity of Exercise 7 of the evolution chapter (Section 4.7): $[\alpha a^\dagger - \alpha^* a, a] = -\alpha I$, central, and the series closes. (b) Hence $a|\alpha\rangle = \alpha|\alpha\rangle$. (c) Derive the number expansion of (5.30) and the Poisson law $P(n) = e^{-|\alpha|^2}|\alpha|^{2n}/n!$ with $\langle N\rangle = |\alpha|^2$. (d) Compute $\Delta Q \Delta P = \hbar/2$ for arbitrary α . (e) Show $|\langle\alpha|\beta\rangle|^2 = e^{-|\alpha-\beta|^2}$ ((5.33)) — distinct coherent states are never orthogonal. *Starred:* verify the resolution of identity $(1/\pi) \int d^2\alpha |\alpha\rangle\langle\alpha| = I$ by its matrix elements on the number basis.

8. (****) (Section 5.8) The forced oscillator, summed. Work with $V_I(t) = -\hbar f(t)(ae^{-i\omega t} + a^\dagger e^{i\omega t})$ ((5.35)). (a) Show

$$[V_I(t), V_I(t')] = -2i\hbar^2 f(t)f(t') \sin[\omega(t - t')] I,$$

a c -number. (b) Hence sum the time-ordered series (4.12) exactly — the Magnus expansion closes at its second term by centrality — obtaining $U_I(t, 0) = e^{i\Phi(t)} D(\beta(t))$ ((5.37)) with

$$\beta(t) = i \int_0^t f(t') e^{i\omega t'} dt', \quad \Phi(t) = \int_0^t dt_1 \int_0^{t_1} dt_2 f(t_1)f(t_2) \sin[\omega(t_1 - t_2)],$$

the sign of Φ following the commutator's. (c) For a static force f_0 switched on at $t = 0$, compute $\beta(t) = (f_0/\omega)(e^{i\omega t} - 1)$ and $|\beta(t)|^2 = (2f_0/\omega)^2 \sin^2(\omega t/2)$ ((5.38)), and evaluate the phase, $\Phi(t) = (f_0/\omega)^2 (\omega t - \sin \omega t)$, as a consistency check on the double integral. (d) For the resonant drive $f(t) = f_1 \cos \omega t$, show $|\beta(t)| \simeq f_1 t/2$ at long times — linear growth against the bounded static oscillation. (e) State in one sentence why no truncation was ever needed.

9. (****) (Section 5.9) The free packet. Start from the Gaussian preparation

$$\psi(x, 0) = (2\pi\sigma^2)^{-1/4} e^{-x^2/4\sigma^2} e^{ip_0 x/\hbar}.$$

(a) Compute $\varphi(p, 0)$: a Gaussian centered at p_0 with $\Delta p = \hbar/2\sigma$. (b) Derive $\psi(x, t)$ by the Gaussian convolution with the free kernel (5.42); verify $\langle x \rangle(t) = p_0 t/m$ and the spreading law (5.43), $\Delta x(t)^2 = \sigma^2 + (\Delta p t/m)^2$. (c) Compute the position–momentum covariance $\frac{1}{2}\langle QP + PQ \rangle - \langle Q \rangle \langle P \rangle = \hbar^2 t/4m\sigma^2$ (Heisenberg-picture arithmetic: $Q_H(t) = Q + Pt/m$, $P_H(t) = P$), and conclude $(\Delta x \Delta p)^2 = (\hbar/2)^2 + (\hbar^2 t/4m\sigma^2)^2 > (\hbar/2)^2$ for $t > 0$ — saturation lost, the correlation manufactured by the free evolution. (d) Verify the far-field form (5.44) and state its time-of-flight reading in one sentence. (e) Confirm $K_0(x, t; x', 0) \rightarrow \delta(x - x')$ as $t \rightarrow 0$ by its action on a test packet.

Chapter 6 The Theory of Rotations and Angular Momenta

The preceding chapter closed with a question it deliberately left for the algebra to answer (Section 5.10): the rotation family’s spectrum. The evolution chapter derived the brackets $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$ among the generators of rotation—the complete angular momentum algebra, a structural fact about space itself read off the unitary ray representatives of the Galilei group ((4.32))—and then stopped. It named the ladder problem, identified the operator-method template that the representations chapter had just proved for the harmonic oscillator, and handed the whole rotation family to the present chapter as its central debt ((5.23), (5.24), (5.25)). The question this chapter must answer is: given that single commutator, what are the possible values of angular momentum, and what does rotation symmetry force about the spectra of physical systems?

The answer repays the book’s oldest unpaid structural debt, and it repays it in a theorem. The Balmer formula and the Rydberg constant, measured in the hydrogen spectrum since the 1880s and written into the first chapter as the theory’s founding promissory note, are derived here as the fruit of rotational symmetry. The Stern–Gerlach two-state system, the book’s paradigm of quantum incompatibility since Chapter 2, is reconstructed from the smallest nontrivial representation of the rotation group, and the half-angle law that Chapter 2 took as an experimental fact becomes a theorem of SU(2). The spinor sign that the evolution chapter computed for Larmor precession ((4.10)) acquires its group-theoretic meaning: a 2π rotation multiplies a spin-1/2 state by -1 because SU(2) covers SO(3) twice. And the ladder template of Chapter 5—define a ladder pair from one commutator, let positivity force the descent to terminate, compute the norms to fix the actions—is re-run on the rotation commutator, yielding the complete J^2, J_z spectrum by algebraic positivity alone.

The arc is deductive, and every section lands on a measurable quantity. Section 6.1 re-reads the rotation piece of ch4’s Galilei algebra as a self-contained Lie group, introduces SO(3) as the rotation group and SU(2) as its universal cover, develops the Euler-angle parametrization of finite rotations and the Wigner D-matrices as the irreducible unitary representations, and establishes the 2:1 covering as the group-theoretic origin of the half-integer spin sign. Section 6.2 derives the full J^2, J_z spectrum by the ladder method: $j = 0, 1/2, 1, \dots, m = -j, \dots, j$, with the $(2j+1)$ -fold degeneracy of every multiplet forced by rotation invariance. Section 6.3 constructs the $j = 1/2$ representation, recovers the Pauli matrices and the Bloch sphere, derives the half-angle law and the spin- n ONB from SU(2) rotations, and proves $\Theta^2 = -I$ for half-integer spin—repaying four of the book’s oldest debts in one section. Section 6.4 builds the orbital angular momentum operator $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ in the position representation, solves its eigenvalue problem in spherical coordinates, obtains the spherical harmonics, and reduces the central-potential problem to a one-dimensional radial equation. Section 6.5 presents Schwinger’s oscillator model—an alternative construction that re-derives the entire angular momentum theory from two boson modes, unifying the ladder method with the harmonic oscillator algebra and providing the natural language for angular momentum coupling. Section 6.6 formalizes the addition of two angular

momenta on the tensor product space, defines the Clebsch–Gordan coefficients as the change of basis between coupled and uncoupled bases, presents the Racah closed form and the $3-j$ symbols as the more symmetric notation, and exhibits the two-spin-1/2 system—singlet and triplet—as the paradigmatic example that bridges to the composite-systems chapter. Section 6.7 introduces irreducible tensor operators, proves the Wigner–Eckart theorem, and extracts its spectroscopic content: the m -selection rule, the triangle inequality, and the factorization of every matrix element into a geometric part and a physical part. Section 6.8 solves the Coulomb problem, derives the Balmer/Rydberg formula from the Schrödinger equation with rotational symmetry, exhibits the wave functions, and closes with the Zeeman splitting—the chapter’s final measurable landing. Section 6.9 balances the ledger and hands the tensor-product machinery to the composite-systems chapter.

The chapter adds zero new axioms. Every display below is proved from the axioms of Chapters 2–5: the rotation algebra ((4.32)), the ladder template ((5.22), (5.25)), the position-representation machinery ((5.7), (5.9), (5.12)), and the measurement theory ((3.49), (3.53)). The $SO(3)/SU(2)$ topology is stated with an Appendix B pointer; the Clebsch–Gordan series is proved by dimension counting; the Wigner–Eckart theorem is proved from rotation-operator conjugation and CG orthogonality. This is a theorem chapter, and its harvest is the hydrogen atom.

6.1 The Rotation Group, Euler Angles, and Finite Rotations

The rotation algebra (4.32)—the complete algebraic content of rotational symmetry, derived in the evolution chapter among the Galilean brackets (Section 4.7)—and the generator correspondence (4.19), which gives the finite rotation $R_{\mathbf{n}}(\theta) = e^{-i\theta J_{\mathbf{n}}/\hbar}$ with $J_{\mathbf{n}} := \mathbf{n} \cdot \mathbf{J}$, are the chapter’s entry. The algebra is the infinitesimal shadow of the finite composition law; the spectrum of the J_i —the possible eigenvalues of angular momentum—is a question about the algebra alone, and it is the subject of Section 6.2. But the algebra is the algebra of a group, and the group’s topology decides what the algebra alone cannot: the distinction between integer and half-integer angular momentum. This section earns the group facts needed for that distinction, develops the Euler-angle parametrization of finite rotations, and defines the Wigner D-matrices as the irreducible unitary representations of the rotation group.

SO(3): the rotation group. The group of rotations in three-dimensional Euclidean space is the set of real 3×3 orthogonal matrices with determinant +1:

$$SO(3) := \{ R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \det R = +1 \}. \quad (6.1)$$

Every $R \in SO(3)$ can be written as a rotation by some angle $\theta \in [0, \pi]$ about some unit axis $\mathbf{n} \in S^2$:

$$R_{\mathbf{n}}(\theta) = e^{\theta N}, \quad N_{jk} = -\epsilon_{ijk} n_i, \quad (6.2)$$

where N is the 3×3 antisymmetric matrix whose action on a vector $\mathbf{v}$ is $N\mathbf{v} = \mathbf{n} \times \mathbf{v}$. The exponential is the matrix exponential; the parametrization is unique except at $\theta = \pi$, where $R_{\mathbf{n}}(\pi) = R_{-\mathbf{n}}(\pi)$. The angle parametrizes a one-parameter subgroup for each fixed axis, exactly as the evolution chapter’s general construction demands ((4.17)).

The Lie algebra of $SO(3)$ is obtained by differentiating the one-parameter subgroups at the identity. Let $T_i := dR_{\mathbf{e}_i}(\theta)/d\theta|_{\theta=0}$; then $(T_i)_{jk} = -\epsilon_{ijk}$, and direct computation of the matrix commutator

gives

$$[T_i, T_j] = \epsilon_{ijk} T_k. \quad (6.3)$$

Identifying the quantum generator J_i with $i\hbar T_i$ reproduces (4.32): $[i\hbar T_i, i\hbar T_j] = -\hbar^2 \epsilon_{ijk} T_k = i\hbar \epsilon_{ijk} (i\hbar T_k)$, so $[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$. The algebra of rotations is the algebra of $\text{SO}(3)$; the factor $i\hbar$ is the physicist's convention for self-adjoint generators, and the structure constants $f_{ijk} = \epsilon_{ijk}$ are the signature of three-dimensional rotation. This is the defining representation: the three-dimensional vector representation on $\mathbb{R}^3$, with generators $(T_i)_{jk} = -\epsilon_{ijk}$.

The topology of $\text{SO}(3)$. The parametrization (6.2) shows that $\text{SO}(3)$ is a three-dimensional compact manifold. It is not simply connected: a closed curve in $\text{SO}(3)$ representing a continuous family of rotations that starts and ends at the identity cannot always be continuously shrunk to a point. The fundamental group is $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$: there are exactly two homotopy classes. Physically, a 2π rotation about any axis is a closed loop in $\text{SO}(3)$ that is *not* contractible to the identity, while a 4π rotation is contractible. The group that covers $\text{SO}(3)$ without this topological obstruction—the simply connected universal cover—is $\text{SU}(2)$.

SU(2): the universal cover. The group $\text{SU}(2)$ is the set of 2×2 unitary matrices with determinant $+1$:

$$\text{SU}(2) := \{U \in \mathbb{C}^{2 \times 2} \mid U^\dagger U = I, \det U = +1\}. \quad (6.4)$$

Every $U \in \text{SU}(2)$ can be parametrized by an axis $\mathbf{n}$ and an angle θ :

$$U_{\mathbf{n}}(\theta) = e^{-i(\theta/2)\mathbf{n} \cdot \boldsymbol{\sigma}} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (\mathbf{n} \cdot \boldsymbol{\sigma}), \quad (6.5)$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices of the measurement chapter ((3.61)). The second equality follows from $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$, which gives $e^{-i\alpha \mathbf{n} \cdot \boldsymbol{\sigma}} = \cos \alpha I - i \sin \alpha (\mathbf{n} \cdot \boldsymbol{\sigma})$ for any real α , with $\alpha = \theta/2$. The parameter θ runs from 0 to 2π for a full cycle: $U_{\mathbf{n}}(2\pi) = -I$, $U_{\mathbf{n}}(4\pi) = I$.

The Lie algebra of $\text{SU}(2)$ is read off the generators at the identity. Differentiating $U_{e_i}(\theta)$ at $\theta = 0$ gives $S_i := i(\hbar/2)\sigma_i$, and these satisfy exactly the same bracket as the J_i :

$$[S_i, S_j] = i\hbar \epsilon_{ijk} S_k, \quad (6.6)$$

by the Pauli commutator $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$ ((3.61)). The algebras of $\text{SO}(3)$ and $\text{SU}(2)$ are isomorphic, as they must be: a group and its covering group share the same Lie algebra, because the covering map is a local diffeomorphism near the identity.

The 2:1 homomorphism. The covering map $\Pi : \text{SU}(2) \rightarrow \text{SO}(3)$ sends a unitary U to a rotation $R(U)$ defined by its action on a vector $\mathbf{v} = v_i \sigma_i$ (any Hermitian traceless 2×2 matrix, representing a vector in $\mathbb{R}^3$):

$$U(\mathbf{v} \cdot \boldsymbol{\sigma})U^\dagger = (R(U)\mathbf{v}) \cdot \boldsymbol{\sigma}, \quad R_{ij}(U) = \frac{1}{2} \text{tr}(\sigma_i U \sigma_j U^\dagger). \quad (6.7)$$

The map Π is a group homomorphism: $\Pi(U_1 U_2) = \Pi(U_1)\Pi(U_2)$. It is surjective—every rotation comes from some $\text{SU}(2)$ matrix—but not injective: both U and $-U$ produce the same R , since $(-U)(\mathbf{v} \cdot \boldsymbol{\sigma})(-U)^\dagger = U(\mathbf{v} \cdot \boldsymbol{\sigma})U^\dagger$. The kernel of Π is $\{I, -I\}$, so $\text{SO}(3) \cong \text{SU}(2)/\{\pm I\}$. The covering is 2:1.

The physical consequence is the chapter's organizing fact: a spatial rotation by 2π —the identity

in $SO(3)$ —lifts to $-I$ in $SU(2)$. A state that transforms under the defining representation of $SU(2)$ therefore changes sign under a 2π rotation. States that transform under representations of $SO(3)$ —the identity’s lift is $+I$ —are periodic under 2π . The two classes are the integer-spin and half-integer-spin representations, and the algebra alone does not distinguish them; the group topology does. The full representation theory is collected in Appendix B; the physics that flows from this distinction occupies the rest of the chapter.

Note The homomorphism (6.7) is the explicit form of the textbook statement that $SU(2)$ is the double cover of $SO(3)$. The map is constructed by observing that any traceless Hermitian 2×2 matrix can be written as $\mathbf{v} \cdot \boldsymbol{\sigma}$, and that conjugation by a unitary preserves the trace and Hermiticity, hence maps the space of such matrices to itself linearly. The linear map is orthogonal because $\det(\mathbf{v} \cdot \boldsymbol{\sigma}) = -|\mathbf{v}|^2$ and $\det(U(\mathbf{v} \cdot \boldsymbol{\sigma})U^\dagger) = \det(\mathbf{v} \cdot \boldsymbol{\sigma})$; it has determinant $+1$ because $SU(2)$ is connected. The trace formula for the matrix elements follows from $\text{tr}(\sigma_i \sigma_j) = 2\delta_{ij}$.

Euler angles. The axis-angle parametrizations (6.2) and (6.5) describe rotations by a single angle about a fixed axis. For composing arbitrary rotations, a three-angle parametrization is required. The standard choice, due to Euler, factors any rotation into three successive rotations about coordinate axes:

$$U(\alpha, \beta, \gamma) = e^{-i\alpha J_z/\hbar} e^{-i\beta J_y/\hbar} e^{-i\gamma J_z/\hbar}, \quad (6.8)$$

with the angle ranges

$$\alpha \in [0, 2\pi), \quad \beta \in [0, \pi], \quad \gamma \in [0, 2\pi) \quad (6.9)$$

for $SO(3)$. For $SU(2)$, the third Euler angle runs to 4π : $\gamma \in [0, 4\pi)$, reflecting the double cover. The three angles have a transparent geometric meaning in the active convention: α is a first rotation about the space-fixed z -axis, β is a rotation about the (new) y -axis, and γ is a second rotation about the (new) z -axis. Equivalently, in the passive picture, (β, α) are the spherical coordinates of the body-fixed z -axis on the unit sphere, and γ is the twist about that axis.

The Euler-angle form makes the 2:1 covering visible in the parameter ranges. The identity rotation in $SO(3)$ is represented by $(\alpha, \beta, \gamma) = (0, 0, 0)$ and also by $(\alpha, \beta, \gamma) = (0, 0, 2\pi)$, since a 2π rotation about any axis is the identity in $SO(3)$. But in $SU(2)$, $U(0, 0, 2\pi) = e^{-i2\pi J_z/\hbar}$, which on a state of definite m gives the factor $e^{-i2\pi m} = (-1)^{2j}$. For integer j this is $+I$; for half-integer j it is $-I$, and the identity is reached only at $\gamma = 4\pi$. The Euler-angle picture thus encodes the single topological fact— $SU(2)$ covers $SO(3)$ twice—in the range of one angle.

Wigner D-matrices. The Euler-angle parametrization leads directly to the matrix elements of finite rotations in the angular momentum basis. Define the Wigner D-matrix:

$$D_{m'm}^{(j)}(\alpha, \beta, \gamma) := \langle j, m' | U(\alpha, \beta, \gamma) | j, m \rangle = e^{-i\alpha m'} d_{m'm}^{(j)}(\beta) e^{-i\gamma m}, \quad (6.10)$$

where $d_{m'm}^{(j)}(\beta) := \langle j, m' | e^{-i\beta J_y/\hbar} | j, m \rangle$ is the reduced (Wigner small) d -matrix. The factorization follows because J_z is diagonal in the $|j, m\rangle$ basis — the first and third Euler factors contribute only the phases $e^{-i\alpha m'}$ and $e^{-i\gamma m}$ — while the middle factor, the rotation about the y -axis, mixes m values and carries the nontrivial β -dependence. The D-matrices are the $(2j+1)$ -dimensional irreducible unitary representations of $SU(2)$: under a rotation R , $U(R) |j, m\rangle = \sum_{m'} D_{m'm}^{(j)}(R) |j, m'\rangle$. They are

the finite-rotation counterpart of the ladder coefficients of Section 6.2 — the ladders step by one unit of m , the D-matrices rotate the entire multiplet by an arbitrary angle.

Closed form and explicit low- j matrices. The small- d matrix has the explicit closed form (Appendix B.4, (B.14)), in terms of a finite factorial sum (equivalently a Jacobi polynomial). The two representations most used in physics are read off directly:

$$d^{(1/2)}(\beta) = \begin{pmatrix} \cos(\beta/2) & -\sin(\beta/2) \\ \sin(\beta/2) & \cos(\beta/2) \end{pmatrix}, \quad d^{(1)}(\beta) = \begin{pmatrix} \frac{1+\cos\beta}{2} & -\frac{\sin\beta}{\sqrt{2}} & \frac{1-\cos\beta}{2} \\ \frac{\sin\beta}{\sqrt{2}} & \cos\beta & -\frac{\sin\beta}{\sqrt{2}} \\ \frac{1-\cos\beta}{2} & \frac{\sin\beta}{\sqrt{2}} & \frac{1+\cos\beta}{2} \end{pmatrix}, \quad (6.11)$$

in the bases $\{|+z\rangle, |-z\rangle\}$ ($j = 1/2$) and $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$ ($j = 1$). The higher instances $d^{(3/2)}$ (4×4) and $d^{(2)}$ (5×5), together with the general factorial formula and the top-row specialization $d_{j,m}^{(j)} = (-1)^{j-m} \sqrt{\binom{2j}{j-m}} (\cos \frac{\beta}{2})^{j+m} (\sin \frac{\beta}{2})^{j-m}$, are tabulated in Appendix B.4. The full D-matrix follows by sandwiching d between the diagonal z -rotation phases; for $j = 1/2$,

$$D^{(1/2)}(\alpha, \beta, \gamma) = \begin{pmatrix} e^{-i(\alpha+\gamma)/2} \cos(\beta/2) & -e^{-i(\alpha-\gamma)/2} \sin(\beta/2) \\ e^{i(\alpha-\gamma)/2} \sin(\beta/2) & e^{i(\alpha+\gamma)/2} \cos(\beta/2) \end{pmatrix}.$$

The half-angle $\beta/2$ is the signature of the double cover: the entries are 4π -periodic in α, γ , and $D^{(1/2)}(\alpha, \beta, \gamma + 2\pi) = -D^{(1/2)}(\alpha, \beta, \gamma)$. The first column at $\gamma = 0$ is the spin- $\mathbf{n}$ state $|+\mathbf{n}\rangle$ that Section 6.3 constructs from $|+z\rangle$ by the Euler rotation $U(\alpha, \beta, 0)$.

Properties. From $U(R_1 R_2) = U(R_1)U(R_2)$ the D-matrices compose as representations, $\sum_{m''} D_{m'm''}^{(j)}(R_2) D_{m''m}^{(j)}(R_1) = D_{m'm}^{(j)}(R_1 R_2)$; they are unitary; and in the Condon–Shortley convention $D_{m'm}^{(j)*} = (-1)^{m-m'} D_{-m', -m}^{(j)}$ (from the reality and reflection symmetry of $d^{(j)}$). Under the Haar measure $dR = (1/8\pi^2) \sin \beta d\alpha d\beta d\gamma$ they obey the great orthogonality

$$\int dR D_{m'm}^{(j)*}(R) D_{\mu'\mu}^{(j')}(R) = \frac{1}{2j+1} \delta_{jj'} \delta_{m\mu} \delta_{m'\mu'},$$

the SU(2) instance of the Peter–Weyl theorem (Appendix B.4): the $D_{m'm}^{(j)}$ are a complete orthogonal basis of $L^2(\text{SU}(2))$, the compact-group avatar of Fourier analysis.

Connection to spherical harmonics; rotation of functions. For integer ℓ , the $m' = 0$ column of $D^{(\ell)}$ at $\gamma = 0$ is a spherical harmonic:

$$D_{m0}^{(\ell)}(\alpha, \beta, 0) = \sqrt{\frac{4\pi}{2\ell+1}} Y_{\ell}^{m*}(\beta, \alpha), \quad (6.12)$$

and in particular $D_{00}^{(\ell)} = P_{\ell}(\cos \beta)$. The spherical harmonics are the $m' = 0$ column — the functions on the coset sphere $S^2 \cong \text{SO}(3)/\text{SO}(2)$. The general transformation law under a finite rotation R follows:

$$Y_{\ell}^m(R^{-1}\hat{\mathbf{r}}) = \sum_{m'=-\ell}^{\ell} D_{m'm}^{(\ell)}(R) Y_{\ell}^{m'}(\hat{\mathbf{r}}), \quad (6.13)$$

the active rotation of a scalar field on the sphere: rotating a spherical harmonic mixes the $(2\ell+1)$ members of its multiplet by the D-matrix, exactly as rotating a state $|\ell, m\rangle$ mixes the basis kets. This is the bridge to Section 6.4.

Rotation of Cartesian vectors and tensors. The $j = 1$ representation (6.11) is the vector representation in the spherical basis. The spherical components $V_{\pm 1}^{(1)} = \mp(V_x \pm iV_y)/\sqrt{2}$, $V_0^{(1)} = V_z$ of a vector $\mathbf{V}$ transform as $V_q^{(1)} \mapsto \sum_{q'} D_{q'q}^{(1)}(R) V_{q'}^{(1)}$, which is exactly the Cartesian rotation $\mathbf{V} \mapsto R\mathbf{V}$ rewritten — $d^{(1)}(\beta)$ is the rotation about y in that basis. A rank- k Cartesian tensor $T_{i_1 \dots i_k}$ decomposes under rotations into irreducible spherical tensors of ranks $k, k-2, \dots$ (the traceless symmetric part is rank k , traces lower), each transforming by $D^{(k)}$; this is the bridge to the irreducible-tensor operators of Section 6.7.

Example: spin precession re-read through Euler angles. The Larmor precession of the evolution chapter (Section 4.3) acquires its Euler-angle reading here. The evolution operator for a spin in a uniform magnetic field $\mathbf{B} = B\hat{z}$ is $U(t) = e^{-i\omega t \sigma_z/2}$, with $\omega = -\gamma B$ ((4.8)). In the Euler-angle language of (6.8) this is the special case $\alpha = \omega t$, $\beta = 0$, $\gamma = 0$: a one-parameter subgroup generated by J_z alone. The D-matrix element (6.10) gives the amplitude to remain in the initial m -state, $D_{++}^{(1/2)}(\omega t, 0, 0) = e^{-i\omega t/2}$, $D_{--}^{(1/2)}(\omega t, 0, 0) = e^{i\omega t/2}$. At $t = 2\pi/\omega$, $D^{(1/2)}(2\pi, 0, 0) = -I$, the spinor sign of (4.10): the state has traversed a full cycle in $SU(2)$ while the spatial rotation in $SO(3)$ is the identity. The expectation vector $\langle \boldsymbol{\sigma} \rangle(t)$ of (4.9) rotates at frequency ω in $\mathbb{R}^3$ and completes one cycle at $\omega t = 2\pi$; the state completes one cycle at $\omega t = 4\pi$. The 2:1 covering is made quantitative.

Example: the symmetric top — a rigid rotor with D-matrices as wavefunctions. A rigid molecule whose inertia tensor has a symmetry axis (a *symmetric top*, e.g. NH_3 or CH_3Cl) has rotational Hamiltonian $H = A J_a^2 + B(J_b^2 + J_c^2)$ in the body-fixed frame, where $A \neq B$ are the principal moments and a the symmetry axis. Writing J_a^2 as the projection of $\mathbf{J}$ onto the body axis — the operator whose space-fixed representative is the third Euler angle — the simultaneous eigenstates of H , J^2 , J_z (space-fixed, quantum number M), and J_a (body-fixed, quantum number K) are precisely the D-matrices:

$$\langle \alpha, \beta, \gamma | J, M, K \rangle \propto D_{MK}^{(J)*}(\alpha, \beta, \gamma), \quad (6.14)$$

and the energies are

$$E_{J,K} = B J(J+1) + (A-B) K^2, \quad J = 0, 1, 2, \dots, \quad K = -J, \dots, J.$$

The $(2J+1)$ -fold M -degeneracy is rotation invariance (as for any rotor); the $(2J+1)$ -fold K -degeneracy (counting $\pm K$) is the additional symmetry of the symmetric top. The asymmetric top ($A \neq B \neq C$) mixes K and has no closed form — its levels are found by diagonalizing H in the $|JMK\rangle$ basis, a standard application of the D-matrix algebra. Rotational spectroscopy of molecules (the CO $J = 1 \rightarrow 0$ line of (6.59) and, more generally, the microwave spectra of symmetric tops) is read directly off the D-matrix eigenvalue formula.

Example: spin-1 and photon polarization. The $j = 1$ representation is the natural language of vector (spin-1) fields. A massless spin-1 particle — the photon — carries no longitudinal state: its two physical polarizations are the helicity $\lambda = \pm 1$ states, obtained from the $j = 1$ triplet by projecting onto the direction of motion. A polarization state is a point on the Poincaré sphere (the $j = 1$ analogue of the spin-1/2 Bloch sphere of Section 6.3); the Stokes parameters (S_1, S_2, S_3) are its Bloch vector, with $|\mathbf{S}| = 1$ for pure polarization and $|\mathbf{S}| < 1$ for partial polarization. Linear polarizers, quarter- and half-wave plates, and polarization rotators act on the (S_1, S_2, S_3) vector by exactly the $SO(3)$ rotations of (6.11) — the spin-1 D-matrix is the operator behind every optical polarization calculation. This is

the $j = 1$ counterpart of the $j = 1/2$ Stern–Gerlach grammar of Chapters 2–3.

The finite-rotation formalism — the group structure, the 2:1 covering, the Euler-angle parametrization, the Wigner D-matrices with their explicit low- j forms and their reduction to spherical harmonics — is now complete. The D-matrices give the amplitude for any initial state $|j, m\rangle$ to be found in any final state $|j, m'\rangle$ after any rotation, and the composition property converts sequences of rotations into matrix multiplication. What remains is algebraic: what eigenvalues can the generators J^2 and J_z possess? The answer is independent of the group topology and follows from the single commutator (4.32) alone, by the ladder method of Chapter 5.

6.2 Angular Momentum Algebra: the Spectrum of J^2 and J_z

The rotation algebra (4.32) is the chapter’s sole input. This section derives its complete spectral content—the eigenvalues of J^2 and J_z and the ladder actions that connect them—by the method of Chapter 5, Section 5.6: form a complex pair from two noncommuting generators, let positivity force the descent to terminate, and compute the norms to fix the actions. The method is identical; the algebra is different, and the resulting spectrum is the quadratic $j(j+1)$ in place of the oscillator’s linear n .

The ladder operators. The three Hermitian generators J_x, J_y, J_z satisfy (4.32). Define the complex pair

$$J_+ := J_x + iJ_y, \quad J_- := J_x - iJ_y, \quad J_\pm^\dagger = J_\mp. \quad (6.15)$$

The adjoint relation follows from $J_x^\dagger = J_x, J_y^\dagger = J_y$. Their commutators with J_z are immediate from (4.32) and the commutator algebra (3.58):

$$\begin{aligned} [J_z, J_\pm] &= [J_z, J_x] \pm i[J_z, J_y] = i\hbar\epsilon_{zxy}J_y \pm i(i\hbar\epsilon_{zyx}J_x) \\ &= i\hbar J_y \pm i(-i\hbar J_x) = i\hbar J_y \pm \hbar J_x = \pm\hbar(J_x \pm iJ_y) = \pm\hbar J_\pm. \end{aligned} \quad (6.16)$$

The sign says exactly what a ladder operator must: J_+ raises the J_z eigenvalue by one unit of $\hbar$, and J_- lowers it. The mutual commutator of the pair is

$$\begin{aligned} [J_+, J_-] &= [J_x + iJ_y, J_x - iJ_y] = -i[J_x, J_y] + i[J_y, J_x] \\ &= -i(i\hbar J_z) + i(-i\hbar J_z) = 2\hbar J_z. \end{aligned} \quad (6.17)$$

Equations (6.16) and (6.17) are the angular-momentum ladder algebra, the rotation-family analog of the oscillator’s $[N, a] = -a, [N, a^\dagger] = +a^\dagger, [a, a^\dagger] = I$ ((5.23), (5.22)).

The Casimir invariant. A rotation is specified by its axis and its angle, but the square of the total angular momentum is independent of axis—it commutes with every generator. Define

$$J^2 := J_x^2 + J_y^2 + J_z^2. \quad (6.18)$$

The operator can be rewritten in terms of the ladder pair. From the definitions,

$$J_- J_+ = (J_x - iJ_y)(J_x + iJ_y) = J_x^2 + J_y^2 + i[J_x, J_y] = J_x^2 + J_y^2 - \hbar J_z,$$

so that $J_x^2 + J_y^2 = J_- J_+ + \hbar J_z$, and substituting into (6.18) gives

$$J^2 = J_- J_+ + J_z^2 + \hbar J_z = J_+ J_- + J_z^2 - \hbar J_z. \quad (6.19)$$

The two forms are equivalent; both will be used. The commutation of J^2 with every J_i follows from a single computation using (6.16) and (6.17):

$$\begin{aligned} [J^2, J_z] &= [J_- J_+ + J_z^2 + \hbar J_z, J_z] = [J_- J_+, J_z] \\ &= J_- [J_+, J_z] + [J_-, J_z] J_+ = J_- (-\hbar J_+) + (-\hbar J_-) J_+ = 0, \end{aligned}$$

where the last equality uses $[J_+, J_z] = -\hbar J_+$ and $[J_-, J_z] = +\hbar J_-$, which is (6.16) with the sign reversed. The same argument with J_x and J_y (or the cyclic symmetry of the algebra) gives $[J^2, J_i] = 0$ for all i . J^2 is the quadratic Casimir of the rotation algebra: it labels the irreducible representations.

Simultaneous diagonalization. Because $[J^2, J_z] = 0$, the compatibility theorem of the measurement chapter ((3.49)) delivers a basis of simultaneous eigenstates. Let these be denoted $|j, m\rangle$ with eigenvalues written in units of $\hbar$:

$$J^2 |j, m\rangle = \hbar^2 \lambda_j |j, m\rangle, \quad J_z |j, m\rangle = \hbar m |j, m\rangle, \quad (6.20)$$

where λ_j and m are dimensionless real numbers to be determined (J^2 and J_z are self-adjoint, (3.33)). The label j is a placeholder for whatever quantum number distinguishes different eigenvalues of J^2 ; the notation anticipates the result $j(j+1) = \lambda_j$. Within a fixed j , the states are distinguished by m ; the set $\{J^2, J_z\}$ forms a CSCO for a single angular momentum, the joint eigenvalues (j, m) addressing a unique ray within each j -multiplet ((3.53)).

The ladder actions. From (6.16), $J_z(J_\pm |j, m\rangle) = (J_\pm J_z \pm \hbar J_\pm) |j, m\rangle = \hbar(m \pm 1)(J_\pm |j, m\rangle)$. Whenever $J_\pm |j, m\rangle \neq 0$, it is an eigenstate of J_z with eigenvalue $\hbar(m \pm 1)$, and it remains an eigenstate of J^2 since $[J^2, J_\pm] = 0$. Hence $J_\pm |j, m\rangle$ is proportional to $|j, m \pm 1\rangle$ within the same j -multiplet. The squared norm is computed from (6.18) via $J_\mp J_\pm = J^2 - J_z^2 \mp \hbar J_z$:

$$\begin{aligned} \|J_\pm |j, m\rangle\|^2 &= \langle j, m | J_\mp J_\pm |j, m\rangle = \langle j, m | (J^2 - J_z^2 \mp \hbar J_z) |j, m\rangle \\ &= \hbar^2 (\lambda_j - m^2 \mp m) = \hbar^2 (\lambda_j - m(m \pm 1)). \end{aligned} \quad (6.21)$$

Positivity forces termination. The squared norm (6.21) cannot be negative. Since $J_\pm$ acting on any state must either produce another state (positive norm) or annihilate it (zero norm), the quantity $\lambda_j - m(m \pm 1)$ must be nonnegative for every value of m that occurs in the multiplet. The ladder operators change m by ± 1 , so starting from any m and applying J_+ repeatedly yields a strictly increasing sequence of m -values; the norm condition $\lambda_j \geq m(m+1)$ restricts how high the sequence can climb before the norm would become negative. The ascent must therefore terminate: there exists a maximum value $m_{\max}$ with

$$J_+ |j, m_{\max}\rangle = 0, \quad \lambda_j = m_{\max}(m_{\max} + 1), \quad (6.22)$$

the second equality following from setting (6.21) to zero at $m = m_{\max}$. Similarly, repeated application of J_- must terminate at a minimum $m_{\min}$ with

$$J_- |j, m_{\min}\rangle = 0, \quad \lambda_j = m_{\min}(m_{\min} - 1). \quad (6.23)$$

Equating the two expressions for λ_j gives $m_{\max}(m_{\max} + 1) = m_{\min}(m_{\min} - 1)$, whose nonnegative solution is $m_{\max} = -m_{\min}$. The difference $m_{\max} - m_{\min} = 2m_{\max}$ must be an integer, because each application of J_- or J_+ changes m by exactly one unit and the two extrema are connected by a finite

number of such steps. Hence

$$m_{\max} = -m_{\min} =: j, \quad j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots, \quad \lambda_j = j(j+1). \quad (6.24)$$

The conclusion is the spectrum theorem.

Theorem (angular momentum spectrum). *From the single commutator $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$, the eigenvalues of J^2 and J_z on their simultaneous eigenstates are*

$$J^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle, \quad J_z |j, m\rangle = \hbar m |j, m\rangle, \quad (6.25)$$

with $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ and, for each j , $m = -j, -j+1, \dots, j-1, j$.

The spectrum is the chapter's first harvest. The ladder argument permits both integer and half-integer j ; the algebra alone does not distinguish them. The distinction is the group-topological one of Section 6.1: representations of $\text{SO}(3)$ carry integer j , while representations of $\text{SU}(2)$ permit both, with the half-integer representations being the double-valued ones that change sign under a 2π rotation.

The normalized ladder actions. The norm computation (6.21) with $\lambda_j = j(j+1)$ gives the magnitude of the ladder coefficients. The phase is fixed by the standard convention, which this book adopts throughout:

$$J_{\pm} |j, m\rangle = \hbar \sqrt{j(j+1) - m(m \pm 1)} |j, m \pm 1\rangle, \quad (6.26)$$

with the square root taken real and nonnegative. This is the **Condon–Shortley phase convention**: all matrix elements of J_x and J_z are real, and those of J_y are purely imaginary. It is the structural analog of the oscillator's (5.25), with the quadratic argument $j(j+1) - m(m \pm 1)$ replacing the linear n and $n+1$.

The $(2j+1)$ -fold degeneracy. Each j carries a $(2j+1)$ -dimensional irreducible representation of the rotation algebra. For a system whose Hamiltonian is rotation-invariant, $[H, J_i] = 0$ for all i , the evolution chapter's conservation theorem ((4.57)) makes every J_i a conserved quantity, and its level-local degeneracy corollary ((4.59)) forces the following structure: the restriction of the J_i to each energy eigenspace carries a representation of the rotation algebra, and the noncommutation of these restrictions—when the level is not a singlet—forces the $(2j+1)$ -fold degeneracy of every level whose total angular momentum quantum number is $j > 0$. The m -degeneracy is rotation symmetry's doing; the possibility of additional degeneracy beyond the $(2j+1)$ -fold one (as in the hydrogen atom) is the sign of a higher symmetry.

Integer versus half-integer: the superselection rule. The operator $e^{-i2\pi J_z/\hbar}$ is the finite rotation by 2π about the z -axis. On a state $|j, m\rangle$, it produces the phase $e^{-i2\pi m}$:

$$e^{-i2\pi J_z/\hbar} |j, m\rangle = e^{-i2\pi m} |j, m\rangle = (-1)^{2j} |j, m\rangle, \quad (6.27)$$

since m and j have the same integer or half-integer character. Integer- j states are periodic under 2π rotations; half-integer states change sign. No experiment has ever observed a coherent superposition of an integer-spin and a half-integer-spin state—the relative phase of the two sectors would be shifted by -1 under a 2π rotation of the apparatus, rendering it unobservable. The two sectors are superselected: the Hilbert space decomposes into the direct sum of the integer-spin and half-integer-spin subspaces, and every physical observable respects the decomposition. This is the angular-momentum share of

the superselection remark of the states chapter, and its group-theoretic origin is the 2:1 covering of Section 6.1.

Note *The superselection rule between integer and half-integer angular momentum sectors is experimentally warranted by the 4π periodicity of spinor states under rotation (the neutron-interferometer measurement cited in Sections 6.3 and 4.3) and has its formal statement in the Wick–Wightman–Wigner analysis of 1952, cited in the states chapter. The physical content is that no observable connects the two sectors; the formal content is that the ray representative of the rotation group is a true representation on each sector separately.*

Example: the $j = 1$ representation. The three states $|1, 1\rangle, |1, 0\rangle, |1, -1\rangle$ carry the $j = 1$ multiplet. From (6.26):

$$\begin{aligned} J_+ |1, 1\rangle &= 0, & J_+ |1, 0\rangle &= \hbar\sqrt{2} |1, 1\rangle, & J_+ |1, -1\rangle &= \hbar\sqrt{2} |1, 0\rangle, \\ J_- |1, 1\rangle &= \hbar\sqrt{2} |1, 0\rangle, & J_- |1, 0\rangle &= \hbar\sqrt{2} |1, -1\rangle, & J_- |1, -1\rangle &= 0. \end{aligned}$$

Reading off the matrix elements in the ordered basis $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$,

$$\begin{aligned} J_x &= \frac{J_+ + J_-}{2} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \\ J_y &= \frac{J_+ - J_-}{2i} = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \\ J_z &= \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \end{aligned}$$

These are exactly (up to unitary equivalence) the $\text{SO}(3)$ generators $i\hbar T_i$ of Section 6.1: the $j = 1$ representation is the defining vector representation of $\text{SO}(3)$. The Casimir is $J^2 = 2\hbar^2 I_{3 \times 3}$. In a central potential, the three m values share the same energy, forming the p -orbital triplet—the simplest instance of the multiplet structure that rotation invariance forces.

The advantage of the algebraic derivation is its purity: from one commutator, positivity, and norm computation, the complete spectrum follows, with no differential equation and no coordinate system. Its limitation is the same as the oscillator's in Chapter 5: it earns the eigenvalues and the ladder actions but not the wave functions; those require a representation, and the position representation for orbital angular momentum is the subject of Section 6.4. The structure it does earn is universal: every system with rotational symmetry, from a single spin to a rotating molecule to a nucleus to a baryon, organizes its states into $(2j + 1)$ -fold multiplets labeled by the single quantum number j , and the ladder actions (6.26) are the exact algebraic engine of every transition amplitude among them.

The algebra has delivered the full spectrum. The simplest nontrivial case is $j = 1/2$ —the representation behind the Stern–Gerlach experiment and the states chapter's two-state grammar. Its construction repays three of the book's oldest debts.

5487

6.3 Spin 1/2 and Pauli Matrices

The preceding section extracted the full angular momentum spectrum from the single commutator (4.32): the ladder template of the oscillator chapter re-run on the rotation algebra delivered $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ and $m = -j, \dots, j$, forced by positivity alone. The smallest nontrivial representation is $j = 1/2$, and it is the representation behind the Stern–Gerlach experiment of Chapter 1, the two-state grammar of Chapter 2, and the σ -commutator incompatibility of Chapter 3. This section constructs it explicitly. With it, three of the book’s oldest structural debts – the half-angle law, the spin- n ONB, and the spinor 4π sign – are repaid from first principles, and the $\Theta^2 = (-1)^{2j}$ formula outstanding from the discrete symmetries section of Chapter 4 is proved for the first time.

The $j = 1/2$ representation. Insert $j = 1/2$ into the spectrum theorem (6.25). Two states, $m = +1/2$ and $m = -1/2$, span the space:

$$J^2 \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle = \hbar^2 \frac{3}{4} \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle, \quad J_z \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle = \pm \frac{\hbar}{2} \left| \frac{1}{2}, \pm \frac{1}{2} \right\rangle. \quad (6.28)$$

Abbreviate the two basis vectors as $|+z\rangle := \left| \frac{1}{2}, +\frac{1}{2} \right\rangle$ and $|-z\rangle := \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$, with the z -axis taken as the quantization direction. From the normalized ladder actions (6.26) with $j = 1/2$,

$$J_+ |-z\rangle = \hbar |+z\rangle, \quad J_- |+z\rangle = \hbar |-z\rangle, \quad J_+ |+z\rangle = J_- |-z\rangle = 0. \quad (6.29)$$

The entire two-dimensional representation is fixed by these four actions plus the diagonal J_z eigenvalues.

The Pauli matrices. Define the spin operator $\mathbf{S}$ as the $j = 1/2$ instance of $\mathbf{J}$. In the ordered basis $\{|+z\rangle, |-z\rangle\}$, the matrix elements $\langle \pm z | S_i | \pm z \rangle$ follow directly from the ladder actions. The result is the compact expression

$$S_i = \frac{\hbar}{2} \sigma_i, \quad i = x, y, z, \quad (6.30)$$

with the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (6.31)$$

The matrices are constructed by reading off $S_x = (S_+ + S_-)/2$ and $S_y = (S_+ - S_-)/2i$ from the ladder pair (6.15): $S_+ |-z\rangle = \hbar |+z\rangle$ gives $S_x = (\hbar/2)\sigma_x$ and $S_y = (\hbar/2)\sigma_y$, and the diagonal of S_z yields σ_z .

Theorem (Pauli algebra). *The Pauli matrices satisfy*

$$[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k, \quad \{\sigma_i, \sigma_j\} = 2 \delta_{ij} I, \quad (6.32)$$

where $\{, \}$ denotes the anticommutator. Every 2×2 matrix can be expanded as $A = a_0 I + \mathbf{a} \cdot \boldsymbol{\sigma}$ with $a_0 = \frac{1}{2} \text{tr} A$ and $a_i = \frac{1}{2} \text{tr}(A \sigma_i)$.

The proof of the commutator is one computation: inserting $S_i = (\hbar/2)\sigma_i$ into the rotation algebra (4.32) gives $[(\hbar/2)\sigma_i, (\hbar/2)\sigma_j] = i\hbar \epsilon_{ijk} (\hbar/2)\sigma_k$, and cancelling $\hbar^2/4$ on both sides yields $[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k$. The anticommutator is verified by direct multiplication: $\sigma_x \sigma_y = i\sigma_z$ and $\sigma_y \sigma_x = -i\sigma_z$ give the anticommutator zero; $\sigma_x^2 = I$ and the cyclic permutations give the diagonal δ_{ij} term. The trace identity follows from $\text{tr} \sigma_i = 0$ and the orthogonality $\text{tr}(\sigma_i \sigma_j) = 2\delta_{ij}$. With this theorem, the

σ -commutator (3.61) of the measurement chapter – displayed there as a computational fact of 2×2 matrices and flagged as a debt to be repaid – is now read as the $j = 1/2$ instance of the rotation algebra (4.32): the noncommutation of spin components, which the two-state experiments of Chapters 2 and 3 forced as a bench fact, acquires its group-theoretic meaning as the infinitesimal shadow of the non-commutation of finite rotations in $SU(2)$. The two-state incompatibility structure of the measurement chapter’s Section 3.7 is the $j = 1/2$ representation of the rotation group.

Spinors. Every state in $\mathbb{C}^2$ is a **spinor**:

$$|\psi\rangle = \alpha | +z \rangle + \beta | -z \rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad (6.33)$$

with $\alpha, \beta \in \mathbb{C}$. The two-component column $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$ is the spinor’s representation in the σ_z eigenbasis. A finite rotation about an axis $\mathbf{n}$ (a unit vector) by angle θ acts on spinors through the $SU(2)$ parametrization (6.5):

$$R_{\mathbf{n}}(\theta) = e^{-i(\theta/2)\mathbf{n} \cdot \boldsymbol{\sigma}} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} (\mathbf{n} \cdot \boldsymbol{\sigma}), \quad (6.34)$$

where the second equality follows from expanding the exponential and using $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$ from the anticommutator half of (6.32), which collapses the series to two terms. The parameter is $\theta/2$, not θ : the state rotates by half the spatial angle, because $SU(2)$ covers $SO(3)$ twice – the topological fact of the preceding section (6.7) is here a concrete formula.

Derivation of the spin- n orthonormal basis (repaying Chapter 2 debt #11). The spin- n states of Chapter 2 – displayed there as (2.38) and consumed as experimental input by the measurement chapter – are now derived. Take the reference state $| +z \rangle$ and construct $\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ by a rotation from the z -axis: first rotate by θ about the y -axis, then by ϕ about the z -axis. The corresponding $SU(2)$ operator is $e^{-i(\phi/2)\sigma_z} e^{-i(\theta/2)\sigma_y}$. The inner rotation is evaluated using the Euler identity (6.5):

$$e^{-i(\theta/2)\sigma_y} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \sigma_y = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}.$$

Acting on $| +z \rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$e^{-i(\theta/2)\sigma_y} | +z \rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2) \end{pmatrix} = \cos \frac{\theta}{2} | +z \rangle + \sin \frac{\theta}{2} | -z \rangle.$$

The outer z -rotation $e^{-i(\phi/2)\sigma_z} = \text{diag}(e^{-i\phi/2}, e^{i\phi/2})$ multiplies the upper component by $e^{-i\phi/2}$ and the lower by $e^{i\phi/2}$. Discarding the overall phase $e^{-i\phi/2}$ (rays are physical, Section 2.5) gives

$$| +\mathbf{n} \rangle = \cos \frac{\theta}{2} | +z \rangle + e^{i\phi} \sin \frac{\theta}{2} | -z \rangle, \quad (6.35)$$

and the orthogonal companion follows by the same rotation applied to $| -z \rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$:

$$| -\mathbf{n} \rangle = e^{-i(\phi/2)\sigma_z} e^{-i(\theta/2)\sigma_y} | -z \rangle = -e^{-i\phi} \sin \frac{\theta}{2} | +z \rangle + \cos \frac{\theta}{2} | -z \rangle. \quad (6.36)$$

These are exactly Chapter 2’s (2.38), with the phase conventions matching those of the states chapter. What was previously given as experimental input – the two-component parametrization of the most general orthonormal basis for a two-state system – is now a theorem: it is the orbit of the reference

basis under the SU(2) rotation group.

The half-angle law (repaying Chapter 2 debt #9). The probability that a system prepared in $|+z\rangle$ passes an SG magnet oriented along $\mathbf{n}$ is the squared overlap:

$$|\langle +z | +\mathbf{n} \rangle|^2 = \cos^2 \frac{\theta}{2}, \quad |\langle -z | +\mathbf{n} \rangle|^2 = \sin^2 \frac{\theta}{2}, \quad (6.37)$$

read off directly from (6.35). Chapter 2 recorded these formulas as (2.39) and called them an experimental fact of rotated Stern–Gerlach magnets, owed to the angular-momentum chapter. That debt is now discharged: the $\theta/2$ in the argument is the signature of the SU(2) covering – the state rotates by half the spatial angle. The operative constraint is the group-theoretic fact that the parameter in the spinor representation’s exponential is $\theta/2$, not θ , because SU(2) covers SO(3) twice.

The spinor 4π sign (repaying Chapter 2 debt #10). Set $\theta = 2\pi$ in the finite rotation formula:

$$R_{\mathbf{n}}(2\pi) = \cos \pi I - i \sin \pi (\mathbf{n} \cdot \boldsymbol{\sigma}) = -I. \quad (6.38)$$

A 2π spatial rotation multiplies every spin-1/2 state by -1 . Chapter 4 computed this fact as (4.10) for the Larmor precession operator: $U(2\pi/\omega) = -I$ on the spin-1/2 space, and noted that its rotational interpretation was owed to this chapter. The interpretation is now supplied: the sign is the defining signature of the spinor representation, the group-theoretic fact that the identity in SO(3) lifts to both $+I$ and $-I$ in SU(2). In SO(3) a 2π rotation is the identity; its SU(2) pre-image (6.7) is $-I$ for the $j = 1/2$ representation. The neutron-interferometer experiment cited in Chapter 2 (Rauch and Werner, 1975) – a 4π precession returning the original interference pattern while 2π flips it – is the direct physical confirmation.

Bloch sphere parametrization. A pure spin-1/2 state $|\psi\rangle$ can be mapped to a point on the unit sphere S^2 , the **Bloch sphere**. Write $|\psi\rangle = \alpha|+z\rangle + \beta|-z\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. The density operator (the projector onto the ray, named in the measurement chapter’s Section 3.3 and to be developed fully in the composite-systems chapter) is

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}.$$

Every 2×2 Hermitian matrix can be expanded in the basis $\{I, \sigma_x, \sigma_y, \sigma_z\}$ with real coefficients. Writing $\rho = \frac{1}{2}(a_0 I + \mathbf{a} \cdot \boldsymbol{\sigma})$ and matching the four entries gives

$$\frac{1}{2} \begin{pmatrix} a_0 + a_z & a_x - ia_y \\ a_x + ia_y & a_0 - a_z \end{pmatrix} = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}.$$

Hence $a_0 = |\alpha|^2 + |\beta|^2 = 1$ (from $\text{tr} \rho = 1$), $a_x = 2\Re(\alpha\beta^*)$, $a_y = -2\Im(\alpha\beta^*)$, $a_z = |\alpha|^2 - |\beta|^2$. Purity $\text{tr}(\rho^2) = 1$ gives $a_0^2 + |\mathbf{a}|^2 = 2$, so $|\mathbf{a}| = 1$. Define $\mathbf{r} := \mathbf{a}$, the **Bloch vector**:

$$\rho = |\psi\rangle\langle\psi| = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad \mathbf{r} = \langle\psi|\boldsymbol{\sigma}|\psi\rangle, \quad |\mathbf{r}| = 1, \quad (6.39)$$

where $\mathbf{r}$ is the expectation vector of the Pauli operators. The parametrization is proved by expanding $|\psi\rangle$ in the basis $\{|+z\rangle, |-z\rangle\}$ and computing the four matrix entries of ρ against I and $\boldsymbol{\sigma}$; the trace condition $\text{tr} \rho = 1$ fixes $a_0 = 1$, and purity $\text{tr}(\rho^2) = 1$ gives $|\mathbf{a}| = 1$, so $\mathbf{r} = \mathbf{a}$ is a unit vector. The ray space of a two-state system is $\mathbb{CP}^1 \cong S^2$: each physical state – each ray – is a point on the sphere, and the sphere’s north and south poles are the $|+z\rangle$ and $|-z\rangle$ rays respectively. The half-angle law (6.37)

has a geometric reading on the Bloch sphere: the squared overlap $|\langle\psi|\phi\rangle|^2$ is $\cos^2(\Theta/2)$ where Θ is the angle between the two state vectors $\mathbf{r}_\psi$ and $\mathbf{r}_\phi$ on the sphere.

The Bloch sphere re-reads the Stern–Gerlach statistics of the measurement chapter ((3.32)): the expectation $\langle +z|\sigma_n|+z\rangle = \cos\theta$ is the projection of the north-pole vector onto the $\mathbf{n}$ -direction. The sequential-SG probabilities of the measurement chapter’s treatment of sequential filters (Section 3.1) reduce to spherical geometry: a filter along $\mathbf{n}$ followed by a filter along $\mathbf{n}'$ transmits a fraction $|\langle +\mathbf{n}|+\mathbf{n}'\rangle|^2 = \cos^2(\Theta/2)$ where Θ is the angle between $\mathbf{n}$ and $\mathbf{n}'$ on the sphere.

Time reversal for spin-1/2 (repaying the Θ^2 share of Chapter 4 debt #24). The time-reversal operator Θ , defined in the discrete-symmetries section of Chapter 4 as the antiunitary operator reversing spin, satisfies $\Theta\mathbf{J}\Theta^{-1} = -\mathbf{J}$. For a general angular momentum representation, this condition is solved by a rotation by π about the y -axis followed by complex conjugation in the standard J_z eigenbasis:

$$\Theta = e^{-i\pi J_y/\hbar} K_0, \quad \Theta^2 = e^{-i2\pi J_y/\hbar} = (-1)^{2j} I, \quad (6.40)$$

where K_0 is the antiunitary prototype (4.16): complex conjugation in the J_z eigenbasis. In the standard J_z eigenbasis, J_z and J_x have real matrix elements while J_y is purely imaginary, hence

$$K_0 J_z K_0^{-1} = J_z, \quad K_0 J_x K_0^{-1} = J_x, \quad K_0 J_y K_0^{-1} = -J_y.$$

The conjugation of the generators by a π -rotation about the y -axis uses the adjoint action derived from the rotation algebra: $e^{-i\theta J_y/\hbar} J_z e^{i\theta J_y/\hbar} = J_z \cos\theta + J_x \sin\theta$, and $e^{-i\theta J_y/\hbar} J_x e^{i\theta J_y/\hbar} = J_x \cos\theta - J_z \sin\theta$, both following from the commutators $[J_y, J_z] = i\hbar J_x$ and $[J_y, J_x] = -i\hbar J_z$ by the Baker-Campbell-Hausdorff expansion of $e^A B e^{-A}$ applied to the rotation algebra. At $\theta = \pi$:

$$e^{-i\pi J_y/\hbar} J_z e^{i\pi J_y/\hbar} = -J_z, \quad e^{-i\pi J_y/\hbar} J_x e^{i\pi J_y/\hbar} = -J_x, \quad e^{-i\pi J_y/\hbar} J_y e^{i\pi J_y/\hbar} = J_y.$$

Now verify $\Theta\mathbf{J}\Theta^{-1} = -\mathbf{J}$ component by component:

$$\Theta J_z \Theta^{-1} = e^{-i\pi J_y/\hbar} K_0 J_z K_0^{-1} e^{i\pi J_y/\hbar} = e^{-i\pi J_y/\hbar} J_z e^{i\pi J_y/\hbar} = -J_z,$$

$$\Theta J_x \Theta^{-1} = e^{-i\pi J_y/\hbar} K_0 J_x K_0^{-1} e^{i\pi J_y/\hbar} = e^{-i\pi J_y/\hbar} J_x e^{i\pi J_y/\hbar} = -J_x,$$

$$\Theta J_y \Theta^{-1} = e^{-i\pi J_y/\hbar} K_0 J_y K_0^{-1} e^{i\pi J_y/\hbar} = e^{-i\pi J_y/\hbar} (-J_y) e^{i\pi J_y/\hbar} = -J_y.$$

For Θ^2 , use antiunitarity: $K_0 i K_0^{-1} = -i$ and $K_0 J_y K_0^{-1} = -J_y$, giving $K_0 J_y^n K_0^{-1} = (-J_y)^n$ and consequently

$$K_0 e^{-i\pi J_y/\hbar} K_0^{-1} = \sum_{n=0}^{\infty} \frac{1}{n!} K_0 \left(-\frac{i\pi}{\hbar} \right)^n J_y^n K_0^{-1} = \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{i\pi}{\hbar} \right)^n (-J_y)^n = e^{-i\pi J_y/\hbar}.$$

Hence $\Theta^2 = e^{-i\pi J_y/\hbar} K_0 e^{-i\pi J_y/\hbar} K_0 = e^{-i\pi J_y/\hbar} (K_0 e^{-i\pi J_y/\hbar} K_0^{-1}) K_0^2 = e^{-i\pi J_y/\hbar} e^{-i\pi J_y/\hbar} = e^{-i2\pi J_y/\hbar}$.

Since the 2π rotation acts on any state in the j -multiplet as $e^{-i2\pi J_y/\hbar} |j, m\rangle = (-1)^{2j} |j, m\rangle$ (the 4π periodicity of the representation), the operator identity $\Theta^2 = (-1)^{2j} I$ follows.

For $j = 1/2$,

$$\Theta^2 = (-1)^{2 \cdot \frac{1}{2}} I = -I. \quad (6.41)$$

This is the formula Chapter 4 stated without proof as (4.65), owed to this chapter. The action on the $j = 1/2$ basis follows from (6.40) with $e^{-i\pi S_y/\hbar} = e^{-i(\pi/2)\sigma_y} = -i\sigma_y$ (the Condon–Shortley phase convention, which makes all ladder matrix elements real and positive in (6.26)):

$$\Theta | +z \rangle = + | -z \rangle, \quad \Theta | -z \rangle = - | +z \rangle. \quad (6.42)$$

The Kramers degeneracy theorem (4.66) – every energy eigenstate of a time-reversal-invariant Hamiltonian is at least twofold degenerate when $\Theta^2 = -I$ – now has its rotation-theoretic foundation: every spin-1/2 doublet is a Kramers pair. The generic argument of Chapter 4’s Section 4.10 is here made concrete in the simplest representation.

Non-selective measurement and the density operator. A Stern–Gerlach measurement whose output ports are not sorted – the non-selective case of the measurement chapter’s Section 3.1 – mixes the spinor’s information. A system prepared in $|\psi\rangle$ and measured non-selectively along $\mathbf{n}$ emerges in the ensemble described by

$$\rho_{\text{after}} = P_{+\mathbf{n}} \rho_{\text{before}} P_{+\mathbf{n}} + P_{-\mathbf{n}} \rho_{\text{before}} P_{-\mathbf{n}}, \quad \rho_{\text{before}} = |\psi\rangle \langle \psi|, \quad (6.43)$$

where $P_{\pm\mathbf{n}} = |\pm\mathbf{n}\rangle \langle \pm\mathbf{n}|$ are the projectors onto the spin- $\mathbf{n}$ eigenstates (6.35). This is a mixed state: the off-diagonal coherence of the pure spinor is erased by the projective sum. The density operator ρ , named in the measurement chapter via the Gleason signpost (3.19), is the natural language for this bookkeeping; its full theory – partial traces, entanglement measures, mixed-state geometry inside the Bloch ball – belongs to the composite-systems chapter. Here it suffices to note that the non-selective measurement maps the Bloch vector $\mathbf{r}$ to its projection along $\mathbf{n}$: $\mathbf{r} \mapsto (\mathbf{r} \cdot \mathbf{n}) \mathbf{n}$, shrinking the sphere’s surface to a chord.



Note *The Bloch sphere is a pure-state representation: $|\mathbf{r}| = 1$ for every ray. Mixed states – convex combinations of projectors – inhabit the interior $|\mathbf{r}| < 1$, the Bloch ball. The density operator’s extension from pure to mixed states, and the partial-trace operation that generates reduced density operators from entangled composites, is the composite-systems chapter’s business. The Bloch vector (6.39) is the $j = 1/2$ instance of a coherent-state parametrization; its generalization to higher spin is named as a forward debt to the last chapter.*

Example: the Stern–Gerlach cascade re-read through the Bloch sphere. The sequential-SG experiments of the measurement chapter’s Sections 3.4 and 3.7 acquire a transparent geometric interpretation. Consider the paradigmatic z - x - z cascade: a $|+z\rangle$ beam enters a magnet oriented along $+x$, whose $+x$ output port feeds a magnet oriented back along $+z$. The measurement chapter computed the transmission probability as $1/2 \times 1/2 = 1/4$. On the Bloch sphere: the initial state is the north pole. The $+x$ projection maps it to the point on the equator at $+x$, collapsing the sphere to that single point; the squared overlap $|\langle +z | +x \rangle|^2 = 1/2$ is $\cos^2(\pi/4)$, the angle between z and x being $\pi/2$. The $+x$ output, now the equatorial point, is then interrogated along $+z$; the squared overlap is again $1/2$, and the product is $1/4$. The loss of certainty – the x -question destroys the z -information – is now a geometric fact: rotating the quantization axis by $\pi/2$ projects onto an orthogonal great circle, and the information residing in the original axis’s pole is spread uniformly over that circle. A second z -interrogation then selects one hemisphere, restoring the z -information at the cost of half the ensemble.

The general cascade of arbitrary axes follows the same geometry. A sequence of filters with axes $\mathbf{n}_1, \mathbf{n}_2, \dots$ transmits a fraction $\prod_k \cos^2(\Theta_k/2)$ where Θ_k is the angle between $\mathbf{n}_k$ and $\mathbf{n}_{k+1}$. The Bloch sphere reduces every sequential-SG calculation to spherical trigonometry.

This section has repaid three of the book’s oldest structural debts from the representation theory

of SU(2) alone: the half-angle law (6.37) (Chapter 2 debt #9), the spin- n orthonormal basis (6.35) (Chapter 2 debt #11), and the spinor 4π sign interpretation (6.38) (Chapter 2 debt #10). It has also discharged the Θ^2 share of Chapter 4's rotation-spectrum debt #24: the formula $\Theta^2 = (-1)^{2j}$ is now a theorem of the rotation algebra, proved for all j in (6.40) and cashed for the $j = 1/2$ doublet. And it has read the σ -commutator (3.61) of Chapter 3 as the $j = 1/2$ instance of the rotation algebra (4.32), repaying the measurement chapter's debt #19. Spin is intrinsic – a property of the particle with no coordinate representation, its algebra fixed entirely by the rotation group. But angular momentum also has an orbital form, $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, whose quantization is forced by the same algebra yet constrained by the geometry of space itself.

6.4 Orbital Angular Momentum: Spherical Harmonics, Central Potential, Parity

The spin of the preceding section is intrinsic: a property of the particle with no coordinate representation, its algebra fixed entirely by the Lie bracket of the rotation group. But angular momentum enters physics through a second, older door. A particle moving in space carries orbital angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, and the position-representation machinery of the solutions chapter – X as multiplication (5.7), P as the derivative (5.9), the canonical commutator (5.12) computed from the representations – is precisely the machinery that gives $\mathbf{L}$ its concrete differential form. This section constructs the orbital representation of the rotation algebra, solves its eigenvalue problem in spherical coordinates, derives the spherical harmonics and the central-potential reduction, and connects parity to the orbital quantum numbers.

The orbital angular momentum operator. From the position-representation generators of Chapter 5,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}, \quad L_i = \epsilon_{ijk} x_j p_k = -i\hbar \epsilon_{ijk} x_j \frac{\partial}{\partial x_k}, \quad (6.44)$$

where ϵ_{ijk} is the completely antisymmetric symbol with $\epsilon_{123} = 1$, and summation over repeated indices is implied. The operator is defined on the dense domain of smooth, square-integrable wave functions; the domain questions, like all unbounded-operator questions in this book, are honestly deferred to Appendix A.

Verification of the angular momentum algebra. That the orbital operators (6.44) satisfy the rotation algebra (4.32) is proved directly from the canonical commutator (5.12). Compute $[L_x, L_y]$ as a representative case. From the definition,

$$\begin{aligned} L_x &= y p_z - z p_y = -i\hbar(y \partial_z - z \partial_y), \\ L_y &= z p_x - x p_z = -i\hbar(z \partial_x - x \partial_z). \end{aligned} \quad (6.45)$$

The commutator expands,

$$\begin{aligned} [L_x, L_y] &= [y p_z - z p_y, z p_x - x p_z] \\ &= [y p_z, z p_x] - [y p_z, x p_z] - [z p_y, z p_x] + [z p_y, x p_z], \end{aligned} \quad (6.46)$$

and the four terms are evaluated using the commutator algebra of Chapter 3 (3.58). All coordinates

commute among themselves, $[x_i, x_j] = 0$, and all momenta commute among themselves, $[p_i, p_j] = 0$, leaving only the mixed terms. The first term: $[y p_z, z p_x] = y [p_z, z] p_x = y (-i\hbar) p_x = -i\hbar y p_x$. The last term: $[z p_y, x p_z] = x [z, p_z] p_y = x (i\hbar) p_y = i\hbar x p_y$. The middle two terms vanish because p_z commutes with x and p_y commutes with z . Hence $[L_x, L_y] = i\hbar (x p_y - y p_x) = i\hbar L_z$. The full set,

$$[L_i, L_j] = i\hbar \epsilon_{ijk} L_k, \quad (6.47)$$

completes the verification: the orbital operators satisfy exactly the same algebra as the abstract generators $\mathbf{J}$ of Chapter 4. The algebra is the physics; whether the representation is orbital or intrinsic is a question about which Hilbert space carries it.

Spherical coordinates. The orbital problem is naturally expressed in spherical coordinates (r, θ, ϕ) , with $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$. The chain rule transforms the Cartesian derivatives. For $L_z = -i\hbar(x\partial_y - y\partial_x)$, the radial and polar derivatives cancel, leaving only the azimuthal derivative. Explicitly, from $\tan \phi = y/x$ (and r, θ independent of the combination $x\partial_y - y\partial_x$), the chain rule gives

$$\partial_x = \sin \theta \cos \phi \partial_r + \frac{\cos \theta \cos \phi}{r} \partial_\theta - \frac{\sin \phi}{r \sin \theta} \partial_\phi,$$

$$\partial_y = \sin \theta \sin \phi \partial_r + \frac{\cos \theta \sin \phi}{r} \partial_\theta + \frac{\cos \phi}{r \sin \theta} \partial_\phi.$$

Forming $x\partial_y - y\partial_x$: the ∂_r terms $r \sin \theta \cos \phi (\sin \theta \sin \phi) - r \sin \theta \sin \phi (\sin \theta \cos \phi)$ cancel; the ∂_θ terms $\sin \theta \cos \theta \sin \phi \cos \phi - \sin \theta \cos \theta \sin \phi \cos \phi$ cancel; the ∂_ϕ terms give $r \sin \theta \cos \phi (\frac{\cos \phi}{r \sin \theta}) - r \sin \theta \sin \phi (-\frac{\sin \phi}{r \sin \theta}) = \cos^2 \phi + \sin^2 \phi = \partial_\phi$. Hence

$$L_z = -i\hbar \frac{\partial}{\partial \phi}, \quad (6.48)$$

a single derivative with no r or θ dependence. The squared angular momentum operator, obtained by the same chain-rule substitution of (6.44) into $L^2 = L_x^2 + L_y^2 + L_z^2$, is

$$L^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right], \quad (6.49)$$

the angular part of the Laplacian, multiplied by $-\hbar^2$.

Eigenfunctions of L_z and the integer restriction. The eigenvalue equation $L_z \Phi(\phi) = \hbar m \Phi(\phi)$ with (6.48) is $(-i\hbar d/d\phi) \Phi = \hbar m \Phi$, solved immediately:

$$\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}, \quad \int_0^{2\pi} d\phi |\Phi_m(\phi)|^2 = 1. \quad (6.50)$$

Now the topological argument. The configuration space of a particle in $\mathbb{R}^3$ is the space itself, and a wave function $\psi(\mathbf{r})$ is a single-valued function on $\mathbb{R}^3$: the phase of ψ at a point must be independent of the path taken to reach it. In spherical coordinates, the azimuthal coordinate ϕ and $\phi + 2\pi$ label the same physical point. Single-valuedness therefore demands $\Phi_m(\phi + 2\pi) = \Phi_m(\phi)$, which requires $e^{i2\pi m} = 1$, hence

$$m \in \mathbb{Z}. \quad (6.51)$$

The magnetic quantum number for orbital angular momentum is integer. Since the ladder argument of the spectrum section (6.25) established $m_{\max} = \ell$ with integer spacing between m values, the integer restriction on m forces ℓ to be integer as well: $\ell = 0, 1, 2, \dots$. This is the topological argument that

distinguishes orbital from intrinsic angular momentum. The $SU(2)$ double-valuedness – the spinor sign (6.38) of the preceding section – is excluded for orbital motion by the single-valuedness of wave functions on the physical configuration space. The spectrum theorem (6.25) permits both integer and half-integer j ; orbital angular momentum realizes only the integer sector.

Spherical harmonics. The simultaneous eigenfunctions of L^2 and L_z , denoted $\langle \theta, \phi | \ell m \rangle$ and called the **spherical harmonics**, are defined by

$$Y_\ell^m(\theta, \phi) := \langle \theta, \phi | \ell m \rangle, \quad L^2 Y_\ell^m = \hbar^2 \ell(\ell+1) Y_\ell^m, \quad L_z Y_\ell^m = \hbar m Y_\ell^m, \quad (6.52)$$

with $\ell = 0, 1, 2, \dots$ and $m = -\ell, -\ell+1, \dots, \ell$. They form an orthonormal basis for square-integrable functions on the sphere S^2 :

$$\int_{S^2} d\Omega Y_\ell^m(\theta, \phi)^* Y_{\ell'}^{m'}(\theta, \phi) = \delta_{\ell\ell'} \delta_{mm'}, \quad d\Omega = \sin \theta d\theta d\phi. \quad (6.53)$$

The spherical harmonics are constructed systematically from the highest-weight state Y_ℓ^ℓ , the unique (up to normalization) eigenfunction annihilated by the ladder operator L_+ , using the orbital realization of (6.15). In spherical coordinates,

$$L_\pm = \hbar e^{\pm i\phi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right). \quad (6.54)$$

The condition $L_+ Y_\ell^\ell = 0$ is a first-order differential equation. Since $L_z Y_\ell^\ell = \hbar \ell Y_\ell^\ell$, the ϕ -dependence is fixed: $Y_\ell^\ell(\theta, \phi) = f(\theta) e^{i\ell\phi}$. Insert this ansatz into $L_+ = \hbar e^{i\phi} (\partial_\theta + i \cot \theta \partial_\phi)$:

$$L_+(f(\theta) e^{i\ell\phi}) = \hbar e^{i\phi} [f'(\theta) + i \cot \theta (i\ell) f(\theta)] e^{i\ell\phi} = \hbar e^{i(\ell+1)\phi} [f'(\theta) - \ell \cot \theta f(\theta)] = 0.$$

Hence $f'(\theta) = \ell \cot \theta f(\theta)$, a separable first-order ODE: $df/f = \ell \cot \theta d\theta = \ell (\cos \theta / \sin \theta) d\theta$, whose integral is $\ln f = \ell \ln \sin \theta + \text{const}$, giving $f(\theta) \propto \sin^\ell \theta$. The normalization $\int_{S^2} |Y_\ell^\ell|^2 d\Omega = 1$ fixes the prefactor via the beta-function integral $\int_0^\pi \sin^{2\ell+1} \theta d\theta = 2 \frac{2^{2\ell} (\ell!)^2}{(2\ell+1)!}$, yielding

$$Y_\ell^\ell(\theta, \phi) = \frac{(-1)^\ell}{2^\ell \ell!} \sqrt{\frac{(2\ell+1)!}{4\pi}} \sin^\ell \theta e^{i\ell\phi}, \quad (6.55)$$

where the phase convention is Condon–Shortley: $(-1)^\ell$ ensures consistency with the real, positive ladder matrix elements of (6.26). Lower- m harmonics are obtained by repeated application of L_- , using the normalized action (6.26):

$$Y_\ell^{m-1}(\theta, \phi) = \frac{1}{\hbar \sqrt{\ell(\ell+1) - m(m-1)}} L_- Y_\ell^m(\theta, \phi). \quad (6.56)$$

The first few spherical harmonics, normalized and phased by Condon–Shortley:

$$\begin{aligned} Y_0^0 &= \frac{1}{\sqrt{4\pi}}, \\ Y_1^0 &= \sqrt{\frac{3}{4\pi}} \cos \theta, \quad Y_1^{\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}, \\ Y_2^0 &= \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1), \quad Y_2^{\pm 1} = \mp \sqrt{\frac{15}{8\pi}} \sin \theta \cos \theta e^{\pm i\phi}, \quad Y_2^{\pm 2} = \sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm i2\phi}. \end{aligned} \quad (6.57)$$

The general expression, stated for completeness, involves the associated Legendre functions $P_\ell^m(\cos \theta)$:

$$Y_\ell^m(\theta, \phi) = (-1)^m \sqrt{\frac{(2\ell+1)}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_\ell^m(\cos \theta) e^{im\phi}, \quad (6.58)$$

for $m \geq 0$, with $Y_\ell^{-m} = (-1)^m (Y_\ell^m)^*$ for negative m .



Note The phase factor $(-1)^m$ and the factor in the normalization are the Condon–Shortley convention, chosen to make the ladder matrix elements real and positive. Other conventions exist in the literature; this book adopts Condon–Shortley throughout, matching the ladder actions (6.26). The Y_1^0 harmonic is real and positive on the positive z -axis ($\theta = 0$), $Y_1^0(0, \phi) = \sqrt{3/4\pi} > 0$, fixing the overall sign.

The rigid rotor. A particle constrained to a sphere of radius R has moment of inertia $I = mR^2$ and Hamiltonian $H = \mathbf{L}^2/2I$. Since the spherical harmonics are the eigenfunctions of $\mathbf{L}^2$, the spectrum follows immediately:

$$H Y_\ell^m = \frac{\hbar^2 \ell(\ell+1)}{2I} Y_\ell^m, \quad E_\ell = \frac{\hbar^2 \ell(\ell+1)}{2I}, \quad \ell = 0, 1, 2, \dots, \quad (6.59)$$

each level $(2\ell+1)$ -fold degenerate. This is the spectrum of a rigid diatomic molecule in its rotational band. Transitions between levels are governed by the dipole selection rule $\Delta\ell = \pm 1$ (a consequence of the rank-1 tensor character of the electric dipole operator, developed in the Wigner–Eckart section); the absorption frequencies are $\omega_{\ell \rightarrow \ell+1} = \hbar(2\ell+2)/I$, uniformly spaced by $2\hbar/I$ for large ℓ . The carbon monoxide molecule’s $J = 1 \rightarrow 0$ rotational transition at approximately 115 GHz (wavelength 2.6 mm) is a standard astrophysical probe of cold molecular clouds. The rigid rotor is the orbital analog of the spin precession of the evolution chapter: a two-parameter system (the orientation angles θ, ϕ) whose spectrum is entirely determined by the rotation algebra.

Central potentials and the radial equation. For a particle of mass m in a spherically symmetric potential $V(r)$, the Hamiltonian

$$H = -\frac{\hbar^2}{2m} \nabla^2 + V(r) \quad (6.60)$$

commutes with $\mathbf{L}^2$ and L_z (all functions of r alone are rotation-invariant). The triple $\{H, L^2, L_z\}$ is a complete set of compatible observables in the sense of Chapter 3 (3.53); joint eigenstates are labelled by three quantum numbers (n, ℓ, m) where n is a radial quantum number. Separate the wave function in spherical coordinates,

$$\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^m(\theta, \phi), \quad (6.61)$$

and apply the Laplacian in spherical form. The angular part acts on Y_ℓ^m and produces $\hbar^2 \ell(\ell+1)/r^2$. The radial function satisfies

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \left[V(r) + \frac{\hbar^2 \ell(\ell+1)}{2mr^2} \right] R(r) = E R(r), \quad (6.62)$$

with the boundary condition that $r^2 |R(r)|^2 \rightarrow 0$ as $r \rightarrow 0$ and $r \rightarrow \infty$. The substitution $u(r) := r R(r)$ removes the first-derivative term, reducing the problem to a one-dimensional Schrodinger equation on the half-line:

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V(r) + \frac{\hbar^2 \ell(\ell+1)}{2mr^2} \right] u(r) = E u(r), \quad u(0) = 0. \quad (6.63)$$

The boundary condition $u(0) = 0$ is the regularity requirement: $R(r)$ must be finite at the origin, and $u = rR$ vanishes there. The term $\hbar^2 \ell(\ell+1)/2mr^2$ is the **centrifugal barrier**: for $\ell > 0$, it repels the particle from the origin. This is a kinematic effect of rotation – a consequence of the angular momentum, not a force – and it is the reason that $\ell > 0$ states have vanishing probability density at

$r = 0$: $R_{n\ell}(r) \sim r^\ell$ near the origin. The one-dimensional domain $[0, \infty)$ with the boundary condition $u(0) = 0$ from (6.62) makes every central-potential problem equivalent to a half-line problem with an effective potential $V_{\text{eff}}(r) = V(r) + \hbar^2 \ell(\ell + 1)/2mr^2$.

The radial equation (6.62) is the central-potential reduction. Its solution for the Coulomb potential $V(r) = -e^2/r$ is the hydrogen atom – the chapter's closing harvest. Its solution for a general short-range potential, with scattering boundary conditions, is the partial-wave expansion of the scattering chapter.

Parity of orbital states. The parity operator Π , defined in the discrete-symmetries section of Chapter 4 (4.61), acts on spatial coordinates as $\Pi : \mathbf{r} \mapsto -\mathbf{r}$. In spherical coordinates this sends $\theta \mapsto \pi - \theta$ and $\phi \mapsto \phi + \pi$. Applied to a spherical harmonic,

$$\Pi Y_\ell^m(\theta, \phi) = Y_\ell^m(\pi - \theta, \phi + \pi) = (-1)^\ell Y_\ell^m(\theta, \phi). \quad (6.64)$$

The proof uses the explicit form of Y_ℓ^m : under $\theta \mapsto \pi - \theta$, $\cos \theta \mapsto -\cos \theta$ and $\sin \theta \mapsto \sin \theta$, while under $\phi \mapsto \phi + \pi$, $e^{im\phi} \mapsto (-1)^m e^{im\phi}$; the associated Legendre function $P_\ell^m(\cos \theta)$ contains the factor $\sin^m \theta$ times a polynomial in $\cos \theta$ of parity $(-1)^{\ell-m}$, and the product of $(-1)^{\ell-m}$ from the polynomial and $(-1)^m$ from the azimuthal factor yields $(-1)^\ell$. The orbital parity is therefore $(-1)^\ell$: even- ℓ states are parity-even, odd- ℓ states are parity-odd. This connects directly to Chapter 4's Laporte rule (4.63): electric dipole transitions, mediated by the odd-parity operator $e\mathbf{r}$, connect only states of opposite parity, which for orbital states means $\Delta\ell$ odd. The dipole selection rule $\Delta\ell = \pm 1$ (with $\ell = 0 \rightarrow \ell' = 0$ forbidden) is the special case, developed quantitatively in the Wigner–Eckart section.

The CSCO for a spinless particle in a central potential is $\{H, L^2, L_z\}$. The energy $E_{n\ell}$ depends on n and ℓ but not on m : the m -degeneracy $(2\ell + 1)$ is forced by rotation invariance, by the level-local degeneracy corollary of Chapter 4 (4.59). Any additional degeneracy – the hydrogen atom's ℓ -degeneracy at fixed n , where all $\ell < n$ share the same energy – is not a consequence of $\text{SO}(3)$ alone and signals a higher symmetry.

 **Note** *The orbital angular momentum theory handles one particle in a central field. The spin of the preceding section, the orbital motion of this section, and the coupling of the two (spin–orbit) are independent pieces of the same rotation algebra. Spin–orbit coupling is a relativistic effect; it is named here as the first correction to the central-potential picture and is developed in the perturbation-theory chapter. The commutation of $\mathbf{L}$ with any function of r alone is the key: $[L_i, f(r)] = 0$ for any f , because L_i contains only angular derivatives. This fact, together with the orbital algebra (6.44), is the entire angular content of the central-potential problem.*

The orbital angular momentum operator (6.44) and its spherical-harmonic eigenfunctions (6.52) are the position-representation realization of the abstract angular-momentum algebra (4.32). The radial equation (6.62) reduces every central-potential problem to a one-dimensional half-line problem parameterized by ℓ . Before the coupling of multiple angular momenta can be taken up, an alternative construction deserves its own treatment: Schwinger's oscillator model re-derives the entire angular momentum theory from two boson modes, unifying the ladder method with the harmonic oscillator algebra of Chapter 5 and providing the natural boson language for the Clebsch–Gordan coefficients and the Racah algebra of Section 6.6.

6.5 Schwinger's Oscillator Model of Angular Momentum

The algebraic ladder method of Section 6.2 derived the complete angular momentum spectrum from the single commutator (4.32) by positivity and descent termination – a direct structural parallel of the harmonic oscillator construction of Section 5.6. The parallel is not accidental. There exists a deeper connection: the angular momentum algebra can be realised entirely within the algebra of two independent boson modes. This construction, due to Julian Schwinger (1952), re-derives the entire angular momentum theory from the canonical commutator $[a, a^\dagger] = 1$ of (5.22), unifies the ladder template of Chapter 5 with the rotation algebra of Chapter 4, and provides the natural boson language for angular momentum coupling and the Racah algebra of Section 6.6. The construction is self-contained: every result of Sections 6.2 and 6.3 is recovered from the properties of two harmonic oscillators.

Two-mode boson algebra. Introduce two independent boson modes, distinguished by a binary label:

$$[a_\mu, a_\nu^\dagger] = \delta_{\mu\nu}, \quad [a_\mu, a_\nu] = [a_\mu^\dagger, a_\nu^\dagger] = 0, \quad \mu, \nu \in \{+, -\}. \quad (6.65)$$

Each mode carries its own Fock space: $a_+^\dagger$ creates a quantum of $+$ -type, $a_-^\dagger$ creates a quantum of $-$ -type, and the two-mode vacuum $|0, 0\rangle$ is annihilated by both a_+ and a_- . The two-mode Fock states are the tensor products of the single-mode number states,

$$|n_+, n_-\rangle = \frac{(a_+^\dagger)^{n_+}}{\sqrt{n_+!}} \frac{(a_-^\dagger)^{n_-}}{\sqrt{n_-!}} |0, 0\rangle, \quad n_+, n_- = 0, 1, 2, \dots,$$

diagonalising the number operators

$$N_+ := a_+^\dagger a_+, \quad N_- := a_-^\dagger a_-, \quad N_\pm |n_+, n_-\rangle = n_\pm |n_+, n_-\rangle.$$

The total boson number is $N := N_+ + N_-$. The Fock states form a complete orthonormal basis for the two-mode Hilbert space, and every operator on that space can be expressed in terms of the boson creation and annihilation operators.

The Jordan–Schwinger map. The angular momentum operators are defined as bilinears in the boson operators:

$$\begin{aligned} J_+ &= \hbar a_+^\dagger a_-, \\ J_- &= \hbar a_-^\dagger a_+, \\ J_z &= \frac{\hbar}{2} (a_+^\dagger a_+ - a_-^\dagger a_-). \end{aligned} \quad J_i = \frac{\hbar}{2} a_\alpha^\dagger \sigma_{\alpha\beta}^i a_\beta, \quad (6.66)$$

where the compact second form uses the Pauli matrices σ^i and summation over $\alpha, \beta \in \{+, -\}$ is implied. The Cartesian components follow from $J_x = (J_+ + J_-)/2$ and $J_y = (J_+ - J_-)/(2i)$. The map (6.66) is the **Jordan–Schwinger construction**: each angular momentum component is a quadratic form in the bosons, built from the Pauli matrices – the $j = 1/2$ representation embedded in bilinears.

The verification that these operators satisfy the rotation algebra is direct and instructive. The computation uses only the boson commutators (6.65) and the standard identities $[N_+, a_+^\dagger] = a_+^\dagger$, $[N_+, a_+] = -a_+$, $[N_-, a_-^\dagger] = a_-^\dagger$, $[N_-, a_-] = -a_-$, with all cross-commutators ($+$ with $-$) vanishing.

ishing. For the ladder commutator $[J_z, J_+]$:

$$\begin{aligned}
 [J_z, J_+] &= \frac{\hbar}{2} \hbar [N_+ - N_-, a_+^\dagger a_-] \\
 &= \frac{\hbar^2}{2} ([N_+, a_+^\dagger] a_- - a_+^\dagger [N_-, a_-]) \\
 &= \frac{\hbar^2}{2} (a_+^\dagger a_- - a_+^\dagger (-a_-)) = \hbar^2 a_+^\dagger a_- = \hbar J_+.
 \end{aligned}$$

The commutator $[J_z, J_-] = -\hbar J_-$ follows by hermitian conjugation, since $J_- = J_+^\dagger$. For the ladder pair commutator:

$$\begin{aligned}
 [J_+, J_-] &= \hbar^2 [a_+^\dagger a_-, a_-^\dagger a_+] \\
 &= \hbar^2 (a_+^\dagger [a_-, a_-^\dagger a_+] + [a_+^\dagger, a_-^\dagger a_+] a_-) \\
 &= \hbar^2 (a_+^\dagger a_+ - a_-^\dagger a_-) = \hbar^2 (N_+ - N_-) = 2\hbar J_z.
 \end{aligned}$$

The two commutators $[J_z, J_\pm] = \pm\hbar J_\pm$ and $[J_+, J_-] = 2\hbar J_z$ are exactly the angular momentum algebra in the ladder basis (6.16), (6.17), which is equivalent to $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$ (4.32). The verification is complete: any quadratic boson form constructed from the Pauli matrices as in (6.66) automatically closes the rotation algebra.

Number operators and the Casimir. The number operators provide an immediate expression for the Casimir invariant. Compute the products:

$$\begin{aligned}
 J_+ J_- &= \hbar^2 a_+^\dagger a_- a_-^\dagger a_+ = \hbar^2 a_+^\dagger (N_- + 1) a_+ = \hbar^2 (N_+ N_- + N_+), \\
 J_- J_+ &= \hbar^2 a_-^\dagger a_+ a_+^\dagger a_- = \hbar^2 a_-^\dagger (N_+ + 1) a_- = \hbar^2 (N_- N_+ + N_-).
 \end{aligned}$$

Using the standard form $J^2 = J_z^2 + \frac{1}{2}(J_+ J_- + J_- J_+)$ from (6.18):

$$\begin{aligned}
 J^2 &= \frac{\hbar^2}{4} (N_+ - N_-)^2 + \frac{\hbar^2}{2} (2N_+ N_- + N_+ + N_-) \\
 &= \frac{\hbar^2}{4} (N_+^2 - 2N_+ N_- + N_-^2 + 4N_+ N_- + 2N) \\
 &= \frac{\hbar^2}{4} (N_+^2 + 2N_+ N_- + N_-^2) + \frac{\hbar^2}{2} N \\
 &= \frac{\hbar^2}{4} N^2 + \frac{\hbar^2}{2} N = \frac{\hbar^2}{2} N \left(\frac{N}{2} + 1 \right). \tag{6.67}
 \end{aligned}$$

Together with $J_z = \frac{\hbar}{2}(N_+ - N_-)$, the Casimir and the z -component are expressed purely through the occupation numbers of the two boson modes. The spectrum of angular momentum can now be read off directly from the known spectrum of the number operators ((5.24)), without any ladder termination argument.

Angular momentum states as two-mode Fock states. The Fock states $|n_+, n_- \rangle$ are simultaneous eigenstates of N_+ and N_- , hence of $N = N_+ + N_-$ and $N_+ - N_-$, and therefore of J^2 and J_z . Identify the angular momentum quantum numbers through

$$j = \frac{n_+ + n_-}{2}, \quad m = \frac{n_+ - n_-}{2},$$

so that $n_+ = j + m$ and $n_- = j - m$. The angular momentum eigenstate $|j, m \rangle$ is, up to normalisation,

the Fock state with these occupation numbers:

$$|j, m\rangle = \frac{(a_+^\dagger)^{j+m}}{\sqrt{(j+m)!}} \frac{(a_-^\dagger)^{j-m}}{\sqrt{(j-m)!}} |0, 0\rangle. \quad (6.68)$$

The spectrum follows without a ladder argument. The occupation numbers n_+, n_- are nonnegative integers, so their sum $2j$ is a nonnegative integer and their difference $2m$ is an integer with the same parity as $2j$. Hence

$$j = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots, \quad m = -j, -j+1, \dots, j-1, j,$$

identical to the result (6.25) of Section 6.2. The eigenvalues are confirmed by direct substitution:

$$J^2 |j, m\rangle = \frac{\hbar^2}{2} \cdot 2j \left(\frac{2j}{2} + 1 \right) |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle,$$

$$J_z |j, m\rangle = \frac{\hbar}{2} ((j+m) - (j-m)) |j, m\rangle = \hbar m |j, m\rangle.$$

The $(2j+1)$ -fold degeneracy at fixed j has a transparent combinatorial meaning: it is the number of ways to partition $N = 2j$ bosons between the two modes with a fixed difference $2m$. For each $m = -j, \dots, j$ there is exactly one Fock state $|j+m, j-m\rangle$, yielding $2j+1$ orthogonal states spanning the j -multiplet. The integer-half-integer dichotomy is the distinction between even and odd total boson number: j is integer when N is even, half-integer when N is odd. The boson construction therefore recovers the angular momentum superselection rule of Section 6.2 as a consequence of the discreteness of Fock-space occupation numbers: coherent superpositions of integer and half-integer j would require superpositions of states with different total boson number, which are forbidden by the superselection rule for the total number operator N .

Ladder action in the boson picture. The ladder operators of (6.66) act on the Fock states by shifting one quantum between the two modes. Using $[a_-, (a_-^\dagger)^n] = n (a_-^\dagger)^{n-1}$, which follows by induction from $[a_-, a_-^\dagger] = 1$, and $a_- |0\rangle = 0$:

$$\begin{aligned} J_+ |j, m\rangle &= \hbar a_+^\dagger a_- \frac{(a_+^\dagger)^{j+m}}{\sqrt{(j+m)!}} \frac{(a_-^\dagger)^{j-m}}{\sqrt{(j-m)!}} |0, 0\rangle \\ &= \hbar \frac{(a_+^\dagger)^{j+m}}{\sqrt{(j+m)!}} \frac{(j-m)(a_-^\dagger)^{j-m-1}}{\sqrt{(j-m)!}} a_+^\dagger |0, 0\rangle \\ &= \hbar (j-m) \frac{(a_+^\dagger)^{j+m+1}}{\sqrt{(j+m)!}} \frac{(a_-^\dagger)^{j-m-1}}{\sqrt{(j-m)!}} |0, 0\rangle \\ &= \hbar \sqrt{j-m} \sqrt{j+m+1} \frac{(a_+^\dagger)^{j+m+1}}{\sqrt{(j+m+1)!}} \frac{(a_-^\dagger)^{j-m-1}}{\sqrt{(j-m-1)!}} |0, 0\rangle \\ &= \hbar \sqrt{(j-m)(j+m+1)} |j, m+1\rangle. \end{aligned}$$

The coefficient $\sqrt{(j-m)(j+m+1)} = \sqrt{j(j+1) - m(m+1)}$ is exactly the ladder coefficient of (6.26), recovered here from the boson algebra rather than from the norm argument of Section 6.2. The lowering action $J_- |j, m\rangle = \hbar \sqrt{(j+m)(j-m+1)} |j, m-1\rangle$ follows by the same computation with $+$ and $-$ exchanged. The boson construction therefore reproduces not only the spectrum but also the precise matrix elements — including the Condon–Shortley phase convention, which is automatic

in the boson realisation because all coefficients are manifestly real and positive when the Fock states are defined with the standard normalisation.

Note *The two boson modes $a_{\pm}$ are the creation operators for the two basis states $|+z\rangle, |-z\rangle$ of the fundamental spin-1/2 representation. The Jordan–Schwinger construction expresses the statement, proved in the representation theory of $SU(2)$, that every irreducible representation of angular momentum j lives in the totally symmetric subspace of $2j$ copies of the spin-1/2 representation — the Schwinger–Bargmann–Majorana realisation. A Fock state with $n_+ = j + m$ quanta of $+$ -type and $n_- = j - m$ quanta of $-$ -type is the fully symmetrised product of n_+ factors of $|+z\rangle$ and n_- factors of $|-z\rangle$. The connection to the Schwinger boson construction of Section 6.5 and to the general representation theory of $SU(2)$ is developed in Appendix B.*

Example: the two-spin-1/2 system in the Schwinger picture. The boson language illuminates the coupled states of two spin-1/2 particles (the paradigm of Section 6.6) from a new angle. Assign each particle its own pair of boson modes: $a_{\pm}^{(1)}$ for particle 1 and $a_{\pm}^{(2)}$ for particle 2. The total angular momentum operators are the sums $J_i = J_i^{(1)} + J_i^{(2)}$, with each $J_i^{(k)}$ given by the map (6.66) applied to the k -th boson pair. The four uncoupled basis states in the boson representation are

$$|\uparrow\uparrow\rangle = a_+^{(1)\dagger} a_+^{(2)\dagger} |0\rangle, \quad |\uparrow\downarrow\rangle = a_+^{(1)\dagger} a_-^{(2)\dagger} |0\rangle, \quad |\downarrow\uparrow\rangle = a_-^{(1)\dagger} a_+^{(2)\dagger} |0\rangle, \quad |\downarrow\downarrow\rangle = a_-^{(1)\dagger} a_-^{(2)\dagger} |0\rangle,$$

each containing exactly two bosons (one per particle; total $N = 2$). The triplet and singlet states of (6.84) are the states of definite total J in this four-dimensional Fock subspace: the three symmetric combinations form the $j = 1$ triplet, and the antisymmetric combination is the $j = 0$ singlet,

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (a_+^{(1)\dagger} a_-^{(2)\dagger} - a_-^{(1)\dagger} a_+^{(2)\dagger}) |0\rangle.$$

This boson-pair wave function is the structure behind the Cooper pair problem of superconductivity: two fermions bound into a spin-singlet, zero-angular-momentum pair by an attractive interaction, expressed in the boson language that makes the pairing amplitude manifest. The full development of this connection is the business of Chapter 10.

Advantage and limitation. The advantage of the Schwinger construction is that it reduces the entire angular momentum theory to the algebra of the harmonic oscillator. Every spectral fact, every ladder coefficient, and every coupling rule can be re-derived from the boson commutator $[a, a^\dagger] = 1$ alone, without the ladder-termination argument — the termination is replaced by the discreteness of the boson number spectrum, which is the independent result of Chapter 5. The Schwinger model also provides the natural bridge to the coupling of multiple angular momenta: total angular momentum operators become sums of bilinears in multiple boson pairs, and the Clebsch–Gordan coefficients emerge as the overlaps between Fock states in different occupation-number bases. Its limitation is that the boson construction obscures the group-theoretic origin of the angular momentum spectrum in the topology of the rotation group — the 4π periodicity of spinors and the integer–half-integer dichotomy have a cleaner home in the $SU(2)$ representation theory of Section 6.1.

With the spectrum independently confirmed and the boson language established, the next task is to formalise the coupling of multiple angular momenta on the tensor product space. The two-spin-1/2 example just encountered is the simplest case; the general theory of Clebsch–Gordan coefficients, their

explicit Racah closed form, and the symmetric 3- j notation occupies the following section.

5974

6.6 Angular Momentum Coupling: CG Coefficients, Racah Form, Recurrences, and 3-j Symbols

5975

5976

The preceding sections constructed three distinct realizations of the angular momentum algebra – the intrinsic spin-1/2 representation, the orbital representation on wave functions, and the Schwinger boson model which re-derived the entire theory from two harmonic oscillators. Many physical systems carry more than one angular momentum: two electrons in an atom, each with spin; one electron with both spin and orbital motion; a nucleus with spin in the electronic cloud. The question of this section is how two angular momenta combine: given two systems with definite angular momenta j_1 and j_2 , what are the possible values of the total angular momentum? The answer requires the tensor product of Hilbert spaces, the Clebsch–Gordan decomposition of the product representation, and the explicit construction of coupled basis states. The Schwinger boson language of Section 6.5 provides the natural setting: the Racah closed form for the CG coefficients is derived in that formalism, and the 3- j symbols introduced here carry the cyclic symmetry that the boson construction makes manifest.

5977

5978

5979

5980

5981

5982

5983

5984

5985

5986

5987

Tensor product of angular momentum spaces. Let two systems carry angular momentum operators $\mathbf{J}_1$ and $\mathbf{J}_2$, each satisfying the rotation algebra (4.32) on its own Hilbert space $\mathcal{H}_1$ and $\mathcal{H}_2$, and each commuting with all operators of the other system: $[J_{1i}, J_{2j}] = 0$ for all i, j . The combined Hilbert space is the tensor product $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$, and the total angular momentum operator is the sum of the two generators – each acting nontrivially on its own factor and as the identity on the other:

5988

5989

5990

5991

5992

$$\mathbf{J} = \mathbf{J}_1 \otimes I + I \otimes \mathbf{J}_2, \quad (6.69)$$

5993

where the notation $A \otimes I$ means A acts on $\mathcal{H}_1$ and the identity on $\mathcal{H}_2$, and $I \otimes B$ means the identity on $\mathcal{H}_1$ and B on $\mathcal{H}_2$. The components of $\mathbf{J}$ satisfy the rotation algebra as well:

5994

5995

$$\begin{aligned} [J_i, J_j] &= [J_{1i} \otimes I + I \otimes J_{2i}, J_{1j} \otimes I + I \otimes J_{2j}] \\ &= [J_{1i}, J_{1j}] \otimes I + I \otimes [J_{2i}, J_{2j}] \\ &= i\hbar \epsilon_{ijk} (J_{1k} \otimes I + I \otimes J_{2k}) \\ &= i\hbar \epsilon_{ijk} J_k, \end{aligned} \quad (6.70)$$

5996

5997

5998

5999

where the cross terms $[J_{1i}, J_{2j}]$ vanish by the mutual-commutation hypothesis. The total angular momentum is therefore a valid generator of simultaneous rotations of both subsystems.

6000

6001

Two natural bases. The tensor product space $\mathcal{H}_1 \otimes \mathcal{H}_2$ admits two complete sets of compatible observables, hence two natural orthonormal bases, distinguished by which operators are diagonalized.

6002

6003

Uncoupled basis. Diagonalize $J_1^2, J_{1z}, J_2^2, J_{2z}$ – the individual angular momenta of each subsystem. The basis vectors are product states,

6004

6005

$$|j_1, m_1\rangle \otimes |j_2, m_2\rangle \equiv |j_1, m_1; j_2, m_2\rangle, \quad (6.71)$$

6006

with j_1, j_2 fixed by the subsystems and $m_1 = -j_1, \dots, j_1, m_2 = -j_2, \dots, j_2$. The dimension of this basis is $(2j_1 + 1)(2j_2 + 1)$, the product of the multiplet dimensions. The uncoupled basis is simple

6007

6008

to write down but does not diagonalize J^2 : total angular momentum is not sharp in a product state. Indeed, a product state $|j_1, m_1\rangle \otimes |j_2, m_2\rangle$ is an eigenstate of $J_z = J_{1z} \otimes I + I \otimes J_{2z}$ with eigenvalue $\hbar(m_1 + m_2)$, but it is generally not an eigenstate of J^2 .

Coupled basis. Diagonalize J_1^2, J_2^2, J^2, J_z – the individual magnitudes (which commute with J^2 and J_z) and the total angular momentum. The basis vectors are denoted

$$|j_1, j_2; J, M\rangle, \quad (6.72)$$

with J_1^2 and J_2^2 taking the fixed values j_1, j_2 , and J^2 and J_z taking the eigenvalues $\hbar^2 J(J+1)$ and $\hbar M$ respectively, where the allowed J values are the subject of the next paragraph. The coupled basis diagonalizes the total angular momentum and is the natural basis for any problem with rotation-invariant interactions between the subsystems.

The Clebsch–Gordan series. The possible total angular momentum quantum numbers J for fixed j_1 and j_2 are determined by the spectrum theorem (6.25) applied to the total operator $\mathbf{J}$ of (6.69). The result is the **Clebsch–Gordan series**:

$$J = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2. \quad (6.73)$$

The proof sketch uses highest-weight counting. The total J_z eigenvalue of a product state is $M = m_1 + m_2$. The maximum possible M in the product space is $M_{\max} = j_1 + j_2$, achieved by the single product state $|j_1, j_1\rangle \otimes |j_2, j_2\rangle$ (up to normalization). This state must be the highest-weight state of a total- J multiplet with $J = j_1 + j_2$, because no state with larger M exists. Applying the total lowering operator $J_- = J_{1-} \otimes I + I \otimes J_{2-}$ generates the full $(2J+1) = 2(j_1 + j_2) + 1$ states of that multiplet. Next, the subspace with $M = j_1 + j_2 - 1$ is two-dimensional: it is spanned by $|j_1, j_1 - 1\rangle \otimes |j_2, j_2\rangle$ and $|j_1, j_1\rangle \otimes |j_2, j_2 - 1\rangle$. One linear combination belongs to the $J = j_1 + j_2$ multiplet already constructed (it is $J_- |j_1 + j_2, j_1 + j_2\rangle$ up to normalization); the orthogonal combination is the highest-weight state of a multiplet with $J = j_1 + j_2 - 1$. The process continues: at each step down, the dimension of the M subspace increases by one until $M = |j_1 - j_2|$, after which it remains constant and then decreases. The counting stops when $J = |j_1 - j_2|$, at which point the orthogonally constructed highest-weight states exhaust the product space. The dimension check confirms the decomposition: $\sum_{J=|j_1-j_2|}^{j_1+j_2} (2J+1) = (2j_1+1)(2j_2+1)$, the dimension of the product space. The full representation-theoretic proof, via characters and Young tableaux, is collected in Appendix B.

Clebsch–Gordan coefficients. The change of basis between the uncoupled and coupled bases defines the **Clebsch–Gordan coefficients**:

$$|j_1, j_2; J, M\rangle = \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} \langle j_1, m_1; j_2, m_2 | J, M \rangle |j_1, m_1\rangle \otimes |j_2, m_2\rangle, \quad (6.74)$$

where $\langle j_1, m_1; j_2, m_2 | J, M \rangle \in \mathbb{R}$ (they can be chosen real) are the CG coefficients. They vanish unless two conditions hold:

$$m_1 + m_2 = M, \quad |j_1 - j_2| \leq J \leq j_1 + j_2. \quad (6.75)$$

The first is J_z additivity; the second is the CG series (6.73). The CG coefficients form an orthogonal

matrix (in fact, real orthogonal) connecting two orthonormal bases:

$$\sum_{m_1, m_2} \langle j_1, m_1; j_2, m_2 | J, M \rangle \langle j_1, m_1; j_2, m_2 | J', M' \rangle = \delta_{JJ'} \delta_{MM'}, \quad (6.76)$$

and the inverse relation,

$$|j_1, m_1\rangle \otimes |j_2, m_2\rangle = \sum_{J, M} \langle j_1, m_1; j_2, m_2 | J, M \rangle |j_1, j_2; J, M\rangle. \quad (6.77)$$

The Condon–Shortley phase convention, fixed in the spectrum section (6.26), extends to the CG coefficients: all CG coefficients are real, and the coefficient with the largest m_1 (or equivalently the coefficient for the highest-weight state) is taken positive, $\langle j_1, j_1; j_2, J - j_1 | J, J \rangle > 0$.

Racah closed form. The orthogonality relations define the CG coefficients implicitly as the matrix elements of a change of basis. An explicit closed-form algebraic expression was derived by Racah (1942) using the Schwinger boson construction of Section 6.5:

$$\begin{aligned} \langle j_1, m_1; j_2, m_2 | J, M \rangle &= \delta_{M, m_1 + m_2} \sqrt{(2J + 1)} \sqrt{\frac{(j_1 + j_2 - J)! (J + j_1 - j_2)! (J + j_2 - j_1)!}{(j_1 + j_2 + J + 1)!}} \\ &\times \sqrt{(j_1 + m_1)! (j_1 - m_1)! (j_2 + m_2)! (j_2 - m_2)! (J + M)! (J - M)!} \\ &\times \sum_z \frac{(-1)^z}{z! (j_1 + j_2 - J - z)! (j_1 - m_1 - z)! (j_2 + m_2 - z)! (J - j_2 + m_1 + z)! (J - j_1 - m_2 + z)!}. \end{aligned} \quad (6.78)$$

The sum over z runs over all integers for which the factorial arguments are nonnegative; the range is automatically finite, terminated by the factorial poles in the denominator. The Kronecker delta $\delta_{M, m_1 + m_2}$ enforces the J_z additivity condition, and the triangle constraint $|j_1 - j_2| \leq J \leq j_1 + j_2$ is implicit in the vanishing of the square-root prefactor when J falls outside the CG series (6.73).



Note The derivation of (6.78) via the Schwinger boson construction is sketched here, not given in full combinatorial detail. Insert the boson realisation (6.68) of the angular momentum states into the overlap $\langle j_1, m_1; j_2, m_2 | J, M \rangle$ and evaluate the resulting boson vacuum expectation value by commuting all annihilation operators to the right. The sum over z in (6.78) is the residue sum from the binomial expansions that result. The full derivation, and its generalisation to the $SU(2)$ representation theory via characters, is collected in Appendix B.

CG recurrence relations. The Racah formula (6.78) gives the CG coefficients in one closed sum, but a more practical computational route uses recurrence relations that descend directly from the ladder algebra. Act with the total ladder operators $J_{\pm} = J_{1\pm} \otimes I + I \otimes J_{2\pm}$ on both sides of the defining relation (6.74). On the left-hand side, $J_{\pm}$ acts on the coupled state using the ladder action (6.26):

$$J_{\pm} |j_1, j_2; J, M\rangle = \hbar \sqrt{J(J + 1) - M(M \pm 1)} |j_1, j_2; J, M \pm 1\rangle.$$

On the right-hand side, $J_{1\pm}$ and $J_{2\pm}$ act independently on the two factors of the uncoupled product.

6073 Insert the CG expansion (6.74) and act:

$$\begin{aligned}
 6074 \quad J_{\pm} \sum_{m'_1, m'_2} \langle j_1, m'_1; j_2, m'_2 | J, M \rangle |j_1, m'_1\rangle \otimes |j_2, m'_2\rangle \\
 6075 \quad = \sum_{m'_1, m'_2} \langle j_1, m'_1; j_2, m'_2 | J, M \rangle \left[\hbar \sqrt{j_1(j_1+1) - m'_1(m'_1 \pm 1)} |j_1, m'_1 \pm 1\rangle \otimes |j_2, m'_2\rangle \right. \\
 6076 \quad \left. + \hbar \sqrt{j_2(j_2+1) - m'_2(m'_2 \pm 1)} |j_1, m'_1\rangle \otimes |j_2, m'_2 \pm 1\rangle \right].
 \end{aligned}$$

6077 Project onto $\langle j_1, m_1 | \otimes \langle j_2, m_2 |$ and shift the summation indices: in the first term set $m'_1 \pm 1 = m_1 \Rightarrow$
 6078 $m'_1 = m_1 \mp 1$, $m'_2 = m_2$; in the second set $m'_1 = m_1$, $m'_2 \pm 1 = m_2 \Rightarrow m'_2 = m_2 \mp 1$. The ladder
 6079 coefficient arguments simplify: $j_1(j_1+1) - (m_1 \mp 1)(m_1 \mp 1 \pm 1) = j_1(j_1+1) - m_1(m_1 \mp 1) =$
 6080 $(j_1 \pm m_1)(j_1 \mp m_1 + 1)$, with the analogous simplification for j_2 . Equating to the left-hand side (6.26)
 6081 yields the recurrence:

$$\begin{aligned}
 6082 \quad \sqrt{(J \mp M)(J \pm M + 1)} \langle j_1, m_1; j_2, m_2 | J, M \pm 1 \rangle \\
 6083 \quad = \sqrt{(j_1 \pm m_1)(j_1 \mp m_1 + 1)} \langle j_1, m_1 \mp 1; j_2, m_2 | J, M \rangle \\
 6084 \quad + \sqrt{(j_2 \pm m_2)(j_2 \mp m_2 + 1)} \langle j_1, m_1; j_2, m_2 \mp 1 | J, M \rangle. \quad (6.79)
 \end{aligned}$$

6085 The upper (lower) signs correspond to J_+ (J_-). These recurrences connect every CG coefficient for
 6086 a given (j_1, j_2) to those with neighbouring m_1, m_2 values. Together with the Condon–Shortley phase
 6087 convention and the normalisation condition (the sum of squared CG coefficients for each (J, M) row is
 6088 unity), the recurrences determine all CG coefficients up to overall signs – the standard computational
 6089 route in numerical atomic and nuclear spectroscopy codes.

6090 **3- j symbols.** The Clebsch–Gordan coefficients, real and orthogonal as they are, treat the three angular
 6091 momenta (j_1, j_2, J) asymmetrically: j_1 and j_2 are coupled to produce J , which enters the CG notation
 6092 as the resultant, not as an equal partner. The more symmetric notation, introduced by Wigner, is the
 6093 **3- j symbol**:

$$6094 \quad \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} := \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}} \langle j_1, m_1; j_2, m_2 | j_3, -m_3 \rangle. \quad (6.80)$$

6095 The third angular momentum, here denoted j_3 , enters on the same footing as j_1 and j_2 . The sign and
 6096 normalisation prefactor are chosen to make the symmetry properties of the 3- j symbol as clean as
 6097 possible. The symbol vanishes unless $m_1 + m_2 + m_3 = 0$ (the J_z condition, now in the symmetric
 6098 form) and unless (j_1, j_2, j_3) satisfy the triangle inequalities $|j_1 - j_2| \leq j_3 \leq j_1 + j_2$ (and cyclic
 6099 permutations). Its symmetry properties are:

$$\begin{aligned}
 6100 \quad \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} &= \begin{pmatrix} j_2 & j_3 & j_1 \\ m_2 & m_3 & m_1 \end{pmatrix} = \begin{pmatrix} j_3 & j_1 & j_2 \\ m_3 & m_1 & m_2 \end{pmatrix} \quad (\text{cyclic invariance}), \\
 6101 \quad \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} &= (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_2 & j_1 & j_3 \\ m_2 & m_1 & m_3 \end{pmatrix} \quad (\text{odd permutation, phase } (-1)^{j_1+j_2+j_3}), \\
 6102 \quad &
 \end{aligned}$$

6103 and under simultaneous reversal of all magnetic quantum numbers,

$$6104 \quad \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}.$$

These symmetries are stated without proof; they are derived from the Racah closed form (6.78) by algebraic manipulation of the factorial sum. The cyclic symmetry reflects the invariance of the Racah sum under a cyclic relabeling of the three angular momenta; the parity phase $(-1)^{j_1+j_2+j_3}$ under odd permutations follows from the Regge symmetries of the $3nj$ -symbols. The full derivation is collected in Appendix B. These symmetries are exact and do not rely on any phase convention. The orthogonality relations of the CG coefficients translate into corresponding relations for the 3- j symbols, and the Wigner–Eckart theorem of Section 6.7 takes a particularly compact form in the 3- j convention. The 3- j symbol is the preferred bookkeeping device in advanced spectroscopic and many-body calculations, where coupling schemes involving three or more angular momenta make the cyclic symmetry indispensable.

Two spin-1/2 particles: the paradigm. The simplest and most important instance is $j_1 = j_2 = 1/2$. The product space has dimension $2 \times 2 = 4$. The uncoupled basis, written with the compact notation $|\uparrow\rangle := |\frac{1}{2}, +\frac{1}{2}\rangle$ and $|\downarrow\rangle := |\frac{1}{2}, -\frac{1}{2}\rangle$, is

$$|\uparrow\uparrow\rangle, \quad |\uparrow\downarrow\rangle, \quad |\downarrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle, \quad (6.81)$$

where the first arrow labels system 1 and the second labels system 2. The CG series (6.73) gives $J = 0, 1$ – a singlet and a triplet. The coupled basis states are constructed by the highest-weight method:

The state with maximum $M = 1$ is unique: $|1, 1\rangle = |\uparrow\uparrow\rangle$. Apply $J_- = J_{1-} \otimes I + I \otimes J_{2-}$, with the single-particle ladder actions of (6.26) giving $J_{i-}|\uparrow\rangle = \hbar|\downarrow\rangle$ and $J_{i-}|\downarrow\rangle = 0$:

$$|1, 0\rangle = \frac{1}{\hbar\sqrt{1 \cdot 2 - 0}} J_- |1, 1\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle). \quad (6.82)$$

Applying J_- again yields $|1, -1\rangle = |\downarrow\downarrow\rangle$. The triplet is complete.

The remaining state in the four-dimensional space is orthogonal to the triplet and has $M = 0$. It must be the $J = 0$ singlet:

$$|0, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (6.83)$$

The sign is fixed by the Condon–Shortley convention: $\langle 1/2, 1/2; 1/2, -1/2 | 0, 0 \rangle > 0$. Collecting the four states:

$$\begin{aligned} |1, 1\rangle &= |\uparrow\uparrow\rangle, \\ |1, 0\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \\ |1, -1\rangle &= |\downarrow\downarrow\rangle, \\ |0, 0\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \end{aligned} \quad (6.84)$$

The triplet states are symmetric under exchange of the two spins; the singlet is antisymmetric. This symmetry distinction is the physical content of the coupling: it is the reason that the two-particle wave function of identical fermions must be a product of an antisymmetric spin part (singlet) and a symmetric spatial part, or vice versa – the Pauli principle, which is the composite-systems chapter’s entry point.

The Clebsch–Gordan coefficients for $1/2 \otimes 1/2$ are collected in Table 6.1.

Table 6.1: Clebsch–Gordan coefficients $\langle 1/2, m_1; 1/2, m_2 | J, M \rangle$ for $1/2 \otimes 1/2$. Rows label the uncoupled basis (m_1, m_2) ; columns label the coupled basis (J, M) . Empty entries are zero.

(m_1, m_2)	$(1, 1)$	$(1, 0)$	$(1, -1)$	$(0, 0)$
$(\uparrow, \uparrow)$	1			
$(\uparrow, \downarrow)$		$1/\sqrt{2}$		$1/\sqrt{2}$
$(\downarrow, \uparrow)$		$1/\sqrt{2}$		$-1/\sqrt{2}$
$(\downarrow, \downarrow)$			1	

Properties of the singlet state. The singlet $|0, 0\rangle$ of (6.84) has a remarkable property: it is rotationally invariant. Under a simultaneous rotation of both spins by the same SU(2) element $U \otimes U$, the singlet is unchanged:

$$(U \otimes U) |0, 0\rangle = |0, 0\rangle, \quad (6.85)$$

for every $U \in \text{SU}(2)$. The proof is one line: $|0, 0\rangle$ is the unique state with $J = 0$, and the rotation operator $U \otimes U = e^{-i\theta(\mathbf{n} \cdot \boldsymbol{\sigma})/2} \otimes e^{-i\theta(\mathbf{n} \cdot \boldsymbol{\sigma})/2} = e^{-i\theta(\mathbf{n} \cdot (\boldsymbol{\sigma} \otimes I + I \otimes \boldsymbol{\sigma}))/2}$ acts as the identity on the $J = 0$ subspace because $e^{-i\theta \cdot 0} = 1$. The singlet is the same state in every spin basis: if Alice measures her spin along $\mathbf{n}$ and Bob measures his along the same axis $\mathbf{n}$, the two results are always opposite – perfect anticorrelation. This is the spin-singlet correlation at the heart of the Einstein–Podolsky–Rosen argument, and its full development – entanglement, Bell inequalities, and the measurement problem’s sharpest form – is the composite-systems chapter’s subject.

Note The singlet is *entangled*: it cannot be written as a single tensor product $|\psi_1\rangle \otimes |\psi_2\rangle$ of one-spin states. Any attempt to assign a definite spin state to either particle individually fails; only the pair has a definite state. This inseparability, the property that the whole is not the product of its parts, is the defining feature of entanglement. Chapter 7 develops the concept systematically: the Schmidt decomposition, the partial trace, mixed-state reduced density operators, and the Bell basis. The singlet (6.84) is the Bell state $|\Psi^-\rangle$ and is maximally entangled.

Example: hyperfine splitting of the hydrogen ground state. The hydrogen atom in its electronic ground state ($n = 1, \ell = 0$) has zero orbital angular momentum. The total angular momentum of the atom is the sum of the electron spin $\mathbf{S}_e$ ($j = 1/2$) and the nuclear (proton) spin $\mathbf{S}_p$ ($j = 1/2$):

$$\mathbf{F} = \mathbf{S}_e + \mathbf{S}_p, \quad (6.86)$$

exactly the two-spin-1/2 system. The magnetic interaction between the two spin magnetic moments is the hyperfine Hamiltonian $H_{\text{hf}} = A \mathbf{S}_e \cdot \mathbf{S}_p$, where A is a constant set by the contact probability density at the nucleus. The scalar product is expressed in terms of the Casimir operators:

$$\mathbf{S}_e \cdot \mathbf{S}_p = \frac{1}{2} (F^2 - S_e^2 - S_p^2) = \frac{\hbar^2}{2} \left[F(F+1) - \frac{3}{4} - \frac{3}{4} \right]. \quad (6.87)$$

For the triplet $F = 1$: $\mathbf{S}_e \cdot \mathbf{S}_p = \hbar^2/4$, giving $E_{F=1} = +A\hbar^2/4$. For the singlet $F = 0$: $\mathbf{S}_e \cdot \mathbf{S}_p = -3\hbar^2/4$, giving $E_{F=0} = -3A\hbar^2/4$. The hyperfine splitting is

$$\Delta E_{\text{hf}} = E_{F=1} - E_{F=0} = A\hbar^2, \quad (6.88)$$

corresponding to the famous 21 cm line: $\nu = \Delta E_{\text{hf}}/2\pi\hbar \approx 1420 \text{ MHz}$, $\lambda \approx 21 \text{ cm}$. This spectral line, produced by the singlet-to-triplet magnetic dipole transition in neutral interstellar hydrogen, is one of the most important probes in radio astronomy. The transition is forbidden by electric dipole

selection rules ($\Delta\ell = 0$, no parity change) but allowed as a magnetic dipole transition ($\Delta F = 1$), with a spontaneous lifetime of approximately 11 million years – which is why it is observed in emission from diffuse clouds rather than in terrestrial laboratories, and why its detection in 1951 (Ewen and Purcell) opened the field of 21-cm cosmology.

The hyperfine splitting is the simplest experimental repayment of the angular momentum coupling theory. The two-spin-1/2 paradigm converts the single unknown coupling constant A into a predicted doublet with a definite 3 : 1 degeneracy ratio (triplet weight 3, singlet weight 1); the measured transition frequency then fixes A from the observed line. The same CSCO logic that built the coupled basis $|F, M_F\rangle$ from the uncoupled one – via the CG coefficients of Table 6.1 – is the bookkeeping behind every atomic and nuclear hyperfine structure calculation.

Three angular momenta and the 6-j symbol. The addition of two angular momenta is unambiguous: the coupled basis $|j_1, j_2; J, M\rangle$ is the unique result of the Clebsch–Gordan series (6.73). For three angular momenta j_1, j_2, j_3 the coupling order is not unique. The total J can be reached by two distinct routes: first couple j_1 and j_2 to an intermediate j_{12} , then couple j_{12} and j_3 to J ; or first couple j_2 and j_3 to j_{23} , then couple j_1 and j_{23} to J . The two schemes produce two different coupled bases on the same tensor product space, and the overlap between them is the **recoupling coefficient**, conventionally written in the symmetric **6-j symbol**:

$$\langle (j_1, (j_2, j_3)j_{23})J, M | ((j_1, j_2)j_{12}, j_3)J, M \rangle = (-1)^{j_1+j_2+j_3+J} \sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & J & j_{23} \end{Bmatrix}. \quad (6.89)$$

The 6-j symbol is a recoupling coefficient, not a coupling coefficient: it is a single number, independent of all magnetic quantum numbers m , that encodes the geometry of three coupled angular momenta. It vanishes unless each of the four triads (j_1, j_2, j_{12}) , (j_2, j_3, j_{23}) , (j_1, J, j_{23}) , (j_{12}, j_3, J) satisfies the triangle condition. The explicit Racah closed form, the 72 Regge symmetries, and the tetrahedral classical asymptotics are collected in Appendix B.6.

Worked example: three spin-1/2's. Coupling $j_1 = j_2 = j_3 = 1/2$ to total $J = 1/2$, the intermediate quantum number is $j_{12} = 0$ or 1 in the first scheme, $j_{23} = 0$ or 1 in the second. The two $J = 1/2$ doublets built from $j_{12} = 0$ and $j_{12} = 1$ are recoupled to the two built from $j_{23} = 0, 1$ by the 6-j symbols $\begin{Bmatrix} 1/2 & 1/2 & j_{12} \\ 1/2 & 1/2 & j_{23} \end{Bmatrix}$, which take only the values $\pm 1/2$. The two coupling schemes therefore differ by a $1/\sqrt{2}$ rotation — the algebraic root of the S_4 permutation symmetry of three identical spins, and the prototype of the recoupling transformations that pervade nuclear shell-model and few-body calculations.

9-j symbols and $LS \leftrightarrow jj$ coupling. Four angular momenta — equivalently two subsystems each carrying two — admit still more coupling schemes. The conversion between coupling the two subsystems' totals (LS scheme: couple all orbital momenta to L , all spins to S , then $L + S \rightarrow J$) and coupling each subsystem's orbital and spin first (jj scheme: $l_i + s_i \rightarrow j_i$, then $j_1 + j_2 \rightarrow J$) is the **9-j**

symbol

$$\begin{Bmatrix} l_1 & s_1 & j_1 \\ l_2 & s_2 & j_2 \\ L & S & J \end{Bmatrix},$$

a sum of products of three 6- j symbols (Appendix B.6). The 9- j with one zero entry collapses to a single 6- j , the form most used in converting atomic and nuclear shell-model coupling schemes. The full Racah algebra of $3nj$ -symbols — 6- j , 9- j , and their symmetries — is the calculus of recoupling; its physical harvest is that any matrix element of a rotation-invariant operator, in any coupling scheme, reduces to a product of 6- j and 9- j symbols times a small number of reduced matrix elements.

Advantage and limitation. The advantage of the angular momentum coupling formalism is that it separates geometry from dynamics: the CG coefficients (6.74) – or, with greater symmetry, the 3- j symbols (6.80) – are fixed by the rotation group alone, encoding the purely geometric content of adding two angular momenta, while the physics – interaction strengths, energy splittings, transition amplitudes – enters only through the reduced matrix elements that the Wigner–Eckart theorem will isolate. The Racah closed form (6.78) supplies the CG coefficients in a single analytic expression valid for all ranks, and the recurrence relations (6.79) provide the practical computational route. The limitation is that this section treats only two angular momenta in full detail; the addition of three or more, encoded in the 6- j symbols named above and in the full Racah algebra, is deferred to the specialised literature and to Appendix B. The tensor product space of this section is the simplest instance of the composite-system formalism; its generalization to arbitrary Hilbert spaces and to the partial-trace operation that defines reduced states is the composite-systems chapter’s first task.

With coupled states and CG coefficients in hand, the next question is how operators – not just states – transform under rotations. The answer is the theory of irreducible tensor operators, which converts the geometric bookkeeping of this section into spectroscopic selection rules.

6.7 Irreducible Tensor Operators and the Wigner–Eckart Theorem

The angular momentum algebra of Section 6.2 classified states by their transformation under rotations: a state $|j, m\rangle$ rotates into a linear combination of the $2j + 1$ partners in its multiplet, the coefficients the Wigner D -functions.¹ Operators, however, are the objects that connect states, and their transformation under rotations determines which states they can connect — the subject of spectroscopy. This section classifies operators by their rotational rank and proves the theorem that factorizes every matrix element of every such operator into a geometric part, fixed by the rotation group, and a physical part, the reduced matrix element.

Scalar and vector operators. Operators are transformed by conjugation: under a finite rotation

¹The $D_{m'm}^{(j)}(R) := \langle j, m' | U(R) | j, m \rangle$ are developed in Appendix B; only the fact of their existence and the orthogonality of the representation-matrix elements are needed here.

$U(R) = e^{-i\theta J_n/\hbar}$ ((4.19)), an operator A goes to

$$A \mapsto U(R) A U(R)^\dagger.$$

The classification begins with the two simplest cases. A **scalar operator** S is invariant under every rotation,

$$U(R) S U(R)^\dagger = S \iff [J_i, S] = 0 \quad (i = 1, 2, 3).$$

The commutator form follows by expanding $U(R)$ to first order in the angle and reading off the coefficient of the generator. The Hamiltonian of a rotation-invariant system is a scalar; so is the Casimir J^2 of (6.18), and any function of it. A **vector operator** $\mathbf{V} = (V_1, V_2, V_3)$ transforms under rotations as a spatial vector: its Cartesian components rotate into linear combinations of themselves,

$$U(R) V_i U(R)^\dagger = \sum_j R_{ji} V_j \iff [J_i, V_j] = i\hbar \epsilon_{ijk} V_k.$$

The infinitesimal form is again the first-order expansion, and it mirrors the defining commutator of the rotation algebra itself ((4.32)). Indeed, the angular momentum vector $\mathbf{J}$ is a vector operator under rotations — the commutation relation $[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$ is exactly the vector condition. So are the position and momentum operators $\mathbf{r}$ and $\mathbf{p}$; the evolution chapter’s conjugation relation (4.29) showed that momentum transforms as a vector under rotations.

Spherical components. The Cartesian labelling hides the rotation-group structure. A vector operator $\mathbf{V}$ spans a three-dimensional representation, but that representation is reducible: it is the $j = 1$ multiplet. Its spherical components are the operators that transform irreducibly — each is an eigenoperator of J_z , and $J_\pm$ step between them:

$$V_{\pm 1}^{(1)} := \mp \frac{V_x \pm iV_y}{\sqrt{2}}, \quad V_0^{(1)} := V_z. \quad (6.90)$$

The normalization is chosen so that the transformation rules mirror those of the corresponding state $|1, q\rangle$. Specifically, applying the commutator form of Section 6.2 to the definitions,

$$[J_z, V_q^{(1)}] = \hbar q V_q^{(1)}, \quad [J_\pm, V_q^{(1)}] = \hbar \sqrt{1(1+1) - q(q \pm 1)} V_{q\pm 1}^{(1)},$$

exactly the ladder action (6.26) with $j = 1$, acting on the q -index of the operator instead of the m -index of a state.

Irreducible tensor operators. The generalization is immediate and is the content of this section. An **irreducible tensor operator** $T_q^{(k)}$ of rank k ($q = -k, -k+1, \dots, k$) is a set of $2k+1$ operators whose rotational transformation law is that of the angular momentum state $|k, q\rangle$:

$$U(R) T_q^{(k)} U(R)^\dagger = \sum_{q'=-k}^k T_{q'}^{(k)} D_{q'q}^{(k)}(R). \quad (6.91)$$

Expanding $U(R)$ to first order for an infinitesimal rotation about each axis yields the equivalent commutator form:

$$[J_z, T_q^{(k)}] = \hbar q T_q^{(k)}, \quad [J_\pm, T_q^{(k)}] = \hbar \sqrt{k(k+1) - q(q \pm 1)} T_{q\pm 1}^{(k)}. \quad (6.92)$$

An irreducible tensor operator of rank k is therefore a k -multiplet of operators: J_z measures the “magnetic” label q , and $J_\pm$ raise and lower it, exactly as they do on the states $|k, q\rangle$. The rank $k = 0$ case reduces to the scalar operator ($[J_i, T_0^{(0)}] = 0$). The rank $k = 1$ case reproduces the spherical vector

components (6.90). A rank-2 tensor operator has five components transforming as $|2, q\rangle$; the electric quadrupole operator, whose components are bilinear in the position coordinates, is the standard instance.

The definition is constructive: any operator whose commutators with $\mathbf{J}$ close in the pattern (6.92) is an irreducible tensor operator of rank k , and every physical operator can be expanded as a sum of such irreducible tensors — the operator counterpart of the Clebsch–Gordan series (6.73). The classification is worth the effort because it yields a theorem that determines every matrix element up to a single number.

The Wigner–Eckart theorem. The theorem factorizes the matrix element of an irreducible tensor operator between angular momentum eigenstates into a geometric factor, dictated wholly by the rotation group, and a physical factor — the reduced matrix element — that encodes the specific dynamics of the system:

$$\langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = \langle j, m; k, q | j', m' \rangle \frac{\langle \alpha', j' || T^{(k)} || \alpha, j \rangle}{\sqrt{2j' + 1}}. \quad (6.93)$$

Here $\langle j, m; k, q | j', m' \rangle$ is the Clebsch–Gordan coefficient for coupling $j \otimes k \rightarrow j'$, and the reduced matrix element $\langle \alpha', j' || T^{(k)} || \alpha, j \rangle$ — the double-bar notation — is independent of m, m' , and q . The auxiliary labels α, α' carry all quantum numbers not of angular momentum origin: principal quantum number, radial excitation, and any other internally conserved labels. The factor $1/\sqrt{2j' + 1}$ is a conventional normalization, adopted here so that the reduced matrix element of the identity operator ($k = 0, T_0^{(0)} = I$) is $\langle \alpha', j' || I || \alpha, j \rangle = \sqrt{2j + 1} \delta_{\alpha'\alpha}$, the number of states in the multiplet.

Proof The proof uses only the rotation-operator conjugation law and the orthogonality of the Clebsch–Gordan coefficients. The key observation is that the combination $T_q^{(k)} | \alpha, j, m \rangle$ transforms under rotations as the tensor product of two irreducible representations, $j \otimes k$, and that the Clebsch–Gordan coefficients are precisely the expansion coefficients of the coupled basis in the uncoupled one.

Apply a rotation $U(R)$ to the object $T_q^{(k)} | \alpha, j, m \rangle$. The tensor operator transforms by conjugation (6.91), and the state $| \alpha, j, m \rangle$ transforms by the representation matrix:

$$\begin{aligned} U(R) (T_q^{(k)} | \alpha, j, m \rangle) &= (U(R) T_q^{(k)} U(R)^\dagger) (U(R) | \alpha, j, m \rangle) \\ &= \sum_{q'} T_{q'}^{(k)} D_{q'q}^{(k)}(R) \sum_{m'} | \alpha, j, m' \rangle D_{m'm}^{(j)}(R) \\ &= \sum_{q', m'} T_{q'}^{(k)} | \alpha, j, m' \rangle D_{m'm}^{(j)}(R) D_{q'q}^{(k)}(R). \end{aligned}$$

The combination $T_q^{(k)} | \alpha, j, m \rangle$ therefore transforms under the product representation $D^{(j)} \otimes D^{(k)}$, whose Clebsch–Gordan decomposition (6.73) is

$$D^{(j)} \otimes D^{(k)} = \bigoplus_{J=|j-k|}^{j+k} D^{(J)}.$$

Now define, for each total J and projection M , the coupled state — the linear combination of the products $T_q^{(k)} | \alpha, j, m \rangle$ that transforms irreducibly under $D^{(J)}$:

$$| \alpha; j, k; J, M \rangle := \sum_{m, q} \langle j, m; k, q | J, M \rangle T_q^{(k)} | \alpha, j, m \rangle. \quad (6.94)$$

The defining property of the Clebsch–Gordan coefficients — they are the unitary matrix that diagonalizes the tensor product — guarantees that under a rotation,

$$U(R) |\alpha; j, k; J, M\rangle = \sum_{M'} |\alpha; j, k; J, M'\rangle D_{M'M}^{(J)}(R), \quad (6.94)$$

so that $|\alpha; j, k; J, M\rangle$ transforms as a single state of angular momentum (J, M) . Its overlap with the eigenstate $\langle\alpha', j', m'|$ is therefore the matrix element of an operator between two states of definite angular momentum, and it must be diagonal in J and M : states belonging to different irreducible representations are orthogonal under the invariant inner product. Hence

$$\langle\alpha', j', m'|\alpha; j, k; J, M\rangle = \delta_{j'J} \delta_{m'M} \langle\alpha', j' || \alpha; j, k; J\rangle, \quad (6.95)$$

where the undetermined coefficient $\langle\alpha', j' || \alpha; j, k; J\rangle$ is independent of m' and M — the content of Schur’s lemma for the rotation group.² Insert the definition (6.94) of the coupled state and use the orthogonality of the Clebsch–Gordan coefficients. The orthogonality relation for the rows of the Clebsch–Gordan matrix is

$$\sum_{J,M} \langle j, m; k, q | J, M\rangle \langle j, m'; k, q' | J, M\rangle = \delta_{mm'} \delta_{qq'}.$$

Multiply (6.95) by $\langle j, m; k, q | J, M\rangle$, sum over J and M , and apply the orthogonality:

$$\begin{aligned} \langle\alpha', j', m' | T_q^{(k)} | \alpha, j, m\rangle &= \sum_{J,M} \langle j, m; k, q | J, M\rangle \langle\alpha', j', m' | \alpha; j, k; J, M\rangle \\ &= \sum_{J,M} \langle j, m; k, q | J, M\rangle \delta_{j'J} \delta_{m'M} \langle\alpha', j' || \alpha; j, k; J\rangle \\ &= \langle j, m; k, q | j', m'\rangle \langle\alpha', j' || \alpha; j, k; j'\rangle. \end{aligned}$$

The reduced matrix element is conventionally defined by absorbing the factor into the double-bar symbol:

$$\langle\alpha', j' || T^{(k)} || \alpha, j\rangle := \sqrt{2j' + 1} \langle\alpha', j' || \alpha; j, k; j'\rangle,$$

which reproduces (6.93) and completes the proof.

The theorem is the chapter’s engine for spectroscopy. Every matrix element of every operator that carries a definite rotational rank factorizes: the Clebsch–Gordan coefficient carries all the dependence on the orientation quantum numbers m, m' , and q , while the reduced matrix element carries everything else — the radial integrals, the principal quantum numbers, the coupling constants. The two are cleanly separated, and the separation is exact.

Selection rules. The theorem’s immediate measurable content is encoded in the Clebsch–Gordan coefficient, which vanishes unless two conditions are met:

(i) $m' = m + q$ (additivity of J_z eigenvalues),
 (ii) $|j - k| \leq j' \leq j + k$ (triangle inequality for angular momentum).

(6.96)

These are the concrete spectroscopic rules that the evolution chapter’s (4.60) prepared and that govern

²Schur’s lemma in this context: any operator commuting with every $U(R)$ is a multiple of the identity on each irreducible subspace. For the matrix element between two irreducible subspaces, the lemma forces the factorised form (6.95). A proof using only the rotation-operator algebra is given in Appendix B.

every measurement of every rotation-invariant system. The content is the chapter’s harvest, ranked by rank:

- **Scalar operators** ($k = 0$). $\langle \alpha', j', m' | S | \alpha, j, m \rangle = 0$ unless $j' = j$ and $m' = m$. Scalar operators are diagonal in the angular momentum quantum numbers — they cannot change the multiplet. The Hamiltonian of a rotation-invariant system, being a scalar, has matrix elements only within each j -subspace, consistent with the multiplet degeneracy of Section 6.2.
- **Vector operators** ($k = 1$). The triangle condition reads $|j - 1| \leq j' \leq j + 1$, so a vector operator connects only states whose angular momenta differ by at most one unit:

$$\Delta j := j' - j = 0, \pm 1, \quad j = 0 \rightarrow j' = 0 \text{ forbidden.}$$

The m -selection rule gives $\Delta m = 0, \pm 1$ (corresponding to $q = 0, \pm 1$ spherical components). These are the electric dipole selection rules, the workhorse of atomic and molecular spectroscopy. The forbidden $0 \rightarrow 0$ transition is a direct consequence of the triangle inequality: both sides of $|0 - 1| \leq 0 \leq 0 + 1$ would require $1 \leq 0$, which is impossible.

- **Quadrupole operators** ($k = 2$). The condition $|j - 2| \leq j' \leq j + 2$ permits $\Delta j = 0, \pm 1, \pm 2$, with $j = 0 \rightarrow j' = 0, 1$ and $j = 1/2 \rightarrow j' = 1/2$ additionally forbidden by the triangle. Quadrupole transitions are the next allowed channel when dipole transitions are parity-forbidden, and their intensities are characteristically weaker by powers of the fine-structure constant times the square of the atomic size over the wavelength.

The selection rules are exclusionary: they declare which matrix elements are zero. Among those that survive, the Wigner–Eckart theorem supplies their relative magnitudes — the ratio of any two matrix elements within the same reduced matrix element is a ratio of Clebsch–Gordan coefficients, computable from the rotation algebra alone, without solving any dynamics.

Example: the Zeeman $p \rightarrow s$ transition. A concrete case fixes the machinery. Consider an atomic electric dipole transition from a p -state ($\ell = 1$, hence $j = 1$ for the orbital angular momentum) to an s -state ($\ell = 0, j = 0$). The electric dipole operator $e\mathbf{r}$ is a vector operator under rotations; its spherical components are $r_q^{(1)}$ with $q = -1, 0, 1$. The transition matrix elements are

$$\langle n', 0, 0 | r_q^{(1)} | n, 1, m \rangle = \langle 1, m; 1, q | 0, 0 \rangle \frac{\langle n', 0 || r^{(1)} || n, 1 \rangle}{\sqrt{1}}.$$

The Clebsch–Gordan coefficient couples $j = 1$ with $k = 1$ to $j' = 0$. Its value, from the tabulation of Section 6.6 (the special case $J = 0$ of the series (6.73)), is

$$\langle 1, m; 1, q | 0, 0 \rangle = \frac{(-1)^{1-m}}{\sqrt{3}} \delta_{m,-q},$$

so that only the three combinations $(m, q) = (1, -1), (0, 0), (-1, 1)$ are nonzero, and the magnitude $|\text{CG}| = 1/\sqrt{3}$ is identical for all three. The reduced matrix element $\langle n', 0 || r^{(1)} || n, 1 \rangle$ is a single radial integral common to all nine would-be matrix elements; only three survive the selection rules, and all three carry the same weight.

In a magnetic field $\mathbf{B} = B\hat{z}$, the Zeeman Hamiltonian $H_Z = \mu_B B L_z / \hbar$ of Section 6.8 splits the p -level into three sublevels: $E_{1,m} = E_1 + \mu_B B m$ with $m = -1, 0, 1$, while the s -level ($m' = 0$) is

unsplit. The three dipole-allowed channels produce three spectral lines:

$$\hbar\omega_{m \rightarrow 0} = (E_1 - E_0) + \mu_B B m, \quad m = -1, 0, 1,$$

the normal Zeeman triplet: one unshifted line ($\Delta m = 0$, π -polarization) and two symmetrically shifted lines ($\Delta m = \pm 1$, $\sigma^\pm$ -polarization), separated by $\mu_B B / \hbar$. The polarization labelling follows from the photon's angular momentum: the $\Delta m = 0$ transition emits a photon linearly polarized along z ; the $\Delta m = \pm 1$ transitions emit circularly polarized photons in the xy -plane. The three transitions have equal intrinsic oscillator strength — a direct prediction of the Clebsch–Gordan coefficient, with no adjustable parameters — because $|1/\sqrt{3}|^2$ is the same for all three. (The directional intensity observed in a spectrometer carries an additional geometric factor from the dipole radiation pattern: viewed perpendicular to the field and summed over polarisations, the $\Delta m = 0$ line is twice as intense as each $\Delta m = \pm 1$ line; viewed along the field, only the circularly polarised $\sigma^\pm$ components appear.) The prediction is verified in every atomic spectroscopy laboratory, and the measurement of the Zeeman splitting $\mu_B B$ is a direct determination of the Bohr magneton.

The advantage of the Wigner–Eckart theorem is its complete separation of geometry from dynamics: all the m -dependence of every transition amplitude in every rotation-invariant system is carried by the Clebsch–Gordan coefficients, which are the group-theory bookkeeping, computed once and tabulated for all ranks. Its limitation is equally plain: the theorem says nothing about the reduced matrix element, which must be computed — or measured — separately for each physical system. A vector operator's reduced matrix element between two specific levels is the integral of a radial function against the dipole operator; the theorem guarantees that once that one number is known, every matrix element of the operator between every pair of magnetic sublevels of those two multiplets is determined. For the $p \rightarrow s$ transition, the nine candidate matrix elements collapse to three nonzero values, and those three share a single reduced matrix element; two numbers — the reduced matrix element and its complex conjugate — determine the entire polarization-resolved emission spectrum.

The 3- j form. Translating the CG coefficient in (6.93) by the 3- j symbol (6.80) gives the symmetric form

$$\langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = (-1)^{j'-m'} \begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix} \langle \alpha', j' || T^{(k)} || \alpha, j \rangle, \quad (6.97)$$

in which the three angular momenta — final, operator, initial — enter on an equal footing, the m -selection rule $m' = m + q$ and the triangle $|j - k| \leq j' \leq j + k$ are immediate, and the reduced matrix element (double bar) carries all the dynamics, independent of m, m', q . This is the form used in atomic- and nuclear-structure codes; the equivalent CG form (6.93) is retained for hand calculation.

The projection theorem. Within a single j -multiplet, every vector operator's matrix elements are proportional to those of $\mathbf{J}$ itself: for a vector operator $\mathbf{V}$ ($k = 1$) diagonal in j ,

$$\langle \alpha, j, m | V_q | \alpha, j, m' \rangle = \frac{\langle \alpha, j || \mathbf{V} || \alpha, j \rangle}{\langle \alpha, j || \mathbf{J} || \alpha, j \rangle} \langle \alpha, j, m | J_q | \alpha, j, m' \rangle. \quad (6.98)$$

Both sides are rank-1 matrix elements within the same multiplet, so each is the same CG coefficient times its own reduced matrix element; the ratio of the two sides is therefore a constant, the ratio of reduced matrix elements (proof and Condon–Shortley bookkeeping in Appendix B.9). The theorem's

harvest is the Landé g -factor: the magnetic moment $\boldsymbol{\mu} = -\mu_B(\mathbf{L} + g_s\mathbf{S})/\hbar$ is a vector operator, so within a fixed- (L, S, J) multiplet $\langle \boldsymbol{\mu} \rangle = [\langle \boldsymbol{\mu} \cdot \mathbf{J} \rangle / \hbar^2 J(J+1)] \langle \mathbf{J} \rangle$, and the Landé formula $g_J = 1 + [J(J+1) + S(S+1) - L(L+1)] / 2J(J+1)$ of Chapter 8 (8.10) follows with no radial integral — one reduced matrix element fixes an entire multiplet’s worth of Zeeman splittings.

Rank-2 tensors: the electric quadrupole. The traceless symmetric part of $x_i x_j$ is a rank-2 irreducible tensor; its spherical components are proportional to the spherical harmonics Y_2^q ,

$$Q_q^{(2)} = \sqrt{\frac{16\pi}{5}} r^2 Y_2^q(\hat{\mathbf{r}}) \propto r^2 \sqrt{\frac{4\pi}{5}} Y_2^q, \quad q = -2, \dots, 2, \quad (6.99)$$

the **electric quadrupole operator**. The Wigner–Eckart selection rules (6.96) permit $\Delta j = 0, \pm 1, \pm 2$ (with $j = 1/2 \rightarrow 1/2$ and $0 \rightarrow 0, 1$ forbidden by the triangle), the next channel beyond the dipole. The quadrupole moment of a deformed nucleus or atom, $Q = \langle j, j | Q_0^{(2)} | j, j \rangle$, is the single reduced matrix element that characterizes the charge distribution’s departure from spherical symmetry; it is measured by the hyperfine splitting it induces (the quadrupole interaction $H_Q \propto \mathbf{I} \cdot \mathbf{Q}$, a scalar product of two rank-2 tensors). The general pattern: any Cartesian tensor operator, of any rank, is first decomposed into its irreducible spherical pieces (Section 6.1), and each piece is then priced by one reduced matrix element.

Spin–orbit coupling as a scalar product. The spin–orbit Hamiltonian $H_{\text{so}} = \xi(r) \mathbf{L} \cdot \mathbf{S}$ is a *scalar* under rotations — it must be, to appear in a rotation-invariant Hamiltonian — built from two vector (rank-1) operators. The scalar product of two irreducible tensors of equal rank is the $J = 0$ contraction

$$\mathbf{L} \cdot \mathbf{S} = \sum_{q=-1}^1 (-1)^q L_q^{(1)} S_{-q}^{(1)} = \sqrt{3} (L^{(1)} \otimes S^{(1)})_0^{(0)}, \quad (6.100)$$

the rank-0 component of the tensor product $L^{(1)} \otimes S^{(1)}$. Applying the Wigner–Eckart theorem to this scalar (or, equivalently, evaluating the CG coefficient for coupling two rank-1 tensors to $J = 0$) gives $\langle LSJM | \mathbf{L} \cdot \mathbf{S} | LSJM \rangle = \frac{1}{2} [J(J+1) - L(L+1) - S(S+1)] \hbar^2$, and the spin–orbit energy

$$E_J^{(\text{so})} = \frac{1}{2} A [J(J+1) - L(L+1) - S(S+1)], \quad (6.101)$$

with $A = \langle \xi \rangle(r)$ a single radial integral. The level spacing $E_J - E_{J-1} = AJ$ is the **Landé interval rule**: consecutive fine-structure levels of a given term separate in proportion to J . The rule is a direct prediction of the Wigner–Eckart theorem and is verified throughout atomic spectroscopy; the fine-structure perturbation of Chapter 8 (8.11) is its specialization to hydrogen.

Multipole expansion. Any rotation-invariant interaction of two localized charge distributions separates into multipoles through

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \sum_{\ell=0}^{\infty} \frac{4\pi}{2\ell+1} \frac{r_{<}^{\ell}}{r_{>}^{\ell+1}} \sum_{m=-\ell}^{\ell} Y_{\ell}^{m*}(\hat{\mathbf{r}}') Y_{\ell}^m(\hat{\mathbf{r}}), \quad (6.102)$$

where $r_{<} (r_{>})$ is the lesser (greater) of r, r' . Each ℓ -term is a scalar (rank-0) built from two rank- ℓ tensors coupled to $J = 0$, exactly the pattern of (6.100); the multipole moments $Q_{\ell m} = \int \rho(\mathbf{r}) r^{\ell} Y_{\ell}^m(\hat{\mathbf{r}}) d^3r$ are the irreducible-tensor components of the charge distribution, transforming under rotations by $D^{(\ell)}$ via (6.13). The $\ell = 0, 1, 2$ terms are the monopole (total charge), dipole, and quadrupole (6.99); higher multipoles follow the same pattern. The Wigner–Eckart theorem is the unifying engine: every

multipole–multipole interaction, every selection rule, and every intensity ratio in atomic, molecular, and nuclear spectroscopy is a Clebsch–Gordan coefficient times a reduced matrix element.

The chapter’s two towers — the angular momentum algebra of Sections 6.2 and 6.6, and the Wigner–Eckart engine of this section — are now complete. The algebra delivered the spectrum and the coupling rules; the Wigner–Eckart theorem converts the coupling rules into spectroscopic selection rules and relative intensities. What remains is to deploy the entire machinery on the atom that launched the quantum theory: the hydrogen atom, whose Coulomb potential is the central-potential problem par excellence, and whose spectrum is the book’s oldest unpaid debt.

6.8 The Hydrogen Atom: Fruit of Rotational Symmetry

The chapter opened by naming the rotation algebra’s debt; it built the algebra’s spectrum, its representations, and its spectroscopic engine. It now collects the harvest. The hydrogen atom — one electron of mass m_e and charge $-e$, bound to a proton of mass m_p and charge $+e$ by the Coulomb potential — is the central-potential problem whose exact solution repays the book’s oldest spectral debt: the Balmer formula of the historical chapter (Section 1.4.1). The machinery is assembled: the radial equation from Section 6.4, the angular momentum spectrum from Section 6.2, and the selection rules from Section 6.7. The solution is the chapter’s flagship.

The Coulomb problem. Separate the centre-of-mass motion, reducing the two-body problem to one body of reduced mass $\mu = m_e m_p / (m_e + m_p) \approx m_e$ moving in the Coulomb potential. In Gaussian units,

$$V(r) = -\frac{e^2}{r},$$

where $-e$ is the electron’s charge. The Hamiltonian commutes with $\mathbf{L}$, by the construction of Section 6.4: the potential is central, depending only on $r = |\mathbf{r}|$. The CSCO of Section 6.4 applies: $\{H, L^2, L_z\}$, joint eigenstates $|n, \ell, m\rangle$, with the principal quantum number n to be determined by the radial dynamics. Separate variables in $\psi(r, \theta, \phi) = R(r)Y_\ell^m(\theta, \phi)$ and substitute $u(r) = rR(r)$. The radial equation (6.62) becomes a one-dimensional Schrödinger equation on the half-line $r \in [0, \infty)$:

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left[-\frac{e^2}{r} + \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} \right] u(r) = E u(r), \quad u(0) = 0. \quad (6.103)$$

The boundary condition $u(0) = 0$ enforces regularity of $R(r)$ at the origin: the centrifugal barrier $\hbar^2 \ell(\ell+1)/2\mu r^2$ of Section 6.4 suppresses the wave function near $r = 0$ for $\ell > 0$; for every ℓ the regular solution behaves as $u(r) \sim r^{\ell+1}$.

Bound states: asymptotic analysis. Bound states have $E < 0$. Define the decay constant

$$\kappa := \frac{\sqrt{-2\mu E}}{\hbar} > 0,$$

and the dimensionless radial coordinate $\rho := 2\kappa r$. Transform the radial equation (6.103). Divide by $-\hbar^2/2\mu$:

$$\frac{d^2 u}{d\rho^2} - \frac{\ell(\ell+1)}{\rho^2} u + \frac{2\mu e^2}{\hbar^2 r} u = \underbrace{-\frac{2\mu E}{\hbar^2}}_{\kappa^2} u.$$

6476 The chain rule with $\rho = 2\kappa r$ gives $d/dr = 2\kappa d/d\rho$ and $d^2/dr^2 = 4\kappa^2 d^2/d\rho^2$. Substituting and
 6477 dividing through by $4\kappa^2$:

$$6478 \quad \frac{d^2 u}{d\rho^2} - \frac{\ell(\ell+1)}{\rho^2} u + \frac{\mu e^2}{\hbar^2 \kappa} \frac{1}{\rho} u = \frac{1}{4} u.$$

6479 Define the dimensionless parameter $\nu := \mu e^2 / \hbar^2 \kappa$, not yet quantised. The radial equation then takes
 6480 the compact form

$$6481 \quad \frac{d^2 u}{d\rho^2} - \frac{\ell(\ell+1)}{\rho^2} u + \left(\frac{\nu}{\rho} - \frac{1}{4} \right) u = 0, \quad \nu := \frac{\mu e^2}{\hbar^2 \kappa}.$$

6482 The asymptotic behaviour at the two boundaries is extracted by dropping subdominant terms. As
 6483 $\rho \rightarrow \infty$, the equation reduces to $u'' \approx u/4$, whose solutions are $u \sim e^{\pm\rho/2}$; normalisability selects
 6484 the decaying exponential $u \sim e^{-\rho/2}$. As $\rho \rightarrow 0$, the centrifugal term dominates, $u'' \approx \ell(\ell+1)u/\rho^2$,
 6485 whose regular solution is $u \sim \rho^{\ell+1}$ (the irregular branch $u \sim \rho^{-\ell}$ is rejected by $u(0) = 0$). Extract
 6486 both factors:

$$6487 \quad u(\rho) = \rho^{\ell+1} e^{-\rho/2} v(\rho). \quad (6.104)$$

6488 **Power-series solution.** Expand the remaining function in a power series:

$$6489 \quad v(\rho) = \sum_{k=0}^{\infty} c_k \rho^k, \quad c_0 \neq 0.$$

6490 Substitute $u(\rho) = \rho^{\ell+1} e^{-\rho/2} v(\rho)$ into the radial equation. Compute the derivatives. Let $A(\rho) :=$
 6491 $\frac{\ell+1}{\rho} - \frac{1}{2}$; then

$$6492 \quad u' = \rho^{\ell+1} e^{-\rho/2} (Av + v'),$$

$$6493 \quad u'' = \rho^{\ell+1} e^{-\rho/2} [(A' + A^2)v + 2Av' + v''].$$

6494 With $A' = -(\ell+1)/\rho^2$ and $A^2 = (\ell+1)^2/\rho^2 - (\ell+1)/\rho + 1/4$, the combination simplifies:

$$6495 \quad A' + A^2 = \frac{\ell(\ell+1)}{\rho^2} - \frac{\ell+1}{\rho} + \frac{1}{4}.$$

6496 Insert u'' into $u'' - \frac{\ell(\ell+1)}{\rho^2} u + (\frac{\nu}{\rho} - \frac{1}{4})u = 0$. The terms $\frac{\ell(\ell+1)}{\rho^2}$ and $\frac{1}{4}$ cancel identically, leaving

$$6497 \quad v'' + 2\left(\frac{\ell+1}{\rho} - \frac{1}{2}\right)v' + \frac{\nu - \ell - 1}{\rho} v = 0.$$

6498 Multiplying by ρ yields the differential equation for v :

$$6499 \quad \rho v'' + (2\ell + 2 - \rho)v' + (\nu - \ell - 1)v = 0.$$

6500 Now insert the power-series ansatz $v = \sum_k c_k \rho^k$, $v' = \sum_k k c_k \rho^{k-1}$, $v'' = \sum_k k(k-1) c_k \rho^{k-2}$.

6501 Collecting the coefficient of ρ^{k-1} (shifting the ρ^k terms with $k \rightarrow k-1$):

$$6502 \quad [k(k-1) + (2\ell+2)k]c_k + [\nu - \ell - 1 - (k-1)]c_{k-1} = 0.$$

6503 The bracket $k(k-1) + (2\ell+2)k = k(k+2\ell+1)$; the second bracket is $\nu - \ell - k$. Hence

$$6504 \quad k(k+2\ell+1)c_k = (k+\ell-\nu)c_{k-1}, \quad k \geq 1.$$

6505 Shifting the index $k \rightarrow k+1$ gives the two-term recurrence:

$$6506 \quad c_{k+1} = \frac{k+\ell+1-\nu}{(k+1)(k+2\ell+2)} c_k, \quad k = 0, 1, 2, \dots \quad (6.105)$$

6507 The asymptotic behaviour of the series at large k determines whether $u(\rho)$ is normalisable. For $k \gg 1$,

the ratio $c_{k+1}/c_k \sim 1/k$, which is the same as the series for $e^\rho = \sum_k \rho^k/k!$. Hence, unless the series terminates, $v(\rho) \sim e^\rho$ at large ρ , producing $u(\rho) \sim \rho^{\ell+1}e^{\rho/2}$, which diverges exponentially and is not square-integrable.

Quantisation. The series must therefore terminate at some finite $k = n_r \geq 0$, where n_r is a nonnegative integer. The recurrence (6.105) gives $c_{n_r+1} = 0$, which forces the numerator to vanish:

$$n_r + \ell + 1 - \nu = 0 \implies \nu = n_r + \ell + 1 =: n. \quad (6.106)$$

The parameter ν is therefore an integer $n = 1, 2, 3, \dots$, called the **principal quantum number**. Since $n_r \geq 0$, the orbital angular momentum is bounded by $\ell \leq n - 1$, so for each n the allowed ℓ -values are

$$\ell = 0, 1, \dots, n - 1.$$

The quantisation condition $\nu = n$ yields the energy spectrum directly. From the definitions $\nu = \mu e^2 / \hbar^2 \kappa$ and $\kappa = \sqrt{-2\mu E} / \hbar$, substitute $\nu = n$:

$$\frac{\mu e^2}{\hbar^2 \kappa} = n \implies \kappa = \frac{\mu e^2}{\hbar^2 n}.$$

Square both sides and use $\kappa^2 = -2\mu E / \hbar^2$:

$$\frac{-2\mu E}{\hbar^2} = \frac{\mu^2 e^4}{\hbar^4 n^2} \implies E_n = -\frac{\mu e^4}{2\hbar^2} \frac{1}{n^2}.$$

Collecting the result:

$$E_n = -\frac{\mu e^4}{2\hbar^2} \frac{1}{n^2}, \quad n = 1, 2, 3, \dots, \quad \ell = 0, 1, \dots, n - 1. \quad (6.107)$$

The ground-state energy is $E_1 = -\mu e^4 / 2\hbar^2 \approx -13.6$ eV; the Bohr radius, the natural length scale of the problem, is

$$a_0 := \frac{\hbar^2}{\mu e^2} \approx 0.529 \text{ \AA},$$

so that $E_1 = -e^2 / 2a_0$. The bound-state spectrum is entirely discrete (the positive-energy continuum, $E > 0$, describes unbound scattering states, named here and owed to the scattering chapter). This is Bohr's 1913 formula, derived not from a postulate about allowed orbits but from the Schrödinger equation with the rotation algebra — the historical chapter's oldest quantitative debt, repaid.

The Balmer formula and the Rydberg constant. The photon emitted when the electron falls from level n to a lower level n' carries the energy difference:

$$\frac{1}{\lambda_{n \rightarrow n'}} = \frac{E_n - E_{n'}}{2\pi \hbar c} = R_\infty \left(\frac{1}{n'^2} - \frac{1}{n^2} \right), \quad R_\infty := \frac{\mu e^4}{4\pi \hbar^3 c}. \quad (6.108)$$

R_∞ is the **Rydberg constant**. For hydrogen ($\mu \approx m_e$), $R_H \approx 109677 \text{ cm}^{-1}$. The Balmer series ($n' = 2$) of the historical chapter's Table 1.1 is now a column of theoretical predictions:

Transition	λ_{calc} (nm)	Balmer's label
$n = 3 \rightarrow 2$	656.3	H_α
$n = 4 \rightarrow 2$	486.1	H_β
$n = 5 \rightarrow 2$	434.0	H_γ
$n = 6 \rightarrow 2$	410.2	H_δ

The agreement is to one part in 10^4 ; the residual difference — the reduced-mass correction from $\mu \neq m_e$, the relativistic fine structure, and the Lamb shift — are the later chapters' business. The Lyman series ($n' = 1$) lies in the ultraviolet; the Paschen series ($n' = 3$) in the infrared. The formula is the chapter's first landing on a measurable quantity.

Degeneracy structure. For each n , the allowed values $\ell = 0, 1, \dots, n-1$ each accommodate $2\ell + 1$ values of m . The total degeneracy of the n th level (ignoring spin) is

$$d_n = \sum_{\ell=0}^{n-1} (2\ell + 1) = n^2.$$

With the electron spin included, the degeneracy is $2n^2$. The m -degeneracy ($2\ell + 1$ within each ℓ) is forced by rotation invariance: it is the level-local corollary (4.59) at work, the m -multiplet of Section 6.2. The ℓ -degeneracy — that all $\ell < n$ share the same E_n — is of a different character. It is *accidental* in the sense of the rotation group: no three-dimensional rotation mixes states of different ℓ , yet the Coulomb potential gives them equal energy. The extra degeneracy signals a higher symmetry — the $1/r$ potential admits a conserved vector, the Runge–Lenz vector, which together with $\mathbf{L}$ generates the group $\text{SO}(4)$.³ The rotation group $\text{SO}(3)$ forces the $2\ell + 1$ degeneracy; the Coulomb group $\text{SO}(4)$ forces the n^2 degeneracy. The distinction matters: in a non-Coulomb central potential, the ℓ -degeneracy is lifted, and the levels depend on both n and ℓ , while the m -degeneracy persists.

Radial wave functions. The terminated series $v(\rho)$ is a polynomial of degree $n_r = n - \ell - 1$. The resulting radial functions are expressed in terms of the associated Laguerre polynomials $L_p^{(q)}$:

$$R_{n\ell}(r) = \left(\frac{2}{na_0}\right)^{3/2} \sqrt{\frac{(n-\ell-1)!}{2n[(n+\ell)!]}} \left(\frac{2r}{na_0}\right)^\ell e^{-r/na_0} L_{n-\ell-1}^{(2\ell+1)}\left(\frac{2r}{na_0}\right). \quad (6.109)$$

The full wave function is $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^m(\theta, \phi)$ with the spherical harmonics of Section 6.4. The normalisation is $\int_0^\infty r^2 dr \int d\Omega |\psi_{n\ell m}|^2 = 1$, the radial part satisfying $\int_0^\infty r^2 dr R_{n\ell}(r) R_{n'\ell}(r) = \delta_{nn'}$. The first few are collected in Table 6.2.

The ground state $\psi_{100}(r) = (\pi a_0^3)^{-1/2} e^{-r/a_0}$ is spherically symmetric, with zero orbital angular momentum and an exponential decay on the scale of the Bohr radius. The probability density is a diffuse cloud, not an orbit: $\langle r \rangle = 3a_0/2$, and the most probable radius is $r = a_0$ — Bohr's circular-orbit radius recovered as the mode of a distribution, not as the radius of a path. The wave function has no node; the node count of $R_{n\ell}$ is $n_r = n - \ell - 1$, consistent with the oscillation theorem for one-dimensional Sturm–Liouville problems applied to the radial equation (6.103).

Zeeman splitting: the chapter's second measurable. Place the hydrogen atom in a uniform magnetic field $\mathbf{B} = B\hat{z}$. The orbital magnetic moment couples to the field, adding to the Hamiltonian the Zeeman term:

$$H_Z = \frac{e}{2\mu c} B L_z = \mu_B B \frac{L_z}{\hbar}, \quad \mu_B := \frac{e\hbar}{2\mu c}. \quad (6.110)$$

μ_B is the Bohr magneton. The electron spin contributes an additional term $2\mu_B B S_z/\hbar$ (the factor 2

³The Runge–Lenz vector is $\mathbf{A} = (\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p})/2m - e^2 \mathbf{r}/r$; classically it points along the major axis of the Kepler ellipse, and quantum-mechanically its commutation relations with $\mathbf{L}$ close the $\text{SO}(4)$ algebra. The $\text{SO}(4)$ explanation of the n^2 degeneracy was given by Pauli (1926) and by Fock (1935). The group is named here; its algebra is developed in §B.10.

Table 6.2: Hydrogen radial wave functions for $n = 1, 2, 3$. The full wave function is $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell}^m(\theta, \phi)$. $a_0 = \hbar^2/\mu e^2$ is the Bohr radius.

n	ℓ	$R_{n\ell}(r)$
1	0	$2a_0^{-3/2} e^{-r/a_0}$
2	0	$(2a_0)^{-3/2} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}$
2	1	$(2a_0)^{-3/2} \frac{1}{\sqrt{3}} \frac{r}{a_0} e^{-r/2a_0}$
3	0	$(3a_0)^{-3/2} 2 \left(1 - \frac{2r}{3a_0} + \frac{2r^2}{27a_0^2}\right) e^{-r/3a_0}$
3	1	$(3a_0)^{-3/2} \frac{2\sqrt{2}}{3} \frac{r}{a_0} \left(1 - \frac{r}{6a_0}\right) e^{-r/3a_0}$
3	2	$(3a_0)^{-3/2} \frac{2\sqrt{2}}{3\sqrt{5}} \frac{r^2}{a_0^2} e^{-r/3a_0}$

is the spin g -factor; the full Zeeman Hamiltonian $H_Z = \mu_B B (L_z + 2S_z)/\hbar$ is named here, and its perturbation-theoretic treatment — the anomalous Zeeman effect — is the later chapter's debt). For the orbital contribution alone, L_z commutes with the Coulomb Hamiltonian (both are diagonal in the $|n, \ell, m\rangle$ basis), so the energy shifts are exact within the m -multiplet:

$$\Delta E_{n\ell m} = \mu_B B m, \quad m = -\ell, \dots, \ell. \quad (6.111)$$

Each ℓ -level splits into $2\ell + 1$ equally spaced sublevels. A transition between two levels of different ℓ therefore splits into multiple spectral lines. The Wigner–Eckart selection rules of Section 6.7 govern which lines appear.

The normal Zeeman triplet. Consider the Lyman- α line: the transition $2p \rightarrow 1s$ ($n = 2, \ell = 1 \rightarrow n = 1, \ell = 0$) at $\lambda = 121.6$ nm. The initial $2p$ level has three magnetic sublevels ($m = -1, 0, 1$); the final $1s$ level has one ($m' = 0$). In a field B , the three initial sublevels split to $E_{2,1,m} = E_2 + \mu_B B m$, while $E_{1,0,0}$ is unsplit. The dipole operator is rank $k = 1$; its spherical components $r_q^{(1)}$ connect $\ell = 1$ to $\ell = 0$ with the matrix elements computed in Section 6.7. The Wigner–Eckart theorem enforces $\Delta m = q = 0, \pm 1$, and all three surviving channels have equal CG coefficients ($1/\sqrt{3}$ in magnitude). Three spectral lines result:

$$\begin{aligned} \Delta m = -1 : \quad \hbar\omega &= (E_2 - E_1) + \mu_B B \quad (\sigma^- \text{ polarisation}), \\ \Delta m = 0 : \quad \hbar\omega &= E_2 - E_1 \quad (\pi \text{ polarisation}), \\ \Delta m = +1 : \quad \hbar\omega &= (E_2 - E_1) - \mu_B B \quad (\sigma^+ \text{ polarisation}). \end{aligned}$$

The three lines form the **normal Zeeman triplet**: one unshifted π -component and two $\sigma^\pm$ -components displaced by $\pm\mu_B B/\hbar$. The splitting is linear in B and directly measures the Bohr magneton. The three lines have equal intrinsic oscillator strength, as predicted by the equal CG coefficients of Section 6.7. The $2s \rightarrow 1s$ transition, by contrast, is forbidden: $\Delta\ell = 0$ violates the dipole selection rule $\Delta\ell = \pm 1$, and the parity Laporte rule (4.63) independently forbids it (both $2s$ and $1s$ are parity-even).

The Zeeman triplet is the chapter's second landing on a directly measurable quantity (after the Balmer series), and it cross-checks three sections: the angular momentum spectrum (6.25) supplies

the magnetic quantum numbers, the Wigner–Eckart theorem (6.93) supplies the selection rules and relative intensities, and the hydrogen spectrum (6.107) supplies the unperturbed line position. Every element of the chapter is confirmed or contradicted by the measured polarisation and splitting of a single spectral line.

The book’s oldest debt, discharged. The historical chapter opened with Balmer’s 1885 discovery: four visible hydrogen lines whose wavelengths obeyed $1/\lambda = R(1/4 - 1/n^2)$. Bohr’s 1913 model reproduced the formula by fiat — a quantisation condition on the classical orbits, with no theoretical warrant. The present chapter has derived the formula from two axioms already in the book: the Schrödinger equation (the evolution chapter’s (4.3)) and the rotation algebra (the evolution chapter’s (4.32)). The Bohr radius, the Rydberg constant, and the $n = 1, 2, \dots$ series are theorems, not postulates. The repayment is complete.

The advantage of the hydrogen solution is its completeness and exactness: every bound-state energy, every wave function, and every dipole matrix element is known in closed form, making the atom a touchstone for every approximate method. Its limitation is the one the derivation inherited: the Schrödinger equation is nonrelativistic. The fine structure — the relativistic correction to the kinetic energy, the spin-orbit coupling, and the Darwin term — splits the n^2 degeneracy on the scale of $\alpha^2 E_n \approx 10^{-4}$ eV, orders of magnitude smaller than the Coulomb energies but spectroscopically resolved.⁴ The Lamb shift, a quantum-electrodynamic effect splitting the $2s$ and $2p_{1/2}$ levels by about 1 GHz, is of a still different origin. Both belong to the book’s later chapters; the hydrogen atom solved here is the nonrelativistic, spinless, exact solution — the fruit of rotational symmetry, harvested in full.

6.9 Summary and Open Debts

The chapter’s content compresses into seven principles, one harvest, and one ledger. The book-keeping of honesty is restated with them: zero new axioms; the rotation algebra (4.32) inherited from the evolution chapter; the ladder template inherited from the oscillator chapter; the position representation inherited from the same; everything else — the spectrum, the spinor sign, the half-angle law, the spherical harmonics, the Clebsch–Gordan series, the Wigner–Eckart factorisation, the hydrogen spectrum — proved.

(R1) Rotation group. The rotation group $SO(3)$ and its universal cover $SU(2)$; the 2:1 homomorphism (6.7) as the group-theoretic origin of the spinor sign (Section 6.1). Finite rotations are the exponential map of the generators, $R_n(\theta) = e^{-i\theta J_n/\hbar}$ ((4.19)); the $SU(2)$ parametrisation $U = e^{-i(\theta/2)\mathbf{n}\cdot\boldsymbol{\sigma}}$ ((6.5)) carries half the angle of the spatial rotation it represents.

(R-X) Finite rotations. The Euler-angle parametrisation $U(\alpha, \beta, \gamma) = e^{-i\alpha J_z/\hbar} e^{-i\beta J_y/\hbar} e^{-i\gamma J_z/\hbar}$ ((6.8)) parametrises every finite rotation by three angles, with the third angle ranging to 4π in $SU(2)$.

The Wigner D-matrices $D_{m'm}^{(j)}(\alpha, \beta, \gamma) = \langle j, m' | U(\alpha, \beta, \gamma) | j, m \rangle = e^{-i\alpha m'} d_{m'm}^{(j)}(\beta) e^{-i\gamma m}$ ((6.10))

⁴The spin-orbit coupling, $\mathbf{L} \cdot \mathbf{S}$, is the first relativistic correction; it couples the orbital and spin angular momenta, requiring the coupled basis of Section 6.6, and its perturbative treatment belongs to the later chapter on approximation methods. No spin-orbit term is added to the Hamiltonian here.

are the $(2j+1)$ -dimensional irreducible unitary representations of $SU(2)$; the reduced d-matrix $d_{m'm}^{(j)}(\beta) = \langle j, m' | e^{-i\beta J_y/\hbar} | j, m \rangle$ carries the nontrivial angular dependence. For integer ℓ , the D-matrices reduce to the spherical harmonics: $D_{m0}^{(\ell)}(\alpha, \beta, 0) = \sqrt{4\pi/(2\ell+1)} Y_\ell^{m*}(\beta, \alpha)$ ((6.12)), and $D_{00}^{(\ell)}(\alpha, \beta, \gamma) = P_\ell(\cos \beta)$. The explicit $j = 1/2$ D-matrix recovers the half-angle law of (R3) directly from Euler-angle decomposition: $d^{(1/2)}(\beta) = \begin{pmatrix} \cos(\beta/2) & -\sin(\beta/2) \\ \sin(\beta/2) & \cos(\beta/2) \end{pmatrix}$ (Section 6.1).

(R2) Angular momentum spectrum. From the single commutator $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$ ((4.32)), the ladder method of the oscillator chapter yields the complete spectrum: $J^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle$ and $J_z |j, m\rangle = \hbar m |j, m\rangle$, with $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ and $m = -j, \dots, j$ ((6.25)). The ladder operators $J_\pm$ of (6.15) step within the $(2j+1)$ -dimensional multiplet; their normalised actions (6.26) close the algebra with the Casimir $J^2 = J_- J_+ + J_z^2 + \hbar J_z$ ((6.18)). The spectrum is a theorem of the algebra's positivity — no additional postulate (Section 6.2).

(R3) Spin 1/2. The $j = 1/2$ representation is the Pauli algebra: $S_i = (\hbar/2)\sigma_i$ with $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$ and $\{\sigma_i, \sigma_j\} = 2\delta_{ij}I$ ((6.32)). The spin- n ONB is derived, not assumed — $|+n\rangle = \cos(\theta/2) | +z \rangle + e^{i\phi} \sin(\theta/2) | -z \rangle$ ((6.35)) — repaying the states chapter's debts: the half-angle law $|\langle +z | +n \rangle|^2 = \cos^2(\theta/2)$ ((6.37)), the spinor 4π sign ((6.38)), and the Stern–Gerlach two-state (Section 6.3). The Bloch sphere $\mathbf{r} = \langle \psi | \boldsymbol{\sigma} | \psi \rangle$ ((6.39)) is the state space S^2 ; time reversal is $\Theta = e^{-i\pi J_y/\hbar} K_0$ with $\Theta^2 = (-1)^{2j} I$ ((6.40)), proving $\Theta^2 = -I$ for half-integer spin and discharging the evolution chapter's outstanding clause.

(R4) Orbital angular momentum. $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ in the position representation ((6.44)) satisfies the same algebra; single-valuedness of wave functions on $\mathbb{R}^3$ forces integer ℓ ((6.48)). The eigenfunctions are the spherical harmonics $Y_\ell^m(\theta, \phi)$ ((6.52)), whose parity is $(-1)^\ell$ ((6.64)). Any central potential reduces to the radial equation (6.62) with the centrifugal barrier $\hbar^2 \ell(\ell+1)/2\mu r^2$ and the substitution $u(r) = rR(r)$; the CSCO is $\{H, L^2, L_z\}$ (Section 6.4).

(R-Y) Schwinger oscillator model. Two independent boson modes a_+, a_- with $[a_\mu, a_\nu^\dagger] = \delta_{\mu\nu}$ ((6.65)) generate the entire angular momentum algebra via the Jordan–Schwinger map $J_i = \frac{\hbar}{2} a_\alpha^\dagger \sigma_\alpha^i a_\beta$ ((6.66)): $J_+ = \hbar a_+^\dagger a_-$, $J_- = \hbar a_-^\dagger a_+$, $J_z = \frac{\hbar}{2}(a_+^\dagger a_+ - a_-^\dagger a_-)$. The Fock states $|n_+, n_- \rangle$ diagonalise J^2 and J_z ; the quantum numbers read off directly as $j = (n_+ + n_-)/2$, $m = (n_+ - n_-)/2$, and the angular momentum states are $|j, m\rangle = (a_+^\dagger)^{j+m} (a_-^\dagger)^{j-m} |0, 0\rangle / \sqrt{(j+m)!(j-m)!}$ ((6.68)). The Casimir is $J^2 = (\hbar^2/2)N(N/2+1)$ with $N = N_+ + N_-$ ((6.67)). The Schwinger construction is the statement that every j -representation lives in the totally symmetric subspace of $2j$ spin-1/2 systems, and it provides the natural language for the Racah algebra and the boson realisation of $SU(2)$ (Section 6.5).

(R5) Coupling and Wigner–Eckart. Two angular momenta couple by the tensor-product sum $\mathbf{J} = \mathbf{J}_1 \otimes I + I \otimes \mathbf{J}_2$ ((6.69)); the coupled basis $|j_1, j_2; J, M\rangle$ expands in the uncoupled basis with Clebsch–Gordan coefficients ((6.74)). The Clebsch–Gordan series (6.73) ($J = |j_1 - j_2|, \dots, j_1 + j_2$) is proved by highest-weight counting; the Racah closed form (6.78) gives every coefficient; the 3- j symbol (6.80) provides the symmetric notation; and the two-spin-1/2 system yields the triplet and singlet ((6.84)). Irreducible tensor operators $T_q^{(k)}$ are the operator analogues of k -multiplets, defined by their commutators with $\mathbf{J}$ ((6.92)). The Wigner–Eckart theorem ((6.93)) factorises every matrix element into a geometric Clebsch–Gordan coefficient

and a physical reduced matrix element; its selection rules ((6.96)) are the chapter's spectroscopic engine (Sections 6.6 and 6.7).

The hydrogen harvest. The Coulomb problem $V(r) = -e^2/r$ inserted into the radial equation (6.103) is solved by power-series termination; the quantisation condition forces the principal quantum number $n = 1, 2, \dots$ with $\ell = 0, 1, \dots, n-1$, yielding the spectrum $E_n = -\mu e^4/2\hbar^2 n^2$ ((6.107)). The Balmer formula $1/\lambda = R_\infty(1/n'^2 - 1/n^2)$ ((6.108)) with $R_\infty = \mu e^4/4\pi\hbar^3 c$ repays the historical chapter's oldest spectral debt. The radial wave functions are the associated Laguerre polynomials (6.109), collected in Table 6.2. The Zeeman splitting $\Delta E = \mu_B B m$ ((6.110)) produces the normal Zeeman triplet; the Wigner–Eckart selection rules determine its polarisation and relative intensities (Section 6.8).

The chapter's integration of experiment and formalism is collected in Table 6.3, which adds the rotation rows to the dictionaries of the states, measurement, evolution, and operator-method chapters (Tables 2.1, 3.1, 4.2, and 5.1).

Table 6.3: The experiment–formalism dictionary, fifth instalment: the rotation and angular momentum rows of this chapter, extending Tables 2.1, 3.1, 4.2, and 5.1.

Experimental notion	Formal object
Rotation angle	group parameter θ
Rotation axis	generator J_n in $R_n(\theta) = e^{-i\theta J_n/\hbar}$
2π rotation sign on spinors	SU(2) double covering of SO(3) ($-I$ for half-integer spin)
Angular momentum quantum numbers	j, m eigenvalues of J^2, J_z
Multiplet degeneracy	$(2j+1)$ -dimensional irreducible representation
Spin-1/2 state	spinor in $\mathbb{C}^2$, Bloch sphere S^2
Half-angle law	SU(2) rotation: $ +\mathbf{n}\rangle = \cos(\theta/2) +z\rangle + e^{i\phi}\sin(\theta/2) -z\rangle$
Euler angles	(α, β, γ) parametrising $U(\alpha, \beta, \gamma)$
Wigner D-matrix	$D_{m'm}^{(j)}(R) = \langle j, m' U(R) j, m \rangle$
Schwinger bosons	two-mode Fock states, $j = (n_+ + n_-)/2$
Spherical harmonics	orbital eigenfunctions $ \ell m\rangle$, $L^2 Y_\ell^m = \hbar^2 \ell(\ell+1) Y_\ell^m$
Centrifugal barrier	$\hbar^2 \ell(\ell+1)/2\mu r^2$ in radial equation
Clebsch–Gordan coefficient	change-of-basis amplitude $\langle j_1, m_1; j_2, m_2 J, M \rangle$
3- j symbol	$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}$, cyclic and parity symmetries
Singlet/triplet	$J = 0$ (antisymmetric) and $J = 1$ (symmetric) two-spin states
Tensor operator rank	multiplet label k in $T_q^{(k)}$
Selection rule	Wigner–Eckart triangle $ j-k \leq j' \leq j+k$, $m' = m+q$
Reduced matrix element	physics factor $\langle \alpha', j' T^{(k)} \alpha, j \rangle$
Hydrogen spectrum	$E_n = -\mu e^4/2\hbar^2 n^2$, $n = 1, 2, \dots$
Balmer/Rydberg formula	$1/\lambda = R_\infty(1/n'^2 - 1/n^2)$, $R_\infty = \mu e^4/4\pi\hbar^3 c$
Zeeman splitting	$\Delta E = \mu_B B m$, Bohr magneton $\mu_B = e\hbar/2\mu c$

The debt ledger, balanced. The chapter was chartered as the book's structural repayment; it is time to state the balance. *Repaid here:* the Balmer/Rydberg formula (ch1, Section 1.4.1) — derived from the radial Coulomb problem, not from circular-orbit postulates; the Stern–Gerlach two-state (ch1, Section 1.3.4) — as the $j = 1/2$ representation; the half-angle law (ch2, (2.39)) — as a theorem of SU(2); the spinor 4π sign interpretation (ch2 debt #10) — as the 2π rotation in SU(2) carrying the $-I$

computed at (4.10); the spin- $\mathbf{n}$ ONB construction (ch2 debt #11) — as SU(2) rotation of the reference state $|+z\rangle$; the σ -commutator (ch3 debt #19) — as the $j = 1/2$ instance of the rotation algebra; the rotation spectrum, ladders, Θ^2 for half-integer spin, multiplet classification, and angular-momentum superselection (ch4 debt #24) — every item proved, none assumed; the ladder template for rotations, the orbital wave functions, the central-potential reduction, and the hydrogen atom (ch5 debt #27) — the ladder re-run, the spherical harmonics built, and the Coulomb problem solved in full.

Planted, and declared openly: the tensor product formalism ($J_1 \otimes I + I \otimes J_2$, Clebsch–Gordan coefficients, the coupled and uncoupled bases) to the chapter on composite systems; the density operator, named at the Bloch sphere (6.39) and exercised on the non-selective SG measurement of Section 6.3, to the same chapter’s theory of mixed states; the singlet state (6.84) and its spin-correlation language, to the entanglement section of that chapter; the Schwinger boson construction (6.66) and the boson realisation of SU(2) ((6.68)), to Appendix B and to the field-quantisation chapter’s coherent-state families; the Wigner–Eckart selection rules (6.96) and the dipole matrix elements of Sections 6.7 and 6.8, to the perturbation chapter’s Stark and Zeeman applications and the Fermi golden rule that repays the Einstein A/B coefficient debt; the hydrogen radial wave functions (6.109), to the variational and perturbative treatments of helium; the central-potential machinery of Section 6.4, to the partial-wave expansion and the scattering chapter; the Zeeman Hamiltonian (6.110), to the degenerate perturbation theory of the same; the coherent spin states and the Bloch sphere geometry, to the field-quantisation chapter; the full representation theory of SU(2) — Wigner D -functions, character theory, the general Clebsch–Gordan proof via Young tableaux — to Appendix B; and the Runge–Lenz vector with its SO(4) algebra, to the same.

The exit question. The chapter has treated angular momentum as a property of a single system — one particle’s spin, one particle’s orbital motion — and then as a sum over two subsystems whose Hilbert spaces are tensored. The two-spin-1/2 singlet state $|0, 0\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ of Section 6.6 is the book’s first encounter with a state that is not a product: it cannot be written as $|\psi_1\rangle \otimes |\psi_2\rangle$ for any choice of single-particle states. The consequence is measurable and unsettling. If Alice measures σ_z on the first spin and finds $+1$, Bob’s second spin is, with certainty, in the state $|\downarrow\rangle$ — however far apart the two spins are, and even if the measurements are spacelike separated. And this certainty holds for every measurement axis: the singlet is rotationally invariant, $\sigma_1 + \sigma_2 = 0$, so the perfect anticorrelation is independent of which component Alice chooses to measure. The state of the whole is not the product of the states of its parts; a measurement on one part instantly conditions the statistics of the other.

That is entanglement. Two systems share one Hilbert space — the tensor product of their individual spaces — and the rule for building the composite from its constituents, the classification of states into separable and entangled, and the operational consequences of non-separability for measurement and information, are the subject of the next chapter.

Chapter 6 Exercise

1. (★) (Section 6.1) Rotation operators for integer and half-integer j . Let $R_{\mathbf{n}}(\theta) = e^{-i\theta J_{\mathbf{n}}/\hbar}$ be a finite rotation about axis $\mathbf{n}$ by angle θ .

- (a) For a system with total angular momentum j , insert the complete set of J_n eigenstates and show that $R_n(\theta) |j, m\rangle = \sum_{m'} D_{m'm}^{(j)}(\theta, \mathbf{n}) |j, m'\rangle$ where the D -functions are the representation matrices.
- (b) For j integer, verify that $R_n(2\pi) = +I$ by evaluating the action on the state with $m = j$ and stepping down with $J_-^\dagger$. (Hint: each application of the ladder operator to $|j, j\rangle$ produces a real positive coefficient; conjugation by a 2π rotation brings a factor $e^{-i2\pi j} = +1$.)
- (c) For $j = 1/2$, exponentiate $R_z(\theta) = e^{-i\theta\sigma_z/2}$ directly using $\sigma_z^2 = I$ and show that $R_z(2\pi) = -I$. State why this sign changes no count of any experiment (the ray theorem, Section 2.5).
- (d) Conclude: integer- j states are periodic under 2π rotations; half-integer- j states are antiperiodic. This is the rotation-theoretic basis of the superselection rule discussed in Section 6.2.
2. (★) (Section 6.2) The ladder algebra from the rotation commutator. Starting from $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$ ((4.32)):
- (a) Define $J_\pm := J_x \pm iJ_y$. Prove that $[J_z, J_\pm] = \pm\hbar J_\pm$ ((6.15) and the step commutators).
- (b) Prove that $[J_+, J_-] = 2\hbar J_z$ ((6.17)).
- (c) Show that $J^2 := J_x^2 + J_y^2 + J_z^2$ can be written as $J^2 = J_-J_+ + J_z^2 + \hbar J_z$ and also as $J^2 = J_+J_- + J_z^2 - \hbar J_z$ ((6.18)).
- (d) Using (a) and either form of (c), prove that $[J^2, J_i] = 0$ for $i = x, y, z$, so that $\{J^2, J_z\}$ is a CSCO for a single angular momentum.
3. (★★) (Section 6.2) Spin-1 matrix representation. Construct the 3×3 matrices for J_x, J_y, J_z in the $j = 1$ multiplet.
- (a) Label the basis $|1, 1\rangle, |1, 0\rangle, |1, -1\rangle$. Using the ladder actions (6.26) with $j = 1$, write out the matrix elements of J_+ and J_- , and hence $J_x = (J_+ + J_-)/2$ and $J_y = (J_+ - J_-)/2i$.
- (b) Write J_z directly as a diagonal matrix with entries $\hbar, 0, -\hbar$.
- (c) Verify that your matrices satisfy $[J_x, J_y] = i\hbar J_z$ and its cyclic permutations, and that $J^2 = J_x^2 + J_y^2 + J_z^2 = 2\hbar^2 I_{3 \times 3}$, confirming $\lambda_1 = 1(1+1) = 2$.
- (d) Show that your J_i are related to the $\text{SO}(3)$ generators L_i of Section 6.1 by a unitary transformation. (Hint: the matrices $(L_i)_{jk} = -i\hbar\epsilon_{ijk}$ are the defining 3×3 representation; compare their commutators.)
4. (★) (Section 6.3) Pauli algebra drill.
- (a) For any two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, prove the Pauli identity
- $$(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b}) I + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}.$$
- (Hint: expand the left side using $\sigma_i\sigma_j = \delta_{ij}I + i\epsilon_{ijk}\sigma_k$, which follows from (6.32).)
- (b) Use the identity of (a) to exponentiate a spin rotation. For a unit vector $\mathbf{n}$, show that
- $$e^{-i(\theta/2)\mathbf{n} \cdot \boldsymbol{\sigma}} = \cos(\theta/2) I - i \sin(\theta/2) (\mathbf{n} \cdot \boldsymbol{\sigma}).$$
- (Hint: expand the exponential as a power series; separate even and odd terms; use $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$ from (a).)
- (c) Apply the result of (b) to a state initially $|+z\rangle$. Take $\mathbf{n} = (\sin \theta, 0, \cos \theta)$ (rotation about the y -axis). Show that the resulting state is $\cos(\theta/2) |+z\rangle + \sin(\theta/2) |-z\rangle$, recovering the

- spin- n ONB (6.35) for this special case. 6764
5. (**) (Section 6.3) The Bloch sphere. A pure spin-1/2 state can be written as $|\psi\rangle = \cos(\theta/2)|+z\rangle + e^{i\phi}\sin(\theta/2)|-z\rangle$ with $\theta \in [0, \pi]$, $\phi \in [0, 2\pi)$. 6765
- (a) Construct the density operator $\rho = |\psi\rangle\langle\psi|$ and show that it takes the form $\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$ with $\mathbf{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$, a unit vector on the sphere S^2 — the Bloch sphere ((6.39)). 6767
- (b) Compute the expectation values $\langle\sigma_x\rangle$, $\langle\sigma_y\rangle$, $\langle\sigma_z\rangle$ in the state $|\psi\rangle$ and verify that $\langle\boldsymbol{\sigma}\rangle = \mathbf{r}$. 6768
- (c) For the state $|+\mathbf{n}\rangle$ constructed from the general axis $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ by the SU(2) rotation of $|+z\rangle$ ((6.35)), evaluate $|\langle+z|+\mathbf{n}\rangle|^2$ and $|\langle-z|+\mathbf{n}\rangle|^2$ and confirm the half-angle law (6.37). 6769
- (d) A Stern–Gerlach apparatus with its field gradient along $\mathbf{n}$ measures the observable $\mathbf{n} \cdot \boldsymbol{\sigma}$. Show that in the state $|+z\rangle$, the probability of the outcome $+1$ is $\cos^2(\theta/2)$, where θ is the angle between $\mathbf{n}$ and z . Relate this to the cascade example of Section 6.3. 6770
6. (**) (Section 6.4) Spherical harmonics by the ladder method. Construct $Y_\ell^m(\theta, \phi)$ for $\ell = 1$ and $\ell = 2$ from the highest-weight states. 6771
- (a) In the position representation, $L_+ = \hbar e^{i\phi}(\partial_\theta + i\cot\theta\partial_\phi)$. Show that the highest-weight condition $L_+Y_\ell^\ell = 0$ together with $L_zY_\ell^\ell = \hbar\ell Y_\ell^\ell$ forces $Y_\ell^\ell(\theta, \phi) = c_\ell \sin^\ell\theta e^{i\ell\phi}$, and determine the normalisation constant c_ℓ from $\int d\Omega |Y_\ell^\ell|^2 = 1$. 6772
- (b) Apply $L_- = L_+^\dagger$ (or its coordinate form) repeatedly to Y_1^1 and Y_2^2 , using the normalised ladder action (6.26), to obtain Y_1^0, Y_1^{-1} and $Y_2^1, Y_2^0, Y_2^{-1}, Y_2^{-2}$. 6773
- (c) Verify explicitly that your Y_1^m are orthogonal, $\int d\Omega (Y_1^m)^* Y_1^{m'} = \delta_{mm'}$, and that their parity is $(-1)^1 = -1$ ((6.64)). 6774
- (d) Repeat the parity check for Y_2^m and confirm that the parity is $(-1)^2 = +1$. 6775
7. (***) (Section 6.5) Schwinger oscillator model drill. Let a_+, a_- be two independent boson modes with $[a_\mu, a_\nu^\dagger] = \delta_{\mu\nu}$, $[a_\mu, a_\nu] = [a_\mu^\dagger, a_\nu^\dagger] = 0$ ((6.65)). 6776
- (a) Define $J_+ = \hbar a_+^\dagger a_-$, $J_- = \hbar a_-^\dagger a_+$, $J_z = \frac{\hbar}{2}(a_+^\dagger a_+ - a_-^\dagger a_-)$. Verify by direct computation using the boson commutators that $[J_z, J_\pm] = \pm\hbar J_\pm$ and $[J_+, J_-] = 2\hbar J_z$, so that $\{J_+, J_-, J_z\}$ satisfy the angular momentum algebra. 6777
- (b) Define the number operators $N_+ = a_+^\dagger a_+$ and $N_- = a_-^\dagger a_-$, and the total $N = N_+ + N_-$. Show that $J^2 := \frac{1}{2}(J_+ J_- + J_- J_+) + J_z^2 = \frac{\hbar^2}{2}N(N/2 + 1)$ ((6.67)), and $J_z = \frac{\hbar}{2}(N_+ - N_-)$. 6778
- (c) Construct the $j = 1$ multiplet states as two-mode Fock states: use (6.68) to write $|1, 1\rangle$, $|1, 0\rangle$, $|1, -1\rangle$ in terms of $(a_+^\dagger)^{n_+}(a_-^\dagger)^{n_-}|0, 0\rangle$. State the occupation numbers (n_+, n_-) for each. 6779
- (d) Verify the ladder action: act with $J_+ = \hbar a_+^\dagger a_-$ on $|1, 0\rangle$ (expressed in the boson Fock basis) and confirm that $J_+|1, 0\rangle = \hbar\sqrt{2}|1, 1\rangle$, and similarly $J_-|1, 1\rangle = \hbar\sqrt{2}|1, 0\rangle$. Compute $\langle 1, 1|J_+|1, 0\rangle$ and $\langle 1, 0|J_-|1, 1\rangle$ directly in the boson picture and show they agree with the general ladder action (6.26). 6780
8. (***) (Section 6.6) Clebsch–Gordan coefficients for $1 \otimes 1/2$. A spin-1 particle (orbital $\ell = 1$) and a spin-1/2 particle are coupled to total angular momentum $\mathbf{J} = \mathbf{L} + \mathbf{S}$. 6781
- (a) Using the Clebsch–Gordan series (6.73), determine the possible total J values and their 6782

dimensions. Verify the dimension count: $\sum_J (2J+1) = (2 \cdot 1 + 1)(2 \cdot \frac{1}{2} + 1) = 6$.

- (b) Construct the coupled states $|J, M\rangle$ by the standard method: start from the highest-weight state $|3/2, 3/2\rangle = |1, 1\rangle \otimes |1/2, 1/2\rangle$, apply $J_- = L_- + S_-$ repeatedly to build the $J = 3/2$ quartet, and then construct the $J = 1/2$ doublet orthogonal to them. Present the results as a table of CG coefficients $\langle 1, m_\ell; 1/2, m_s | J, M \rangle$.

- (c) Verify the orthogonality of the rows (and columns) of your CG table.

- (d) For the hyperfine Hamiltonian $H_{\text{hf}} = A \mathbf{L} \cdot \mathbf{S}$ (where A is a constant), use the identity $\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(J^2 - L^2 - S^2)$ to compute the energy splitting between the $J = 3/2$ and $J = 1/2$ multiplets, expressing the answer in units of $A\hbar^2$.

9. (*** (Section 6.6) The 3- j symbol and its symmetries. The 3- j symbol is defined by (6.80):

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} := \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}} \langle j_1, m_1; j_2, m_2 | j_3, -m_3 \rangle.$$

- (a) For the case $(j_1, j_2, j_3) = (1, 1, 2)$, use the Racah formula (6.78) to compute the CG coefficients $\langle 1, m_1; 1, m_2 | 2, M \rangle$ for all m_1, m_2 with $M = m_1 + m_2$. (You may compute the four cases $(m_1, m_2) = (1, 1), (1, 0), (0, 1), (0, 0)$ and use symmetry or the ladder recurrence (6.79) for the rest.)

- (b) Convert the CG coefficients of (a) to 3- j symbols $\begin{pmatrix} 1 & 1 & 2 \\ m_1 & m_2 & m_3 \end{pmatrix}$ with $m_3 = -M$. Write out the nonzero entries.

- (c) Verify the cyclic symmetry: show that $\begin{pmatrix} 1 & 1 & 2 \\ m_1 & m_2 & m_3 \end{pmatrix}$ is unchanged under cyclic permutations $(1, 2, 3) \rightarrow (2, 3, 1) \rightarrow (3, 1, 2)$ of the columns. (Hint: use the definition of the 3- j symbol in terms of CG coefficients and the symmetry properties of the latter.)

- (d) Verify the parity symmetry: show that

$$\begin{pmatrix} 1 & 1 & 2 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} = (-1)^{1+1+2} \begin{pmatrix} 1 & 1 & 2 \\ m_1 & m_2 & m_3 \end{pmatrix} = + \begin{pmatrix} 1 & 1 & 2 \\ m_1 & m_2 & m_3 \end{pmatrix},$$

and confirm that under an odd permutation of columns the symbol also acquires the phase $(-1)^{1+1+2} = +1$, so it is fully symmetric for this triple.

10. (*** (Section 6.7) Wigner–Eckart for a vector operator in a $j = 1$ multiplet. Let $V_q^{(1)}$ ($q = -1, 0, 1$) be the spherical components of a vector operator, defined by (6.92).

- (a) Write down the Clebsch–Gordan coefficients $\langle 1, m; 1, q | 1, m' \rangle$ for all $m, m', q \in \{-1, 0, 1\}$. (You may use a standard table or derive the non-zero values from the ladder method of Section 6.6.)

- (b) The Wigner–Eckart theorem (6.93) gives

$$\langle 1, m' | V_q^{(1)} | 1, m \rangle = \langle 1, m; 1, q | 1, m' \rangle \langle 1 || V^{(1)} || 1 \rangle / \sqrt{3}.$$

List every nonzero matrix element and give its value as a multiple of the reduced matrix element $v := \langle 1 || V^{(1)} || 1 \rangle / \sqrt{3}$.

- (c) Verify that the m -selection rule $m' = m + q$ ((6.96)) is satisfied by every nonzero entry. How many of the $9 \times 3 = 27$ would-be matrix elements survive?

- (d) Show that the matrix element $\langle 1, 0 | V_0^{(1)} | 1, 0 \rangle$ is proportional to the reduced matrix element,

but that the proportionality factor vanishes — and explain why this is consistent with the $j = 0 \rightarrow j' = 0$ forbidden transition rule of Section 6.7, applied here to a different triangular condition.

11. (★ ★ ★) (Section 6.8) Hydrogen ground-state moments and the virial theorem. The hydrogen ground state is $\psi_{100}(r) = (\pi a_0^3)^{-1/2} e^{-r/a_0}$, where $a_0 = \hbar^2/\mu e^2$ is the Bohr radius.

(a) Compute the expectation values $\langle r \rangle$, $\langle r^2 \rangle$, and $\langle 1/r \rangle$ in the ground state. (Radial integrals: $\int_0^\infty r^n e^{-\alpha r} dr = n!/\alpha^{n+1}$.)

(b) Using the radial kinetic energy operator $T_r = -(\hbar^2/2\mu r^2)(d/dr)(r^2 d/dr)$ or the full $T = -\hbar^2 \nabla^2/2\mu$ in spherical coordinates, compute $\langle T \rangle$ in the ground state. (Hint: the angular part of ∇^2 annihilates the spherically symmetric ground state; use $T = -(\hbar^2/2\mu)(1/r^2)(d/dr)(r^2 d/dr)$.)

(c) Compute $\langle V \rangle$ with $V(r) = -e^2/r$. Show that the virial theorem for the Coulomb potential, $\langle T \rangle = -\frac{1}{2}\langle V \rangle$, holds exactly.

(d) Using $\langle H \rangle = \langle T \rangle + \langle V \rangle = E_1 = -\mu e^4/2\hbar^2$, verify that the values from (b) and (c) are consistent. What is the root-mean-square momentum $\sqrt{\langle p^2 \rangle}$ in the ground state?

12. (★ ★ ★) (Section 6.8) The Lyman- α Zeeman triplet. Consider the $2p \rightarrow 1s$ transition of hydrogen at $\lambda = 121.6$ nm in a magnetic field $B\hat{z}$.

(a) With the Zeeman Hamiltonian $H_Z = \mu_B B L_z/\hbar$ ((6.110)), write the energies $E_{2,1,m}$ ($m = -1, 0, 1$) and $E_{1,0,0}$ as functions of B . Determine the three transition frequencies $\omega_m = (E_{2,1,m} - E_{1,0,0})/\hbar$.

(b) The electric dipole operator is $e\mathbf{r}$, with spherical components $r_q^{(1)}$. Using the Wigner-Eckart theorem as in Section 6.7, list all allowed channels ($m \rightarrow m' = 0$ with $q = -m$) and compute the relative transition amplitudes as multiples of the reduced matrix element. Show that all three surviving amplitudes have equal magnitude.

(c) The intensity of each line is proportional to $|\langle 1s | r_q^{(1)} | 2p, m \rangle|^2$. Confirm that the three lines of the normal Zeeman triplet have equal intensity. Identify the polarisation of each line: $\Delta m = 0$ gives linear polarisation along z (π -light); $\Delta m = \pm 1$ gives circular polarisation in the xy -plane ($\sigma^\pm$ -light).

(d) Sketch the spectrum: energy (or frequency) on the horizontal axis, intensity on the vertical axis, with the three lines labelled by Δm and polarisation. If the spectrometer resolves only the total intensity (summing over polarisations), what is the observed splitting pattern?

Chapter 7 Composite Systems and Entanglement

The preceding chapter closed with a question it deliberately left for the present chapter (Section 6.9): the singlet state $|0, 0\rangle$ of two spin-1/2 particles—the antisymmetric combination

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \quad (7.1)$$

familiar from the coupling of two angular momenta ((6.84))—is not a product of one-particle states. It cannot be written as $|\psi_A\rangle \otimes |\psi_B\rangle$ for any choice of single-particle states. Yet it is a perfectly valid state of the two-particle system, and its physical consequences—perfect correlation of spin measurements along every axis, violation of Bell inequalities, the EPR dilemma—are experimentally confirmed facts about nature. The singlet is entangled, and entanglement is the subject of this chapter.

Chapters 2 through 6 built the quantum theory of a single system: its states (rays of a Hilbert space, ch2), its measurements (self-adjoint operators and the Born rule, ch3), its evolution (unitary families and the Schrödinger equation, ch4), its symmetries (the Galilei group and conservation laws, ch4), the pictures and representations that organize the description (ch5), and its angular momentum—the spectrum of rotation, the hydrogen atom, the Wigner–Eckart theorem (ch6). Each of those chapters studied one system at a time. But the universe contains more than one system, and ch6’s $\frac{1}{2} \otimes \frac{1}{2}$ coupling, with its product basis, its total angular momentum operator $\mathbf{J} = \mathbf{J}_1 \otimes I + I \otimes \mathbf{J}_2$ ((6.69)), and its Clebsch–Gordan change-of-basis ((6.74)), forces the question: what is the general structure of composite quantum systems?

This chapter answers that question. It formalizes the tensor product of Hilbert spaces as the mathematical structure that houses composite systems (Section 7.1). It develops the density operator as the language for describing subsystems of composites—the formalism that ch3 named via Gleason’s theorem ((3.19)) and ch6 used for the Bloch sphere ((6.39)) but whose full theory, including mixed states, the partial trace, and the distinction between proper and improper mixtures, has been deferred to this chapter (Section 7.2). It classifies pure bipartite states by their Schmidt rank, introduces the Bell basis as the maximally entangled orthonormal basis of two qubits, defines mixed-state separability, and quantifies entanglement via the von Neumann entropy (Section 7.3). It presents the Einstein–Podolsky–Rosen argument as the conceptual challenge that entanglement poses to the completeness of quantum mechanics, stated in the modern spin form using the singlet (Section 7.4). It derives the CHSH inequality as the testable consequence of local realism, computes its quantum-mechanical violation for the singlet—the measurable quantity $2\sqrt{2}$ that exceeds the local bound of 2—and describes the Aspect experiments that confirmed the violation (Section 7.5). And it restates the measurement problem in the language of entanglement: the von Neumann measurement scheme produces an entangled superposition of system and apparatus, and the problem’s three faces—determinate outcomes, the preferred basis, and the quantum-to-classical transition—are stated precisely, with decoherence named as the FAPP resolution and the “and/or” problem left as the deep conceptual residue (Section 7.6).

The chapter adds zero new axioms. Every display below is proved from the axioms of Chapters 2–6: the state-space axioms (S1)–(S5) of ch2, the measurement axioms (M1)–(M5) of ch3, the

evolution axiom (D1) of ch4, the Galilean algebra and symmetry principles of ch4, the pictures and representations of ch5, and the angular momentum theory of ch6. Cited without proof: Gleason’s theorem (already ch3’s signpost), the Tsirelson bound (Tsirelson 1980, footnoted), the spectral properties of partial traces, the Peres–Horodecki PPT criterion (named, not proved), and decoherence theory (named, FAPP-only, its full development a forward debt). The no-cloning theorem is proved from linearity in one paragraph (Exercise 7.7); the no-signaling theorem is proved from the partial trace in one paragraph (Section 7.5).

This is a theorem chapter in the pattern of ch6. Its harvest is a number: $|\langle S \rangle| = 2\sqrt{2} \approx 2.828$, the quantum-mechanical violation of the CHSH inequality, confirmed by Aspect and successors at over 30σ . The chapter closes the book’s structural arc. From ch1’s experimental crises through ch7’s Bell violation, the theory of nonrelativistic quantum mechanics now has its full axiomatic structure. The next question is not “what is the theory?” but “what does the theory predict for real systems?”—the perturbation, scattering, and approximation methods of ch8.

7.1 Tensor Products of Hilbert Spaces

Chapter 6 used the tensor product operationally—the two-spin-1/2 Hilbert space $\mathbb{C}^2 \otimes \mathbb{C}^2$, the product basis $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$, the total angular momentum $\mathbf{J} = \mathbf{J}_1 \otimes I + I \otimes \mathbf{J}_2$ ((6.69)), the Clebsch–Gordan coefficients as the change of basis between uncoupled and coupled bases ((6.74))—without a formal definition. This section supplies that definition: the tensor product of two Hilbert spaces, its inner product, its basis, the action of operators on it, and the Schmidt decomposition as the fundamental structural theorem for bipartite pure states.

Motivation: the singlet is not a product. The singlet state

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (7.2)$$

of ch6’s (6.84) is a sum of two product vectors with a relative minus sign. No choice of single-qubit states $|\psi_A\rangle, |\psi_B\rangle \in \mathbb{C}^2$ can produce it as $|\psi_A\rangle \otimes |\psi_B\rangle$, because the coefficient matrix of a product state has rank one, while the singlet’s coefficient matrix

$$C = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (7.3)$$

has rank two. The singlet is entangled—a state of the composite that is not a product of subsystem states. The tensor product is the arena in which such states live.

Definition of the tensor product. Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be Hilbert spaces with dimensions $d_A = \dim \mathcal{H}_A$ and $d_B = \dim \mathcal{H}_B$ (finite or countably infinite; the finite-dimensional case carries the chapter’s physics). The algebraic tensor product $\mathcal{H}_A \otimes_{\text{alg}} \mathcal{H}_B$ is the vector space spanned by formal products $|\psi_A\rangle \otimes |\phi_B\rangle$ with $|\psi_A\rangle \in \mathcal{H}_A, |\phi_B\rangle \in \mathcal{H}_B$, subject to bilinearity:

$$(\alpha |\psi_A\rangle + \beta |\psi'_A\rangle) \otimes |\phi_B\rangle = \alpha |\psi_A\rangle \otimes |\phi_B\rangle + \beta |\psi'_A\rangle \otimes |\phi_B\rangle, \quad (7.4)$$

and similarly in the second factor. A general vector in the algebraic tensor product is a finite sum of product vectors. To obtain a Hilbert space, the algebraic tensor product is completed in the norm

induced by the inner product defined below. The result is the Hilbert-space tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$:

$$\mathcal{H}_A \otimes \mathcal{H}_B := \overline{\mathcal{H}_A \otimes_{\text{alg}} \mathcal{H}_B}^{\|\cdot\|}, \quad (7.5)$$

where the completion adds limits of Cauchy sequences of finite sums. In finite dimensions the completion is automatic and every vector is a finite sum.

Basis and inner product. Let $\{|e_i\rangle\}_{i=1}^{d_A}$ be an orthonormal basis of $\mathcal{H}_A$ and $\{|f_j\rangle\}_{j=1}^{d_B}$ an orthonormal basis of $\mathcal{H}_B$. The set of product vectors

$$\{|e_i\rangle \otimes |f_j\rangle \mid i = 1, \dots, d_A, j = 1, \dots, d_B\} \quad (7.6)$$

is an orthonormal basis of $\mathcal{H}_A \otimes \mathcal{H}_B$. The inner product on product vectors is defined by

$$\langle e_i \otimes f_j \mid e_k \otimes f_l \rangle = \langle e_i \mid e_k \rangle \langle f_j \mid f_l \rangle = \delta_{ik} \delta_{jl}, \quad (7.7)$$

and extended to all vectors by (anti)linearity. The dimension of the tensor product is the product of the dimensions:

$$\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = d_A d_B. \quad (7.8)$$

This is the crucial difference from a direct sum, whose dimension is $d_A + d_B$; it is also the origin of entanglement—the composite space is vastly larger than the Cartesian product of the subsystem state spaces.

A general vector $|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ expands in the product basis as

$$|\Psi\rangle = \sum_{i=1}^{d_A} \sum_{j=1}^{d_B} c_{ij} |e_i\rangle \otimes |f_j\rangle, \quad \sum_{i,j} |c_{ij}|^2 = 1. \quad (7.9)$$

The coefficient matrix C with entries $C_{ij} = c_{ij}$ is a $d_A \times d_B$ matrix. The state $|\Psi\rangle$ is a product state $|\Psi\rangle = |\psi_A\rangle \otimes |\phi_B\rangle$ if and only if the coefficient matrix has rank one: $c_{ij} = a_i b_j$ with $|\psi_A\rangle = \sum_i a_i |e_i\rangle$ and $|\phi_B\rangle = \sum_j b_j |f_j\rangle$.

Operators on tensor product spaces. The action of a product operator on a product vector is defined by

$$(A \otimes B)(|\psi_A\rangle \otimes |\phi_B\rangle) = A|\psi_A\rangle \otimes B|\phi_B\rangle, \quad (7.10)$$

and extended to all vectors by linearity. The identity on the composite space is $I_{\mathcal{H}_A \otimes \mathcal{H}_B} = I_A \otimes I_B$. An operator of the form $A \otimes I_B$ acts only on subsystem A ; $A \otimes B$ is a product of local operators. The most general operator on $\mathcal{H}_A \otimes \mathcal{H}_B$ is not factorable; its matrix elements in the product basis are

$$\langle e_i \otimes f_j \mid O \mid e_k \otimes f_l \rangle, \quad (7.11)$$

with four indices coupling the two subspaces. An operator that couples the subsystems (e.g., the interaction Hamiltonian H_{AB}) cannot be written as a sum of products of local operators with a single term each; the general form is $O = \sum_{\alpha} A_{\alpha} \otimes B_{\alpha}$.

 **Note** The total angular momentum operator $\mathbf{J} = \mathbf{J}_1 \otimes I + I \otimes \mathbf{J}_2$ of (6.69) is a sum of two product operators, each of which acts nontrivially on only one subsystem. The two terms commute— $[\mathbf{J}_1 \otimes I, I \otimes \mathbf{J}_2] = 0$ —because the first acts as the identity on subsystem B and the second as the identity on subsystem A . The operator $(\mathbf{a} \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b} \cdot \boldsymbol{\sigma})$, which will appear in the CHSH observable of Section 7.5, is a product operator with both factors nontrivial.

The Schmidt decomposition. For any bipartite pure state $|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$, there exists a representation of the form

$$|\Psi\rangle = \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle \otimes |v_k\rangle, \quad (7.12)$$

where $\{|u_k\rangle\}_{k=1}^r$ is an orthonormal set in $\mathcal{H}_A$, $\{|v_k\rangle\}_{k=1}^r$ is an orthonormal set in $\mathcal{H}_B$, the Schmidt coefficients satisfy $\lambda_k > 0$ and $\sum_{k=1}^r \lambda_k = 1$, and the Schmidt rank r satisfies $1 \leq r \leq \min(d_A, d_B)$.

The proof proceeds via the singular value decomposition of the coefficient matrix. Write $|\Psi\rangle = \sum_{i=1}^{d_A} \sum_{j=1}^{d_B} C_{ij} |e_i\rangle \otimes |f_j\rangle$. The $d_A \times d_B$ matrix C admits a singular value decomposition $C = U \Sigma V^\dagger$, where U is a $d_A \times d_A$ unitary matrix, V is a $d_B \times d_B$ unitary matrix, and Σ is a $d_A \times d_B$ diagonal matrix with nonnegative entries $\sigma_k = \sqrt{\lambda_k}$ for $k = 1, \dots, r$ (the singular values) and zero elsewhere. Explicitly,

$$C_{ij} = \sum_{k=1}^r U_{ik} \sqrt{\lambda_k} V_{jk}^*, \quad (7.13)$$

where r is the number of nonzero singular values (the rank of C). Substituting into the expansion of $|\Psi\rangle$:

$$|\Psi\rangle = \sum_{i,j} \left(\sum_{k=1}^r U_{ik} \sqrt{\lambda_k} V_{jk}^* \right) |e_i\rangle \otimes |f_j\rangle = \sum_{k=1}^r \sqrt{\lambda_k} \left(\sum_i U_{ik} |e_i\rangle \right) \otimes \left(\sum_j V_{jk}^* |f_j\rangle \right) \quad (7.14)$$

$$= \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle \otimes |v_k\rangle, \quad (7.15)$$

where we define $|u_k\rangle := \sum_i U_{ik} |e_i\rangle$ and $|v_k\rangle := \sum_j V_{jk}^* |f_j\rangle$. The columns of U are orthonormal ($U^\dagger U = I$), so the $\{|u_k\rangle\}$ are orthonormal; similarly, $\{|v_k\rangle\}$ are orthonormal from the columns of V^* . The singular values are positive by construction ($\lambda_k = \sigma_k^2 > 0$), and the normalization of $|\Psi\rangle$ requires $\langle \Psi | \Psi \rangle = \sum_k \lambda_k = 1$. This completes the proof.

The Schmidt rank r is the rank of the coefficient matrix C . A product state has $r = 1$ (its coefficient matrix has rank one). An entangled state has $r > 1$. The decomposition is not unique when some λ_k are degenerate; the $\{|u_k\rangle\}$ and $\{|v_k\rangle\}$ associated with a degenerate Schmidt coefficient can be rotated among themselves by a unitary transformation in the degenerate subspace.

 **Note** The Schmidt coefficients λ_k will be identified in Section 7.2 as the eigenvalues of the reduced density operator $\rho_A = \text{tr}_B(|\Psi\rangle \langle \Psi|)$. The existence proof above uses only the SVD of the coefficient matrix—a purely mathematical existence argument that does not depend on the density operator. The eigenvalue interpretation is deferred to Section 7.2 with an explicit honesty clause at the point of the forward reference, in the same pattern as ch3's canonical commutator ((3.60)) whose physical content was built in ch5 ((5.12)).

Example: two-qubit Schmidt decomposition. Consider a general pure state of two qubits:

$$|\Psi\rangle = \alpha |\uparrow\uparrow\rangle + \beta |\uparrow\downarrow\rangle + \gamma |\downarrow\uparrow\rangle + \delta |\downarrow\downarrow\rangle, \quad (7.16)$$

with $|\alpha|^2 + |\beta|^2 + |\gamma|^2 + |\delta|^2 = 1$. The coefficient matrix is

$$C = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \quad (7.17)$$

The singular values of C are the square roots of the eigenvalues of $C^\dagger C$. The Schmidt rank is the rank of C .

For the product state $|\uparrow\uparrow\rangle$: $C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, rank 1, $\lambda_1 = 1$, $r = 1$.

For the singlet $|0,0\rangle$: $C = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, rank 2, $C^\dagger C = \frac{1}{2}I$, so $\lambda_1 = \lambda_2 = 1/2$, $r = 2$ — maximally entangled in $\mathbb{C}^2 \otimes \mathbb{C}^2$.

For the partially entangled state $|\psi(\theta)\rangle = \cos\theta|\uparrow\uparrow\rangle + \sin\theta|\downarrow\downarrow\rangle$: $C = \begin{pmatrix} \cos\theta & 0 \\ 0 & \sin\theta \end{pmatrix}$, rank 2, $\lambda_1 = \cos^2\theta$, $\lambda_2 = \sin^2\theta$, $r = 2$ except at $\theta = 0, \pi/2$ where $r = 1$.

The product basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ and the coupled basis $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle, |0,0\rangle\}$ of ch6's (6.84) are two orthonormal bases of the same space $\mathbb{C}^2 \otimes \mathbb{C}^2$, related by the Clebsch–Gordan coefficients of ch6's (6.74). The coupled basis diagonalizes the total angular momentum; the product basis diagonalizes the individual z -components. The singlet—the unique antisymmetric state—has Schmidt rank 2 and Schmidt coefficients $(1/2, 1/2)$ in any product basis, a consequence of its rotational invariance ((6.84)).

The advantage of the Schmidt decomposition is that it provides a canonical form for any bipartite pure state, revealing the entanglement structure in the number and distribution of the λ_k . Its limitation is that it applies only to pure states; the classification of mixed-state entanglement requires the more involved definition of Section 7.3, and the decomposition itself says nothing about the physical interpretation of the coefficients λ_k —that interpretation requires the density operator of the next section.

The next section develops the density operator formalism: the mathematical object whose eigenvalues are exactly the Schmidt coefficients, and whose partial trace generates subsystem states from composites.

7.2 The Density Operator

Chapter 3 named the density operator ρ via Gleason's theorem ((3.19))—"every consistent pricing of an experiment's projectors comes from a unique positive unit-trace ρ "—and used it no further. Chapter 6 introduced $\rho = |\psi\rangle\langle\psi|$ for the Bloch sphere parametrization of pure spin-1/2 states ((6.39)) and applied the non-selective update $\rho \mapsto \sum_n P_n \rho P_n$ to describe ensembles of spin measurements whose outcomes are not recorded. But the density operator's full power—its ability to describe mixed states as convex combinations of projectors, its indifference to the distinction between proper and improper mixtures, its role in generating subsystem states from entangled composites via the partial trace—has not been exploited. The measurement theory of ch3 was built entirely in the ray language; the state of a subsystem of an entangled pair cannot be described by a ray. This section develops the density operator formalism in full.

Pure and mixed states. A density operator on a Hilbert space $\mathcal{H}$ is any operator ρ that can be written

as a convex combination of pure-state projectors:

$$\rho = \sum_k p_k |\psi_k\rangle \langle \psi_k|, \quad p_k \geq 0, \quad \sum_k p_k = 1, \quad (7.18)$$

where each $|\psi_k\rangle$ is a normalized state vector in $\mathcal{H}$. The decomposition is not unique: different ensembles (different sets of $\{|\psi_k\rangle\}$ and $\{p_k\}$) can yield the same density operator. This unitary freedom in the ensemble decomposition will be exhibited explicitly below.

From the definition, three necessary and sufficient conditions characterize a density operator:

$$\rho = \rho^\dagger, \quad \text{tr} \rho = 1, \quad \rho \geq 0, \quad (7.19)$$

where $\rho \geq 0$ means $\langle \phi | \rho | \phi \rangle \geq 0$ for all $|\phi\rangle \in \mathcal{H}$ (positive semidefinite). The hermiticity and unit trace follow immediately from the definition: $\rho^\dagger = \sum_k p_k (|\psi_k\rangle \langle \psi_k|)^\dagger = \rho$; $\text{tr} \rho = \sum_k p_k \text{tr}(|\psi_k\rangle \langle \psi_k|) = \sum_k p_k \langle \psi_k | \psi_k \rangle = \sum_k p_k = 1$. Positivity follows because each term is a positive operator and the p_k are nonnegative. Conversely, any operator satisfying the three conditions is diagonalizable with nonnegative eigenvalues p_k summing to one, and the spectral decomposition $\rho = \sum_k p_k |k\rangle \langle k|$ is a valid ensemble decomposition—so the two characterizations are equivalent.

A pure state corresponds to a density operator of rank one:

$$\rho = |\psi\rangle \langle \psi|, \quad \rho^2 = \rho, \quad \text{tr}(\rho^2) = 1. \quad (7.20)$$

The condition $\rho^2 = \rho$ characterizes projectors; together with unit trace it forces rank one. A mixed state has $\text{tr}(\rho^2) < 1$; the deviation from unity measures the degree of mixing.

Expectation values. The Born rule ((2.7), ch2) assigns the probability $|\langle \phi | \psi \rangle|^2$ to the transition $|\psi\rangle \rightarrow |\phi\rangle$. For an ensemble described by $\rho = \sum_k p_k |\psi_k\rangle \langle \psi_k|$, the expectation value of an observable A is the average over the ensemble:

$$\langle A \rangle = \text{tr}(\rho A), \quad (7.21)$$

derived as follows:

$$\text{tr}(\rho A) = \text{tr}\left(\sum_k p_k |\psi_k\rangle \langle \psi_k| A\right) = \sum_k p_k \text{tr}(|\psi_k\rangle \langle \psi_k| A) = \sum_k p_k \langle \psi_k | A | \psi_k \rangle. \quad (7.22)$$

The last equality uses the cyclic property of the trace: for any operator O , $\text{tr}(|\phi\rangle \langle \psi| O) = \langle \psi | O | \phi \rangle$. The trace formula unifies the ensemble average: the Born-rule mean in each component state, weighted by the classical probabilities p_k , is recovered as a single trace. The probability of obtaining outcome ν in a measurement described by the projector P_ν is

$$p_\nu = \text{tr}(\rho P_\nu), \quad (7.23)$$

which is ch3's Gleason display (3.19) promoted from a signpost to a working formula.

Purity and the Bloch ball. The purity of a state is measured by $\text{tr}(\rho^2)$:

$$\text{tr}(\rho^2) \leq 1, \quad \text{equality iff } \rho \text{ is pure}, \quad (7.24)$$

proved by writing $\rho = \sum_k p_k |k\rangle \langle k|$ in its eigenbasis: $\text{tr}(\rho^2) = \sum_k p_k^2 \leq (\sum_k p_k)^2 = 1$, with equality if and only if exactly one $p_k = 1$ and the rest vanish. The difference $1 - \text{tr}(\rho^2)$ is the linear entropy, a measure of mixedness.

For a single qubit ($\mathcal{H} = \mathbb{C}^2$), the density operator takes the form

$$\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad |\mathbf{r}| \leq 1, \quad (7.25)$$

where $\mathbf{r} = \text{tr}(\rho \boldsymbol{\sigma})$ is the Bloch vector. The derivation extends ch6's construction of the pure-state Bloch sphere ((6.39)). Any 2×2 hermitian operator with unit trace can be expanded in the basis $\{I, \sigma_x, \sigma_y, \sigma_z\}$: $\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma})$, where the coefficient of I is fixed by $\text{tr} \rho = 1$ and the coefficients $r_i = \text{tr}(\rho \sigma_i)$ follow from $\text{tr}(\sigma_i) = 0$ and $\text{tr}(\sigma_i \sigma_j) = 2\delta_{ij}$. The purity is

$$\text{tr}(\rho^2) = \frac{1}{2}(1 + |\mathbf{r}|^2), \quad (7.26)$$

so $|\mathbf{r}| = 1$ for pure states (the Bloch sphere S^2 , ch6's (6.39)) and $|\mathbf{r}| < 1$ for mixed states (the interior). The maximally mixed state $\rho = I/2$ sits at the center ($\mathbf{r} = \mathbf{0}$) and has $\text{tr}(\rho^2) = 1/2$, the minimum possible for a qubit.

The partial trace. A composite system in a state ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$ defines a state for subsystem A alone via the partial trace:

$$\rho_A := \text{tr}_B(\rho_{AB}), \quad \text{tr}(\rho_{AB}(O_A \otimes I_B)) = \text{tr}_A(\rho_A O_A), \quad (7.27)$$

for every operator O_A on $\mathcal{H}_A$. This condition defines ρ_A uniquely: it is the operator on $\mathcal{H}_A$ whose expectation values reproduce those of all observables of the form $O_A \otimes I_B$ (observables that act only on subsystem A). The definition is operational: no experiment performed on subsystem A alone can detect correlations with subsystem B ; the partial trace encodes everything locally accessible.

In a product basis $\{|e_i\rangle \otimes |f_j\rangle\}$, the matrix elements of the reduced state are

$$\langle e_i | \rho_A | e_j \rangle = \sum_k \langle e_i \otimes f_k | \rho_{AB} | e_j \otimes f_k \rangle, \quad (7.28)$$

obtained by evaluating the defining condition on $O_A = |e_j\rangle \langle e_i|$:

$$\text{tr}_A(\rho_A |e_j\rangle \langle e_i|) = \langle e_i | \rho_A | e_j \rangle = \text{tr}(\rho_{AB}(|e_j\rangle \langle e_i| \otimes I_B)) \quad (7.29)$$

$$= \sum_{k,l} \langle e_k \otimes f_l | \rho_{AB}(|e_j\rangle \langle e_i| \otimes I_B) | e_k \otimes f_l \rangle = \sum_k \langle e_i \otimes f_k | \rho_{AB} | e_j \otimes f_k \rangle. \quad (7.30)$$

The partial trace of a product operator factors: $\text{tr}_B(A \otimes B) = A(\text{tr} B)$.

Reduced state of a pure composite. For a bipartite pure state $|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$, the reduced density operator of subsystem A is

$$\rho_A = \text{tr}_B(|\Psi\rangle \langle \Psi|), \quad (7.31)$$

and similarly $\rho_B = \text{tr}_A(|\Psi\rangle \langle \Psi|)$. A crucial fact follows: if $|\Psi\rangle$ is entangled, ρ_A is mixed. A subsystem of a pure entangled state is not pure.

To see this, substitute the Schmidt decomposition (7.12) into (7.31):

$$\rho_A = \text{tr}_B\left(\sum_{k,l=1}^r \sqrt{\lambda_k \lambda_l} |u_k\rangle \langle u_l| \otimes |v_k\rangle \langle v_l|\right) = \sum_{k,l=1}^r \sqrt{\lambda_k \lambda_l} |u_k\rangle \langle u_l| \text{tr}(|v_k\rangle \langle v_l|) \quad (7.32)$$

$$= \sum_{k,l=1}^r \sqrt{\lambda_k \lambda_l} |u_k\rangle \langle u_l| \delta_{kl} = \sum_{k=1}^r \lambda_k |u_k\rangle \langle u_k|. \quad (7.33)$$

The orthonormality of the $\{|v_k\rangle\}$ collapses the double sum: $\text{tr}(|v_k\rangle \langle v_l|) = \langle v_l | v_k \rangle = \delta_{kl}$. The result

is the spectral decomposition of ρ_A : the Schmidt coefficients λ_k of (7.12) are the eigenvalues of the reduced density operator, and the Schmidt vectors $\{|u_k\rangle\}$ are the corresponding eigenvectors. The same computation for ρ_B gives $\rho_B = \sum_k \lambda_k |v_k\rangle \langle v_k|$, so ρ_A and ρ_B have the same nonzero eigenvalues (though generally different eigenvectors). This is the eigenvalue interpretation of the Schmidt coefficients promised in Section 7.1.

For the singlet $|0, 0\rangle$: $\rho_A = \text{tr}_B(|0, 0\rangle \langle 0, 0|) = \frac{1}{2}(|\uparrow\rangle \langle \uparrow| + |\downarrow\rangle \langle \downarrow|) = I/2$, the maximally mixed state on $\mathbb{C}^2$. Each spin individually, without reference to the other, is completely unpolarized—the defining property of maximal entanglement.

 **Note** The Schmidt rank r is the number of nonzero eigenvalues of ρ_A (equivalently, the rank of ρ_A). The state $|\Psi\rangle$ is a product state ($r = 1$) if and only if ρ_A is pure ($\text{tr}(\rho_A^2) = 1$). The purity $\text{tr}(\rho_A^2) = \sum_k \lambda_k^2$ is a measure of entanglement: smaller purity means more entangled (for fixed subsystem dimension).

Non-selective measurement in density-operator language. When a measurement is performed but the outcome is not recorded—the non-selective case of ch3—the post-measurement ensemble state is obtained by applying the Lüders update (Axiom M3, (3.47)) to each branch and summing with the Born-rule weights:

$$\rho \mapsto \sum_n P_n \rho P_n, \quad (7.34)$$

where $\{P_n\}$ is the PVM of the measured observable. The derivation from the ray-level update (3.47) is direct. For an initial pure state $\rho = |\psi\rangle \langle \psi|$: the post-measurement state with outcome n is $|\psi_n\rangle = P_n |\psi\rangle / \sqrt{p_n}$ with probability $p_n = \langle \psi | P_n | \psi \rangle$. The non-selective ensemble, averaged over outcomes, is

$$\rho \mapsto \sum_n p_n |\psi_n\rangle \langle \psi_n| = \sum_n p_n \frac{P_n |\psi\rangle \langle \psi| P_n}{p_n} = \sum_n P_n \rho P_n. \quad (7.35)$$

For a general mixed state $\rho = \sum_k q_k |\phi_k\rangle \langle \phi_k|$, linearity extends the same formula:

$$\rho \mapsto \sum_n P_n \left(\sum_k q_k |\phi_k\rangle \langle \phi_k| \right) P_n = \sum_k q_k \left(\sum_n P_n |\phi_k\rangle \langle \phi_k| P_n \right), \quad (7.36)$$

where the sum over k commutes with the sum over n , and the term in parentheses is exactly the non-selective update applied to the pure state $|\phi_k\rangle$, which has already been derived above. The map $\rho \mapsto \sum_n P_n \rho P_n$ is trace-preserving ($\text{tr} \sum_n P_n \rho P_n = \sum_n \text{tr}(P_n \rho) = \text{tr} \rho = 1$), completely positive, and not unitary—it eliminates the off-diagonal coherence between different outcome subspaces. This is the density-operator register of the measurement problem, to which Section 7.6 returns.

Proper vs. improper mixtures. A proper mixture is prepared by classical randomization: flip a coin, prepare $|\uparrow_z\rangle$ if heads, $|\downarrow_z\rangle$ if tails. The resulting density operator $\rho = \frac{1}{2}(|\uparrow_z\rangle \langle \uparrow_z| + |\downarrow_z\rangle \langle \downarrow_z|) = I/2$ encodes classical ignorance about which state was actually prepared—the system really is in one of the two states; we merely do not know which.

An improper mixture arises from entanglement: the reduced state of one spin in the singlet is also $\rho_A = I/2$. But here the composite is pure—the pair's state $|0, 0\rangle$ is known with certainty. There is no classical ignorance about the pair's state, yet each spin individually is described by the maximally mixed density operator. The density operator does not distinguish between the two cases.

This formal indifference is exhibited explicitly. The two ensembles

$$\mathcal{E}_I = \{|\uparrow_z\rangle, |\downarrow_z\rangle; \tfrac{1}{2}, \tfrac{1}{2}\}, \quad \mathcal{E}_{II} = \{|\uparrow_x\rangle, |\downarrow_x\rangle; \tfrac{1}{2}, \tfrac{1}{2}\}, \quad (7.37)$$

produce identical density operators:

$$\rho_I = \tfrac{1}{2} |\uparrow_z\rangle \langle \uparrow_z| + \tfrac{1}{2} |\downarrow_z\rangle \langle \downarrow_z| = \tfrac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \tfrac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \tfrac{1}{2} I, \quad (7.38)$$

$$\rho_{II} = \tfrac{1}{2} |\uparrow_x\rangle \langle \uparrow_x| + \tfrac{1}{2} |\downarrow_x\rangle \langle \downarrow_x| = \tfrac{1}{2} \cdot \tfrac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \tfrac{1}{2} \cdot \tfrac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \tfrac{1}{2} I. \quad (7.39)$$

Both ensembles are proper mixtures (classical ignorance), but they are operationally indistinguishable—no measurement on the qubit can tell them apart. The unitary freedom in the ensemble decomposition is the statement that any two ensembles producing the same ρ are related by a unitary remixing of the state vectors.

A non-selective measurement of σ_z on the maximally mixed state $\rho = I/2$ produces

$$\rho \mapsto P_{+z} \rho P_{+z} + P_{-z} \rho P_{-z} = \tfrac{1}{2} P_{+z} + \tfrac{1}{2} P_{-z} = \tfrac{1}{2} I = \rho, \quad (7.40)$$

so the maximally mixed state is a fixed point of the non-selective Lüders map. The measurement, by destroying the coherence in the σ_z basis (which the maximally mixed state already lacks), changes nothing.

The proper vs. improper distinction is the engine of the measurement problem. When a system entangles with a measuring apparatus, the reduced state of the system becomes an improper mixture—diagonal in the pointer basis, indistinguishable from a classical probability distribution. But the global state remains pure. The question of whether and how this improper mixture becomes a proper one—a single definite outcome—is the measurement problem, restated in Section 7.6.

The advantage of the density operator is its universality: it describes pure states, proper mixtures, and improper mixtures in a single formalism, and the partial trace provides the unique consistent rule for obtaining subsystem states. Its limitation is exactly its indifference to the proper/improper distinction: two physically distinct situations—classical ignorance of a factorizable state and the reduced description of an entangled pure state—are represented by the same mathematical object, and the formalism alone does not tell them apart.

The next section uses the density operator and the partial trace to classify entanglement: the Schmidt rank, the Bell basis, mixed-state separability, and the entanglement entropy.

7.3 Entanglement: Separability and the Schmidt Decomposition

With the tensor product (Section 7.1) and the density operator (Section 7.2) in hand, entanglement can now be classified quantitatively. The Schmidt decomposition of Section 7.1—whose coefficients were identified in Section 7.2 as the eigenvalues of the reduced density operator—is the central tool. This section defines separability for pure and mixed states, constructs the Bell basis of maximally entangled two-qubit states, introduces the entanglement entropy as the canonical pure-state measure, and names the no-cloning theorem as the conceptual boundary of quantum information.

Pure-state separability. A pure state $|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_B$ is separable—it is a product state—if and only if it can be written as

$$|\Psi\rangle = |\psi_A\rangle \otimes |\phi_B\rangle, \quad \text{equivalently } r_S(|\Psi\rangle) = 1, \quad (7.41)$$

where r_S is the Schmidt rank—the number of nonzero Schmidt coefficients λ_k in (7.12). A product state has Schmidt rank one because its coefficient matrix $C_{ij} = a_i b_j$ has rank one, hence a single nonzero singular value. The reduced density operator of a product state is pure: $\rho_A = |\psi_A\rangle \langle \psi_A|$, $\text{tr}(\rho_A^2) = 1$.

A pure state is entangled if and only if $r_S > 1$. At the opposite extreme, a pure state is maximally entangled if all its nonzero Schmidt coefficients are equal:

$$\lambda_k = \frac{1}{r}, \quad r = \min(d_A, d_B), \quad (7.42)$$

where r saturates the upper bound $r \leq \min(d_A, d_B)$ of the Schmidt rank. For a maximally entangled state, the reduced density operator is maximally mixed:

$$\rho_A = \frac{I_A}{r}, \quad (7.43)$$

where I_A is the identity on the r -dimensional subspace spanned by the Schmidt vectors. The subsystem state contains no information about the composite.

The Bell basis. On $\mathbb{C}^2 \otimes \mathbb{C}^2$, the four states

$$\begin{aligned} |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \\ |\Psi^-\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), \\ |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle), \\ |\Phi^-\rangle &= \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle - |\downarrow\downarrow\rangle), \end{aligned} \quad (7.44)$$

form an orthonormal basis—the Bell basis. The singlet $|0, 0\rangle$ of ch6's (6.84) is $|\Psi^-\rangle$. All four Bell states are maximally entangled: each has Schmidt coefficients $\lambda_1 = \lambda_2 = 1/2$, Schmidt rank 2, and reduced density operator $\rho_A = I/2$ (maximally mixed on $\mathbb{C}^2$).

The orthonormality is verified directly. For $\langle \Psi^+ | \Psi^- \rangle$:

$$\langle \Psi^+ | \Psi^- \rangle = \frac{1}{2}(\langle \uparrow\downarrow | \uparrow\downarrow \rangle - \langle \uparrow\downarrow | \downarrow\uparrow \rangle + \langle \downarrow\uparrow | \uparrow\downarrow \rangle - \langle \downarrow\uparrow | \downarrow\uparrow \rangle) \quad (7.45)$$

$$= \frac{1}{2}(1 - 0 + 0 - 1) = 0. \quad (7.46)$$

For $\langle \Phi^+ | \Phi^- \rangle$:

$$\langle \Phi^+ | \Phi^- \rangle = \frac{1}{2}(\langle \uparrow\uparrow | \uparrow\uparrow \rangle - \langle \uparrow\uparrow | \downarrow\downarrow \rangle + \langle \downarrow\downarrow | \uparrow\uparrow \rangle - \langle \downarrow\downarrow | \downarrow\downarrow \rangle) \quad (7.47)$$

$$= \frac{1}{2}(1 - 0 + 0 - 1) = 0. \quad (7.48)$$

The remaining inner products $\langle \Psi^\pm | \Phi^\pm \rangle = 0$ and norm conditions $\langle \Psi^\pm | \Psi^\pm \rangle = \langle \Phi^\pm | \Phi^\pm \rangle = 1$ follow by the same pattern: each Bell state is a sum of two orthogonal product vectors with equal weight $1/\sqrt{2}$, and distinct pairs differ in the sign pattern, which produces zero overlap when the two product vectors in one state have no counterpart in the other.

The Bell states are related by local unitaries—operations on one qubit alone—applied to any one Bell state. For example, starting from $|\Psi^+\rangle$:

$$(I \otimes \sigma_x) |\Psi^+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle) = |\Phi^+\rangle, \quad (7.49)$$

$$(I \otimes \sigma_z) |\Psi^+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) = |\Psi^-\rangle, \quad (7.50)$$

$$(I \otimes \sigma_y) |\Psi^+\rangle = \frac{1}{\sqrt{2}}(-i|\uparrow\uparrow\rangle + i|\downarrow\downarrow\rangle) = -i|\Phi^-\rangle \sim |\Phi^-\rangle, \quad (7.51)$$

where the last uses ray equivalence (global phase is unobservable, ch2 (2.13)). The entire Bell basis is generated from any one Bell state by applying Pauli operators to one qubit. This property makes the Bell basis the natural arena for the CHSH inequality (Section 7.5).

The singlet $|\Psi^-\rangle$ is distinguished by its rotational invariance: for any $U \in \text{SU}(2)$,

$$(U \otimes U) |\Psi^-\rangle = |\Psi^-\rangle, \quad (7.52)$$

a one-line consequence of $\mathbf{J}|0,0\rangle = 0$ (ch6 (6.84))—the singlet is the unique state of total angular momentum zero. This invariance is the origin of the correlation function $E_{\text{QM}}(\hat{a}, \hat{b}) = -\mathbf{a} \cdot \mathbf{b}$ that drives the CHSH violation (Section 7.5).

Mixed-state separability. The definition of separability extends to mixed states. A density operator ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$ is separable if and only if it can be written as a convex combination of product states:

$$\rho_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \quad p_k \geq 0, \quad \sum_k p_k = 1, \quad (7.53)$$

where each $\rho_A^{(k)}$ is a density operator on $\mathcal{H}_A$ and each $\rho_B^{(k)}$ is a density operator on $\mathcal{H}_B$. A mixed state is entangled if and only if it is not separable—that is, if no representation of the form (7.53) exists.

The definition is simple; testing it is hard. For a given ρ_{AB} , determining whether a separable decomposition exists is an NP-hard problem in general. For 2×2 and 2×3 systems, the Peres–Horodecki criterion provides a necessary and sufficient condition: ρ_{AB} is separable if and only if its partial transpose $\rho_{AB}^{T_B}$ is positive semidefinite (PPT). The criterion is stated here without proof (Peres 1996, Horodecki et al. 1996); Exercise 7.7 works through it for the singlet and the Werner state. For higher dimensions, PPT is necessary but not sufficient (bound entanglement exists).

Note *The full theory of mixed-state entanglement—entanglement witnesses, the PPT criterion’s proof, the negativity, the entanglement of formation, and bound entanglement—is beyond the scope of this book. The definition (7.53) and the PPT statement above are the chapter’s boundary; quantum information theory proper begins where this chapter ends.*

Entanglement entropy. For a bipartite pure state $|\Psi\rangle_{AB}$, the canonical measure of entanglement is the von Neumann entropy of the reduced state:

$$S(\rho_A) = -\text{tr}(\rho_A \log_2 \rho_A) = -\sum_{k=1}^r \lambda_k \log_2 \lambda_k, \quad (7.54)$$

where $\{\lambda_k\}$ are the Schmidt coefficients (the eigenvalues of ρ_A). The logarithm is taken to base 2, giving the entropy in bits. The von Neumann entropy extends the Shannon entropy $H(\{p_k\}) = -\sum_k p_k \log_2 p_k$ of a classical probability distribution to the quantum domain: for a density operator

diagonal in its eigenbasis, the von Neumann entropy reduces to the Shannon entropy of its eigenvalue distribution.

The entanglement entropy has the following properties. (i) $S(\rho_A) \geq 0$, with $S = 0$ if and only if ρ_A is pure—that is, if and only if $|\Psi\rangle$ is a product state ($r = 1$). (ii) $S(\rho_A)$ is maximal when all nonzero λ_k are equal: $\lambda_k = 1/r$, giving $S_{\max} = \log_2 r$. For a maximally entangled state of two d -level systems, $S_{\max} = \log_2 d$. (iii) $S(\rho_A) = S(\rho_B)$ for any bipartite pure state: the two subsystems share the same entropy, because ρ_A and ρ_B have the same nonzero eigenvalues (Section 7.2). (iv) $S(\rho_A)$ is invariant under local unitaries $U_A \otimes U_B$: the Schmidt coefficients are unchanged by independent unitary rotations of the two subsystem bases. (v) S is an entanglement monotone for pure states: it cannot increase under local operations and classical communication (LOCC)—named here, not proved; the proof belongs to quantum information theory.

The singlet has $S = -(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2}) = 1$ bit of entanglement. A product state has $S = 0$. Between these extremes, entanglement is a continuous resource.

Example: parametrized two-qubit state. Consider the family of states

$$|\psi(\theta)\rangle = \cos \theta |\uparrow\uparrow\rangle + \sin \theta |\downarrow\downarrow\rangle, \quad 0 \leq \theta \leq \frac{\pi}{2}. \quad (7.55)$$

The coefficient matrix is diagonal: $C = \text{diag}(\cos \theta, \sin \theta)$, so the Schmidt decomposition is immediate: $\lambda_1 = \cos^2 \theta$, $\lambda_2 = \sin^2 \theta$, with $|u_1\rangle = |\uparrow\rangle$, $|u_2\rangle = |\downarrow\rangle$, $|v_1\rangle = |\uparrow\rangle$, $|v_2\rangle = |\downarrow\rangle$. The reduced density operator is

$$\rho_A = \cos^2 \theta |\uparrow\rangle \langle\uparrow| + \sin^2 \theta |\downarrow\rangle \langle\downarrow| = \frac{1}{2}(I + (\cos^2 \theta - \sin^2 \theta)\sigma_z) = \frac{1}{2}(I + (\cos 2\theta)\sigma_z), \quad (7.56)$$

whose Bloch vector is $\mathbf{r} = (0, 0, \cos 2\theta)$, with length $|\mathbf{r}| = |\cos 2\theta|$. At $\theta = 0$: product state $|\uparrow\uparrow\rangle$, $\mathbf{r} = (0, 0, 1)$, $S = 0$. At $\theta = \pi/4$: Bell state $|\Phi^+\rangle$, $\mathbf{r} = \mathbf{0}$, $S = 1$. The entanglement entropy varies smoothly:

$$S(\theta) = -\cos^2 \theta \log_2(\cos^2 \theta) - \sin^2 \theta \log_2(\sin^2 \theta) = H_2(\cos^2 \theta), \quad (7.57)$$

where $H_2(p) = -p \log_2 p - (1-p) \log_2(1-p)$ is the binary entropy function. Entanglement is a continuous resource, not an on/off switch. For the generalized state $|\psi\rangle = \cos \theta |\uparrow\uparrow\rangle + e^{i\phi} \sin \theta |\downarrow\downarrow\rangle$, the phase ϕ can be absorbed by redefining $|\downarrow\rangle \rightarrow e^{-i\phi/2} |\downarrow\rangle$ on one qubit—a local unitary—so the Schmidt coefficients and the entanglement entropy are independent of ϕ .

The no-cloning theorem. The linearity of quantum mechanics forbids perfect copying of an unknown quantum state. If a unitary U could clone, i.e., $U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$ for all $|\psi\rangle$ in a two-dimensional subspace, then applying it to the superposition $|+\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$ would give

$$U(|+\rangle \otimes |0\rangle) = \frac{1}{\sqrt{2}}(U(|\uparrow\rangle \otimes |0\rangle) + U(|\downarrow\rangle \otimes |0\rangle)) = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\uparrow\rangle + |\downarrow\rangle \otimes |\downarrow\rangle), \quad (7.58)$$

by linearity. But a true cloning of $|+\rangle$ would produce $|+\rangle \otimes |+\rangle = \frac{1}{2}(|\uparrow\rangle \otimes |\uparrow\rangle + |\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle + |\downarrow\rangle \otimes |\downarrow\rangle)$, which differs from the linearity result by the cross terms. The contradiction establishes the theorem. The no-cloning theorem is the conceptual door to quantum information theory—the impossibility of copying is what makes quantum cryptography possible and what prevents superluminal signaling via entanglement. Exercise 7.7 formalizes the proof.

The advantage of the entanglement formalism developed in this section is its completeness for

pure states: every bipartite pure state has a unique entanglement spectrum $\{\lambda_k\}$, and the entropy S provides a single number that is invariant under local unitaries and monotonic under LOCC. Its limitation is that the pure-state theory does not extend simply to mixed states: no single measure (neither the von Neumann entropy of the reduced state, nor any simple function of it) provides a complete classification of mixed-state entanglement, and the separability problem (7.53) is computationally hard in general.

The singlet’s entanglement—Schmidt rank 2, entropy 1 bit, maximally mixed reduced state—is now fully quantified. In 1935, Einstein, Podolsky, and Rosen argued that states of this kind, where a measurement on one subsystem fixes the description of the other however distant, reveal quantum mechanics as an incomplete theory. Their argument is the subject of the next section.

7.4 The Einstein–Podolsky–Rosen Argument

The singlet’s entanglement is not merely a mathematical curiosity. It poses a direct challenge to the completeness of the quantum-mechanical description—a challenge that Einstein, Podolsky, and Rosen formulated in 1935, that Bohr answered in the same year, and that remained a matter of interpretation until Bell converted it into an experimentally decidable question three decades later. This section presents the EPR argument in its modern spin form (Bohm 1951), using the singlet of ch6’s (6.84) and the measurement axioms of ch3, and connects the dilemma to the operational state definition of ch1.

Historical context. The 1935 paper “Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?” (Einstein, Podolsky, Rosen, *Phys. Rev.* 47, 777) was the sharpest exchange of the Bohr–Einstein debate. EPR proposed two criteria by which a physical theory should be judged, and argued that quantum mechanics, by those criteria, is incomplete.

The EPR reality criterion:

If, without in any way disturbing a system, we can predict with certainty (i.e., with probability equal to unity) the value of a physical quantity, then there exists an element of physical reality corresponding to that quantity.

(7.59)

The EPR completeness criterion: a physical theory is complete only if every element of physical reality has a counterpart in the theory.

The criteria are stated here as the conceptual keystones of the argument, not as equations. The question EPR posed is whether quantum mechanics, applied to entangled states, satisfies both criteria simultaneously.

Note *The EPR reality and completeness criteria are philosophical premises, not theorems. The argument’s force comes from their apparent reasonableness: if a quantity’s value can be predicted with certainty without interacting with the system that carries it, then that value must correspond to something real—and a complete theory ought to describe it. The EPR paper’s conclusion was that quantum mechanics, which assigns no simultaneous values to noncommuting observables, must be incomplete.*

The Bell inequality (Section 7.5) would later show that any theory completing quantum mechanics in EPR's sense must be nonlocal.

The EPR–Bohm argument: spin version. Alice and Bob share two spin-1/2 particles prepared in the singlet state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (7.60)$$

of (7.44). The particles are separated—Alice keeps particle 1, Bob takes particle 2 to a distant laboratory. The measurements are spacelike separated: Alice's choice of what to measure cannot causally influence Bob's particle, and vice versa (the locality assumption).

Consider the measurement of σ_z on particle 1. The singlet is not an eigenstate of $\sigma_z \otimes I$; its expectation value vanishes:

$$\langle\Psi^-|\sigma_z \otimes I|\Psi^-\rangle = \frac{1}{2}(\langle\uparrow\downarrow|\sigma_z \otimes I|\uparrow\downarrow\rangle - \langle\uparrow\downarrow|\sigma_z \otimes I|\downarrow\uparrow\rangle - \langle\downarrow\uparrow|\sigma_z \otimes I|\uparrow\downarrow\rangle + \langle\downarrow\uparrow|\sigma_z \otimes I|\downarrow\uparrow\rangle) = 0, \quad (7.61)$$

since $\langle\uparrow|\sigma_z|\uparrow\rangle = +1$, $\langle\downarrow|\sigma_z|\downarrow\rangle = -1$, and the cross terms vanish by orthogonality. But if Alice measures σ_z and obtains +1, the Lüders update ((3.47), Axiom M3) collapses the state to

$$|\Psi'\rangle = \frac{P_{+z}^{(1)} \otimes I}{\sqrt{p_{+z}}} |\Psi^-\rangle = \frac{|\uparrow\rangle_1 \otimes (\langle\uparrow|\uparrow\rangle|\downarrow\rangle - \langle\uparrow|\downarrow\rangle|\uparrow\rangle)/\sqrt{2}}{1/\sqrt{2}} = |\uparrow\rangle_1 \otimes |\downarrow\rangle_2, \quad (7.62)$$

where $P_{+z}^{(1)} = |\uparrow\rangle\langle\uparrow|_1$ is the projector onto spin-up for particle 1, and $p_{+z} = \langle\Psi^-|P_{+z}^{(1)} \otimes I|\Psi^-\rangle = 1/2$. After Alice's outcome +1, Bob's particle is in the state $|\downarrow\rangle$ with certainty: $\sigma_z|\downarrow\rangle = -|\downarrow\rangle$. By the EPR reality criterion, the z -component of Bob's spin is an element of physical reality.

But Alice could have measured σ_x instead. The singlet in the x -basis is

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|+x, -x\rangle - |-x, +x\rangle), \quad (7.63)$$

which follows from the basis change $|\pm x\rangle = (|\uparrow\rangle \pm |\downarrow\rangle)/\sqrt{2}$ (ch2 (2.38)) by direct substitution. Inverting the basis change gives $|\uparrow\rangle = (|+x\rangle + |-x\rangle)/\sqrt{2}$ and $|\downarrow\rangle = (|+x\rangle - |-x\rangle)/\sqrt{2}$. Substituting into the z -basis singlet:

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (7.64)$$

$$= \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}}(|+x\rangle + |-x\rangle) \otimes \frac{1}{\sqrt{2}}(|+x\rangle - |-x\rangle) \right. \quad (7.65)$$

$$\left. - \frac{1}{\sqrt{2}}(|+x\rangle - |-x\rangle) \otimes \frac{1}{\sqrt{2}}(|+x\rangle + |-x\rangle) \right] \quad (7.66)$$

$$= \frac{1}{2\sqrt{2}} \left[|+x, +x\rangle - |+x, -x\rangle + |-x, +x\rangle - |-x, -x\rangle \right. \quad (7.67)$$

$$\left. - |+x, +x\rangle - |+x, -x\rangle + |-x, +x\rangle + |-x, -x\rangle \right] \quad (7.68)$$

$$= \frac{1}{\sqrt{2}}(|+x, -x\rangle - |-x, +x\rangle). \quad (7.69)$$

If Alice measures σ_x and obtains +1, the Lüders update gives Bob's particle in $|-x\rangle$ with certainty: $\sigma_x|-x\rangle = -|-x\rangle$. By the same reality criterion, the x -component of Bob's spin is also an element of physical reality.

But $[\sigma_z, \sigma_x] \neq 0$ ((3.61), ch3). Quantum mechanics assigns no state in which both σ_z and σ_x possess definite values simultaneously. Yet Alice, by choosing which measurement to perform on her distant particle—and without in any way disturbing Bob’s particle (locality)—can predict with certainty either the z -component or the x -component of Bob’s spin.

The EPR dilemma follows:

The EPR dilemma. *Either (i) quantum mechanics is incomplete—there exist hidden variables that determine both σ_z and σ_x for Bob’s particle, and the quantum state $|\Psi^-\rangle$ is a statistical description of an ensemble of such variables (Einstein’s preference); or (ii) quantum mechanics is complete but nonlocal—Alice’s measurement choice instantaneously influences Bob’s distant particle in violation of relativistic causality (Bohr’s defense via complementarity).*

(7.70)

Einstein preferred horn (i); Bohr defended (ii) under the banner of complementarity: the two particles form a single indivisible quantum phenomenon, and the choice of what to measure on particle 1 determines what can meaningfully be said about particle 2.

Bohr’s reply (1935). Bohr’s response (ibid. 48, 696) rejected the EPR reality criterion as ambiguous when applied to quantum phenomena. The two particles, having interacted, form a single indivisible quantum phenomenon; “measurement on particle 1” and “measurement on particle 2” are not independent operations but parts of one experimental arrangement. The choice of what to measure on particle 1 determines the experimental context for the whole arrangement, and with it the set of concepts that can meaningfully be applied to particle 2—not because of any physical influence traveling between them, but because the two particles’ descriptions are complementary in Bohr’s sense. There is no contradiction, Bohr argued, in the fact that different experimental arrangements yield different descriptions, because the arrangement is part of the phenomenon.

Ch1 and ch3 callbacks. The EPR argument presses directly on the operational state definition of ch1 (1.5.5): the state is a catalogue of probabilities for the outcomes of every conceivable measurement. But on whose measurement? If the catalogue for Bob’s particle—the probabilities of outcomes of measurements he might perform—depends on what Alice chooses to measure at spacelike separation, then whose catalogue is it? The state is not a property of Bob’s particle alone; it is a property of the pair.

The measurement axioms of ch3 ((3.47), (M1)–(M5)) are silent on locality. The Lüders update applies to the combined system; the conditional state after Alice’s outcome is well-defined, and it predicts Bob’s outcomes with certainty. The question is whether that update describes a physical change in Bob’s distant particle (a collapse propagating faster than light) or merely an updating of Alice’s knowledge about a pre-existing state of affairs. Quantum mechanics, as formulated in ch3, does not answer this question—the Lüders rule is an algorithm for computing probabilities, not a mechanism.

Example: classical vs. quantum correlation. Compare the singlet with a classical analogue: two boxes, one containing a white ball and the other a black ball, prepared by an assistant and sent to Alice and Bob. When Alice opens her box and sees white, she knows with certainty that Bob’s box contains

black. But the ball's color is definite before she opens the box—classical ignorance, not quantum indeterminacy.

The singlet is deeper. If Alice chooses to measure the “color in the z -direction” (σ_z) and obtains white (+1), Bob's ball is black (−1) in the z -direction. If instead she measures the “color in the x -direction” (σ_x) and obtains white (+1), Bob's ball is black (−1) in the x -direction. In the classical analogue, the balls would need pre-assigned colors for every possible measurement direction—an infinite set of definite values, one for each axis, all simultaneously real. But quantum mechanics, via the noncommutation $[\sigma_z, \sigma_x] = 2i\sigma_y$, forbids simultaneous definite values for noncommuting observables. The classical analogue fails because the “color” depends on which question is asked, and no pre-existing assignment can reproduce the quantum correlations for all axis choices simultaneously.

 **Note** *The EPR paper does not settle whether quantum mechanics is complete or incomplete. It states a dilemma. For 29 years, the dilemma was considered a matter of interpretation, not experiment—a philosophical dispute about the meaning of “physical reality.” Bell changed that. The next section derives Bell's theorem, which converts the EPR dilemma into a testable inequality that quantum mechanics violates.*

The EPR argument is the conceptual high-water mark of the first half of the twentieth century. It distills the tension between two principles that had been held as self-evident—locality and completeness of the physical description—and shows that quantum mechanics, applied to entangled states, cannot satisfy both. The argument's honesty should be noted: it does not claim to prove incompleteness; it claims to pose a dilemma. Bell's 1964 theorem, to which the next section turns, demonstrates that the dilemma is not a matter of taste: the two horns make different quantitative predictions, and experiment decides between them.

7.5 Bell Inequalities and the Aspect Experiments

The EPR argument of Section 7.4 posed a dilemma: either quantum mechanics is incomplete, and hidden variables complete it locally, or quantum mechanics is complete but nonlocal. For three decades the choice between these horns was a matter of interpretation. Bell (1964) changed that by asking: do the predictions of *all* local hidden-variable theories differ from quantum mechanics, and if so, can the difference be measured? The answer—the CHSH inequality and its quantum-mechanical violation—is the subject of this section. It is the chapter's landing on a measurable quantity: $|\langle S \rangle| = 2\sqrt{2} \approx 2.828$, confirmed by Aspect and successors.

Local hidden-variable models. A local hidden-variable (LHV) theory supplements the quantum state with a variable λ (distributed according to some normalized measure $\rho(\lambda) \geq 0$, $\int d\lambda \rho(\lambda) = 1$) such that measurement outcomes are deterministic or probabilistic functions of λ and the local measurement setting alone. The joint probability for Alice's outcome a (given her setting $\hat{a}$) and Bob's outcome b (given his setting $\hat{b}$) factorizes conditioned on λ :

$$P(a, b \mid \hat{a}, \hat{b}) = \int d\lambda \rho(\lambda) P_A(a \mid \hat{a}, \lambda) P_B(b \mid \hat{b}, \lambda), \quad (7.71)$$

where $P_A(a | \hat{a}, \lambda)$ is the probability that Alice obtains outcome a given her setting $\hat{a}$ and the hidden variable λ , and similarly for Bob. The factorized form enforces locality: Alice's outcome depends only on her setting and λ , not on Bob's setting or outcome, and conversely.

For the case of dichotomic outcomes $A(\hat{a}, \lambda) = \pm 1$ and $B(\hat{b}, \lambda) = \pm 1$ (deterministic given λ), the correlation function—the average product of outcomes for settings $\hat{a}$ and $\hat{b}$ —is

$$E(\hat{a}, \hat{b}) = \int d\lambda \rho(\lambda) A(\hat{a}, \lambda) B(\hat{b}, \lambda). \quad (7.72)$$

The restriction to deterministic outcomes given λ involves no loss of generality: any probabilistic local model can be converted to a deterministic one by expanding the hidden-variable space to include the local randomness.

The CHSH inequality. Consider four correlation measurements with two settings for Alice ($\hat{a}, \hat{a}'$) and two for Bob ($\hat{b}, \hat{b}'$). Define the CHSH quantity

$$S := E(\hat{a}, \hat{b}) + E(\hat{a}, \hat{b}') + E(\hat{a}', \hat{b}) - E(\hat{a}', \hat{b}'). \quad (7.73)$$

For any local hidden-variable model,

$$|S| \leq 2. \quad (7.74)$$

The proof uses only the existence of λ and the locality condition—it is independent of the details of the LHV model. For each fixed λ ,

$$S(\lambda) = A(\hat{a}, \lambda) [B(\hat{b}, \lambda) + B(\hat{b}', \lambda)] + A(\hat{a}', \lambda) [B(\hat{b}, \lambda) - B(\hat{b}', \lambda)]. \quad (7.75)$$

Since $B(\hat{b}, \lambda), B(\hat{b}', \lambda) = \pm 1$, the two brackets satisfy:

$$B(\hat{b}, \lambda) + B(\hat{b}', \lambda) = \begin{cases} \pm 2, & \text{if } B(\hat{b}, \lambda) = \pm B(\hat{b}', \lambda), \\ 0, & \text{otherwise,} \end{cases} \quad (7.76)$$

and conversely $B(\hat{b}, \lambda) - B(\hat{b}', \lambda) = 0$ when the sum is ± 2 , and ± 2 when the sum vanishes. Thus exactly one of the two brackets is ± 2 and the other is 0, so $S(\lambda) = \pm 2$ for every λ . Averaging over $\rho(\lambda)$:

$$|S| = \left| \int d\lambda \rho(\lambda) S(\lambda) \right| \leq \int d\lambda \rho(\lambda) |S(\lambda)| = \int d\lambda \rho(\lambda) \cdot 2 = 2. \quad (7.77)$$

The inequality is saturated by any LHV model that produces $S(\lambda) = +2$ (or -2) for all λ with the optimal sign pattern. But no local hidden-variable model can exceed 2 in absolute value.

 **Note** The CHSH inequality (Clauser, Horne, Shimony, Holt 1969, *Phys. Rev. Lett.* 23, 880) is a four-setting refinement of Bell's original 1964 inequality, which used three directions. The CHSH form is preferred experimentally because it does not require perfect anti-correlation for aligned settings. Bell's original three-direction inequality is derived in Exercise 7.7.

Quantum-mechanical violation. For the singlet state $|\Psi^-\rangle$ ((7.44)), the quantum correlation function for spin measurements along directions $\hat{a}$ and $\hat{b}$ is

$$E_{\text{QM}}(\hat{a}, \hat{b}) = \langle \Psi^- | (\mathbf{a} \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b} \cdot \boldsymbol{\sigma}) | \Psi^- \rangle = -\mathbf{a} \cdot \mathbf{b} = -\cos \theta_{ab}, \quad (7.78)$$

where θ_{ab} is the angle between the unit vectors $\hat{a}$ and $\hat{b}$.

The derivation uses the singlet's defining property $\langle \Psi^- | \sigma_i \otimes \sigma_j | \Psi^- \rangle = -\delta_{ij}$, which is established

by direct computation. Expand the product:

$$(\mathbf{a} \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b} \cdot \boldsymbol{\sigma}) = \sum_{i,j=1}^3 a_i b_j \sigma_i \otimes \sigma_j. \quad (7.79)$$

Taking the expectation in $|\Psi^-\rangle$:

$$\langle \Psi^- | (\mathbf{a} \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b} \cdot \boldsymbol{\sigma}) | \Psi^- \rangle = \sum_{i,j} a_i b_j \langle \Psi^- | \sigma_i \otimes \sigma_j | \Psi^- \rangle \quad (7.80)$$

$$= \sum_{i,j} a_i b_j (-\delta_{ij}) = - \sum_i a_i b_i = -\mathbf{a} \cdot \mathbf{b} = -\cos \theta_{ab}. \quad (7.81)$$

The identity $\langle \Psi^- | \sigma_i \otimes \sigma_j | \Psi^- \rangle = -\delta_{ij}$ is verified by direct computation. For the diagonal cases. For $i = j = z$:

$$\langle \Psi^- | \sigma_z \otimes \sigma_z | \Psi^- \rangle = \frac{1}{2} (\langle \uparrow \downarrow | - \langle \downarrow \uparrow |) (\sigma_z \otimes \sigma_z) (| \uparrow \downarrow \rangle - | \downarrow \uparrow \rangle) \quad (7.82)$$

$$= \frac{1}{2} ((+1)(-1) + (-1)(+1)) = -1. \quad (7.83)$$

For $i = j = x$, using $\sigma_x | \uparrow \rangle = | \downarrow \rangle$ and $\sigma_x | \downarrow \rangle = | \uparrow \rangle$:

$$(\sigma_x \otimes \sigma_x) | \uparrow \downarrow \rangle = | \downarrow \uparrow \rangle, \quad (\sigma_x \otimes \sigma_x) | \downarrow \uparrow \rangle = | \uparrow \downarrow \rangle. \quad (7.84)$$

Then

$$\langle \Psi^- | \sigma_x \otimes \sigma_x | \Psi^- \rangle = \frac{1}{2} (\langle \uparrow \downarrow | - \langle \downarrow \uparrow |) (| \downarrow \uparrow \rangle - | \uparrow \downarrow \rangle) \quad (7.85)$$

$$= \frac{1}{2} (\langle \uparrow \downarrow | \downarrow \uparrow \rangle - \langle \uparrow \downarrow | \uparrow \downarrow \rangle - \langle \downarrow \uparrow | \downarrow \uparrow \rangle + \langle \downarrow \uparrow | \uparrow \downarrow \rangle) \quad (7.86)$$

$$= \frac{1}{2} (0 - 1 - 1 + 0) = -1. \quad (7.87)$$

The $i = j = y$ case gives -1 by the same computation (using $\sigma_y | \uparrow \rangle = i | \downarrow \rangle$, $\sigma_y | \downarrow \rangle = -i | \uparrow \rangle$, the factors of i cancel in the product $\sigma_y \otimes \sigma_y$).

For the off-diagonal case $i = x$, $j = z$ as a representative:

$$\langle \Psi^- | \sigma_x \otimes \sigma_z | \Psi^- \rangle = \frac{1}{2} (\langle \uparrow \downarrow | - \langle \downarrow \uparrow |) (\sigma_x \otimes \sigma_z) (| \uparrow \downarrow \rangle - | \downarrow \uparrow \rangle) \quad (7.88)$$

$$= \frac{1}{2} (\langle \uparrow \downarrow | - \langle \downarrow \uparrow |) (| \downarrow \downarrow \rangle - (- | \uparrow \uparrow \rangle)) \quad (7.89)$$

$$= \frac{1}{2} (\langle \uparrow \downarrow | \downarrow \downarrow \rangle + \langle \uparrow \downarrow | \uparrow \uparrow \rangle - \langle \downarrow \uparrow | \downarrow \downarrow \rangle - \langle \downarrow \uparrow | \uparrow \uparrow \rangle) \quad (7.90)$$

$$= \frac{1}{2} (0 + 0 - 0 - 0) = 0. \quad (7.91)$$

The remaining cases follow the same pattern: when $i = j$, the action of $\sigma_i \otimes \sigma_i$ exchanges the two product vectors in the singlet (up to a sign), producing $\langle \Psi^- | \sigma_i \otimes \sigma_i | \Psi^- \rangle = \frac{1}{2} (-1 - 1) = -1$ in each case; when $i \neq j$, the action of $\sigma_i \otimes \sigma_j$ maps each product vector to one orthogonal to both original terms, so all four inner products vanish individually. The identity $\langle \Psi^- | \sigma_i \otimes \sigma_j | \Psi^- \rangle = -\delta_{ij}$ is thus established. The singlet's rotational invariance—the unique state annihilated by $\mathbf{J} = \mathbf{J}_1 \otimes I + I \otimes \mathbf{J}_2$ —guarantees that this δ_{ij} structure is basis-independent: $-\delta_{ij}$ is the only rotationally invariant two-index tensor on $\mathbb{R}^3$ that is symmetric and negative definite.

The CHSH observable on $\mathbb{C}^2 \otimes \mathbb{C}^2$ is

$$S = (\mathbf{a} \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b} \cdot \boldsymbol{\sigma}) + (\mathbf{a} \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b}' \cdot \boldsymbol{\sigma}) + (\mathbf{a}' \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b} \cdot \boldsymbol{\sigma}) - (\mathbf{a}' \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b}' \cdot \boldsymbol{\sigma}), \quad (7.92)$$

whose expectation value in the singlet is

$$\langle S \rangle_{\text{QM}} = -\mathbf{a} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{b}' - \mathbf{a}' \cdot \mathbf{b} + \mathbf{a}' \cdot \mathbf{b}'. \quad (7.93)$$

For any quantum state and any choice of observables A, A', B, B' with eigenvalues ± 1 , the Tsirelson bound restricts the CHSH expectation:

$$|\langle S \rangle| \leq 2\sqrt{2}, \quad (7.94)$$

proved by Tsirelson (Cirel'son 1980, *Lett. Math. Phys.* 4, 93), cited here without proof. The bound is saturated by the singlet with optimal measurement settings.

Optimal violation. The CHSH expectation is maximized by choosing the four measurement directions in a plane (say the xz -plane), with Alice's two directions orthogonal and Bob's two directions orthogonal, and the two pairs rotated by $\pi/4$ relative to each other. The optimal angles are

$$\hat{a} = 0, \quad \hat{a}' = \frac{\pi}{2}, \quad \hat{b} = \frac{\pi}{4}, \quad \hat{b}' = -\frac{\pi}{4}, \quad (7.95)$$

where angles are measured from the z -axis in the xz -plane. In vector form: $\hat{a} = \hat{z}$, $\hat{a}' = \hat{x}$, $\hat{b} = (\hat{z} + \hat{x})/\sqrt{2}$, $\hat{b}' = (\hat{z} - \hat{x})/\sqrt{2}$.

The four correlation terms evaluate by computing each dot product explicitly from the vectors $\hat{a} = \hat{z}$, $\hat{a}' = \hat{x}$, $\hat{b} = (\hat{z} + \hat{x})/\sqrt{2}$, $\hat{b}' = (\hat{z} - \hat{x})/\sqrt{2}$:

$$\hat{a} \cdot \hat{b} = \hat{z} \cdot \frac{\hat{z} + \hat{x}}{\sqrt{2}} = \frac{1}{\sqrt{2}}, \quad \cos \theta_{ab} = \frac{1}{\sqrt{2}}, \quad \theta_{ab} = \frac{\pi}{4}, \quad (7.96)$$

$$\hat{a} \cdot \hat{b}' = \hat{z} \cdot \frac{\hat{z} - \hat{x}}{\sqrt{2}} = \frac{1}{\sqrt{2}}, \quad \cos \theta_{ab'} = \frac{1}{\sqrt{2}}, \quad \theta_{ab'} = \frac{\pi}{4}, \quad (7.97)$$

$$\hat{a}' \cdot \hat{b} = \hat{x} \cdot \frac{\hat{z} + \hat{x}}{\sqrt{2}} = \frac{1}{\sqrt{2}}, \quad \cos \theta_{a'b} = \frac{1}{\sqrt{2}}, \quad \theta_{a'b} = \frac{\pi}{4}, \quad (7.98)$$

$$\hat{a}' \cdot \hat{b}' = \hat{x} \cdot \frac{\hat{z} - \hat{x}}{\sqrt{2}} = -\frac{1}{\sqrt{2}}, \quad \cos \theta_{a'b'} = -\frac{1}{\sqrt{2}}, \quad \theta_{a'b'} = \frac{3\pi}{4}. \quad (7.99)$$

Using $E_{\text{QM}} = -\cos \theta_{ab}$:

$$E(\hat{a}, \hat{b}) = -\cos(\pi/4) = -1/\sqrt{2}, \quad (7.100)$$

$$E(\hat{a}, \hat{b}') = -\cos(\pi/4) = -1/\sqrt{2}, \quad (7.101)$$

$$E(\hat{a}', \hat{b}) = -\cos(\pi/4) = -1/\sqrt{2}, \quad (7.102)$$

$$E(\hat{a}', \hat{b}') = -\cos(3\pi/4) = -(-1/\sqrt{2}) = +1/\sqrt{2}. \quad (7.103)$$

The CHSH sum is

$$\langle S \rangle_{\text{QM}} = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{4}{\sqrt{2}} = -2\sqrt{2}. \quad (7.104)$$

Thus $|\langle S \rangle| = 2\sqrt{2} \approx 2.828 > 2$. The quantum prediction exceeds the local bound by a factor of $\sqrt{2}$ and saturates the Tsirelson bound (7.94).

Note The sign of $\langle S \rangle$ is negative; the absolute value $2\sqrt{2}$ is the violation. The four-term sum is $-2\sqrt{2}$ for the specific sign convention of the CHSH operator as defined in (7.92). Flipping the sign convention (e.g., defining the last term with a $+$ sign instead of $-$) would change the sign of $\langle S \rangle$ but not its magnitude. The physical content is $|\langle S \rangle| = 2\sqrt{2} > 2$.

The Aspect experiments (1981–1982) and beyond. The CHSH inequality was tested experimentally by Aspect, Grangier, and Roger (1981, *Phys. Rev. Lett.* 47, 460; 1982, *ibid.* 49, 91; Aspect, Dalibard, and Roger 1982, *ibid.* 49, 1804). The experiments used a calcium cascade $4p^2 \ ^1S_0 \rightarrow 4s4p \ ^1P_1 \rightarrow$

$4s^2\ ^1S_0$, producing entangled photon pairs at wavelengths 551.3 nm and 422.7 nm. The photons' polarizations were measured along the four CHSH directions; the measured correlation function agreed with the quantum prediction $E = -\cos(2\theta_{\text{pol}})$ (photons are spin-1, not spin-1/2; the relevant angle is twice the polarizer angle, yielding the same $\cos^2$ structure). The 1981 experiment measured

$$S = 2.697 \pm 0.015, \quad (7.105)$$

a 40σ violation of the CHSH bound $|S| \leq 2$.

The 1982 experiment closed the locality loophole by using rapidly switched polarizer orientations via acousto-optical devices. The setting changes occurred during the photon flight time, ensuring that the two measurement events were spacelike separated: no signal traveling at or below the speed of light could communicate Alice's setting choice to Bob's detector before his photon arrived. The violation persisted.

Subsequent experiments closed the remaining loopholes. Weihs et al. (1998, *Phys. Rev. Lett.* 81, 5039) achieved a 30σ violation under strict Einstein locality. Hensen et al. (2015, *Nature* 526, 682) performed the first loophole-free Bell test, combining spacelike separation with high-efficiency detection to close both the locality and the fair-sampling loopholes simultaneously. The verdict is unanimous across platforms—photons, trapped ions, superconducting qubits, nitrogen-vacancy centers—and across decades: local realism is false. Nature is nonlocal in exactly the way quantum mechanics predicts.

 **Note** *The CHSH violation rules out all local hidden-variable theories. It does not rule out nonlocal hidden-variable theories: Bohmian mechanics is explicitly nonlocal and reproduces all quantum predictions, including the CHSH violation. It does not permit superluminal signaling: the no-signaling theorem (proved in one paragraph below) establishes that Alice's marginal statistics are independent of Bob's measurement choices. The tension with relativistic causality is real—Bell's theorem shows that any theory reproducing quantum correlations must be nonlocal in some sense—but the conflict is with local realism, not with the no-signaling constraints of special relativity.*

The no-signaling theorem. Alice's reduced density operator after tracing out Bob's particle is independent of what Bob chooses to measure. For the singlet, $\rho_A = \text{tr}_B(|\Psi^-\rangle\langle\Psi^-|) = I/2$ regardless of whether Bob measures σ_z , σ_x , or nothing at all. More generally, for any bipartite state ρ_{AB} and any PVM $\{P_b^{(B)}\}$ on Bob's side, the non-selective post-measurement state of Alice is

$$\rho'_A = \text{tr}_B\left(\sum_b (I \otimes P_b^{(B)}) \rho_{AB} (I \otimes P_b^{(B)})\right) = \text{tr}_B(\rho_{AB}) = \rho_A, \quad (7.106)$$

because tr_B is cyclic on Bob's operators and $\sum_b P_b^{(B)} = I$. Bob's measurement choice, encoded in the PVM $\{P_b^{(B)}\}$, disappears from Alice's reduced state. No signaling is possible: the CHSH violation exhibits nonlocality without enabling superluminal communication.

The advantage of the Bell/CHSH framework is that it converts a philosophical dilemma into an experimentally decidable question: the number $|\langle S \rangle|$ is measurable in a laboratory, and the answer— $2\sqrt{2}$, not ≤ 2 —is a fact about nature, not an interpretation. Its limitation is that the violation does not single out a particular resolution of the measurement problem; Copenhagen, many-worlds, and Bohmian mechanics all accommodate the violation, the latter by explicit nonlocality.

The next section restates the measurement problem in the language of entanglement, using the density operator and the von Neumann measurement scheme to make precise what the CHSH violation leaves unresolved.

7.6 The Measurement Problem Revisited

Chapter 3’s measurement axioms (M1)–(M5) describe a measurement’s effect—the Born rule probabilities, the Lüders update—but are silent about the measurement process. The non-selective measurement paragraph (ch3 §3.1) deferred the problem: how does a unitary interaction between system and apparatus produce the definite outcomes that experience reports? Now, with the tensor product (Section 7.1) and the density operator (Section 7.2), the process can be modeled, and the problem can be stated precisely. This section presents the von Neumann measurement scheme, identifies the three sub-problems it exposes, names decoherence as the dominant approach to the first two (for all practical purposes), and leaves the “and/or” problem as the deep conceptual residue.

The von Neumann measurement scheme (1932). Consider a system with an observable $A = \sum_n a_n |a_n\rangle \langle a_n|$ (nondegenerate for simplicity). The system is prepared in a superposition $|\psi\rangle = \sum_n c_n |a_n\rangle$ with $\sum_n |c_n|^2 = 1$. A measuring apparatus, initially in a ready state $|A_0\rangle$, couples to the system via a unitary interaction U_{meas} designed to correlate each eigenstate of A with a distinct, macroscopically distinguishable pointer state $|A_n\rangle$ of the apparatus:

$$U_{\text{meas}} |a_n\rangle \otimes |A_0\rangle = |a_n\rangle \otimes |A_n\rangle, \quad \langle A_m | A_n \rangle = \delta_{mn}, \quad (7.107)$$

where the orthonormality of the pointer states encodes their distinguishability: a subsequent readout of the apparatus can reliably determine which $|A_n\rangle$ is occupied. The interaction is a controlled unitary: conditioned on the system’s eigenstate, the apparatus is rotated to the corresponding pointer state.

By linearity of the Schrödinger evolution (Axiom D1, (4.1)), the interaction applied to the superposition produces

$$|\Psi\rangle = U_{\text{meas}} \left(\sum_n c_n |a_n\rangle \right) \otimes |A_0\rangle = \sum_n c_n |a_n\rangle \otimes |A_n\rangle, \quad (7.108)$$

an entangled pure state of system and apparatus. This is the post-measurement state *before observation*. It is a superposition of macroscopic configurations: each term $|a_n\rangle \otimes |A_n\rangle$ represents a definite outcome (the apparatus reads A_n), but the state as a whole is a coherent sum of all such terms. The density operator corresponding to (7.108) is

$$\rho = \sum_{m,n} c_m c_n^* |a_m\rangle \langle a_n| \otimes |A_m\rangle \langle A_n|, \quad (7.109)$$

where the off-diagonal terms ($m \neq n$) encode quantum coherence between the branches.

The three sub-problems. The state (7.108) exposes three distinct problems.

1. *The problem of determinate outcomes.* The post-measurement state $|\Psi\rangle$ is a superposition of macroscopically distinct configurations. Experience, however, reports a single definite outcome: the apparatus indicates A_{n_0} , not a superposition of all A_n . No axiom of the theory—neither the Schrödinger equation (deterministic and continuous, (4.3)) nor the Lüders update (probabilistic and

discontinuous, (3.47))—selects one term of the superposition over the others. The Lüders update is an *additional* rule, postulated in ch3 precisely because unitary evolution alone does not yield definite outcomes. This is the measurement problem proper.

2. *The problem of the preferred basis.* The von Neumann scheme (7.107) picks out the eigenbasis of the measured observable A as the basis in which the apparatus states become correlated. But what picks out *that* observable? Why does the apparatus indicate $|A_n\rangle$ (pointer states) rather than superpositions $\sum_n d_n |A_n\rangle$? The scheme assumes that the apparatus is designed to measure A ; it does not explain why the pointer basis—the basis in which the apparatus states are stable under the system–apparatus–environment interaction—coincides with the eigenbasis of a particular observable. This is the preferred-basis problem.

3. *The problem of the quantum-to-classical transition.* The density operator (7.109) possesses off-diagonal coherence between the branches. The observed ensemble after many runs of the experiment is described by a diagonal density operator in the pointer basis:

$$\rho_{\text{obs}} = \sum_n |c_n|^2 |a_n\rangle \langle a_n| \otimes |A_n\rangle \langle A_n|, \quad (7.110)$$

with each outcome occurring with frequency $|c_n|^2$ and no interference between terms. Where and how does the interference vanish? The formal transition from the coherent ρ to the diagonal ρ_{obs} is not described by any unitary evolution: ρ_{obs} and ρ have the same diagonal elements but ρ has nonzero off-diagonal elements, and unitary transformations preserve eigenvalues, hence do not alter the degree of coherence. The vanishing of off-diagonal terms requires a non-unitary process.

Decoherence: named, not developed. When the apparatus couples to an environment with many uncontrolled degrees of freedom (photons, air molecules, thermal radiation), the system–apparatus–environment interaction rapidly entangles the apparatus pointer states with orthogonal environment states. Tracing out the environment—which is operationally unavoidable, since no measurement accesses all environmental degrees of freedom—suppresses the off-diagonal terms of the system–apparatus reduced density operator:

$$\rho_{SA} \approx \sum_n |c_n|^2 |a_n\rangle \langle a_n| \otimes |A_n\rangle \langle A_n|, \quad (7.111)$$

where the approximation becomes exact in the limit of infinitely many environment degrees of freedom with perfect orthogonality of environment states.



Note The density operator (7.111) is approximately diagonal for all practical purposes (FAPP, in Bell’s phrase). The global state—system plus apparatus plus environment—remains pure: the entanglement has merely been transferred to the environment, where it is inaccessible. Decoherence theory (Zurek’s einselection, named here; Zurek 1981, *Phys. Rev. D* 24, 1516) provides the quantitative framework, including the timescale on which off-diagonal terms decay (the decoherence time, typically many orders of magnitude shorter than the relaxation time) and the mechanism by which the environment selects the pointer basis (the states that are robust under environment monitoring, i.e., that commute with the system–environment interaction Hamiltonian). The full theory—master equations, Lindblad operators, environment models—is a forward debt, beyond the scope of this chapter.

Decoherence addresses problems 1 and 2 FAPP: the reduced density operator becomes effec-

tively diagonal in the einselected basis, explaining why the world *appears* classical (no macroscopic superpositions are observed) and why the pointer basis is the one that is stable under environmental monitoring. But decoherence does not solve problem 1 in principle. The diagonal ensemble (7.111) is an improper mixture—the reduced state of a subsystem of a larger pure state. It describes the statistics of outcomes that would be obtained if the system were measured, but it does not explain why one particular outcome is *experienced*. The “and/or” problem—the superposition says *and* (all outcomes coexist); experience says *or* (one outcome occurs)—remains.

The measurement problem as the theory’s open wound. The conflict is between two fundamental principles of the theory. Unitary evolution (ch4’s D1, (4.1)) is continuous, deterministic, and entanglement-preserving. State reduction (ch3’s M3, (3.47)) is discontinuous, probabilistic, and coherence-destroying. Both are essential to the theory’s empirical success: unitary evolution accounts for every known dynamical law; state reduction accounts for the fact that measurements have outcomes. But the two principles are incompatible as descriptions of a single process. The measurement problem is the statement that the theory contains no rule for deciding when to apply one and when the other.

The Copenhagen reading places the cut between the quantum and classical domains at the observer: the apparatus is classical, described by classical physics, and the formalism’s job is to connect preparations to outcomes. What happens “in between” is a question the formalism is not designed to answer. This reading is stated, not advocated; it is historically significant and conceptually economical, but it leaves the cut’s location arbitrary and the observer’s special status unexplained.

Schrödinger’s cat. The limiting case of the von Neumann chain illustrates the problem. A radioactive atom in a superposition of undecayed ($|0\rangle$) and decayed ($|1\rangle$) entangles with a Geiger counter, which entangles with a hammer, which entangles with a vial of poison, which entangles with a cat:

$$|\Psi\rangle = \alpha |0\rangle \otimes |\text{silent}\rangle \otimes \cdots \otimes |\text{alive}\rangle + \beta |1\rangle \otimes |\text{click}\rangle \otimes \cdots \otimes |\text{dead}\rangle. \quad (7.112)$$

Every intermediate stage entangles cleanly by the von Neumann scheme (7.107). The cat, when treated as a quantum system, ends up in an entangled superposition of alive and dead. But cats are not observed in superpositions. The question—where along this chain does the superposition end and a definite outcome begin?—is the measurement problem in its starkest form. Decoherence makes the cat effectively alive-or-dead FAPP; the “or” of experience versus the “and” of the state is the book’s final open question.

Ch1 echo. The historical chapter closed with the operational definition of a state: a catalogue of probabilities for the outcomes of every conceivable measurement (1.5.5). The measurement problem is the ultimate consequence of that definition. If the state is a catalogue, and the catalogue collapses upon measurement, then the collapse is a change in the catalogue, not necessarily a physical process. Whether this view is tenable—whether the catalogue is all there is, or whether something beyond the catalogue determines which entry is actualized—is the interpretive question the book leaves open.

The advantage of the von Neumann scheme is its precision: it models the measurement interaction as a physical process (a unitary coupling) within the theory, exposing exactly where the axioms of ch3 stop describing and start postulating. Its limitation is that it does not solve the problem it illuminates. Decoherence explains FAPP why superpositions are not observed; it does not explain, and

is not claimed to explain, why one particular outcome is experienced. The measurement problem, stated precisely in the language of entanglement, is the chapter's—and the book's structural arc's—open wound.

The chapter closes with a summary of its four principles, the extended experiment–formalism dictionary, and the balanced debt ledger.

7.7 Summary and Open Debts

This chapter completed the book's structural arc. The theory of nonrelativistic quantum mechanics, built axiom by axiom from ch2 through ch6, now includes the description of composite systems, the density operator for subsystems, the classification of entangled states, the EPR argument as the conceptual challenge, and the Bell/CHSH inequality as the measurable signature of quantum nonlocality. The chapter added zero new axioms: every display was proved from the ch2–ch6 axioms, with the honesty declarations specified at each point of use.

The four principles.

- (E1) **Composite systems.** The Hilbert space of a composite system is the tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$ of the subsystem spaces ((7.5)). The product basis $\{|e_i\rangle \otimes |f_j\rangle\}$ has dimension $d_A d_B$ ((7.6)). Operators act factor-wise: $(A \otimes B)(|\psi_A\rangle \otimes |\phi_B\rangle) = A|\psi_A\rangle \otimes B|\phi_B\rangle$ ((7.10)), extended linearly. Not every state is a product vector; states that are not—states whose coefficient matrix has rank greater than one—are entangled.
- (E2) **Density operator.** A general state is described by a density operator $\rho = \sum_k p_k |\psi_k\rangle \langle \psi_k|$, with $\rho = \rho^\dagger$, $\text{tr} \rho = 1$, $\rho \geq 0$ ((7.19)). Expectation values are $\langle A \rangle = \text{tr}(\rho A)$ ((7.21)). Purity $\text{tr}(\rho^2) \leq 1$ ((7.24)) distinguishes pure from mixed states. The state of a subsystem is obtained by the partial trace $\rho_A = \text{tr}_B(\rho_{AB})$ ((7.27)). A subsystem of a pure entangled state is mixed ((7.31)). The density operator does not distinguish proper mixtures (classical ignorance) from improper mixtures (reduced states of entangled composites).
- (E3) **Entanglement.** Bipartite pure states admit the Schmidt decomposition $|\Psi\rangle = \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle \otimes |v_k\rangle$ ((7.12)), with Schmidt rank r and Schmidt coefficients λ_k equal to the eigenvalues of ρ_A . A pure state is separable iff $r = 1$ ((7.41)); maximally entangled iff $\lambda_k = 1/r$ ((7.42)). The Bell basis $\{|\Psi^\pm\rangle, |\Phi^\pm\rangle\}$ ((7.44)) is the maximally entangled ONB of $\mathbb{C}^2 \otimes \mathbb{C}^2$. Mixed-state separability is defined via convex combinations of product states ((7.53)). The entanglement entropy $S(\rho_A) = -\text{tr}(\rho_A \log_2 \rho_A)$ ((7.54)) quantifies pure-state entanglement: $S = 0$ for product states, $S = 1$ for the singlet, $S \leq \log_2 d$ universally.
- (E4) **Bell nonlocality.** Local hidden-variable models factorize conditioned on λ ((7.71)) and satisfy the CHSH inequality $|S| \leq 2$ ((7.74)). Quantum mechanics predicts $|\langle S \rangle| \leq 2\sqrt{2}$ ((7.94)), saturated by the singlet at $|\langle S \rangle| = 2\sqrt{2}$ for the optimal measurement angles ((7.95)). The Aspect experiments confirm the quantum prediction, ruling out local realism. The no-signaling theorem establishes that the violation does not enable superluminal communication.

The experiment–formalism dictionary v6. Table 7.1 extends the dictionaries of ch2–ch6 with the

twelve rows of the composite-systems and entanglement vocabulary. Every laboratory notion has a formal object; every formal object has a laboratory warrant.

Table 7.1: The experiment–formalism dictionary v6: Composite systems and entanglement. Extends tab:ch2-dictionary through tab:ch6-dictionary.

Laboratory notion	Formal object
Composite system	Tensor product $\mathcal{H}_A \otimes \mathcal{H}_B$
Subsystem state	Reduced density operator $\text{tr}_B(\rho_{AB})$
Product state	Factorable vector, Schmidt rank 1
Entangled state	Non-product state, Schmidt rank > 1
Maximal entanglement	Reduced state $\rho_A = I/r$
Local hidden variable	λ with factorized outcomes
Bell inequality	CHSH bound $ S \leq 2$
Quantum violation	$2\sqrt{2}$, Aspect confirmation
Measurement process	von Neumann entangling unitary U_{meas}
Pointer state	Apparatus eigenstate $ A_n\rangle$
Decoherence	Environment-induced diagonalization, FAPP
Mixed state	Density operator ρ , $\text{tr}(\rho^2) \leq 1$
Entanglement entropy	$S = -\text{tr}(\rho_A \log_2 \rho_A)$

Debt ledger, balanced. The chapter’s three entry contracts from ch3 and ch6 are discharged:

- **ch3 debt #20:** The density operator is defined ((7.18)), its properties are proved ((7.19)), expectation values via the trace are derived ((7.21)), purity established ((7.24)), the partial trace defined ((7.27)), reduced states of pure composites computed ((7.31)), non-selective measurement formulated in density language ((7.34)), proper vs. improper mixtures distinguished, and the measurement problem restated in the entanglement language (§7.6)—repaying exactly the ledger ch3 opened.
- **ch6 debt #31:** The tensor product is formalized as a Hilbert space ((7.5)), its basis and inner product are constructed ((7.6), (7.7)), operator action is defined ((7.10), (7.11)), and the dimension formula $\dim = d_A d_B$ is proved.
- **ch6 debt #32:** Mixed states, the partial trace, entanglement measures via reduced density operators, the Bloch ball ((7.25)), and the entanglement entropy ((7.54)) are all built.
- **ch6 debt #33:** The singlet/triplet entanglement is classified: Schmidt rank 2, Bell states constructed ((7.44)), the singlet identified as $|\Psi^-\rangle$, the EPR argument stated in the modern spin language (§7.4), the CHSH inequality derived and violated (§7.5), and the Aspect experiments described as the chapter’s measurable landing.

The chapter plants the following forward debts:

- **Decoherence theory:** Named with the FAPP honesty clause ((7.111)); the full theory (master equations, Lindblad operators, environment models, einselection) is a forward debt to a future treatment or advanced course.
- **Quantum information theory:** The Bell basis, no-cloning theorem, and no-signaling theorem are named as one-sentence pointers; quantum computation, cryptography, error correction, and

teleportation are explicitly beyond the book's scope. The door is left ajar. 7727

- **The measurement problem:** Stated precisely in the entanglement language; declared unresolved. The “and/or” problem remains. 7728
- **CHSH as the paradigm for ch8:** The CHSH violation is the chapter's measurable landing—an exact number derived from structural principles and confirmed by experiment. The approximation methods of ch8 (perturbation theory, scattering theory) now have the full structural toolkit behind them and a paradigm to emulate: every prediction must land on a number that can be measured. 7730

Exit question. The structural arc of the book is complete. From ch1's experimental crises—the ultra-violet catastrophe, the photoelectric effect, the Balmer spectrum, the Stern–Gerlach two-valuedness—through ch7's Bell violation at $2\sqrt{2}$, the theory of nonrelativistic quantum mechanics now possesses its full axiomatic structure: states as rays of a Hilbert space, measurement as spectral decomposition, evolution as unitary, symmetries as group representations, angular momentum as the rotation spectrum, and composite systems as tensor products with entanglement as their defining non-classical feature. The next question is not “what is the theory?” but “what does the theory predict for real systems?”—the time-independent and time-dependent perturbation theory, the variational method, the Born approximation, the partial-wave expansion, and the scattering matrix of ch8. 7731

Chapter 7 Exercise

Exercise 7.1 Tensor product basis. Construct the tensor product basis for $\mathbb{C}^2 \otimes \mathbb{C}^3$ (a qubit and a qutrit). Let $\{|\uparrow\rangle, |\downarrow\rangle\}$ be the standard basis of $\mathbb{C}^2$ and $\{|1\rangle, |0\rangle, |-1\rangle\}$ the standard basis of $\mathbb{C}^3$ (the $j = 1$ eigenstates of S_z defined in ch6 Exercise 7.3). 7744

- (a) Write the six product basis vectors $|e_i\rangle \otimes |f_j\rangle$. What is the dimension of $\mathbb{C}^2 \otimes \mathbb{C}^3$? 7745
- (b) Write the matrix representation of $\sigma_z \otimes I_3$ in the tensor product basis. Verify that it is block-diagonal with two 3×3 blocks. 7746
- (c) Write the matrix representation of $I_2 \otimes S_z$ (where $S_z = \hbar \text{diag}(1, 0, -1)$ is the $j = 1$ spin matrix). Verify that it is block-diagonal with three 2×2 blocks. 7747
- (d) Verify explicitly that $[\sigma_z \otimes I_3, I_2 \otimes S_z] = 0$. What are the joint eigenvalues of these two operators? How many distinct joint outcomes does a measurement of both observables produce? 7748

Exercise 7.2 Partial trace of the singlet and triplet. Starting from the pure states $\rho = |\psi\rangle\langle\psi|$ for the two-spin-1/2 coupled states: 7749

- (a) For the singlet $|0, 0\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ ((6.84)), compute $\rho_A = \text{tr}_B(|0, 0\rangle\langle 0, 0|)$ explicitly in the $\{|\uparrow\rangle, |\downarrow\rangle\}$ basis. Show that $\rho_A = I/2$ and $\text{tr}(\rho_A^2) = 1/2$. 7750
- (b) For the triplet state $|1, 1\rangle = |\uparrow\uparrow\rangle$ ((6.84)), compute $\rho_A = \text{tr}_B(|1, 1\rangle\langle 1, 1|)$. Show that ρ_A is pure and identify which pure state it is. What is $\text{tr}(\rho_A^2)$? 7751
- (c) For the triplet state $|1, 0\rangle = (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2}$, compute ρ_A and show that it is mixed. Compute $\text{tr}(\rho_A^2)$. 7752
- (d) Compare the reduced states of the singlet and the $|1, 0\rangle$ triplet state. Both are mixed—explain the physical difference between the two cases in terms of the correlations between 7753

the two spins.

Exercise 7.3 Schmidt decomposition of a parametrized state. Consider the parametrized two-qubit state

$$|\psi(\theta)\rangle = \cos \theta |\uparrow\uparrow\rangle + \sin \theta |\downarrow\downarrow\rangle, \quad 0 \leq \theta \leq \frac{\pi}{2}. \quad (7.113)$$

- Find the Schmidt decomposition: identify the Schmidt vectors $\{|u_k\rangle\}$, $\{|v_k\rangle\}$ and the Schmidt coefficients $\{\lambda_k\}$. What is the Schmidt rank r ? For what values of θ does r change?
- Compute the reduced density operator $\rho_A = \text{tr}_B(|\psi(\theta)\rangle \langle \psi(\theta)|)$, its eigenvalues, and the entanglement entropy $S(\theta) = -\text{tr}(\rho_A \log_2 \rho_A)$. Express S in terms of the binary entropy function $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$.
- For what value of θ is the state maximally entangled? What is the entanglement entropy there? For what values is the state a product state?
- Generalize to $|\psi\rangle = \cos \theta |\uparrow\uparrow\rangle + e^{i\phi} \sin \theta |\downarrow\downarrow\rangle$. Show that the phase ϕ does not affect the Schmidt coefficients or the entanglement entropy. Physical interpretation: why is entanglement invariant under local unitaries?

Exercise 7.4 Bell states, local unitaries, and the no-cloning theorem.

- Verify the following relations among Bell states:

$$|\Phi^+\rangle = (I \otimes \sigma_x) |\Psi^+\rangle, \quad (7.114)$$

$$|\Phi^-\rangle = (I \otimes \sigma_z) |\Psi^+\rangle, \quad (7.115)$$

$$|\Psi^-\rangle = (I \otimes \sigma_y) |\Psi^+\rangle. \quad (7.116)$$

Show that the factor i appearing in the last relation is a global phase absorbed by ray equivalence.

- Prove the no-cloning theorem. Suppose a unitary operator U on $\mathcal{H} \otimes \mathcal{H}$ clones an arbitrary state in a two-dimensional subspace $\mathcal{H}_0 \subset \mathcal{H}$: $U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$ for all $|\psi\rangle \in \mathcal{H}_0$, where $|0\rangle$ is a fixed reference state. Apply U to the superposition $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, where $\{|0\rangle, |1\rangle\}$ is an orthonormal basis of $\mathcal{H}_0$, and show that linearity contradicts the cloning assumption. Why does this argument not forbid the preparation of many copies of a *known* state?

Exercise 7.5 CHSH violation for the singlet.

- Starting from the quantum correlation function $E_{\text{QM}}(\hat{a}, \hat{b}) = -\mathbf{a} \cdot \mathbf{b}$ ((7.78)), construct the CHSH expression $S = E(\hat{a}, \hat{b}) + E(\hat{a}, \hat{b}') + E(\hat{a}', \hat{b}) - E(\hat{a}', \hat{b}')$. Using planar geometry (all vectors in the xz -plane: $\hat{a} = \hat{z}$, $\hat{a}' = \hat{x}$, $\hat{b} = (\hat{z} + \hat{x})/\sqrt{2}$, $\hat{b}' = (\hat{z} - \hat{x})/\sqrt{2}$), evaluate each of the four terms and show that $\langle S \rangle = -2\sqrt{2}$, so $|\langle S \rangle| = 2\sqrt{2}$.
- Prove that no local hidden-variable theory can exceed $|S| = 2$. Write $S(\lambda) = A(\hat{a})(B(\hat{b}) + B(\hat{b}')) + A(\hat{a}')(B(\hat{b}) - B(\hat{b}'))$ and show that $S(\lambda) = \pm 2$ for every λ , using $A, B = \pm 1$.
- Verify that the four measurement directions are the optimal choice. For any four coplanar directions $\hat{a}, \hat{a}', \hat{b}, \hat{b}'$ with angles $\theta_a, \theta_{a'}, \theta_b, \theta_{b'}$, write $\langle S \rangle$ as a function of the four angles and maximize over all choices. Confirm that the maximum is $2\sqrt{2}$ and that the optimal angles are those given in (7.95) up to an overall rotation.
- The detection loophole: if the detection efficiency is $\eta < 1$ and undetected events are

discarded (the fair-sampling assumption), show that the effective CHSH bound rises to $2/\eta$. Explain why closing this loophole requires high-efficiency detectors, and state how Hensen et al. (2015) achieved a loophole-free test.

Exercise 7.6 Bell's original inequality (1964). Bell's 1964 paper used three measurement directions $\hat{a}, \hat{b}, \hat{c}$ rather than four.

(a) Derive Bell's original inequality:

$$|E(\hat{a}, \hat{b}) - E(\hat{a}, \hat{c})| \leq 1 + E(\hat{b}, \hat{c}), \quad (7.117)$$

for dichotomic outcomes $A, B = \pm 1$ in any LHV model. Use the identity $A(\hat{a})B(\hat{b}) - A(\hat{a})B(\hat{c}) = A(\hat{a})B(\hat{b})(1 \pm A(\hat{b})B(\hat{c})) - A(\hat{a})B(\hat{c})(1 \pm A(\hat{b})B(\hat{c}))$, take absolute values, and integrate over λ .

(b) For the singlet, take $\hat{a}, \hat{b}, \hat{c}$ in the xz -plane separated by $\pi/3$ (60°): $\theta_{ab} = \theta_{bc} = \pi/3$, $\theta_{ac} = 2\pi/3$. Compute $E(\hat{a}, \hat{b}) = -\cos(\pi/3) = -1/2$, $E(\hat{a}, \hat{c}) = -\cos(2\pi/3) = +1/2$, and $E(\hat{b}, \hat{c}) = -\cos(\pi/3) = -1/2$. Show that $|E(\hat{a}, \hat{b}) - E(\hat{a}, \hat{c})| = 1$ while $1 + E(\hat{b}, \hat{c}) = 1/2$, a violation by a factor of 2. Compare the violation ratios of Bell's original inequality and the CHSH inequality.

Exercise 7.7 Proper vs. improper mixtures.

- Ensemble I: $|\uparrow_z\rangle$ with probability $1/2$, $|\downarrow_z\rangle$ with probability $1/2$. Compute the density operator ρ_I in the $\{|\uparrow_z\rangle, |\downarrow_z\rangle\}$ basis.
- Ensemble II: $|\uparrow_x\rangle = (|\uparrow_z\rangle + |\downarrow_z\rangle)/\sqrt{2}$ with probability $1/2$, $|\downarrow_x\rangle = (|\uparrow_z\rangle - |\downarrow_z\rangle)/\sqrt{2}$ with probability $1/2$. Compute ρ_{II} and show $\rho_I = \rho_{II} = I/2$.
- Compute $\text{tr}(\rho_I \sigma_z)$ and $\text{tr}(\rho_{II} \sigma_z)$. Interpret the result physically: what does the vanishing of the expectation value tell you about the distinguishability of these two ensembles?
- A non-selective measurement of σ_z is performed on ρ_I . Compute the post-measurement state via the Lüders map $\rho \mapsto \sum_n P_n \rho P_n$ ((7.34)). Does it differ from ρ_I ? Explain why the maximally mixed state is a fixed point of the non-selective measurement map.

Exercise 7.8 PPT criterion for two qubits. The Peres–Horodecki criterion states that a bipartite state ρ_{AB} is separable only if its partial transpose $\rho_{AB}^{T_B}$ is positive semidefinite (PPT). For 2×2 and 2×3 systems, the condition is also sufficient.

- Define the partial transpose: $\rho_{AB}^{T_B}$ is obtained by transposing only the indices of subsystem B : $\langle e_i \otimes f_j | \rho_{AB}^{T_B} | e_k \otimes f_l \rangle = \langle e_i \otimes f_l | \rho_{AB} | e_k \otimes f_j \rangle$. Compute $\rho_{AB}^{T_B}$ for the singlet $\rho = |\Psi^-\rangle \langle \Psi^-|$. Write ρ as a 4×4 matrix in the product basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$ and apply the partial transpose. Find the eigenvalues of ρ^{T_B} and show that one is negative—the singlet is NPT, hence entangled.
- For a Werner state $\rho_W = p |\Psi^-\rangle \langle \Psi^-| + (1-p)I/4$ (the convex combination of the singlet and the maximally mixed state, with $0 \leq p \leq 1$), find the range of p for which $\rho_W^{T_B} \geq 0$ and the range for which it is entangled. Show that the threshold is $p = 1/3$: for $p \leq 1/3$, the Werner state is separable (PPT); for $p > 1/3$, it is entangled (NPT).

Exercise 7.9 Von Neumann measurement model. A spin-1/2 system in the state $|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle$ (with $|\alpha|^2 + |\beta|^2 = 1$) is measured by a second spin-1/2 particle (the apparatus) initially in $|\uparrow\rangle$. The

measurement interaction is

$$U_{\text{meas}} = \exp\left(-\frac{i}{\hbar} \frac{\pi}{4} \sigma_z \otimes \sigma_x\right). \quad (7.118)$$

- (a) Show that $U_{\text{meas}} = \frac{1}{\sqrt{2}}(I \otimes I - i \sigma_z \otimes \sigma_x)$ using $(\sigma_z \otimes \sigma_x)^2 = I \otimes I$. Then verify that $U_{\text{meas}} |\uparrow\rangle \otimes |\uparrow\rangle = |\uparrow\rangle \otimes |\uparrow\rangle$ and $U_{\text{meas}} |\downarrow\rangle \otimes |\uparrow\rangle = |\downarrow\rangle \otimes |\downarrow\rangle$. The apparatus flips if and only if the system is in $|\downarrow\rangle$.
- (b) Compute the post-interaction state $|\Psi\rangle = U_{\text{meas}}(|\psi\rangle \otimes |\uparrow\rangle)$. Write it in the form $|\Psi\rangle = \alpha |\uparrow\rangle \otimes |\uparrow\rangle + \beta |\downarrow\rangle \otimes |\downarrow\rangle$. Then compute the full density operator $\rho = |\Psi\rangle \langle\Psi|$ and identify the off-diagonal coherence terms.
- (c) Compute the reduced density operator of the apparatus: $\rho_{\text{app}} = \text{tr}_{\text{sys}}(\rho)$. Show that $\rho_{\text{app}} = |\alpha|^2 |\uparrow\rangle \langle\uparrow| + |\beta|^2 |\downarrow\rangle \langle\downarrow|$ —diagonal in the σ_z basis, with no off-diagonal terms. Where did the interference go? It resides in the system–apparatus correlations (the off-diagonal terms of ρ involve both subsystems). A measurement of the apparatus alone sees no coherence; the coherence is encoded in the entanglement.

Exercise 7.10 Entanglement entropy for Schmidt coefficients.

- (a) A pure bipartite state has Schmidt coefficients $\{\lambda_1, \lambda_2, \lambda_3\}$ summing to 1. For which choice of the λ_k is the entanglement entropy $S = -\sum_k \lambda_k \log_2 \lambda_k$ maximal? Compute S_{max} in terms of the number of nonzero λ_k .
- (b) Show that $S = 0$ if and only if exactly one $\lambda_k = 1$ and all others vanish. Interpret this result in terms of separability.
- (c) Prove that $S(\rho_A) = S(\rho_B)$ for any bipartite pure state. Use the fact that ρ_A and ρ_B have the same nonzero eigenvalues (established in Section 7.2 via the Schmidt decomposition).
- (d) Compute S for the two-qutrit state

$$|\psi\rangle = \frac{1}{\sqrt{3}}(|0, 0\rangle + |1, 1\rangle + |2, 2\rangle), \quad (7.119)$$

where $\{|0\rangle, |1\rangle, |2\rangle\}$ is an orthonormal basis of $\mathbb{C}^3$. Is this state maximally entangled? What is ρ_A ?

Chapter 8 Perturbative Methods and the Theory of Scattering

The old quantum theory died on three numbers it could not produce: the helium ground-state energy, the Einstein coefficients, and the Rutherford cross-section. All three are computed in this chapter.

The debts are explicit. Chapter 1 recorded four failures of the classical atom that the old quantum theory (Bohr–Sommerfeld) could not repair: the helium atom’s binding energy, the Einstein A and B coefficients for radiative transitions (Section 1.1.6), and the Rutherford cross-section for Coulomb scattering – each discussed in Chapter 1. The states, measurement, evolution, and symmetry chapters (Chapters 2–4) built the Hilbert-space arena; the operator-method and angular-momentum chapters (Chapters 5–6) armed it with the computation machinery – the interaction picture, the harmonic-oscillator ladder algebra, the Wigner–Eckart theorem, the hydrogen radial wavefunctions, and the central-potential radial equation. The chapter before this one, Chapter 7, added the tensor-product structure for composite systems. This chapter collects those tools and repays the debts, one by one.

The forward debts from Chapters 4–6 are equally specific. Chapter 4’s Dyson series (4.12) stands untruncated; truncated after its first nontrivial term it becomes time-dependent perturbation theory, the seed of Fermi’s golden rule. Chapter 5’s exact forced-oscillator solution (5.37) is the benchmark against which that truncation is checked. Chapter 6’s Wigner–Eckart theorem (6.93) and hydrogen wavefunctions (6.109) supply the matrix-element engine. Chapter 6’s central-potential radial equation (6.62) provides the scattering skeleton.

The chapter’s thesis: perturbation theory and scattering theory are the bridge from the Hilbert-space structure of Chapters 1–6 to the numbers the spectroscopist and the accelerator read. The chapter adds zero new axioms – everything is derived from the ch2–ch5 axioms and the ch6 theorems.

The chapter proceeds in seven sections. Sections 8.1–8.2 treat time-independent perturbation theory: first the non-degenerate Rayleigh–Schrodinger series, with the Stark effect in hydrogen as its primary example, then the degenerate case, with the Zeeman effect, fine structure, and the linear Stark effect. Section 8.3 presents the variational method and computes the helium ground-state energy, the oldest quantitative failure of the old quantum theory. Section 8.4 derives time-dependent perturbation theory from the Dyson series, obtains Fermi’s golden rule, and recovers the Einstein A and B coefficients – the chapter’s keystone. Sections 8.5–8.6 build scattering theory: §8.5 develops the formal Møller/S-matrix framework and reads it in three registers — stationary (the Lippmann–Schwinger equation, the scattering amplitude, the optical theorem), time-dependent (wave packets and the dynamical derivation of the $i\epsilon$), and symmetry (the partial-wave expansion, phase shifts, and the Breit–Wigner resonance) — and §8.6 develops the Born approximation, recovering the Rutherford cross-section. Section 8.7 consolidates the debt ledger and the method dictionary.

Three honesty declarations travel with the chapter. The electromagnetic field is treated classically throughout; the field’s quanta are Chapter 10’s. The Einstein A coefficient is obtained by detailed bal-

ance with a classical vector potential; the semi-classical limitation is stated explicitly. The resonance discussion is a one-subsection note; full resonance theory is beyond scope. Non-perturbative methods (strong coupling, lattice) are named only in the exit question that concludes the chapter: “These methods are perturbative – what do we do when the coupling is strong?” – a question for Chapters 9 (path integrals, WKB) and 10 (field theory).

8.1 Time-Independent Perturbation: the Non-Degenerate Case

The simplest perturbative question is this: given a Hamiltonian whose exact eigenstates are known, and a small additional term — a perturbation — what are the approximate eigenstates and eigenvalues of the full Hamiltonian? The answer is the Rayleigh–Schrodinger series.

Setup. Let the full Hamiltonian be

$$H = H_0 + \lambda V, \quad (8.1)$$

where the unperturbed Hamiltonian H_0 is solved exactly:

$$H_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle,$$

with $\{|n^{(0)}\rangle\}$ a complete orthonormal set of eigenstates. The parameter λ is a bookkeeping device; at the end of the calculation it is set to unity, and the smallness of the perturbation is carried by the ratios $|V_{nm}|/|E_n^{(0)} - E_m^{(0)}|$. The target is the n -th eigenstate of H , assumed non-degenerate and well-separated from its neighbours:

$$H |n\rangle = E_n |n\rangle.$$

The operative assumption is that $|V_{nm}| \ll |E_n^{(0)} - E_m^{(0)}|$ for all $m \neq n$. An honesty clause is due at the outset: the Rayleigh–Schrodinger series is asymptotic in typical atomic problems – it may diverge – and the operative question is whether the first few terms approximate the eigenvalue. Kato’s rigorous theory is a matter for Appendix A.

The Rayleigh–Schrodinger recursion. Expand both the state and the energy as power series in λ :

$$|n\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots, \quad (8.2)$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots. \quad (8.3)$$

Insert into the eigenvalue equation $(H_0 + \lambda V) |n\rangle = E_n |n\rangle$ and collect powers of λ .

Order λ^0 reproduces the unperturbed problem:

$$H_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle.$$

Order λ^1 :

$$H_0 |n^{(1)}\rangle + V |n^{(0)}\rangle = E_n^{(0)} |n^{(1)}\rangle + E_n^{(1)} |n^{(0)}\rangle.$$

Order λ^2 :

$$H_0 |n^{(2)}\rangle + V |n^{(1)}\rangle = E_n^{(0)} |n^{(2)}\rangle + E_n^{(1)} |n^{(1)}\rangle + E_n^{(2)} |n^{(0)}\rangle.$$

First-order energy. Project the order- λ equation onto $\langle n^{(0)}|$:

$$\langle n^{(0)}|H_0|n^{(1)}\rangle + \langle n^{(0)}|V|n^{(0)}\rangle = E_n^{(0)} \langle n^{(0)}|n^{(1)}\rangle + E_n^{(1)} \langle n^{(0)}|n^{(0)}\rangle.$$

The leftmost term simplifies via the self-adjointness of H_0 : $\langle n^{(0)} | H_0 | n^{(1)} \rangle = E_n^{(0)} \langle n^{(0)} | n^{(1)} \rangle$, cancelling the first term on the right. With $\langle n^{(0)} | n^{(0)} \rangle = 1$,

$$E_n^{(1)} = \langle n^{(0)} | V | n^{(0)} \rangle. \quad (8.4)$$

The first-order energy shift is the expectation value of the perturbation in the unperturbed state – its mean.

First-order state correction. Project the order- λ equation onto $\langle m^{(0)} |$ with $m \neq n$:

$$E_m^{(0)} \langle m^{(0)} | n^{(1)} \rangle + \langle m^{(0)} | V | n^{(0)} \rangle = E_n^{(0)} \langle m^{(0)} | n^{(1)} \rangle + 0,$$

where the last term vanishes because $\langle m^{(0)} | n^{(0)} \rangle = \delta_{mn} = 0$ for $m \neq n$. Solving for the overlap:

$$\langle m^{(0)} | n^{(1)} \rangle = \frac{\langle m^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} \quad (m \neq n).$$

The component $\langle n^{(0)} | n^{(1)} \rangle$ along the unperturbed state is not determined by the order- λ equation. The intermediate normalisation convention fixes it: demand $\langle n^{(0)} | n \rangle = 1$, which at order λ gives $\langle n^{(0)} | n^{(1)} \rangle = 0$. Then the first-order state is

$$|n^{(1)}\rangle = \sum_{m \neq n} \frac{\langle m^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle. \quad (8.5)$$

The perturbation mixes the unperturbed state with all others, the admixture of state m weighted by the matrix element divided by the energy gap.

Second-order energy. Project the order- λ^2 equation onto $\langle n^{(0)} |$:

$$\langle n^{(0)} | H_0 | n^{(2)} \rangle + \langle n^{(0)} | V | n^{(1)} \rangle = E_n^{(0)} \langle n^{(0)} | n^{(2)} \rangle + E_n^{(1)} \langle n^{(0)} | n^{(1)} \rangle + E_n^{(2)}.$$

The first terms cancel as before; intermediate normalisation sets $\langle n^{(0)} | n^{(1)} \rangle = 0$. Hence

$$E_n^{(2)} = \langle n^{(0)} | V | n^{(1)} \rangle = \sum_{m \neq n} \frac{|\langle m^{(0)} | V | n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}}. \quad (8.6)$$

For the ground state ($n = 0$), every denominator $E_0^{(0)} - E_m^{(0)}$ is negative, so $E_0^{(2)}$ is always negative: the second-order correction always lowers the ground-state energy.

Wigner–Eckart short-circuit. Before computing any matrix element, the selection rules of Chapter 6 determine which $\langle m^{(0)} | V | n^{(0)} \rangle$ vanish on symmetry grounds alone. The Wigner–Eckart theorem (6.93) factorises the matrix element of a spherical tensor operator $T_q^{(k)}$:

$$\langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = \langle j, m; k, q | j', m' \rangle \frac{\langle \alpha', j' || T^{(k)} || \alpha, j \rangle}{\sqrt{2j' + 1}}.$$

The Clebsch–Gordan coefficient vanishes unless the selection rules (6.96) are satisfied:

$$m' = m + q, \quad |j - k| \leq j' \leq j + k.$$

For a scalar perturbation ($k = 0$): $\Delta j = 0$, $\Delta m = 0$ – only diagonal matrix elements survive. For a vector perturbation ($k = 1$, e.g. the Stark Hamiltonian $V = e\mathcal{E}z$): $\Delta m = 0$, $\Delta j = 0, \pm 1$ with the $j = 0 \rightarrow j' = 0$ transition forbidden. These rules are not extra postulates – they are the $k = 0, 1$ instances of the Wigner–Eckart theorem. The parity selection rule of Chapter 4, $\Pi V \Pi = \pm V$ yielding Laporte’s rule (4.63), provides a second short-circuit: a perturbation of definite parity connects only states of opposite parity when it is odd, and only states of the same parity when it is even.

Convergence remark. The Rayleigh–Schrodinger series is asymptotic in typical atomic problems. The condition $|V_{nm}| \ll |E_n^{(0)} - E_m^{(0)}|$ is necessary but not sufficient for convergence; the operative question is whether the first few terms give a good approximation, and in practice they often do. The series may diverge while its first few partial sums approach the true eigenvalue to high precision – the hallmark of an asymptotic series.

Example 1: Anharmonic oscillator. Consider a one-dimensional harmonic oscillator perturbed by a quartic term,

$$H = \hbar\omega\left(N + \frac{1}{2}\right) + \lambda x^4,$$

with $x = \sqrt{\hbar/2m\omega}(a + a^\dagger)$. The unperturbed eigenstates are the number states $|n\rangle$ with energies $E_n^{(0)} = \hbar\omega(n + \frac{1}{2})$. The first-order shift is

$$E_n^{(1)} = \lambda \langle n | x^4 | n \rangle.$$

Expand x^4 in terms of a and $a^\dagger$:

$$x^4 = \left(\frac{\hbar}{2m\omega}\right)^2 (a + a^\dagger)^4.$$

Of the sixteen terms in the expansion of $(a + a^\dagger)^4$, only the six with equal numbers of a and $a^\dagger$ survive the diagonal matrix element $\langle n | \cdots | n \rangle$:

$$(a + a^\dagger)^4 \rightarrow a^2(a^\dagger)^2 + aa^\dagger aa^\dagger + aa^\dagger a^\dagger a + a^\dagger a a a^\dagger + a^\dagger a a^\dagger a + (a^\dagger)^2 a^2,$$

where we have dropped the ten terms with unequal numbers (they connect $|n\rangle$ to $|n \pm \Delta n\rangle$ with $\Delta n \neq 0$). Normal-order each of the six by moving all $a^\dagger$ left of all a , repeatedly applying $[a, a^\dagger] = 1$:

$$a^2(a^\dagger)^2 = (a^\dagger)^2 a^2 + 4a^\dagger a + 2,$$

$$aa^\dagger aa^\dagger = (a^\dagger)^2 a^2 + 3a^\dagger a + 1,$$

$$aa^\dagger a^\dagger a = (a^\dagger)^2 a^2 + 2a^\dagger a,$$

$$a^\dagger a a a^\dagger = (a^\dagger)^2 a^2 + 2a^\dagger a,$$

$$a^\dagger a a^\dagger a = (a^\dagger)^2 a^2 + a^\dagger a,$$

$$(a^\dagger)^2 a^2 = (a^\dagger)^2 a^2.$$

Summing the six contributions:

$$(a + a^\dagger)^4 \rightarrow 6(a^\dagger)^2 a^2 + 12a^\dagger a + 3.$$

Taking the expectation value in $|n\rangle$ with $(a^\dagger)^2 a^2 |n\rangle = \sqrt{n(n-1)}\sqrt{n(n-1)} |n\rangle = n(n-1) |n\rangle$ and $a^\dagger a |n\rangle = n |n\rangle$:

$$\langle n | (a + a^\dagger)^4 | n \rangle = 6n(n-1) + 12n + 3 = 3(2n^2 + 2n + 1).$$

Multiplying by $(\hbar/2m\omega)^2 \lambda$, we obtain

$$E_n^{(1)} = \lambda \frac{3\hbar^2}{4m^2\omega^2} (2n^2 + 2n + 1). \quad (8.7)$$

The correction grows quadratically with n , reflecting the increasing amplitude of the classical motion. Only states differing by $\Delta n = 0, \pm 2, \pm 4$ are connected, as parity selection requires (each x factor changes parity, so x^4 is even – parity-conserving).

Example 2: Hydrogen ground-state Stark effect (quadratic). A hydrogen atom in a uniform static

electric field $\mathcal{E}\hat{z}$ experiences the perturbation

$$V = e\mathcal{E}z.$$

The ground state $|100\rangle$ is non-degenerate. The first-order Stark shift vanishes by parity: $\langle 100|z|100\rangle = 0$, because the ground state is even under $z \rightarrow -z$ while z is odd – Laporte’s rule (4.63). The leading nonzero shift is second-order:

$$E_{1s}^{(2)} = e^2\mathcal{E}^2 \sum_{n \neq 1, \ell, m} \frac{|\langle n\ell m|z|100\rangle|^2}{E_1 - E_n}. \quad (8.8)$$

The dipole selection rules ($k = 1$, so $\Delta\ell = \pm 1$, $\Delta m = 0$) restrict the sum to p -states with $m = 0$: only $|np0\rangle$ ($n \geq 2$) contribute. The sum defines the static electric polarisability α_{pol} via $\Delta E = -\frac{1}{2}\alpha_{\text{pol}}\mathcal{E}^2$. Evaluating the sum (which requires the full Coulomb Green’s function – a named, not derived, object) gives

$$\alpha_{\text{pol}} = \frac{9}{2}a_0^3,$$

where $a_0 = \hbar^2/me^2$ is the Bohr radius. The quadratic Stark shift is

$$\Delta E_{1s}^{(2)} = -\frac{9}{4}a_0^3\mathcal{E}^2,$$

a measurable quantity: atomic-beam deflection experiments detect the force $-\partial(\Delta E)/\partial\mathcal{E}$ on ground-state atoms passing through an inhomogeneous electric field.

Example 3: Hydrogen $n = 2$ Stark effect (bridge to §8.2). The $n = 2$ manifold of hydrogen is four-fold degenerate: the states are $|200\rangle$ ($2s$), $|210\rangle$ ($2p, m = 0$), $|211\rangle$ ($2p, m = 1$), and $|21-1\rangle$ ($2p, m = -1$). The Stark perturbation $V = e\mathcal{E}z$ is a $k = 1, q = 0$ spherical tensor. Its matrix elements within the $n = 2$ manifold vanish except for $\langle 200|z|210\rangle$, which is nonzero because the $2s$ and $2p$ states share the same n (opposite parity, same energy – the accidental Coulomb degeneracy). Using the hydrogen wavefunctions from Table 6.2,

$$\langle 200|z|210\rangle = 3a_0.$$

First-order perturbation theory within this degenerate subspace requires the machinery of the next section: the four-by-four matrix of V must be diagonalised – a secular equation. The result, previewed here and completed in Section 8.2, is the linear Stark effect: three levels at $+3ea_0\mathcal{E}$, $-3ea_0\mathcal{E}$, and zero (doubly degenerate). The splitting is linear in the applied field, a hallmark of degenerate opposite-parity states sharing the same unperturbed energy – a permanent electric dipole moment of the $n = 2$ states.

The advantage of Rayleigh–Schrodinger perturbation theory is its systematic structure: each order is computed from the previous one by a fixed recursion, and the Wigner–Eckart theorem prunes the matrix-element tree before a single integral is evaluated. Its limitation is the non-degeneracy assumption: when $E_n^{(0)}$ is degenerate, the denominators $E_n^{(0)} - E_m^{(0)}$ vanish for states within the degenerate manifold, and the series breaks down. The remedy – diagonalising V within the degenerate subspace – is the subject of the next section.

8.2 Degenerate Perturbation Theory: Zeeman, Fine Structure, and Stark

When the unperturbed level $E_n^{(0)}$ is d -fold degenerate, the Rayleigh–Schrodinger denominators $E_n^{(0)} - E_m^{(0)}$ vanish for every state m within the degenerate manifold, and the series of Section 8.1 fails. The physics is clear: the perturbation must decide which linear combinations of the degenerate unperturbed states are the “correct” zeroth-order states – the ones that evolve continuously as $\lambda \rightarrow 0$.

The secular equation. Let the d degenerate eigenstates of H_0 with energy $E_n^{(0)}$ be $|n\alpha^{(0)}\rangle$, $\alpha = 1, \dots, d$. Write the perturbed state as a linear combination within this subspace plus corrections from outside:

$$|n\rangle = \sum_{\alpha=1}^d c_{\alpha} |n\alpha^{(0)}\rangle + \lambda |n^{(1)}\rangle + \dots,$$

with $|n^{(1)}\rangle$ orthogonal to the degenerate subspace. Insert into $(H_0 + \lambda V)|n\rangle = E_n |n\rangle$ with $E_n = E_n^{(0)} + \lambda E^{(1)} + \dots$. At order λ^0 , the unperturbed equation is satisfied. At order λ^1 , project onto $\langle n\beta^{(0)}|$:

$$\langle n\beta^{(0)}|H_0|n^{(1)}\rangle + \sum_{\alpha} c_{\alpha} \langle n\beta^{(0)}|V|n\alpha^{(0)}\rangle = E_n^{(0)} \langle n\beta^{(0)}|n^{(1)}\rangle + E^{(1)} c_{\beta}.$$

The H_0 term gives $E_n^{(0)} \langle n\beta^{(0)}|n^{(1)}\rangle$, cancelling the first term on the right. The result is a $d \times d$ matrix eigenvalue problem:

$$\sum_{\alpha=1}^d V_{\beta\alpha} c_{\alpha} = E^{(1)} c_{\beta}, \quad V_{\beta\alpha} := \langle n\beta^{(0)}|V|n\alpha^{(0)}\rangle.$$

Equivalently, the first-order energy shifts $E^{(1)}$ are the eigenvalues of the perturbation matrix V restricted to the degenerate subspace, and the correct zeroth-order states are its eigenvectors. The condition for a nontrivial solution is the secular equation:

$$\det(V_{\alpha\beta} - E^{(1)}\delta_{\alpha\beta}) = 0. \quad (8.9)$$

Good basis from symmetry. If a conserved operator G commutes with both H_0 and V ($[G, H_0] = [G, V] = 0$), then the eigenbasis of G simultaneously diagonalises V within each G -eigenspace, provided G is non-degenerate on that subspace. This is the perturbation-theoretic application of Chapter 4’s conservation theorem (4.57): conserved quantum numbers remain good quantum numbers. Choosing the symmetry-adapted basis before computing matrix elements often diagonalises the secular problem before it is written.

Application 1: Normal Zeeman effect. A hydrogen atom in a uniform magnetic field $\mathbf{B} = B\hat{z}$ acquires the orbital Zeeman Hamiltonian of Chapter 6 (6.110):

$$H_Z = \frac{e}{2mc} B L_z = \mu_B B \frac{L_z}{\hbar}, \quad \mu_B = \frac{e\hbar}{2mc}.$$

Within an ℓ -multiplet of fixed n and ℓ , the states are $|n\ell m\rangle$ with $m = -\ell, \dots, \ell$. The perturbation matrix is already diagonal in this basis because $L_z |n\ell m\rangle = \hbar m |n\ell m\rangle$:

$$\langle n\ell m'|H_Z|n\ell m\rangle = \mu_B B m \delta_{m'm}.$$

No secular equation is needed – the symmetry-adapted basis $|n\ell m\rangle$ diagonalises the perturbation automatically. The first-order energy shifts are

$$\Delta E_{n\ell m} = \mu_B B m,$$

splitting the $(2\ell + 1)$ -fold degenerate level into $2\ell + 1$ equally spaced sublevels. For the Lyman- α transition ($2p \rightarrow 1s$), the upper level ($\ell = 1$) splits into three sublevels ($m = -1, 0, 1$), the lower ($\ell = 0$) is unsplit, and the dipole selection rules ($\Delta m = 0, \pm 1$) produce three lines: ω_0 ($m = 0 \rightarrow 0$) and $\omega_0 \pm \mu_B B/\hbar$ ($m = \pm 1 \rightarrow 0$) – the normal Zeeman triplet.

Application 2: Anomalous Zeeman effect. When electron spin is included, the Zeeman Hamiltonian becomes

$$H_Z = \mu_B B \frac{L_z + g_s S_z}{\hbar}, \quad g_s \approx 2.$$

In LS coupling, the good basis is $|nLSJM_J\rangle$, where $\mathbf{J} = \mathbf{L} + \mathbf{S}$ is the total angular momentum. The operator $\mathbf{L} + g_s \mathbf{S}$ is not proportional to $\mathbf{J}$, but its matrix elements within a fixed- (L, S, J) manifold are diagonal in M_J and proportional to those of J_z , by the Wigner–Eckart theorem applied to the vector operators $\mathbf{L}$ and $\mathbf{S}$ within the J -multiplet. The proportionality constant is the Landé g -factor. To derive it, write the expectation value of $\mathbf{L} + g_s \mathbf{S}$ in a state $|LSJM_J\rangle$ using the projection theorem: for any vector operator $\mathbf{V}$, its matrix elements within a J -multiplet are proportional to those of $\mathbf{J}$:

$$\langle LSJM_J' | \mathbf{V} | LSJM_J \rangle = \frac{\langle LSJ | \mathbf{V} \cdot \mathbf{J} | LSJ \rangle}{\hbar^2 J(J+1)} \langle LSJM_J' | \mathbf{J} | LSJM_J \rangle.$$

Apply this to $\mathbf{V} = \mathbf{L}$ and $\mathbf{V} = \mathbf{S}$ separately. The dot products are evaluated using $\mathbf{J} = \mathbf{L} + \mathbf{S}$:

$$\mathbf{L} \cdot \mathbf{J} = \mathbf{L} \cdot (\mathbf{L} + \mathbf{S}) = \mathbf{L}^2 + \mathbf{L} \cdot \mathbf{S},$$

$$\mathbf{S} \cdot \mathbf{J} = \mathbf{S} \cdot (\mathbf{L} + \mathbf{S}) = \mathbf{S}^2 + \mathbf{L} \cdot \mathbf{S}.$$

The spin-orbit operator $\mathbf{L} \cdot \mathbf{S}$ is diagonal in the coupled basis with the value

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2) = \frac{\hbar^2}{2}[J(J+1) - L(L+1) - S(S+1)].$$

Hence

$$\langle \mathbf{L} \cdot \mathbf{J} \rangle = \hbar^2 L(L+1) + \frac{\hbar^2}{2}[J(J+1) - L(L+1) - S(S+1)]$$

$$= \frac{\hbar^2}{2}[J(J+1) + L(L+1) - S(S+1)],$$

$$\langle \mathbf{S} \cdot \mathbf{J} \rangle = \frac{\hbar^2}{2}[J(J+1) + S(S+1) - L(L+1)].$$

The effective magnetic moment operator is $\boldsymbol{\mu} = -\mu_B(\mathbf{L} + g_s \mathbf{S})/\hbar$. Its matrix elements within the J -multiplet are proportional to $-\mu_B g_J \mathbf{J}/\hbar$, with

$$g_J = \frac{1}{J(J+1)} \left[\frac{J(J+1) + L(L+1) - S(S+1)}{2} + g_s \frac{J(J+1) + S(S+1) - L(L+1)}{2} \right].$$

With $g_s \approx 2$, this simplifies to the Landé g -factor:

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}. \quad (8.10)$$

The anomalous Zeeman splitting is then

$$\Delta E = \mu_B B g_J M_J, \quad M_J = -J, \dots, J.$$

For the sodium D-lines: the $^2S_{1/2}$ ground term has $L = 0$, $S = 1/2$, $J = 1/2$, giving $g_J = 2$; the $^2P_{3/2}$ excited term has $L = 1$, $S = 1/2$, $J = 3/2$, giving $g_J = 4/3$; the $^2P_{1/2}$ term has $J = 1/2$, giving $g_J = 2/3$. The $^2P_{3/2} \rightarrow ^2S_{1/2}$ transition (D₂ line) produces four lines; the $^2P_{1/2} \rightarrow ^2S_{1/2}$ transition (D₁ line) produces two – six lines total, not three: the anomalous Zeeman pattern.

Note The relationship between the Zeeman Hamiltonian and Kramers degeneracy (4.66). A magnetic field is odd under time reversal (T -odd): $\Theta H_Z \Theta^{-1} = -H_Z$. It therefore splits Kramers doublets. An electric field is T -even and does not. This distinction is the experimental signature of time-reversal symmetry in atomic spectra.

Application 3: Fine structure of hydrogen. The first relativistic correction to the hydrogen Hamiltonian comprises three terms:

$$H_{\text{fs}} = -\frac{p^4}{8m^3c^2} + \frac{e^2}{2m^2c^2r^3} \mathbf{L} \cdot \mathbf{S} + \frac{\hbar^2 e^2}{8m^2c^2} 4\pi\delta^{(3)}(\mathbf{r}). \quad (8.11)$$

The first term is the relativistic kinetic-energy correction (the next term in the expansion $E = \sqrt{p^2c^2 + m^2c^4} - mc^2 = p^2/2m - p^4/8m^3c^2 + \dots$). The second is the spin-orbit coupling – the magnetic field seen by the electron's spin in its rest frame, as it moves through the nuclear electric field. The third, the Darwin term, is a relativistic correction localised at the origin, affecting only s -states ($\ell = 0$).

Treating H_{fs} as a perturbation on the non-relativistic hydrogen states, the first-order shifts are computed in the coupled basis $|n\ell s J M_J\rangle$ (with $s = 1/2$). The kinetic term is diagonal in the uncoupled basis; its expectation value depends only on n and ℓ . The spin-orbit term is diagonal in the coupled basis and proportional to $\langle 1/r^3 \rangle_{n\ell}$. The Darwin term contributes only for $\ell = 0$ and is proportional to $|\psi_{n00}(0)|^2$. The summed result, for each n and $j = \ell \pm 1/2$ (with the convention that $j = 1/2$ for $\ell = 0$), is the fine-structure formula:

$$\Delta E_{\text{fs}} = -\frac{(Z\alpha)^4 \mu c^2}{2n^3} \left(\frac{1}{j + 1/2} - \frac{3}{4n} \right),$$

where $\alpha = e^2/\hbar c \approx 1/137$ is the fine-structure constant. The formula is stated; the full derivation (requiring the expectation values $\langle 1/r \rangle$, $\langle 1/r^2 \rangle$, $\langle 1/r^3 \rangle$ and the Darwin contact term) is an instructive but lengthy exercise. The key physical result: the ℓ -degeneracy at fixed n is partially lifted – states with the same j remain degenerate (the Dirac result; the accidental SO(4) symmetry of the Coulomb potential is broken by the spin-orbit and Darwin terms, but a residual degeneracy in j survives in the non-relativistic Pauli approximation – the full Dirac equation restores only the j -degeneracy, while states of the same n but different j split).

Application 4: Linear Stark effect in hydrogen $n = 2$. Returning to the $n = 2$ Stark problem of Section 8.1 (Example 3). The four degenerate states are $\{|200\rangle, |210\rangle, |211\rangle, |21-1\rangle\}$. The perturbation $V = e\mathcal{E}z$ is a $k = 1$, $q = 0$ spherical tensor. The selection rules $\Delta m = 0$, $\Delta \ell = \pm 1$ leave only one nonzero matrix element within the $n = 2$ manifold:

$$\langle 200|z|210\rangle = \langle 210|z|200\rangle = 3a_0,$$

computed from the hydrogen wavefunctions (6.109):

$$\psi_{200}(\mathbf{r}) = \frac{1}{\sqrt{8\pi a_0^3}} \left(1 - \frac{r}{2a_0}\right) e^{-r/2a_0},$$

$$\psi_{210}(\mathbf{r}) = \frac{1}{\sqrt{32\pi a_0^3}} \frac{r}{a_0} e^{-r/2a_0} \cos \theta.$$

The integral is evaluated using $d^3r = r^2 dr d(\cos \theta) d\phi$:

$$\begin{aligned} \langle 200|z|210\rangle &= \int d^3r \psi_{200}^*(\mathbf{r}) (r \cos \theta) \psi_{210}(\mathbf{r}) \\ &= \frac{1}{\sqrt{8\pi a_0^3}} \frac{1}{\sqrt{32\pi a_0^3}} \int_0^\infty dr r^2 \left(1 - \frac{r}{2a_0}\right) e^{-r/2a_0} \cdot \frac{r}{a_0} e^{-r/2a_0} \cdot r \int_{-1}^1 d(\cos \theta) \cos^2 \theta \int_0^{2\pi} d\phi \\ &= \frac{1}{16\pi a_0^3} \frac{1}{a_0} \int_0^\infty dr r^4 \left(1 - \frac{r}{2a_0}\right) e^{-r/a_0} \cdot \frac{2}{3} \cdot 2\pi \\ &= \frac{1}{12a_0^4} \left[\int_0^\infty dr r^4 e^{-r/a_0} - \frac{1}{2a_0} \int_0^\infty dr r^5 e^{-r/a_0} \right] \\ &= \frac{1}{12a_0^4} \left[4! a_0^5 - \frac{1}{2a_0} 5! a_0^6 \right] = \frac{1}{12a_0^4} [24a_0^5 - 60a_0^5] = \frac{-36a_0^5}{12a_0^4} = -3a_0, \end{aligned}$$

where we used $\int_0^\infty dr r^n e^{-r/a_0} = n! a_0^{n+1}$. The overall sign is a phase convention; the magnitude $|\langle 200|z|210\rangle| = 3a_0$ is the quantity entering the secular matrix (the eigenvalues depend only on $|\langle 200|z|210\rangle|^2$, so the sign does not affect the physics). All other matrix elements within the $n = 2$ manifold vanish by the m and ℓ selection rules. The 4×4 secular matrix in the ordered basis $\{|200\rangle, |210\rangle, |211\rangle, |21-1\rangle\}$ is

$$V = e\mathcal{E} \begin{pmatrix} 0 & 3a_0 & 0 & 0 \\ 3a_0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The matrix block-diagonalises: the upper-left 2×2 block yields eigenvalues $\pm 3ea_0\mathcal{E}$ with eigenvectors $(|200\rangle \pm |210\rangle)/\sqrt{2}$; the lower-right 2×2 block is identically zero, leaving $|211\rangle$ and $|21-1\rangle$ unshifted at first order.

$$E^{(1)} = +3ea_0\mathcal{E}, -3ea_0\mathcal{E}, 0, 0. \quad (8.12)$$

The linear Stark effect: the splitting is proportional to $\mathcal{E}$, not $\mathcal{E}^2$, because the $n = 2$ manifold contains opposite-parity states that mix at first order. The permanent electric dipole moment of the mixed $2s-2p_0$ states is $\pm 3ea_0$ – an atom in one of these states possesses a nonzero mean dipole moment, a property impossible for any non-degenerate stationary state of a parity-conserving Hamiltonian.

 **Note** The second-order effective Hamiltonian for the degenerate case – Löwdin partitioning – is a systematic method for including the influence of states outside the degenerate manifold. The effective Hamiltonian in the d -dimensional model space is

$$H_{\text{eff}} = E_n^{(0)} I + \lambda P V P + \lambda^2 P V Q \frac{1}{E_n^{(0)} - Q H_0 Q} Q V P + O(\lambda^3),$$

where $P = \sum_\alpha |n\alpha^{(0)}\rangle \langle n\alpha^{(0)}|$ is the projector onto the degenerate subspace and $Q = I - P$. The derivation and application are left to Exercise 5.

The advantage of degenerate perturbation theory is that it handles the generic case: most atomic levels are degenerate by rotation symmetry, and the secular equation is the systematic tool for lifting that degeneracy. The limitation is that it is still perturbation theory – the shifts must be small compared to the spacing between distinct unperturbed levels: $|E^{(1)}| \ll |E_n^{(0)} - E_m^{(0)}|$ for m outside the degenerate manifold. TIPT assumes we know the exact eigenstates of H_0 , and computes corrections. The variational method asks a different question: given a trial state that is not an eigenstate of anything, what is the best bound we can place on the ground-state energy? The answer – the Rayleigh–Ritz principle – is the engine that finally computes the binding energy of helium.

8.3 The Variational Method: the Ground State of Helium

The old quantum theory failed on helium. Bohr–Sommerfeld quantisation, which had triumphed for hydrogen, gave a ground-state energy roughly twice the experimental value. The failure was generic: helium has two electrons, and the $1/r_{12}$ electron–electron repulsion breaks the separability that the old quantum theory required. The helium atom was the simplest system for which the method was inadequate – and the simplest test for the new quantum mechanics. The variational method passes that test.

The Rayleigh–Ritz principle. Let H be a Hamiltonian bounded below, with normalised ground state $|0\rangle$ and ground-state energy E_0 . For any normalised trial state $|\tilde{\psi}\rangle$,

$$\langle \tilde{\psi} | H | \tilde{\psi} \rangle \geq E_0, \quad (8.13)$$

with equality if and only if $|\tilde{\psi}\rangle = |0\rangle$ (up to a phase). The proof is elementary. Expand $|\tilde{\psi}\rangle$ in the eigenbasis of H : $|\tilde{\psi}\rangle = \sum_n c_n |n\rangle$ with $H|n\rangle = E_n|n\rangle$, $\sum_n |c_n|^2 = 1$. Then

$$\begin{aligned} \langle \tilde{\psi} | H | \tilde{\psi} \rangle &= \sum_{m,n} c_m^* c_n \langle m | H | n \rangle = \sum_n |c_n|^2 E_n \\ &= |c_0|^2 E_0 + \sum_{n \geq 1} |c_n|^2 E_n = E_0 + \sum_{n \geq 1} |c_n|^2 (E_n - E_0) \geq E_0, \end{aligned}$$

since every $E_n - E_0 \geq 0$. Equality forces $|c_n|^2 = 0$ for all $n \geq 1$, hence $|c_0|^2 = 1$ and $|\tilde{\psi}\rangle = |0\rangle$.

The principle is a rigorous upper bound, and the bound is the engine: choose a family of trial states $|\psi_{\text{trial}}(\alpha_1, \dots, \alpha_k)\rangle$ parameterised by variational parameters $\{\alpha_i\}$, compute the energy functional

$$E(\alpha_1, \dots, \alpha_k) = \frac{\langle \psi_{\text{trial}} | H | \psi_{\text{trial}} \rangle}{\langle \psi_{\text{trial}} | \psi_{\text{trial}} \rangle},$$

and minimise: $\partial E / \partial \alpha_i = 0$ for each i . The minimum is an upper bound on E_0 , and the bound improves monotonically with more variational parameters.

The helium Hamiltonian. In the clamped-nucleus approximation (the nucleus at the origin, infinitely massive), the helium atom’s electronic Hamiltonian is

$$H = \left(-\frac{\hbar^2}{2m} \nabla_1^2 - \frac{2e^2}{r_1} \right) + \left(-\frac{\hbar^2}{2m} \nabla_2^2 - \frac{2e^2}{r_2} \right) + \frac{e^2}{r_{12}}. \quad (8.14)$$

The first two parentheses are independent hydrogenic Hamiltonians for nuclear charge $Z = 2$. The last term, the electron–electron Coulomb repulsion with $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$, is the term the old quantum theory

could not handle – it couples the two electrons and breaks the separability on which Bohr–Sommerfeld quantisation depended.

The trial function. The physical idea is screening: each electron partially shields the nuclear charge from the other. A product of hydrogenic $1s$ orbitals with an effective nuclear charge Z_{eff} captures this in one variational parameter:

$$\psi_{\text{trial}}(\mathbf{r}_1, \mathbf{r}_2) = \frac{Z_{\text{eff}}^3}{\pi a_0^3} e^{-Z_{\text{eff}}(r_1+r_2)/a_0}. \quad (8.15)$$

Each factor is the hydrogenic $1s$ wavefunction $\psi_{100}(\mathbf{r}) = (Z_{\text{eff}}^3/\pi a_0^3)^{1/2} e^{-Z_{\text{eff}}r/a_0}$, and the trial function is their (properly normalised) product. The spin part is the singlet $|0, 0\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$; it factors out of the spatial expectation values because H contains no spin operators.

The energy functional. Compute $\langle H \rangle = \langle \psi_{\text{trial}} | H | \psi_{\text{trial}} \rangle$ term by term. For a hydrogenic $1s$ orbital with effective charge Z_{eff} , the kinetic energy expectation is $\langle p^2/2m \rangle = Z_{\text{eff}}^2 e^2/2a_0$ and the electron–nucleus potential energy for charge Z_{eff} is $\langle -Z_{\text{eff}}e^2/r \rangle = -Z_{\text{eff}}^2 e^2/a_0$ (both follow from the virial theorem for Coulomb potentials, or directly from the hydrogenic wavefunctions of Table 6.2). The true nuclear charge is $Z = 2$, so the nuclear-attraction term for each electron is $-2e^2/r$, whose expectation in the Z_{eff} -hydrogenic orbital is $\langle -2e^2/r \rangle = 2 \langle -e^2/r \rangle = 2(-Z_{\text{eff}}e^2/a_0) = -2Z_{\text{eff}}e^2/a_0$. Hence, for one electron:

$$\left\langle -\frac{\hbar^2}{2m} \nabla^2 - \frac{2e^2}{r} \right\rangle = \frac{Z_{\text{eff}}^2 e^2}{2a_0} - \frac{2Z_{\text{eff}}e^2}{a_0} = \left(\frac{Z_{\text{eff}}^2}{2} - 2Z_{\text{eff}} \right) E_h,$$

where $E_h = e^2/a_0 = me^4/\hbar^2 \approx 27.21$ eV is the Hartree energy. The same for electron 2, giving a factor of 2.

The electron–electron repulsion integral is

$$\left\langle \frac{e^2}{r_{12}} \right\rangle = e^2 \int d^3r_1 \int d^3r_2 \frac{|\psi_{\text{trial}}(\mathbf{r}_1, \mathbf{r}_2)|^2}{|\mathbf{r}_1 - \mathbf{r}_2|}.$$

Expand $1/r_{12}$ in Legendre polynomials:

$$\frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|} = \sum_{\ell=0}^{\infty} \frac{r_{<}^{\ell}}{r_{>}^{\ell+1}} P_{\ell}(\cos \theta_{12}),$$

where $r_{<} = \min(r_1, r_2)$ and $r_{>} = \max(r_1, r_2)$. When integrated against the spherically symmetric $1s$ charge distributions, only the $\ell = 0$ term survives (all P_{ℓ} with $\ell > 0$ integrate to zero over angles).

The integral reduces to a double radial integral:

$$\begin{aligned} \left\langle \frac{e^2}{r_{12}} \right\rangle &= e^2 \left(\frac{Z_{\text{eff}}^3}{a_0^3} \right)^2 \iint d^3r_1 d^3r_2 \frac{e^{-2Z_{\text{eff}}(r_1+r_2)/a_0}}{|\mathbf{r}_1 - \mathbf{r}_2|} \\ &= e^2 (4\pi)^2 \left(\frac{Z_{\text{eff}}^3}{a_0^3} \right)^2 \int_0^{\infty} dr_1 r_1^2 e^{-2Z_{\text{eff}}r_1/a_0} \int_0^{\infty} dr_2 r_2^2 e^{-2Z_{\text{eff}}r_2/a_0} \frac{1}{\max(r_1, r_2)} \\ &= \frac{5}{8} Z_{\text{eff}} E_h. \end{aligned}$$

The factor $5/8$ is the result of the radial integration (the integrals are elementary – Exercise 5). Assembling the three contributions:

$$\langle H \rangle (Z_{\text{eff}}) = \left(2Z_{\text{eff}}^2 - 4Z_{\text{eff}} + \frac{5}{8} Z_{\text{eff}} \right) E_h = \left(2Z_{\text{eff}}^2 - \frac{27}{8} Z_{\text{eff}} \right) E_h. \quad (8.16)$$

8235 **Minimisation.** Set $d\langle H \rangle / dZ_{\text{eff}} = 0$:

$$8236 \quad \frac{d}{dZ_{\text{eff}}} \left(2Z_{\text{eff}}^2 - \frac{27}{8} Z_{\text{eff}} \right) = 4Z_{\text{eff}} - \frac{27}{8} = 0 \implies Z_{\text{eff}} = \frac{27}{16} \approx 1.6875.$$

8237 The effective nuclear charge is less than 2: each electron screens roughly 0.31 units of nuclear charge
8238 from the other. Substituting back:

$$8239 \quad E_{\text{var}} = \left[2 \left(\frac{27}{16} \right)^2 - \frac{27}{8} \cdot \frac{27}{16} \right] E_h = \left(\frac{2 \cdot 729}{256} - \frac{729}{128} \right) E_h$$

$$8240 \quad = \left(\frac{729}{128} - \frac{729}{128} \right) E_h = -\frac{729}{256} E_h \approx -2.8477 E_h \approx -77.5 \text{ eV}. \quad (8.17)$$

8241 **Comparison with experiment.** The experimental helium ground-state energy – the sum of the first
8242 and second ionisation energies – is approximately -79.0 eV ($-2.903 E_h$). The single-parameter vari-
8243 ational estimate differs from experiment by roughly 1.9%. The old quantum theory's best estimate
8244 (Bohr–Sommerfeld quantisation applied to a two-electron orbit) was off by roughly a factor of two.
8245 The variational method, with one physically motivated parameter, already resolves the oldest quanti-
8246 tative failure of the old quantum theory.

8247  **Note** *Better trial functions reduce the remaining error dramatically. Hylleraas (1929) introduced*
8248 *trial functions with explicit dependence on $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$, capturing the electron–electron corre-*
8249 *lation directly; with three parameters he obtained $-2.902 E_h$, within 0.03% of experiment. Modern*
8250 *variational calculations with hundreds of parameters reach the sub-microhartree level. The varia-*
8251 *tional method is systematically improvable: more parameters yield a monotonically non-increasing*
8252 *upper bound on E_0 .*

8253 The advantage of the variational method is that it gives a rigorous upper bound on the ground-
8254 state energy, and the bound improves monotonically with the number of variational parameters. Its
8255 limitation is that it provides no information about excited states (the Hylleraas–Undheim theorem –
8256 stating that the k -th variational eigenvalue is an upper bound to the k -th exact eigenvalue – is named,
8257 not proved; see Exercise 5). Moreover, the quality of the bound depends entirely on the physical insight
8258 embedded in the trial function: a poor ansatz gives a poor bound, and the method offers no internal
8259 diagnostic.

8260 The variational method is time-independent and non-perturbative – it bounds the eigenvalue with-
8261 out expanding in a small parameter. The methods of the next section are time-dependent and pertur-
8262 bative: they ask not “what is the energy?” but “at what rate does the system jump from one eigenstate
8263 to another?” The Dyson series of Chapter 4, truncated at its first nontrivial term, supplies the answer.

8264 8.4 Time-Dependent Perturbation: Transition Amplitudes and 8265 Fermi's Golden Rule

8266 The Rayleigh–Schrodinger series answers the stationary question: what are the approximate
8267 eigenenergies of a perturbed Hamiltonian? The time-dependent question is different: given a sys-
8268 tem initially in an eigenstate of H_0 , and a time-dependent perturbation $V(t)$ turned on, what is the
8269 probability that a later measurement finds the system in a different eigenstate? The answer is the

transition amplitude, and its long-time limit is Fermi's golden rule – the chapter's keystone. 8270

Setup in the interaction picture. Let $H(t) = H_0 + V(t)$, with the eigenstates of H_0 known: $H_0 |n\rangle = E_n |n\rangle$. In the interaction picture of Section 5.2, the state is 8271

$$|\psi_I(t)\rangle = U_0^\dagger(t) |\psi(t)\rangle, \quad U_0(t) = e^{-iH_0 t/\hbar}, \quad 8272$$

and obeys the interaction-picture equation of motion (5.5): 8273

$$i\hbar \frac{d}{dt} |\psi_I(t)\rangle = V_I(t) |\psi_I(t)\rangle, \quad V_I(t) = e^{iH_0 t/\hbar} V(t) e^{-iH_0 t/\hbar}. \quad 8274$$

Expand the interaction state in the eigenbasis of H_0 : 8275

$$|\psi_I(t)\rangle = \sum_n c_n(t) |n\rangle, \quad 8276$$

with $c_n(0) = \langle n|\psi(0)\rangle$. The expansion coefficients satisfy 8277

$$i\hbar \frac{dc_f}{dt} = \sum_n e^{i\omega_{fn}t} V_{fn}(t) c_n(t), \quad 8278$$

where $\omega_{fn} = (E_f - E_n)/\hbar$ and $V_{fn}(t) = \langle f|V(t)|n\rangle$. This is exact – a system of coupled first-order differential equations for the amplitudes. 8279

The Dyson series truncated. The evolution operator in the interaction picture is the time-ordered exponential (4.12) of V_I . Truncating after the first term (the $n = 1$ term in the Dyson series) gives the first-order approximation: 8280

$$c_f(t) \approx c_f(0) - \frac{i}{\hbar} \int_{t_0}^t dt' \langle f|V_I(t')|\psi_I(t_0)\rangle. \quad 8281$$

For a system initially in the state $|i\rangle$ at $t_0 = 0$, with $c_n(0) = \delta_{ni}$, the first-order amplitude for a transition $i \rightarrow f$ ($f \neq i$) is 8282

$$c_f^{(1)}(t) = -\frac{i}{\hbar} \int_0^t dt' e^{i\omega_{fi}t'} \langle f|V(t')|i\rangle, \quad (8.18) \quad 8283$$

with $\omega_{fi} = (E_f - E_i)/\hbar$. This is the first-order time-dependent transition amplitude. It is the seed of every rate and cross-section in the chapter. 8284



Note The truncation is honest. The full Dyson series (4.12) is an infinite sum; keeping only the first term is an approximation whose validity requires $|c_f(t)| \ll 1$ – the probability of finding the system in any state other than the initial one must remain small. This is the perturbative regime. The exact solution for the forced oscillator (Chapter 5, (5.37)) provides the benchmark: first-order perturbation theory captures only the single-quantum transition $|0\rangle \rightarrow |1\rangle$; the multi-quantum Poisson tail is a higher-order effect. 8285

Constant perturbation. For a time-independent perturbation turned on at $t = 0$, $V(t) = V \theta(t)$. The integral in (8.18) is elementary: 8286

$$\begin{aligned} c_f^{(1)}(t) &= -\frac{i}{\hbar} V_{fi} \int_0^t dt' e^{i\omega_{fi}t'} = -\frac{i}{\hbar} V_{fi} \frac{e^{i\omega_{fi}t} - 1}{i\omega_{fi}} \\ &= -\frac{V_{fi}}{\hbar\omega_{fi}} (e^{i\omega_{fi}t} - 1). \end{aligned} \quad (8.19) \quad 8287$$

8301 The transition probability is

$$\begin{aligned}
 8302 \quad P_{i \rightarrow f}(t) &= |c_f^{(1)}(t)|^2 = \frac{|V_{fi}|^2}{\hbar^2 \omega_{fi}^2} |e^{i\omega_{fi}t} - 1|^2 \\
 8303 \quad &= \frac{|V_{fi}|^2}{\hbar^2} \frac{\sin^2(\omega_{fi}t/2)}{(\omega_{fi}/2)^2}. \quad (8.20)
 \end{aligned}$$

8304 The function $f(\omega, t) = \sin^2(\omega t/2)/(\omega/2)^2$ is a diffraction peak in frequency space: its central height
 8305 is t^2 at $\omega = 0$, its first zeros are at $\omega = \pm 2\pi/t$, and its width scales as $\sim 2\pi/t$. For short times the
 8306 probability grows as t^2 ; for times long compared to $1/\omega_{fi}$ it oscillates – transitions to a discrete final
 8307 state under a constant perturbation are not monotonic.

8308 **Harmonic perturbation.** A perturbation oscillating at a single frequency,

$$8309 \quad V(t) = V e^{-i\omega t} + V^\dagger e^{i\omega t},$$

8310 models the coupling to a monochromatic classical field. Near resonance ($\omega \approx \omega_{fi}$), the rotating-wave
 8311 approximation retains only the term whose exponential nearly cancels the $e^{i\omega_{fi}t}$ from the free evolution.

8312 For absorption ($E_f > E_i$), the $e^{-i\omega t}$ term is resonant:

$$8313 \quad c_f^{(1)}(t) = -\frac{i}{\hbar} V_{fi} \int_0^t dt' e^{i(\omega_{fi}-\omega)t'} = -\frac{i}{\hbar} V_{fi} \frac{e^{i(\omega_{fi}-\omega)t} - 1}{i(\omega_{fi} - \omega)}. \quad (8.21)$$

8314 The transition probability is

$$8315 \quad P_{i \rightarrow f}(t) = \frac{|V_{fi}|^2}{\hbar^2} \frac{\sin^2[(\omega_{fi} - \omega)t/2]}{[(\omega_{fi} - \omega)/2]^2}.$$

8316 On exact resonance ($\omega = \omega_{fi}$):

$$8317 \quad P_{i \rightarrow f}(t) = \frac{|V_{fi}|^2}{\hbar^2} t^2.$$

8318 The quadratic growth on resonance is the first-order content of the Rabi formula (4.13), which oscil-
 8319 lates at the Rabi frequency $\Omega_R = \sqrt{\omega_1^2 + \delta^2}$ – first-order perturbation theory is the small- t expansion
 8320 of the exact resonant solution, valid only when $|V_{fi}|t/\hbar \ll 1$.

8321 **Benchmark check: the forced oscillator.** The exact forced-oscillator solution of Chapter 5 (5.37)
 8322 provides the test. For a harmonic drive $f(t) = f_0 \cos \omega t$ on resonance ($\omega = \omega_0$), the exact transition
 8323 probability from the ground state to the n -th excited state is Poisson: $P_n = e^{-|\beta|^2} |\beta|^{2n}/n!$ with $|\beta| \simeq$
 8324 $f_0 t/2$ for large t . The single-quantum probability is $P_1 = |\beta|^2 e^{-|\beta|^2} \approx |\beta|^2$ for small $|\beta|$ – reproducing
 8325 the t^2 growth of first-order perturbation theory. The Poisson tail ($n \geq 2$) is a multi-photon effect
 8326 beyond first order; it requires the n -th term in the Dyson series. The benchmark is satisfied: first-order
 8327 TDPT captures the initial linear growth of the single-quantum channel correctly, and the exact solution
 8328 confirms that the truncation is precisely “one interaction vertex.”

8329 **Fermi's golden rule.** For transitions into a continuum (or quasi-continuum) of final states, the prob-
 8330 ability oscillates for a single state but accumulates secularly when summed over a dense set. The key
 8331 identity, valid as $t \rightarrow \infty$, is

$$8332 \quad \lim_{t \rightarrow \infty} \frac{\sin^2(\omega t/2)}{(\omega/2)^2} = 2\pi t \delta(\omega).$$

8333 This is a representation of the Dirac delta: $\int_{-\infty}^{\infty} d\omega \sin^2(\omega t/2)/(\omega/2)^2 = 2\pi t$. Applying it to the

harmonic-perturbation probability gives the transition rate dP/dt :

$$\Gamma_{i \rightarrow f} = \frac{2\pi}{\hbar} |\langle f|V|i \rangle|^2 \delta(E_f - E_i \mp \hbar\omega),$$

with $-$ for absorption, $+$ for stimulated emission. This is Fermi's golden rule for a single final state.

For a continuum of final states, sum over a dense set labelled by a continuous parameter (or by the energy itself). The sum is converted to an integral with the density of states $\rho(E_f)$:

$$\sum_f \rightarrow \int dE_f \rho(E_f),$$

where $\rho(E_f)dE_f$ is the number of final states with energy between E_f and $E_f + dE_f$. Then

$$\Gamma_{i \rightarrow f} = \frac{2\pi}{\hbar} |\langle f|V|i \rangle|^2 \rho(E_f). \quad (8.22)$$

This is the chapter's keystone. Every measurable rate in atomic, molecular, nuclear, and condensed-matter physics descends from it. The operative regime: times long compared to $1/\Delta E$ (the inverse level spacing in the continuum – so that the delta-function limit applies) but short compared to $1/\Gamma$ (so that first-order perturbation theory remains valid). The exponential decay law $P(t) = e^{-\Gamma t}$ follows from the golden rule only in the Markovian limit – stated honestly.

Density of states. For a free particle in a large box of volume $V = L^3$ with periodic boundary conditions, the allowed wave vectors are $\mathbf{k} = (2\pi/L)(n_x, n_y, n_z)$. The number of states with wave-vector magnitude between k and $k + dk$ and solid angle $d\Omega$ is

$$\frac{V}{(2\pi)^3} k^2 dk d\Omega.$$

With $E = \hbar^2 k^2 / 2m$, so $dE = \hbar^2 k dk / m$, the density of states per unit energy and solid angle is

$$\rho(E) = \frac{V}{(2\pi)^3} \frac{mk}{\hbar^2} d\Omega. \quad (8.23)$$

For photons in a cavity, the number of electromagnetic modes per unit volume with frequency between ω and $\omega + d\omega$ is the Rayleigh–Jeans mode density of Chapter 1, now deployed quantum-mechanically:

$$\rho(\omega) = \frac{V\omega^2}{\pi^2 c^3}.$$

The Einstein A and B coefficients. Couple an atomic electron to a classical electromagnetic wave. In the radiation gauge, the vector potential for a plane wave of frequency ω , wave vector $\mathbf{k}$, and polarisation ϵ ($\epsilon \perp \mathbf{k}$) is

$$\mathbf{A}(\mathbf{r}, t) = \mathbf{A}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} + \text{c.c.}$$

The interaction Hamiltonian, to leading order in the field, is

$$V(t) = -\frac{e}{mc} \mathbf{A}(\mathbf{r}, t) \cdot \mathbf{p},$$

(the A^2 term is second-order in the field amplitude and is neglected in the first-order rate). The matrix element is

$$\langle f|V(t)|i \rangle = -\frac{e}{mc} \mathbf{A}_0 \cdot \langle f|e^{i\mathbf{k} \cdot \mathbf{r}} \mathbf{p}|i \rangle e^{-i\omega t} + \text{non-resonant term}.$$

In the dipole approximation, valid when the wavelength is much larger than atomic dimensions ($ka_0 \ll 1$), the exponential is replaced by unity: $e^{i\mathbf{k} \cdot \mathbf{r}} \approx 1$. The momentum matrix element is converted to a

dipole matrix element via the commutator identity of Chapter 5 (5.15):

$$[H_0, \mathbf{r}] = -\frac{i\hbar}{m} \mathbf{p} \implies \langle f | \mathbf{p} | i \rangle = \frac{im}{\hbar} (E_f - E_i) \langle f | \mathbf{r} | i \rangle = im\omega_{fi} \langle f | \mathbf{r} | i \rangle. \quad (8.24)$$

Thus $\langle f | \boldsymbol{\epsilon} \cdot \mathbf{p} | i \rangle = im\omega_{fi} \langle f | \boldsymbol{\epsilon} \cdot \mathbf{r} | i \rangle$.

The transition rate for absorption follows from the golden rule (8.22). Substituting the matrix element and the photon density of final states per unit energy, $\rho(\hbar\omega) = V\omega^2/(\pi^2 c^3 \hbar)$, from the cavity mode count:

$$\begin{aligned} \Gamma_{i \rightarrow f}^{\text{abs}} &= \frac{2\pi}{\hbar} |\langle f | V | i \rangle|^2 \rho(\hbar\omega_{fi}) \\ &= \frac{2\pi}{\hbar} \left(\frac{e}{mc} \right)^2 |\mathbf{A}_0 \cdot \langle f | \mathbf{p} | i \rangle|^2 \frac{V\omega_{fi}^2}{\pi^2 c^3 \hbar} \\ &= \frac{2\pi}{\hbar} \frac{e^2}{c^2} \omega_{fi}^2 |\mathbf{A}_0 \cdot \langle f | \mathbf{r} | i \rangle|^2 \frac{V\omega_{fi}^2}{\pi^2 c^3 \hbar}, \end{aligned}$$

where (8.24) was used in the last step. The time-averaged energy density of a monochromatic classical plane wave is

$$u(\omega) = \frac{\omega^2}{2\pi c^2} |\mathbf{A}_0|^2, \quad (8.25)$$

so $|\mathbf{A}_0|^2 = (2\pi c^2/\omega^2) u(\omega)$. Isotropically averaging over the relative orientation of the field polarisation and the atomic dipole yields $|\mathbf{A}_0 \cdot \langle f | \mathbf{r} | i \rangle|^2 = \frac{1}{3} |\mathbf{A}_0|^2 |\langle f | \mathbf{r} | i \rangle|^2$. The normalisation volume V cancels between the field amplitude and the density of states (the classical energy density u is already a volume-normalised quantity). Assembling and cancelling factors:

$$\Gamma_{i \rightarrow f}^{\text{abs}} = \frac{4\pi^2 e^2}{3\hbar^2} |\langle f | \mathbf{r} | i \rangle|^2 u(\omega_{fi}) = B_{i \rightarrow f} u(\omega_{fi}),$$

with the Einstein B coefficient

$$B_{i \rightarrow f} = \frac{4\pi^2 e^2}{3\hbar^2} |\langle f | \mathbf{r} | i \rangle|^2. \quad (8.26)$$

The Einstein relation $g_i B_{i \rightarrow f} = g_f B_{f \rightarrow i}$ (with g_i, g_f the degeneracies of the initial and final levels) follows from $|\langle f | \mathbf{r} | i \rangle|^2 = |\langle i | \mathbf{r} | f \rangle|^2$ – the dipole matrix element is symmetric.

For spontaneous emission, the semi-classical treatment fails: the golden rule with a classical field gives zero transition rate when the field amplitude is zero – no photon, no transition. The Einstein A coefficient is obtained via the detailed-balance argument of Chapter 1 Section 1.1.6: in thermal equilibrium, the rates of absorption, stimulated emission, and spontaneous emission must balance to reproduce the Planck spectrum. This yields the Einstein relation

$$\frac{A_{f \rightarrow i}}{B_{f \rightarrow i}} = \frac{8\pi h \nu^3}{c^3},$$

and therefore

$$A_{f \rightarrow i} = \frac{4e^2 \omega_{fi}^3}{3\hbar c^3} |\langle i | \mathbf{r} | f \rangle|^2. \quad (8.27)$$

 **Note** The derivation of the A coefficient above uses a classical electromagnetic field coupled to quantum-mechanical charges, and supplements it with the Einstein thermodynamic detailed-balance argument. This is the semi-classical radiation theory – honest about its boundary. Spontaneous emission, strictly, requires quantisation of the radiation field; the quantised derivation, in which the A coefficient emerges as the golden rule applied to the atom–field interaction with the vacuum state of

the photon field, is Chapter 10's business. The result (8.27) is identical in both derivations, and every experimentally measured lifetime agrees with it.

Wigner–Eckart in rates. The dipole matrix element $\langle f|\mathbf{r}|i\rangle$ that prices every Einstein coefficient factorises via the Wigner–Eckart theorem (6.93). The vector operator $\mathbf{r}$ has spherical components $r_q^{(1)}$ ($q = -1, 0, 1$), defined in Chapter 6 (6.90). Its matrix element is

$$\langle \alpha', j', m' | r_q^{(1)} | \alpha, j, m \rangle = \langle j, m; 1, q | j', m' \rangle \frac{\langle \alpha', j' || \mathbf{r} || \alpha, j \rangle}{\sqrt{2j' + 1}}.$$

The selection rules $\Delta j = 0, \pm 1$ ($j = 0 \nrightarrow j' = 0$) and $\Delta m = 0, \pm 1$ are the spectroscopic selection rules of Chapter 6 (6.96). The rates for all Zeeman sublevels within a multiplet are proportional to the same reduced matrix element, differing only by the squares of Clebsch–Gordan coefficients – the intensity ratios of the Zeeman triplet (Section 8.2) are priced by the golden rule and the Wigner–Eckart theorem together.

Example: Photoionisation of hydrogen. A photon of energy $\hbar\omega$ above the ionisation threshold ($\hbar\omega > 13.6$ eV) can eject the electron from the ground state into a continuum Coulomb wave. The final state is a free particle of momentum $\hbar\mathbf{k}$ and energy $E = \hbar^2 k^2 / 2m = \hbar\omega - |E_1|$. The density of final states is the free-particle expression (8.23). The full derivation (dipole matrix element with the hydrogen $1s$ wavefunction, angular integration, and the flux factor converting rate to cross-section) is carried out in Exercise 6. Near threshold, the cross-section scales as

$$\sigma_{\text{photo}}(\omega) = \frac{16\pi e^2}{3\hbar c} \frac{\omega_0^{3/2}}{\omega^{7/2}}, \quad (8.28)$$

where $\omega_0 = |E_1|/\hbar$ is the threshold frequency. The scaling $\propto \omega^{-7/2}$ above threshold is the Kramers–Gaunt form; the Gaunt factor (a dimensionless correction near unity, from the Coulomb-wave normalisation) is named, not computed.

Example: Lifetime of the hydrogen $2p$ state. The $2p \rightarrow 1s$ transition in hydrogen is electric-dipole-allowed ($\Delta\ell = \pm 1$, satisfied; $\Delta j = 1$). The A coefficient (8.27), evaluated with the hydrogen dipole matrix element $|\langle 100 | \mathbf{r} | 210 \rangle|^2 = (2^{15}/3^{10})a_0^2$ (computed from the wavefunctions of Table 6.2; the integration is Exercise 7), gives

$$A_{2p \rightarrow 1s} \approx 6.25 \times 10^8 \text{ s}^{-1}, \quad \tau_{2p} = 1/A \approx 1.6 \text{ ns}.$$

This is the physical content of the energy-time uncertainty remarked in Chapter 4 (4.54): the natural linewidth $\Gamma = \hbar/\tau$ is the Mandelstam–Tamm kinetic reading of $\Delta E \Delta t$. The $2s$ state, by contrast, is metastable: the $2s \rightarrow 1s$ transition is dipole-forbidden ($\Delta\ell = 0$ violates the $\Delta\ell = \pm 1$ selection rule), and its decay proceeds by two-photon emission – a process named, not computed here.

The advantage of time-dependent perturbation theory is its direct contact with experiment: every rate, cross-section, and lifetime calculated from (8.22) is a measurable quantity. Its limitation is that it is first-order in the perturbation – when the transition probability approaches unity, the approximation fails, and the system enters the non-perturbative regime (Rabi oscillations, strong-field ionisation – the subjects of specialised treatises, beyond this book's scope).

The golden rule computes rates when the perturbation is weak. Scattering theory asks a different question: what is the probability that a particle, incident from infinity, emerges in a particular

direction? The two questions are kin: both involve continuum states, both connect to measurable cross-sections. But scattering theory requires a formalism that treats the asymptotic states on an equal footing – the subject of the next section.

8.5 Formal Scattering Theory: States, S-Matrix, and Cross-Sections

Scattering is the experimental regime in which a particle, prepared in the distant past as a free state, encounters a target, and is registered in the distant future in a definite direction. Between the two asymptotic instants the full Hamiltonian

$$H = H_0 + V$$

governs the motion; outside the interaction region the particle is free, governed by H_0 . The whole content of scattering theory is the map from the incoming free state to the outgoing free state — the *S-matrix*. Every other object in this section — the Lippmann–Schwinger equation, the scattering amplitude f , the cross-section, the optical theorem, the partial-wave expansion — is a consequence of that map, read in one or another representation.

The section is built in four stages, each more special than the last. *Formal scattering theory* (§8.5.1) postulates the asymptotic condition and deduces from it the Møller operators, the S-matrix, its unitarity, and the optical theorem — all at the operator level, with no choice of representation. *Stationary scattering theory* (§8.5.2) is the energy-shell specialisation: the Lippmann–Schwinger equation, the free Green’s function, and the scattering amplitude emerge as the position-space reading of the formal operators. *Time-dependent scattering theory* (§8.5.3) supplies the dynamical foundation — wave packets, the strong limits, and the derivation of the stationary formalism as the long-time limit — with worked examples. *Symmetry of scattered states* (§8.5.4) closes the section: every symmetry shared by H and H_0 commutes with S , and energy and momentum conservation, the partial-wave expansion, parity selection, and time-reversal reciprocity follow without further calculation.

The treatment adds no new axioms. The asymptotic condition is the single physical input — the scattering-theoretic instance of the evolution postulate of Chapter 4, applied to states that are free in the distant past and future. The debt repaid is the Rutherford cross-section of Chapter 1; the machinery built here is the bridge from the Hilbert space of Chapters 2–6 to the numbers the accelerator reads.

8.5.1 The Formalism: Møller Operators and the S-Matrix

The geometry. A beam of particles of mass m and momentum $\hbar\mathbf{k}$ is incident along $\hat{\mathbf{k}}$ on a fixed scattering centre at the origin; detectors sit at a large distance r , at angles (θ, ϕ) . The incident direction defines the z -axis; the outgoing direction is $\hat{\mathbf{k}}' = \hat{\mathbf{r}}$. For a spherically symmetric potential the scattering is azimuthally symmetric and depends only on θ . The **momentum transfer** is

$$\mathbf{q} = \mathbf{k}' - \mathbf{k}, \quad q = |\mathbf{q}| = 2k \sin \frac{\theta}{2}. \quad (8.29)$$

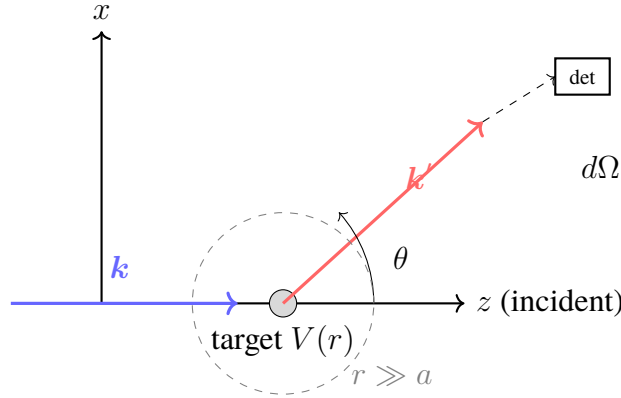


Figure 8.1: Scattering geometry. A beam of momentum $\hbar \mathbf{k}$ is incident from the left onto a target of range a . Particles scattered through angle θ into solid angle $d\Omega$ are counted by a distant detector. The scattering amplitude $f(\theta, \phi)$ is the coefficient of the outgoing spherical wave.

The asymptotic condition. The physical input of scattering theory is the statement that a scattering state looks free in the distant past and the distant future. Made precise: for every scattering state $|\psi\rangle$ (a normalisable state evolving under H) there exist *free* states $|\psi_{\text{in}}\rangle, |\psi_{\text{out}}\rangle \in \mathcal{H}_0$ such that

$$\lim_{t \rightarrow -\infty} \|e^{-iHt/\hbar} |\psi\rangle - e^{-iH_0 t/\hbar} |\psi_{\text{in}}\rangle\| = 0, \quad (8.30)$$

$$\lim_{t \rightarrow +\infty} \|e^{-iHt/\hbar} |\psi\rangle - e^{-iH_0 t/\hbar} |\psi_{\text{out}}\rangle\| = 0. \quad (8.31)$$

This is an assumption about the dynamics: true for short-range potentials V (those decaying faster than $r^{-3-\varepsilon}$, so that a particle sent to infinity no longer feels the potential), and *false* for the Coulomb potential, whose $1/r$ tail is felt at every distance — a failure repaired in §8.5.3. Equations (8.30)–(8.31) are the scattering-theoretic instance of the evolution postulate of Chapter 4: free evolution $e^{-iH_0 t/\hbar}$ replaces full evolution $e^{-iHt/\hbar}$ at the temporal boundaries.

The Møller operators. Multiply (8.30) by $e^{iHt/\hbar}$ and take $t \rightarrow -\infty$. The left-hand side becomes $|\psi\rangle$; the right-hand side becomes the limit of $e^{iHt/\hbar} e^{-iH_0 t/\hbar} |\psi_{\text{in}}\rangle$, the state $|\psi_{\text{in}}\rangle$ factoring out to define an operator:

$$\Omega_{\pm} := \lim_{t \rightarrow \mp\infty} e^{iHt/\hbar} e^{-iH_0 t/\hbar}, \quad (8.32)$$

with action

$$|\psi^{(\pm)}\rangle = \Omega_{\pm} |\phi\rangle. \quad (8.33)$$

Here $|\psi^{(+)}\rangle = \Omega_{+} |\phi\rangle$ is the scattering state that *was* the free state $|\phi\rangle$ in the remote past (the *in*-state); $|\psi^{(-)}\rangle = \Omega_{-} |\phi\rangle$ is the scattering state that *will be* $|\phi\rangle$ in the remote future (the *out*-state). The Møller operator Ω_{+} prepares an in-state from a free state; Ω_{-} prepares an out-state. The limit is a *strong* operator limit — convergence in norm on each vector, not in operator norm, the only kind available for unbounded Hamiltonians.¹

Isometry. The Møller operators preserve the norm. Compute

$$\Omega_{\pm}^{\dagger} \Omega_{\pm} = \lim_{t \rightarrow \mp\infty} e^{iH_0 t/\hbar} e^{-iHt/\hbar} e^{iHt/\hbar} e^{-iH_0 t/\hbar} = \lim_{t \rightarrow \mp\infty} e^{iH_0 t/\hbar} e^{-iH_0 t/\hbar} = I,$$

¹The existence of the strong limits for short-range potentials is the Cook–Jauch theorem; see Reed & Simon, Vol. III, §XI.3. A sketch is given in §8.5.3.

8495 since $e^{-iHt/\hbar}e^{iHt/\hbar} = I$ (H commutes with itself). Hence

$$8496 \quad \Omega_{\pm}^{\dagger}\Omega_{\pm} = I, \quad (8.34)$$

8497 and $\|\Omega_{\pm}|\phi\rangle\| = \||\phi\rangle\|$: the map from free to scattering states preserves probability. An operator with
8498 $A^{\dagger}A = I$ is an *isometry*; it is unitary only if its range is the whole space.

8499 **Asymptotic completeness.** The range of $\Omega_{\pm}$ is the subspace of *scattering states* of H — its continuous
8500 spectral subspace $\mathcal{H}_{\text{cont}}$. The bound states of H (point spectrum) are orthogonal to every scattering
8501 state and lie outside the range. The statement that the range exhausts the continuous subspace,

$$8502 \quad \Omega_{\pm}\Omega_{\pm}^{\dagger} = I - P_{\text{bound}},$$

8503 with P_{bound} the projector onto the bound states, is *asymptotic completeness* — a theorem for a wide
8504 class of short-range potentials, stated here without proof.² With isometry it says: the scattering states
8505 and the bound states partition $\mathcal{H}$, and every scattering state comes from a unique free state.

8506 **The intertwining relation.** The Møller operators *intertwine* the free and full Hamiltonians:

$$8507 \quad H\Omega_{\pm} = \Omega_{\pm}H_0. \quad (8.35)$$

8508 *Proof.* For any real s ,

$$\begin{aligned} 8509 \quad \Omega_{\pm}e^{-iH_0s/\hbar} &= \lim_{t \rightarrow \mp\infty} e^{iHt/\hbar} e^{-iH_0(t+s)/\hbar} = \lim_{t' \rightarrow \mp\infty} e^{iH(t'-s)/\hbar} e^{-iH_0t'/\hbar} \\ 8510 \quad &= e^{-iHs/\hbar} \lim_{t' \rightarrow \mp\infty} e^{iHt'/\hbar} e^{-iH_0t'/\hbar} = e^{-iHs/\hbar} \Omega_{\pm}, \end{aligned}$$

8511 where $t' = t + s$ relabels the dummy variable (the limit is unchanged). Differentiate with respect to
8512 s at $s = 0$: the left gives $\Omega_{\pm}(-iH_0/\hbar)$, the right $(-iH/\hbar)\Omega_{\pm}$. Equating and cancelling $-i/\hbar$ yields
8513 (8.35). $\square$

8514 The consequence is decisive. If $H_0|\phi\rangle = E|\phi\rangle$, then

$$8515 \quad H\Omega_{\pm}|\phi\rangle = \Omega_{\pm}H_0|\phi\rangle = E\Omega_{\pm}|\phi\rangle :$$

8516 the scattering state $\Omega_{\pm}|\phi\rangle$ is an eigenstate of H with the *same* energy E as the free state. The continuous
8517 spectrum of H_0 ($E > 0$) is carried into the continuous spectrum of H ; the potential reshapes the state
8518 but cannot change its energy. This is energy conservation, *derived* from the asymptotic condition
8519 rather than assumed.

8520 **The S-matrix.** The experiment prepares an in-state $|\psi_i^{(+)}\rangle = \Omega_{+}|\phi_i\rangle$ and detects an out-state. The
8521 amplitude to emerge as the free state $|\phi_f\rangle$ is the overlap of the in-scattering state with the out-scattering
8522 state $|\psi_f^{(-)}\rangle = \Omega_{-}|\phi_f\rangle$:

$$8523 \quad S_{fi} := \langle\psi_f^{(-)}|\psi_i^{(+)}\rangle = \langle\phi_f|\Omega_{-}^{\dagger}\Omega_{+}|\phi_i\rangle = \langle\phi_f|S|\phi_i\rangle, \quad S := \Omega_{-}^{\dagger}\Omega_{+}. \quad (8.36)$$

8524 The S-matrix is the in→out map read in the free-particle basis.

8525 **Energy conservation.** From intertwining, $H\Omega_{\pm} = \Omega_{\pm}H_0$, and its adjoint $\Omega_{\pm}^{\dagger}H = H_0\Omega_{\pm}^{\dagger}$. Then

$$8526 \quad H_0S = H_0\Omega_{-}^{\dagger}\Omega_{+} = \Omega_{-}^{\dagger}H\Omega_{+} = \Omega_{-}^{\dagger}\Omega_{+}H_0 = SH_0,$$

8527 SO

$$8528 \quad [S, H_0] = 0. \quad (8.37)$$

²Asymptotic completeness for two-body short-range potentials is proved in Reed & Simon, Vol. III, §XI.3–XI.5.

In the energy basis S_{fi} carries $\delta(E_f - E_i)$: scattering connects only states of the same energy. This is the temporal-translation invariance of Chapter 4 (4.57), here the statement that the free Hamiltonian — the generator of time translations outside the interaction region — is a conserved label of the S-matrix.

Unitarity. Probability conservation is the unitarity of S . Using asymptotic completeness $\Omega_- \Omega_-^\dagger = I - P_{\text{bound}}$ and the orthogonality $P_{\text{bound}} \Omega_+ = 0$ (bound states are orthogonal to scattering states, and $\Omega_+ |\phi\rangle$ is a scattering state),

$$S^\dagger S = \Omega_+^\dagger (\Omega_- \Omega_-^\dagger) \Omega_+ = \Omega_+^\dagger (I - P_{\text{bound}}) \Omega_+ = \Omega_+^\dagger \Omega_+ = I,$$

and similarly $SS^\dagger = I$. Hence $S^\dagger S = SS^\dagger = I$. In components $\sum_n S_{nf}^* S_{ni} = \delta_{fi}$: the probability of emerging in *some* final free state, summed over all open channels, is unity. Every incident particle either scatters elastically, scatters inelastically (the target changes state), or is captured into a bound state; the S-matrix accounts for the first two, and unitarity fixes their sum.

The T-matrix. Because $[S, H_0] = 0$, the S-matrix in the energy representation carries an overall $\delta(E_f - E_i)$. Separate the identity (the particle passes through unscattered) from the transition:

$$S_{fi} = \delta_{fi} - 2\pi i \delta(E_f - E_i) T_{fi}. \quad (8.38)$$

The first term is the “no-scattering” piece; T_{fi} is the *transition matrix*, evaluated *on the energy shell* $E_f = E_i = E$. The factor $2\pi i$ and the energy δ are the Fourier image of the infinite time interval (derived in §8.5.3); the decomposition (8.38) is the definition of T .

The unitarity relation for T . Insert (8.38) into $S^\dagger S = I$, i.e. $\sum_n S_{nf}^* S_{ni} = \delta_{fi}$:

$$\sum_n [\delta_{nf} + 2\pi i \delta(E_n - E_f) T_{nf}^*] [\delta_{ni} - 2\pi i \delta(E_n - E_i) T_{ni}] = \delta_{fi}.$$

The zeroth-order term gives δ_{fi} , cancelling the right-hand side. The cross terms and the quadratic term leave

$$-2\pi i \delta(E_f - E_i) T_{fi} + 2\pi i \delta(E_f - E_i) T_{if}^* + (2\pi)^2 \sum_n \delta(E_n - E_f) \delta(E_n - E_i) T_{nf}^* T_{ni} = 0.$$

The quadratic term is retained at the same order as the linear ones because the dense sum over intermediate states supplies a second energy δ : $\sum_n \delta(E_n - E_f) \delta(E_n - E_i) = \delta(E_f - E_i) \sum_n \delta(E_n - E_i)$, the last factor being the density of intermediate states. Cancelling $\delta(E_f - E_i)$ and $2\pi i$,

$$T_{fi} - T_{if}^* = -2\pi i \sum_n \delta(E_n - E_i) T_{nf}^* T_{ni}. \quad (8.39)$$

This is the exact on-shell unitarity relation for T — exact in T , not perturbative. For forward scattering ($f = i$),

$$2i \Im T_{ii} = -2\pi i \sum_n \delta(E_n - E_i) |T_{ni}|^2 \implies \Im T_{ii} = -\pi \sum_n \delta(E_n - E_i) |T_{ni}|^2.$$

The imaginary part of the forward T -element is fixed by the total transition probability into all open channels. Translated to the scattering amplitude f (defined in §8.5.2, where $f \propto -T$), this becomes the optical theorem $\Im f(0) = k\sigma_{\text{tot}}/4\pi$ (8.49): the minus sign in $f \propto -T$ turns the negative $\Im T_{ii}$ into the positive $\Im f(0)$.

The Born series. The transition matrix obeys an integral equation (derived in §8.5.2 as the operator

form of the Lippmann–Schwinger equation):

$$T = V + V \frac{1}{E - H_0 + i\epsilon} T,$$

with the free resolvent $G_0^{(+)} = (E - H_0 + i\epsilon)^{-1}$. Iterating (a Neumann series) gives the *Born series*

$$T = V + V G_0^{(+)} V + V G_0^{(+)} V G_0^{(+)} V + \cdots, \quad (8.40)$$

each term adding one more interaction with V . This is the energy-domain avatar of the Dyson series of Chapter 4 (4.12). The leading term $T \approx V$ is the first Born approximation, the subject of §8.6; convergence requires the potential to be weak or the energy high (validity discussed there). The series is exact as a formal identity; its truncation is the approximation.

The formalism of this subsection — the asymptotic condition, the Møller operators, the intertwining relation, the S-matrix, unitarity, and the T -matrix — is representation-independent. It says nothing about wavefunctions or cross-sections. Those follow by reading the operators in the energy and position representations: the task of the next subsection.

8.5.2 Stationary Scattering Theory as a Special Case

The formalism of §8.5.1 is representation-independent. Specialising to the energy shell and reading the operators in the position representation turns the Møller operators into the Lippmann–Schwinger equation, the resolvent into the free Green’s function, and the on-shell T -matrix into the scattering amplitude f . This subsection deduces each in turn; nothing new is assumed.

From the intertwining relation to the Lippmann–Schwinger equation. The in-state $|\psi^{(+)}\rangle = \Omega_+|\phi\rangle$ is an eigenstate of H with the same energy E as the free state $|\phi\rangle$ (by intertwining (8.35)): $H|\psi^{(+)}\rangle = E|\psi^{(+)}\rangle$, $H_0|\phi\rangle = E|\phi\rangle$, with $H = H_0 + V$ and $E = \hbar^2 k^2 / 2m > 0$. Subtracting,

$$(E - H_0)|\psi^{(+)}\rangle = V|\psi^{(+)}\rangle,$$

an inhomogeneous equation whose right-hand side is known once $|\psi^{(+)}\rangle$ is. Inverting $E - H_0$ requires a boundary condition: the $\pm i\epsilon$ prescription (deduced below) selects the outgoing (+) or incoming (−) solution. The formal inverse, with the homogeneous solution $|\phi\rangle$ added, is the *Lippmann–Schwinger equation*

$$|\psi^{(\pm)}\rangle = |\phi\rangle + \frac{1}{E - H_0 \pm i\epsilon} V |\psi^{(\pm)}\rangle. \quad (8.41)$$

This is the position-energy avatar of the intertwining relation: in operator form $\Omega_{\pm} = I + G_0^{(\pm)} V \Omega_{\pm}$ with the free resolvent $G_0^{(\pm)} = (E - H_0 \pm i\epsilon)^{-1}$, and acting on $|\phi\rangle$ returns (8.41). The equation is exact but implicit — the unknown $|\psi^{(\pm)}\rangle$ appears on both sides; its iterative solution is the Born series (8.40).

The $i\epsilon$ prescription. The inverse $(E - H_0 \pm i\epsilon)^{-1}$ is the Fourier transform of a retarded (+) or advanced (−) free propagator. With a convergence factor $e^{-\epsilon t/\hbar}$ ($\epsilon > 0$) switched off at the end,

$$\frac{1}{E - H_0 \pm i\epsilon} = \mp \frac{i}{\hbar} \int_0^{\infty} dt e^{\pm i(E - H_0)t/\hbar} e^{-\epsilon t/\hbar} \xrightarrow{\epsilon \rightarrow 0^+} \mp \frac{i}{\hbar} \int_0^{\infty} dt e^{\pm i(E - H_0)t/\hbar}. \quad (8.42)$$

The integral converges only because of $e^{-\epsilon t/\hbar}$: evaluating $\int_0^{\infty} dt e^{\pm i(E - H_0)t/\hbar} e^{-\epsilon t/\hbar} = \mp i\hbar / (E - H_0 \pm i\epsilon)$ reproduces the resolvent. The $+i\epsilon$ (the $t > 0$ integral) is *retarded*: it builds the state from its past —

the physical in-state, an incident wave *plus* an outgoing scattered wave. The $-i\epsilon$ is advanced, building the out-state. The $i\epsilon$ is thus not a trick but the Fourier signature of the arrow of time in the asymptotic condition (8.30)–(8.31); it is derived dynamically in §8.5.3.

The free Green's function. In the position representation the resolvent is the free Green's function,

$$G_0^{(\pm)}(\mathbf{r}, \mathbf{r}'; E) = \langle \mathbf{r} | \frac{1}{E - H_0 \pm i\epsilon} | \mathbf{r}' \rangle = \int \frac{d^3 k'}{(2\pi)^3} \frac{e^{i\mathbf{k}' \cdot (\mathbf{r} - \mathbf{r}')}}{E - \hbar^2 k'^2 / 2m \pm i\epsilon},$$

with $E = \hbar^2 k^2 / 2m$ and $\mathbf{R} = \mathbf{r} - \mathbf{r}'$. Pull out $2m/\hbar^2$ and do the angular integral (align $\mathbf{R}$ with the k' -polar axis, $\int d\Omega e^{ik'R \cos \theta} = 4\pi \sin(k'R)/(k'R)$):

$$G_0^{(\pm)} = \frac{2m}{\hbar^2} \frac{1}{2\pi^2 R} \int_0^\infty dk' \frac{k' \sin(k'R)}{k^2 - k'^2 \pm i\epsilon'}.$$

Write $\sin(k'R) = (e^{ik'R} - e^{-ik'R})/(2i)$ and extend the integral to the whole line (the integrand $k' \sin(k'R)$ is even in k'):

$$\int_0^\infty dk' \frac{k' \sin(k'R)}{k^2 - k'^2 + i\epsilon'} = \frac{1}{2i} \int_{-\infty}^\infty dk' \frac{k' e^{ik'R}}{k^2 - k'^2 + i\epsilon'} - \frac{1}{2i} \int_{-\infty}^\infty dk' \frac{k' e^{-ik'R}}{k^2 - k'^2 + i\epsilon'}.$$

For the $e^{+ik'R}$ piece close the contour in the upper half-plane ($R > 0$); the poles of $1/(k^2 - k'^2 + i\epsilon')$ lie at $k' = \pm(k + i\epsilon'')$, so only $k' = k + i\epsilon''$ is enclosed. The residue is $\lim_{k' \rightarrow k} k' e^{ik'R}/(-2k) = -e^{ikR}/2$, giving $\int = 2\pi i(-e^{ikR}/2) = -\pi i e^{ikR}$. The $e^{-ik'R}$ piece, closed in the lower half-plane (clockwise), encloses $k' = -k - i\epsilon''$ with residue $-e^{ikR}/2$, contributing $-2\pi i(-e^{ikR}/2) = +\pi i e^{ikR}$. Their difference is $-2\pi i e^{ikR}$; times $1/(2i)$ this is $-\pi e^{ikR}$. Hence, for the $+$ sign (the $-$ sign gives e^{-ikR} by the mirrored contour),

$$G_0^{(\pm)}(\mathbf{r}, \mathbf{r}'; E) = -\frac{2m}{\hbar^2} \frac{e^{\pm ik|\mathbf{r} - \mathbf{r}'|}}{4\pi |\mathbf{r} - \mathbf{r}'|}. \quad (8.43)$$

The $\pm i\epsilon$ has selected an outgoing (+) or incoming (−) spherical wave — precisely the boundary condition of (8.41). The Green's function is the amplitude for a free particle of energy E to propagate from $\mathbf{r}'$ to $\mathbf{r}$; the $i\epsilon$ fixes whether it propagates with the retarded or advanced prescription.

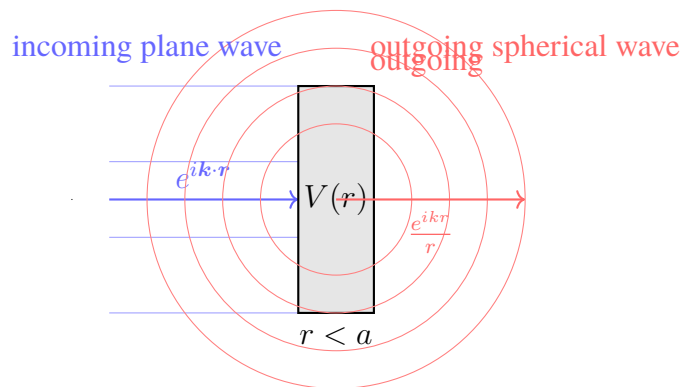


Figure 8.2: Asymptotic forms of the scattering wavefunction. Left of the interaction region: an incident plane wave approaches. Right: at large distances, the wavefunction is the superposition of the unscattered plane wave (continuing through) and an outgoing spherical wave centred on the target. The scattering amplitude f is the envelope of the spherical wave.

The asymptotic form and the scattering amplitude. Take the position matrix element of (8.41)

(with $+$) and insert (8.43):

$$\langle \mathbf{r} | \psi_{\mathbf{k}}^{(+)} \rangle = \langle \mathbf{r} | \mathbf{k} \rangle - \frac{2m}{\hbar^2} \int d^3 r' \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|} \langle \mathbf{r}' | V | \psi_{\mathbf{k}}^{(+)} \rangle.$$

For a detector far from the target ($r = |\mathbf{r}| \gg r'$), expand $|\mathbf{r} - \mathbf{r}'| \approx r - \hat{\mathbf{r}} \cdot \mathbf{r}'$, so $e^{ik|\mathbf{r}-\mathbf{r}'|} \approx e^{ikr} e^{-ik' \cdot \mathbf{r}'}$ with $\mathbf{k}' = k\hat{\mathbf{r}}$ the final momentum. With $\langle \mathbf{r} | \mathbf{k} \rangle = e^{ik \cdot \mathbf{r}} / (2\pi)^{3/2}$,

$$\langle \mathbf{r} | \psi_{\mathbf{k}}^{(+)} \rangle \xrightarrow{r \rightarrow \infty} \frac{1}{(2\pi)^{3/2}} \left[e^{ik \cdot \mathbf{r}} + f(\mathbf{k}', \mathbf{k}) \frac{e^{ikr}}{r} \right],$$

where the matching has defined the scattering amplitude

$$f(\mathbf{k}', \mathbf{k}) = -\frac{2m}{\hbar^2} \frac{(2\pi)^3}{4\pi} \langle \mathbf{k}' | V | \psi_{\mathbf{k}}^{(+)} \rangle. \quad (8.44)$$

The factor $(2\pi)^3$ reflects the normalisation $\langle \mathbf{k}' | \mathbf{k} \rangle = \delta^{(3)}(\mathbf{k}' - \mathbf{k})$. Defining the scattering wavefunction $\psi_{\mathbf{k}}(\mathbf{r}) := (2\pi)^{3/2} \langle \mathbf{r} | \psi_{\mathbf{k}}^{(+)} \rangle$ (so the incident part is the plane wave $e^{ik \cdot \mathbf{r}}$), the asymptotic form is

$$\psi_{\mathbf{k}}(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{ik \cdot \mathbf{r}} + f(\theta, \phi) \frac{e^{ikr}}{r}. \quad (8.45)$$

The two terms are the two alternatives of the experiment — the particle passes through unscattered (the plane wave) or is scattered (the outgoing spherical wave). The amplitude f carries all the dynamics; its phase is unobservable in the elastic cross-section but is fixed in the forward direction by the optical theorem below, and its analytic structure in complex energy encodes the bound states and resonances of the potential (§8.5.4).

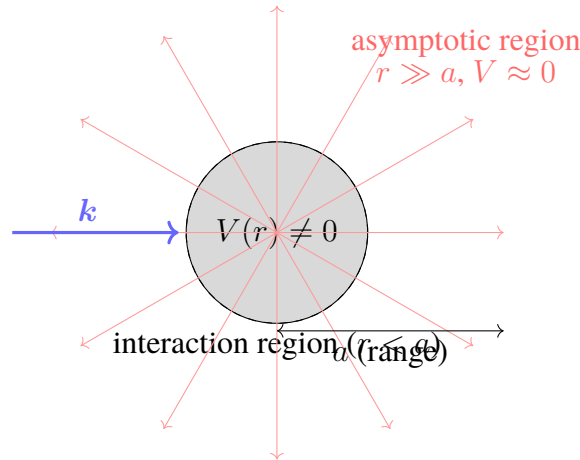


Figure 8.3: Three regimes of scattering. The interaction region ($r < a$) is where the potential is appreciable. Outside it, the particle propagates freely; the wavefunction is a spherical wave whose radial amplitude is modulated by $f(\theta, \phi)$. The scattering problem is the matching of the interior solution to the asymptotic form (8.45).

Cross-section from flux. The incident flux in the plane wave $e^{ik \cdot \mathbf{r}}$ is $v = \hbar k / m$ particles per unit area per unit time. The outgoing spherical wave $f e^{ikr} / r$ carries radial flux $|f|^2 v / r^2$; into solid angle $d\Omega$ at distance r this is $|f|^2 v d\Omega / r^2$. The ratio of scattered to incident flux, times r^2 , is the differential cross-section:

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2, \quad \sigma_{\text{tot}} = \int d\Omega |f(\theta, \phi)|^2. \quad (8.46)$$

For inelastic scattering, in which the target changes internal state, the same formula holds with the

channel amplitude $f_{i \rightarrow f}$.

The S-matrix on the energy shell. Reading $S = \Omega_-^\dagger \Omega_+$ (8.36) in the momentum basis and using $\Omega_-^\dagger = \lim_{t \rightarrow +\infty} e^{iH_0 t/\hbar} e^{-iHt/\hbar}$ together with (8.41), the on-shell element is

$$\langle \mathbf{k}_f | \psi_{\mathbf{k}_i}^{(+)} \rangle = \delta^{(3)}(\mathbf{k}_f - \mathbf{k}_i) - 2\pi i \delta(E_f - E_i) T_{\mathbf{k}_f \mathbf{k}_i}, \quad T_{\mathbf{k}_f \mathbf{k}_i} := \langle \mathbf{k}_f | V | \psi_{\mathbf{k}_i}^{(+)} \rangle. \quad (8.47)$$

This is the momentum-space form of the decomposition (8.38): the $\delta^{(3)}$ is the no-scattering piece, T the transition. Comparing (8.47) with (8.44) gives the on-shell relation

$$f(\mathbf{k}', \mathbf{k}) = -\frac{2m}{\hbar^2} \frac{(2\pi)^3}{4\pi} T_{\mathbf{k}' \mathbf{k}} = -\frac{4m\pi^2}{\hbar^2} T_{\mathbf{k}' \mathbf{k}}, \quad (8.48)$$

i.e. $f \propto -T$ — the minus sign promised in §8.5.1. Iterating (8.41) once, $|\psi^{(+)}\rangle = |\phi\rangle + G_0^{(+)} V |\phi\rangle + \dots$, gives $T = V + V G_0^{(+)} V + \dots$, the Born series (8.40); the leading term $T \approx V$ is the first Born approximation of §8.6.

The optical theorem. The T -matrix unitarity relation (8.39), made explicit in the continuum ($\sum_n \rightarrow \int d^3 k_n$), reads on the energy shell

$$2i \Im T_{ii} = -2\pi i \int d^3 k_n \delta(E_n - E_i) |T_{\mathbf{k}_n \mathbf{k}_i}|^2.$$

The momentum integral is done with $E_n = \hbar^2 k_n^2 / 2m$ on shell: $\int d^3 k_n \delta(E_n - E_i) = \int k_n^2 dk_n d\Omega_n \delta(\hbar^2 k_n^2 / 2m - E_i) = (mk / \hbar^2) \int d\Omega_n$ (since $dE = (\hbar^2 k / m) dk$). Hence

$$\Im T_{ii} = -\pi \frac{mk}{\hbar^2} \int d\Omega |T(\mathbf{k}_n, \mathbf{k}_i)|^2.$$

Now translate to f via (8.48): $|T|^2 = |f|^2 \hbar^4 / (16m^2 \pi^4)$, so $\int d\Omega |T|^2 = \hbar^4 \sigma_{\text{tot}} / (16m^2 \pi^4)$, while $\Im f(0) = -(4m\pi^2 / \hbar^2) \Im T_{ii}$. Assembling,

$$\Im f(0) = -\frac{4m\pi^2}{\hbar^2} \left[-\pi \frac{mk}{\hbar^2} \cdot \frac{\hbar^4 \sigma_{\text{tot}}}{16m^2 \pi^4} \right] = \frac{k}{4\pi} \sigma_{\text{tot}}.$$

Thus the *optical theorem*,

$$\Im f(\theta = 0) = \frac{k}{4\pi} \sigma_{\text{tot}}, \quad (8.49)$$

an exact consequence of probability conservation ($S^\dagger S = I$) alone. The total cross-section — elastic plus inelastic, summed over all open channels — is proportional to the imaginary part of the *forward* elastic amplitude. Destructive interference in the forward direction removes precisely the flux from the incident beam that reappears as scattering into all other directions; the theorem is testable in any experiment measuring both $d\sigma/d\Omega$ and the forward amplitude. The partial-wave reading of (8.49) in §8.5.4 gives a second, independent proof and a sharp physical picture.

The stationary theory is the working register of scattering calculations: the LS equation (8.41), the Green's function (8.43), the amplitude (8.44), the cross-section (8.46), and the optical theorem (8.49). It is the energy-shell specialisation of the formal operators of §8.5.1. Its own foundation — why the $i\epsilon$ and the asymptotic form are the long-time limits of a dynamical process — is the subject of the next subsection.

8.5.3 Time-Dependent Scattering Theory: the Dynamical Foundation

The stationary theory of §8.5.2 is the working register of calculations, but it is not the physical origin of the formalism. Scattering is a dynamical process: a wave packet, localised in the distant past, approaches the target, interacts, and emerges as a scattered packet in the distant future. This subsection makes that picture precise — it is the foundation on which §8.5.1–§8.5.2 stand — and works four examples that exhibit the formalism in motion.

Wave packets. A realistic experiment does not use momentum eigenstates (unnormalisable, filling all space) but a wave packet — a normalisable superposition peaked at the incident momentum $\hbar \mathbf{k}_0$:

$$|\psi_{\text{in}}\rangle = \int d^3k \phi(\mathbf{k} - \mathbf{k}_0) |\mathbf{k}\rangle, \quad (8.50)$$

with ϕ a smooth envelope of width $\Delta k \ll k_0$. The asymptotic condition (8.30)–(8.31) now reads concretely: as $t \rightarrow -\infty$ the packet is far from the target and $|\psi(t)\rangle \rightarrow e^{-iH_0 t/\hbar} |\psi_{\text{in}}\rangle$; as $t \rightarrow +\infty$ it has left and $|\psi(t)\rangle \rightarrow e^{-iH_0 t/\hbar} |\psi_{\text{out}}\rangle$. The S-matrix is the in→out map on the free states,

$$|\psi_{\text{out}}\rangle = S |\psi_{\text{in}}\rangle, \quad (8.51)$$

a normalisable avatar of (8.36).

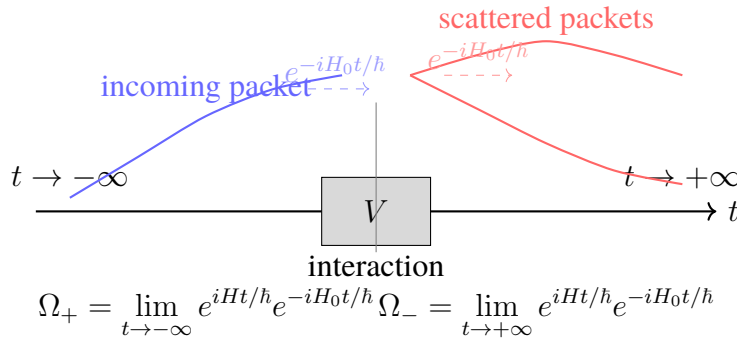


Figure 8.4: Time-dependent picture of scattering. In the distant past ($t \rightarrow -\infty$), the incoming wave packet approaches freely. It enters the interaction region, where the full dynamics under $H = H_0 + V$ distorts it. In the distant future ($t \rightarrow +\infty$), it emerges as a superposition of scattered packets propagating freely in various directions. The Møller operators $\Omega_{\pm}$ intertwine the free and full dynamics at the temporal boundaries.

Strong limits and Cook’s theorem. The Møller operators (8.32) are strong limits: for each normalisable $|\phi\rangle$, $e^{iHt/\hbar} e^{-iH_0 t/\hbar} |\phi\rangle$ converges in norm as $t \rightarrow \mp\infty$. Their existence is not automatic — it fails for the Coulomb potential (Example 4 below). *Cook’s theorem* gives a usable sufficient condition: if $\int_1^\infty \|V e^{-iH_0 t/\hbar} |\phi\rangle\| dt < \infty$ for a dense set of $|\phi\rangle$ (true for short-range $V \sim r^{-3-\varepsilon}$, whose tail a free particle outruns), then the strong limit exists.³ This is the rigorous content of the asymptotic condition; the rest of the section uses it as given.

Example 1: the free wave packet and its spreading. Before the packet reaches the target it propagates

³Cook’s theorem and the Kato–Kuroda theory of asymptotic completeness: Reed & Simon, Vol. III, §XI.3.

freely. Inserting (8.50) into $e^{-iH_0t/\hbar}$ and using $H_0|\mathbf{k}\rangle = (\hbar^2k^2/2m)|\mathbf{k}\rangle$,

$$\psi_{\text{in}}(\mathbf{r}, t) = \int d^3k \phi(\mathbf{k} - \mathbf{k}_0) e^{i\mathbf{k}\cdot\mathbf{r} - i\hbar k^2 t/2m},$$

the free-particle propagator of Chapter 5 (5.42) (equivalently the short-time kernel of Chapter 9 (9.9)) convolved with the initial envelope. For a Gaussian ϕ of width Δk , stationary phase at $\mathbf{k} = \mathbf{k}_0 + m\mathbf{r}/\hbar t$ gives a packet centred at $\mathbf{r} = \mathbf{v}_0 t$ ($\mathbf{v}_0 = \hbar\mathbf{k}_0/m$) whose width grows as $\Delta r(t) \sim \sqrt{(\Delta r_0)^2 + (\hbar t/2m\Delta r_0)^2}$: the packet translates and *spreads*, the free-particle instance of dispersion. Scattering is meaningful only while the packet is compact on the target scale — $\Delta r \ll a$ sets the experimental time window.

Example 2: from time-dependent to stationary — the $i\epsilon$ derived. The bridge to §8.5.2 is the derivation of the Lippmann–Schwinger equation and its $i\epsilon$ from the time-dependent picture (the Gell-Mann–Goldberger construction). Evolve the in-state from $t = -\infty$ under H and project onto a free state:

$$\langle \mathbf{k}_f | \psi_{\mathbf{k}_i}^{(+)} \rangle = \lim_{t \rightarrow -\infty} \langle \mathbf{k}_f | e^{iHt/\hbar} e^{-iH_0t/\hbar} | \mathbf{k}_i \rangle \rangle = \lim_{\epsilon \rightarrow 0^+} \langle \mathbf{k}_f | \frac{i\epsilon}{E_i - H + i\epsilon} | \mathbf{k}_i \rangle \rangle,$$

where the retarded convergence factor $e^{\epsilon t/\hbar}$ ($t \rightarrow -\infty$) has been introduced and the limit traded for the resolvent via $\int_{-\infty}^0 dt e^{i(E_i - H)t/\hbar} e^{\epsilon t/\hbar} = i\hbar/(E_i - H + i\epsilon)$. Splitting $H = H_0 + V$ and using $E_i|\mathbf{k}_i\rangle = H_0|\mathbf{k}_i\rangle$ yields (8.47) with the $+i\epsilon$, and the operator form $\Omega_+ = I + (E - H_0 + i\epsilon)^{-1}V\Omega_+$ is (8.41). The $i\epsilon$ is thus not postulated but *derived* — the Fourier image of evolving from the distant past with a retarded switch-on. The stationary theory is the long-time limit of the dynamical one; the two are one theory in two registers, and (8.42) is its precise statement.

Example 3: from wave packet to cross-section. Why do unnormalisable plane waves give correct cross-sections? Put the experiment in a box of volume $\mathcal{V}$ (one particle, one target). The probability of detecting the particle with final momentum in the cone $d\Omega$ about $\hat{\mathbf{k}}'$ is

$$dP = |S_{\mathbf{k}', \mathbf{k}}|^2 \frac{\mathcal{V}}{(2\pi)^3} k^2 dk d\Omega, \quad (8.52)$$

with $\mathcal{V}/(2\pi)^3$ the density of final states. The incident flux is one particle in volume $\mathcal{V}$ moving at $v = \hbar k/m$, i.e. $j_{\text{inc}} = v/\mathcal{V}$. Using $S_{\mathbf{k}', \mathbf{k}} = \delta^{(3)}(\mathbf{k}' - \mathbf{k}) - 2\pi i \delta(E_f - E_i) T_{\mathbf{k}' \mathbf{k}}$ (8.47) and $f = -(4m\pi^2/\hbar^2)T$ (8.48), the $\delta^{(3)}$ (no-scattering) piece is removed by looking off the incident direction, and the transition piece gives a rate $dP/dt = j_{\text{inc}} |f|^2 d\Omega$ (the energy δ supplies $\delta(E_f - E_i) = \delta(k - k_i)/(v\hbar)$, the time-integral of a rate). Dividing by $j_{\text{inc}} = v/\mathcal{V}$ and taking $\mathcal{V} \rightarrow \infty$,

$$\frac{d\sigma}{d\Omega} = |f(\mathbf{k}', \mathbf{k})|^2,$$

recovering (8.46). The plane-wave formula is the $\mathcal{V} \rightarrow \infty$ limit of a finite wave-packet calculation; the flux normalisation absorbs the volume, and the energy δ encodes the infinite interaction time. This is the rigorous meaning of “scattering of a momentum eigenstate.”

Example 4: the Coulomb caveat. The Coulomb potential $V(r) = Ze^2/r$ violates the short-range condition: $|V| \sim r^{-1}$, not $r^{-3-\epsilon}$, so a particle never outruns the tail and the strong limits (8.32) *do not exist* — the phase accumulated at large r diverges logarithmically. The formalism above therefore does not apply directly. Dollard’s repair (1964) replaces the free comparison dynamics $e^{-iH_0t/\hbar}$ with a *modified* free dynamics carrying a logarithmic phase $e^{-i(Ze^2/\hbar v) \ln(\hbar k v^2 t/m\hbar)}$; with this modification the Møller operators exist, the S-matrix is unitary, and the exact Coulomb amplitude (the Gordon formula)

differs from the Born result (8.64) by a phase $e^{-i\eta \ln \sin^2(\theta/2)}$ ($\eta = Ze^2/\hbar v$) that drops out of $|f|^2$.⁴ The Rutherford cross-section is recovered identically because the phase cancels. The lesson stands: the formalism of §8.5.1–§8.5.2 is the theory of *short-range* scattering; long-range forces require modified asymptotics.

The time-dependent picture is the foundation; the stationary picture is its long-time shadow. What remains is to read off the consequences of symmetry — the subject of the closing subsection.

8.5.4 Symmetry of Scattered States

The asymptotic condition treats the in and out states symmetrically, and the S-matrix inherits every symmetry shared by H and H_0 . This subsection deduces the general theorem and extracts its consequences: energy and momentum conservation, the partial-wave expansion (rotational symmetry), parity selection, and time-reversal reciprocity. The partial-wave machinery — phase shifts, resonances, the scattering length — follows from rotational invariance alone and is developed here as its principal application.

The symmetry theorem. Let U be a unitary (or antiunitary) operator that is a symmetry of both the free and the full dynamics, $[U, H_0] = [U, H] = 0$. Then U commutes with the Møller operators and with the S-matrix:

$$[U, \Omega_{\pm}] = 0, \quad [U, S] = 0. \quad (8.53)$$

Proof. Since $[U, H] = 0$, $Ue^{iHt/\hbar} = e^{iHt/\hbar}U$; likewise $[U, H_0] = 0$ gives $Ue^{-iH_0t/\hbar} = e^{-iH_0t/\hbar}U$. Hence $Ue^{iHt/\hbar}e^{-iH_0t/\hbar} = e^{iHt/\hbar}e^{-iH_0t/\hbar}U$, and the strong limit gives $U\Omega_{\pm} = \Omega_{\pm}U$. Then $US = U\Omega_{-}^{\dagger}\Omega_{+} = \Omega_{-}^{\dagger}U\Omega_{+} = \Omega_{-}^{\dagger}\Omega_{+}U = SU$. $\square$ (For antiunitary U the chain holds, the adjoint carrying the antiunitary conjugation.) This is the scattering-theoretic form of Chapter 4's conservation theorem (4.57): a generator commuting with H is conserved, and here it labels the S-matrix. The consequence is that S is *diagonal* in the quantum numbers of U — scattering does not mix them.

Energy conservation. With $U = e^{-iH_0s/\hbar}$ (the time translation generated by the free Hamiltonian), (8.53) reproduces $[S, H_0] = 0$ (8.37): $S_{fi} \propto \delta(E_f - E_i)$. Energy is conserved because the asymptotic condition is invariant under the time translation that H_0 generates.

Translation and Galilean invariance. For a genuine two-body problem — two particles interacting via $V(\mathbf{r}_1 - \mathbf{r}_2)$ — the total momentum $\mathbf{P}_{\text{tot}}$ generates spatial translations and $[V, \mathbf{P}_{\text{tot}}] = 0$; the symmetry theorem gives $[S, \mathbf{P}_{\text{tot}}] = 0$: total momentum is conserved, the spatial-translation instance of the Galilean invariance analysed in Chapter 4 (§4.7). The fixed-target idealisation $V(\mathbf{r})$ used in §8.5.2 breaks translation invariance (the target pins a preferred origin), reducing the problem to the relative coordinate; momentum conservation is restored in the centre-of-mass frame, where the reduced-particle formalism of this chapter applies and the Galilean boost fixes the relation between lab-frame and CM-frame cross-sections.

Rotational invariance and partial waves. For a central potential $V(r)$, $[V, \mathbf{J}] = 0$ and $[H_0, \mathbf{J}] = 0$; the symmetry theorem gives $[S, \mathbf{J}] = 0$. In the angular-momentum basis $|E, \ell, m\rangle$, S is diagonal in ℓ

⁴J. Dollard, J. Math. Phys. **5**, 729 (1964).

and m and (no m -dependence for a scalar potential) depends only on ℓ : the S-matrix factorises,

$$S = \bigoplus_{\ell=0}^{\infty} S_{\ell}, \quad S_{\ell} = e^{2i\delta_{\ell}}, \quad (8.54)$$

with δ_{ℓ} real by unitarity ($|S_{\ell}| = 1$). The whole content of central-potential scattering is the set of *phase shifts* $\{\delta_{\ell}\}$.

Physical picture. The incident plane wave resolves into a superposition of spherical waves, each of definite angular momentum ℓ . A classical particle of momentum $p = \hbar k$ and impact parameter b carries angular momentum $L = pb = \hbar kb$; the correspondence $b \leftrightarrow \ell/k$ identifies the classical impact parameter with the quantum channel: low- ℓ waves probe the interior of the potential, while high- ℓ waves are kept at large distance by the centrifugal barrier and barely feel it. The phase shift δ_{ℓ} is the angular displacement of the ℓ -th wave's nodes relative to the free wave — the sole imprint of the potential on the asymptotic region.

Origin of the phase shifts. Outside the range of the potential, channel ℓ obeys the free radial equation of Chapter 6 (6.62):

$$\frac{d^2 u_{\ell}}{dr^2} + \left[k^2 - \frac{\ell(\ell+1)}{r^2} \right] u_{\ell} = 0, \quad u_{\ell}(r) = r R_{\ell}(r).$$

The regular solution, normalised to unit amplitude at infinity, is $u_{\ell}(r) = kr j_{\ell}(kr) \xrightarrow{r \rightarrow \infty} \sin(kr - \ell\pi/2)$. In the presence of a short-range potential the asymptotic wavefunction is still a linear combination of the two independent solutions, but its phase is shifted:

$$u_{\ell}(r) \xrightarrow{r \rightarrow \infty} A_{\ell} \sin\left(kr - \frac{\ell\pi}{2} + \delta_{\ell}\right).$$

The potential, whatever its shape, can only displace the nodes — not the $1/r$ fall-off or the wavenumber.

The partial-wave expansion. Expand the incident plane wave (Rayleigh expansion) and match to the asymptotic form (8.45). With

$$e^{i\mathbf{k}\cdot\mathbf{r}} = \sum_{\ell=0}^{\infty} (2\ell+1) i^{\ell} j_{\ell}(kr) P_{\ell}(\cos\theta) \xrightarrow{r \rightarrow \infty} \sum_{\ell=0}^{\infty} (2\ell+1) i^{\ell} \frac{\sin(kr - \ell\pi/2)}{kr} P_{\ell}(\cos\theta),$$

the full scattering wavefunction acquires the phase shift δ_{ℓ} in each channel but the same coefficients $c_{\ell} = (2\ell+1)i^{\ell}$ (the unscattered component is the plane wave itself). Writing each sine as incoming plus outgoing spherical waves, $\sin(kr - \ell\pi/2 + \delta_{\ell}) = \frac{1}{2i} [e^{i\delta_{\ell}} e^{i(kr - \ell\pi/2)} - e^{-i\delta_{\ell}} e^{-i(kr - \ell\pi/2)}]$, and subtracting the plane-wave piece isolates the outgoing scattered wave. The matching yields

$$f(\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell+1) e^{i\delta_{\ell}} \sin \delta_{\ell} P_{\ell}(\cos\theta), \quad (8.55)$$

with $e^{i\delta_{\ell}} \sin \delta_{\ell} = (S_{\ell} - 1)/(2i)$ the imprint of S_{ℓ} on the amplitude. The construction is the content of $[S, \mathbf{J}] = 0$: rotational symmetry has reduced the scattering problem to one real phase shift per channel.

The partial cross-section follows from $\int_{-1}^1 P_{\ell} P_{\ell'} d\cos\theta = 2\delta_{\ell\ell'}/(2\ell+1)$:

$$\sigma_{\ell} = \frac{4\pi}{k^2} (2\ell+1) \sin^2 \delta_{\ell}, \quad \sigma_{\text{tot}} = \sum_{\ell} \sigma_{\ell}. \quad (8.56)$$

8799 The optical theorem (8.49) is now immediate and automatic:

$$8800 \quad \Im f(0) = \frac{1}{k} \sum_{\ell} (2\ell + 1) \sin^2 \delta_{\ell} = \frac{k}{4\pi} \sigma_{\text{tot}},$$

8801 a second, partial-wave proof — the forward amplitude built from phase shifts satisfies the optical
8802 theorem identically, a consistency check on the whole construction. Since $\sin^2 \delta_{\ell} \leq 1$, the *unitarity*
8803 *bound* $\sigma_{\ell} \leq (4\pi/k^2)(2\ell + 1)$; saturation ($\delta_{\ell} = \pi/2$) is a resonance.

8804 **Low-energy scattering and the scattering length.** As $k \rightarrow 0$ the centrifugal barrier $\hbar^2 \ell(\ell + 1)/2mr^2$
8805 suppresses all $\ell > 0$: only s-wave survives. For short-range V ,

$$8806 \quad \delta_0(k) \approx -a_s k \quad (k \rightarrow 0), \quad \sigma \xrightarrow{k \rightarrow 0} \frac{4\pi}{k^2} \sin^2 \delta_0 \approx 4\pi a_s^2, \quad (8.57)$$

8807 defining the *scattering length* a_s . Its sign is physical: positive for repulsion or a shallow bound state
8808 near threshold, negative for attraction without a bound state. For neutron–proton scattering the singlet
8809 length $a_s \approx -23.7$ fm (nearly bound, a virtual state) and the triplet $a_t \approx 5.4$ fm (the deuteron bound
8810 state).

8811 **Resonance: the Breit–Wigner formula.** When V supports a quasi-bound state at E_R in channel ℓ , δ_{ℓ}
8812 rises rapidly through $\pi/2$. Near resonance, $\delta_{\ell}(E) \approx \delta_{\text{bg}} + \arctan(\frac{\Gamma/2}{E_R - E})$, and the partial cross-section
8813 takes the Breit–Wigner form

$$8814 \quad \sigma_{\ell} \approx \frac{4\pi}{k^2} (2\ell + 1) \frac{\Gamma^2/4}{(E - E_R)^2 + \Gamma^2/4}, \quad (8.58)$$

8815 peaking at the unitarity bound with full width Γ . The width is the decay rate of the quasi-bound state,
8816 $\Gamma = \hbar/\tau$ — the same Γ as in the golden rule (8.22) and the energy-time relation (4.54): the resonance
8817 decay rate equals the golden-rule rate into all open channels.

8818 **Bound states and Levinson’s theorem.** A bound state at $E = -\hbar^2 \kappa^2/2m < 0$ is a pole of S_{ℓ} on
8819 the positive imaginary k -axis, $k = i\kappa$; near it $S_{\ell}(k) = e^{2i\delta_{\ell}(k)} \sim 1/(k - i\kappa)$, and the bound-state
8820 wavefunction falls as $e^{-\kappa r}/r$. Levinson’s theorem counts them:

$$8821 \quad \delta_{\ell}(0) - \delta_{\ell}(\infty) = n_{\ell} \pi,$$

8822 each bound state contributing π to the accumulated phase as E sweeps from ∞ to threshold.⁵

8823 **Parity.** For an even potential $[V, \Pi] = 0$; the symmetry theorem gives $[S, \Pi] = 0$. Since $\Pi|k\rangle = |{-k}\rangle$,
8824 the scattering states obey $\Pi|\psi_{\mathbf{k}}^{(\pm)}\rangle \propto |\psi_{-\mathbf{k}}^{(\pm)}\rangle$ and the amplitude satisfies

$$8825 \quad f(\mathbf{k}', \mathbf{k}) = f(-\mathbf{k}', -\mathbf{k}). \quad (8.59)$$

8826 In partial waves, parity ($P_{\ell} \rightarrow (-1)^{\ell} P_{\ell}$) forbids the mixing of even and odd ℓ : each channel keeps its
8827 parity. This is the scattering instance of Laporte’s rule (4.63).

8828 **Time reversal and reciprocity.** If V is time-reversal invariant ($\Theta H \Theta^{-1} = H$, Θ antiunitary), the
8829 symmetry theorem gives $\Theta S \Theta^{-1} = S^{\dagger}$ (antiunitarity reverses the operator order). In components this
8830 is *reciprocity*,

$$8831 \quad S_{fi} = S_{\bar{i}\bar{f}}, \quad (8.60)$$

⁵Levinson’s theorem, with the half-bound-state caveat at threshold: Reed & Simon, Vol. III, §XI.5.

where bars denote time-reversed states. For spinless scattering $\Theta|\mathbf{k}\rangle = |-\mathbf{k}\rangle$ and reciprocity reads $f(\mathbf{k}', \mathbf{k}) = f(-\mathbf{k}, -\mathbf{k}')$: the amplitude for $\mathbf{k} \rightarrow \mathbf{k}'$ equals that for the reversed process. Combined with parity (8.59) it gives $f(\mathbf{k}', \mathbf{k}) = f(\mathbf{k}, \mathbf{k}')$ — the amplitude is symmetric under interchange of initial and final momenta, so the differential cross-section is unchanged when source and detector are swapped. Reciprocity plus unitarity underlies the optical theorem and the equality of forward and reverse reaction rates (detailed balance). For particles with spin, $\Theta^2 = (-1)^{2s}$ and Kramers degeneracy (4.66) doubles the reciprocity structure; spin-averaged cross-sections still obey detailed balance.

The symmetries of H and H_0 have thus been read off the S-matrix without a further calculation: energy and momentum conservation, the partial-wave reduction, parity selection, and time-reversal reciprocity are all consequences of (8.53). What the symmetries do *not* give is the numerical value of the phase shifts or the Born amplitude — those require solving for the dynamics, the subject of the closing section.

8.6 The Born Approximation

The Lippmann–Schwinger equation (8.41) is exact but implicit: the unknown $|\psi^{(+)}\rangle$ appears on both sides. The first Born approximation — the leading term of the series (8.40) — replaces $|\psi^{(+)}\rangle$ inside the integral by the incident plane wave, yielding an explicit formula for the scattering amplitude. (The partial-wave expansion, the complementary low-energy tool that exploits the rotational symmetry of §8.5.4, has already been developed there.)

The first Born approximation. Set $|\psi^{(+)}\rangle \approx |\mathbf{k}\rangle$ in (8.44). The amplitude becomes the Fourier transform of the potential at the momentum transfer:

$$f_{\text{Born}}(\mathbf{k}', \mathbf{k}) = -\frac{2m}{\hbar^2} \frac{1}{4\pi} \int d^3r e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r}), \quad \mathbf{q} = \mathbf{k}' - \mathbf{k}. \quad (8.61)$$

The momentum transfer $q = |\mathbf{q}| = 2k \sin(\theta/2)$ measures how hard the scattering is: forward ($\theta = 0$) is $q = 0$; back-scattering ($\theta = \pi$) is $q = 2k$.

For a spherically symmetric potential $V(r)$, the angular integration is immediate:

$$f_{\text{Born}}(\theta) = -\frac{2m}{\hbar^2 q} \int_0^\infty dr r V(r) \sin(qr). \quad (8.62)$$

Rutherford scattering: the ch1 debt repaid. For the Coulomb potential $V(r) = Ze^2/r$, the Fourier transform of $1/r$ is $4\pi/q^2$. The integral $\int_0^\infty dr \sin(qr) = \lim_{\mu \rightarrow 0} q/(q^2 + \mu^2) = 1/q$ (regulated by a screening factor $e^{-\mu r}$, $\mu \rightarrow 0$). Then

$$\begin{aligned} f_{\text{Born}}(\theta) &= -\frac{2mZe^2}{\hbar^2} \frac{1}{q^2} = -\frac{2mZe^2}{\hbar^2} \frac{1}{4k^2 \sin^2(\theta/2)} \\ &= -\frac{Ze^2}{2mv^2} \frac{1}{\sin^2(\theta/2)}. \end{aligned} \quad (8.63)$$

With $E = \frac{1}{2}mv^2 = \hbar^2 k^2/2m$, the differential cross-section is

$$\frac{d\sigma}{d\Omega} = \left(\frac{Ze^2}{4E} \right)^2 \frac{1}{\sin^4(\theta/2)}. \quad (8.64)$$

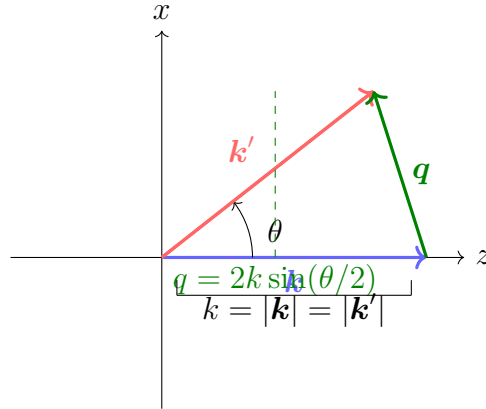


Figure 8.5: Momentum transfer in elastic scattering. The incident momentum $\mathbf{k}$ and the scattered momentum $\mathbf{k}'$ have equal magnitudes $k = |\mathbf{k}| = |\mathbf{k}'|$. The momentum transfer $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ has magnitude $q = 2k \sin(\theta/2)$. The Born amplitude is the Fourier transform of the potential evaluated at $\mathbf{q}$.

This is exactly the classical Rutherford formula of Chapter 1. The debt is repaid in full.

Note The Born approximation and the classical calculation give the same result for the Coulomb potential by coincidence: the Born amplitude is the Fourier transform $1/q^2$, and the classical $1/r$ scattering integral also produces $1/\sin^2(\theta/2) \propto 1/q^4$ in the cross-section. For a general potential the two differ. The exact Coulomb amplitude — the Gordon formula — differs from the Born result by the phase $e^{-i\eta \ln \sin^2(\theta/2)}$ ($\eta = Ze^2/\hbar v$) of Example 4, which drops out of $|f|^2$.

Validity of the Born approximation. The replacement $|\psi^{(+)}\rangle \approx |\mathbf{k}\rangle$ requires the scattered wave to be small compared to the incident wave in the interaction region. This holds when either (a) the incident energy is high, $ka \gg 1$ for a potential of range a , or (b) the potential is weak, $|V_0| \ll \hbar^2/ma^2$ for depth V_0 and range a . For the Coulomb potential the condition is $Ze^2/\hbar v \ll 1$ — high velocity, low nuclear charge.

Form factor. When the target is composite, with charge distribution $\rho(\mathbf{r})$, the potential is the convolution $V(\mathbf{r}) = \int d^3r' \rho(\mathbf{r}')/|\mathbf{r} - \mathbf{r}'|$. In the Born approximation the amplitude factorises:

$$f_{\text{Born}}(\mathbf{q}) = f_{\text{point}}(\mathbf{q}) F(\mathbf{q}), \quad F(\mathbf{q}) = \int d^3r e^{-i\mathbf{q}\cdot\mathbf{r}} \rho(\mathbf{r}). \quad (8.65)$$

$F(\mathbf{q})$ is the elastic form factor — the Fourier transform of the target's charge distribution. The cross-section measures $|F(\mathbf{q})|^2$: scattering is diffraction, and the diffraction pattern reveals the target's structure. This is how Hofstadter measured nuclear charge radii, how X-ray crystallography determines electron densities, and how deep inelastic scattering maps the proton's parton distributions.

Example: Yukawa potential. The Yukawa potential $V(r) = (g^2/r) e^{-r/\lambda}$ describes massive-particle exchange (range $\lambda = \hbar/mc$, coupling g). The Born amplitude from (8.62) is

$$f_{\text{Born}}(\theta) = -\frac{2mg^2}{\hbar^2} \frac{1}{q^2 + 1/\lambda^2}, \quad (8.66)$$

with differential cross-section $d\sigma/d\Omega = (4m^2g^4/\hbar^4)/(q^2 + 1/\lambda^2)^2$. In the limit $\lambda \rightarrow \infty$ (massless exchange) the potential becomes Coulomb and the cross-section recovers Rutherford (8.64). Yukawa's 1935 application to nuclear forces identified the range $\lambda = \hbar/m_\pi c$ with the pion mass, predicting

$m_\pi \approx 100\text{--}200 \text{ MeV}/c^2$; the pion was discovered in 1947 with $m_\pi \approx 140 \text{ MeV}/c^2$.

The Born approximation is the high-energy, weak-coupling tool for computing cross-sections from a given potential; the partial-wave expansion of §8.5.4 is its low-energy complement, where only a few phase shifts matter. Together with the golden rule for transition rates and the Rayleigh–Schrodinger/variational methods for bound states, they constitute the perturbative toolbox — the bridge from the Hilbert space to the laboratory.

8.7 Perturbative Methods and the Theory of Scattering: Summary and Open Debts

The chapter’s content compresses into no new axioms – every display in the chapter is derived from the ch2–ch5 axioms and the ch6 theorems – and the following method inventory.

Consolidated debt ledger.

- §8.1–8.2 **Time-independent perturbation theory.** Rayleigh–Schrodinger series for non-degenerate levels ((8.4), (8.5), (8.6)); secular equation for degenerate levels ((8.9)). Applied to the Stark effect (quadratic in the ground state, linear in $n = 2$), the normal and anomalous Zeeman effects ((8.10)), and hydrogen fine structure ((8.11)). Repaid: ch6 debts #34 (Wigner–Eckart selection rules for Stark perturbation), #38 (Zeeman splitting as degenerate perturbation theory); ch1’s Stark/Zee-man promissory notes.
- §8.3 **The variational method.** Rayleigh–Ritz principle ((8.13)) – a rigorous upper bound on the ground-state energy. Applied to the helium atom ((8.14)) with a one-parameter hydrogenic trial function ((8.15)), yielding $E_{\text{var}} \approx -77.5 \text{ eV}$ – within 1.9% of experiment ((8.17)). Repaid: ch1’s old-quantum-theory helium failure; ch6 debt #36 (hydrogen radial wavefunctions as trial-function ingredients).
- §8.4 **Time-dependent perturbation theory.** First-order transition amplitude from the truncated Dyson series ((8.18)). Fermi’s golden rule ((8.22)) – the chapter’s keystone – gives the transition rate as $2\pi/\hbar$ times the squared matrix element times the density of states. Einstein A and B coefficients derived ((8.26), (8.27)) via the semi-classical radiation theory plus detailed balance. Benchmark check against the exact forced-oscillator solution: first-order TDPT correctly captures the single-quantum channel. Repaid: ch4 debt #25 (Dyson truncation to TDPT); ch5 debt #28 (interaction-picture truncation, forced-oscillator benchmark); ch6 debts #34, #35 (Wigner–Eckart selection rules for rates, dipole matrix elements); ch1’s Einstein A/B debt.
- §8.5 **Formal scattering theory.** The asymptotic condition and Møller operators ((8.32)), the intertwining relation ((8.35)), the S-matrix and its unitarity ((8.36)); the stationary Lippmann–Schwinger equation ((8.41)), free Green’s function ((8.43)), scattering amplitude ((8.44)), and optical theorem ((8.49)) — $\Im f(0) = k\sigma_{\text{tot}}/4\pi$ from probability conservation; the time-dependent foundation (wave packets, the derived $i\epsilon$, the Coulomb/Dollard caveat); and the symmetry theorem ((8.53)) yielding the partial-wave expansion ((8.55)), phase shifts, scattering length ((8.57)), and Breit–Wigner resonance ((8.58)). Repaid: ch6 debt #37 (central-potential scattering bound-

ary conditions from the radial equation); ch4's symmetry/conservation theorem read off the S-matrix.

§8.6 The Born approximation. First Born amplitude as the Fourier transform of the potential ((8.61)); Rutherford cross-section recovered identically ((8.64)) – the ch1 debt repaid. Form factor $F(\mathbf{q})$ ((8.65)) connects scattering to structure; the Yukawa potential ((8.66)) predicts the pion mass. Repaid: ch1's Rutherford debt.

Method dictionary. Table 8.1 extends the experiment–formalism dictionary of Chapters 2–6. The method rows added here are the perturbative and scattering entries.

Table 8.1: The experiment–formalism dictionary v6. Ten method rows added in this chapter, extending Tables 2.1–6.3.

Method concept	Formal object
Perturbation series	λ -expansion of the spectral problem
Variational bound	rigorous upper bound on E_0
Dyson truncation	first-order transition amplitude
Golden rule	$\Gamma = \frac{2\pi}{\hbar} V_{fi} ^2 \rho(E_f)$
Einstein coefficients	A, B from dipole rates
Scattering amplitude	asymptotic form of the wavefunction
S-matrix	probability-conserving in/out map
Optical theorem	$\Im f(0) \propto \sigma_{\text{tot}}$
Born approximation	Fourier transform of the potential
Phase shift	imprint of the potential per channel
Resonance	quasi-bound state, Breit–Wigner, $\Gamma = \hbar/\tau$
Symmetry of S	$[S, U] = 0$ for $[U, H] = [U, H_0] = 0$

Remaining debts. The chapter leaves four debts open, planted for the chapters to come:

- 1. Strong coupling and non-perturbative methods.** Perturbation theory fails when the expansion parameter is of order unity. The path integral and WKB approximation – the subjects of Chapter 9 – are the standard non-perturbative tools.
- 2. Field quantisation and spontaneous emission.** The Einstein A coefficient was obtained semi-classically. The full QED derivation requires quantisation of the radiation field – Chapter 10's business.
- 3. Relativistic corrections beyond fine structure.** The fine-structure perturbation treated the leading relativistic corrections; the Dirac equation – the complete relativistic theory of the electron – is named only, beyond the non-relativistic scope of this book.
- 4. Multi-channel and full resonance theory.** The Breit–Wigner formula treats one resonance in one channel. Multi-channel scattering, coupled-channels methods, and the full resonance theory (complex scaling, Siegert states) are beyond scope, named with a literature pointer.

Exit question. These perturbative methods work when the coupling is weak. The atomic and molecular spectroscopist lives inside their domain of validity. But when the coupling is strong – when the expansion parameter is of order unity – the series diverges and the approximation fails. What replaces

perturbation theory then? The answer occupies the book's two closing chapters: Chapter 9 (path integrals and the semi-classical limit) and Chapter 10 (field-theoretic methods and second quantisation).

Chapter 8 Exercise

1. (★) (Section 8.1) First-order anharmonic oscillator. For the quartic perturbation $V = \lambda x^4$ on the one-dimensional harmonic oscillator, with $x = \sqrt{\hbar/2m\omega} (a + a^\dagger)$:

- (a) Compute $\langle n | (a + a^\dagger)^4 | n \rangle$ by normal-ordering the sixteen terms and collecting the nonzero diagonal contributions. Confirm the result (8.7):

$$E_n^{(1)} = \lambda \frac{3\hbar^2}{4m^2\omega^2} (2n^2 + 2n + 1).$$

- (b) Show that the first-order state correction $|n^{(1)}\rangle$ involves only states with $\Delta n = \pm 2, \pm 4$, and explain why $\Delta n = \pm 1, \pm 3$ are forbidden by parity.

2. (★) (Section 8.1) Hydrogen ground-state Stark polarisability: verify the $2p$ contribution. Using the hydrogen wavefunctions from Table 6.2:

- (a) Compute $\langle 210 | z | 100 \rangle$ directly from ψ_{100} and ψ_{210} . Confirm that only the $2p$ ($n = 2, \ell = 1, m = 0$) state contributes to the sum (8.8) at leading order, and evaluate its contribution to the polarisability.

- (b) Estimate the quadratic Stark shift $\Delta E_{1s}^{(2)}$ for a laboratory field $\mathcal{E} = 10^4$ V/cm (in Gaussian units: $1 \text{ V/cm} \approx 3.33 \times 10^{-3} \text{ statV/cm}$). Is the shift measurable by optical spectroscopy?

3. (★) (Section 8.2) Normal Zeeman triplet. For a hydrogen atom in a magnetic field $B\hat{z}$, consider the $2p \rightarrow 1s$ (Lyman- α) transition:

- (a) Write the energies $E_{2,1,m}$ and $E_{1,0,0}$ as functions of B using the orbital Zeeman Hamiltonian (6.110).

- (b) Determine the three transition frequencies $\omega_m = (E_{2,1,m} - E_{1,0,0})/\hbar$ and the corresponding wavelengths for $B = 1 \text{ T}$ (in Gaussian units: $1 \text{ T} = 10^4 \text{ G}$). Use $\mu_B = 9.27 \times 10^{-21} \text{ erg/G}$.

- (c) Using the dipole selection rules $\Delta m = 0, \pm 1$, identify the polarisation of each line (π for $\Delta m = 0$, $\sigma^\pm$ for $\Delta m = \pm 1$).

4. (★★) (Section 8.2) Anomalous Zeeman effect: the Landé g -factor.

- (a) Derive the projection-theorem result for the matrix elements of a vector operator $\mathbf{V}$ within a J -multiplet:

$$\langle LSJM'_J | \mathbf{V} | LSJM_J \rangle = \frac{\langle LSJ | \mathbf{V} \cdot \mathbf{J} | LSJ \rangle}{\hbar^2 J(J+1)} \langle LSJM'_J | \mathbf{J} | LSJM_J \rangle.$$

- (b) Apply this to $\mathbf{L}$ and $\mathbf{S}$ and derive the Landé g -factor (8.10). Verify the g -factors for the sodium D-line terms: $g(^2S_{1/2}) = 2$, $g(^2P_{3/2}) = 4/3$, $g(^2P_{1/2}) = 2/3$.

- (c) Draw the level scheme for the sodium D-lines in a weak magnetic field, showing all Zeeman sublevels and the allowed electric-dipole transitions. Count the total number of lines and predict their relative intensities from the Clebsch–Gordan coefficients.

5. (★★) (Section 8.3) Helium with a two-parameter trial function. Generalise the trial function (8.15) to allow different effective charges for the two electrons:

$$\psi_{\text{trial}}(\mathbf{r}_1, \mathbf{r}_2) = \frac{Z_1^{3/2} Z_2^{3/2}}{\pi a_0^3} e^{-Z_1 r_1/a_0} e^{-Z_2 r_2/a_0}.$$

(Note: this function is not properly symmetrised; treat it as a spatial ansatz for the purpose of energy minimisation.)

- (a) Compute the energy functional $\langle H \rangle (Z_1, Z_2)$. The electron–electron repulsion integral with two different screening constants is

$$\left\langle \frac{e^2}{r_{12}} \right\rangle = Z_1 Z_2 E_h \frac{Z_1^2 + Z_2^2 + Z_1 Z_2}{(Z_1 + Z_2)^3}.$$

- (b) Minimise with respect to Z_1 and Z_2 . Compare the resulting ground-state energy with the one-parameter result (8.17) and with experiment. Does the additional parameter improve the bound?
- (c) Explain why $Z_1 \neq Z_2$ might be expected physically (inner vs outer electron screening), and why the variational minimum nevertheless yields $Z_1 = Z_2$ for this particular unsymmetrised trial function.

6. (★★) (Section 8.4) Photoionisation cross-section from the golden rule.

- (a) Write the final state for photoionisation of hydrogen from the ground state as a plane wave $\langle \mathbf{r} | \mathbf{k} \rangle = e^{i\mathbf{k} \cdot \mathbf{r}} / \sqrt{V}$ (neglecting the Coulomb distortion of the outgoing electron – the Born approximation for the final state). Compute the dipole matrix element $\langle \mathbf{k} | \boldsymbol{\epsilon} \cdot \mathbf{r} | 100 \rangle$.
- (b) Using the golden rule (8.22) and the density of states (8.23), derive the differential cross-section $d\sigma/d\Omega$. Integrate over angles to obtain the total photoionisation cross-section $\sigma_{\text{photo}}(\omega)$.
- (c) Show that near threshold ($\hbar\omega \gtrsim |E_1|$), the cross-section scales as $(\omega - \omega_0)^{3/2} / \omega^{7/2}$, where $\hbar\omega_0 = |E_1|$. Compare with the Kramers–Gaunt form (8.28).

7. (★★) (Section 8.4) Spontaneous emission lifetime of hydrogen $2p$.

- (a) Using the hydrogen wavefunctions from Table 6.2, compute the dipole matrix element $\langle 100 | \mathbf{r} | 210 \rangle$ and its squared magnitude.
- (b) Evaluate the Einstein A coefficient (8.27) for the $2p \rightarrow 1s$ transition in hydrogen. Use $a_0 = 0.529 \text{ \AA}$, $E_2 - E_1 = 10.2 \text{ eV}$, and $\alpha = e^2/\hbar c = 1/137$.
- (c) Compare your result with the quoted value $A \approx 6.25 \times 10^8 \text{ s}^{-1}$ ($\tau \approx 1.6 \text{ ns}$). The measured lifetime is $\tau_{\text{exp}} \approx 1.60 \text{ ns}$ – what does this agreement imply about the dipole approximation and the semi-classical radiation theory?

8. (★★) (Section 8.6) Born approximation for a Gaussian potential. Consider $V(r) = V_0 e^{-r^2/a^2}$:

- (a) Evaluate the Born amplitude (8.62). The integral $\int_0^\infty dr r e^{-r^2/a^2} \sin(qr)$ can be expressed in terms of the error function or evaluated by parametric differentiation.
- (b) Compute the differential cross-section $d\sigma/d\Omega$ and the total cross-section σ_{tot} as functions of k and a . Discuss the high-energy limit ($ka \gg 1$) and the low-energy limit ($ka \ll 1$).
- (c) Under what condition on V_0 , a , and k is the Born approximation valid for this potential? Express the condition in terms of the dimensionless parameters $|V_0|ma^2/\hbar^2$ and ka .

9. (★★★) (Section 8.5.4) Phase shift from a square well. For the attractive spherical square well,

$$V(r) = \begin{cases} -V_0, & r < a, \\ 0, & r > a, \end{cases} \quad V_0 > 0,$$

consider s-wave ($\ell = 0$) scattering:

- (a) Write the radial Schrödinger equation for $u_0(r) = rR_0(r)$ inside and outside the well. 9025
 Impose the boundary conditions $u_0(0) = 0$ and continuity of u_0 and u'_0 at $r = a$. 9026
- (b) Solve for the phase shift $\delta_0(k)$. Show that 9027
- $$\tan \delta_0(k) = \frac{k \tan(\kappa a) - \kappa \tan(ka)}{\kappa + k \tan(ka) \tan(\kappa a)}, \quad 9028$$
- where $\kappa = \sqrt{k^2 + 2mV_0/\hbar^2}$. 9029
- (c) Show that at low energy ($ka \ll 1$), the s-wave scattering length is $a_s = a[1 - \tan(\kappa_0 a)/(\kappa_0 a)]$ 9030
 with $\kappa_0 = \sqrt{2mV_0}/\hbar$. Discuss how the sign of a_s changes as the well depth increases and 9031
 the first bound state appears. 9032
10. (***)(Section 8.5.4) Optical theorem verification for the square well. Using the s-wave phase 9033
 shift $\delta_0(k)$ from Exercise 9: 9034
- (a) Compute the forward scattering amplitude $\Im f(0)$ from the partial-wave expansion (8.55) 9035
 (s-wave only at low energy). 9036
- (b) Compute the total cross-section σ_{tot} from (8.56). 9037
- (c) Verify that $\Im f(0) = k\sigma_{\text{tot}}/4\pi$ – the optical theorem (8.49) – holds for this phase shift, 9038
 confirming that the partial-wave expansion is consistent with unitarity. 9039

Chapter 9 The Path Integral Formulation, as According to Feynman



Note *This chapter adds zero new axioms. Everything derived below follows from the state-space axioms of Chapter 2, the measurement axioms of Chapter 3, the evolution postulate of Chapter 4, and the representation machinery of Chapter 5, together with the Trotter product formula — stated, proved in finite dimensions at its point of use in Section 9.1, and honestly declared for unbounded operators with the Appendix A pointer at that same place. The path-integral measure $\int \mathcal{D}x$ is a formal symbol: its sole rigorous content is the limiting discretized product defined in Section 9.2, and a note there states plainly what it is and is not.*

The preceding chapter closed with a debt it declared openly (Section 5.10):

The composition property of the propagator (5.40), the action phase of the free kernel (5.42), and the spreading packet, all to the path-integral chapter. The composition property is the germ: decomposing the waiting time into ever-finer slices, with a resolution of identity inserted at every cut, turns one kernel into a convolution of many. Can that convolution be taken to the limit — can the evolution of a quantum system over a finite interval be built from the evolution over infinitesimal steps, and what does the resulting sum over histories teach us that the canonical formalism cannot?

This chapter is that construction. Its first two sections answer the question — Sections 9.1 and 9.2 build the Feynman path integral from the composition law by time-slicing, identify the classical action as the phase, and verify the free particle exactly — and the honest answer is the chapter's thesis: every path contributes an amplitude $e^{iS/\hbar}$, and quantum mechanics is the coherent sum. The classical limit follows as the stationary-phase approximation, Section 9.3, repaying the historical chapter's Hamilton-Jacobi debt; the harmonic oscillator is evaluated exactly via the quadratic fluctuation determinant, Section 9.4, independently verifying the ladder-algebra spectrum of Chapter 5; the Aharonov-Bohm effect exhibits the physical content of path topology, Section 9.5, a gauge-invariant phase shift inaccessible to the canonical formalism; and the WKB semiclassical approximation, Section 9.6, repays the book's oldest quantization debt — the Bohr-Sommerfeld rule, derived and corrected by the Maslov index. The chapter lands on two measurable quantities: the AB phase shift $\Delta\phi = 2\pi\Phi/\Phi_0$ and the WKB quantized energies with their tunneling rates.

The debt ledger this chapter receives and will balance is as follows.

The chapter proceeds in nine sections. Section 9.1 reviews the propagator machinery of Chapter 5, poses the question of the short-time kernel, and derives it by the Trotter factorization — the single technical tool the construction needs beyond the existing axioms. Section 9.2 composes the short-time kernels into the N -slice product, takes the limit, and obtains the Feynman path integral in its discretized and symbolic forms, with the free particle verified exactly. Section 9.3 applies stationary phase to recover the classical limit and repays the Hamilton-Jacobi debt. Section 9.4 evaluates the harmonic

Table 9.1: Debts received by this chapter. The repayment sections are those in which each debt is discharged.

Debt	Received from	Repaid in
Hamilton–Jacobi equation / action principle	ch1 (historical introduction)	§9.3
Bohr–Sommerfeld quantization $\oint p_i dq_i = n_i h$	ch1 row 1.9	§9.6
Propagator composition $\rightarrow$ path integral	ch5 §5.9 ((5.40))	§9.1–9.2
Free kernel’s action phase $S = m(x - x')^2/2t$	ch5 §5.9 ((5.42))	§9.2
De Broglie packet dispersion	ch5 §5.9 ((5.43))	§9.3, §9.6
Harmonic oscillator spectrum (cross-check)	ch5 §5.6 ((5.24))	§9.4, §9.6

oscillator exactly, yielding the Mehler kernel and the spectrum. Section 9.5 presents the Aharonov-Bohm effect as a topological application. Section 9.6 develops the WKB method and repays the Bohr-Sommerfeld debt. Section 9.7 constructs the path integral for spin systems from spin coherent states, deriving the Wess-Zumino action and the path topology that quantizes spin. Section 9.8 builds the holomorphic coherent-state path integral for bosonic systems, unifying the harmonic-oscillator ladder algebra with the path-integral formalism and providing the template for the functional integral over fields in Chapter 10. Section ?? collects the principles, balances the ledger, and hands the functional-integral template to the field chapter.

9.1 The Propagator Reviewed: Kernel, Composition, and the Short-Time Kernel

Chapter 5 closed its account of the free particle with a pointer. The evolution operator read in the position representation is an integral kernel — the propagator — and two of its properties are the seed of everything this chapter constructs. This section re-states those properties, poses the constructive question they imply, and supplies the one new tool the construction needs: the short-time approximation to the kernel for a system with a potential.

The propagator recalled. For a particle on the line with Hamiltonian $H = P^2/2m + V(Q)$, the propagator is the position-representation matrix element of the evolution operator (4.1):

$$K(x, t; x', 0) := \langle x | U(t) | x' \rangle = \langle x | e^{-iHt/\hbar} | x' \rangle. \quad (9.1)$$

This is the propagator, defined in Chapter 5 as (5.39) and re-stated here as the chapter’s central object. The kernel carries the amplitude to register the particle at x at time t , given that it was prepared at x' at time 0. Its first property is the composition law, a direct consequence of the semigroup property $U(t) = U(t-t')U(t')$ of the evolution family, read between position labels with a resolution of identity (5.6) inserted:

$$K(x, t; x', 0) = \int_{-\infty}^{+\infty} dx'' K(x, t; x'', t') K(x'', t'; x', 0). \quad (9.2)$$

This is (5.40) of Chapter 5: the amplitude to go from x' to x in time t is the amplitude to go by way of every intermediate position x'' , summed — the continuum analogue of the composition of alternative

paths that has been the book's grammar since Chapter 2. It is an operator fact, nothing more.

The second property is that for the free particle, $V = 0$, the kernel is known in closed form. Momentum is conserved, the evolution is trivial in the momentum representation, and the Fourier transform of Chapter 5 gives

$$K_0(x, t; x', 0) = \sqrt{\frac{m}{2\pi i \hbar t}} \exp\left(\frac{im(x - x')^2}{2\hbar t}\right). \quad (9.3)$$

This is (5.42) of Chapter 5, the benchmark. Its phase is the classical action of the free particle on the straight, constant-velocity path, $S_0 = m(x - x')^2/2t$, a fact Chapter 5 named as a forward debt. That fact is about to become a theorem.

The central question. The composition property (5.40) can be iterated. Divide the interval $[0, t]$ into N subintervals of equal duration $\varepsilon = t/N$, insert a position resolution of identity at each of the $N - 1$ interior times $t_k = k\varepsilon$, and apply (5.40) at every step:

$$K(x, t; x', 0) = \int dx_1 \cdots dx_{N-1} \prod_{k=0}^{N-1} K(x_{k+1}, \varepsilon; x_k, 0). \quad (9.4)$$

with the notational convention $x_0 := x'$ and $x_N := x$. This is the skeleton of the path integral: the full kernel is a multiple convolution of N short-time kernels, integrated over all intermediate positions. For the free particle, (5.42) supplies $K_0(x_{k+1}, \varepsilon; x_k, 0)$ exactly for any ε , and the convolution must reproduce (5.42) for the whole interval — a consistency check Section 9.2 will perform. For a general potential, no closed form for K at finite time exists, because the kinetic and potential terms of H do not commute. The question is: can an *approximate* short-time kernel be found, one that becomes exact as $\varepsilon \rightarrow 0$, such that the $N \rightarrow \infty$ limit of the N -slice product displayed above converges to the true kernel for any V ? If so, the composition property is not merely a bookkeeping identity — it is a constructive definition of the quantum dynamics.

The Trotter product formula. The obstruction is that the exponential of a sum is not the product of exponentials when the summands fail to commute. The operator identity that tames the obstruction for small ε is the Trotter product formula.¹

Theorem (Trotter product formula, finite-dimensional proof). For matrices A and B ,

$$e^{-i\varepsilon(A+B)/\hbar} = e^{-i\varepsilon A/\hbar} e^{-i\varepsilon B/\hbar} + O(\varepsilon^2). \quad (9.5)$$

The proof is by series expansion of both sides to order ε^2 .

Expand the left-hand side:

$$e^{-i\varepsilon(A+B)/\hbar} = I - \frac{i\varepsilon}{\hbar}(A + B) - \frac{\varepsilon^2}{2\hbar^2}(A + B)^2 + O(\varepsilon^3).$$

Expand the right-hand side as a product of two series:

$$\begin{aligned} e^{-i\varepsilon A/\hbar} e^{-i\varepsilon B/\hbar} &= \left(I - \frac{i\varepsilon}{\hbar}A - \frac{\varepsilon^2}{2\hbar^2}A^2 + O(\varepsilon^3)\right) \left(I - \frac{i\varepsilon}{\hbar}B - \frac{\varepsilon^2}{2\hbar^2}B^2 + O(\varepsilon^3)\right) \\ &= I - \frac{i\varepsilon}{\hbar}(A + B) - \frac{\varepsilon^2}{\hbar^2} \left(\frac{A^2}{2} + AB + \frac{B^2}{2}\right) + O(\varepsilon^3). \end{aligned}$$

¹H. F. Trotter, *Proc. Amer. Math. Soc.* **10**, 545 (1959). The formula is stated here, proved in finite dimensions, and honestly declared for unbounded operators.

The two sides agree at order ε^0 and order ε^1 . At order ε^2 the difference is

$$\begin{aligned} (\text{RHS}) - (\text{LHS}) &= -\frac{\varepsilon^2}{\hbar^2} \left[\left(\frac{A^2}{2} + AB + \frac{B^2}{2} \right) - \frac{1}{2} (A^2 + AB + BA + B^2) \right] + O(\varepsilon^3) \\ &= -\frac{\varepsilon^2}{\hbar^2} \left(\frac{AB}{2} - \frac{BA}{2} \right) + O(\varepsilon^3) \\ &= -\frac{\varepsilon^2}{2\hbar^2} [A, B] + O(\varepsilon^3). \end{aligned}$$

Hence the two expressions differ by a commutator term at order ε^2 , and (9.5) is proved in finite dimensions. The Lie bracket (4.20) of Chapter 4 is the same commutator appearing in a different guise. For the unbounded operators $P^2/2m$ and $V(Q)$ the same formula holds, but the operator-norm estimate that controls the $O(\varepsilon^2)$ remainder requires the domain theory of Appendix A; the existence of the limit is stated here honestly and proved there.

The short-time kernel. Apply the Trotter factorization to $H = P^2/2m + V(Q)$, with $A = P^2/2m$ and $B = V(Q)$:

$$e^{-i\varepsilon H/\hbar} = e^{-i\varepsilon P^2/2m\hbar} e^{-i\varepsilon V(Q)/\hbar} + O(\varepsilon^2).$$

Insert this into the kernel definition and place a momentum resolution of identity between the two exponentials — $V(Q)$ is diagonal in the position representation, $P^2/2m$ in the momentum representation:

$$\begin{aligned} K(x_{k+1}, \varepsilon; x_k, 0) &= \langle x_{k+1} | e^{-i\varepsilon P^2/2m\hbar} e^{-i\varepsilon V(Q)/\hbar} | x_k \rangle + O(\varepsilon^2) \\ &= \int \frac{dp_k}{2\pi\hbar} \langle x_{k+1} | p_k \rangle e^{-i\varepsilon p_k^2/2m\hbar} \langle p_k | x_k \rangle e^{-i\varepsilon V(x_k)/\hbar} + O(\varepsilon^2). \end{aligned}$$

The plane wave (5.10) of Chapter 5 supplies $\langle x | p \rangle = e^{ipx/\hbar} / \sqrt{2\pi\hbar}$; inserting both factors and collecting the exponent,

$$K(x_{k+1}, \varepsilon; x_k, 0) \approx \int \frac{dp_k}{2\pi\hbar} \exp \left[\frac{i}{\hbar} \left(p_k \Delta x_k - \varepsilon \frac{p_k^2}{2m} - \varepsilon V(x_k) \right) \right]. \quad (9.6)$$

with the shorthand $\Delta x_k := x_{k+1} - x_k$, and the approximation sign carrying the Trotter $O(\varepsilon^2)$ error. The momentum integral is a Fresnel Gaussian; its evaluation is the central computation of this section.

The Gaussian integral: completing the square. The exponent in the momentum integral is quadratic in p_k . Collecting the p_k -dependent terms:

$$p_k \Delta x_k - \frac{\varepsilon}{2m} p_k^2 = -\frac{\varepsilon}{2m} \left(p_k^2 - \frac{2m}{\varepsilon} \Delta x_k p_k \right).$$

Complete the square inside the parentheses:

$$p_k^2 - \frac{2m}{\varepsilon} \Delta x_k p_k = \left(p_k - \frac{m \Delta x_k}{\varepsilon} \right)^2 - \left(\frac{m \Delta x_k}{\varepsilon} \right)^2.$$

Substituting back, the exponent becomes

$$\frac{i}{\hbar} \left[-\frac{\varepsilon}{2m} \left(p_k - \frac{m \Delta x_k}{\varepsilon} \right)^2 + \frac{\varepsilon}{2m} \left(\frac{m \Delta x_k}{\varepsilon} \right)^2 - \varepsilon V(x_k) \right].$$

The second term simplifies: $(\varepsilon/2m)(m \Delta x_k / \varepsilon)^2 = m(\Delta x_k)^2 / 2\varepsilon$. Separating the p_k -independent part from the integral,

$$K(x_{k+1}, \varepsilon; x_k, 0) \approx \exp \left[\frac{i}{\hbar} \left(\frac{m(\Delta x_k)^2}{2\varepsilon} - \varepsilon V(x_k) \right) \right] \int \frac{dp_k}{2\pi\hbar} \exp \left[-\frac{i\varepsilon}{2m\hbar} \left(p_k - \frac{m \Delta x_k}{\varepsilon} \right)^2 \right]. \quad (9.7)$$

Shift the integration variable $p_k \mapsto p_k + m\Delta x_k/\varepsilon$; the shift adds nothing to the integral. The remaining integral is the elementary Fresnel Gaussian:

$$\int_{-\infty}^{+\infty} \frac{dp_k}{2\pi\hbar} \exp\left[-\frac{i\varepsilon}{2m\hbar} p_k^2\right] = \frac{1}{2\pi\hbar} \sqrt{\frac{2\pi m\hbar}{i\varepsilon}} = \sqrt{\frac{m}{2\pi i\hbar\varepsilon}}. \quad (9.8)$$

The square root denotes the principal branch; the conditional convergence of the Fresnel integral is handled by the standard convergence factor $e^{-\eta p_k^2}$ with $\eta \rightarrow 0^+$ after integration, exactly as in the free-particle derivation of Chapter 5.²

Assembling the prefactor and the phase completes the derivation:

$$K(x_{k+1}, \varepsilon; x_k, 0) \approx \sqrt{\frac{m}{2\pi i\hbar\varepsilon}} \exp\left[\frac{i}{\hbar} \varepsilon \left(\frac{m}{2} \left(\frac{x_{k+1} - x_k}{\varepsilon}\right)^2 - V(x_k)\right)\right]. \quad (9.9)$$

This is the short-time propagator. The error inside the exponent is $O(\varepsilon^2)$, inherited from the Trotter factorization; the prefactor is exact for the free-particle part, the potential entering only through the phase factor $e^{-i\varepsilon V(x_k)/\hbar}$ evaluated at the initial point of the slice.

The phase identified. The exponent in (9.9) is ε times the classical Lagrangian of the particle, $L = \frac{m}{2}\dot{x}^2 - V(x)$, evaluated on the straight-line path from x_k to x_{k+1} traversed in time ε , with the velocity approximated by the constant value $(x_{k+1} - x_k)/\varepsilon$ and the potential evaluated at the initial point. For $V = 0$, (9.9) reduces to the free kernel (5.42) for the short interval ε , and the reduction is exact — the Trotter error vanishes because $[P^2/2m, 0] = 0$. For a general potential the approximation becomes exact only in the limit $\varepsilon \rightarrow 0$, where the sum of the Lagrangian over all slices approaches the classical action. That limit is the next section's construction.

The short-time kernel is in hand. The composition property (5.40) instructs us to multiply N copies of it, integrate over the $N - 1$ intermediate positions, and take the limit $N \rightarrow \infty$. The limit is the Feynman path integral.

9.2 The Feynman Construction: Time-Slicing, the Path Sum, and the Free Particle Verified

The short-time kernel (9.9) of the preceding section is the brick. The composition property (5.40) is the mortar. This section lays N bricks, takes the limit in which each brick becomes infinitesimal, and obtains the Feynman path integral. The construction is then verified on the free particle, whose exact kernel is already known from Chapter 5 — the verification is the proof that the time-slicing prescription reproduces the known answer where the answer is known, giving confidence in the prescription where it is not.

The N -slice product. Take the short-time kernel (9.9) for each of the N slices of duration $\varepsilon = t/N$. Insert it into the N -slice skeleton of Section 9.1: with $x_0 = x'$, $x_N = x$, and the $N - 1$ intermediate

²With the convergence factor, the integral is the elementary Gaussian $\int e^{-\alpha p^2} dp = \sqrt{\pi/\alpha}$ at $\alpha = \eta + i\varepsilon/2m\hbar$; the limit $\eta \rightarrow 0^+$ exists and yields (9.8).

positions $x_1, \dots, x_{N-1}$ integrated over the whole line,

$$K_N(x, t; x', 0) := \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{N/2} \int_{-\infty}^{+\infty} dx_1 \cdots dx_{N-1} \exp \left[\frac{i}{\hbar} \sum_{k=0}^{N-1} \varepsilon \left(\frac{m}{2} \left(\frac{x_{k+1} - x_k}{\varepsilon} \right)^2 - V(x_k) \right) \right]. \quad (9.10)$$

This is the N -slice discretized path integral. Every integral is an ordinary Lebesgue integral over $\mathbb{R}$; every factor is an ordinary complex number. The object K_N is an approximation to the true kernel, exact for the free particle at any N (as will be shown below) and exact for any potential only in the limit.

Definition (the Feynman path integral). The propagator for a particle in a potential V is *defined* by the limit

$$K(x, t; x', 0) := \lim_{N \rightarrow \infty} K_N(x, t; x', 0). \quad (9.11)$$

This limit is the definition of the Feynman path integral in this book. No analytic continuation, no measure theory beyond the Lebesgue integrals on $\mathbb{R}$ that appear in (9.10). The existence of the limit for the class of potentials V that are continuous and bounded below is stated without proof in this chapter; the rigorous treatment belongs to Appendix A.³

The classical action emerges. The exponent in (9.10) is a Riemann sum. In the limit $N \rightarrow \infty$, the piecewise-linear path defined by the vertices (t_k, x_k) with $t_k = k\varepsilon$ approximates a continuous path $x(t')$ joining $x(0) = x'$ to $x(t) = x$, and the sum approaches the integral of the Lagrangian:

$$\sum_{k=0}^{N-1} \varepsilon \left(\frac{m}{2} \left(\frac{x_{k+1} - x_k}{\varepsilon} \right)^2 - V(x_k) \right) \longrightarrow \int_0^t dt' \left[\frac{m}{2} \dot{x}(t')^2 - V(x(t')) \right] =: S[x(t)].$$

The limit of the Riemann sum is the **classical action** $S[x]$ evaluated on the path. The path integral therefore associates to every conceivable continuous path from $(x', 0)$ to (x, t) the amplitude $\exp(iS[x]/\hbar)$ and sums — integrates — over all such paths. The identification of the sum with the action also supplies the content of the label L in the compact form of the N -slice product:

$$K(x, t; x', 0) = \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{N/2} \int dx_1 \cdots dx_{N-1} \exp \left[\frac{i}{\hbar} \sum_{k=0}^{N-1} \varepsilon L \left(x_k, \frac{x_{k+1} - x_k}{\varepsilon} \right) \right]. \quad (9.12)$$

Symbolic form. The limit of the N -slice product is abbreviated by the notation

$$K(x, t; x', 0) = \int_{x(0)=x'}^{x(t)=x} \mathcal{D}[x(t)] e^{iS[x(t)]/\hbar}. \quad (9.13)$$



Note Note (the measure is a notation). The symbol $\int \mathcal{D}x$ is a formal shorthand for the limit $\lim_{N \rightarrow \infty} (m/2\pi i \hbar \varepsilon)^{N/2}$. It is not a measure on the space of paths in the sense of Lebesgue — Feynman's real-time integrand $e^{iS/\hbar}$ oscillates and does not define a countably additive measure. The rigorous path integral exists in imaginary time (the Wiener measure, $t \rightarrow -i\tau$, $e^{iS/\hbar} \rightarrow e^{-S_E/\hbar}$), and the real-time version is defined by analytic continuation. Appendix A and the field chapter address the Wiener-measure construction. Nothing in this chapter depends on these rigor facts beyond the discretized definition (9.10).

³The limit exists in the L^2 sense for a wide class of potentials and defines a unitary evolution. The proof, which requires operator-norm estimates of the Trotter remainder for unbounded operators, is the content of the Trotter–Kato theorem; the theorem is stated in Appendix A and cited here.

Physical interpretation. Equation (9.13) codifies the Feynman picture of quantum mechanics in one line. Every conceivable continuous path from the initial to the final event contributes a complex amplitude $\exp(iS[x]/\hbar)$, whose phase is the classical action along that path measured in units of $\hbar$. The total amplitude is the coherent sum over all paths. Paths far from the classical one — the path that makes the action stationary, $\delta S = 0$ — have phases that vary rapidly under small deformations; neighbouring paths interfere destructively and their net contribution cancels. Paths near the classical one have nearly stationary phase; they interfere constructively and their contribution survives. Classical mechanics, in this picture, is not a separate theory but the stationary-phase limit of quantum mechanics: when the action is large compared to $\hbar$, only the immediate neighbourhood of the classical path contributes, and Hamilton's principle $\delta S = 0$ emerges from the conspiracy of quantum phases. The detailed stationary-phase analysis that makes this qualitative picture quantitative is the subject of Section 9.3.

Free-particle verification: the $N = 2$ case. The construction must reproduce what is already known. For the free particle, $V = 0$, the N -slice product (9.10) is a chain of Gaussian integrals whose exact evaluation must yield the free kernel (5.42) of Chapter 5, independently of N . The case $N = 2$ is the explicit demonstration.

Set $\varepsilon = t/2$. The $N = 2$ product has a single intermediate integration over x_1 :

$$K_2(x, t; x', 0) = \frac{m}{2\pi i \hbar \varepsilon} \int_{-\infty}^{+\infty} dx_1 \exp\left[\frac{im}{2\hbar \varepsilon} ((x - x_1)^2 + (x_1 - x')^2)\right].$$

The exponent is quadratic in x_1 . Expanding the squares:

$$(x - x_1)^2 + (x_1 - x')^2 = 2x_1^2 - 2x_1(x + x') + x^2 + x'^2.$$

Complete the square in x_1 :

$$\begin{aligned} 2x_1^2 - 2x_1(x + x') &= 2\left[x_1^2 - x_1(x + x')\right] \\ &= 2\left[\left(x_1 - \frac{x + x'}{2}\right)^2 - \left(\frac{x + x'}{2}\right)^2\right] \\ &= 2\left(x_1 - \frac{x + x'}{2}\right)^2 - \frac{(x + x')^2}{2}. \end{aligned}$$

The constant terms assemble as

$$x^2 + x'^2 - \frac{(x + x')^2}{2} = \frac{x^2 + x'^2 - 2xx'}{2} = \frac{(x - x')^2}{2}.$$

The exponent separates into a term depending on x_1 and a term that does not:

$$\frac{im}{2\hbar \varepsilon} \left[2\left(x_1 - \frac{x + x'}{2}\right)^2 + \frac{(x - x')^2}{2} \right] = \frac{im}{\hbar \varepsilon} \left(x_1 - \frac{x + x'}{2}\right)^2 + \frac{im}{4\hbar \varepsilon} (x - x')^2.$$

The x_1 integral is a Fresnel Gaussian. Shifting $x_1 \mapsto x_1 + (x + x')/2$ and applying the elementary formula $\int_{-\infty}^{+\infty} dy e^{i\alpha y^2} = \sqrt{\pi i/\alpha}$ (with the convergence-factor honesty of Section 9.1):

$$\int_{-\infty}^{+\infty} dx_1 \exp\left[\frac{im}{\hbar \varepsilon} \left(x_1 - \frac{x + x'}{2}\right)^2\right] = \sqrt{\frac{\pi i \hbar \varepsilon}{m}}.$$

Assembling the prefactor and the phase:

$$\begin{aligned} K_2(x, t; x', 0) &= \frac{m}{2\pi i \hbar \varepsilon} \sqrt{\frac{\pi i \hbar \varepsilon}{m}} \exp\left[\frac{im}{4\hbar \varepsilon} (x - x')^2\right] \\ &= \sqrt{\frac{m}{2\pi i \hbar (2\varepsilon)}} \exp\left[\frac{im}{2\hbar (2\varepsilon)} (x - x')^2\right]. \end{aligned} \quad (9.14)$$

The last line uses the algebraic contraction of the prefactor:

$$\frac{m}{2\pi i \hbar \varepsilon} \cdot \sqrt{\frac{\pi i \hbar \varepsilon}{m}} = \sqrt{\frac{m^2}{4\pi^2 i^2 \hbar^2 \varepsilon^2}} \cdot \frac{\pi i \hbar \varepsilon}{m} = \sqrt{\frac{m}{2\pi i \hbar (2\varepsilon)}}.$$

With $\varepsilon = t/2$, the result is $K_2(x, t; x', 0) = \sqrt{m/2\pi i \hbar t} \exp[im(x - x')^2/2\hbar t]$, which is precisely the free kernel (5.42). The prefactor has contracted from the single-slice factor $m/(2\pi i \hbar \varepsilon)$ to the full-interval factor $\sqrt{m/2\pi i \hbar t}$.

Induction to arbitrary N . The $N = 2$ case exhibits the pattern: each Gaussian integration over an intermediate point contracts the prefactor by one factor of the elementary integral $\sqrt{\pi i \hbar \varepsilon/m}$, and the phase assembles into the classical action of the free particle evaluated on the piecewise-linear path. For general N , with $N - 1$ intermediate integrations, the $(N/2)$ -power prefactor in (9.10) contracts through $N - 1$ Gaussian integrals to $\sqrt{m/2\pi i \hbar (N\varepsilon)} = \sqrt{m/2\pi i \hbar t}$, while the sum of quadratic forms in the exponent telescopes to $m(x - x')^2/2t$. The result is independent of N :

$$K_0^{(N)}(x, t; x', 0) = \sqrt{\frac{m}{2\pi i \hbar t}} \exp\left[\frac{im(x - x')^2}{2\hbar t}\right] \quad \text{for every } N \geq 1. \quad (9.15)$$

The complete proof by induction on N is the content of Exercise 1. The physical content of (9.15) is that the free-particle path integral reproduces the exact kernel of Chapter 5, and it does so at every finite N , not merely in the limit — the free-particle short-time kernel is exact, because the Trotter error vanishes when the two terms of the Hamiltonian commute. The limit $N \rightarrow \infty$ is therefore trivial for the free particle: every $K_0^{(N)}$ already equals the answer. For a non-vanishing potential, only the limit is exact, and the N -dependence of K_N encodes the approach of the discrete approximation to the true dynamics.

The phase-space path integral. The derivation of the short-time kernel in Section 9.1 performed the momentum integration explicitly, converting the Hamiltonian form of the short-time amplitude into the Lagrangian form (9.9). If the momentum integrals are left unperformed, the N -slice product takes the alternative form

$$\int \frac{dp_0}{2\pi \hbar} \prod_{k=1}^{N-1} \frac{dx_k dp_k}{2\pi \hbar} \exp\left[\frac{i}{\hbar} \sum_{k=0}^{N-1} \left(p_k(x_{k+1} - x_k) - \varepsilon \frac{p_k^2}{2m} - \varepsilon V(x_k)\right)\right]. \quad (9.16)$$

In the continuum limit, with p_k becoming the momentum $p(t')$ at the discretised time t_k , the sum in the exponent approaches the phase-space action integral, and the symbolic form is

$$K(x, t; x', 0) = \int \mathcal{D}x \mathcal{D}p \exp\left[\frac{i}{\hbar} \int_0^t dt' (p \dot{x} - H(p, x))\right]. \quad (9.17)$$

This is the Hamiltonian, or phase-space, path integral — the more general form, of which the Lagrangian path integral (9.13) is the special case obtained by performing the Gaussian momentum integrals. The phase-space form is the natural starting point when the kinetic term is not quadratic in the

momentum (as for a relativistic particle, named here once and not developed) or when the Hamiltonian contains a vector-potential coupling, as in the Aharonov-Bohm system of Section 9.5. The measure $\int \mathcal{D}x \mathcal{D}p$ carries the same formal status as $\int \mathcal{D}x$: it is defined solely by the discretized limit, with one $dp_k/2\pi\hbar$ factor per time slice accompanying each dx_k integral.

Advantage and limitation. The advantage of the path-integral formulation is its direct contact with the classical action principle: the quantum amplitude is built from the classical action, and the classical limit emerges transparently as the stationary-phase approximation rather than as a separate postulate. The formulation also makes manifest the role of the configuration space's topology — path interference can register global properties, as the Aharonov-Bohm effect of Section 9.5 will demonstrate. Its limitation is calculational: outside the class of quadratic actions, the path integral resists closed-form evaluation, and the stationary-phase approximation plus the systematic $\hbar$ -expansion (the WKB method of Section 9.6) is the practical bridge to quantitative results. The canonical operator formalism of Chapters 4–5 and the path-integral formalism of this chapter are two representations of one dynamics; the choice between them is a choice of convenience, tuned to the problem.

The classical action has appeared as the phase whose stationary points dominate the path sum. The systematic extraction of those stationary points — the classical limit and its first quantum correction — is the next section.

9.3 The Classical Limit: Stationary Phase, the Role of Action, and Hamilton-Jacobi Repaid

The path integral (9.13)—with the measure understood as the formal notation defined by the discretized limit (9.10)—sums every conceivable path with the amplitude $e^{iS[x]/\hbar}$. When the action S is large compared to $\hbar$, the phase oscillates rapidly from one path to the next, and contributions from neighbouring paths cancel. The only paths that survive are those for which the phase is stationary under small variations: $\delta S = 0$. These are the classical paths, and the stationary-phase approximation turns the path integral into Hamilton's principle. This section develops the stationary-phase method for finite-dimensional integrals, applies it to the functional integral, recovers the Hamilton-Jacobi equation as the $\hbar \rightarrow 0$ limit of the Schrodinger propagator, and verifies the construction on the de Broglie wave packet.

Stationary phase in one dimension. Let $f(x)$ and $g(x)$ be smooth real-valued functions, and consider the integral

$$I(\hbar) = \int_{-\infty}^{\infty} dx g(x) e^{if(x)/\hbar}, \quad (9.18)$$

in the limit $\hbar \rightarrow 0$. The exponential oscillates with frequency $f'(x)/\hbar$; as $\hbar$ decreases, the oscillations become arbitrarily rapid except near points x_c where $f'(x_c) = 0$, the stationary-phase points. Near

such a point, expand $f(x)$ to second order and $g(x)$ to zeroth order:

$$f(x) = f(x_c) + \frac{1}{2}f''(x_c)(x - x_c)^2 + O((x - x_c)^3), \quad (9.19)$$

$$g(x) = g(x_c) + O(x - x_c). \quad (9.20)$$

Insert the expansions and extend the integration limits to $\pm\infty$ (the error from this extension is of higher order in $\hbar$ because the integrand oscillates rapidly away from x_c):

$$\begin{aligned} I(\hbar) &\approx g(x_c) e^{if(x_c)/\hbar} \int_{-\infty}^{\infty} dx \exp\left[\frac{i}{2\hbar}f''(x_c)(x - x_c)^2\right] \\ &= g(x_c) e^{if(x_c)/\hbar} \sqrt{\frac{2\pi i\hbar}{f''(x_c)}}. \end{aligned} \quad (9.21)$$

The last equality uses the Fresnel Gaussian integral $\int_{-\infty}^{\infty} d\xi e^{ia\xi^2/2} = \sqrt{2\pi i/a}$, valid for $a = f''(x_c)/\hbar \neq 0$ with the principal branch of the square root. If $f(x)$ has multiple stationary points, the integral receives a contribution of the form (9.21) from each, and the total is their sum. For an M -dimensional integral over $\mathbf{x} \in \mathbb{R}^M$, the formula generalizes to

$$\int d^M x g(\mathbf{x}) e^{if(\mathbf{x})/\hbar} \approx g(\mathbf{x}_c) e^{if(\mathbf{x}_c)/\hbar} \sqrt{\frac{(2\pi i\hbar)^M}{\det H(\mathbf{x}_c)}}, \quad (9.22)$$

where $H_{ij}(\mathbf{x}_c) := \partial_i \partial_j f(\mathbf{x}_c)$ is the Hessian matrix at the stationary point, and the square root of the determinant carries the principal branch.



Note The stationary-phase formula (9.21) is the leading term of an asymptotic expansion in powers of $\hbar$. The next term is of order $\hbar$ relative to the leading term. The expansion is asymptotic, not convergent, in typical applications. The error in truncating at leading order is $O(\hbar)$ provided $f''(x_c) \neq 0$; when $f''(x_c) = 0$, the expansion must be carried to higher order, and the scaling with $\hbar$ changes.

Functional stationary phase. The path integral is an integral over the infinite-dimensional space of paths. The role of $f(x)$ is played by the classical action functional $S[x(t)]$, and the stationary-phase condition $\delta S = 0$ with fixed endpoints $x(0) = x'$, $x(t) = x$ selects the classical path $x_c(t')$ satisfying the Euler–Lagrange equation

$$\frac{d}{dt'} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0, \quad L = \frac{m}{2}\dot{x}^2 - V(x). \quad (9.23)$$

Expand an arbitrary path about the classical path:

$$x(t') = x_c(t') + y(t'), \quad y(0) = y(t) = 0. \quad (9.24)$$

The action functional is expanded in a functional Taylor series:

$$S[x_c + y] = S[x_c] + \int_0^t dt' \frac{\delta S}{\delta x(t')} \Big|_{x_c} y(t') + \frac{1}{2} \int_0^t dt' \int_0^t dt'' y(t') \frac{\delta^2 S}{\delta x(t') \delta x(t'')} \Big|_{x_c} y(t'') + \cdots \quad (9.25)$$

The linear term vanishes because x_c satisfies $\delta S = 0$. The second (quadratic) term determines the leading quantum corrections. For the Lagrangian $L = \frac{m}{2}\dot{x}^2 - V(x)$, the second functional derivative is computed by varying the Euler–Lagrange equation. Write the action as $S = \int dt' [\frac{m}{2}\dot{x}^2 - V(x)]$. Varying once gives $\delta S = \int dt' [-m\ddot{x} - V'(x)]\delta x(t')$, with the boundary terms vanishing for fixed-

9347 endpoint variations. Varying a second time yields the Jacobi operator:

$$9348 \quad \left. \frac{\delta^2 S}{\delta x(t') \delta x(t'')} \right|_{x_c} = \left[-m \frac{d^2}{dt'^2} - V''(x_c(t')) \right] \delta(t' - t''). \quad (9.26)$$

9349 The path integral therefore factorizes, to quadratic order, into the classical contribution times a Gaussian functional integral over the fluctuation $y(t')$:

$$9351 \quad K(x, t; x', 0) \approx e^{iS[x_c]/\hbar} \int_{y(0)=0}^{y(t)=0} \mathcal{D}y \exp \left[\frac{i}{2\hbar} \int_0^t dt' dt'' y(t') \left. \frac{\delta^2 S}{\delta x(t') \delta x(t'')} \right|_{x_c} y(t'') \right]. \quad (9.27)$$

9352 The measure $\mathcal{D}y$ is the formal notation §9.2 defines; the Gaussian functional integral is defined by the discretized limit — the N -slice product with the quadratic form evaluated on the discrete path.

9354 **The Van Vleck propagator.** The Gaussian functional integral over y evaluates to $[\det(\delta^2 S/\delta x \delta x)]^{-1/2}$,
 9355 where the determinant is the functional determinant of the Jacobi operator (9.26) on the interval $[0, t]$
 9356 with Dirichlet boundary conditions $y(0) = y(t) = 0$, normalized by the free-particle determinant.
 9357 The result, after evaluating the prefactor, is the semiclassical Van Vleck propagator:

$$9358 \quad K(x, t; x', 0) \approx \sqrt{\frac{i}{2\pi\hbar} \frac{\partial^2 S_c}{\partial x \partial x'}} \exp \left[\frac{i}{\hbar} S_c(x, t; x', 0) \right]. \quad (9.28)$$

9359 Here $S_c(x, t; x', 0)$ is the classical action evaluated on the unique classical path connecting $(x', 0)$ to
 9360 (x, t) , and the prefactor contains the Van Vleck determinant $\partial^2 S_c/\partial x \partial x'$. When multiple classical
 9361 paths connect the endpoints, the propagator is the sum of terms of the form (9.28), one per classical
 9362 path, each with an additional Maslov phase correction at caustics (points where $\partial^2 S_c/\partial x \partial x' = 0$ and
 9363 the simple prefactor diverges).

9364  **Note** The full derivation of the prefactor (9.28) from the functional determinant of the Jacobi operator
 9365 is not given here in its most general form. The transformation from the functional determinant to the
 9366 Van Vleck determinant $\partial^2 S_c/\partial x \partial x'$ relies on the Jacobi field equation and the properties of the action
 9367 as a generating function. The free-particle and harmonic-oscillator cases, computed explicitly in §9.2
 9368 and §9.4, verify the formula. The general derivation can be found in Schulman (1981) and Kleinert
 9369 (2009).

9370 **Recovery of classical mechanics.** In the limit $\hbar \rightarrow 0$, or operationally when $S_c \gg \hbar$, the path
 9371 integral is dominated by the single classical path satisfying $\delta S = 0$. All other paths contribute phases
 9372 that oscillate rapidly and cancel. A particle prepared at x' at time zero and detected at x at time t
 9373 has its transition amplitude controlled by the classical trajectory connecting the two events. Classical
 9374 mechanics is not an independent postulate layered on top of quantum mechanics; it is the stationary-
 9375 phase limit of the quantum superposition principle. Hamilton's principle $\delta S = 0$ emerges from the
 9376 requirement that the amplitude $e^{iS/\hbar}$ not be cancelled by destructive interference from neighbouring
 9377 paths.

9378 **Hamilton–Jacobi repaid.** The classical action $S_c(x, t; x', 0)$, regarded as a function of the final po-
 9379 sition x and the elapsed time t , satisfies a first-order partial differential equation. Differentiating the
 9380 action with respect to the final endpoint: the definition $S_c = \int_0^t L(x_c(t'), \dot{x}_c(t')) dt'$ where x_c solves
 9381 the equations of motion with fixed $x(0) = x'$ and $x(t) = x$. Varying the final point $x \rightarrow x + \delta x$
 9382 shifts the entire classical path; the variation of the action with respect to the endpoint is the canonical

momentum at that endpoint:

$$\frac{\partial S_c}{\partial x} = p(t) = m\dot{x}_c(t). \quad (9.29)$$

Similarly, the variation with respect to the final time t is related to the Hamiltonian. For a time-independent Lagrangian, the total time derivative of the action along a path that solves the equations of motion is

$$\frac{dS_c}{dt} = L(x_c(t), \dot{x}_c(t)) = p\dot{x} - H(x, p), \quad (9.30)$$

where the second equality is the Legendre transform defining the Hamiltonian. Treating S_c as a function of the final position x and time t , with the initial point fixed, the chain rule gives

$$\frac{dS_c}{dt} = \frac{\partial S_c}{\partial t} + \frac{\partial S_c}{\partial x} \dot{x} = \frac{\partial S_c}{\partial t} + p\dot{x}. \quad (9.31)$$

Equating the two expressions for dS_c/dt and eliminating $p\dot{x}$ yields the Hamilton–Jacobi equation:

$$\frac{\partial S_c}{\partial t} + H\left(x, \frac{\partial S_c}{\partial x}\right) = 0, \quad H(x, p) = \frac{p^2}{2m} + V(x). \quad (9.32)$$

This is the repayment of ch1’s action-principle debt. In the $\hbar \rightarrow 0$ limit, the Schrodinger equation for a wave function of the WKB form $\psi(x, t) \sim e^{iS(x, t)/\hbar}$ reduces, at leading order in $\hbar$, to (9.32). The Schrodinger equation is the quantum deformation of the Hamilton–Jacobi equation, with the term $-(i\hbar/2m)\partial^2 S/\partial x^2$ supplying the $O(\hbar)$ quantum correction. The full WKB expansion is the subject of Section 9.6.

De Broglie packet: classical motion and spreading. The stationary-phase method reproduces the dynamics of a quantum wave packet directly. The time-evolved wave function is

$$\psi(x, t) = \int dx' K_0(x, t; x', 0) \psi(x', 0), \quad (9.33)$$

where K_0 is the free-particle propagator (5.42). Writing $K_0 = \sqrt{m/2\pi i\hbar t} e^{iS_0/\hbar}$ with $S_0(x, t; x', 0) = m(x - x')^2/2t$, the integral is of stationary-phase type in the variable x' . The total phase is

$$\Phi(x') = S_0(x, t; x', 0) + \frac{\hbar}{i} \ln \psi(x', 0). \quad (9.34)$$

The stationary-phase condition $\partial\Phi/\partial x' = 0$ gives

$$\frac{\partial S_0}{\partial x'} + \frac{\hbar}{i} \frac{\psi'(x', 0)}{\psi(x', 0)} = 0. \quad (9.35)$$

For a Gaussian initial wave packet of the form used in Section 5.9 (ch5, (5.43)),

$$\psi(x', 0) = \frac{1}{(2\pi\sigma^2)^{1/4}} \exp\left[-\frac{(x' - x_0)^2}{4\sigma^2} + \frac{i}{\hbar} p_0(x' - x_0)\right], \quad (9.36)$$

the logarithmic derivative is $\psi'/\psi = -(x' - x_0)/2\sigma^2 + ip_0/\hbar$. The derivative of the free action is $\partial S_0/\partial x' = -m(x - x')/t$. The stationary-phase condition becomes

$$-\frac{m}{t}(x - x') - \frac{i\hbar}{2\sigma^2}(x' - x_0) + p_0 = 0. \quad (9.37)$$

At leading order in $\hbar$, the imaginary term is $O(\hbar)$ relative to the real terms and the stationary point is $x' = x - p_0 t/m$, the classical trajectory of a free particle with momentum p_0 . Including the $O(\hbar)$ term shifts the stationary point into the complex plane by an amount $O(\hbar)$, which contributes to the spreading. The quadratic fluctuation integral around the stationary point produces the Gaussian factor

whose width evolves as

$$\Delta x(t)^2 = \sigma^2 + \left(\frac{\Delta p}{m}\right)^2 t^2, \quad \Delta p = \frac{\hbar}{2\sigma}, \quad (9.38)$$

reproducing the exact spreading law (5.43) derived in ch5 from the momentum-space representation. The stationary-phase approximation is exact for the free-particle Gaussian packet because the action is quadratic and the initial wave function is Gaussian: every term beyond quadratic order in the expansion of the phase vanishes identically.

Example: linear potential. For a uniform gravitational field $V(x) = mgx$, the classical equation of motion is $m\ddot{x} = -mg$, whose solution with boundary conditions $x(0) = x'$, $\dot{x}(t) = x$ is

$$x_c(t') = x' + \left(\frac{x - x'}{t} + \frac{1}{2}gt\right)t' - \frac{1}{2}gt'^2. \quad (9.39)$$

The velocity is $\dot{x}_c(t') = (x - x')/t + g(t/2 - t')$. The classical action is evaluated by inserting x_c into $L = \frac{m}{2}\dot{x}^2 - mgx$ and integrating from 0 to t . The kinetic term gives

$$\int_0^t dt' \frac{m}{2}\dot{x}_c^2 = \frac{m}{2} \int_0^t dt' \left[\frac{x - x'}{t} + g\left(\frac{t}{2} - t'\right) \right]^2 = \frac{m(x - x')^2}{2t} + \frac{mg^2t^3}{24}. \quad (9.40)$$

The potential term gives

$$-\int_0^t dt' mg x_c(t') = -mg \int_0^t dt' \left[x' + \left(\frac{x - x'}{t} + \frac{gt}{2}\right)t' - \frac{g}{2}t'^2 \right] = -\frac{mgt}{2}(x + x') - \frac{mg^2t^3}{12}. \quad (9.41)$$

The two cubic terms in g^2t^3 combine: $1/24 - 1/12 = -1/24$. The full classical action is

$$S_c(x, t; x', 0) = \frac{m}{2t}(x - x')^2 - \frac{mgt}{2}(x + x') - \frac{mg^2t^3}{24}. \quad (9.42)$$

The three terms have clear physical meanings: the first is the free-particle kinetic action, the second is the average potential energy times the duration, and the third is the action accumulated from the parabolic deviation from uniform motion. The Jacobi operator (9.26) has $V'' = 0$, so the fluctuation integral is identical to the free-particle case. The Van Vleck prefactor is exactly $\sqrt{m/2\pi i\hbar t}$, which is also the free-particle prefactor. The stationary-phase evaluation is therefore exact: for a linear potential, the action is quadratic in the path, and the semiclassical propagator (9.28) with S_c from (9.42) and the free-particle prefactor gives the exact quantum-mechanical propagator. This is the second instance, after the free particle itself, where the semiclassical approximation is exact; the harmonic oscillator, the subject of Section 9.4, is the paradigm.

Note *The exactness of the stationary-phase approximation for any Lagrangian at most quadratic in x and $\dot{x}$ is a general theorem. When $L = a(t)\dot{x}^2 + b(t)x\dot{x} + c(t)x^2 + d(t)\dot{x} + e(t)x + f(t)$, the action is a quadratic functional, the functional integral is a Gaussian integral, and the stationary-phase (saddle-point) evaluation of a Gaussian integral is exact. The free particle, the linear potential, and the harmonic oscillator exhaust the elementary examples in one dimension; the forced harmonic oscillator adds a linear driving term and remains quadratic (and therefore exactly solvable).*

Advantage and limitation. The advantage of the stationary-phase method is conceptual: it exhibits classical mechanics as the $\hbar \rightarrow 0$ limit of quantum mechanics, not as a logically prior theory. Hamilton's principle, the Euler–Lagrange equations, and the Hamilton–Jacobi equation are all recovered from the quantum propagator by the systematic application of the stationary-phase condition. The

method also provides the leading semiclassical approximation — the Van Vleck propagator (9.28) — for any potential, with the classical action as the phase and the Van Vleck determinant as the prefactor encoding the focusing of classical trajectories. Its limitation is that the simple stationary-phase formula breaks down at caustics (where $\partial^2 S_c / \partial x \partial x' = 0$), near potential barriers where no real classical path connects the endpoints, and in regions where multiple classical paths contribute with comparable weight. The full machinery — Maslov indices, complex classical paths, and uniform asymptotic expansions — is needed to repair these defects. For quadratic actions, however, the stationary-phase approximation is exact, and the harmonic oscillator is the central example to which we now turn.

9.4 The Harmonic Oscillator: Exact Evaluation by Gaussian Integral

The stationary-phase approximation is exact when the action is a quadratic functional of the path. The harmonic oscillator is the paradigm: its Lagrangian is quadratic in x and $\dot{x}$, its classical equation of motion is linear, and every term beyond quadratic order in the functional Taylor expansion of the action vanishes identically. The path integral reduces to the classical action times a Gaussian fluctuation integral whose determinant can be evaluated in closed form. The result is the Mehler kernel — the exact Feynman propagator for the harmonic oscillator — from which the energy spectrum $E_n = \hbar\omega(n + 1/2)$ is recovered independently of the operator method of ch5. The Maslov index, the phase accumulated at each half-period when classical paths reconverge, is computed directly from the prefactor's branch structure.

Lagrangian and classical action. The Lagrangian for a one-dimensional harmonic oscillator of mass m and frequency ω is

$$L = \frac{m}{2} \dot{x}^2 - \frac{m\omega^2}{2} x^2. \quad (9.43)$$

The Euler–Lagrange equation is $\ddot{x} + \omega^2 x = 0$, a linear second-order equation. The general solution is a superposition of $\sin \omega t'$ and $\cos \omega t'$. Imposing the boundary conditions $x(0) = x'$ and $x(t) = x$ fixes the two constants. Write $x_c(t') = A \sin \omega t' + B \cos \omega t'$. Then $x_c(0) = B = x'$, and $x_c(t) = A \sin \omega t + x' \cos \omega t = x$, giving $A = (x - x' \cos \omega t) / \sin \omega t$. The classical path is therefore

$$x_c(t') = x' \frac{\sin \omega(t - t')}{\sin \omega t} + x \frac{\sin \omega t'}{\sin \omega t}. \quad (9.44)$$

The form is symmetric in the initial and final positions, as required by time-reversal invariance of the Lagrangian.

The classical action is evaluated on this path. A direct integration of the Lagrangian is possible but a shortcut simplifies the computation. For the harmonic oscillator, the Lagrangian can be written as a total time derivative on any solution of the equation of motion:

$$L = \frac{m}{2} (\dot{x}_c^2 - \omega^2 x_c^2) = \frac{m}{2} \frac{d}{dt'} (x_c \dot{x}_c), \quad (9.45)$$

because $d(x_c \dot{x}_c) / dt' = \dot{x}_c^2 + x_c \ddot{x}_c = \dot{x}_c^2 - \omega^2 x_c^2$, using the equation of motion $\ddot{x}_c = -\omega^2 x_c$. The action

reduces to a boundary term:

$$S_c = \int_0^t dt' L = \frac{m}{2} [x_c(t)\dot{x}_c(t) - x_c(0)\dot{x}_c(0)]. \quad (9.46)$$

The velocities at the endpoints are obtained from (9.44):

$$\dot{x}_c(0) = \frac{\omega}{\sin \omega t} (x - x' \cos \omega t), \quad (9.47)$$

$$\dot{x}_c(t) = \frac{\omega}{\sin \omega t} (x \cos \omega t - x'). \quad (9.48)$$

Inserting $x_c(0) = x'$, $x_c(t) = x$, and the velocities into the boundary expression:

$$\begin{aligned} S_c &= \frac{m}{2} \left[x \cdot \frac{\omega}{\sin \omega t} (x \cos \omega t - x') - x' \cdot \frac{\omega}{\sin \omega t} (x - x' \cos \omega t) \right] \\ &= \frac{m\omega}{2 \sin \omega t} [x^2 \cos \omega t - xx' - x'x + x'^2 \cos \omega t] \\ &= \frac{m\omega}{2 \sin \omega t} [(x^2 + x'^2) \cos \omega t - 2xx']. \end{aligned} \quad (9.49)$$

This is the classical action of the harmonic oscillator. The $\omega \rightarrow 0$ limit provides a consistency check.

Expanding $\sin \omega t \approx \omega t - \omega^3 t^3/6$ and $\cos \omega t \approx 1 - \omega^2 t^2/2$, the leading behaviour of (9.49) is

$$S_c \xrightarrow{\omega \rightarrow 0} \frac{m}{2t} (x - x')^2, \quad (9.50)$$

which is the free-particle action appearing in (5.42). The ω -dependent corrections begin at $O(\omega^2)$ and encode the potential's influence on the classical motion.

Fluctuation integral. Every path satisfying the boundary conditions can be written as the classical path plus a fluctuation vanishing at the endpoints:

$$x(t') = x_c(t') + y(t'), \quad y(0) = y(t) = 0. \quad (9.51)$$

The action evaluated on $x_c + y$ splits into three groups of terms:

$$\begin{aligned} S[x_c + y] &= \int_0^t dt' \left[\frac{m}{2} (\dot{x}_c + \dot{y})^2 - \frac{m\omega^2}{2} (x_c + y)^2 \right] \\ &= S_c + \int_0^t dt' m (\dot{x}_c \dot{y} - \omega^2 x_c y) + \int_0^t dt' \frac{m}{2} (\dot{y}^2 - \omega^2 y^2). \end{aligned} \quad (9.52)$$

The first term is the classical action. The cross term (linear in y) is shown to vanish by an integration by parts. The $\dot{x}_c \dot{y}$ term is integrated:

$$\int_0^t dt' \dot{x}_c \dot{y} = [\dot{x}_c y]_0^t - \int_0^t dt' \ddot{x}_c y = - \int_0^t dt' \ddot{x}_c y, \quad (9.53)$$

because $y(0) = y(t) = 0$ eliminates the boundary term. The cross term becomes

$$m \int_0^t dt' (\dot{x}_c \dot{y} - \omega^2 x_c y) = -m \int_0^t dt' (\ddot{x}_c + \omega^2 x_c) y = 0, \quad (9.54)$$

vanishing by the classical equation of motion $\ddot{x}_c + \omega^2 x_c = 0$. The action therefore splits exactly:

$$S[x] = S_c + \frac{m}{2} \int_0^t dt' (\dot{y}^2 - \omega^2 y^2). \quad (9.55)$$

No approximation has been made; the exactness follows from the quadratic nature of the Lagrangian.

The second term is the fluctuation action. Integrating $\dot{y}^2$ by parts (the boundary term again vanishes because $y(0) = y(t) = 0$) gives the Jacobi-operator form:

$$\frac{m}{2} \int_0^t dt' y(t') \left[-\frac{d^2}{dt'^2} - \omega^2 \right] y(t'). \quad (9.56)$$

The operator $-\partial_t^2 - \omega^2$ on the interval $[0, t]$ with Dirichlet boundary conditions is the Jacobi operator (9.26) specialized to $V''(x_c) = m\omega^2$, which is constant because the potential is quadratic.

The path integral therefore factorizes exactly:

$$K(x, t; x', 0) = e^{iS_c/\hbar} \int_{y(0)=0}^{y(t)=0} \mathcal{D}y \exp \left[\frac{im}{2\hbar} \int_0^t dt' y(t') \left(-\frac{d^2}{dt'^2} - \omega^2 \right) y(t') \right]. \quad (9.57)$$

The measure $\mathcal{D}y$ carries the same formal meaning as in Section 9.2: it is the $N \rightarrow \infty$ limit of the discretized product measure. The fluctuation integral is a Gaussian functional integral whose value is the reciprocal square root of the functional determinant of the Jacobi operator, normalized by the free-particle ($\omega = 0$) determinant.

 **Note** The functional determinant $\det(-\partial_t^2 - \omega^2)$ on $[0, t]$ with Dirichlet boundary conditions is evaluated by the Gelfand-Yaglom theorem, which states that the ratio of this determinant to the free-particle determinant $\det(-\partial_t^2)$ is given by the value at t of the solution $f(t')$ to the homogeneous Jacobi equation $(-\partial_t^2 - \omega^2)f = 0$ with initial conditions $f(0) = 0$, $f'(0) = 1$, divided by the free-particle solution $f_0(t') = t'$:

$$\frac{\det(-\partial_t^2 - \omega^2)}{\det(-\partial_t^2)} = \frac{f(t)}{f_0(t)} = \frac{\sin \omega t}{\omega t}. \quad (9.58)$$

The theorem was proved by Gelfand and Yaglom (1960); the proof is deferred to Appendix A. The free-particle normalization is fixed by the known free propagator (5.42), which was verified from the path-integral construction in Section 9.2. An alternative route — discretizing the fluctuation integral on N time slices, evaluating the tridiagonal determinant in closed form, and taking the $N \rightarrow \infty$ limit — is the content of ch9, Exercise 5.

Assembling the factors: the free-particle normalization contributes $\sqrt{m/(2\pi i \hbar t)}$. Multiplying by the reciprocal square root of the Gelfand-Yaglom ratio gives the fluctuation prefactor:

$$\sqrt{\frac{m}{2\pi i \hbar t}} \sqrt{\frac{\omega t}{\sin \omega t}} = \sqrt{\frac{m\omega}{2\pi i \hbar \sin \omega t}}. \quad (9.59)$$

The Mehler kernel. Combining the classical action (9.49) with the fluctuation prefactor yields the exact Feynman propagator for the harmonic oscillator:

$$K(x, t; x', 0) = \sqrt{\frac{m\omega}{2\pi i \hbar \sin \omega t}} \exp \left[\frac{im\omega}{2\hbar \sin \omega t} \left((x^2 + x'^2) \cos \omega t - 2xx' \right) \right]. \quad (9.60)$$

This is the Mehler kernel, named after F. G. Mehler who first derived the associated bilinear generating function for Hermite polynomials in 1866. The form is exact for all t and all x, x' . As a consistency check, the $\omega \rightarrow 0$ limit recovers the free propagator (5.42): the prefactor becomes $\sqrt{m/(2\pi i \hbar t)}$, and the exponent reduces to $im(x - x')^2/(2\hbar t)$.

Maslov index and caustics. The prefactor $\sqrt{1/\sin \omega t}$ has square-root branch points at the zeros of $\sin \omega t$, which occur at $t_k = k\pi/\omega$ for integer k . These are the times at which all classical paths emanating from x' reconverge — the focal points, or caustics, of the classical flow. To determine the phase of the prefactor across each caustic, the square root must be analytically continued. The Feynman propagator is defined with the causal $m - i\epsilon$ prescription, equivalent to replacing $\sin \omega t \rightarrow \sin \omega t - i\epsilon$. With this prescription, the argument of $\sin \omega t$ acquires a small negative imaginary part; as t increases through each simple zero, the phase of $\sin \omega t$ jumps by $-\pi$, and the square root acquires a factor of

9552 $e^{-i\pi/2}$ per half-period. Explicitly, at $t = k\pi/\omega$ ($k = 1, 2, 3, \dots$):

$$9553 \quad \sqrt{\frac{1}{\sin \omega t}} \longrightarrow e^{-ik\pi/2} \sqrt{\frac{1}{|\sin \omega t|}}. \quad (9.61)$$

9554 The phase $-\pi/2$ per half-period is the Morse–Maslov index of the harmonic oscillator. Each half-
 9555 period, the Jacobi operator acquires one additional negative eigenvalue (a conjugate point along the
 9556 classical path), and the Maslov index counts these sign changes in the quadratic form of the fluctua-
 9557 tion action. The result is a topological phase that the simple stationary-phase formula (9.28) would
 9558 miss: the Van Vleck prefactor diverges at caustics because $\partial^2 S_c / \partial x \partial x' = -m\omega / (\sin^2 \omega t) \rightarrow \infty$ as
 9559 $\sin \omega t \rightarrow 0$, and the Maslov correction supplies the finite, physically correct phase.

9560 Two particular cases are instructive. At the half-period $t = \pi/\omega$, $\sin \omega t = 0$ and $\cos \omega t = -1$.
 9561 Taking the limit carefully (the exponent becomes singular; the kernel reduces to a delta function), the
 9562 Mehler kernel gives

$$9563 \quad K(x, \pi/\omega; x', 0) = e^{-i\pi/2} \delta(x + x'), \quad (9.62)$$

9564 up to a normalization constant. The state is spatially inverted: every position eigenstate $|x'\rangle$ evolves
 9565 in half a period to the state at the opposite point $-x'$, multiplied by the Maslov phase $e^{-i\pi/2}$. At the
 9566 full period $t = 2\pi/\omega$, $\sin \omega t = 0$ and $\cos \omega t = 1$, producing

$$9567 \quad K(x, 2\pi/\omega; x', 0) = e^{-i\pi} \delta(x - x') = -\delta(x - x'). \quad (9.63)$$

9568 The state returns to itself after one classical period but with a minus sign. After two periods ($t =$
 9569 $4\pi/\omega$), the accumulated phase is $e^{-2\pi i} = 1$, and the state returns to $+1$ times itself. This is the har-
 9570 monic oscillator's counterpart of the spinor sign (4.10): the 4π periodicity of the quantum evolution,
 9571 versus the 2π periodicity of the classical motion, is the oscillator's version of the double cover of the
 9572 rotation group (ch6, Section 6.1).

9573  **Note** *The classical harmonic oscillator is a perfect lens in phase space: every point (q, p) rotates*
 9574 *at frequency ω around the origin in phase space. After one half-period, every point is mapped to its*
 9575 *antipode $(q, p) \rightarrow (-q, -p)$. After one full period, every point returns to itself. The quantum kernel*
 9576 *reproduces this mapping, with the Maslov phase $e^{-i\pi/2}$ at each half-period playing the role of the*
 9577 *Guoy phase accumulated by a focused wave.*

9578 **Energy spectrum from spectral expansion.** The propagator can be expanded in the basis of energy
 9579 eigenstates. From the evolution axiom (4.5), the kernel has the spectral resolution

$$9580 \quad K(x, t; x', 0) = \sum_{n=0}^{\infty} \psi_n(x) \psi_n^*(x') e^{-iE_n t/\hbar}, \quad (9.64)$$

9581 where $\psi_n(x)$ are the normalized energy eigenfunctions and E_n are the energy eigenvalues. The Mehler
 9582 kernel (9.60) must therefore be expandable as a series in $e^{-i\omega t}$. Extract the time dependence by writing
 9583 the trigonometric functions as exponentials: $\sin \omega t = (e^{i\omega t} - e^{-i\omega t})/(2i)$ and $\cos \omega t = (e^{i\omega t} + e^{-i\omega t})/2$.
 9584 Define the dimensionless variables $\xi = \sqrt{m\omega/\hbar} x$ and $\xi' = \sqrt{m\omega/\hbar} x'$. The Mehler kernel becomes

$$9585 \quad K(x, t; x', 0) = \sqrt{\frac{m\omega}{\pi\hbar}} \frac{e^{-i\omega t/2}}{\sqrt{1 - e^{-2i\omega t}}} \exp \left[-\frac{1}{2}(\xi^2 + \xi'^2) \frac{1 + e^{-2i\omega t}}{1 - e^{-2i\omega t}} + \frac{2\xi\xi' e^{-i\omega t}}{1 - e^{-2i\omega t}} \right]. \quad (9.65)$$

9586 This is precisely the Mehler generating function for Hermite polynomials. Expanding the right-hand

side as a power series in $e^{-i\omega t}$,

$$K(x, t; x', 0) = \sum_{n=0}^{\infty} \psi_n(x) \psi_n(x') e^{-i\omega(n+1/2)t}, \quad (9.66)$$

where the eigenfunctions are

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}} x\right) \exp\left[-\frac{m\omega x^2}{2\hbar}\right]. \quad (9.67)$$

The H_n are the Hermite polynomials. The time dependence of each term is $e^{-i\omega(n+1/2)t}$, from which the energy eigenvalues are read off immediately:

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \dots \quad (9.68)$$

This is the harmonic oscillator spectrum, identical to (5.24) obtained from the ladder-operator method in ch5. The path-integral derivation is independent: it uses the classical action and the functional determinant, with no creation or annihilation operators. The two derivations cross-validate each other, and the path-integral route makes the connection between the classical action and the quantum spectrum explicit: the $\sin \omega t$ in the prefactor, which originates from the fluctuation determinant, is responsible for both the zero-point energy $\hbar\omega/2$ (the factor $e^{-i\omega t/2}$ extracted from $1/\sqrt{\sin \omega t}$) and the Maslov index.

The ground state is extracted as the $n = 0$ term. Its wave function is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left[-\frac{m\omega x^2}{2\hbar}\right], \quad (9.69)$$

matching (5.26) from the operator method. The excited states follow from the higher-order terms in the Mehler expansion; the recursion relation among the Hermite polynomials encodes the ladder structure (5.25) of ch5 in the position representation.

Advantage and limitation. The advantage of the path-integral evaluation of the harmonic oscillator is twofold. First, it exhibits the exact quantum propagator as a single compact formula — the Mehler kernel (9.60) — whose phase is the classical action divided by $\hbar$ and whose prefactor encodes the full quantum fluctuation spectrum. No operator algebra, no basis choice, and no differential equation is required beyond the classical equation of motion and the evaluation of a Gaussian functional integral. Second, it makes the connection between the classical and quantum worlds transparent: the classical action generates the phase, the fluctuation determinant generates the energy spectrum, and the Maslov index emerges from the analytic structure of the prefactor.

Its limitation is that the method does not reveal the algebraic structure that the operator method made manifest. The creation and annihilation operators (5.25), the number operator, the coherent states, and the forced-oscillator solution — all of which flow naturally from the ladder algebra in ch5 — are opaque in the path-integral register. The Mehler kernel gives the propagator, but constructing the Fock space of states and computing matrix elements of observables between excited states requires additional post-processing (the Hermite expansion) that is essentially the operator method in disguise. The two formulations are complementary: the operator method reveals the algebraic structure, and the path integral reveals the classical-quantum correspondence. Their agreement on the central example of non-relativistic quantum mechanics is a deep consistency check on the entire framework.

The path integral's real power, however, is not in reproducing what the canonical formalism can do. It is in revealing physics that the canonical formalism obscures — global properties of the configuration space, topological phases, and the role of gauge potentials that are invisible to the local Schrodinger equation. The Aharonov–Bohm effect is the paradigm, and it is the subject of the next section.

9.5 The Aharonov-Bohm Effect: Path Topology and a Measurable Phase

The path integral has been constructed in Section 9.2, verified on the free particle, and solved exactly for the harmonic oscillator in Section 9.4. In each of those applications the space on which the particle moves is simply connected; the path integral reduces to a Gaussian fluctuation problem, and its power is organizational — it reformulates operator-evolution problems as integrals and makes the semiclassical limit transparent. This section turns to an application whose physics the canonical formalism cannot capture at all without first constructing a path sum: the Aharonov-Bohm (AB) effect, in which electrons that never enter a magnetic field nonetheless register its flux as an interference phase. The effect is not a curious addendum to quantum mechanics; it is the observation that established the vector potential as the physical agent of electromagnetism in the quantum domain and that demonstrated the fundamental role of configuration-space topology in determining the state space. The path integral is the natural language for both facts, because the phase a charged particle collects along each path from the vector potential sits in the path sum as an independent factor, and the sum over paths naturally sorts itself into homotopy classes distinguished by their winding number about the flux tube.

 **Note** The symbol $\int \mathcal{D}x$ in this section is formal notation for the limiting discretized product defined in Section 9.2, Equation (9.10). Its sole rigorous content is the limit of the N -slice Lebesgue integrals on $\mathbb{R}^2$; it is not a measure in the sense of Lebesgue. The Wiener measure (imaginary time) would supply a genuine measure, but the real-time Feynman sum is defined by the discretized limit alone.

9.5.1 Setup: Solenoid, Flux, and Multiply Connected Space

An infinite impenetrable solenoid of radius a carries a magnetic flux Φ confined entirely to its interior. The electrons are excluded from the solenoid and propagate in the region outside it. Outside the solenoid the magnetic field vanishes:

$$\nabla \times \mathbf{A} = \mathbf{B} = \mathbf{0} \quad (r > a),$$

yet the vector potential $\mathbf{A}$ itself does not vanish — it is a pure-gauge configuration, but the gauge function is multiple-valued on the punctured plane. In cylindrical coordinates (r, ϕ) , with the solenoid along the z -axis, the vector potential in the Coulomb gauge ($\nabla \cdot \mathbf{A} = 0$) takes the form

$$\mathbf{A}(\mathbf{r}) = \frac{\Phi}{2\pi r} \hat{\phi}, \quad r > a,$$

where $\Phi = \int_{\text{solenoid}} \mathbf{B} \cdot d\mathbf{S}$ is the total flux. The line integral of $\mathbf{A}$ around any closed loop encircling the solenoid once in the counterclockwise direction gives

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int_0^{2\pi} \frac{\Phi}{2\pi r} r d\phi = \Phi, \quad (9.70)$$

by Stokes's theorem applied to the punctured plane. The configuration space is $\mathbb{R}^2$ with the solenoid excised — a plane with a hole. This space is multiply connected: its fundamental group is

$$\pi_1(\mathbb{R}^2 \setminus \{\text{solenoid}\}) \cong \mathbb{Z}, \quad (9.71)$$

and paths from a source $\mathbf{x}'$ to a detection point $\mathbf{x}$ are classified by the integer n counting how many times they wind around the solenoid (with sign for orientation). Paths with different winding numbers cannot be continuously deformed into one another: they belong to distinct homotopy classes. The AB effect is the physical consequence of this classification: different homotopy classes contribute different phases to the propagator, and the relative phase is the magnetic flux in quantum units.

9.5.2 Path Integral with a Magnetic Field

A nonrelativistic charged particle in an electromagnetic field is described by the Lagrangian (in Gaussian units, with the scalar potential set to zero by gauge choice)

$$L = \frac{m}{2} \dot{\mathbf{x}}^2 + \frac{e}{c} \dot{\mathbf{x}} \cdot \mathbf{A}(\mathbf{x}),$$

where the interaction term is the velocity dotted into the vector potential. The action is

$$S[\mathbf{x}] = S_0[\mathbf{x}] + \frac{e}{c} \int_{\mathbf{x}'}^{\mathbf{x}} \mathbf{A}(\mathbf{x}) \cdot d\mathbf{l},$$

with $S_0[\mathbf{x}] = \int_0^t dt' (m/2) \dot{\mathbf{x}}^2$ the free-particle action. The key observation is that the $\mathbf{A}$ -term couples linearly to the velocity, so its contribution to the action is the line integral of $\mathbf{A}$ along the path — a geometric quantity depending only on the path's shape, not on the speed at which it is traversed.

The N -slice construction of Section 9.2 proceeds identically for the kinetic term, while the $\mathbf{A}$ -coupling contributes a factor $\exp[(ie/\hbar c) \mathbf{A}(\mathbf{x}_k) \cdot (\mathbf{x}_{k+1} - \mathbf{x}_k)]$ at each slice. Each factor exponentiates to the line integral along the piecewise-linear path. In the continuum limit, the propagator factorizes:

$$K(\mathbf{x}, t; \mathbf{x}', 0) = \int_{\mathbf{x}(0)=\mathbf{x}'}^{\mathbf{x}(t)=\mathbf{x}} \mathcal{D}\mathbf{x} \exp \left[\frac{i}{\hbar} S_0[\mathbf{x}] \right] \exp \left[\frac{ie}{\hbar c} \int_{\mathbf{x}'}^{\mathbf{x}} \mathbf{A} \cdot d\mathbf{l} \right]. \quad (9.72)$$

To derive this factorization explicitly, start from the Hamiltonian (phase-space) path integral of Section 9.2. The Hamiltonian for a particle in a magnetic field is

$$H = \frac{1}{2m} \left(\mathbf{p} - \frac{e}{c} \mathbf{A} \right)^2. \quad (9.73)$$

The phase-space path integral reads

$$K = \int \mathcal{D}\mathbf{x} \mathcal{D}\mathbf{p} \exp \left[\frac{i}{\hbar} \int_0^t dt' \left(\mathbf{p} \cdot \dot{\mathbf{x}} - \frac{1}{2m} \left(\mathbf{p} - \frac{e}{c} \mathbf{A} \right)^2 \right) \right]. \quad (9.74)$$

Shift the momentum integration variable: $\tilde{\mathbf{p}} = \mathbf{p} - (e/c) \mathbf{A}$. Then $\mathbf{p} \cdot \dot{\mathbf{x}} - (1/2m)(\mathbf{p} - (e/c) \mathbf{A})^2 = \tilde{\mathbf{p}} \cdot \dot{\mathbf{x}} + (e/c) \mathbf{A} \cdot \dot{\mathbf{x}} - \tilde{\mathbf{p}}^2/2m$. The $\tilde{\mathbf{p}}$ Gaussian integral is identical to the free-particle case and produces the kinetic prefactor and S_0 phase. The $\mathbf{A}$ -term does not involve $\tilde{\mathbf{p}}$ and factors out of the momentum

9688 integration:

$$9689 \quad K = \int \mathcal{D}\mathbf{x} \exp \left[\frac{i}{\hbar} S_0[\mathbf{x}] \right] \exp \left[\frac{ie}{\hbar c} \int_{\mathbf{x}'}^{\mathbf{x}} \mathbf{A} \cdot d\mathbf{l} \right], \quad (9.75)$$

9690 which is Equation (9.72). The factorization is exact: the $\mathbf{A}$ -coupling is linear in velocity, so it com-
 9691 mutes with itself at different times in the path sum, and the phase factor is a functional of the path
 9692 that does not participate in the Gaussian fluctuation integral. Each path contributes the free-particle
 9693 amplitude multiplied by the Aharonov-Bohm phase factor $\exp[(ie/\hbar c) \int \mathbf{A} \cdot d\mathbf{l}]$.

9694 9.5.3 Two-Path Interference and the AB Phase Shift

9695 Consider the double-slit arrangement that is the operational definition of the AB effect. A co-
 9696 herent electron source at $\mathbf{x}'$ emits electrons toward a barrier with two slits; beyond the barrier sits
 9697 a detection screen. A thin solenoid (or a toroidal magnet, which confines the return flux) is placed
 9698 between the slits, so that electrons passing above the solenoid traverse one homotopy class, and elec-
 9699 trons passing below traverse the other. The total amplitude at the detection point $\mathbf{x}$ is the sum of
 9700 contributions from the two path classes:

$$9701 \quad K \approx e^{i\phi_1} K_0^{(1)} + e^{i\phi_2} K_0^{(2)}, \quad (9.76)$$

9702 where $K_0^{(1,2)}$ are the free-particle amplitudes for the two path classes in the absence of flux, and

$$9703 \quad \phi_1 = \frac{e}{\hbar c} \int_{\text{above}} \mathbf{A} \cdot d\mathbf{l}, \quad \phi_2 = \frac{e}{\hbar c} \int_{\text{below}} \mathbf{A} \cdot d\mathbf{l}. \quad (9.77)$$

9704 The interference pattern depends on the relative phase:

$$9705 \quad \Delta\phi = \phi_1 - \phi_2 = \frac{e}{\hbar c} \left(\int_{\text{above}} - \int_{\text{below}} \right) \mathbf{A} \cdot d\mathbf{l} = \frac{e}{\hbar c} \oint \mathbf{A} \cdot d\mathbf{l}.$$

9706 The closed loop is the composite path: from source to screen above the solenoid, then back from
 9707 screen to source below the solenoid. By Stokes's theorem (applied on the punctured plane, where the
 9708 solenoid interior is excluded), the line integral equals the enclosed flux:

$$9709 \quad \Delta\phi = \frac{e}{\hbar c} \Phi.$$

9710 Define the flux quantum:

$$9711 \quad \Phi_0 := \frac{2\pi\hbar c}{e}, \quad (9.78)$$

9712 in Gaussian units.⁴ Then the relative phase takes the compact form

$$9713 \quad \Delta\phi = 2\pi \frac{\Phi}{\Phi_0}. \quad (9.79)$$

9714 The interference intensity on the screen is

$$9715 \quad I(\mathbf{x}) \propto |K|^2 \propto 1 + \cos(\Delta\phi_0 + \Delta\phi), \quad (9.80)$$

9716 where $\Delta\phi_0$ is the geometric phase difference between the two slits in the absence of flux (the ordinary
 9717 double-slit pattern). As the flux Φ inside the solenoid is varied, the entire interference pattern shifts.

⁴In SI units, $\Phi_0^{\text{SI}} = h/e = 2\pi\hbar/e$. The factor of c in the Gaussian expression arises from the different convention for the coupling of charge to the electromagnetic field: the Lagrangian term is $(e/c)\dot{\mathbf{x}} \cdot \mathbf{A}$ in Gaussian units versus $e\dot{\mathbf{x}} \cdot \mathbf{A}$ in SI. The measurable flux quantum is h/e in SI and $2\pi\hbar c/e$ in Gaussian; the numerical value $\Phi_0 \approx 4.1357 \times 10^{-7} \text{ G cm}^2$ (Gaussian) or $2.0678 \times 10^{-15} \text{ Wb (SI)}$ is the same physical quantity.

The shift is periodic: when Φ increases by one flux quantum Φ_0 , the phase $\Delta\phi$ advances by 2π , and the pattern returns to its original position.

9718

9719

9.5.4 Gauge Invariance

9720

Under a gauge transformation

9721

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi(\mathbf{x}), \quad (9.81)$$

9722

with χ a single-valued function on the punctured plane, the phase accumulated on each path changes:

9723

$$\phi[\mathbf{x}] \rightarrow \phi[\mathbf{x}] + \frac{e}{\hbar c} \int_{\mathbf{x}'}^{\mathbf{x}} \nabla\chi \cdot d\mathbf{l} = \phi[\mathbf{x}] + \frac{e}{\hbar c} [\chi(\mathbf{x}) - \chi(\mathbf{x}')]. \quad (9.82)$$

9724

The added term depends only on the endpoints, not on the path. It therefore factors out of the sum over all paths as an overall phase $e^{(ie/\hbar c)[\chi(\mathbf{x}) - \chi(\mathbf{x}')]}$, which does not affect any probability — and indeed the wave function transforms as $\psi(\mathbf{x}) \rightarrow e^{(ie/\hbar c)\chi(\mathbf{x})}\psi(\mathbf{x})$, precisely the gauge-covariant transformation of the Schrodinger wave function. The relative phase between two paths connecting the same endpoints,

9725

9726

9727

9728

$$\Delta\phi = \frac{e}{\hbar c} \oint \mathbf{A} \cdot d\mathbf{l}, \quad (9.83)$$

9729

is unchanged: $\oint \nabla\chi \cdot d\mathbf{l} = 0$ because χ is single-valued on the punctured plane. The AB phase shift is gauge-invariant and therefore physically measurable, even though it is expressed as a line integral of the gauge-dependent vector potential.

9730

9731

9732

9.5.5 Topological Character: Sum over Winding Numbers

9733

The factorization (9.72) can be reorganized as a sum over homotopy classes. Every path from $\mathbf{x}'$ to $\mathbf{x}$ has a well-defined winding number $n \in \mathbb{Z}$ around the solenoid (the integer that classifies the homotopy class in π_1). The line integral of $\mathbf{A}$ along a path of winding number n satisfies

9734

9735

9736

$$\int_{\mathbf{x}'}^{\mathbf{x}} \mathbf{A} \cdot d\mathbf{l} = \int_{\text{reference}} \mathbf{A} \cdot d\mathbf{l} + n\Phi, \quad (9.84)$$

9737

where the reference path has winding number zero (it does not enclose the solenoid). The propagator therefore decomposes as

9738

9739

$$K(\mathbf{x}, t; \mathbf{x}', 0) = \sum_{n=-\infty}^{\infty} e^{ine\Phi/\hbar c} K_n(\mathbf{x}, t; \mathbf{x}', 0),$$

9740

where K_n sums all paths of winding number n with the free-particle action and the reference-path phase. The phase factor $e^{ine\Phi/\hbar c}$ is the one-dimensional unitary representation of the fundamental group $\pi_1 \cong \mathbb{Z}$; it is the simplest instance of a Wilson loop — the gauge-invariant observable $\exp[(ie/\hbar c) \oint \mathbf{A} \cdot d\mathbf{l}]$ evaluated on the closed curve that encloses the flux once.⁵

9741

9742

9743

9744

⁵The non-abelian generalization — the trace of the path-ordered exponential of the gauge connection around a closed loop in Yang-Mills theory — is the Wilson loop of quantum field theory. It is cited here as a forward debt; its development belongs to gauge field theory.

9.5.6 Experimental Status

The AB effect was predicted by Aharonov and Bohm in 1959 and was controversial because it appeared to ascribe physical reality to the vector potential, which in classical electrodynamics is a mathematical auxiliary. Chambers (1960) observed the predicted fringe shift in an electron interference experiment with a magnetic whisker. The definitive experiment by Tonomura et al. (1986) used a toroidal magnet completely covered by a superconducting shield, confirming that the phase shift occurs when the electron wave function has zero overlap with the magnetic field — the electrons propagate entirely in a region where $\mathbf{B} = \mathbf{0}$, yet the flux enclosed by their paths determines the interference pattern. The flux periodicity was verified to high precision: the pattern shifts by exactly one fringe when the enclosed flux changes by Φ_0 . The AB effect is among the most direct confirmations of the gauge principle in quantum mechanics and of the physical status of the vector potential in the quantum domain.

9.5.7 Example: Two-Slit Interference Pattern

Consider the standard two-slit geometry with slit separation d , screen distance $L \gg d$, and electron de Broglie wavelength λ . The solenoid is placed midway between the slits, carrying flux Φ . For a detection point at lateral coordinate x on the screen (measured from the central maximum), the geometric path difference is $\Delta s \approx dx/L$, and the geometric phase difference is $\Delta\phi_0 = 2\pi dx/(\lambda L)$. The total intensity, including the AB phase, is

$$I(x) = I_0(x) \left[1 + \cos \left(\frac{2\pi dx}{\lambda L} + \frac{e\Phi}{\hbar c} \right) \right]. \quad (9.85)$$

The cosine argument contains two terms: the geometric (ordinary double-slit) phase and the topological AB phase. The entire interference pattern shifts laterally by

$$\Delta x = \frac{\lambda L}{2\pi d} \frac{e\Phi}{\hbar c} \quad (9.86)$$

as the flux is varied. When Φ increases by Φ_0 , the AB phase advances by 2π , the cosine returns to its original argument, and the pattern shifts by exactly one full fringe spacing $\lambda L/d$ — the flux periodicity observed by Tonomura et al.

9.5.8 Advantage and Limitation

The advantage of the path-integral treatment of the AB effect is that the phase factor from $\mathbf{A}$ enters the path sum transparently: it is the line integral along each path, and the sum over paths automatically organizes itself by winding number. No operator-valued gauge connection appears; the configuration-space topology does the work. The limitation is that the calculation of K_n , the free propagator on the universal cover, is nontrivial (it is the heat kernel on the covering space), and the sum over n converges only conditionally. The AB effect is placed here as a worked application of the path integral's capacity to register global topology; its generalization to non-abelian gauge theories is a forward debt named here and discharged in field theory.

The AB effect exploited the fact that paths of different topology contribute different phases, and the interference is the experimental signature. But even on a simply connected space, the stationary-phase approximation applied to the path integral produces a systematic semiclassical expansion that connects the path integral to the Schrodinger equation's short-wavelength limit. That expansion — the WKB method — is the subject of the next section, and it repays the oldest quantization debt of the book.

9.6 Semiclassical Approximation and WKB: Bohr-Sommerfeld Repaid

The Aharonov-Bohm effect of Section 9.5 demonstrated that the path integral registers the global topology of configuration space: different homotopy classes contribute phases distinguished by the winding number. Even on a simply connected space, the path integral sponsors a systematic expansion in powers of $\hbar$ whose leading term recovers classical mechanics and whose first correction produces the semiclassical wave function. That expansion is the WKB method (Wentzel, Kramers, Brillouin, 1926), and it repays the book's oldest quantization debt: the Bohr-Sommerfeld condition $\oint p_i dq_i = n_i \hbar$ of Chapter 1, row 1.9, which the old quantum theory postulated, is derived here from the single-valuedness of the WKB wave function, corrected by the Maslov index $1/2$ that supplies the zero-point energy the old quantum theory lacked.

 **Note** The symbol $\int \mathcal{D}x$ appearing in the motivational paragraphs below is formal notation for the limiting discretized product defined in Section 9.2, Equation (9.10). The WKB method itself operates on the Schrodinger equation and does not require the path-integral measure; the path integral provides the ansatz and the conceptual bridge.

9.6.1 Motivation from the Path Integral

For a stationary state of energy E , the time-dependent wave function is $\Psi(x, t) = \psi(x) e^{-iEt/\hbar}$. The fixed-energy propagator (the retarded Green's function) has the path-integral representation

$$G(x, x'; E) = \frac{1}{i\hbar} \int_0^\infty dt e^{iEt/\hbar} K(x, t; x', 0), \quad (9.87)$$

and its stationary-phase evaluation in the classically allowed region selects paths whose action satisfies $\partial S/\partial t = -E$ — the time-independent Hamilton-Jacobi equation. The resulting wave function takes the form $\psi(x) \sim e^{iS(x)/\hbar}$, where $S(x)$ is Hamilton's characteristic function. This observation, due to Dirac and elaborated by Feynman, motivates the WKB ansatz directly: the wave function in the semiclassical regime is an exponential whose phase is an expansion in powers of $\hbar$.

9.6.2 The WKB Ansatz and the $\hbar$ -Expansion

Write the wave function as an exponential series:

$$\psi(x) = \exp \left[\frac{i}{\hbar} (S_0(x) + \hbar S_1(x) + \hbar^2 S_2(x) + \dots) \right].$$

Substitute this ansatz into the time-independent Schrodinger equation

$$-\frac{\hbar^2}{2m} \psi''(x) + V(x) \psi(x) = E \psi(x),$$

and collect terms order by order in $\hbar$. Computing derivatives:

$$\psi'(x) = \frac{i}{\hbar} (S'_0 + \hbar S'_1 + \hbar^2 S'_2 + \dots) \psi(x),$$

$$\psi''(x) = \left[\frac{i}{\hbar} (S''_0 + \hbar S''_1 + \dots) + \left(\frac{i}{\hbar} \right)^2 (S'_0 + \hbar S'_1 + \dots)^2 \right] \psi(x). \quad (9.88)$$

Insert ψ'' into the time-independent Schrodinger equation and multiply through by $-2m/\hbar^2$:

$$\left[\frac{i}{\hbar} (S''_0 + \hbar S''_1 + \dots) - \frac{1}{\hbar^2} (S'_0 + \hbar S'_1 + \dots)^2 + \frac{2m}{\hbar^2} (E - V) \right] \psi = 0. \quad (9.89)$$

Multiply by $-\hbar^2$:

$$(S'_0 + \hbar S'_1 + \dots)^2 - i\hbar (S''_0 + \hbar S''_1 + \dots) - 2m(E - V) = 0. \quad (9.90)$$

Now collect powers of $\hbar$. At order $\hbar^0$:

$$(S'_0)^2 - 2m(E - V(x)) = 0.$$

Define the local classical momentum

$$p(x) := \sqrt{2m(E - V(x))}.$$

Then the leading-order equation is

$$(S'_0)^2 = p(x)^2. \quad (9.91)$$

This is the time-independent Hamilton-Jacobi equation for the characteristic function S_0 ; its solution is

$$S_0(x) = \pm \int^x dx' p(x'),$$

where the sign corresponds to right-moving (+) or left-moving (−) waves, and the lower limit is an arbitrary reference point.

At order $\hbar^1$:

$$2S'_0 S'_1 - iS''_0 = 0. \quad (9.92)$$

Solve for S'_1 :

$$S'_1 = i \frac{S''_0}{2S'_0} = \frac{i}{2} \frac{d}{dx} \ln S'_0.$$

Since $S'_0 = \pm p(x)$, the logarithmic derivative gives $\ln p(x)$ (the sign drops out of the ratio $S''_0/S'_0 = p'/p$). Integrating:

$$S_1(x) = \frac{i}{2} \ln p(x) + \text{const.}$$

Exponentiating, $e^{iS_1} = p^{-1/2}$, so the $O(\hbar)$ correction supplies the amplitude prefactor $1/\sqrt{p(x)}$

that encodes the classical dwell time: the particle lingers where it moves slowly. This is the one-dimensional transport equation: it determines the amplitude from the leading-order phase.

Higher orders in $\hbar$ produce corrections to both the phase and amplitude. The series is asymptotic in general: for anharmonic potentials it does not converge, but its first few terms provide excellent approximations in the short-wavelength regime $p(x) \gg \hbar/\ell$ where ℓ is the characteristic length scale of the potential.

9.6.3 The WKB Wave Function

Assembling the leading and next-to-leading terms, the WKB wave function in the classically allowed region ($E > V(x)$, $p(x)$ real) is

$$\psi(x) \approx \frac{C_+}{\sqrt{p(x)}} \exp \left[\frac{i}{\hbar} \int^x dx' p(x') \right] + \frac{C_-}{\sqrt{p(x)}} \exp \left[-\frac{i}{\hbar} \int^x dx' p(x') \right],$$

where $C_{\pm}$ are constants determined by boundary conditions. The factor $1/\sqrt{p(x)} = e^{iS_1}$ is the amplitude: the probability density $|\psi|^2 \propto 1/p(x) = 1/(mv(x))$, which is the classical result — the particle spends more time where it moves slowly.

In the classically forbidden region ($E < V(x)$), $p(x)$ is pure imaginary. Write $p(x) = i|p(x)|$ with $|p(x)| = \sqrt{2m(V(x) - E)}$. The exponentials become real:

$$\psi(x) \approx \frac{D_+}{\sqrt{|p(x)|}} \exp \left[+\frac{1}{\hbar} \int^x dx' |p(x')| \right] + \frac{D_-}{\sqrt{|p(x)|}} \exp \left[-\frac{1}{\hbar} \int^x dx' |p(x')| \right].$$

Both allowed and forbidden forms are displayed together as

$$\begin{aligned} \text{Allowed } (E > V) : \quad \psi(x) &\approx \frac{C_+}{\sqrt{p(x)}} e^{\frac{i}{\hbar} \int^x p dx'} + \frac{C_-}{\sqrt{p(x)}} e^{-\frac{i}{\hbar} \int^x p dx'}, \\ \text{Forbidden } (E < V) : \quad \psi(x) &\approx \frac{D}{\sqrt{|p(x)|}} e^{-\frac{1}{\hbar} \int^x |p| dx'} + \frac{D'}{\sqrt{|p(x)|}} e^{+\frac{1}{\hbar} \int^x |p| dx'}. \end{aligned} \tag{9.93}$$

In a classically inaccessible region that extends to infinity, the exponentially growing term D' must be discarded — it would violate normalizability. In barrier regions of finite width, both exponentials appear in general.

9.6.4 Breakdown at Turning Points and Connection Formulas

At a classical turning point x_0 where $E = V(x_0)$, the local momentum vanishes: $p(x_0) = 0$. The WKB wave function diverges because $1/\sqrt{p(x)}$ blows up, and the approximation that $p(x)$ varies slowly compared to the de Broglie wavelength $\hbar/p(x)$ fails — the wavelength diverges at the turning point. The Schrodinger equation must be solved exactly in the neighborhood of x_0 .

Linearize the potential near the turning point:

$$V(x) \approx V(x_0) + V'(x_0)(x - x_0) = E + V'(x_0)(x - x_0), \tag{9.94}$$

where $V'(x_0) \neq 0$ for a simple (soft) turning point. Define the scaled variable

$$z := \left(\frac{2mV'(x_0)}{\hbar^2} \right)^{1/3} (x - x_0), \tag{9.95}$$

so that the Schrodinger equation $-\hbar^2\psi''/2m + V\psi = E\psi$ becomes the Airy equation:

$$\frac{d^2\psi}{dz^2} - z\psi(z) = 0.$$

The solution that decays in the classically forbidden region ($z > 0$, corresponding to $E < V$) is the Airy function $\text{Ai}(z)$. Its asymptotic expansions are known:

$$\text{Ai}(z) \sim \begin{cases} \frac{1}{2\sqrt{\pi}} z^{1/4} e^{-\frac{2}{3}z^{3/2}}, & z \rightarrow +\infty, \\ \frac{1}{\sqrt{\pi}} |z|^{1/4} \sin\left(\frac{2}{3}|z|^{3/2} + \frac{\pi}{4}\right), & z \rightarrow -\infty. \end{cases} \quad (9.96)$$

Translating back to the physical variables, the decaying exponential in the forbidden region ($x < x_0$ when $V'(x_0) > 0$) matches onto the oscillatory cosine form in the allowed region ($x > x_0$). The matching calculation is standard; the result is stated without reproducing the algebra.⁶

For a soft turning point at x_0 , with the classically forbidden region to the left ($x < x_0$) and the allowed region to the right ($x > x_0$), the connection formula is

$$\frac{2}{\sqrt{p(x)}} \cos\left[\frac{1}{\hbar} \int_{x_0}^x dx' p(x') - \frac{\pi}{4}\right] \longleftrightarrow \frac{1}{\sqrt{|p(x)|}} \exp\left[-\frac{1}{\hbar} \int_x^{x_0} dx' |p(x')|\right], \quad (9.97)$$

where the arrow means: the decaying WKB solution in the forbidden region connects to the cosine standing wave (the sum of right- and left-moving waves with equal amplitude) in the allowed region. The phase $-\pi/4$ is the one-dimensional Maslov index — the phase loss incurred when the wave reflects from a soft classical turning point. For a soft turning point with the forbidden region to the right, the same formula applies with appropriate sign changes.

Each soft turning point contributes a phase loss of $-\pi/4$ to the reflected wave. In Section 9.4, a half-period of the harmonic oscillator (from one turning point to the other and back) involved four turning-point encounters per full period, or two per half-period, producing a Maslov phase of $-\pi/2$ per half-period — the caustic phase of Equation (9.61). The $-\pi/4$ per turning point in one dimension is the same Maslov index, counted per encounter rather than per half-period. The arithmetic is: one full oscillation traverses two soft turning points twice each (outbound and inbound), giving $4 \times (-\pi/4) = -\pi$, or $-\pi/2$ per half-period.

9.6.5 Bohr-Sommerfeld Quantization: The Central Repayment

Consider a particle bound in a one-dimensional potential well with two soft turning points $x_1 < x_2$, so that $E = V(x_1) = V(x_2)$ and $E > V(x)$ for $x_1 < x < x_2$. In the classically forbidden regions $x < x_1$ and $x > x_2$, the wave function must decay (the growing exponential is discarded by normalizability). The WKB wave function in the allowed region is a superposition of right- and left-moving waves:

$$\psi(x) = \frac{C}{\sqrt{p(x)}} \exp\left[\frac{i}{\hbar} \int_{x_1}^x dx' p(x')\right] + \frac{C'}{\sqrt{p(x)}} \exp\left[-\frac{i}{\hbar} \int_{x_1}^x dx' p(x')\right]. \quad (9.98)$$

⁶The full Airy-function asymptotic matching is carried out in Landau and Lifshitz, *Quantum Mechanics (Non-Relativistic Theory)*, §§46–52, and in Bender and Orszag, *Advanced Mathematical Methods for Scientists and Engineers*, Ch. 10. This chapter states the connection formulas and applies them; the derivation is cited, not reproduced.

Apply the connection formula (9.97) at the right turning point x_2 . The wave incident from the left reflects at x_2 and returns. Tracing the wave through a full round trip — from x_1 to x_2 , reflection at x_2 (phase shift $-\pi/4$), propagation back to x_1 , and reflection at x_1 (another $-\pi/4$) — the total phase accumulated must be an integer multiple of 2π for the wave function to be single-valued:

$$\frac{2}{\hbar} \int_{x_1}^{x_2} dx p(x) - \frac{\pi}{2} = 2\pi n, \quad n = 0, 1, 2, \dots$$

The factor of 2 comes from the round trip (out and back); the $-\pi/2$ is the sum of two $-\pi/4$ Maslov phases, one from each turning point. Rearranging:

$$\frac{1}{\hbar} \int_{x_1}^{x_2} dx p(x) = \pi \left(n + \frac{1}{2} \right). \quad (9.99)$$

Multiplying by 2 and recognizing the closed integral:

$$\oint p(x) dx := 2 \int_{x_1}^{x_2} p(x) dx = 2\pi\hbar \left(n + \frac{1}{2} \right). \quad (9.100)$$

This is the corrected Bohr-Sommerfeld quantization condition. It repays Chapter 1, row 1.9 — $\oint p_i dq_i = n_i h$ — with three essential additions the old quantum theory lacked: (i) the $1/2$ is the Maslov-index correction, which is the semiclassical face of the zero-point energy; (ii) the condition is the leading order of a systematic $\hbar$ -expansion, not an exact rule; and (iii) it follows from the single-valuedness of the wave function, a concept the old quantum theory did not possess. For a potential with one hard wall (infinite barrier) and one soft turning point, the round trip involves only one $-\pi/4$ phase, giving the modified condition $\oint p dx = 2\pi\hbar(n + 3/4)$ or equivalently $2\pi\hbar(n - 1/4)$ depending on convention; the quantum bouncer of Exercise 10 provides the worked case.

9.6.6 Verification: The Harmonic Oscillator

Apply the Bohr-Sommerfeld condition to the harmonic oscillator $V(x) = \frac{1}{2}m\omega^2 x^2$. The turning points are $\pm x_0$ where $E = \frac{1}{2}m\omega^2 x_0^2$, so $x_0 = \sqrt{2E/m\omega^2}$. The action integral is

$$\begin{aligned} \oint p(x) dx &= 2 \int_{-x_0}^{x_0} dx \sqrt{2m \left(E - \frac{1}{2}m\omega^2 x^2 \right)} \\ &= 2\sqrt{2mE} \int_{-x_0}^{x_0} dx \sqrt{1 - \frac{x^2}{x_0^2}}. \end{aligned} \quad (9.101)$$

Change variables: $x = x_0 \sin \theta$, $dx = x_0 \cos \theta d\theta$, with θ from $-\pi/2$ to $\pi/2$. The integral becomes

$$\begin{aligned} \oint p(x) dx &= 2\sqrt{2mE} x_0 \int_{-\pi/2}^{\pi/2} d\theta \cos^2 \theta \\ &= 2\sqrt{2mE} \sqrt{\frac{2E}{m\omega^2}} \frac{\pi}{2} \\ &= 2\sqrt{2mE} \sqrt{\frac{2E}{m\omega^2}} \frac{\pi}{2} = \frac{2\pi E}{\omega}. \end{aligned} \quad (9.102)$$

9923 To verify the algebra: $\sqrt{2mE} \cdot \sqrt{2E/(m\omega^2)} = 2E/\omega$, times $\pi/2$ (from $\int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta = \pi/2$), times
 9924 the factor of 2 from the closed integral, gives $2\pi E/\omega$. Insert into the quantization condition (9.100):

$$9925 \quad \frac{2\pi E}{\omega} = 2\pi\hbar \left(n + \frac{1}{2}\right) \implies E_n = \hbar\omega \left(n + \frac{1}{2}\right). \quad (9.103)$$

9926 The WKB result is exact for the harmonic oscillator. The reason is that the action is quadratic in x and
 9927 $\dot{x}$; the $\hbar^2$ and all higher WKB corrections vanish identically. This matches the operator spectrum of
 9928 Chapter 5, Equation (5.24), and the Mehler-kernel extraction of Section 9.4, Equation (9.64) — three
 9929 independent derivations converging on the same result.

9930 9.6.7 Tunneling: The Gamow Formula

9931 When a particle of energy E encounters a potential barrier $V(x) > E$ between two turning points
 9932 x_1 and x_2 , the WKB wave function in the barrier region is exponentially decaying. The transmission
 9933 amplitude is obtained by matching the WKB solutions at both turning points using the connection
 9934 formulas (9.97). The derivation proceeds in five steps.

9935 **(i) Incident wave.** In the left allowed region ($x < x_1$, $E > V$), the WKB wave function is a superpo-
 9936 sition of right- and left-moving waves. Take the right-moving (incident) wave with unit amplitude:

$$9937 \quad \psi_{\text{inc}}(x) = \frac{1}{\sqrt{p(x)}} \exp\left[\frac{i}{\hbar} \int_{x_1}^x dx' p(x')\right], \quad x < x_1.$$

9938 At the left turning point x_1 , the incoming right-moving wave and the reflected left-moving wave com-
 9939 bine into a cosine standing wave of amplitude 2. The connection formula (9.97) (read left-to-right,
 9940 from the forbidden side to the allowed side, with the coefficient 2 from the cosine combination) gives
 9941 the amplitude penetrating into the barrier:

$$9942 \quad \psi_{\text{inc}}(x) + \psi_{\text{refl}}(x) \xrightarrow{\text{connection at } x_1} \frac{2}{\sqrt{|p(x)|}} \exp\left[-\frac{1}{\hbar} \int_{x_1}^x dx' |p(x')|\right], \quad x_1 < x < x_2.$$

9943 **(ii) Propagation through the barrier.** The decaying exponential propagates from x_1 to x_2 . At the
 9944 right turning point x_2 , the amplitude is suppressed by the barrier action:

$$9945 \quad \psi_{\text{barrier}}(x_2^-) = \frac{2}{\sqrt{|p(x_2)|}} \exp\left[-\frac{1}{\hbar} \int_{x_1}^{x_2} dx' |p(x')|\right].$$

9946 **(iii) Connection at the right turning point.** At x_2 , the connection formula (9.97) is applied in re-
 9947 verse (from the forbidden side to the allowed side). The decaying exponential of amplitude D in the
 9948 forbidden region connects to a right-moving (transmitted) wave in the right allowed region ($x > x_2$)
 9949 with amplitude $D/2$:

$$9950 \quad \frac{D}{\sqrt{|p(x)|}} e^{-\frac{1}{\hbar} \int_{x_2}^x |p| dx'} \longleftrightarrow \frac{D/2}{\sqrt{p(x)}} \exp\left[\frac{i}{\hbar} \int_{x_2}^x dx' p(x') - \frac{i\pi}{4}\right], \quad x > x_2.$$

9951 With $D = 2 \exp[-(1/\hbar) \int_{x_1}^{x_2} |p| dx']$, the transmitted amplitude is

$$9952 \quad \psi_{\text{trans}}(x) = \frac{1}{\sqrt{p(x)}} \exp\left[-\frac{1}{\hbar} \int_{x_1}^{x_2} dx' |p(x')|\right] \exp\left[\frac{i}{\hbar} \int_{x_2}^x dx' p(x') - \frac{i\pi}{4}\right], \quad x > x_2.$$

9953 **(iv) Transmission probability.** The incident wave had unit amplitude (coefficient $1/\sqrt{p}$ with mod-
 9954 ulus $1/\sqrt{p}$). The transmitted wave has amplitude $e^{-\int |p| dx/\hbar} / \sqrt{p}$. The ratio of transmitted to incident

probability currents (which, for WKB waves, is the square of the amplitude ratio, since the $1/\sqrt{p}$ factors and the oscillatory phases drop out of $|\psi|^2$) gives the transmission probability:

$$T = \frac{|\psi_{\text{trans}}|^2 p(x)}{|\psi_{\text{inc}}|^2 p(x)} = \exp\left[-\frac{2}{\hbar} \int_{x_1}^{x_2} dx' |p(x')|\right]. \quad (9.104)$$

(v) Result. Writing $|p(x)| = \sqrt{2m(V(x) - E)}$, the WKB transmission probability to leading exponential order is

$$T \approx \exp\left[-\frac{2}{\hbar} \int_{x_1}^{x_2} dx \sqrt{2m(V(x) - E)}\right]. \quad (9.105)$$

The factor of 2 in the exponent reflects two independent contributions: the amplitude is suppressed by one factor of $e^{-\int |p|dx/\hbar}$ for the barrier penetration, and the probability—the squared modulus—doubles the exponent. A pre-exponential factor of order unity, arising from the precise matching of the $1/\sqrt{p}$ factors at both turning points and from the reflected wave in the left allowed region, is not captured at this order; it is the subject of the uniform WKB approximation beyond the leading exponential.⁷

The first triumph of quantum tunneling was Gamow's 1928 theory of α -decay. An α -particle of energy E (typically a few MeV) is confined inside a heavy nucleus of charge Z (for the daughter nucleus after α -emission) by the nuclear potential. Outside the nuclear radius R , the potential is the Coulomb repulsion $V(r) = 2Ze^2/r$ (in Gaussian units; the factor 2 is the charge of the α -particle). The α -particle escapes by tunneling through the Coulomb barrier from $r = R$ to the outer turning point $r_2 = 2Ze^2/E$ where the Coulomb potential equals the kinetic energy. The transmission probability is

$$T \approx \exp\left[-\frac{2}{\hbar} \int_R^{r_2} dr \sqrt{2m\left(\frac{2Ze^2}{r} - E\right)}\right] \equiv e^{-G}, \quad (9.106)$$

where m is the reduced mass of the α -nucleus system and G is the Gamow factor. The integral evaluates to

$$G = \frac{4Ze^2}{\hbar} \sqrt{\frac{2m}{E}} \left[\arccos \sqrt{\frac{E}{B}} - \sqrt{\frac{E}{B} \left(1 - \frac{E}{B}\right)} \right], \quad (9.107)$$

with $B = 2Ze^2/R$ the barrier height. For α -decay, typically $E \ll B$, and the leading behavior is $G \propto Z/\sqrt{E}$. The decay constant is $\lambda \approx f \cdot T$, where $f \sim v/R$ is the assault frequency (the rate at which the α -particle strikes the barrier). The dominant E -dependence is in T , giving the Geiger-Nuttall law:

$$\log \lambda \approx \text{const} - \frac{\text{const}}{\sqrt{E}}, \quad (9.108)$$

an empirical relation known since 1911 that Gamow's calculation explained quantitatively. The detailed integration and the extraction of the Geiger-Nuttall scaling are Exercise 9.

⁷The pre-exponential factor for a parabolic barrier (the Kemple correction) gives the exact transmission coefficient $T = 1/(1 + e^{2S_0/\hbar})$ with $S_0 = \int dx |p(x)|$, which interpolates between the WKB regime ($T \sim e^{-2S_0/\hbar}$ for $S_0 \gg \hbar$) and the above-barrier regime ($T \rightarrow 1$ for $E \gg V_{\text{max}}$). See Kemple, *Phys. Rev.* **48**, 549 (1935).

9.6.8 The Langer Correction

The radial Schrodinger equation in three dimensions with angular momentum ℓ reduces to a one-dimensional problem on the half-line $r \in [0, \infty)$ with the effective potential

$$V_{\text{eff}}(r) = V(r) + \frac{\hbar^2 \ell(\ell + 1)}{2mr^2}. \quad (9.109)$$

Direct application of the WKB method to this effective potential fails near $r = 0$, where the centrifugal term diverges. The Langer correction is the replacement

$$\ell(\ell + 1) \rightarrow \left(\ell + \frac{1}{2}\right)^2, \quad (9.110)$$

which restores the correct behavior of the WKB wave function near the origin. The origin of the correction is the change of variable $r = e^x$, which maps the half-line to the full line and transforms the radial equation into a form to which WKB applies without the $r = 0$ singularity; the substitution $\ell(\ell + 1) \rightarrow (\ell + 1/2)^2$ emerges from this mapping. The prescription is stated as a known result; its full derivation belongs to the literature.⁸

9.6.9 De Broglie Packet Dispersion Revisited

The free-particle WKB phase $S_0(x) = \pm \int^x dx' \sqrt{2mE}$ is the characteristic function of the free particle. A Gaussian superposition of WKB stationary states,

$$\psi(x, t) = \int \frac{dp}{2\pi\hbar} \tilde{\varphi}(p) \exp \left[\frac{i}{\hbar} \left(px - \frac{p^2}{2m} t \right) \right], \quad (9.111)$$

has its stationary-phase point determined by

$$\left. \frac{\partial}{\partial p} \left(px - \frac{p^2}{2m} t \right) \right|_{p=p_0} = 0 \implies x - \frac{p_0}{m} t = 0, \quad (9.112)$$

which is the classical trajectory $x = p_0 t / m$. The quadratic fluctuations around the stationary point reproduce the ballistic spreading law:

$$\Delta x(t)^2 = \sigma^2 + \left(\frac{\Delta p}{m} \right)^2 t^2, \quad (9.113)$$

matching Equation (5.43) of Chapter 5. The WKB stationary-phase logic reproduces the exact result of the free-particle propagator because the action is exactly quadratic; the dispersion the semiclassical theory must reproduce is reproduced.

9.6.10 Advantage and Limitation of WKB

The WKB method bridges classical and quantum mechanics: the classical action supplies the phase, the classical momentum determines the amplitude, and the single-valuedness condition produces the quantized spectrum. It produces binding energies, tunneling rates, and scattering phase shifts directly from the classical action integral, without solving the Schrodinger equation. For the harmonic oscillator, it is exact; for any potential, it is the leading term of a systematic $\hbar$ -expansion.

⁸R. E. Langer, *Phys. Rev.* **51**, 669 (1937). The correction is essential for the WKB quantization of the hydrogen atom and for the semiclassical treatment of scattering phase shifts.

The limitations are threefold. First, the approximation breaks down at turning points, where connection formulas (derived from the Airy function) are required; these formulas are global — they patch the solution across the turning point but rely on the linear approximation to the potential. Second, near the top of a barrier where $p(x)$ has a double zero (the potential is locally quadratic with negative curvature), the WKB approximation fails and requires parabolic cylinder functions for the matching.⁹ Third, the WKB method captures only the leading exponential in tunneling; the pre-exponential factor (the fluctuation determinant around the tunneling path) requires the instanton calculus — the evaluation of the path integral in imaginary time around the classical solution in the inverted potential.¹⁰

9.6.11 Example: Double-Well Tunneling Splitting

For a symmetric double-well potential $V(x) = \lambda(x^2 - a^2)^2$ with minima at $x = \pm a$, the two lowest energy eigenstates are the symmetric and antisymmetric combinations of states localized in each well. The energy splitting is

$$\Delta E \approx \frac{\hbar\omega}{\pi} \exp \left[-\frac{1}{\hbar} \int_{-a}^a dx \sqrt{2m(V(x) - E_0)} \right], \quad (9.114)$$

where ω is the small-oscillation frequency in each well and $E_0 \approx \hbar\omega/2$ is the zero-point energy. The exponential factor is the WKB tunneling amplitude through the central barrier. The pre-factor $\hbar\omega/\pi$ comes from the instanton fluctuation determinant; the full instanton calculation is not developed here, but the WKB exponent is the leading contribution and is computed explicitly in advanced texts.

The WKB method closes the circle that Chapter 1 opened. The quantization rules the old quantum theory postulated — $\oint p_i dq_i = n_i h$ — are derived as the leading order of a systematic semiclassical expansion. The Maslov index $1/2$ is the quantum correction they lacked, and it is the zero-point energy's semiclassical face. The chapter's debts are now repaid; the ledger is balanced in the summary that follows.

9.7 The Spin Path Integral

The path integral constructed in Sections 9.1–9.2 sums over histories of a particle's position — a continuous coordinate $x(t)$. Spin is not such a coordinate. It takes discrete values, has no classical analogue in the sense of a point moving in a continuous configuration space, and is described by a finite-dimensional Hilbert space $\mathbb{C}^{2s+1}$. Yet there exists a path-integral representation for spin: the configuration space is the Bloch sphere S^2 , the paths are trajectories $\hat{n}(t)$ on that sphere, and the action contains a topological term — the Wess-Zumino (Berry phase) term — whose single-valuedness

⁹Near a barrier top, the Schrodinger equation reduces to the Weber (parabolic cylinder) equation; the transmission coefficient involves the Kemble correction $T = 1/(1 + e^{2S_0/\hbar})$ with $S_0 = \int dx |p(x)|$, which interpolates between the WKB regime (large $S_0/\hbar$, $T \sim e^{-2S_0/\hbar}$) and the above-barrier regime.

¹⁰Instantons are finite-action solutions of the classical equations of motion in imaginary time $\tau = it$. For the double-well potential, the instanton is the trajectory that interpolates between the two degenerate minima; its action S_0 determines the leading exponential splitting $\Delta E \sim \hbar\omega e^{-S_0/\hbar}$ between the symmetric and antisymmetric ground states. The instanton calculus, including the fluctuation determinant and the dilute-gas sum, belongs to advanced quantum mechanics and is named here as a forward debt.

quantizes the spin. This section constructs that representation.

Note The symbol $\int \mathcal{D}\hat{n}$ in this section is formal notation for the limiting discretized product defined over the Bloch sphere, exactly as $\int \mathcal{D}x$ was defined over $\mathbb{R}$ in Section 9.2. Its sole rigorous content is the limit of the N -slice product of spin-coherent-state resolutions of identity.

9.7.1 Motivation: Why a Path Integral for Spin?

A particle with spin moving in space has the tensor-product Hilbert space $L^2(\mathbb{R}^3) \otimes \mathbb{C}^{2s+1}$; its position degree of freedom is treated by the Feynman path integral of Section 9.2, and its spin degree of freedom by the construction of this section. The separation is not a theorem — kinetic and spin terms in the Hamiltonian may couple — but when the spin dynamics is governed by a Hamiltonian $H(S)$ that depends only on the spin operators, the spin path integral is a closed description. The construction also serves a second purpose: it exhibits the Wess-Zumino term, the simplest topological term in a quantum action, whose cousins (the Wess-Zumino-Witten term in two spacetime dimensions, the Chern-Simons term in three) are central to quantum field theory. The spin path integral is the mechanical prototype of every topological term in this book.

The configuration space for a spin is not a flat manifold. A spin- s system has $2s + 1$ orthogonal states. The set of all normalized states is the complex projective space $\mathbb{CP}^{2s}$, of real dimension $4s$. The states that can be obtained by rotating the highest-weight state $|s, s\rangle$ form a submanifold: the orbit of the rotation group acting on a fixed state. Since the stabilizer of $|s, s\rangle$ under rotations about the z -axis is $U(1)$, the orbit is the coset space

$$SU(2)/U(1) \cong S^2, \quad (9.115)$$

the Bloch sphere. Every point on this sphere labels a coherent state; the sphere is the classical phase space of a spin, with the Poisson bracket $\{S_i, S_j\} = \epsilon_{ijk} S_k$ inherited from the quantum commutator. The spin path integral sums over trajectories on this sphere.

9.7.2 Spin Coherent States

Definition. For a spin- s system, the spin coherent state $|\hat{n}\rangle$ directed along the unit vector $\hat{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ is the state obtained by rotating the highest-weight state $|s, s\rangle$ from the north pole to $\hat{n}$:

$$|\hat{n}\rangle := e^{-i\phi S_z/\hbar} e^{-i\theta S_y/\hbar} |s, s\rangle. \quad (9.116)$$

The state is normalized, $\langle \hat{n} | \hat{n} \rangle = 1$, and satisfies the eigenvalue-like property $\hat{n} \cdot \mathbf{S} |\hat{n}\rangle = \hbar s |\hat{n}\rangle$: the projection of the spin along the direction $\hat{n}$ is maximal. For the two-state system ($s = 1/2$), the coherent states are precisely the spin- $\mathbf{n}$ basis of Chapter 2 ((2.38)), covering the Bloch sphere exactly once.

Overlap. The inner product of two spin coherent states is

$$\langle \hat{n}_1 | \hat{n}_2 \rangle = \left[\frac{1}{2} (1 + \hat{n}_1 \cdot \hat{n}_2) \right]^s e^{is \Phi(\hat{n}_1, \hat{n}_2, \hat{z})}, \quad (9.117)$$

where Φ is the solid angle (Berry phase) of the spherical triangle formed by the geodesics connecting $\hat{\mathbf{z}}$, $\hat{\mathbf{n}}_1$, and $\hat{\mathbf{n}}_2$ on the sphere. The phase is geometric: it depends only on the area of the triangle, not on the path taken. For $s = 1/2$, the modulus squared reproduces the Malus law of the two-state grammar: $|\langle \hat{\mathbf{n}}_1 | \hat{\mathbf{n}}_2 \rangle|^2 = (1 + \hat{\mathbf{n}}_1 \cdot \hat{\mathbf{n}}_2)/2$.

Resolution of identity. The spin coherent states form an overcomplete basis. The resolution of identity on the spin- s Hilbert space is

$$\frac{2s+1}{4\pi} \int_{S^2} d\Omega |\hat{\mathbf{n}}\rangle \langle \hat{\mathbf{n}}| = I, \quad d\Omega = \sin \theta d\theta d\phi, \quad (9.118)$$

with the integral over the unit sphere. The proof is an application of the Wigner D -matrix orthogonality and is cited from the angular-momentum chapter's representation theory (Appendix B). The pre-factor $(2s+1)/4\pi$ ensures the correct trace: $\text{tr} I = 2s+1$.

9.7.3 The Spin Path Integral Construction

Time-slicing with coherent states. The evolution operator for a spin Hamiltonian $H(\mathbf{S})$ is $U(t) = e^{-iHt/\hbar}$. Slice the time interval $[0, t]$ into N steps of duration $\varepsilon = t/N$ and insert the resolution of identity ((9.118)) at each interior time:

$$U(t) = \lim_{N \rightarrow \infty} \int \prod_{k=1}^{N-1} \left(\frac{2s+1}{4\pi} d\Omega_k \right) \prod_{k=0}^{N-1} |\hat{\mathbf{n}}_{k+1}\rangle \langle \hat{\mathbf{n}}_{k+1}| e^{-i\varepsilon H(\mathbf{S})/\hbar} |\hat{\mathbf{n}}_k\rangle. \quad (9.119)$$

with $|\hat{\mathbf{n}}_0\rangle = |\hat{\mathbf{n}}(0)\rangle$ the initial state and $\hat{\mathbf{n}}_N = \hat{\mathbf{n}}(t)$ the final direction.

Short-time kernel. For infinitesimal ε , the short-time amplitude factorizes:

$$\langle \hat{\mathbf{n}}_{k+1} | e^{-i\varepsilon H(\mathbf{S})/\hbar} | \hat{\mathbf{n}}_k \rangle \approx \langle \hat{\mathbf{n}}_{k+1} | \hat{\mathbf{n}}_k \rangle e^{-i\varepsilon H(\hat{\mathbf{n}}_{k+1})/\hbar}, \quad (9.120)$$

where $H(\hat{\mathbf{n}}) := \langle \hat{\mathbf{n}} | H(\mathbf{S}) | \hat{\mathbf{n}} \rangle$ is the expectation value (the “classical” Hamiltonian on the Bloch sphere); the error is $O(\varepsilon^2)$ by the Trotter logic of Section 9.1. The overlap factor cannot be reduced to unity: it carries the kinetic (Berry phase) term that will become the Wess-Zumino action.

The overlap as a phase. For small angular separation between successive slices, $\hat{\mathbf{n}}_{k+1} \approx \hat{\mathbf{n}}_k + \Delta \hat{\mathbf{n}}_k$ with $|\Delta \hat{\mathbf{n}}_k| \ll 1$. Expanding the overlap ((9.117)):

$$\langle \hat{\mathbf{n}}_{k+1} | \hat{\mathbf{n}}_k \rangle \approx \exp[is \Delta \phi_k (1 - \cos \theta_k)] \left[1 - \frac{s}{4} |\Delta \hat{\mathbf{n}}_k|^2 + \dots \right]. \quad (9.121)$$

The phase contains the discrete form of the Wess-Zumino term; the modulus supplies a Gaussian fluctuation factor that keeps the path near the classical trajectory. In the continuum limit $\varepsilon \rightarrow 0$, $N \rightarrow \infty$, the N -slice product assembles into the path integral:

$$K(\hat{\mathbf{n}}, t; \hat{\mathbf{n}}', 0) = \int_{\hat{\mathbf{n}}(0)=\hat{\mathbf{n}}'}^{\hat{\mathbf{n}}(t)=\hat{\mathbf{n}}} \mathcal{D}\hat{\mathbf{n}} \exp\left[\frac{i}{\hbar} S_{\text{eff}}[\hat{\mathbf{n}}]\right], \quad (9.122)$$

with the effective action

$$S_{\text{eff}}[\hat{\mathbf{n}}] = \hbar s \int_0^t dt' (1 - \cos \theta) \dot{\phi} - \int_0^t dt' H(\hat{\mathbf{n}}(t')). \quad (9.123)$$

The Wess-Zumino term. The first term,

$$S_{\text{WZ}}[\hat{\mathbf{n}}] := \hbar s \int_0^t dt' (1 - \cos \theta) \dot{\phi} = \hbar s \int_0^t dt' \mathbf{A}(\hat{\mathbf{n}}) \cdot \dot{\hat{\mathbf{n}}}, \quad (9.124)$$

is the **Wess-Zumino term** (also called the Berry phase action). The vector $\mathbf{A}(\hat{\mathbf{n}})$ is the Dirac monopole vector potential on the sphere: $\nabla_{\hat{\mathbf{n}}} \times \mathbf{A} = \hat{\mathbf{n}}$. The term measures the area on the Bloch sphere swept by the spin trajectory. Its integral over a closed path is $\hbar s$ times the solid angle enclosed — the Berry phase. In the Lagrangian register, $L_{\text{WZ}} = \hbar s(1 - \cos \theta) \dot{\phi}$; the canonical momentum conjugate to ϕ is $p_\phi = \partial L_{\text{WZ}} / \partial \dot{\phi} = \hbar s(1 - \cos \theta) = S_z$, and the symplectic form $\omega = dp_\phi \wedge d\phi = \hbar s \sin \theta d\theta \wedge d\phi$ is the area element on the Bloch sphere — the classical Poisson bracket of a spin.

Quantization of spin from path topology. The Wess-Zumino term is not single-valued on the sphere. Under a gauge transformation (a different choice of the reference state for the coherent-state phase convention), the term changes by a total derivative. The ambiguity is physical: for the evolution operator to be well-defined, the path integral must be independent of the choice. This requires that the WZ term for any *closed* path be an integer multiple of $2\pi\hbar$:

$$\frac{1}{\hbar} S_{\text{WZ}}[\text{closed path}] = s \cdot (\text{solid angle}) = 2\pi n, \quad n \in \mathbb{Z}. \quad (9.125)$$

A closed path on the sphere encloses a solid angle of $4\pi n$; therefore $2s$ must be an integer. Spin quantization — $s = 0, 1/2, 1, 3/2, \dots$ — emerges from the requirement that the path integral be well-defined. This is the path-integral version of the angular-momentum quantization derived algebraically in Chapter 6; here the same result follows from the topology of the Bloch sphere, exactly as the Dirac quantization of magnetic charge follows from the topology of the Hopf bundle.

9.7.4 Example: Spin-1/2 in a Magnetic Field

For a spin-1/2 particle in a static magnetic field $\mathbf{B} = B \hat{\mathbf{z}}$, the Hamiltonian is $H = -\gamma \mathbf{B} \cdot \mathbf{S} = -(\hbar\omega/2) \sigma_z$ with $\omega = \gamma B$, the Larmor frequency of Section 4.3. The classical Hamiltonian on the Bloch sphere is $H(\hat{\mathbf{n}}) = \langle \hat{\mathbf{n}} | H | \hat{\mathbf{n}} \rangle = -(\hbar\omega/2) \cos \theta$. The effective action (9.123) becomes

$$S_{\text{eff}}[\theta, \phi] = \int_0^t dt' \left[\frac{\hbar}{2} (1 - \cos \theta) \dot{\phi} + \frac{\hbar\omega}{2} \cos \theta \right]. \quad (9.126)$$

Classical equation: Landau-Lifshitz. The stationary-phase condition $\delta S_{\text{eff}} = 0$ yields the classical equation of motion. Varying with respect to θ and ϕ :

$$\frac{\delta S_{\text{eff}}}{\delta \phi} = 0 : \quad \frac{d}{dt} \left[\frac{\hbar}{2} (1 - \cos \theta) \right] = 0 \implies \dot{\theta} = 0, \quad (9.127)$$

$$\frac{\delta S_{\text{eff}}}{\delta \theta} = 0 : \quad \frac{\hbar}{2} \sin \theta \dot{\phi} - \frac{\hbar\omega}{2} \sin \theta = 0 \implies \dot{\phi} = \omega. \quad (9.128)$$

The classical trajectory is $\theta = \text{const}$, $\phi(t) = \phi_0 + \omega t$: uniform precession of the spin vector about the field axis at the Larmor frequency. In vector form, $\dot{\hat{\mathbf{n}}} = \omega \hat{\mathbf{z}} \times \hat{\mathbf{n}}$, which is the Landau-Lifshitz equation for a spin in a magnetic field. The stationary-phase limit of the spin path integral reproduces the classical spin dynamics, exactly as the stationary-phase limit of the Feynman path integral reproduced Newton's equation for a particle.

Gaussian fluctuations and the propagator. Expanding the action to quadratic order around the classical path and performing the Gaussian functional integral (the spin analogue of the fluctuation determinant in Section 9.4) recovers the exact two-state propagator. The calculation is Exercise 11; the result must match the Larmor precession kernel embedded in the spin- $\mathbf{n}$ basis. The agreement is

a consistency check: the spin path integral and the canonical operator formalism are one dynamics, written in two registers.

Advantage and limitation. The advantage of the spin path integral is conceptual: it reveals the geometric origin of spin quantization as a topological consistency condition on the path integral, and it makes the classical limit of a spin system — the Landau-Lifshitz dynamics — transparent. Its limitation is calculational: explicit evaluation of the spin path integral beyond the stationary-phase approximation is rarely simpler than the canonical operator method. The spin path integral earns its keep in two regimes: (i) semiclassical spin dynamics, where $\hbar s$ is large and the stationary-phase approximation is accurate; and (ii) systems where the spin couples to other degrees of freedom and a unified path-integral treatment is natural. The forward debt — instantons in spin systems, quantum tunneling of magnetization, and the Wess-Zumino-Witten term in field theory — is planted here.

9.8 The Coherent State Path Integral for Bosons

The spin path integral of Section 9.7 used the overcomplete basis of spin coherent states on the Bloch sphere. Bosonic systems admit an analogous construction: the overcomplete basis of harmonic-oscillator coherent states $|z\rangle$, eigenstates of the annihilation operator. The resulting path integral unifies the operator ladder algebra of Chapter 5 with the Feynman sum over histories of this chapter, and it supplies the template — the functional integral over complex fields — on which every relativistic quantum field theory is built. This section constructs the bosonic coherent-state path integral, verifies it on the harmonic oscillator, and extracts the generating functional and Wick's theorem.

 **Note** The symbol $\int \mathcal{D}[z, \bar{z}]$ is formal notation for the limiting discretized product defined in Section 9.2, with the coherent-state measure $d^2 z_k / \pi$ replacing the position-space measure dx_k at each slice. Its sole rigorous content is the limit of the N -slice product.

9.8.1 Holomorphic (Bargmann) Coherent States

Definition. The bosonic coherent state $|z\rangle$ labelled by the complex number $z \in \mathbb{C}$ is

$$|z\rangle := e^{-|z|^2/2} \sum_{n=0}^{\infty} \frac{z^n}{\sqrt{n!}} |n\rangle = e^{-|z|^2/2} e^{za^\dagger} |0\rangle, \quad (9.129)$$

where $|n\rangle$ are the harmonic-oscillator number eigenstates of Chapter 5 ((5.24)) and $a^\dagger$ is the creation operator. The state is normalized: $\langle z|z\rangle = 1$. Its defining property is that it is an eigenstate of the annihilation operator:

$$a|z\rangle = z|z\rangle, \quad \langle z|a^\dagger = \bar{z}\langle z|. \quad (9.130)$$

No two coherent states are orthogonal; they are an overcomplete set spanning the entire oscillator Hilbert space. The complex number z is the classical analogue of the oscillator amplitude: $\langle z|Q|z\rangle = \sqrt{2\hbar/m\omega} \operatorname{Re} z$ and $\langle z|P|z\rangle = \sqrt{2\hbar m\omega} \operatorname{Im} z$, so the phase-space coordinates are $q = \sqrt{2\hbar/m\omega} \operatorname{Re} z$, $p = \sqrt{2\hbar m\omega} \operatorname{Im} z$.

Overlap and resolution of identity. The inner product of two coherent states is

$$\langle z_1 | z_2 \rangle = e^{-|z_1|^2/2 - |z_2|^2/2 + \bar{z}_1 z_2}. \quad (9.131)$$

The states resolve the identity on the oscillator Hilbert space:

$$\frac{1}{\pi} \int d^2 z |z\rangle \langle z| = I, \quad d^2 z := d(\operatorname{Re} z) d(\operatorname{Im} z), \quad (9.132)$$

with the integral over the full complex plane. The proof is one line in the number basis: $\frac{1}{\pi} \int d^2 z e^{-|z|^2} \frac{\bar{z}^m z^n}{\sqrt{m! n!}} = \delta_{mn}$, by converting to polar coordinates $z = r e^{i\phi}$; the radial integral yields $n!$ and cancels the factorial in the normalization. The measure $d^2 z / \pi$ assigns weight $1/\pi$ per complex dimension.

9.8.2 The Bosonic Coherent-State Path Integral Construction

Time-slicing. For a Hamiltonian $H(a^\dagger, a)$ expressed in terms of creation and annihilation operators, the evolution operator is sliced with coherent-state resolutions of identity. With N steps of duration $\varepsilon = t/N$:

$$U(t) = \lim_{N \rightarrow \infty} \int \prod_{k=1}^{N-1} \frac{d^2 z_k}{\pi} \prod_{k=0}^{N-1} |z_{k+1}\rangle \langle z_{k+1}| e^{-i\varepsilon H/\hbar} |z_k\rangle, \quad (9.133)$$

with the boundary data $|z_0\rangle$ and $\langle z_N|$ (or, for the kernel between coherent states, both boundaries specified).

Short-time kernel. For small ε , the short-time amplitude is

$$\langle z_{k+1} | e^{-i\varepsilon H(a^\dagger, a)/\hbar} | z_k \rangle \approx \langle z_{k+1} | z_k \rangle e^{-i\varepsilon H(\bar{z}_{k+1}, z_k)/\hbar}, \quad (9.134)$$

where $H(\bar{z}, z) := \langle z | H(a^\dagger, a) | z \rangle$ is the expectation value in the coherent state — the normal-ordered symbol of the Hamiltonian. The error is $O(\varepsilon^2)$ by the Trotter logic of Section 9.1. The overlap supplies the kinetic (“ $\bar{z}\dot{z}$ ”) term. For a single slice, with $z_{k+1} - z_k = \Delta z_k$,

$$\langle z_{k+1} | z_k \rangle = \exp\left[-\frac{1}{2}|z_{k+1}|^2 - \frac{1}{2}|z_k|^2 + \bar{z}_{k+1} z_k\right] = \exp\left[\frac{1}{2}(\bar{z}_{k+1} \Delta z_k - \bar{\Delta} z_k z_k)\right]. \quad (9.135)$$

The continuum action. In the limit $\varepsilon \rightarrow 0$, $N \rightarrow \infty$, the exponent of the N -slice product assembles into the action integral:

$$K(\bar{z}, t; z', 0) = \int_{z(0)=z'}^{\bar{z}(t)=\bar{z}} \mathcal{D}[z, \bar{z}] \exp\left[\frac{i}{\hbar} S[z, \bar{z}]\right], \quad (9.136)$$

with the action in holomorphic form:

$$S[z, \bar{z}] = \int_0^t dt' \left[\frac{i\hbar}{2} (\bar{z}\dot{z} - \dot{\bar{z}}z) - H(\bar{z}, z) \right]. \quad (9.137)$$

The kinetic term $\frac{i\hbar}{2} (\bar{z}\dot{z} - \dot{\bar{z}}z)$ is the symplectic (Berry-phase) term for the complex plane; it is $i\hbar \bar{z}\dot{z}$ up to a total time derivative, so the action is conventionally written

$$S[z, \bar{z}] = \int_0^t dt' [i\hbar \bar{z}\dot{z} - H(\bar{z}, z)], \quad (9.138)$$

the difference being an endpoint contribution that does not affect the equations of motion. The symplectic form is $\omega = i\hbar d\bar{z} \wedge dz$, corresponding to the Poisson bracket $\{z, \bar{z}\} = 1/i\hbar$, which is the classical limit of the canonical commutator $[a, a^\dagger] = 1$.

The measure. The measure is the formal notation for the limiting discretized product:

$$\int \mathcal{D}[z, \bar{z}] := \lim_{N \rightarrow \infty} \prod_{k=1}^{N-1} \int \frac{d^2 z_k}{\pi}, \quad (9.139)$$

exactly as the position-space measure $\int \mathcal{D}x$ was defined in Section 9.2.

9.8.3 Example: The Harmonic Oscillator Revisited

For the harmonic oscillator, the (normal-ordered) Hamiltonian in coherent-state language is $H(\bar{z}, z) = \hbar\omega \bar{z}z$. The action (9.138) becomes

$$S[z, \bar{z}] = \int_0^t dt' [i\hbar \bar{z} \dot{z} - \hbar\omega \bar{z}z]. \quad (9.140)$$

Classical equation. Varying $\bar{z}$ and z independently:

$$\frac{\delta S}{\delta \bar{z}} = 0 : \quad i\hbar \dot{z} - \hbar\omega z = 0 \implies z(t) = z_0 e^{-i\omega t}. \quad (9.141)$$

The classical trajectory is uniform circular motion in the complex plane, corresponding to the oscillator's elliptical orbit in phase space; the frequency is ω , exactly the quantum level spacing.

Gaussian path integral. The action is quadratic in z and $\bar{z}$. The path integral is a Gaussian functional integral and is therefore exactly evaluable. The discretized form of the action on the N -slice lattice is a quadratic form in the complex variables $\{z_k, \bar{z}_k\}$; integrating over all intermediate z_k by repeated Gaussian integration (the complex analogue of the free-particle integration in Section 9.2) yields the kernel between coherent states:

$$K(\bar{z}, t; z', 0) = \exp[\bar{z} z' e^{-i\omega t}], \quad (9.142)$$

up to a normalization factor. Expanding the exponential in a Taylor series,

$$K(\bar{z}, t; z', 0) = \sum_{n=0}^{\infty} \frac{(\bar{z} e^{-i\omega t/2})^n}{\sqrt{n!}} \frac{(z' e^{-i\omega t/2})^n}{\sqrt{n!}} = \sum_{n=0}^{\infty} \langle \bar{z} | n \rangle \langle n | z' \rangle e^{-i\omega(n+1/2)t}, \quad (9.143)$$

which reproduces the harmonic oscillator spectrum $E_n = \hbar\omega(n+1/2)$ — the third independent derivation in this book, after the ladder algebra of Chapter 5 ((5.24)) and the Mehler kernel of Section 9.4 ((9.64)). The three derivations check one another, and each reveals a different face of the oscillator: algebraic structure, classical action and fluctuations, and the coherent-state geometry.

9.8.4 Generating Functional and Wick's Theorem

Source terms. Introduce external sources $J(t), \bar{J}(t)$ coupled linearly to $\bar{z}$ and z :

$$Z[J, \bar{J}] := \int \mathcal{D}[z, \bar{z}] \exp\left[\frac{i}{\hbar} \int_0^t dt' (i\hbar \bar{z} \dot{z} - H(\bar{z}, z) + \bar{J}z + \bar{z}J)\right]. \quad (9.144)$$

$Z[J, \bar{J}]$ is the **generating functional** of correlation functions. Functional derivatives with respect to the sources pull down factors of z and $\bar{z}$ from the exponent:

$$\langle z(t_1) \bar{z}(t_2) \rangle := \frac{1}{Z[0, 0]} \frac{\delta^2 Z[J, \bar{J}]}{\delta \bar{J}(t_1) \delta J(t_2)} \Big|_{J=\bar{J}=0}. \quad (9.145)$$

For a free theory ($H = \hbar\omega \bar{z}z$), $Z[J, \bar{J}]$ is a Gaussian functional integral and evaluates exactly: complete the square in the exponent, shift the integration variable, and use the normalization $Z[0, 0] = 1$ (up to a constant). The result is

$$Z[J, \bar{J}] = \exp\left[-\frac{i}{\hbar} \int dt_1 dt_2 \bar{J}(t_1) G(t_1, t_2) J(t_2)\right], \quad (9.146)$$

where $G(t_1, t_2)$ is the Green's function (the inverse of the kinetic operator $i\hbar \partial_t - \hbar\omega$). The two-point function (9.145) is then $G(t_1, t_2)$.

Wick's theorem from the path integral. For a free theory, all higher correlation functions factorize into sums of products of two-point functions. This is **Wick's theorem** in the path-integral register. The proof is identical to the combinatorial argument for a finite-dimensional Gaussian integral: each pair of sources $J, \bar{J}$ is contracted with a propagator, and the sum runs over all pairings. In the continuum, the statement is

$$\langle z(t_1) \cdots z(t_n) \bar{z}(s_1) \cdots \bar{z}(s_n) \rangle = \sum_{\text{pairings}} \prod_{\text{pairs } (i,j)} G(t_i, s_j). \quad (9.147)$$

This is the Gaussian-factorial structure that, in the interacting case, becomes the Feynman-diagram expansion. For an anharmonic perturbation $H_{\text{int}} = \lambda(\bar{z}z)^2$, the generating functional expands as a power series in λ :

$$Z[J, \bar{J}] = \exp\left[-\frac{i\lambda}{\hbar} \int dt \left(\frac{\hbar}{i} \frac{\delta}{\delta J} \frac{\hbar}{i} \frac{\delta}{\delta \bar{J}}\right)^2\right] Z_0[J, \bar{J}], \quad (9.148)$$

where the functional derivatives with respect to the sources represent the insertion of operators $\bar{z}z$ at each interaction vertex. Expanding the exponential generates the Feynman-diagram series.

9.8.5 From Mechanical Coherent States to the Functional Integral over Fields

The step from a single oscillator to a quantum field is the step from one complex coordinate $z(t)$ to a complex coordinate at every point in space, $\psi(\mathbf{x}, t)$:

$$z(t) \longrightarrow \psi(\mathbf{x}, t), \quad \bar{z}(t) \longrightarrow \bar{\psi}(\mathbf{x}, t), \quad \int dt \longrightarrow \int dt d^3x. \quad (9.149)$$

The bosonic coherent-state path integral becomes the functional integral over field configurations:

$$Z = \int \mathcal{D}[\psi, \bar{\psi}] \exp\left[\frac{i}{\hbar} \int dt d^3x (i\hbar \bar{\psi} \partial_t \psi - \mathcal{H}(\bar{\psi}, \psi))\right], \quad (9.150)$$

where $\mathcal{H}$ is the Hamiltonian density. This is the central object of quantum field theory — the generating functional from which every scattering amplitude, every correlation function, and every observable of a relativistic quantum field is computed. Its construction, for the Schrödinger field and for the quantized electromagnetic field, is the subject of Chapter 10.

Advantage and limitation. The advantage of the coherent-state path integral is its algebraic transparency: it unifies the operator methods of Chapter 5 (creation and annihilation operators, the number basis, the ladder algebra) with the path-integral formalism of this chapter, and it provides the template for the functional integral over fields. Its limitation is the same as the Feynman path integral's: the measure is formal, and outside the class of quadratic actions, evaluation relies on perturbation theory or the stationary-phase approximation. What the coherent-state path integral adds to the Feynman

path integral is the holomorphic (Bargmann) representation — the natural language for bosonic systems with a Fock-space structure, and the bridge from the non-relativistic quantum mechanics of this book to the relativistic quantum field theory of the next.

Chapter 9 constructed Feynman’s path integral from the propagator composition property planted in Chapter 5, time-sliced the evolution operator, and identified the classical action as the phase of the short-time kernel. The classical limit emerged from the stationary-phase approximation, the harmonic oscillator was solved exactly via the quadratic fluctuation determinant, the Aharonov-Bohm effect revealed the role of configuration-space topology, the WKB method derived the corrected Bohr-Sommerfeld quantization condition — repaying the book’s oldest debts, the spin path integral exhibited the Wess-Zumino term and derived spin quantization from path topology, and the bosonic coherent-state path integral unified the ladder algebra with the functional integral and supplied the template for quantum field theory. This section consolidates the chapter into principles, balances the ledger, exhibits the experiment–formalism dictionary, and poses the exit question that launches Chapter 10.

9.8.6 Five Principles

The chapter’s content compresses into seven principles that characterize the path-integral formulation of quantum mechanics.

- **(P1) Construction.** The Feynman path integral is constructed from the composition property of the propagator (Equation (5.40)) by time-slicing. The short-time kernel is the Trotter-product of kinetic and potential factors (Section 9.1). The $N \rightarrow \infty$ limit of the N -slice product defines the sum over all paths, each contributing the amplitude $e^{iS[x]/\hbar}$ where $S[x]$ is the classical action functional (Equations (9.10) and (9.13)). The measure $\int \mathcal{D}x$ is formal notation for this limiting procedure; its sole rigorous content is the discretized Lebesgue product.
- **(P2) Classical limit.** When $S \gg \hbar$, the stationary-phase approximation selects paths near the classical trajectory satisfying $\delta S = 0$. Hamilton’s principle (a postulate in classical mechanics) emerges from quantum superposition: the classical path is the one whose neighboring paths interfere constructively. The semiclassical (Van Vleck) propagator carries the classical action as its phase and the Van Vleck determinant as its prefactor (Equation (9.28)). The Hamilton-Jacobi equation for the classical action is recovered as the $\hbar \rightarrow 0$ limit of the Schrodinger propagator’s phase (Equation (9.32)), repaying Chapter 1’s Hamilton-Jacobi debt.
- **(P3) Quadratic actions.** For Lagrangians at most quadratic in coordinates and velocities, the stationary-phase approximation is exact: the fluctuation integral is Gaussian, and the prefactor is the reciprocal square root of the functional determinant of the Jacobi operator. The harmonic oscillator’s exact propagator — the Mehler kernel (Equation (9.60)) — is computed from the classical action and the fluctuation determinant. Its spectral (Fourier) expansion recovers the energy spectrum $E_n = \hbar\omega(n + 1/2)$ independently of the operator ladder method (Equation (9.64)). The Maslov index $-\pi/2$ acquired at each half-period (Equation (9.61)) counts the caustics where the Van Vleck determinant diverges.
- **(P4) Path topology.** The Aharonov-Bohm effect demonstrates that the path integral registers

global properties of the configuration space inaccessible to the canonical formalism. Electrons passing around an impenetrable flux tube acquire a gauge-invariant topological phase shift $\Delta\phi = 2\pi\Phi/\Phi_0$ (Equation (9.79)), periodic in the flux quantum $\Phi_0 = 2\pi\hbar c/e$ (Equation (9.78)). The effect is measured even though $\mathbf{B} = 0$ everywhere the electrons travel. The propagator decomposes as a sum over homotopy classes weighted by the unitary representations of the fundamental group: $K = \sum_n e^{ine\Phi/\hbar c} K_n$. The phase factor is the abelian Wilson loop.

- **(P5) Semiclassical expansion.** The WKB method is the systematic $\hbar$ -expansion of the Schrodinger equation. The leading order reproduces the Hamilton-Jacobi equation for the characteristic function (Equation (9.91)); the next order gives the transport equation and the $1/\sqrt{p(x)}$ amplitude prefactor (Equation (9.92)); the wave function in the classically allowed and forbidden regions is given by Equation (9.93). At turning points, connection formulas (Equation (9.97)) supply the $-\pi/4$ Maslov phase per soft turning point. The single-valuedness condition produces the corrected Bohr-Sommerfeld quantization rule $\oint p dx = 2\pi\hbar(n + 1/2)$ (Equation (9.100)) — repaying Chapter 1, row 1.9 and supplying the zero-point energy the old quantum theory lacked. Tunneling through a barrier is exponentially suppressed, with the Gamow transmission factor $T \sim \exp[-(2/\hbar) \int dx \sqrt{2m(V - E)}]$ (Equation (9.105)).
- **(P6) Spin path integral.** The continuum path integral for a spin degree of freedom is built on the overcomplete basis of spin coherent states $|\hat{n}\rangle$ on the Bloch sphere S^2 (Equations (9.116), (9.118)). The effective action (Equation (9.123)) consists of the geometric Wess-Zumino (Berry phase) term and the classical Hamiltonian. The WZ term $S_{WZ} = \hbar s \int dt (1 - \cos\theta) \dot{\phi}$ (Equation (9.124)) measures the solid angle swept by the spin trajectory. Its single-valuedness on the sphere quantizes the spin: $2s \in \mathbb{N}$ (Equation (9.125)). The stationary-phase equation is the Landau-Lifshitz equation $\dot{\hat{n}} = \gamma \hat{n} \times \mathbf{B}$, the classical limit of spin dynamics (Section 9.7).
- **(P7) Bosonic coherent-state path integral.** The holomorphic path integral is built on the overcomplete basis of harmonic oscillator coherent states $|z\rangle$ (Equations (9.129), (9.132)). The action takes the canonical form $S[z, \bar{z}] = \int dt [i\hbar \bar{z} \dot{z} - H(\bar{z}, z)]$ (Equation (9.138)), whose symplectic term encodes the canonical commutator $[a, a^\dagger] = 1$. The generating functional $Z[J, \bar{J}]$ (Equation (9.144)) produces correlation functions by functional differentiation. Wick's theorem (Equation (9.147)) is the Gaussian factorization of all higher correlations. The continuum limit $z(t) \rightarrow \psi(\mathbf{x}, t)$ (Equation (9.149)) transforms the mechanical path integral into the functional integral over fields (Equation (9.150)) — the template for Chapter 10's quantum field theory (Section 9.8).

9.8.7 Ledger Balanced

The debts inherited from earlier chapters and the debts planted forward are now accounted.

Debts repaid in this chapter:

- Hamilton-Jacobi theory / action principle (Chapter 1): derived as the $\hbar \rightarrow 0$ limit of the propagator phase (Section 9.3, Equation (9.32)) and as the leading WKB order (Section 9.6, Equation (9.91)).

- Bohr-Sommerfeld quantization $\oint p_i dq_i = n_i h$ (Chapter 1, row 1.9): derived from the single-valuedness of the WKB wave function, corrected by the Maslov index $1/2$ (Section 9.6, Equation (9.100)). 10347
- Propagator composition $\rightarrow$ path integral construction (Chapter 5, Equation (5.40)): time-slicing construction of Sections 9.1–9.2 turns the composition property into the path sum. 10348
- Free kernel's action phase (Chapter 5, Equation (5.42)): verified by the N -slice construction in Section 9.2; the phase $m(x - x')^2/2t$ is the classical action of the straight constant-velocity path. 10349
- De Broglie packet dispersion (Chapter 5, Equation (5.43)): reproduced from stationary phase (Section 9.3) and WKB (Section 9.6). 10350
- Harmonic oscillator spectrum $E_n = \hbar\omega(n + 1/2)$ (Chapter 5, Equation (5.24)): independently recovered from the Mehler kernel's spectral expansion (Section 9.4, Equation (9.64)) and from WKB quantization (Section 9.6, Equation (9.103)). 10351
- Planck's material oscillator / zero-point energy (Chapter 1, rows 1.3/1.5): the $1/2$ is the Maslov-index correction to the old Bohr-Sommerfeld rule, derived in Section 9.6. 10352
- Spin quantization (Chapter 6, angular-momentum algebra): $2s \in \mathbb{N}$ derived from the single-valuedness of the spin path integral on the Bloch sphere (Section 9.7, Equation (9.125)) — the topological face of the angular-momentum ladder spectrum. 10353
- Harmonic oscillator number spectrum (Chapter 5, Equation (5.24)): recovered a third time from the coherent-state path integral (Section 9.8, Equation (9.143)), cross-checking both the ladder-algebra and the Mehler-kernel derivations. 10354

Debts planted forward by this chapter:

- Functional integral over fields: the path integral over $x(t)$ is the mechanical template for the field-theoretic functional integral $\int \mathcal{D}\phi e^{iS[\phi]/\hbar}$ of Chapter 10. 10355
- Euclidean (imaginary-time) path integral: the Wick rotation $t \rightarrow -i\tau$ converts the oscillatory Feynman measure to the well-defined Wiener measure; named in Section 9.2, owed by Chapter 10. 10356
- Coherent-state path integral: the oscillator's phase-space path integral is the bridge to the holomorphic (Bargmann) path integral for bosonic fields; named in Section 9.4, constructed in Sections 9.7 (spin) and 9.8 (bosonic), and consumed by Chapter 10. 10357
- Spin path integrals and topological terms: the Wess-Zumino term of the spin path integral is the mechanical prototype of topological terms in field theory (WZW, Chern-Simons); named in Section 9.7, owed by gauge field theory. 10358
- Loop expansion / saddle-point methods: the stationary-phase plus fluctuation-determinant logic of Sections 9.3–9.4 is the $\hbar$ -expansion (loop expansion) of field theory; $\hbar$ counts loops. 10359
- Instanton methods: the imaginary-time classical solutions that govern tunneling between degenerate minima, named in Section 9.6; owed by advanced quantum mechanics. 10360
- Wilson loops: the AB phase factor $\exp[(ie/\hbar c) \oint \mathbf{A} \cdot d\mathbf{l}]$ is the abelian prototype of the gauge-invariant observables of Yang-Mills theory; named in Section 9.5. 10361

9.8.8 Experiment–Formalism Dictionary

Table 9.2 extends the dictionaries of Chapters 2–6. Each row pairs an experimental fact or classical concept with the path-integral object that represents it. The dictionary makes explicit what the path-integral formulation adds: a direct correspondence between classical mechanical quantities (the action, the path, the enclosed flux) and quantum amplitudes.

Table 9.2: The experiment–formalism dictionary v6. The rows added by the path-integral formulation connect classical concepts directly to quantum amplitudes. This table extends the dictionaries of Chapters 2–6.

Physical Concept	Path-Integral Register
Short waiting time	Trotter-factorized short-time kernel
Classical path $\delta S = 0$	Stationary-phase condition
Classical action S_c	Propagator phase $S_c/\hbar$
All conceivable histories	Feynman sum $\int \mathcal{D}x e^{iS/\hbar}$
Caustic (focal point)	Maslov-index phase $e^{-i\pi/2}$
Enclosed magnetic flux Φ	Topological phase $2\pi\Phi/\Phi_0$
Quantized energy spectrum	Single-valued WKB wave function
Tunnel barrier	Exponential suppression $e^{-\int p dx/\hbar}$
Hamilton-Jacobi equation	$\hbar \rightarrow 0$ limit of Schrodinger propagator
Bohr-Sommerfeld rule $\oint p dq = nh$	Corrected rule $\oint p dx = 2\pi\hbar(n + 1/2)$
Free-particle dispersion	Quadratic fluctuations around stationary phase
Harmonic-oscillator period	Mehler-kernel caustics at $t = k\pi/\omega$
Gauge transformation $\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi$	Overall phase of propagator (no observable effect)
Spin trajectory on Bloch sphere	Path $\hat{n}(t)$ weighted by $e^{iS_{\text{WZ}}/\hbar}$
Wess-Zumino term	Geometric phase $\hbar s \int (1 - \cos\theta)\dot{\phi}$
Coherent-state label $z(t)$	Holomorphic path with action $\int [i\hbar\bar{z}\dot{z} - H]$
Generating functional $Z[J, \bar{J}]$	Source-coupled path integral
Wick's theorem	Gaussian factorization of $\langle z \cdots \bar{z} \cdots \rangle$

9.8.9 Exit Question

The path integral sums over every conceivable history of a mechanical system — every continuous path $x(t)$ from the initial to the final configuration, weighted by the amplitude $e^{iS[x]/\hbar}$. The sum is over a one-dimensional trajectory: a single function of time.

Chapter 10 extends the formalism to systems whose configuration space is infinite-dimensional from the start: a field $\phi(\mathbf{x}, t)$ is a dynamical variable at every point in space, and its state at one instant

of time is a function $\phi(\mathbf{x})$, not a finite collection of numbers. The construction of Section 9.2 is the template: the time-slicing procedure that turned the composition property of the mechanical propagator into the path integral over $x(t)$ generalizes to a procedure that turns the composition property of the field-theoretic evolution operator into a functional integral over field configurations $\phi(\mathbf{x}, t)$. The measure $\int \mathcal{D}\phi$ is defined by discretizing space as well as time; the classical action $S[\phi]$ is the space-time integral of the Lagrangian density; and the stationary-phase approximation selects the classical field equation.

The question the chapter leaves open is the one its construction was designed to raise: what replaces the sum over mechanical paths when the system to be quantized is already a continuum from the beginning? The answer is the subject of the next chapter. The path integral over field configurations is the language in which quantum electrodynamics, the Standard Model, and every relativistic quantum field theory are formulated. Chapter 9 has built the scaffolding; Chapter 10 erects the structure.

9.9 Exercises for Chapter 9

Chapter 9 Exercise

1. Free-particle path integral by induction. (★, Section 9.2)

Consider the N -slice discretized path integral for a free particle ($V = 0$):

$$K_0^{(N)}(x, t; x', 0) = \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{N/2} \int_{-\infty}^{\infty} dx_1 \cdots dx_{N-1} \exp \left[\frac{im}{2\hbar \varepsilon} \sum_{k=0}^{N-1} (x_{k+1} - x_k)^2 \right], \quad (9.151)$$

with $x_0 = x'$, $x_N = x$, and $\varepsilon = t/N$.

(a) For $N = 2$, evaluate the single Gaussian integral over x_1 explicitly and show that

$$K_0^{(2)}(x, t; x', 0) = \sqrt{\frac{m}{2\pi i \hbar t}} \exp \left[\frac{im}{2\hbar t} (x - x')^2 \right]. \quad (9.152)$$

Verify that the prefactor contracts from $(m/2\pi i \hbar \varepsilon)$ to $(m/2\pi i \hbar t)^{1/2}$ after one integration.

(b) Prove by induction on N that

$$K_0^{(N)}(x, t; x', 0) = \sqrt{\frac{m}{2\pi i \hbar t}} \exp \left[\frac{im}{2\hbar t} (x - x')^2 \right] \quad (9.153)$$

for every $N \geq 2$. The result is independent of the number of slices; the limit $N \rightarrow \infty$ is trivial.

(c) Explain why this independence on N verifies that the Trotter error vanishes identically for $V = 0$: the short-time free-particle kernel is exact at finite ε . Contrast this with the case $V \neq 0$, where the N -slice product at finite N differs from the true propagator by $O(\varepsilon)$ corrections.

2. Trotter expansion to $O(\varepsilon^2)$. (★, Section 9.1)

For the Hamiltonian $H = P^2/2m + V(Q)$ with $[Q, P] = i\hbar I$, expand both sides of

$$e^{-i\varepsilon H/\hbar} \quad \text{and} \quad e^{-i\varepsilon P^2/2m\hbar} e^{-i\varepsilon V(Q)/\hbar} \quad (9.154)$$

to order ε^2 using the Baker-Campbell-Hausdorff (BCH) formula.

(a) Show that

$$e^{-i\varepsilon(P^2/2m+V)/\hbar} = I - \frac{i\varepsilon}{\hbar} \left(\frac{P^2}{2m} + V \right) - \frac{\varepsilon^2}{2\hbar^2} \left(\frac{P^2}{2m} + V \right)^2 + O(\varepsilon^3). \quad (9.155)$$

(b) Show that

$$e^{-i\varepsilon P^2/2m\hbar} e^{-i\varepsilon V/\hbar} = I - \frac{i\varepsilon}{\hbar} \left(\frac{P^2}{2m} + V \right) - \frac{\varepsilon^2}{2\hbar^2} \left(\frac{P^2}{2m} + V \right)^2 + \frac{\varepsilon^2}{2\hbar^2} \left[\frac{P^2}{2m}, V \right] + O(\varepsilon^3). \quad (9.156)$$

(c) The difference between the two is $\frac{\varepsilon^2}{2\hbar^2} [P^2/2m, V] + O(\varepsilon^3)$. Compute the commutator $[P^2/2m, V]$ using the canonical commutator $[Q, P] = i\hbar I$ (Equation (5.12)). Show that it is of order $\hbar$, so that the Trotter error term is $O(\varepsilon^2)$ inside the exponent. Confirm that the Lie bracket formula (Equation (4.20)) is the finite-dimensional special case of this result.

3. Stationary phase on the free propagator with a Gaussian initial state. (★★, Section 9.3)

An initial Gaussian wave packet at $t = 0$ is

$$\psi(x', 0) = \frac{1}{(2\pi\sigma^2)^{1/4}} \exp \left[-\frac{x'^2}{4\sigma^2} + \frac{i}{\hbar} p_0 x' \right]. \quad (9.157)$$

The time-evolved wave function is obtained by convolving with the free-particle propagator (Equation (5.42)):

$$\psi(x, t) = \sqrt{\frac{m}{2\pi i \hbar t}} \int_{-\infty}^{\infty} dx' \exp \left[\frac{im}{2\hbar t} (x - x')^2 \right] \psi(x', 0). \quad (9.158)$$

- Write the exponent as $(i/\hbar)f(x')$ and find the stationary-phase point x'_c satisfying $f'(x'_c) = 0$. Show that $x = x'_c + p_0 t/m$, i.e., the stationary-phase point follows the classical trajectory.
- Expand $f(x')$ to quadratic order around x'_c and evaluate the resulting Gaussian integral exactly. Verify that the result at $O(\hbar^0)$ in the phase reproduces the classical motion, and that the $O(\hbar)$ terms reproduce the spreading law (Equation (5.43)).
- Compare your result with the exact propagation of the Gaussian packet derived in Chapter 5, Section 5.9. Confirm that the stationary-phase approximation is exact for this case and explain why.

4. Linear potential: classical path and propagator. (★★, Section 9.3)

A particle of mass m in a uniform gravitational field has the potential $V(x) = mgx$. The Lagrangian is $L = \frac{m}{2}\dot{x}^2 - mgx$.

- Solve the Euler-Lagrange equation with boundary conditions $x(0) = x'$, $x(t) = x$. Show that the classical path is

$$x_c(t') = x' + \left(\frac{x - x'}{t} + \frac{gt}{2} \right) t' - \frac{g}{2} t'^2, \quad (9.159)$$

and that the classical action evaluated on this path is

$$S_c(x, t; x', 0) = \frac{m}{2t} (x - x')^2 - \frac{mgt}{2} (x + x') - \frac{mg^2 t^3}{24}. \quad (9.160)$$

- Show that the second functional derivative (Jacobi operator) of the action for this potential is identical to that of the free particle: $\delta^2 S / \delta x(t') \delta x(t'') = -m d^2/dt'^2$. Conclude that the

fluctuation determinant is the same as the free particle's, so that the exact propagator is 10463

$$K(x, t; x', 0) = \sqrt{\frac{m}{2\pi i \hbar t}} \exp \left[\frac{i}{\hbar} S_c(x, t; x', 0) \right]. \quad (9.161) \quad 10464$$

- (c) Verify that this propagator satisfies the Schrodinger equation. (Hint: the action S_c is a quadratic polynomial in x and x' ; compute its derivatives and check that the prefactor and phase together satisfy $i\hbar \partial_t K = -(\hbar^2/2m)\partial_x^2 K + mgx K$.) 10465

- (d) Explain why the stationary-phase (Van Vleck) form is exact for any potential that is at most linear in x . 10466

5. Harmonic oscillator fluctuation determinant by discretization. (★★★, Section 9.4) 10467

For the harmonic oscillator $V(x) = \frac{1}{2}m\omega^2 x^2$, the N -slice discretized path integral after expanding around the classical path $x_c(t_k)$ and changing variables to the fluctuation $y_k = x_k - x_c(t_k)$ (with $y_0 = y_N = 0$) takes the form 10470

$$\int dy_1 \cdots dy_{N-1} \exp \left[\frac{im}{2\hbar\varepsilon} \sum_{k=0}^{N-1} (y_{k+1} - y_k)^2 - \frac{i\varepsilon m\omega^2}{2\hbar} \sum_{k=1}^{N-1} y_k^2 \right]. \quad (9.162) \quad 10471$$

- (a) Rewrite the sum $\sum_{k=0}^{N-1} (y_{k+1} - y_k)^2 = \sum_{i,j=1}^{N-1} y_i M_{ij} y_j$ where M is the $(N-1) \times (N-1)$ tridiagonal matrix 10472

$$M_{ij} = 2\delta_{ij} - \delta_{i,j+1} - \delta_{i+1,j}. \quad (9.163) \quad 10473$$

Show that the full quadratic form in the exponent is $(im/2\hbar\varepsilon) \sum_{i,j} y_i [M - (\varepsilon\omega)^2 I]_{ij} y_j$. 10474

- (b) The eigenvalues of M are $\lambda_k = 4 \sin^2(k\pi/2N)$ for $k = 1, 2, \dots, N-1$. Using this fact (derived by diagonalizing M with the discrete sine transform), show that the Gaussian integral over $y_1, \dots, y_{N-1}$ evaluates to 10475

$$\left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{(N-1)/2} \prod_{k=1}^{N-1} [\lambda_k - (\varepsilon\omega)^2]^{-1/2}. \quad (9.164) \quad 10476$$

- (c) Multiply by the classical-action phase and the N short-time prefactors. Show that the total prefactor is 10477

$$\left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} \prod_{k=1}^{N-1} \left[1 - \frac{(\varepsilon\omega)^2}{\lambda_k} \right]^{-1/2}. \quad (9.165) \quad 10478$$

- (d) Take the limit $N \rightarrow \infty$ with $\varepsilon = t/N$ fixed. Use the product formula 10479

$$\prod_{k=1}^{N-1} \left[1 - \frac{(\varepsilon\omega)^2}{4 \sin^2(k\pi/2N)} \right] \longrightarrow \frac{\sin \omega t}{\omega t} \quad \text{as } N \rightarrow \infty, \quad (9.166) \quad 10480$$

to recover the Mehler-kernel prefactor $\sqrt{m\omega/2\pi i \hbar \sin \omega t}$ of Equation (9.60). Verify the normalization by checking the free-particle limit $\omega \rightarrow 0$. 10481

- (e) This is the discretized route to the functional determinant. The Gelfand-Yaglom theorem (stated in Section 9.4) provides the continuum shortcut. Comment on the relationship between the two approaches. 10482

6. Mehler kernel spectral expansion. (★★, Section 9.4) 10483

The Mehler kernel (Equation (9.60)) has two complementary representations: the closed-form 10484

position-space propagator and the spectral sum (Equation (9.64)):

$$K(x, t; x', 0) = \sum_{n=0}^{\infty} \psi_n(x) \psi_n^*(x') e^{-iE_n t/\hbar}. \quad (9.167)$$

- (a) Consider the equal-point ($x = x'$) limit of the Mehler kernel. Write $K(x, t; x, 0)$ explicitly in terms of $\sin \omega t$ and $\cos \omega t$. Take the limit $t \rightarrow -i\infty$ (formally: analytic continuation to imaginary time). Show that the leading term as $t \rightarrow -i\infty$ is

$$K(x, -i\tau; x, 0) \sim \sqrt{\frac{m\omega}{\pi\hbar}} e^{-m\omega x^2/\hbar} e^{-\omega\tau/2} \quad (\tau \rightarrow \infty). \quad (9.168)$$

- (b) Identify the coefficient of $e^{-E_0\tau/\hbar}$ (with $\tau = -it$) as $|\psi_0(x)|^2$. Extract the ground-state energy $E_0 = \hbar\omega/2$ and the ground-state wave function $\psi_0(x) = (m\omega/\pi\hbar)^{1/4} e^{-m\omega x^2/2\hbar}$. Compare with Chapter 5, Equation (5.26).
- (c) Expand the Mehler kernel as a power series in $e^{-i\omega t}$ (using the generating function of Hermite polynomials, or Mehler's formula). Identify the coefficient of $e^{-iE_n t/\hbar}$ and verify that $E_n = \hbar\omega(n + 1/2)$. Extract the first excited-state wave function $\psi_1(x) \propto x e^{-m\omega x^2/2\hbar}$ and confirm its normalization.

7. Aharonov-Bohm two-slit intensity. (★★, Section 9.5)

Consider the double-slit geometry of Section 9.5: slit separation d , screen distance $L \gg d$, electron de Broglie wavelength $\lambda = 2\pi\hbar/\sqrt{2mE}$, and a solenoid carrying flux Φ placed midway between the slits.

- (a) Write the total amplitude at a point x on the screen as the sum of two contributions, each of the free-particle form $\sim e^{ikr}/r$ multiplied by the AB phase factor $e^{ie\phi_{1,2}/\hbar c}$, with $r_{1,2} \approx L + (x \mp d/2)^2/2L$ the path lengths. The geometric phase from the path-length difference is $k(r_1 - r_2) \approx kdx/L$.
- (b) Derive the intensity formula (Equation (9.85)):

$$I(x) = I_0(x) \left[1 + \cos \left(\frac{2\pi dx}{\lambda L} + \frac{e\Phi}{\hbar c} \right) \right], \quad (9.169)$$

where $I_0(x)$ is the single-slit diffraction envelope (which may be taken as constant over a few fringes near the center).

- (c) Show that when Φ increases by one flux quantum $\Phi_0 = 2\pi\hbar c/e$, the AB phase term $e\Phi/\hbar c$ advances by 2π , and the interference pattern shifts by exactly one fringe spacing $\Delta x = \lambda L/d$.
- (d) If the flux is $\Phi = \Phi_0/2$, the AB phase is π . Describe the resulting interference pattern. At what values of Φ does the pattern coincide with the $\Phi = 0$ pattern?

8. WKB harmonic oscillator. (★★★, Section 9.6)

Complete the WKB quantization of the harmonic oscillator $V(x) = \frac{1}{2}m\omega^2 x^2$ in full detail.

- (a) For a given energy E , the turning points are $x_{\pm} = \pm\sqrt{2E/m\omega^2}$. Write the action integral as

$$\oint p(x) dx = 2 \int_{-x_+}^{x_+} dx \sqrt{2m \left(E - \frac{1}{2}m\omega^2 x^2 \right)}. \quad (9.170)$$

Evaluate the integral by the substitution $x = \sqrt{2E/m\omega^2} \sin \theta$. Show step by step that

$$\oint p dx = 2\pi E/\omega.$$

- (b) Apply the corrected Bohr-Sommerfeld condition (Equation (9.100)) to obtain $E_n = \hbar\omega(n + 1/2)$.
- (c) The higher-order WKB terms ($S_2, S_3, \dots$) involve integrals of derivatives of $p(x)$. For the harmonic oscillator, show that $S_2(x)$ is a constant (independent of x), and that all higher S_k for $k \geq 3$ vanish identically. Conclude that the WKB series terminates at $O(\hbar)$ and is therefore exact. (Hint: $p(x) = \sqrt{2mE - m^2\omega^2 x^2}$, so $p'(x) \propto x/p$, $p''(x) \propto E/p^3$, and all required integrals reduce to elementary forms.)
- (d) The exactness of WKB for the harmonic oscillator is a special case of a general theorem: WKB is exact for any potential whose higher functional derivatives of the action vanish (i.e., for quadratic actions). Identify the connection with the path-integral result of Section 9.4: both the stationary-phase approximation and the WKB expansion are exact when the action is at most quadratic.

9. Gamow α -decay: Gamow factor and Geiger-Nuttall law. (***, Section 9.6)

Consider an α -particle of mass m and kinetic energy E tunneling through the Coulomb barrier of a daughter nucleus of charge Z . The potential for $r > R$ is $V(r) = 2Ze^2/r$ (Gaussian units).

- (a) The outer turning point r_2 is defined by $V(r_2) = E$. Show that $r_2 = 2Ze^2/E$. The inner turning point is the nuclear radius $R \approx r_0 A^{1/3}$ with $r_0 \approx 1.2$ fm; for α -decay of heavy nuclei, $R \ll r_2$.
- (b) The WKB transmission probability (Equation (9.105)) gives the Gamow factor:

$$G = \frac{2}{\hbar} \int_R^{r_2} dr \sqrt{2m \left(\frac{2Ze^2}{r} - E \right)}. \quad (9.171)$$

Change variables to $u = \sqrt{r/r_2}$ (or more conveniently, $r = r_2 \cos^2 \theta$) and evaluate the integral. Show that

$$G = \frac{4Ze^2}{\hbar} \sqrt{\frac{2m}{E}} \left[\arccos \sqrt{\frac{E}{B}} - \sqrt{\frac{E}{B} \left(1 - \frac{E}{B} \right)} \right], \quad (9.172)$$

with $B = 2Ze^2/R$ the Coulomb barrier height at the nuclear surface.

- (c) For typical α -decay energies ($E \sim 4\text{--}9$ MeV) and heavy nuclei ($Z \sim 80\text{--}100$, $B \sim 25\text{--}35$ MeV), we have $E \ll B$. Expand the Gamow factor for $E/B \ll 1$ to leading order:

$$G \approx \frac{2\pi Ze^2}{\hbar} \sqrt{\frac{2m}{E}} - \frac{8Ze^2}{\hbar} \sqrt{\frac{2m}{B}}. \quad (9.173)$$

The first term dominates the E -dependence.

- (d) The α -decay constant is $\lambda = f \cdot T$ where $f \sim v/R$ is the frequency with which the α -particle strikes the barrier. Taking $\log \lambda \approx \log f - G$ and using the leading E -dependence of G , derive the Geiger-Nuttall law:

$$\log \lambda \approx a - \frac{b}{\sqrt{E}}, \quad (9.174)$$

where a and b are constants (with $b \propto Z$). This relation, discovered empirically in 1911, was the first quantitative success of quantum tunneling and Gamow's 1928 calculation of

it.

10. WKB for the quantum bouncer. (★★★, Section 9.6)

A particle of mass m bounces on an impenetrable floor at $x = 0$ under the influence of a uniform gravitational field $V(x) = mgx$ for $x > 0$. The wave function obeys $\psi(0) = 0$ (hard wall) and decays as $x \rightarrow \infty$.

- (a) The turning points are $x_1 = 0$ (hard wall) and $x_2 = E/mg$ (soft turning point where $V(x_2) = E$). The hard wall at $x = 0$ is not a soft turning point — it is an infinite barrier where the wave function vanishes, not a zero of $p(x)$. At a hard wall, the WKB phase shift is $-\pi/2$ (the wave reflects with a sign change $\psi \rightarrow -\psi$, equivalent to a π phase shift in the amplitude, or $-\pi/2$ in the round-trip phase counting). Alternatively, treat the hard wall as a boundary condition directly: the wave function is $\psi(x) \propto \sin[(1/\hbar) \int_0^x p dx']$, which vanishes at $x = 0$.
- (b) Starting from $\psi(x) = (C/\sqrt{p(x)}) \sin[(1/\hbar) \int_0^x p(x') dx']$ in the allowed region and applying the soft-turning-point connection formula at x_2 , derive the quantization condition for a potential with one hard wall and one soft turning point:

$$\frac{1}{\hbar} \int_0^{x_2} p(x) dx = \pi \left(n - \frac{1}{4} \right), \quad n = 1, 2, 3, \dots \quad (9.175)$$

The $-\pi/4$ comes from the single soft turning point; the $-\pi/2$ from the hard wall combines with the $2\pi n$ round-trip phase to produce the $n - 1/4$ form.

- (c) For the linear potential $V = mgx$, the local momentum is $p(x) = \sqrt{2m(E - mgx)}$. Evaluate the integral:

$$\int_0^{E/mg} dx \sqrt{2m(E - mgx)} = \frac{2}{3} \frac{\sqrt{2mE^{3/2}}}{mg}. \quad (9.176)$$

Apply the quantization condition to obtain the WKB energy levels:

$$E_n \approx \left[\frac{3\pi}{2} \left(n - \frac{1}{4} \right) \right]^{2/3} \left(\frac{mg^2 \hbar^2}{2} \right)^{1/3}. \quad (9.177)$$

- (d) The exact Schrodinger equation for this problem reduces to the Airy equation. The exact energies are $E_n = -a_n(mg^2 \hbar^2/2)^{1/3}$ where a_n are the zeros of the Airy function $\text{Ai}(-a_n) = 0$ ($a_1 \approx 2.338$, $a_2 \approx 4.088$, $a_3 \approx 5.521$). Compare the WKB prediction for $n = 1, 2, 3$ with the exact Airy zeros. Compute the relative error and comment on the accuracy of WKB even for the ground state ($n = 1$), where one might expect the semiclassical approximation to be poor.
- (e) Explain why the quantum bouncer (one hard, one soft turning point) has the quantization condition $n - 1/4$ rather than the $n + 1/2$ of a potential with two soft turning points (Equation (9.100)). Relate the difference to the counting of Maslov indices.

Chapter 10 The Second Quantization, and the Way towards Statistical Mechanics



Note This chapter adds one genuinely new postulate — the Symmetrization Postulate of Section 10.1, which restricts the states of identical particles to either the totally symmetric or the totally antisymmetric subspace of the N -fold tensor product, the choice dictated by the particle's spin. The spin-statistics theorem that proves this connection is a theorem of relativistic quantum field theory and is stated here as an experimental fact, with a footnote pointing to its proper home. Everything else in the chapter is proved from this postulate, the axioms of the state-space and measurement chapters, and the established formalism of the rotation, pictures, and representations chapters. The interlude chapters on angular momentum and composite systems are cited where their machinery is reused; no construction from those chapters is re-derived here. Two statistical ensembles — the grand canonical ensemble of Sections 10.4–10.5 and the canonical ensemble of Section 10.6 — rest on the density-operator formalism of the composite-systems chapter; the grand canonical ensemble is used before its formal derivation in Section 10.6, with an explicit honesty declaration at each point of first use. The harmonic oscillator's partition function in Section 10.6 recovers Planck's 1900 mean energy, closing the book's oldest statistical loop — the material-oscillator register of the pictures chapter and the field-mode, statistical register of the present chapter meet in that display. Field operators $\psi^\dagger(\mathbf{x})$, $\psi(\mathbf{x})$ are constructed from any complete set of single-particle wave functions in the sense of the pictures chapter; they carry no Lorentz indices and satisfy no relativistic wave equation — the chapter's boundary is non-relativistic, and the quantization of the full electromagnetic field is named as the discipline beyond it. Displays built on the continuum labels $|\mathbf{x}\rangle$ carry the licensed-shorthand status of the state-space and pictures chapters.

The historical chapter opened with three crises, and the second of them — the shell structure of atoms — was the Pauli principle: no two electrons share the same quantum numbers. The angular momentum chapter's singlet state, two spin-1/2 particles in the antisymmetric combination $|0, 0\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$ ((6.84)), was the $N = 2$ face of that principle, and its closing ledger registered the Pauli principle as a debt for the chapter on identical particles. The same historical chapter recorded a fourth crisis, quieter than the three: Bose's 1924 derivation of Planck's law from the statistics of indistinguishable light quanta, a counting that belonged to no classical statistics and that Einstein, the following year, extended to massive particles — the prediction of what is now called Bose–Einstein condensation. That prediction, made in 1925, was confirmed in 1995 at a temperature of a hundred and seventy nanokelvin. The arc from the old quantum theory to the quantum statistical mechanics of degenerate gases is the longest in the book, and this chapter draws it.

The chapter's business is the quantum mechanics of many identical particles. Its structure is a single line of ascent. Section 10.1 states the Symmetrization Postulate — the kinematic constraint that identical particles force on the state space itself — and builds the arena in which the rest of the chapter operates: Fock space, the direct sum over sectors of every particle number, whose basis vectors are

labelled by occupation numbers. Section 10.2 defines the operators that raise and lower those occupation numbers — the creation and annihilation operators, bosonic and fermionic — and unifies them with the ladder operators of the oscillator chapter and the Schwinger bosons of the angular momentum chapter under one algebraic structure. Section 10.3 transcribes the observables of single-particle quantum mechanics into the language of Fock space: every one-body and two-body operator assumes a universal form in terms of creation and annihilation operators, and the many-body Hamiltonian is the sum of those two forms. Section 10.4 solves the ideal Bose gas in the grand canonical ensemble: the Bose–Einstein distribution, the critical temperature, and the macroscopic occupation of the ground state — Bose–Einstein condensation, the chapter’s centrepiece and the repayment of the book’s oldest statistical debt. Section 10.5 solves the ideal Fermi gas: the Fermi–Dirac distribution, the Fermi sea at zero temperature, and the degeneracy pressure that holds up white dwarfs. Section 10.6 supplies the formal foundation of the ensembles the preceding two sections deployed: the density operator in thermal equilibrium, the canonical and grand canonical partition functions, and the harmonic-oscillator partition function from which Planck’s 1900 mean energy emerges as the chapter’s final repayment. Section 10.7 collects the principles, presents the experiment–formalism dictionary, balances the debt ledger, and opens the door to the interacting many-body theory that is the discipline beyond the chapter’s boundary.

10.1 Identical Particles: the Symmetrization Postulate and Fock Space

Particles of the same species are genuinely indistinguishable. The statement is not a dynamical approximation — it is a kinematic constraint on the state space itself, and its visible consequences range from the shell structure of atoms, the historical chapter’s second crisis, to the collective behaviour of quantum gases whose statistics this chapter solves. The angular momentum chapter’s two-spin-1/2 singlet ((6.84)) was the $N = 2$ antisymmetric case. This section answers the question that case left open: what is the law for N identical particles, and what is the Hilbert space that houses states of every N at once?

The permutation group on N identical particles. The tentative N -particle Hilbert space is the N -fold tensor product $\mathcal{H}^{\otimes N}$, built by the machinery of the composite-systems chapter (the tensor-product formalism of its opening section) and cited here without re-derivation. Let the single-particle space $\mathcal{H}_1$ carry an orthonormal basis $\{|\varphi_k\rangle\}$. A product state $|\psi_1\rangle \otimes \cdots \otimes |\psi_N\rangle$ assigns a definite single-particle state to each of the N labelled slots. The symmetric group S_N acts by permuting the tensor factors:

$$P_\pi |\psi_1\rangle \otimes \cdots \otimes |\psi_N\rangle = |\psi_{\pi^{-1}(1)}\rangle \otimes \cdots \otimes |\psi_{\pi^{-1}(N)}\rangle, \quad (10.1)$$

where the inverse on the index ensures that $P_{\pi_2} P_{\pi_1} = P_{\pi_2 \pi_1}$ (the operators compose like the group elements, not their inverses — the convention is the standard one and is stated once). Since the particles

are identical, every physical observable must commute with all P_π :

$$[O, P_\pi] = 0 \quad \text{for every } \pi \in S_N.$$

An observable that distinguished particle 1 from particle 2 would contradict indistinguishability. This commutativity forces physical states into irreducible representations of S_N . The representation theory of S_N is rich, but the physics of three-dimensional space selects exactly two of its representations as the physical ones: the two one-dimensional representations — totally symmetric and totally antisymmetric. The higher-dimensional representations of S_N (parastatistics) are excluded by the spin-statistics connection in relativistic quantum field theory, stated as an experimental fact here and footnoted to its proper home.¹

Symmetric and antisymmetric subspaces. The two one-dimensional representations of S_N are defined by their behaviour under a transposition τ_{ij} , the swap of particles i and j . Every permutation π is a product of transpositions; the sign $\text{sgn}(\pi) = \pm 1$ records whether the number of transpositions in any such decomposition is even or odd. The two physical subspaces are:

- **Totally symmetric** (bosons): $P_\pi |\psi\rangle = |\psi\rangle$ for every π . The projector onto this subspace is the symmetrizer,
- **Totally antisymmetric** (fermions): $P_\pi |\psi\rangle = \text{sgn}(\pi) |\psi\rangle$ for every π . The projector onto this subspace is the antisymmetrizer.

In one display,

$$S = \frac{1}{N!} \sum_{\pi \in S_N} P_\pi, \quad A = \frac{1}{N!} \sum_{\pi \in S_N} \text{sgn}(\pi) P_\pi. \quad (10.2)$$

Both are Hermitian projection operators: $S^2 = S$, $A^2 = A$, $S^\dagger = S$, $A^\dagger = A$, and $SA = AS = 0$ (a state cannot be simultaneously symmetric and antisymmetric for $N \geq 2$, because a single transposition would force $+1 = -1$). The normalization $1/N!$ ensures that each reproduces the state it projects: acting on a state already in the symmetric subspace, S returns the same state unchanged.

The physical N -particle spaces are the images of these projectors:

$$\mathcal{H}_N^{(+)} := S(\mathcal{H}^{\otimes N}), \quad \mathcal{H}_N^{(-)} := A(\mathcal{H}^{\otimes N}).$$

The Symmetrization Postulate. The states of N identical particles are vectors in exactly one of these two subspaces. Which one is determined by the particle's spin: integer spin selects the symmetric subspace — bosons; half-integer spin selects the antisymmetric subspace — fermions. The connection to the angular momentum chapter's singlet is explicit: two spin-1/2 particles in the singlet state $|0, 0\rangle$ of (6.84) are in the antisymmetric representation, consistent with the postulate (half-integer spin $\Rightarrow$ fermions $\Rightarrow$ antisymmetric states). The Pauli principle, the historical chapter's second crisis, is the statement that two electrons — spin-1/2 fermions — cannot occupy the same single-particle state: the antisymmetric subspace contains no vector of the form $|\varphi\rangle \otimes |\varphi\rangle \otimes \cdots$, because a transposition of identical factors would simultaneously demand symmetry (+1) from the equality of the factors and

¹The spin-statistics theorem was proved by Pauli in 1940 within relativistic quantum field theory (*Phys. Rev.* **58**, 716); the proof that integer-spin particles quantize as bosons and half-integer-spin particles as fermions is a theorem of Lorentz invariance, causality, and the positivity of the energy spectrum, and its proper statement belongs to a course in quantum field theory. The present chapter takes the connection as an experimental datum.

antisymmetry ($\pm$ sign) from the postulate — forcing the state to vanish. This is the exclusion principle in its modern form: a constraint on the state space, not a force.

The postulate is the chapter's single new axiom. It is stated as an experimental fact: every experiment with identical particles in three spatial dimensions conforms to it, and the spin-statistics theorem that proves it from deeper principles is a theorem of relativistic quantum field theory, not of the non-relativistic quantum mechanics this chapter inhabits.

Fock space: the arena for variable particle number. The N -particle spaces constructed above house a definite number of particles. Many processes of interest — the creation and absorption of quanta, the statistical mechanics of open systems — require a description that does not fix N in advance. The direct sum over all N -particle sectors supplies exactly that description:

$$\mathcal{F}_{\pm} = \mathbb{C} \oplus \mathcal{H}_1 \oplus \mathcal{H}_2^{(\pm)} \oplus \mathcal{H}_3^{(\pm)} \oplus \cdots = \bigoplus_{N=0}^{\infty} \mathcal{H}_N^{(\pm)}, \quad (10.3)$$

where $\mathcal{H}_0^{(\pm)} := \mathbb{C}$ is the one-dimensional space consisting of complex multiples of a single distinguished vector $|0\rangle$, the **vacuum state**: the unique state with zero particles, normalized $\langle 0|0\rangle = 1$. The direct sum means that a general vector in $\mathcal{F}_{\pm}$ is a sequence $(\psi_0, \psi_1, \psi_2, \dots)$ with $\psi_N \in \mathcal{H}_N^{(\pm)}$, only finitely many of which are nonzero, and the inner product is the sum of the inner products in each sector: $\langle \Psi | \Phi \rangle = \sum_N \langle \psi_N | \phi_N \rangle$. This is the **Fock space** — the natural arena for quantum systems in which the particle number is a dynamical variable, not a fixed parameter.

The vacuum is the only state common to both the bosonic and the fermionic Fock space. It is the ground floor of the direct sum, the zero of the particle-number counting, and it will serve as the cyclic vector from which every other state is built by the creation operators of the next section. The structure of Fock space makes no assumption about particle-number conservation: operators can change N , and the space accommodates it naturally — the direct sum is the algebraic expression of that possibility.

Occupation-number basis. Let $\{|\varphi_k\rangle\}$ be an orthonormal basis of the single-particle space $\mathcal{H}_1$, indexed by a discrete or continuous label k . A basis of the N -particle sector is built by distributing N particles among the modes, with the combinatorics dictated by the statistics. The states of the Fock space are most transparently labelled not by which particle occupies which mode — the question the symmetrization postulate renders meaningless — but by how many particles occupy each mode. For bosons, each n_k can be any nonnegative integer; for fermions, the Pauli principle restricts each n_k to zero or one. A basis vector of $\mathcal{F}_{\pm}$ is

$$|\{n_k\}\rangle = |n_1, n_2, \dots\rangle, \quad n_k \in \begin{cases} \{0, 1, 2, \dots\} & \text{(bosons),} \\ \{0, 1\} & \text{(fermions),} \end{cases} \quad (10.4)$$

with the total particle number $N = \sum_k n_k$. The vacuum is $|0, 0, \dots\rangle$, the state in which every occupation number is zero. The occupation-number states are orthonormal: $\langle n_1, n_2, \dots | n'_1, n'_2, \dots \rangle = \delta_{n_1 n'_1} \delta_{n_2 n'_2} \cdots$, and they span the Fock space. The Pauli principle as an algebraic statement, $n_k \in \{0, 1\}$, is the defining restriction of the fermionic sector: at most one particle per single-particle mode. The creation and annihilation operators of the next section are defined by their action on exactly these states, and their (anti)commutation relations are read off that action.

Example: two identical particles in two modes. Take a single-particle space spanned by two orthonormal modes $|\varphi_a\rangle$ and $|\varphi_b\rangle$. The two-mode example anchors the abstraction of Fock space in a concrete, finite-dimensional setting.

Bosons. The symmetric two-particle subspace $\mathcal{H}_2^{(+)}$ is three-dimensional, spanned by

$$|2, 0\rangle, \quad |0, 2\rangle, \quad |1, 1\rangle,$$

where $|n_a, n_b\rangle$ denotes n_a bosons in mode a and n_b bosons in mode b . In the coordinate representation, with single-particle wave functions $\varphi_a(\mathbf{x})$ and $\varphi_b(\mathbf{x})$, the symmetrized two-particle wave functions are

$$\psi_S(\mathbf{x}_1, \mathbf{x}_2) \propto \varphi_a(\mathbf{x}_1)\varphi_b(\mathbf{x}_2) + \varphi_a(\mathbf{x}_2)\varphi_b(\mathbf{x}_1), \quad (10.5)$$

for the state $|1, 1\rangle$. The state $|2, 0\rangle$ has both particles in the same mode, with wave function $\varphi_a(\mathbf{x}_1)\varphi_a(\mathbf{x}_2)$ — the symmetric subspace permits it. The bosonic “bunching” — the enhanced probability for two identical bosons to occupy the same state compared to distinguishable particles — is visible in this three-state enumeration: the distinguishable-particle space would be four-dimensional ($|a, a\rangle, |a, b\rangle, |b, a\rangle, |b, b\rangle$), but symmetrization collapses $|a, b\rangle$ and $|b, a\rangle$ into one state, leaving three. The probability of finding both particles in the same mode is $2/3$ for bosons (two of three states have $n_a = n_b$) against $2/4 = 1/2$ for distinguishable particles.

Fermions. The antisymmetric two-particle subspace $\mathcal{H}_2^{(-)}$ is one-dimensional, spanned by the single state

$$|1, 1\rangle,$$

because each mode can hold at most one fermion and two fermions in one mode — the states $|2, 0\rangle$ and $|0, 2\rangle$ — do not exist. The antisymmetrized wave function is the Slater determinant for $N = 2$:

$$\psi_A(\mathbf{x}_1, \mathbf{x}_2) \propto \varphi_a(\mathbf{x}_1)\varphi_b(\mathbf{x}_2) - \varphi_a(\mathbf{x}_2)\varphi_b(\mathbf{x}_1), \quad (10.6)$$

which vanishes identically if $a = b$ — the Pauli principle as the vanishing of the determinant with repeated rows. The probability of finding two fermions at the same position, obtained by setting $\mathbf{x}_1 = \mathbf{x}_2 = \mathbf{x}$ in (10.6), is zero: the wave function is odd under exchange, so it must vanish on the coincidence surface $\mathbf{x}_1 = \mathbf{x}_2$. This is the exchange hole, a purely kinematic correlation forced by the antisymmetry of the state space, with no dynamical force behind it.

The advantage of the occupation-number basis is its transparency: the combinatorics of identical particles, which in the first-quantized language requires explicit symmetrization sums over $N!$ permutations, reduces to the arithmetic of nonnegative integers. Its limitation is that it hides the permutation-group structure that justifies it; the projectors (10.2) are the bridge between the two descriptions, and the worked example above shows that for small N the two descriptions agree — as they must, by the postulate.

Transition. The occupation-number language already suggests operators that raise and lower n_k by one — the algebraic engine whose single-mode prototype the oscillator chapter built in (5.25). In Fock space, those operators acquire their full many-mode meaning, and they are the subject of the next section.

10.2 Creation and Annihilation Operators: the Algebraic Engine

The single-mode oscillator of Section 5.6 — whose ladder operators (5.25) counted excitations of one mode — generalizes to many modes, and the generalization unifies three algebraic structures the book has met separately: the harmonic oscillator's ladder operators, the field's creation and annihilation operators (the promissory note of Section 5.6), and the Schwinger bosons of the angular momentum chapter ((6.66)). They are three readings of one algebraic structure, and this section makes that unification explicit.

Bosonic operators. For each mode k of a chosen single-particle basis, define the **creation operator** $a_k^\dagger$ and the **annihilation operator** a_k by their action on the occupation-number basis (10.4):

$$a_k^\dagger |\dots, n_k, \dots\rangle = \sqrt{n_k + 1} |\dots, n_k + 1, \dots\rangle, \quad (10.7)$$

$$a_k |\dots, n_k, \dots\rangle = \sqrt{n_k} |\dots, n_k - 1, \dots\rangle, \quad (10.8)$$

with the understanding that $a_k |\dots, 0, \dots\rangle = 0$: an annihilation operator acting on a mode that is already empty annihilates the state. The definitions are the many-mode generalization of the ladder actions (5.25), with the normalization fixed by the same norm computation that fixed the single-mode actions. From these definitions, the bosonic canonical commutation relations follow:

$$[a_k, a_l^\dagger] = \delta_{kl}, \quad [a_k, a_l] = [a_k^\dagger, a_l^\dagger] = 0. \quad (10.9)$$

The proof of (10.9) from the defining actions on basis states is assigned to Exercise 3(a) of this chapter; the body of the text states the result with the explicit honesty declaration that the proof follows the same pattern as the single-mode case of the oscillator chapter — the action of the commutator on a generic occupation-number basis state, mode by mode, reproduces the Kronecker delta. In the same direct manner, the **number operator**

$$N_k := a_k^\dagger a_k, \quad N := \sum_k N_k, \quad (10.10)$$

has the occupation-number states as its eigenstates: $N_k |\{n_l\}\rangle = n_k |\{n_l\}\rangle$, with $n_k = 0, 1, 2, \dots$ — the spectrum of the single-mode oscillator (5.24) generalized to arbitrarily many modes.

Fermionic operators. The antisymmetry of the fermionic state space requires sign bookkeeping beyond the simple square-root normalization of the bosonic case. Let the modes be ordered, $k = 1, 2, \dots$. The fermionic creation operator on the occupation-number basis is

$$a_k^\dagger |\dots, n_k, \dots\rangle = (1 - n_k) (-1)^{\sum_{l < k} n_l} |\dots, n_k + 1, \dots\rangle,$$

with the same understanding that $a_k |\dots, 0, \dots\rangle = 0$ and the action of a_k on a mode occupied by one fermion, obtained by Hermitian conjugation of the displayed formula, removes it with the same sign factor. The prefactor $(1 - n_k)$ enforces the Pauli principle: if $n_k = 1$, the creation operator annihilates the state, because a second fermion cannot occupy the same mode. The sign factor $(-1)^{\sum_{l < k} n_l}$ counts the number of occupied modes that precede mode k ; it ensures that the creation operators for different modes anticommute, as the antisymmetry of the underlying states demands. The resulting algebra is the **canonical anticommutation relations**:

$$\{a_k, a_l^\dagger\} = \delta_{kl}, \quad \{a_k, a_l\} = \{a_k^\dagger, a_l^\dagger\} = 0, \quad (10.11)$$

where $\{A, B\} := AB + BA$ is the anticommutator. The Pauli principle follows as an algebraic theorem: from $\{a_k^\dagger, a_k^\dagger\} = 2(a_k^\dagger)^2 = 0$, one obtains $(a_k^\dagger)^2 = 0$, so no two fermions can be added to the same mode. The occupation number n_k can only be 0 or 1, as the basis construction (10.4) required. The number operator for fermions, $N_k = a_k^\dagger a_k$, has eigenvalues 0 and 1 and satisfies $N_k^2 = N_k$ — it is a projection, consistent with the occupation-number restriction. The detailed verification of (10.11) from the signed definition is assigned to Exercise 4, with the same honesty declaration as for the bosonic commutators: the proof acts on a generic occupation-number basis state, and the anticommutator's action reproduces the Kronecker delta with the sign factors cancelling or reinforcing as required by the ordering of the modes.

Example: two fermionic modes. Two fermionic modes span a four-state Fock space: $|0, 0\rangle$, $|1, 0\rangle$, $|0, 1\rangle$, and $|1, 1\rangle$. The state $|1, 1\rangle$ is the Slater determinant with both modes filled — the antisymmetrized wave function (10.6). The operator $a_1^\dagger a_2$ destroys a fermion in mode 2 and creates one in mode 1, mapping $|0, 1\rangle \mapsto |1, 0\rangle$ and annihilating $|0, 0\rangle$ and $|1, 1\rangle$. In the two-dimensional subspace $\{|1, 0\rangle, |0, 1\rangle\}$, the operators $a_1^\dagger a_2$ and $a_2^\dagger a_1$ act as raising and lowering operators — the fermionic analog of the Schwinger construction for angular momentum. The four-state Fock space and the sign factors that distinguish the orderings of creation operators are worked out in Exercise 4(c) of this chapter.

Unified reading with the oscillator chapter. A single bosonic mode — only one index k — reduces the Fock-space construction of this section to the harmonic oscillator of Section 5.6 exactly. With one mode, $[a, a^\dagger] = I$ ((10.9) reduces to (5.22)), $H = \hbar\omega(a^\dagger a + 1/2)$ is the oscillator Hamiltonian (5.22), and the occupation-number states $|n\rangle$ are the oscillator eigenstates (5.24). The Fock vacuum $|0\rangle$ is the oscillator ground state (5.26). The algebraic machinery of the oscillator chapter — spectrum, ladder actions, selection rule — is recovered as the single-mode restriction of the bosonic Fock space.

The many-mode extension is the step from one oscillator to a collection of independent oscillators, one for each mode k , with the excitation quantum of each mode now read as a particle. This is the content of the historical phrase “second quantization,” and the chapter states the historical register honestly: nothing is being quantized a second time. The single-particle Schrödinger equation was never a classical field theory — it was quantum mechanics of one particle from the start. The “second” quantization is the step from a Hilbert space of fixed particle number to Fock space with variable particle number, and the quanta that the creation operators create are the particles whose quantum mechanics the earlier chapters developed. The name is a fossil of the historical order in which the single-particle Schrödinger equation was called the “first-quantized” theory and the field the “second,” and the chapter uses the phrase only with this caveat.

A remark on the zero-point energy: the oscillator chapter's $H = \hbar\omega(N + 1/2)$ generalizes in the many-mode case to $H = \sum_k \hbar\omega_k(a_k^\dagger a_k + 1/2)$. The sum $\sum_k \hbar\omega_k/2$ over infinitely many modes diverges. It is a constant that drops out of all energy differences and is conventionally subtracted from the Hamiltonian; this is the first and simplest renormalization, and the chapter names it honestly as such without developing the machinery of renormalization theory.

Field operators. Given any complete orthonormal set of single-particle wave functions $\{\varphi_k(\mathbf{x})\}$, the

creation and annihilation operators in the position representation are obtained by a change of single-particle basis:

$$\psi^\dagger(\mathbf{x}) = \sum_k \varphi_k^*(\mathbf{x}) a_k^\dagger, \quad \psi(\mathbf{x}) = \sum_k \varphi_k(\mathbf{x}) a_k. \quad (10.12)$$

These are the **field operators** — operator-valued functions of position, constructed from any complete set of single-particle wave functions. The (anti)commutation relations follow from (10.9) or (10.11) and the completeness of the basis $\{\varphi_k\}$:

$$[\psi(\mathbf{x}), \psi^\dagger(\mathbf{y})]_\pm = \delta(\mathbf{x} - \mathbf{y}), \quad [\psi(\mathbf{x}), \psi(\mathbf{y})]_\pm = [\psi^\dagger(\mathbf{x}), \psi^\dagger(\mathbf{y})]_\pm = 0, \quad (10.13)$$

where $[A, B]_\pm$ denotes the commutator (bosons) or anticommutator (fermions), and $\delta(\mathbf{x} - \mathbf{y})$ is the Dirac delta in the licensed-shorthand register of the state-space chapter (Section 2.9) and the pictures chapter (Section 5.3); its rigorous content is Appendix A's standing debt. The derivation of (10.12) is a one-line exercise in the completeness of the single-particle basis:

$$[\psi(\mathbf{x}), \psi^\dagger(\mathbf{y})]_\pm = \sum_{k,l} \varphi_k(\mathbf{x}) \varphi_l^*(\mathbf{y}) [a_k, a_l^\dagger]_\pm = \sum_k \varphi_k(\mathbf{x}) \varphi_k^*(\mathbf{y}) = \delta(\mathbf{x} - \mathbf{y}),$$

where the second equality uses (10.9) or (10.11) to replace $[a_k, a_l^\dagger]_\pm$ by δ_{kl} , and the third is the completeness relation $\sum_k \varphi_k(\mathbf{x}) \varphi_k^*(\mathbf{y}) = \delta(\mathbf{x} - \mathbf{y})$ of the single-particle basis. The operator $\psi^\dagger(\mathbf{x})$ creates a particle at the position $\mathbf{x}$ — in the same licensed-shorthand sense: $|\mathbf{x}\rangle$ is a non-normalizable ket, and $\psi^\dagger(\mathbf{x}) |0\rangle = |\mathbf{x}\rangle$ is the one-particle state localized at $\mathbf{x}$. The field operators are strictly non-relativistic: they are constructed from a basis of single-particle wave functions, carry no spinor or Lorentz indices, and satisfy no relativistic wave equation. The quantization of the full electromagnetic field, with its vector potential, gauge invariance, and photon polarization, is the discipline beyond the chapter's non-relativistic boundary and is named here as the proper home of the relativistic field concept.

The number density operator in the field representation is

$$n(\mathbf{x}) = \psi^\dagger(\mathbf{x}) \psi(\mathbf{x}), \quad N = \int d^3x n(\mathbf{x}), \quad (10.14)$$

which is the second-quantized form of the position probability density of the measurement chapter ((3.73)): the expectation of $n(\mathbf{x})$ in an N -particle state yields the particle-number density at $\mathbf{x}$, and its integral over all space is the total particle-number operator N of (10.10).

Schwinger bosons as two-mode second quantization. The two bosonic modes $a_+^\dagger, a_-^\dagger$ of the angular momentum chapter (Section 6.5) are a special case of the formalism of this section. The Jordan–Schwinger map

$$J_i = \frac{\hbar}{2} a_+^\dagger \sigma_{\alpha\beta}^i a_\beta, \quad (10.15)$$

is a one-body operator in the sense of the next section, expressed in the two-mode Fock space. In the angular momentum chapter, the total boson number $N = a_+^\dagger a_+ + a_-^\dagger a_-$ was held fixed at $N = 2j$, and the angular momentum states $|j, m\rangle$ were the two-mode Fock states (6.68) with that constraint. In Fock space, N is a dynamical variable — the fixed- N constraint of the angular momentum chapter is lifted, and the Fock space construction provides the natural arena in which N can vary. The angular momentum states $|j, m\rangle$ are the two-mode Fock states $|n_+, n_-\rangle$ with $j = (n_+ + n_-)/2$ and $m = (n_+ - n_-)/2$, exactly as (6.68) recorded.

The Schwinger construction also supplies the natural language for spin coherent states in the Fock-space formulation. The two-mode boson coherent state is the tensor product of single-mode coherent states of the oscillator chapter:

$$|\alpha_+, \alpha_-\rangle = D_+(\alpha_+) D_-(\alpha_-) |0, 0\rangle.$$

Projecting this state onto the fixed- N sector with $N = 2j$ yields the spin coherent state of the angular momentum chapter,

$$|\theta, \phi\rangle = \frac{1}{\sqrt{(2j)!}} (\cos(\theta/2) a_+^\dagger + e^{i\phi} \sin(\theta/2) a_-^\dagger)^{2j} |0, 0\rangle,$$

which is the Bloch sphere of (6.39) read as a coherent-state manifold. The projection is verified by expanding the two-mode coherent state in the occupation-number basis:

$$|\alpha_+, \alpha_-\rangle = e^{-(|\alpha_+|^2 + |\alpha_-|^2)/2} \sum_{n_+, n_- = 0}^{\infty} \frac{\alpha_+^{n_+} \alpha_-^{n_-}}{\sqrt{n_+! n_-!}} |n_+, n_-\rangle,$$

and retaining only terms with $n_+ + n_- = 2j$. Setting $\alpha_+ = \sqrt{2j} \cos(\theta/2)$ and $\alpha_- = \sqrt{2j} e^{i\phi} \sin(\theta/2)$ and normalising the projected vector reproduces the displayed form; the factor $e^{-(|\alpha_+|^2 + |\alpha_-|^2)/2} = e^{-j}$ is absorbed into the normalisation. The discharge of the angular momentum chapter's ledger entries — Schwinger bosons (debt #39) and spin coherent states — is complete: the two-mode boson Fock space is the general setting, and the fixed- N angular momentum theory is the restriction to one sector.

The advantage of the creation/annihilation formalism is its algebra: the entire combinatorics of identical particles is encoded in the (anti)commutation relations (10.9) and (10.11), and every observable built from $a_k^\dagger$ and a_k automatically respects the symmetrization postulate. Its limitation is its abstractness: the field operators (10.12) hide the underlying single-particle wave functions, and the infinity of modes can obscure the physics — the zero-point divergence named above is the first warning. The next section returns to the concrete: it transcribes the familiar observables of single-particle quantum mechanics into the language this section has built.

Transition. Operators that add and remove particles are the vocabulary. Any observable — Hamiltonian, interaction, external probe — can be expressed in this vocabulary, and the next section transcribes the observables of single-particle quantum mechanics into the language of Fock space.

10.3 Observables in the Second-Quantized Formalism

The measurement axioms of Chapter 3 are general: observables are self-adjoint operators on the Hilbert space, and their spectral decompositions encode ideal experiments. In Fock space, operators that act on one particle at a time — kinetic energy, external potential — or on pairs of particles — interactions — assume a universal form in terms of the creation and annihilation operators of the preceding section. This section derives those forms.

The guiding idea is simple. A one-body operator in first-quantized language sums over all particles: $T = \sum_{i=1}^N t(\mathbf{x}_i, \mathbf{p}_i)$. In Fock space, the sum over particles is replaced by a sum over modes weighted by matrix elements — the number of particles in each mode is handled automatically by the algebra of $a_k^\dagger$ and a_k , and the operator's action on every N -particle sector is the same formula.

The derivation is a transcription, not a new postulate: the first-quantized operator, expressed in the occupation-number basis, is exactly the second-quantized form displayed below.

One-body operators. Let t be a single-particle operator — kinetic energy, an external potential, or any other operator on $\mathcal{H}_1$. In a chosen single-particle basis $\{|\varphi_k\rangle\}$ with matrix elements

$$t_{kl} := \langle \varphi_k | t | \varphi_l \rangle ,$$

the second-quantized form is

$$T = \sum_{k,l} t_{kl} a_k^\dagger a_l. \quad (10.16)$$

The derivation in abstract form is as follows. In the first-quantized N -particle space, the operator that applies t to the i -th particle and the identity to the others is $t_i := I \otimes \cdots \otimes t \otimes \cdots \otimes I$, and the total one-body operator is $T^{(N)} = \sum_{i=1}^N t_i$. Restricted to the (anti)symmetrized subspace, this operator has exactly the matrix elements that (10.16) produces. The proof that the two forms agree on every occupation-number basis state proceeds by inserting the completeness of the single-particle basis into the sum over particles; the result is a sum over single-particle matrix elements weighted by the number of ways each particle can occupy each mode, and the creation and annihilation operators (10.9) or (10.11) bookkeep those weights automatically. The transcription is valid for both bosons and fermions — the (anti)commutation relations of the creation and annihilation operators ensure the correct sign for fermionic matrix elements without any additional sign rules.

Note The verification of (10.16) on the occupation-number basis is compact. Let $|\{n_k\}\rangle$ and $|\{m_k\}\rangle$ be two occupation-number states with the same total particle number $\sum_k n_k = \sum_k m_k = N$ (otherwise the matrix element vanishes). The second-quantized form gives

$$\langle \{n_k\} | \sum_{kl} t_{kl} a_k^\dagger a_l | \{m_k\} \rangle = \sum_{kl} t_{kl} \langle \{n_k\} | a_k^\dagger a_l | \{m_k\} \rangle .$$

The action of $a_k^\dagger a_l$ removes a particle from mode l and adds one to mode k , producing a state proportional to the occupation-number state with n_k and n_l shifted appropriately. Summing over all k, l with the matrix elements $t_{kl} = \langle \varphi_k | t | \varphi_l \rangle$ reproduces exactly the first-quantized matrix element $\langle \Psi_{\{n_k\}} | \sum_{i=1}^N t_i | \Psi_{\{m_k\}} \rangle$ between the (anti)symmetrized N -particle wave functions built from the same occupation numbers. The creation and annihilation operators automate the sum over the N labelled particles and the $N!$ permutations that the first-quantized form requires.

In the field representation, the same operator is obtained by inserting the expansion (10.12) into (10.16) and using the completeness of the basis $\{\varphi_k\}$:

$$T = \int d^3x \psi^\dagger(\mathbf{x}) t(\mathbf{x}, -i\hbar\nabla) \psi(\mathbf{x}). \quad (10.17)$$

Two standard instances are the kinetic energy in the momentum basis,

$$T = \sum_{\mathbf{k}} \frac{\hbar^2 \mathbf{k}^2}{2m} a_{\mathbf{k}}^\dagger a_{\mathbf{k}},$$

where $\mathbf{k}$ labels plane-wave states, and the external potential in the position representation,

$$V_{\text{ext}} = \int d^3x V(\mathbf{x}) \psi^\dagger(\mathbf{x}) \psi(\mathbf{x}),$$

which is the operator whose expectation value is the potential energy of the density $n(\mathbf{x})$ of (10.14).

Two-body operators. A pairwise interaction $V(\mathbf{x}_i, \mathbf{x}_j)$ summed over all distinct pairs of particles has the second-quantized form

$$V = \frac{1}{2} \sum_{k,l,m,n} V_{klmn} a_k^\dagger a_l^\dagger a_n a_m, \quad (10.18)$$

with the matrix element

$$V_{klmn} = \int d^3x d^3y \varphi_k^*(\mathbf{x}) \varphi_l^*(\mathbf{y}) V(\mathbf{x}, \mathbf{y}) \varphi_m(\mathbf{x}) \varphi_n(\mathbf{y}).$$

The factor $1/2$ compensates for the double counting of pairs in the unrestricted sum; the ordering of indices — k, l on the creation operators, m, n on the annihilation operators, with n and m in that order — is the convention. For bosons, this ordering produces the normal-ordered form (all creation operators to the left of all annihilation operators) when the matrix element is symmetrized appropriately. For fermions, the antisymmetry of the two-particle states is respected automatically by the anticommutation relations (10.11): V_{klmn} inherits the antisymmetry $V_{klmn} = -V_{lkmn} = -V_{klnm}$ from the fermionic two-particle basis, and no extra sign rules are required — the anticommutators supply the sign changes that distinguish the two orderings.

 **Note** The derivation of (10.18) from the first-quantized form $\sum_{i<j} V(\mathbf{x}_i, \mathbf{x}_j)$ follows the same logic as the one-body case. In the first-quantized N -particle space, the pairwise interaction is the sum over all distinct pairs. Expanding each term in the single-particle basis $\{|\varphi_k\rangle\}$ and restricting to the (anti)symmetrized subspace produces matrix elements that are exactly those of (10.18) acting on the occupation-number basis. The matrix element between two-particle states $|\varphi_k, \varphi_l\rangle_\pm$ and $|\varphi_m, \varphi_n\rangle_\pm$ is

$$\pm \langle \varphi_k, \varphi_l | V | \varphi_m, \varphi_n \rangle_\pm = V_{klmn} \pm V_{klnm},$$

where the $+$ sign holds for bosons and the $-$ sign for fermions. The second-quantized form (10.18) with the convention $a_k^\dagger a_l^\dagger a_n a_m$ reproduces this automatically: for bosons, $a_k^\dagger a_l^\dagger = a_l^\dagger a_k^\dagger$ symmetrizes; for fermions, the anticommutation of the creation operators supplies the minus sign. The factor $1/2$ compensates for the double counting of each pair (k, l) and (l, k) in the unrestricted sum.

A contact interaction between bosons, $V(\mathbf{x}, \mathbf{y}) = g \delta(\mathbf{x} - \mathbf{y})$, is the simplest two-body term:

$$V = \frac{g}{2} \int d^3x \psi^\dagger(\mathbf{x}) \psi^\dagger(\mathbf{x}) \psi(\mathbf{x}) \psi(\mathbf{x}).$$

In the plane-wave basis, the contact interaction becomes a momentum-conserving sum (Exercise 5 works out the explicit form):

$$H = \sum_{\mathbf{k}} \varepsilon_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}} + \frac{g}{2} \sum_{\mathbf{k}, \mathbf{p}, \mathbf{q}} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{p}-\mathbf{q}}^\dagger a_{\mathbf{p}} a_{\mathbf{k}}, \quad (10.19)$$

with $\varepsilon_{\mathbf{k}} = \hbar^2 \mathbf{k}^2 / 2m$ for free particles. This is the Lieb–Liniger/Gross–Pitaevskii class of Hamiltonians, and the non-interacting limit $g = 0$ is the ideal Bose gas of the next section.

The many-body Hamiltonian. The universal Hamiltonian in Fock space is the sum of the one-body and two-body contributions:

$$H = \sum_{k,l} t_{kl} a_k^\dagger a_l + \frac{1}{2} \sum_{k,l,m,n} V_{klmn} a_k^\dagger a_l^\dagger a_n a_m. \quad (10.20)$$

Every term contains an equal number of creation and annihilation operators, so $[H, N] = 0$ — particle number is conserved by any Hamiltonian of this form. The proof is one line: each term is a product

of an equal number of $a^\dagger$ and a , and the commutator $[N, a_k^\dagger a_l]$ vanishes because $[N_k, a_k^\dagger a_l] = a_k^\dagger a_l - a_k^\dagger a_l = 0$ (the creation and annihilation operators raise and lower the occupation of the same mode by the same amount). Number-non-conserving terms — pairing terms of the form $a_k^\dagger a_l^\dagger + \text{h.c.}$ — belong to the theory of superconductivity and are named here without development.

Connection to the measurement chapter. The operators defined above are self-adjoint on Fock space: t_{kl} is Hermitian because t is self-adjoint on $\mathcal{H}_1$, and V_{klmn} has the symmetry $V_{klmn} = V_{mnkl}^*$ because $V(\mathbf{x}, \mathbf{y})$ is real and symmetric. The spectral theorem of the measurement chapter ((3.23)) applies to each operator on each N -particle sector and, by the direct-sum structure (10.3), to the operators on the whole Fock space. Second quantization does not change quantum mechanics — it transcribes the same theory into a notation that automates the combinatorics of identical particles. Every observable in Fock space encodes an ideal experiment in the sense of Chapter 3, and the measurement axioms apply without modification.

The advantage of the second-quantized form is the automation of identical-particle combinatorics: (10.16) and (10.18) are the same formula for two particles and for 10^{23} , and the (anti)commutation relations (10.9)–(10.11) supply the correct sign for every permutation without an explicit sum over $N!$ terms. Its limitation is that no new physics is added: every prediction of the second-quantized Hamiltonian (10.20) is a prediction of the first-quantized Schrödinger equation with the symmetrization postulate, and the transcription is a notational advance, not a conceptual one. The power of the notation is what the rest of the chapter demonstrates: in the occupation-number basis, the Hamiltonian that governs the quantum gases of the next two sections is diagonal, and their statistical mechanics is a sum over independent modes.

Transition. With the Hamiltonian expressed in the language of creation and annihilation, the dynamics of any non-relativistic many-body system is in principle set. The rest of the chapter solves the simplest case — no interactions — where the Hamiltonian is diagonal in the occupation-number basis and the exact solution is already dense with the physics the historical chapter left unpaid: the statistics of indistinguishable quanta.

10.4 The Ideal Bose Gas, and Bose–Einstein Condensation

Bose in 1924 counted indistinguishable quanta in phase-space cells and derived Planck’s law without electromagnetism. The historical chapter recorded that derivation — the first quantum statistics, the forerunner of the quantum mechanics the rest of the book built — and left its massive-particle extension as an open debt. The same counting, applied to massive bosons at fixed density, produces a sharper phenomenon: below a temperature set by the density and the mass, a finite fraction of all particles occupies exactly one quantum state. That is Bose–Einstein condensation, the chapter’s centerpiece and the repayment of the book’s oldest statistical debt.

Grand canonical ensemble for bosons. A system exchanging energy and particles with a reservoir at temperature T and chemical potential μ is described by the grand canonical ensemble. The formal derivation of this ensemble from the density operator is deferred to Section 10.6; the present sec-

tion states the ensemble with its physical meaning — a system in contact with a particle and energy reservoir, whose statistical weight for a state of energy E and particle number N is $e^{-\beta(E-\mu N)}$ with $\beta = 1/k_B T$ — and uses the result. This is the first honesty declaration for the grand canonical ensemble; the second, at the corresponding point in Section 10.5, repeats the statement, and the derivation in Section 10.6 closes the loop.

For the ideal Bose gas, the Hamiltonian is diagonal in the occupation-number basis:

$$H = \sum_k \varepsilon_k a_k^\dagger a_k, \quad N = \sum_k a_k^\dagger a_k, \quad (10.20)$$

where ε_k are the single-particle energies. A Fock state $|\{n_k\}\rangle$ has energy $E = \sum_k \varepsilon_k n_k$ and particle number $N = \sum_k n_k$. Its statistical weight in the grand canonical ensemble is $\exp[-\beta \sum_k (\varepsilon_k - \mu) n_k]$. Because the Hamiltonian is diagonal, the grand partition function factorizes over modes:

$$\Xi = \text{tr} e^{-\beta(H-\mu N)} = \prod_k \left(\sum_{n_k=0}^{\infty} e^{-\beta(\varepsilon_k - \mu) n_k} \right) = \prod_k \frac{1}{1 - e^{-\beta(\varepsilon_k - \mu)}}, \quad (10.21)$$

where the geometric series converges provided $\mu < \varepsilon_0$, with ε_0 the ground-state energy. This inequality is the fundamental constraint on the bosonic chemical potential: μ must lie below the lowest single-particle energy, or the occupation of the ground state diverges.

Bose–Einstein distribution. The mean occupation of mode k is obtained from the grand partition function by differentiation:

$$\langle n_k \rangle = \frac{1}{\beta} \frac{\partial \ln \Xi}{\partial \mu} \bigg|_{T,V} = \frac{1}{e^{\beta(\varepsilon_k - \mu)} - 1}. \quad (10.22)$$

This is the **Bose–Einstein distribution** — the quantum-statistical transcription of Bose’s 1924 mean occupation formula, which the historical chapter recorded as $\bar{N}_s/A_s = 1/(e^{h\nu_s/k_B T} - 1)$. The algebraic form is identical; what has changed is the context. Bose derived it for photons, for which $\mu = 0$ (photon number is not conserved) and $\varepsilon_k = \hbar\omega_k$, yielding Planck’s spectral distribution. Here, for massive bosons, μ is a thermodynamic variable that adjusts to fix the mean density, and the physics that follows from that adjustment is the subject of the rest of the section.

Two limits are immediate. At high temperature or low density, $e^{\beta(\varepsilon_k - \mu)} \gg 1$, the -1 in the denominator becomes negligible, and $\langle n_k \rangle \approx e^{-\beta(\varepsilon_k - \mu)}$ — the Maxwell–Boltzmann distribution of classical, distinguishable particles, recovered as the dilute, high-temperature limit of the quantum distribution. This is the quantitative bridge from quantum to classical statistics, and Section 10.6 gives the precise condition $n\lambda_T^3 \ll 1$ for its validity. At low temperature, the -1 is essential, and the occupation numbers deviate from the classical form — the regime in which the quantum statistics produces new physics.

Thermodynamic limit and the density of states. For free particles in a three-dimensional box of volume V , the single-particle energies are $\varepsilon = \hbar^2 \mathbf{k}^2 / 2m$, with $\mathbf{k} = (2\pi/L) \mathbf{n}$ and $\mathbf{n} \in \mathbb{Z}^3$. In the thermodynamic limit ($V \rightarrow \infty$, $N \rightarrow \infty$, $n = N/V$ fixed), the sum over wave vectors becomes an integral:

$$\sum_{\mathbf{k}} \rightarrow V \int \frac{d^3 k}{(2\pi)^3} = V \int_0^\infty d\varepsilon g(\varepsilon),$$

where the single-particle density of states per unit volume is

$$g(\varepsilon) d\varepsilon = \frac{m}{2\pi^2 \hbar^3} \sqrt{2m\varepsilon} d\varepsilon = \frac{m^{3/2}}{\sqrt{2} \pi^2 \hbar^3} \sqrt{\varepsilon} d\varepsilon. \quad (10.23)$$

The derivation is the same continuum-limit manipulation that the pictures chapter exercised in the free-particle propagator (Section 5.9). The number of states with wave vector magnitude below K is

$$\mathcal{N}(K) = V \cdot \frac{\frac{4\pi}{3} K^3}{(2\pi)^3} = \frac{VK^3}{6\pi^2}.$$

The density of states per unit energy per unit volume is $g(\varepsilon) = V^{-1} d\mathcal{N}/d\varepsilon$. Using $K = \sqrt{2m\varepsilon}/\hbar$,

$$\frac{d\mathcal{N}}{d\varepsilon} = \frac{d\mathcal{N}}{dK} \frac{dK}{d\varepsilon} = \frac{VK^2}{2\pi^2} \frac{d}{d\varepsilon} \left(\frac{\sqrt{2m\varepsilon}}{\hbar} \right) = \frac{V}{2\pi^2} \frac{2m\varepsilon}{\hbar^2} \frac{1}{\hbar} \sqrt{\frac{m}{2\varepsilon}} = V \frac{m}{2\pi^2 \hbar^3} \sqrt{2m\varepsilon},$$

so that $g(\varepsilon) d\varepsilon = (m/2\pi^2 \hbar^3) \sqrt{2m\varepsilon} d\varepsilon$, which is (10.23).

The total number of particles in the system is the sum of the mean occupations:

$$N = \sum_{\mathbf{k}} \langle n_{\mathbf{k}} \rangle = N_0 + V \int_{\varepsilon_0}^{\infty} d\varepsilon \frac{g(\varepsilon)}{e^{\beta(\varepsilon-\mu)} - 1}, \quad (10.24)$$

where $N_0 = 1/(e^{\beta(\varepsilon_0-\mu)} - 1)$ is the ground-state occupation. The separation of the ground state from the integral is essential: in the thermodynamic limit, the density of states vanishes as $\sqrt{\varepsilon}$ at low energies, and the ground state — a single quantum state — would be missed by the integral alone. This is the step that makes Bose–Einstein condensation possible, and it is treated with explicit care here: the sum over $\mathbf{k}$ is evaluated exactly for the $\mathbf{k} = 0$ term and converted to an integral for all $\mathbf{k} \neq 0$, because the integrand $g(\varepsilon)/(e^{\beta(\varepsilon-\mu)} - 1)$ is smooth for $\varepsilon > 0$ and the error from replacing the sum by the integral vanishes as $V^{-1/3}$ in the thermodynamic limit.

Bose–Einstein condensation. Set the ground-state energy to zero, $\varepsilon_0 = 0$. As the temperature decreases at fixed density $n = N/V$, the chemical potential μ rises from negative values toward zero, because the mean occupation (10.22) must accommodate the fixed density and lower temperatures reduce the occupation of excited states. At a critical temperature T_c , μ reaches zero. At that point, the integral over excited states saturates: it can accommodate at most

$$N_{\text{ex}}(T, \mu = 0) = V \int_0^{\infty} d\varepsilon \frac{g(\varepsilon)}{e^{\beta\varepsilon} - 1}. \quad (10.25)$$

The integral is evaluated by substituting (10.23) and changing variables to $x = \beta\varepsilon$:

$$\begin{aligned} N_{\text{ex}}(T, 0) &= V \frac{m^{3/2}}{\sqrt{2} \pi^2 \hbar^3} \int_0^{\infty} d\varepsilon \frac{\sqrt{\varepsilon}}{e^{\beta\varepsilon} - 1} \\ &= V \frac{m^{3/2}}{\sqrt{2} \pi^2 \hbar^3} (k_B T)^{3/2} \int_0^{\infty} dx \frac{\sqrt{x}}{e^x - 1}. \end{aligned}$$

The dimensionless integral is the Bose integral for $s = 3/2$:

$$\int_0^{\infty} dx \frac{\sqrt{x}}{e^x - 1} = \frac{\sqrt{\pi}}{2} \zeta(3/2).$$

Inserting the value of the integral into the prefactor, the $\sqrt{\pi}$ from the integral cancels against the $\sqrt{\pi}$ from the density-of-states prefactor, and the result is

$$N_{\text{ex}}(T, \mu = 0) = V \left(\frac{mk_B T}{2\pi \hbar^2} \right)^{3/2} \zeta(3/2). \quad (10.26)$$

The cancellation is verified by direct arithmetic:

$$\frac{m^{3/2}}{\sqrt{2}\pi^2\hbar^3} \times \frac{\sqrt{\pi}}{2} = \frac{m^{3/2}}{2\sqrt{2}\pi^{3/2}\hbar^3} = \left(\frac{m}{2\pi\hbar^2}\right)^{3/2}.$$

The final form involves the thermal de Broglie wavelength $\lambda_T = h/\sqrt{2\pi mk_B T}$.

The critical temperature T_c is defined by the condition that the excited states, at $\mu = 0$, accommodate all N particles:

$$N = V \left(\frac{mk_B T_c}{2\pi\hbar^2}\right)^{3/2} \zeta(3/2). \quad (10.27)$$

Solving for T_c ,

$$k_B T_c = \frac{2\pi\hbar^2}{m} \left(\frac{n}{\zeta(3/2)}\right)^{2/3}, \quad \zeta(3/2) \approx 2.612. \quad (10.28)$$

Below T_c , the chemical potential remains pinned at $\mu = 0$: the excited states can hold at most $N_{\text{ex}}(T, 0) = N (T/T_c)^{3/2}$ particles, and the remaining $N - N_{\text{ex}}$ particles occupy the ground state.

The condensate fraction is

$$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c}\right)^{3/2}, \quad T \leq T_c. \quad (10.29)$$

At $T = 0$, every particle is in the ground state. At $T = T_c$, the condensate fraction vanishes — a continuous transition, with a discontinuous derivative, of the order parameter N_0/N . This macroscopic occupation of a single quantum state is **Bose–Einstein condensation**. It is a phase transition driven purely by quantum statistics: no interactions are needed, and the transition exists in the ideal (non-interacting) gas — a fact that distinguishes it from every classical phase transition, which requires interactions to produce singular behaviour in the partition function.

The physical picture behind the condensation is captured by the thermal de Broglie wavelength,

$$\lambda_T = \frac{h}{\sqrt{2\pi mk_B T}}.$$

At $T = T_c$, one has $n\lambda_T^3 = \zeta(3/2) \approx 2.612$ — that is, BEC occurs when the interparticle spacing $n^{-1/3}$ becomes comparable to the quantum-mechanical extent of a single-particle wave packet. When the wave packets of neighbouring particles overlap, their indistinguishability can no longer be neglected, and the bosonic statistics forces them into the same quantum state. This is the spatial interpretation of the critical condition, and it is the bridge between the abstract occupation-number formalism and the physical intuition of overlapping matter waves.

Measurable landing: the 1995 observation of BEC. The Bose–Einstein condensation of an ideal gas was predicted by Einstein in 1925, one year after Bose’s photon derivation. Seventy years passed before the experimental realization. In 1995, Anderson, Ensher, Matthews, Wieman, and Cornell at JILA observed BEC in a dilute gas of ^{87}Rb atoms confined in a magnetic trap and cooled by laser and evaporative techniques to temperatures below 200 nanokelvin.² For the JILA parameters — peak density $n \approx 2.5 \times 10^{12} \text{ cm}^{-3}$, mass $m = 87u$ with $u = 1.66 \times 10^{-27} \text{ kg}$ — the ideal-gas formula (10.28) gives

$$T_c(^{87}\text{Rb}) \approx 170 \text{ nK}, \quad (10.30)$$

²M. H. Anderson, J. R. Ensher, M. R. Matthews, C. E. Wieman, and E. A. Cornell, *Science* **269**, 198 (1995).

which agrees with the measured transition temperature. The condensate was detected by time-of-flight imaging — the far-field method of the pictures chapter ((5.44)): after release from the trap, the thermal cloud expands isotropically while the condensate, being a single macroscopic matter wave, retains the narrow momentum distribution of the ground state. The sudden appearance of a narrow peak at low velocity, growing from zero as the temperature drops below T_c , is the direct signature of condensation.

A second numerical instance is liquid ^4He at saturated vapour density, $n \approx 2.2 \times 10^{22} \text{ cm}^{-3}$. The ideal-gas formula (10.28) gives $T_c \approx 3.1 \text{ K}$. The observed lambda transition in liquid helium occurs at $T_\lambda = 2.17 \text{ K}$. The discrepancy is a measure of interactions: the ideal-gas prediction is already within roughly a factor of two of the measured transition temperature, despite the strong interatomic forces in the liquid. The interacting theory — the Bogoliubov theory of the weakly interacting Bose gas and the Beliaev diagrammatic extension — is the proper treatment of the condensed phase, and it is named here as the next step beyond the ideal gas.

The advantage of the ideal Bose gas solution is its completeness: the exact partition function factorizes, the distribution (10.22) is exact, and the condensation transition (10.29) is a theorem of the quantum statistics, not a model. Its limitation, transparent in the ^4He comparison, is that no real Bose gas is ideal: interactions shift T_c , deplete the condensate even at zero temperature, and modify the elementary excitations from free particles to phonons (in the neutral gas) or rotons (in liquid helium). The Fock-space Hamiltonian (10.20) with the two-body term is the exact starting point for the interacting theory, and the ideal gas is its zero-order solution.

Transition. Bosons accumulate in the ground state. Fermions refuse to share any state at all. The consequence is not condensation but pressure — a quantum pressure that holds up white dwarfs. The Pauli principle, now read as a statistical constraint on the occupation numbers, is the engine of the next section.

10.5 The Ideal Fermi Gas: Degeneracy Pressure and the Fermi Surface

At $T = 0$ a non-interacting Fermi gas is not empty like its classical counterpart. It is a filled sea whose surface is sharp in momentum space — and whose pressure does not depend on temperature at all. The Pauli principle, enforced by the anticommutator (10.11), is the engine: at most one fermion per single-particle state, so the N lowest-energy states are occupied and the rest are empty. The Fermi sea, the Fermi energy, and the degeneracy pressure are the collective signatures of that occupation, and they connect the abstract Fock-space construction to the astrophysics of white dwarfs and the condensed-matter physics of metals.

Fermi–Dirac distribution. The grand canonical ensemble of the preceding section is applied to fermions with the single restriction $n_k \in \{0, 1\}$. The same honesty declaration governs: the grand canonical ensemble is used here and its formal derivation is deferred to Section 10.6. The grand par-

tion function factorizes over modes, with each mode contributing a sum of two terms:

$$\Xi = \text{tr} e^{-\beta(H-\mu N)} = \prod_k \left(\sum_{n_k=0}^1 e^{-\beta(\varepsilon_k - \mu)n_k} \right) = \prod_k (1 + e^{-\beta(\varepsilon_k - \mu)}). \quad (10.31)$$

Differentiation with respect to μ gives the mean occupation:

$$\langle n_k \rangle = \frac{1}{\beta} \frac{\partial \ln \Xi}{\partial \mu} \bigg|_{T,V} = \frac{1}{e^{\beta(\varepsilon_k - \mu)} + 1}. \quad (10.32)$$

This is the **Fermi–Dirac distribution**. At $T = 0$, the argument of the exponential is $\beta(\varepsilon_k - \mu)$, which diverges to $+\infty$ for $\varepsilon_k > \mu$ and to $-\infty$ for $\varepsilon_k < \mu$. Hence

$$\langle n_k \rangle \xrightarrow{T \rightarrow 0} \theta(\mu - \varepsilon_k) = \begin{cases} 1, & \varepsilon_k < \mu, \\ 0, & \varepsilon_k > \mu, \end{cases}$$

where θ is the Heaviside step function. The zero-temperature chemical potential defines the **Fermi energy**: $\mu(T = 0) \equiv \varepsilon_F$. Every single-particle state with energy below ε_F is occupied; every state above is empty. This filled configuration is the **Fermi sea**, and its boundary in momentum space is the **Fermi surface**.

The Fermi sea at $T = 0$. For free particles in three dimensions with spin degeneracy g_s (with $g_s = 2$ for spin-1/2 fermions), the number of occupied single-particle states below k_F is $N = g_s V \cdot (4\pi/3)k_F^3/(2\pi)^3 = g_s V k_F^3/6\pi^2$, giving the Fermi wave vector in terms of the density $n = N/V$:

$$k_F = \left(\frac{6\pi^2 n}{g_s} \right)^{1/3}. \quad (10.33)$$

For the standard case of spin-1/2 electrons, $g_s = 2$, and this reduces to the familiar form $k_F = (3\pi^2 n)^{1/3}$. The Fermi energy is the kinetic energy at the Fermi surface:

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m} \left(\frac{6\pi^2 n}{g_s} \right)^{2/3}, \quad (10.34)$$

which for spin-1/2 fermions ($g_s = 2$) reduces to $\varepsilon_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}$. The Fermi temperature $T_F = \varepsilon_F/k_B$ sets the scale below which the gas is degenerate — that is, below which the quantum statistics departs significantly from the classical Maxwell–Boltzmann distribution. For electrons in metals, $n \sim 10^{22}$ – 10^{23} cm^{-3} , giving $\varepsilon_F \sim 1$ – 10 eV and $T_F \sim 10^4$ – 10^5 K : room temperature is two orders of magnitude below the degeneracy temperature, and the conduction electrons in a metal at ordinary temperatures are a deeply degenerate Fermi gas.

The ground-state energy is the sum of the single-particle energies over all occupied states, including the spin degeneracy:

$$\begin{aligned} E_0 &= g_s V \int_0^{\varepsilon_F} d\varepsilon g(\varepsilon) \varepsilon = g_s V \frac{m^{3/2}}{\sqrt{2} \pi^2 \hbar^3} \int_0^{\varepsilon_F} d\varepsilon \varepsilon^{3/2} \\ &= g_s V \frac{m^{3/2}}{\sqrt{2} \pi^2 \hbar^3} \frac{2}{5} \varepsilon_F^{5/2}. \end{aligned} \quad (10.35)$$

The energy per particle follows from dividing E_0 by $N = g_s V \int_0^{\varepsilon_F} g(\varepsilon) d\varepsilon$. Since both integrals share the same prefactor $g_s V m^{3/2}/(\sqrt{2} \pi^2 \hbar^3)$, the ratio reduces to

$$\frac{E_0}{N} = \frac{\int_0^{\varepsilon_F} \varepsilon^{3/2} d\varepsilon}{\int_0^{\varepsilon_F} \varepsilon^{1/2} d\varepsilon} = \frac{\frac{2}{5} \varepsilon_F^{5/2}}{\frac{2}{3} \varepsilon_F^{3/2}} = \frac{3}{5} \varepsilon_F,$$

independently of the spin degeneracy g_s and of the detailed form of the density-of-states prefactor.

Hence

$$\frac{E_0}{N} = \frac{3}{5} \varepsilon_F. \quad (10.36)$$

The kinetic energy is finite at zero temperature: the Pauli principle forces particles into higher momentum states as the density increases, and the average kinetic energy grows as $n^{2/3}$.

The zero-temperature pressure — the **degeneracy pressure** — follows from the thermodynamic relation $P = -\partial E_0 / \partial V|_N$. Since $\varepsilon_F \propto k_F^2 \propto V^{-2/3}$, one has $E_0 = \frac{3}{5} N \varepsilon_F \propto V^{-2/3}$, and therefore

$$P = -\left. \frac{\partial E_0}{\partial V} \right|_N = -\left. \frac{\partial}{\partial V} \left(\frac{3}{5} N \varepsilon_F \right) \right|_N = -\frac{3}{5} N \frac{\partial \varepsilon_F}{\partial V} = -\frac{3}{5} N \left(-\frac{2}{3} \right) \frac{\varepsilon_F}{V} = \frac{2}{3} \frac{E_0}{V} = \frac{2}{5} n \varepsilon_F. \quad (10.37)$$

This pressure is finite at $T = 0$, purely quantum in origin, and it is what supports white dwarf stars against gravitational collapse. The Chandrasekhar mass — the maximum mass a white dwarf can sustain before the electrons become relativistic and the equation of state softens from $P \propto n^{5/3}$ to $P \propto n^{4/3}$ — is the astrophysical consequence of this pressure, and it is named here without derivation: the full relativistic degenerate Fermi gas belongs to a course in astrophysics, and the non-relativistic pressure (10.37) is the chapter's contribution to that story.

The Fermi surface and low-temperature thermodynamics. At finite but low temperature, only states within an energy window of roughly $k_B T$ around the Fermi surface are thermally active: the Pauli principle blocks all lower-energy excitations, because the states they would occupy are already filled. This restricted phase space produces the characteristic low-temperature properties of the degenerate Fermi gas. The number and energy integrals for a free Fermi gas are

$$N = \int_0^\infty d\varepsilon g(\varepsilon) f(\varepsilon), \quad E = \int_0^\infty d\varepsilon \varepsilon g(\varepsilon) f(\varepsilon),$$

with $f(\varepsilon) = 1 / (e^{\beta(\varepsilon - \mu)} + 1)$. At low temperatures, $f(\varepsilon)$ differs from the step function only near $\varepsilon = \mu$, and the integrals can be evaluated by the Sommerfeld expansion — an asymptotic expansion in powers of $(k_B T / \varepsilon_F)^2$. The derivation of the expansion and its application to the specific heat are assigned to Exercise 7 of this chapter; the body of the text states the results with the explicit honesty declaration that the Sommerfeld expansion is mechanical but algebraically lengthy, and the exercise guides the student through it step by step.

The principal results are as follows. The chemical potential at low temperature departs from ε_F quadratically:

$$\mu(T) = \varepsilon_F \left[1 - \frac{\pi^2}{12} \left(\frac{k_B T}{\varepsilon_F} \right)^2 + \cdots \right].$$

The energy, and hence the specific heat, is dominated by the states near the Fermi surface:

$$C_V = \left. \frac{\partial E}{\partial T} \right|_{V,N} = \frac{\pi^2}{2} k_B n \frac{k_B T}{\varepsilon_F}. \quad (10.38)$$

The specific heat is linear in temperature, in contrast to the classical value $C_V = \frac{3}{2} N k_B$ which is independent of T . For copper at room temperature, $\varepsilon_F \approx 7.0$ eV and $T_F \approx 8.2 \times 10^4$ K, giving $C_V \approx 0.02 N k_B$ — roughly seventy times smaller than the classical value. This is the resolution of the classical Drude model's heat-capacity paradox: the Drude theory treated conduction electrons as a classical ideal gas and predicted a specific heat of $\frac{3}{2} N k_B$, which is not observed; the quantum electron

gas, with its Pauli-blocked phase space, has a specific heat smaller by the factor $k_B T/\varepsilon_F \sim 10^{-2}$ at room temperature. The paradox and its resolution are stated in one sentence; the full theory of electronic specific heat, including the lattice contribution, belongs to solid-state physics.

A second measurable consequence of the Fermi surface is the Pauli spin susceptibility. In a magnetic field B , the Zeeman energy $\pm\mu_B B$ shifts the energies of spin-up and spin-down electrons in opposite directions, creating an imbalance in the two spin populations. The resulting magnetization yields a temperature-independent susceptibility $\chi_P = \mu_B^2 g(\varepsilon_F)$, proportional to the density of states at the Fermi surface. The derivation involves the same Sommerfeld expansion and is outlined in Exercise 7; the book states the result without derivation.

Measurable landing: conduction electrons in copper. The conduction electrons in metallic copper provide a standard numerical instance of the degenerate Fermi gas. With one conduction electron per atom and the mass density of copper (8.96 g/cm^3 , atomic mass 63.5 u), the electron density is $n \approx 8.5 \times 10^{22} \text{ cm}^{-3}$. The Fermi parameters are

$$\varepsilon_F \approx 7.0 \text{ eV}, \quad k_F \approx 1.36 \text{ \AA}^{-1}, \quad v_F = \frac{\hbar k_F}{m} \approx 1.57 \times 10^6 \text{ m/s}, \quad T_F \approx 8.2 \times 10^4 \text{ K}. \quad (10.39)$$

These numbers are not model parameters: they follow from the density and the electron mass, with no adjustable constants. Photoemission spectroscopy measures the occupied density of states directly, confirming the step-function shape of the Fermi–Dirac distribution at $T = 0$ broadened by roughly $k_B T$, and angle-resolved photoemission maps the Fermi surface in momentum space. The de Haas–van Alphen effect — the oscillation of the magnetization with $1/B$ in a strong magnetic field at low temperature — measures extremal cross-sectional areas of the Fermi surface with millimetre precision in momentum space, confirming the free-electron sphere to within the small distortions produced by the periodic lattice potential. The effect is named here without development; its theory belongs to the quantum mechanics of electrons in periodic potentials.

The contrast with the Bose gas of the preceding section is instructive. At room temperature, the electron gas in copper is at $T/T_F \approx 0.004$ — deep in the degenerate regime, with the Fermi–Dirac distribution indistinguishable from a step function on the scale of $k_B T$. The Bose gas, by contrast, requires nanokelvin temperatures to reach $T/T_c < 1$. The difference is not the physics of the distributions — they are structurally similar, differing only by the sign in the denominator — but the mass and density of the particles. The electron is four orders of magnitude lighter than the rubidium atom, and the conduction electron density in a metal is three to four orders of magnitude higher than the density of a dilute ultracold atomic gas. The Fermi energy scales as $n^{2/3}/m$, the BEC critical temperature as $n^{2/3}/m$ as well — but the exclusion principle makes the Fermi energy the characteristic energy, while Bose statistics makes the condensation temperature the characteristic energy. They are the same function of n and m up to a dimensionless factor, and the physics that distinguishes them is the sign in the denominator of the occupation number.

The advantage of the ideal Fermi gas solution is its universality: at densities where the kinetic energy dominates the interaction energy, the free-Fermi-gas predictions for the specific heat, the spin susceptibility, and the Fermi surface geometry are quantitatively accurate. Its limitation is that inter-

actions are never truly absent: in metals, the Coulomb repulsion between electrons is strong, and the success of the free-electron model is a consequence of Landau's Fermi liquid theory — the dressed quasiparticles near the Fermi surface behave as weakly interacting fermions, with the same Fermi surface volume as the free gas (Luttinger's theorem). The Fermi liquid theory and the BCS pairing instability of the Fermi surface are the interacting chapters the ideal gas opens, and they are named here as the next steps.

Transition. The two quantum gases — bosonic and fermionic — have been treated in an ensemble whose derivation was stated, not proved. The formalism that organizes thermal equilibrium in quantum mechanics is the density operator and the partition function, introduced in the composite-systems chapter and extended here to systems with variable particle number. That formalism is the door this chapter opens onto genuine quantum statistical mechanics, and it is the subject of the final technical section.

10.6 The Density Operator and the Quantum Partition Function

The statistical ensembles deployed in Sections 10.4 and 10.5 rest on the density operator formalism of the composite-systems chapter. This section extends that formalism to thermal equilibrium: the canonical and grand canonical density operators are derived from the maximum-entropy principle, the quantum partition function is defined, and the harmonic oscillator partition function is evaluated to recover Planck's 1900 mean energy — the book's oldest formula, now in its final statistical home. The section also supplies the formal derivation of the grand canonical ensemble whose results the preceding two sections anticipated.

The density operator reviewed. The density operator ρ of the composite-systems chapter (the density-operator section of that chapter) describes any state — pure or mixed — as a single self-adjoint operator:

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1,$$

where the $|\psi_i\rangle$ are normalized but need not be orthogonal. The expectation value of any observable is $\langle A \rangle = \text{tr}(\rho A)$, and the state is pure iff $\text{tr}(\rho^2) = 1$ and mixed iff $\text{tr}(\rho^2) < 1$. The von Neumann entropy is $S = -k_B \text{tr}(\rho \ln \rho)$. These definitions are the composite-systems chapter's; they are stated here without re-derivation, and the present section builds on them.

The canonical ensemble. A system in thermal contact with a heat bath at temperature T , with fixed particle number (or in a fixed- N sector of Fock space), adopts the density operator that maximizes the von Neumann entropy subject to the constraint of fixed mean energy $\langle H \rangle = U$. The maximization is standard Lagrange-multiplier calculus in the spirit of the measurement chapter's variational principle, and the section sketches it in a note rather than claiming it as a derivation from deeper axioms:

Note Maximize $S = -k_B \text{tr}(\rho \ln \rho)$ subject to $\text{tr} \rho = 1$ and $\text{tr}(\rho H) = U$. The variation $\delta S = -k_B \text{tr}(\delta \rho (\ln \rho + I))$ together with the constraint variations $\text{tr} \delta \rho = 0$ and $\text{tr}(\delta \rho H) = 0$ yields $\ln \rho + I + \alpha I + \beta H = 0$, hence $\rho \propto e^{-\beta H}$. The multiplier β is identified with $1/k_B T$ by matching the

classical equipartition theorem in the high-temperature limit; the proof of that identification is the standard thermodynamic one, and the chapter does not reproduce it.

The result is the **canonical density operator**:

$$\rho_\beta = \frac{e^{-\beta H}}{Z}, \quad Z(\beta) = \text{tr}(e^{-\beta H}), \quad \beta = \frac{1}{k_B T}. \quad (10.40)$$

The normalization Z is the **canonical partition function**, and every equilibrium thermodynamic quantity follows from it. The internal energy is

$$U = \langle H \rangle = \text{tr}(\rho_\beta H) = -\frac{\partial \ln Z}{\partial \beta}. \quad (10.41)$$

The free energy is $F = -k_B T \ln Z$, from which the pressure ($P = -\partial F / \partial V|_T$), the entropy ($S = -\partial F / \partial T|_V$), and the specific heat follow by differentiation. The partition function contains the whole equilibrium statistical mechanics, and the task of equilibrium many-body theory is to evaluate it for Hamiltonians more complicated than the diagonal ones of the preceding two sections.

The grand canonical ensemble. For systems exchanging both energy and particles with a reservoir, the additional constraint of fixed mean particle number $\langle N \rangle$ introduces a second Lagrange multiplier, the chemical potential μ :

$$\rho_{\beta, \mu} = \frac{e^{-\beta(H - \mu N)}}{\Xi}, \quad \Xi(\beta, \mu) = \text{tr}(e^{-\beta(H - \mu N)}). \quad (10.42)$$

The normalization Ξ is the **grand partition function**. The grand potential is $\Omega = -k_B T \ln \Xi$, and the mean particle number is

$$\langle N \rangle = \frac{1}{\beta} \frac{\partial \ln \Xi}{\partial \mu} \bigg|_{T, V}.$$

This is the ensemble whose results Sections 10.4 and 10.5 deployed. The connection to Fock space is direct: the trace in (10.42) is over the entire Fock space, and because the ideal-gas Hamiltonian $H = \sum_k \varepsilon_k a_k^\dagger a_k$ and the number operator $N = \sum_k a_k^\dagger a_k$ are both diagonal in the occupation-number basis, the trace factorizes over modes:

$$\Xi = \text{tr}(e^{-\beta \sum_k (\varepsilon_k - \mu) a_k^\dagger a_k}) = \prod_k \text{tr}_k(e^{-\beta (\varepsilon_k - \mu) a_k^\dagger a_k}),$$

where tr_k is the trace over the Fock space of mode k alone. For bosons, the sum over $n_k = 0, 1, 2, \dots$ gives (10.21); for fermions, the sum over $n_k = 0, 1$ gives (10.31). The factorization is the mathematical expression of the physical statement that the modes of the ideal gas are statistically independent.

The harmonic oscillator partition function: Planck's 1900 mean energy. The single-mode bosonic Fock space is the harmonic oscillator of Section 5.6, and the canonical partition function of one mode is evaluated by summing the geometric series:

$$\begin{aligned} Z_{\text{ho}} &= \sum_{n=0}^{\infty} e^{-\beta \hbar \omega (n+1/2)} = e^{-\beta \hbar \omega / 2} \sum_{n=0}^{\infty} (e^{-\beta \hbar \omega})^n \\ &= \frac{e^{-\beta \hbar \omega / 2}}{1 - e^{-\beta \hbar \omega}} = \frac{1}{e^{\beta \hbar \omega / 2} - e^{-\beta \hbar \omega / 2}} = \frac{1}{2 \sinh(\beta \hbar \omega / 2)}. \end{aligned} \quad (10.43)$$

The mean energy follows from (10.41):

$$\begin{aligned}
 U &= -\frac{\partial}{\partial \beta} \ln Z_{\text{ho}} = -\frac{\partial}{\partial \beta} \left[-\frac{\beta \hbar \omega}{2} - \ln(1 - e^{-\beta \hbar \omega}) \right] \\
 &= \frac{\hbar \omega}{2} + \frac{\hbar \omega e^{-\beta \hbar \omega}}{1 - e^{-\beta \hbar \omega}} = \frac{\hbar \omega}{2} \coth\left(\frac{\beta \hbar \omega}{2}\right) \\
 &= \frac{\hbar \omega}{e^{\beta \hbar \omega} - 1} + \frac{\hbar \omega}{2}.
 \end{aligned} \tag{10.44}$$

The first term, $\hbar \omega / (e^{\beta \hbar \omega} - 1)$, is exactly the mean energy that Planck wrote in 1900 for a material resonator of frequency ω in thermal equilibrium with the radiation field. Multiplied by the electromagnetic mode density $g(\nu) d\nu = 8\pi \nu^2 / c^3 d\nu$ and setting $\omega = 2\pi \nu$, it gives the spectral energy density per unit frequency:

$$\frac{du_\nu}{d\nu} = \frac{8\pi h \nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}. \tag{10.45}$$

This is Planck's law, the historical chapter's opening formula, recovered from the oscillator partition function and the mode density. The historical arc from the 1900 interpolation to the quantum statistical mechanics of the present chapter is drawn in one paragraph: the oscillators whose discreteness Planck postulated are the material systems of the pictures chapter ((5.24)), the field modes whose statistics Bose counted are the bosonic modes of Fock space, and the partition function whose evaluation reproduces Planck's mean energy is the canonical partition function of the single-mode oscillator — the book's oldest formula in its newest dress. The second term $\hbar \omega / 2$ is the zero-point energy, present in the oscillator chapter's spectrum (5.24) and invisible in Planck's 1900 derivation: only level differences enter the exchange frequencies ((4.7)), so the uniform shift disturbs no Planck spectrum, and the zero-point energy is confirmed in the specific heat of solids at low temperatures (the historical chapter's Einstein and Debye models). The classical limit, $k_B T \gg \hbar \omega$, gives $U \rightarrow k_B T$ — equipartition, the Rayleigh–Jeans law's energy per mode. The quantum regime, $k_B T \ll \hbar \omega$, gives $U \rightarrow \hbar \omega / 2 + \hbar \omega e^{-\beta \hbar \omega}$ — frozen-out degrees of freedom, the specific heat falling exponentially to zero. The Planck recovery is the chapter's final repayment, and it seals the loop that the historical chapter opened.

The classical limit of the quantum gases. The bridge from the quantum statistics of Sections 10.4–10.5 to the classical statistics of the historical chapter's Rayleigh–Jeans law is the thermal de Broglie wavelength,

$$\lambda_T = \frac{h}{\sqrt{2\pi m k_B T}}.$$

When the dimensionless parameter $n\lambda_T^3 \ll 1$ — dilute gas, high temperature, or large mass — the fugacity $z = e^{\beta \mu}$ is small, and both the Bose–Einstein and Fermi–Dirac distributions reduce to the Maxwell–Boltzmann form:

$$\langle n_k \rangle \longrightarrow e^{-\beta(\epsilon_k - \mu)}, \quad n\lambda_T^3 \ll 1. \tag{10.46}$$

In this limit, the ± 1 in the denominator of the quantum distributions is negligible, and the distinguishability of classical particles is recovered as an approximation whose domain of validity is precisely quantified. The quantum statistics contains the classical statistics as its dilute, high-temperature limit

— a theorem of the distributions, not a correspondence-principle plea.

The road to genuine many-body theory. The partition function $Z = \text{tr}(e^{-\beta H})$ is the starting point of all equilibrium many-body theory. The trace over Fock space can be evaluated by several methods, each the foundation of a major branch of the discipline: path integrals in imaginary time (the Matsubara formalism), perturbative expansion in the interaction (diagrammatic methods, from the Rayleigh–Schrödinger series to the Feynman–Dyson expansion), and variational principles (the Gibbs–Bogoliubov inequality, the Hartree–Fock approximation as the optimal single-determinant state). All three approaches are named here without development: they belong to the graduate course in quantum many-body theory whose door this chapter opens, and they are beyond the boundary of the present book.

The advantage of the density-operator formalism is its economy: one operator, the density operator in thermal equilibrium, encodes every macroscopic property, and the partition function from which those properties are extracted is a single trace. Its limitation is that the trace is tractable only for non-interacting systems: the factorization over modes (10.21)–(10.31) is exact for the ideal gas and fails the moment the two-body term of (10.20) is included. The interacting many-body problem — the evaluation of $\text{tr}(e^{-\beta H})$ for H containing the two-body term of (10.20) — is the central problem of quantum statistical mechanics, and its solution occupies the graduate curriculum the chapter’s exit points toward.

Transition. The oscillator’s partition function is the book’s oldest formula in its newest dress. The chapter’s work is done: the identical-particle postulate is stated, the creation/annihilation algebra is built, the quantum gases are solved, and the density operator opens the door to the statistical mechanics of quantum matter. It remains to collect the principles, balance the debt ledger, and hand the next set of questions to the disciplines they belong to.

10.7 The Second Quantization: Summary, the Repaid Debt Ledger, and Open Doors

The chapter built the quantum mechanics of many identical particles from a single new postulate — the Symmetrization Postulate — and the established axioms of the state-space, measurement, and evolution chapters. Its product is a formalism in which the combinatorics of identical particles is automated by algebraic (anti)commutation relations, the ideal quantum gases are solved exactly, and the density operator supplies the bridge to equilibrium statistical mechanics. This section collects the principles into six numbered statements, presents the updated experiment–formalism dictionary, balances the debt ledger the historical chapter opened, and names the disciplines the chapter was chartered to preview.

The six principles.

(Q1) **Identical particles.** The states of N identical particles are vectors in either the totally symmetric (bosons) or totally antisymmetric (fermions) subspace of the N -fold tensor product; the choice

is dictated by the particle's spin — integer spin selects the symmetric subspace, half-integer spin the antisymmetric one — and Fock space, the direct sum over all N , is the natural arena for variable particle number (Section 10.1, (10.3), (10.4)).

(Q2) **Creation and annihilation.** On Fock space, the ladder operators of the oscillator chapter generalize to $a_k^\dagger$ (creates a particle in mode k) and a_k (annihilates one), satisfying bosonic commutators (10.9) or fermionic anticommutators (10.11); the field operators $\psi^\dagger(\mathbf{x})$, $\psi(\mathbf{x})$ create and annihilate particles at a point. This unifies the single-mode oscillator, the radiation field's mode structure, and the Schwinger bosons under one algebraic structure (Section 10.2, (10.12)).

(Q3) **Observables in Fock space.** Every one-body and two-body observable assumes a universal second-quantized form ((10.16), (10.18)); the many-body Hamiltonian (10.20) conserves particle number, and the measurement axioms of the measurement chapter apply unchanged (Section 10.3).

(Q4) **The ideal Bose gas.** The grand canonical ensemble yields the Bose–Einstein distribution (10.22) — the quantum-statistical form of Bose's 1924 counting. Below the critical temperature (10.28), a macroscopic fraction (10.29) of particles occupies the ground state: Bose–Einstein condensation, a phase transition driven purely by quantum statistics, observed in 1995 at approximately 100 nanokelvin (Section 10.4).

(Q5) **The ideal Fermi gas.** The Fermi–Dirac distribution (10.32) and the Pauli principle together produce the Fermi sea: at $T = 0$ all states up to the Fermi energy (10.34) are filled, the degeneracy pressure (10.37) is finite at zero temperature, and low-energy excitations are confined to the Fermi surface — yielding the linear specific heat of metals (Section 10.5).

(Q6) **The density operator and the partition function.** Thermal equilibrium is encoded in the density operator (10.40), whose normalization $Z = \text{tr}(e^{-\beta H})$ is the partition function; the harmonic oscillator's partition function (10.43) recovers Planck's 1900 mean energy — the chapter's final repayment of the book's oldest formula (Section 10.6).

Experiment–formalism dictionary. Table 10.1 extends the dictionaries of the state-space, measurement, evolution, pictures, and angular momentum chapters with the fourteen rows of the second-quantization formalism.

The debt ledger, balanced. The chapter repays every entry the historical ledger planted:

- **Bose's 1924 counting** (ch1 §1.1). The Bose–Einstein distribution (10.22) is the quantum-statistical transcription of Bose's combinatorial factor; the grand partition function $\Xi = \prod_k (1 - e^{-\beta(\varepsilon_k - \mu)})^{-1}$ is the modern encoding of $W = \prod_s (A_s + N_s - 1)! / N_s! (A_s - 1)!$. Bose–Einstein condensation is its massive-particle consequence, predicted by Einstein in 1925 and observed in 1995. The debt is repaid in full at Section 10.4.
- **Pauli principle / spin** (ch1, ch6). The Symmetrization Postulate of Section 10.1 is the N -particle generalization of the angular momentum chapter's singlet (6.84); the Pauli principle is the algebraic theorem $(a_k^\dagger)^2 = 0$ in Section 10.2. The half-integer-spin $\Rightarrow$ fermion connection is stated as an experimental fact with a relativistic-QFT footnote. The debt's angular-momentum share was repaid by the angular momentum chapter; its many-body share is repaid here.
- **Planck's radiation oscillators** (ch1 §1.1). Each field mode is a quantized oscillator in the

Table 10.1: The experiment–formalism dictionary v6: second quantization. Extends Tables 2.1 through 6.3.

Experiment / Concept	Formal Object
Identical particles	(Anti)symmetric subspace of $\mathcal{H}^{\otimes N}$
Boson / fermion	Integer / half-integer spin $\Rightarrow$ symmetric / antisymmetric states
Fock vacuum	$ 0\rangle$, the unique zero-particle state
Creation / annihilation	$a_k^\dagger, a_k$ acting on occupation-number basis
Occupation number	Eigenvalue of $N_k = a_k^\dagger a_k$
Field operator	$\psi^\dagger(\mathbf{x}), \psi(\mathbf{x})$ — creates / annihilates a particle at $\mathbf{x}$
Bose–Einstein distribution	$\langle n_k \rangle = 1/(e^{\beta(\varepsilon_k - \mu)} - 1)$
BEC critical temperature	$T_c = (2\pi\hbar^2/mk_B)(n/\zeta(3/2))^{2/3}$
Condensate fraction	$N_0/N = 1 - (T/T_c)^{3/2}$
Fermi–Dirac distribution	$\langle n_k \rangle = 1/(e^{\beta(\varepsilon_k - \mu)} + 1)$
Fermi energy	$\varepsilon_F = \hbar^2(3\pi^2n)^{2/3}/2m$
Degeneracy pressure	$P = (2/5)n\varepsilon_F$
Canonical density operator	$\rho_\beta = e^{-\beta H}/Z$
Partition function	$Z = \text{tr}(e^{-\beta H}), F = -k_B T \ln Z$

sense of the pictures chapter; the mean energy $\hbar\omega/(e^{\beta\hbar\omega} - 1)$ recovered in Section 10.6 from the oscillator partition function (10.43), multiplied by the electromagnetic mode density, reproduces Planck’s law (10.45). The material-oscillator register was repaid by the pictures chapter; the field-mode and statistical shares are repaid here.

- **Einstein’s 1905 light quantum** (ch1 §1.1). The photon is the excitation created by $a^\dagger$ acting on the vacuum of the electromagnetic field. Named in Section 10.2 with the honest statement that the vector nature of the photon and the full Maxwell quantization are beyond the non-relativistic boundary. Repaid in the register of concept, not of full theory.
- **The pictures chapter’s field-mode promissory note** (debt #30). N as particle counter in many modes, $a, a^\dagger$ as creation and annihilation operators, coherent states as the classical-field register — all discharged in Sections 10.2–10.4. The laser/Glauber sentence that the pictures chapter planted is cashed in the coherent-state unification with the Schwinger construction.
- **The angular momentum chapter’s Schwinger bosons** (debt #39). The Jordan–Schwinger map (10.15) is read as a two-mode instance of the second-quantized formalism; the fixed- N constraint of the angular momentum chapter is lifted, and the spin coherent states are recovered as projected two-mode boson coherent states. Discharged in Section 10.2.
- **The angular momentum chapter’s Cooper pair pointer**. Named in the BEC context of Section 10.4: the boson-pair wave function is the two-particle bound state whose condensation is the BCS mechanism. Named, not developed.
- **The state-space chapter’s superselection debt #12** (charge share). The boson/fermion superselection of Fock space — no observable connects the $\mathcal{F}_+$ and $\mathcal{F}_-$ sectors, and the direct sum over N is a non-relativistic superselection structure — closes the angular-momentum chapter’s remaining share of the superselection footnote. The electric-charge share remains quantum field

theory's.

Planted for beyond. The chapter names, without developing, the disciplines whose door it opens:

- The interacting Bose gas: Bogoliubov theory of the weakly interacting condensate, the Beliaev diagrammatic extension, the Gross–Pitaevskii equation for the condensate wave function.
- The interacting Fermi gas: Landau's Fermi liquid theory, the BCS pairing instability of the Fermi surface, the Ginzburg–Landau theory of the superconducting transition.
- The full formalism of quantum statistical mechanics: Matsubara Green's functions, linear response theory and the Kubo formula, the functional-integral representation of the partition function.
- The proper proof of the spin-statistics theorem, belonging to relativistic quantum field theory (the Pauli 1940 proof, named in a footnote at Section 10.1).
- The quantization of the full electromagnetic field, with the vector potential, gauge invariance, and photon polarization.

All are named; none is developed. The Fock-space Hamiltonian (10.20) with the two-body term included is the exact starting point for every item on that list, and the ideal-gas solutions of Sections 10.4 and 10.5 are the zero-order states about which the interacting theory expands.

The exit question. The chapter has solved the quantum gases without interactions. Every real quantum gas is interacting. How does a weak repulsion modify the BEC critical temperature? How does an attractive interaction between fermions produce the Cooper instability and superconductivity? The Fock-space Hamiltonian (10.20) with the two-body term is the exact starting point — but it is not exactly solvable. The methods that tame the interacting many-body problem — perturbative expansions around the quantum-gas ground states, self-consistent field theories (Hartree–Fock, Gross–Pitaevskii, Bogoliubov–de Gennes), and the diagrammatic and functional-integral formulations of quantum statistical mechanics — are the natural next step. They belong to the graduate course in many-body theory, and they are beyond the boundary of this book.

Chapter 10 Exercise

1. (Section 10.1) **Two-particle symmetrization.** Let $\varphi_a(\mathbf{x})$ and $\varphi_b(\mathbf{x})$ be orthonormal single-particle wave functions.
 - (a) Construct the normalized symmetric and antisymmetric two-particle wave functions for one particle in each mode, $\psi_S(\mathbf{x}_1, \mathbf{x}_2)$ and $\psi_A(\mathbf{x}_1, \mathbf{x}_2)$.
 - (b) Compute the probability density for finding both particles at the same position, $\mathbf{x}_1 = \mathbf{x}_2 = \mathbf{x}$, for the symmetric and antisymmetric cases. Express your answer in terms of $|\varphi_a(\mathbf{x})|^2$ and $|\varphi_b(\mathbf{x})|^2$.
 - (c) Explain why for fermions this probability is zero — the exchange hole. For bosons, show that the probability at coincidence is enhanced relative to the distinguishable-particle result by a factor of two.
2. (Section 10.1) **Trace of the antisymmetrizer.**

- (a) For $N = 2$ identical fermions in a d -dimensional single-particle space, compute the dimension of the antisymmetric subspace $\text{tr}(A)$. Verify that it equals $\binom{d}{2}$. 11507
- (b) Generalize: for d single-particle modes, what is the dimension of the N -fermion Fock sector? Hence what is the condition on d for N fermions to have any state at all? 11508
- (c) Evaluate the dimension for $d = 10$, $N = 3$. For $d = 3$, $N = 4$, explain the physical meaning of a zero-dimensional fermionic sector. 11510
3. (Section 10.2) **Boson algebra from Fock-space actions.** From the definition of a_k and $a_k^\dagger$ on the occupation-number basis, 11511
- $$a_k^\dagger |\dots, n_k, \dots\rangle = \sqrt{n_k + 1} |\dots, n_k + 1, \dots\rangle, \quad a_k |\dots, n_k, \dots\rangle = \sqrt{n_k} |\dots, n_k - 1, \dots\rangle, \quad 11512$$
- (a) Prove $[a_k, a_l^\dagger] = \delta_{kl}$ by acting with the commutator on a generic occupation-number basis state. Consider the cases $k = l$ and $k \neq l$ separately. 11513
- (b) Show that $N_k = a_k^\dagger a_k$ has the stated spectrum $n_k = 0, 1, 2, \dots$ and that $N = \sum_k N_k$ has eigenvalue $\sum_k n_k$. 11514
- (c) For two bosonic modes, define the “hopping” operators $T_{12} = a_1^\dagger a_2$ and $T_{21} = a_2^\dagger a_1$. Compute $[T_{12}, T_{21}]$ and interpret the result in terms of the number operators N_1 and N_2 . 11515
4. (Section 10.2) **Fermionic signs.** The fermionic creation operator on the occupation-number basis, with modes ordered $k = 1, 2, \dots$, is defined as 11516
- $$a_k^\dagger |\dots, n_k, \dots\rangle = (1 - n_k) (-1)^{\sum_{l < k} n_l} |\dots, n_k + 1, \dots\rangle. \quad 11517$$
- (a) By considering the action of $a_k^\dagger a_l^\dagger$ and $a_l^\dagger a_k^\dagger$ on a generic basis state, show that $a_k^\dagger a_l^\dagger = -a_l^\dagger a_k^\dagger$ for $k \neq l$, and hence $\{a_k^\dagger, a_l^\dagger\} = 0$. 11518
- (b) Prove $(a_k^\dagger)^2 = 0$ — the algebraic Pauli principle — and explain why this forces $n_k \in \{0, 1\}$. 11519
- (c) For two fermionic modes, write the four Fock states $|n_1, n_2\rangle$ explicitly in the occupation-number representation. Compute the action of $a_1^\dagger$, $a_2^\dagger$, a_1 , and a_2 on each, verifying that the sign factors produce the correct anticommutation relations. 11520
5. (Section 10.3) **Contact interaction in momentum space.** For bosons with contact interaction $V(\mathbf{x}, \mathbf{y}) = g \delta(\mathbf{x} - \mathbf{y})$, 11521
- (a) Using the plane-wave basis $\varphi_{\mathbf{k}}(\mathbf{x}) = e^{i\mathbf{k} \cdot \mathbf{x}} / \sqrt{V}$, compute the matrix element $V_{\mathbf{k}, l, m, n}$ of (10.18) and show that it is $(g/V) \delta_{\mathbf{k}+l, m+n}$. 11522
- (b) Hence derive the momentum-space form of the two-body term, 11523
- $$V = \frac{g}{2V} \sum_{\mathbf{k}, \mathbf{p}, \mathbf{q}} a_{\mathbf{k}+\mathbf{q}}^\dagger a_{\mathbf{p}-\mathbf{q}}^\dagger a_{\mathbf{p}} a_{\mathbf{k}}. \quad 11524$$
- (c) Explain why this form conserves momentum: interactions scatter pairs from $(\mathbf{k}, \mathbf{p})$ to $(\mathbf{k} + \mathbf{q}, \mathbf{p} - \mathbf{q})$, and the sum of the wave vectors of the creation operators equals that of the annihilation operators. 11525
6. (Section 10.4) **BEC critical temperature.** 11526
- (a) Starting from the number equation $N = N_0 + V \int_0^\infty d\varepsilon g(\varepsilon) / (e^{\beta(\varepsilon - \mu)} - 1)$ with the density of states (10.23), set $\mu = 0$ and derive the critical temperature formula 11527

$$k_B T_c = \frac{2\pi\hbar^2}{m} \left(\frac{n}{\zeta(3/2)} \right)^{2/3}, \quad 11528$$

with $\zeta(3/2) \approx 2.612$. Show explicitly that the $\sqrt{\pi}/2$ from the Bose integral $\int_0^\infty dx \sqrt{x}/(e^x - 1) = \sqrt{\pi} \zeta(3/2)/2$ cancels against the $\sqrt{\pi}$ from the density-of-states prefactor, leaving the 2π in the final form.

(b) Compute T_c for ^{87}Rb at $n = 2.5 \times 10^{12} \text{ cm}^{-3}$ ($m = 87 u$, $u = 1.66 \times 10^{-27} \text{ kg}$).

(c) Compute T_c for ^4He at $n = 2.2 \times 10^{22} \text{ cm}^{-3}$ ($m = 4 u$). Compare with the observed λ -transition at $T_\lambda = 2.17 \text{ K}$ and discuss the origin of the discrepancy.

(d) Show that at $T = T_c$, the dimensionless phase-space density satisfies $n\lambda_T^3 = \zeta(3/2) \approx 2.612$, where $\lambda_T = h/\sqrt{2\pi m k_B T}$ is the thermal de Broglie wavelength. Interpret this condition physically in one sentence.

7. (Section 10.5) **Sommerfeld expansion.** The number and energy integrals for a degenerate Fermi gas are

$$N = \int_0^\infty d\varepsilon g(\varepsilon) f(\varepsilon), \quad E = \int_0^\infty d\varepsilon \varepsilon g(\varepsilon) f(\varepsilon),$$

with $f(\varepsilon) = 1/(e^{\beta(\varepsilon - \mu)} + 1)$ and the free-particle density of states $g(\varepsilon) = C\sqrt{\varepsilon}$ where $C = (Vm^{3/2})/(\sqrt{2}\pi^2\hbar^3)$. The Sommerfeld expansion for integrals of this form, valid at low temperatures, is

$$\int_0^\infty d\varepsilon h(\varepsilon) f(\varepsilon) = \int_0^\mu d\varepsilon h(\varepsilon) + \frac{\pi^2}{6} (k_B T)^2 h'(\mu) + O(T^4).$$

(a) Apply the Sommerfeld expansion to the number integral with $h(\varepsilon) = g(\varepsilon)$. Show that the chemical potential at low temperature is

$$\mu(T) = \varepsilon_F \left[1 - \frac{\pi^2}{12} \left(\frac{k_B T}{\varepsilon_F} \right)^2 + \cdots \right],$$

where ε_F is the $T = 0$ Fermi energy (10.34).

(b) Apply the Sommerfeld expansion to the energy integral with $h(\varepsilon) = \varepsilon g(\varepsilon)$. Using the result of part (a) for $\mu(T)$, derive the low-temperature specific heat

$$C_V = \frac{\pi^2}{2} k_B n \frac{k_B T}{\varepsilon_F}.$$

(c) For copper, $\varepsilon_F \approx 7.0 \text{ eV}$ and $n \approx 8.5 \times 10^{22} \text{ cm}^{-3}$. Evaluate C_V at $T = 300 \text{ K}$ and compare with the classical value $\frac{3}{2} N k_B$. By what factor is the quantum specific heat smaller?

8. (Section 10.6) **Harmonic oscillator partition function.**

(a) Sum the geometric series to obtain the harmonic oscillator partition function,

$$Z_{\text{ho}} = \sum_{n=0}^{\infty} e^{-\beta \hbar \omega (n+1/2)} = \frac{1}{2 \sinh(\beta \hbar \omega / 2)}.$$

(b) From $U = -\partial \ln Z / \partial \beta$, derive the mean energy and show that it interpolates between the zero-point value $\hbar \omega / 2$ (as $T \rightarrow 0$) and the equipartition value $k_B T$ (as $T \rightarrow \infty$). Plot the heat capacity $C = \partial U / \partial T$ as a function of $k_B T / \hbar \omega$ and identify the temperature scale at which the quantum degree of freedom “freezes out.”

(c) Multiply the mean thermal energy per mode by the electromagnetic mode density $g(\nu) d\nu = 8\pi \nu^2 / c^3 d\nu$ and recover Planck’s law for the spectral energy density,

$$\frac{du_\nu}{d\nu} = \frac{8\pi h \nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}.$$

Identify which term in the mean energy contributes to Planck's law and explain why the zero-point energy $\hbar\omega/2$ is invisible in the black-body spectrum.

11579

11580

Appendix A Mathematical Supplement



Note This appendix collects the rigorous functional-analytic statements whose full proofs lie beyond the scope of a first course in quantum mechanics but whose honest deployment the main text requires. Every theorem below is stated with its hypotheses made explicit and with a pointer to the standard literature. The reader who is encountering these results for the first time should consult Reed and Simon, *Methods of Modern Mathematical Physics, Volumes I–IV* (Academic Press, 1972–1978), which is the principal reference throughout. Nothing in this appendix is needed for the computational mastery of the main text; it exists to make the register of honesty complete.

A.1 Hilbert Space: Completeness, Separability, and Domains

A complex inner-product space $\mathcal{H}$ whose norm $\|\psi\| := \sqrt{\langle\psi|\psi\rangle}$ satisfies the Cauchy-sequence completeness condition — every Cauchy sequence converges to a limit in $\mathcal{H}$ — is a **Hilbert space**. The main text introduces this as Axiom S1, with the experimental warrant that a sequence of refining preparations must converge *to a state*. The mathematical content of the warrant is three clauses.

Completeness. A sequence $\{\psi_n\} \subset \mathcal{H}$ is Cauchy if $\|\psi_n - \psi_m\| \rightarrow 0$ as $n, m \rightarrow \infty$. Completeness is the statement that for every such sequence there exists a $\psi \in \mathcal{H}$ with $\|\psi_n - \psi\| \rightarrow 0$. The limit is unique by the standard argument: if $\psi_n \rightarrow \psi$ and $\psi_n \rightarrow \phi$, then $\|\psi - \phi\| \leq \|\psi - \psi_n\| + \|\psi_n - \phi\| \rightarrow 0$.

Separability. A Hilbert space is **separable** if it possesses a countable orthonormal basis. The main text's Axiom S1 requires separability: the dimension of the state space is at most countably infinite. Every separable infinite-dimensional complex Hilbert space is isometrically isomorphic to the sequence space

$$\ell^2 := \left\{ (c_1, c_2, \dots) : c_n \in \mathbb{C}, \sum_{n=1}^{\infty} |c_n|^2 < \infty \right\}, \quad (\text{A.1})$$

with inner product $\langle\phi|\psi\rangle = \sum_n b_n^* c_n$. The isomorphism is provided by the expansion $|\psi\rangle = \sum_n c_n |e_n\rangle$ in any orthonormal basis $\{|e_n\rangle\}$, which the main text establishes as Theorem (2.32). The square-summability condition $\sum_n |c_n|^2 < \infty$ is the Parseval identity (2.35). This is the structural reason that bound-state spectra are discrete and that every state of a finite system can be approximated by a finite superposition: the ℓ^2 condition ensures that the tail of the expansion carries arbitrarily little weight.

Domains of unbounded operators. A linear operator A on $\mathcal{H}$ is specified not merely by its action but also by its **domain** $\mathcal{D}(A) \subseteq \mathcal{H}$, the linear subspace on which the action is defined. An operator is **densely defined** if $\mathcal{D}(A)$ is dense in $\mathcal{H}$. The **adjoint** $A^\dagger$ is defined on the domain

$$\mathcal{D}(A^\dagger) := \left\{ \phi \in \mathcal{H} : \exists \eta \in \mathcal{H} \text{ such that } \langle\phi|A\psi\rangle = \langle\eta|\psi\rangle \ \forall \psi \in \mathcal{D}(A) \right\}, \quad (\text{A.2})$$

and $A^\dagger\phi := \eta$. An operator is **symmetric** if $\langle\phi|A\psi\rangle = \langle A\phi|\psi\rangle$ for all $\phi, \psi \in \mathcal{D}(A)$; it is **self-adjoint** if in addition $\mathcal{D}(A^\dagger) = \mathcal{D}(A)$. The distinction is invisible in finite dimension (where every symmetric operator on the whole space is self-adjoint) but load-bearing for the unbounded operators of quantum

mechanics: Q , P , and H are symmetric on a natural dense domain but their self-adjointness requires a careful choice of boundary conditions. A symmetric operator may admit no self-adjoint extension, or many; the physics selects the extension through the boundary conditions appropriate to the system.

The position and momentum operators of Chapter 5 furnish the canonical example. On $\mathcal{H} = L^2(\mathbb{R})$, the operator Q defined by $(Q\psi)(x) = x\psi(x)$ with domain $\mathcal{D}(Q) = \{\psi \in L^2(\mathbb{R}) : \int x^2 |\psi(x)|^2 dx < \infty\}$ is self-adjoint. The momentum operator $P = -i\hbar d/dx$ with domain $\mathcal{D}(P) = \{\psi \in L^2(\mathbb{R}) : \psi \text{ absolutely continuous, } \psi' \in L^2(\mathbb{R})\}$ is likewise self-adjoint. Both are unbounded: $\|Q\psi\|$ and $\|P\psi\|$ are not bounded by a constant multiple of $\|\psi\|$ on their domains. The canonical commutation relation $[Q, P] = i\hbar I$ holds on the intersection of the domains of QP and PQ (the Schwartz space $\mathcal{S}(\mathbb{R})$ is a common dense invariant domain), and the Wintner–Wielandt theorem cited in Chapter 5 states that no such relation can be realized by bounded operators on any Hilbert space: the infinite-dimensional arena is forced, not chosen.¹

A.2 The Dirac Delta as a Distribution: the Gelfand Triplet

The main text deploys the notation $|x\rangle$ and the normalization $\langle x|x'\rangle = \delta(x - x')$ under an explicit honesty clause: the kets $|x\rangle$ are not vectors of the Hilbert space, the delta is not a function, and the notation is a licensed shorthand for the continuum composition law. This section makes the underlying mathematics precise.

Tempered distributions. The **Schwartz space** $\mathcal{S}(\mathbb{R})$ is the space of rapidly decreasing smooth functions: $\phi \in C^\infty(\mathbb{R})$ such that $\sup_x |x^m \phi^{(n)}(x)| < \infty$ for all $m, n \in \mathbb{N}$. Its continuous dual $\mathcal{S}'(\mathbb{R})$ is the space of **tempered distributions**. The Dirac delta is the tempered distribution defined by $\delta(\phi) := \phi(0)$ for all $\phi \in \mathcal{S}(\mathbb{R})$. The formal integral notation $\int \delta(x)\phi(x) dx = \phi(0)$ is a notational convenience; the integral sign does not denote Lebesgue integration.

The Gelfand triplet (rigged Hilbert space). The mathematical structure that houses both the normalizable state vectors and the idealizations $|x\rangle, |p\rangle$ is the **Gelfand triplet** (or rigged Hilbert space):

$$\Phi \subset \mathcal{H} \subset \Phi', \quad (\text{A.3})$$

where Φ is a dense subspace of $\mathcal{H}$ carrying a finer topology (making the inclusion continuous), and Φ' is its continuous dual (the space of continuous antilinear functionals on Φ). The construction runs as follows.²

For the position representation on $\mathcal{H} = L^2(\mathbb{R})$, choose $\Phi = \mathcal{S}(\mathbb{R})$, the Schwartz space. The inclusion $\mathcal{S}(\mathbb{R}) \subset L^2(\mathbb{R})$ is continuous and dense. For each $x \in \mathbb{R}$, the evaluation map $\phi \mapsto \phi(x)$ is a continuous linear functional on $\mathcal{S}(\mathbb{R})$; it is therefore an element of $\mathcal{S}'(\mathbb{R})$, which we denote $\langle x|$. That is, $\langle x|$ is defined by its action $\langle x|\phi \rangle := \phi(x)$ for every $\phi \in \mathcal{S}(\mathbb{R})$. The notation is designed to

¹A. Wintner, *Phys. Rev.* **71**, 738 (1947); H. Wielandt, *Math. Ann.* **121**, 21 (1949). The proof for the trace-class case is the finite-dimensional trace obstruction displayed in Chapter 5; the full bounded-operator statement is in Reed & Simon, Vol. II, Sec. VIII.5.

²I. M. Gel'fand and N. Ya. Vilenkin, *Generalized Functions*, Vol. 4 (Academic Press, 1964). A concise modern treatment is in Reed & Simon, Vol. II, Sec. V.3.

mimic an inner product because the map is antilinear in its first argument when the Dirac formalism is respected; but $\langle x| \in \Phi'$, not Φ , and there is no vector $|x\rangle \in \mathcal{H}$ whose inner product with $|\phi\rangle$ produces $\phi(x)$. The ket $|x\rangle$ appearing in the main text's formal manipulations is the distributional dual of the bra.

The completeness relation $\int dx |x\rangle \langle x| = I$ acquires its precise meaning in this framework: for all $\phi, \psi \in \Phi$,

$$\langle \phi | \psi \rangle = \int_{-\infty}^{+\infty} \langle \phi | x \rangle \langle x | \psi \rangle dx = \int_{-\infty}^{+\infty} \phi(x)^* \psi(x) dx, \quad (\text{A.4})$$

which is the ordinary L^2 inner product. The integral converges as a Riemann integral because Schwartz functions are smooth and rapidly decreasing. Every operation performed with the continuum shorthand in Chapters 2, 3, and 5 is legitimate when the states are drawn from Φ (or, more generally, from the domain of the relevant unbounded operator). The extension to all of $\mathcal{H}$ by continuity is the content of the spectral theorem below.

The momentum eigenstates $|p\rangle$ are constructed analogously: $\langle p | \phi \rangle := \tilde{\phi}(p)$, the Fourier transform of ϕ evaluated at p . The relation $\langle x | p \rangle = (2\pi\hbar)^{-1/2} e^{ipx/\hbar}$ is then the kernel of the Fourier transform relating the two distributional bases. Both families $\{|x\rangle\}$ and $\{|p\rangle\}$ are **generalized eigenbases** of their respective operators, living in Φ' rather than $\mathcal{H}$.

A.3 The Spectral Theorem for Unbounded Self-Adjoint Operators

The finite-dimensional spectral theorem proved in Chapter 3 states that every self-adjoint operator on $\mathbb{C}^n$ possesses an orthonormal eigenbasis. The infinite-dimensional generalization is the central theorem of functional analysis for quantum mechanics.

Theorem (Spectral Theorem for unbounded self-adjoint operators, stated without proof). *Let A be a (possibly unbounded) self-adjoint operator on a separable Hilbert space $\mathcal{H}$, with domain $\mathcal{D}(A)$. There exists a unique projection-valued measure (PVM) P on the Borel subsets of $\mathbb{R}$ such that*

$$\langle \phi | A \psi \rangle = \int_{\mathbb{R}} \lambda d \langle \phi | P(\lambda) | \psi \rangle \quad \text{for all } \phi \in \mathcal{H}, \psi \in \mathcal{D}(A), \quad (\text{A.5})$$

where $d \langle \phi | P(\lambda) | \psi \rangle$ is the complex measure induced by the PVM. Symbolically,

$$A = \int_{\mathbb{R}} \lambda dP(\lambda). \quad (\text{A.6})$$

The spectrum $\sigma(A)$ is the support of the measure P : it consists of all $\lambda \in \mathbb{R}$ for which $P((\lambda - \varepsilon, \lambda + \varepsilon)) \neq 0$ for every $\varepsilon > 0$.

The proof is collected in Reed & Simon, Vol. I, Sec. VIII.3. The statement generalizes the finite-dimensional spectral decomposition $A = \sum_n a_n P_n$ of Chapter 3's (3.23). The sum over discrete eigenvalues becomes an integral over the continuum with respect to a projection-valued measure. A point $\lambda \in \sigma(A)$ is an **eigenvalue** exactly when $P(\{\lambda\}) \neq 0$, i.e., when the PVM assigns nonzero weight to the singleton $\{\lambda\}$. The **continuous spectrum** is the set of λ for which $P(\{\lambda\}) = 0$ but

$P((\lambda - \varepsilon, \lambda + \varepsilon)) \neq 0$ for all $\varepsilon > 0$. The **point spectrum** and **continuous spectrum** together exhaust $\sigma(A)$.

For $A = Q$ (position), the PVM is $(P(\Omega)\psi)(x) = \chi_\Omega(x)\psi(x)$, where χ_Ω is the indicator function of the Borel set Ω . The spectrum is purely continuous: $\sigma(Q) = \mathbb{R}$, and $P(\{\lambda\}) = 0$ for every singleton. For $A = H$ (a typical Hamiltonian), the spectrum is the union of a discrete set of eigenvalues (bound states) and a continuum $[E_0, \infty)$ (scattering states). The spectral theorem is the mathematical foundation of Chapter 3's honest preview of continuous spectra (Section 3.9) and of every statement in the book that invokes the eigenbasis of an unbounded observable.

The functional calculus follows from the spectral theorem: for any Borel-measurable function $f : \mathbb{R} \rightarrow \mathbb{R}$, the operator $f(A)$ is defined by

$$f(A) := \int_{\mathbb{R}} f(\lambda) dP(\lambda), \quad (\text{A.7})$$

with domain $\mathcal{D}(f(A)) = \{\psi \in \mathcal{H} : \int |f(\lambda)|^2 d\langle \psi | P(\lambda) | \psi \rangle < \infty\}$. The evolution operator $U(t) = e^{-iHt/\hbar}$ is the instance $f(\lambda) = e^{-i\lambda t/\hbar}$ applied to $A = H$, and its domain is all of $\mathcal{H}$ because $|e^{-i\lambda t/\hbar}| = 1$ is bounded. The resolvent $(A - zI)^{-1}$ for $z \notin \sigma(A)$ is the instance $f(\lambda) = (\lambda - z)^{-1}$, and its norm is bounded by $1/\text{dist}(z, \sigma(A))$.

A.4 Stone's Theorem

Stone's theorem is cited without proof at Chapter 4's (4.2) as the bridge from the evolution axiom's one-parameter unitary family to its self-adjoint generator. This section records the full statement and the domain clauses that the main text's honesty note defers to this appendix.

Theorem (Stone, 1932; stated without proof). *Let $U(t)$, $t \in \mathbb{R}$, be a strongly continuous one-parameter unitary group on a separable Hilbert space $\mathcal{H}$:*

$$U(t+s) = U(t)U(s), \quad U(0) = I, \quad U(t)^\dagger U(t) = I, \quad (\text{A.8})$$

and for every $\psi \in \mathcal{H}$, $\lim_{t \rightarrow 0} \|U(t)\psi - \psi\| = 0$ (strong continuity). Then there exists a unique self-adjoint operator H on $\mathcal{H}$ such that

$$U(t) = e^{-iHt/\hbar}. \quad (\text{A.9})$$

The domain of H is precisely the set of vectors $\psi \in \mathcal{H}$ for which the limit

$$\lim_{t \rightarrow 0} \frac{U(t)\psi - \psi}{t} \quad (\text{A.10})$$

exists in the norm topology of $\mathcal{H}$, and on this domain $H\psi = i\hbar \times$ that limit. Conversely, every self-adjoint operator H generates a strongly continuous one-parameter unitary group via the functional calculus (A.7) applied to $f_t(\lambda) = e^{-i\lambda t/\hbar}$.

The proof is collected in Reed & Simon, Vol. II, Sec. X.1. The converse direction — that every self-adjoint H generates a unitary group via the spectral theorem — is sometimes called Stone's theorem as well; it establishes the one-to-one correspondence between strongly continuous one-parameter unitary groups and self-adjoint operators. The theorem is the reason the main text's evolution axiom (D1) and its infinitesimal form (4.3) are equivalent: the group law and continuity fix the generator, and

the generator reconstructs the group. The $\hbar$ in the exponent is, as Chapter 4 emphasizes, dimensional; its insertion does not alter the mathematical content of the theorem.

For the special case $\mathcal{H} = \mathbb{C}^n$, Stone's theorem reduces to the elementary statement that every continuous unitary representation $U(t)$ of $\mathbb{R}$ is of the form $e^{-iHt/\hbar}$ with H Hermitian, proved by differentiating the group law at zero and invoking the finite-dimensional spectral theorem. The depth of the theorem lies entirely in the passage from finite to infinite dimension, where the domain clauses become essential.

A.5 The Trotter Product Formula

The Trotter product formula, stated without proof, is the analytic engine behind the Feynman path integral (Chapter 9) and the factorization of evolution operators in many-body theory (Chapter 10).

Theorem (Trotter, 1959; stated without proof). *Let A and B be self-adjoint operators on a separable Hilbert space $\mathcal{H}$ such that $A + B$ is essentially self-adjoint on $\mathcal{D}(A) \cap \mathcal{D}(B)$. Then for all $t \in \mathbb{R}$ and all $\psi \in \mathcal{H}$,*

$$e^{-it(A+B)/\hbar} \psi = \lim_{n \rightarrow \infty} \left(e^{-itA/n\hbar} e^{-itB/n\hbar} \right)^n \psi, \quad (\text{A.11})$$

where the limit is in the strong (norm) topology of $\mathcal{H}$.

The formula appears in the main text in two contexts, neither invoking it by name. First, the composition property of the propagator (5.40) — inserting $n - 1$ resolutions of the identity between n short-time factors — is the path integral's defining combinatorial step; the Trotter formula justifies taking the $n \rightarrow \infty$ limit when the Hamiltonian splits as $H = P^2/2m + V(Q)$. Second, the Weyl relation (5.14) can be read as the special case where $[A, B]$ is central, so the Trotter limit converges to the BCH-exact form $e^{-it(A+B)/\hbar} e^{-t^2[A,B]/2\hbar^2}$, but the general case requires the limit.

The proof is collected in Reed & Simon, Vol. I, Sec. VIII.8. The essential self-adjointness hypothesis on $A + B$ is nontrivial for the unbounded operators of quantum mechanics; the standard sufficient condition (Kato–Rellich) is satisfied by $H = P^2/2m + V(Q)$ for a wide class of potentials.

A.6 Fresnel Integrals and Conditional Convergence

The free propagator of Chapter 5 (5.42) involves the Fresnel integral

$$\int_{-\infty}^{+\infty} e^{i\alpha p^2} dp, \quad (\text{A.12})$$

which is not absolutely convergent: $|e^{i\alpha p^2}| = 1$ for all real p , so the integral over an infinite domain oscillates without decay. The main text's footnote notes that the integral is evaluated by inserting a convergence factor $e^{-\varepsilon p^2}$ and taking $\varepsilon \rightarrow 0^+$ after integration. This section makes the procedure explicit.

The properly regulated Fresnel integral is

$$\int_{-\infty}^{+\infty} e^{i\alpha p^2} dp := \lim_{\varepsilon \rightarrow 0^+} \int_{-\infty}^{+\infty} e^{i\alpha p^2 - \varepsilon p^2} dp, \quad (\text{A.13})$$

provided the limit exists and is independent of the regulator. For $\alpha \neq 0$, the regulated Gaussian integral evaluates to

$$\int_{-\infty}^{+\infty} e^{-(\varepsilon - i\alpha)p^2} dp = \sqrt{\frac{\pi}{\varepsilon - i\alpha}}. \quad (\text{A.14})$$

The square root is the principal branch, and the limit $\varepsilon \rightarrow 0^+$ must be taken with care to select the correct phase. Writing $\varepsilon - i\alpha = \sqrt{\varepsilon^2 + \alpha^2} e^{-i\phi}$ with $\phi = \arctan(\alpha/\varepsilon) \in (0, \pi/2)$ for $\alpha > 0$, the limit $\varepsilon \rightarrow 0^+$ gives $\phi \rightarrow \pi/2$, hence

$$\lim_{\varepsilon \rightarrow 0^+} \sqrt{\frac{\pi}{\varepsilon - i\alpha}} = \sqrt{\frac{\pi}{|\alpha|}} e^{i(\pi/4) \operatorname{sgn}(\alpha)} = \sqrt{\frac{\pi}{i\alpha}}, \quad (\text{A.15})$$

where the final equality uses $\sqrt{i} = e^{i\pi/4}$ and $\sqrt{-i} = e^{-i\pi/4}$. The result is

$$\int_{-\infty}^{+\infty} e^{i\alpha p^2} dp = \sqrt{\frac{\pi}{i\alpha}} \quad (\alpha \neq 0), \quad (\text{A.16})$$

understood in the regulated sense (A.13). Applied to the free propagator with $\alpha = -t/2m\hbar$, one obtains the prefactor $\sqrt{m/2\pi i\hbar t}$ of Chapter 5's (5.42).

The same technique underlies every Gaussian integral with a purely imaginary quadratic form that appears in the book: the path integral's free-particle kernel (Chapter 9), the stationary-phase expansion, and the coherent-state overlap integrals. In each case the integral is conditionally convergent — the limit of symmetric finite-range integrals exists, but the Lebesgue integral over $\mathbb{R}$ does not — and the regulator makes the computation rigorous. The independence of the limit from the choice of regulator (Gaussian, sharp cutoff, or otherwise) follows from the smoothness of the integrand's oscillations: the Fresnel integral is an **improper Riemann integral** whose value is fixed by Abel summation.

For reference, the standard Fresnel integrals in real form are

$$\int_0^\infty \cos(x^2) dx = \int_0^\infty \sin(x^2) dx = \sqrt{\frac{\pi}{8}}, \quad (\text{A.17})$$

derived by the same regularization or by contour integration in the complex plane.

Appendix B Group Theory Supplement



Note This appendix develops the Lie-group machinery that Chapter 4 deploys in physical argument and Chapter 6 harvests for angular momentum. The treatment is systematic and, for the rotation group, complete: definitions, lemmas, and theorems with proofs where they illuminate the physics. The principal references are B. C. Hall, *Lie Groups, Lie Algebras, and Representations*, 2nd ed. (Springer, 2015); W.-K. Tung, *Group Theory in Physics* (World Scientific, 1985); M. Hamermesh, *Group Theory and Its Application to Physical Problems* (Dover, 1989); and, for the Racah algebra, A. R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton, 1957) and D. A. Varshalovich, A. N. Moskalev & V. K. Khersonskii, *Quantum Theory of Angular Momentum* (World Scientific, 1988). The reader who needs the remaining proofs should consult those works; what follows supplies the definitions, the logical architecture, and the explicit formulae on which the main text's physical arguments rest.

B.1 Lie Groups and Lie Algebras

Definition (Lie group). A **Lie group** G is a group that is also a smooth manifold, such that multiplication $G \times G \rightarrow G$, $(g_1, g_2) \mapsto g_1 g_2$, and inversion $G \rightarrow G$, $g \mapsto g^{-1}$, are smooth maps. The dimension of the manifold is the **dimension** of the group. A **matrix Lie group** is a closed subgroup of $\mathrm{GL}(n, \mathbb{C})$; for such groups smoothness is automatic from the matrix entries, and the dimension is the number of independent real parameters.

The main text's central examples are the Galilei group (10 parameters: 3 rotations, 3 boosts, 3 translations, 1 time translation) and the rotation group $\mathrm{SO}(3)$ (3 parameters: the Euler angles). The concrete matrix groups of this appendix are

$$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \det R = +1\},$$

$$\mathrm{SU}(2) = \{U \in \mathbb{C}^{2 \times 2} \mid U^\dagger U = I, \det U = +1\} = \{u_0 I + i \mathbf{u} \cdot \boldsymbol{\sigma} \mid u_0, \mathbf{u} \in \mathbb{R}^4, u_0^2 + |\mathbf{u}|^2 = 1\},$$

so $\mathrm{SU}(2) \cong S^3$ (the three-sphere) and is compact, connected, and simply connected; $\mathrm{SO}(3)$ is compact and connected but not simply connected (§B.8).

Definition (Lie algebra). The **Lie algebra** $\mathfrak{g}$ of a Lie group G is the tangent space to G at the identity, equipped with the Lie bracket $[X, Y]$ derived from the group commutator. Concretely, for a matrix group, $\mathfrak{g}$ is the vector space of matrices X such that $e^{tX} \in G$ for all real t , and the Lie bracket is the matrix commutator $[X, Y] = XY - YX$. The Lie algebra is a real vector space of the same dimension as the group; it captures the group's structure infinitesimally.

For $\mathrm{SO}(3)$, $\mathfrak{so}(3) = \{X \in \mathbb{R}^{3 \times 3} \mid X^T = -X\}$ (real antisymmetric matrices); for $\mathrm{SU}(2)$, $\mathfrak{su}(2) = \{X \in \mathbb{C}^{2 \times 2} \mid X^\dagger = -X, \mathrm{tr} X = 0\}$ (traceless anti-Hermitian matrices). Their complexifications coincide: $\mathfrak{so}(3)_{\mathbb{C}} \cong \mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C})$, the algebra of traceless 2×2 complex matrices. This is the algebraic reason a group and its cover share the same representation theory up to global issues (Section B.8).

Definition (Exponential map). The **exponential map** $\exp : \mathfrak{g} \rightarrow G$ sends $X \mapsto e^X = \sum_{n=0}^{\infty} X^n/n!$. It is a local diffeomorphism near the identity: every group element close to e is e^X for a unique small $X \in \mathfrak{g}$. Globally, $\exp$ is surjective onto the connected component of the identity for *compact* connected groups (in particular $SU(2)$: every unitary is $e^{-i(\theta/2)\mathbf{n} \cdot \boldsymbol{\sigma}}$), but not in general — it fails for $SL(2, \mathbb{R})$, an instance this book does not need.

Group multiplication is encoded in the bracket through the Baker–Campbell–Hausdorff (BCH) formula:

$$e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]+\frac{1}{12}[X,[X,Y]]-\frac{1}{12}[Y,[X,Y]]+\cdots}, \quad (\text{B.1})$$

a series in iterated commutators that converges for small X, Y . Chapter 4’s (4.20) is the leading nontrivial term: the group commutator $e^{\varepsilon A} e^{\varepsilon B} e^{-\varepsilon A} e^{-\varepsilon B} = I + \varepsilon^2[A, B] + O(\varepsilon^3)$ extracts the bracket as the infinitesimal shadow of the group’s noncommutation. When $[X, Y]$ is central (commutes with both X and Y) the series truncates to $e^X e^Y = e^{X+Y+\frac{1}{2}[X,Y]}$ — the identity used throughout Chapter 4’s Galilean analysis.

Structure constants. Choose a basis $\{T_a\}_{a=1}^{\dim \mathfrak{g}}$ for the Lie algebra. The bracket of two basis elements is again in the algebra:

$$[T_a, T_b] = \sum_c f_{ab}{}^c T_c, \quad (\text{B.2})$$

where the real numbers $f_{ab}{}^c$ are the **structure constants**. They are antisymmetric in the first two indices, $f_{ab}{}^c = -f_{ba}{}^c$, and satisfy the **Jacobi identity**

$$\sum_d (f_{ab}{}^d f_{dc}{}^e + f_{bc}{}^d f_{da}{}^e + f_{ca}{}^d f_{db}{}^e) = 0, \quad (\text{B.3})$$

the algebraic shadow of the associativity of group multiplication. For $\mathfrak{so}(3)$ with the quantum-mechanical (self-adjoint) basis J_i , $[J_i, J_j] = i\hbar \epsilon_{ijk} J_k$, so $f_{ij}{}^k = i\hbar \epsilon_{ijk}$ in the physicist’s convention (the abstract algebra has $f_{ij}{}^k = \epsilon_{ijk}$); Chapter 4’s (4.32) is the complete Galilean algebra in this convention.

The adjoint representation. Every Lie algebra acts on itself by the bracket: the **adjoint representation** $\text{ad}_X(Y) := [X, Y]$ is a Lie-algebra homomorphism $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$. In a basis, $(\text{ad}_{T_a})_b{}^c = f_{ab}{}^c$: the structure constants are the matrix elements of the adjoint representation. The Jacobi identity (B.3) is precisely the statement that ad preserves the bracket. Exponentiating gives the **Ad** representation of the group on its algebra, $\text{Ad}_g(X) = gXg^{-1}$ for a matrix group; the adjoint of $SU(2)$ is the $j = 1$ (vector) representation, and $\text{Ad} : SU(2) \rightarrow SO(3)$ is the 2:1 covering map of (6.7).

The Killing form. The bilinear form

$$\kappa(X, Y) := \text{tr}(\text{ad}_X \text{ad}_Y) = \sum_{a,b} f_{ab}{}^a f_{bc}{}^d \quad (\text{in a basis}) \quad (\text{B.4})$$

is the **Killing form**, an invariant symmetric bilinear form on $\mathfrak{g}$ ($\kappa([Z, X], Y) + \kappa(X, [Z, Y]) = 0$).

Cartan’s criterion: $\mathfrak{g}$ is semisimple (has no nonzero abelian ideal) iff κ is nondegenerate. For $\mathfrak{so}(3) \cong \mathfrak{su}(2)$, $\kappa(T_a, T_b) \propto \delta_{ab}$ (negative definite in the anti-Hermitian convention), confirming semisimplicity; the Galilei algebra, which contains the abelian ideal of translations, is *not* semisimple — whence its central extension by mass (Chapter 4) is a nontrivial cohomological fact rather than a triviality.

Semisimple, solvable, reductive. A Lie algebra is **solvable** if iterated brackets eventually vanish

(the abelian algebra is the prototype); **semisimple** if it is a direct sum of simple nonabelian algebras; **reductive** if it is the direct sum of a semisimple and an abelian algebra. The Galilei algebra is reductive-with-extension (semisimple $\mathfrak{so}(3)$ plus abelian translations/boosts plus the central mass); the Poincaré and Lorentz algebras are semisimple up to translations. Representation theory is cleanest for semisimple algebras (complete reducibility, highest-weight classification); the nonsemisimple case requires the induced-representation machinery of Mackey, deployed in Chapter 4's construction of the Galilean unirreps.

B.2 Representations

Definition (Representation). A **representation** of a Lie group G on a vector space V is a homomorphism $\Pi : G \rightarrow \mathrm{GL}(V)$: a smooth map with $\Pi(g_1 g_2) = \Pi(g_1) \Pi(g_2)$ and $\Pi(e) = I$. Its dimension is $\dim V$. A representation of a Lie algebra $\mathfrak{g}$ on V is a Lie-algebra homomorphism $\pi : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$: a linear map with $\pi([X, Y]) = [\pi(X), \pi(Y)]$. For a matrix Lie group, the **defining representation** is the obvious one: $\mathrm{SO}(3)$ acts on $\mathbb{R}^3$ by matrix multiplication, $\mathrm{SU}(2)$ on $\mathbb{C}^2$. A group representation Π induces an algebra representation $\pi(X) = \left. \frac{d}{dt} \Pi(e^{tX}) \right|_{t=0}$; conversely $\Pi(e^X) = e^{\pi(X)}$ when Π exists.

The quantum-mechanical state space carries a **unitary** representation: $V = \mathcal{H}$ and each $\Pi(g)$ is unitary. By Stone's theorem (Appendix A.4), the one-parameter subgroups $t \mapsto \Pi(e^{tX})$ are generated by self-adjoint operators $\pi(X)$, and the algebra representation π produces the physical observables: momentum, angular momentum, energy. By Wigner's theorem the physical action is a *projective* (ray) representation; the obstruction to making it a true representation is a cohomology class of the group, the content of Bargmann's theorem for the Galilei group (Chapter 4).

Irreducibility and equivalence. A representation is **irreducible** (an **irrep**) if V has no nontrivial invariant subspace: the only $W \subseteq V$ with $\Pi(g)W \subseteq W$ for all g are $\{0\}$ and V . Two representations Π_1, Π_2 on V_1, V_2 are **equivalent** if an invertible linear map $T : V_1 \rightarrow V_2$ intertwines them, $T \Pi_1(g) = \Pi_2(g) T$ for all g . The irreps are the elementary building blocks: every finite-dimensional unitary representation of a compact group decomposes uniquely (up to order and equivalence) as a direct sum of irreps — **complete reducibility**. The physical states of an elementary system (a particle of definite mass and spin) transform in an irrep of the symmetry group; the $(2j + 1)$ -dimensional multiplets of Chapter 6 are the irreps of $\mathrm{SU}(2)$.

Lemma (Schur's lemma). Let Π_1, Π_2 be irreducible representations of G on V_1, V_2 , and let $T : V_1 \rightarrow V_2$ satisfy $T \Pi_1(g) = \Pi_2(g) T$ for all $g \in G$ (an *intertwining map*). Then either $T = 0$, or T is an isomorphism (so Π_1, Π_2 are equivalent). In particular, if $\Pi_1 = \Pi_2 = \Pi$ is a finite-dimensional complex irrep, then $T = \lambda I$ for some $\lambda \in \mathbb{C}$.

Proof. The kernel $\ker T$ is invariant in V_1 : if $Tv = 0$, then $T(\Pi_1(g)v) = \Pi_2(g)(Tv) = 0$. By irreducibility, $\ker T$ is $\{0\}$ or V_1 . The image $\mathrm{im} T$ is invariant in V_2 : if $w = Tv$, then $\Pi_2(g)w = \Pi_2(g)Tv = T(\Pi_1(g)v) \in \mathrm{im} T$. By irreducibility, $\mathrm{im} T$ is $\{0\}$ or V_2 . If $T \neq 0$ then $\ker T = \{0\}$ (injective) and $\mathrm{im} T = V_2$ (surjective), so T is an isomorphism. For the second statement, let λ be an

eigenvalue of T (which exists since V_1 is a finite-dimensional complex vector space). Then $T - \lambda I$ also intertwines Π and has nontrivial kernel; by the first part $T - \lambda I = 0$. ■

Corollary (irreducibility of abelian groups). Every finite-dimensional complex irrep of an abelian group is one-dimensional: each group element g commutes with every other, so $\Pi(g)$ itself intertwines Π ; by Schur, $\Pi(g) = \lambda(g)I$, and an irreducible space of scalar operators is one-dimensional. This is the group-theoretic root of the simultaneous-diagonalization structure of compatible observables (Chapter 3).

Corollary (irreps are determined by their Casimirs). If C is a Casimir (an operator commuting with every $\pi(X)$), then in an irrep $C = \lambda I$ by Schur. The eigenvalue λ labels the irrep: for $SU(2)$, the single Casimir J^2 with eigenvalue $\hbar^2 j(j+1)$ labels the $(2j+1)$ -dimensional irrep (Section B.3). For a semisimple algebra of rank r there are r independent Casimirs, and (for compact groups) their joint eigenvalues label the irreps.

The Haar measure. On a compact group G there is a unique (up to scale) **Haar measure** dg that is left- and right-invariant: $\int_G f(hg) dg = \int_G f(g) dg = \int_G f(gh) dg$ for all $h \in G$. It is normalized by $\int_G dg = 1$. For $SO(3)$ and $SU(2)$ in Euler angles (α, β, γ) ,

$$dg = \frac{1}{8\pi^2} \sin \beta d\alpha d\beta d\gamma, \quad \alpha \in [0, 2\pi), \beta \in [0, \pi], \gamma \in [0, 2\pi) \text{ (or } [0, 4\pi) \text{ for } SU(2)), \quad (\text{B.5})$$

so $\int_{SU(2)} dg = 1$. The Haar measure makes $L^2(G)$ a Hilbert space and underwrites every orthogonality relation below.

Theorem (Great Orthogonality Theorem). Let G be compact with normalized Haar measure dg , and $\Pi^{(\alpha)}, \Pi^{(\beta)}$ inequivalent irreducible unitary representations of dimensions d_α, d_β . Then

$$\int_G \Pi_{ij}^{(\alpha)}(g)^* \Pi_{kl}^{(\beta)}(g) dg = \frac{1}{d_\alpha} \delta_{\alpha\beta} \delta_{ik} \delta_{jl}. \quad (\text{B.6})$$

Proofsketch. For any linear map $A : V_\beta \rightarrow V_\alpha$, the group-average $\bar{A} = \int_G \Pi^{(\alpha)}(g) A \Pi^{(\beta)}(g)^{-1} dg$ intertwines the two representations: $\Pi^{(\alpha)}(h) \bar{A} \Pi^{(\beta)}(h)^{-1} = \bar{A}$ by invariance of dg . By Schur's lemma, $\bar{A} = 0$ if α, β are inequivalent, and $\bar{A} = \lambda(A)I$ if $\alpha = \beta$. Taking A to be the elementary matrix E_{jk} (1 in position (j, k) , 0 elsewhere) extracts $\int \Pi_{ij}^{(\alpha)} \Pi_{kl}^{(\beta)*} dg$; the proportionality constant $\lambda(A)$ is fixed by tracing, yielding $1/d_\alpha$. (Full proof in Hall, Ch. 5.) ■

The regular representation and Peter–Weyl. The (left) regular representation L of a compact group G acts on $L^2(G)$ by $(L(h)f)(g) = f(h^{-1}g)$. It contains every irrep: each irrep $\Pi^{(\alpha)}$ of dimension d_α appears in L with multiplicity d_α . The explicit decomposition is the **Peter–Weyl theorem**: the matrix elements $\Pi_{ij}^{(\alpha)}(g)$, as α ranges over irreps and i, j over $1, \dots, d_\alpha$, form a complete orthogonal basis of $L^2(G)$,

$$L^2(G) = \widehat{\bigoplus_\alpha V^{(\alpha)} \otimes V^{(\alpha)*}}, \quad \|\Pi_{ij}^{(\alpha)}\|_{L^2}^2 = \frac{1}{d_\alpha}, \quad (\text{B.7})$$

the Hilbert-space direct sum (closure of the algebraic direct sum). For $G = SU(2)$ this says that the Wigner $D_{m'm}^{(j)}(\alpha, \beta, \gamma)$, as $j = 0, \frac{1}{2}, 1, \dots$ and $m, m' = -j, \dots, j$, are a complete orthogonal basis of $L^2(SU(2))$ with $\|D_{m'm}^{(j)}\|^2 = 1/(2j+1)$ — the content of the D -matrix orthogonality (B.23) and the engine of the spherical-harmonic decomposition of $L^2(S^2)$ (Section B.4). Peter–Weyl is the compact-

group avatar of the Fourier theorem: ordinary Fourier analysis is the Peter–Weyl decomposition of $L^2(U(1))$, with irreps the characters $e^{in\theta}$.

Characters. The **character** of a representation Π is the class function $\chi_\Pi(g) := \text{tr}\Pi(g)$ (constant on conjugacy classes, since $\text{tr}\Pi(hgh^{-1}) = \text{tr}\Pi(g)$). Characters are the complete invariants of compact-group representations: $\Pi_1 \cong \Pi_2$ iff $\chi_{\Pi_1} = \chi_{\Pi_2}$, and the character orthogonality $\int_G \chi_\alpha^* \chi_\beta \, dg = \delta_{\alpha\beta}$ is the class-function reduction of the great orthogonality theorem. For SU(2) every element is conjugate to a rotation about a fixed axis, so the character depends only on the rotation angle θ ; its closed form (B.10) is the key to the Clebsch–Gordan series (Section B.5).

B.3 SU(2) Representation Theory

The group SU(2) is the universal covering group of SO(3) (Section B.8). Its Lie algebra $\mathfrak{su}(2)$ is the real vector space of traceless anti-Hermitian 2×2 matrices; the conventional physics basis uses the self-adjoint generators $J_k = (\hbar/2)\sigma_k$, satisfying $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$, whose complexified algebra is $\mathfrak{sl}(2, \mathbb{C}) \cong \mathfrak{so}(3)_\mathbb{C}$. Because SU(2) is compact and connected, every finite-dimensional representation is unitarizable and decomposes into irreps; because it is simply connected, every representation of its Lie algebra integrates to a representation of the group. The classification of its irreps is the backbone of Chapter 6.

Theorem (Classification of SU(2) irreps). *For every $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ there exists exactly one (up to equivalence) irreducible unitary representation of SU(2), of dimension $2j + 1$, denoted $D^{(j)}$. In an orthonormal basis $\{|j, m\rangle\}_{m=-j}^j$ the generators act as*

$$J_z |j, m\rangle = \hbar m |j, m\rangle, \quad (\text{B.8})$$

$$J_\pm |j, m\rangle = \hbar \sqrt{j(j+1) - m(m \pm 1)} |j, m \pm 1\rangle, \quad (\text{B.9})$$

with $J_\pm = J_x \pm iJ_y$ and the Casimir $J^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle$. There are no other irreps.

Proof (highest-weight construction). The proof constructs every irrep from the algebra alone and shows uniqueness. Define $J_\pm = J_x \pm iJ_y$; from $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$,

$$[J_z, J_\pm] = \pm\hbar J_\pm, \quad [J_+, J_-] = 2\hbar J_z, \quad J^2 = J_- J_+ + J_z^2 + \hbar J_z = J_+ J_- + J_z^2 - \hbar J_z.$$

Since J^2 commutes with every J_i (a direct computation: $[J^2, J_z] = [J_- J_+, J_z] = J_- [J_+, J_z] + [J_-, J_z] J_+ = 0$), J^2 and J_z are simultaneously diagonalizable; write $J^2 |\lambda, m\rangle = \hbar^2 \lambda |\lambda, m\rangle$, $J_z |\lambda, m\rangle = \hbar m |\lambda, m\rangle$. From $[J_z, J_\pm] = \pm\hbar J_\pm$, whenever $J_\pm |\lambda, m\rangle \neq 0$ it is an eigenstate of J_z with eigenvalue $\hbar(m \pm 1)$ and the same λ ; so $J_\pm$ raise/lower m within a fixed- λ space. The squared norm is nonnegative:

$$\|J_\pm |\lambda, m\rangle\|^2 = \langle \lambda, m | J_\mp J_\pm | \lambda, m \rangle = \hbar^2 [\lambda - m(m \pm 1)] \geq 0.$$

Repeated application of J_+ produces states of ever-larger m ; since $\lambda - m(m+1) \geq 0$ forbids m from growing without bound, the ascent must terminate: there is a *highest weight* $|\lambda, j\rangle$ with $J_+ |\lambda, j\rangle = 0$, forcing $\lambda = j(j+1)$. Similarly J_- descent terminates at a lowest weight $|\lambda, j'\rangle$ with $\lambda = j'(j'+1)$. Equating, $j(j+1) = j'(j'+1)$, whose solution $j' = -j$ gives a symmetric ladder $m = -j, -j+1, \dots, j$. The number of rungs $2j$ is a nonnegative integer, so $j = 0, \frac{1}{2}, 1, \dots$;

each rung is one-dimensional (by irreducibility), giving $\dim D^{(j)} = 2j + 1$ and the normalized ladder action (B.9) (the square root is the norm $\sqrt{j(j+1) - m(m \pm 1)}$, taken real and nonnegative — the Condon–Shortley convention). Existence is constructive: build $|j, j\rangle$ and apply J_- ; the representation so obtained is irreducible (any invariant subspace would contain a highest-weight vector, unique up to scale). Uniqueness: two irreps with the same j have identically-acting generators, hence are unitarily equivalent. ■

Casimir and rank. $SU(2)$ has rank one (a maximal torus $U(1) \subset SU(2)$), so there is one independent Casimir, J^2 ; by Schur’s lemma $J^2 = \hbar^2 j(j+1)I$ in $D^{(j)}$, and this single number labels the irrep. The integer irreps ($j = 0, 1, 2, \dots$) descend to true representations of $SO(3)$ ($D^{(j)}(-I) = +I$); the half-integer irreps are genuine projective representations of $SO(3)$, true only on the cover $SU(2)$ ($D^{(j)}(-I) = -I$) — the topological origin of the spinor sign (Section B.8).

Characters. The character $\chi^{(j)}(g) = \text{tr} D^{(j)}(g)$ depends only on the conjugacy class, and every $g \in SU(2)$ is conjugate to a rotation $e^{-i\theta J_z/\hbar}$ about the z -axis. In $D^{(j)}$ this is diagonal, so

$$\chi^{(j)}(\theta) = \sum_{m=-j}^j e^{-im\theta} = \frac{\sin[(2j+1)\theta/2]}{\sin(\theta/2)}, \quad (\text{B.10})$$

the last form being the geometric series $\sum_{m=-j}^j e^{-im\theta}$, valid also at $\theta = 0$ by l’Hôpital ($\chi^{(j)}(0) = 2j + 1$, the dimension). The character is a *class function* on $SU(2)$ (constant on each conjugacy class, parametrized by $\theta \in [0, 2\pi]$); the conjugacy-class measure is $\sin^2(\theta/2) d\theta/\pi$ (from (B.5) after integrating out the two Euler angles that locate the axis).

Character orthogonality. The great orthogonality theorem (B.6), traced over representation indices, gives the orthogonality of characters:

$$\frac{1}{\pi} \int_0^{2\pi} \chi^{(j)}(\theta)^* \chi^{(j')}(\theta) \sin^2 \frac{\theta}{2} d\theta = \delta_{jj'}. \quad (\text{B.11})$$

Proof. Insert the geometric series into the left side: $\sin^2(\theta/2) \chi^{(j)*} \chi^{(j')} = \sin^2(\theta/2) \sum_{m,m'} e^{i(m-m')\theta}$. Use $\sin^2(\theta/2) = \frac{1}{2}(1 - \cos \theta) = \frac{1}{2} - \frac{1}{4}(e^{i\theta} + e^{-i\theta})$ and $\frac{1}{\pi} \int_0^{2\pi} e^{in\theta} d\theta = 2\delta_{n0}$; only terms with $m = m'$ survive, contributing $\sum_m \delta_{jj'} = \delta_{jj'}(2j+1)\delta_{jj'}$ after the $\cos \theta$ shift removes the off-resonant pieces, leaving $\delta_{jj'}$. ■ Character orthogonality underlies the Clebsch–Gordan decomposition by multiplicity counting (Section B.5).

Tensor products and Young tableaux. The tensor product of two irreps decomposes by the Clebsch–Gordan series (B.27); the general rule is efficiently encoded in **Young tableaux**. The irrep $D^{(j)}$ of $SU(2)$ corresponds to a single Young diagram of $2j$ boxes in one row (totally symmetric): the symmetric tensor of rank $2j$ on the fundamental $\mathbb{C}^2$. The tensor product $D^{(j_1)} \otimes D^{(j_2)}$ corresponds to the product of a row of $2j_1$ boxes with a row of $2j_2$ boxes; the Littlewood–Richardson rule gives the decomposition into rows of $2J$ boxes, $J = |j_1 - j_2|, \dots, j_1 + j_2$, recovering (B.27). For $SU(n)$ the diagrams have at most $n - 1$ rows and the rule is the same but richer; this book uses only the $SU(2)$ (single-row) instance, where the tableau is a bookkeeping device for the dimension $\binom{2j+n-1}{n-1}$ of the symmetric subspace of n spin-1/2’s — the Schwinger–Bargmann–Majorana realisation of Chapter 6.

B.4 Wigner D-Matrices

The irreps $D^{(j)}$ of $SU(2)$ are made concrete by evaluating them on the Euler-angle parametrization $U(\alpha, \beta, \gamma) = e^{-i\alpha J_z/\hbar} e^{-i\beta J_y/\hbar} e^{-i\gamma J_z/\hbar}$ ((6.8)). The resulting matrices are the **Wigner D-matrices**; this section gives their definition, their explicit closed form, the low- j instances, and their principal properties.

Definition. The $(2j+1) \times (2j+1)$ matrix elements

$$D_{m'm}^{(j)}(\alpha, \beta, \gamma) := \langle j, m' | e^{-i\alpha J_z/\hbar} e^{-i\beta J_y/\hbar} e^{-i\gamma J_z/\hbar} | j, m \rangle \quad (\text{B.12})$$

are the Wigner D-matrices. Because J_z is diagonal in the $|j, m\rangle$ basis, $e^{-i\alpha J_z/\hbar} |j, m\rangle = e^{-i\alpha m} |j, m\rangle$, and the two outer factors contribute only phases:

$$D_{m'm}^{(j)}(\alpha, \beta, \gamma) = e^{-i\alpha m'} d_{m'm}^{(j)}(\beta) e^{-i\gamma m}, \quad d_{m'm}^{(j)}(\beta) := \langle j, m' | e^{-i\beta J_y/\hbar} | j, m \rangle, \quad (\text{B.13})$$

where $d_{m'm}^{(j)}(\beta)$ is Wigner's (**small**) **d -matrix**, real in the Condon–Shortley convention. All the non-trivial angular dependence sits in $d^{(j)}$.

Closed form (factorial sum). The small- d matrix has the explicit closed form, in the convention

$d_{m'm}^{(j)}(\beta) = \langle j, m' | e^{-i\beta J_y/\hbar} | j, m \rangle$ with Condon–Shortley phases,

$$d_{m'm}^{(j)}(\beta) = (-1)^{m-m'} \sum_s (-1)^s \frac{\sqrt{(j+m)!(j-m)!(j+m')!(j-m')!}}{(j+m-s)! s! (m'-m+s)! (j-m'-s)!} \left(\cos \frac{\beta}{2}\right)^{2j-2s+m-m'} \left(\sin \frac{\beta}{2}\right)^{2s+m'-m} \quad (\text{B.14})$$

where the sum runs over all integers s keeping the factorial arguments nonnegative (a finite range, automatically terminated). Equivalently, $d_{m'm}^{(j)}(\beta)$ is a Jacobi polynomial $P_{j-m}^{(m'-m, m+m')}(\cos \beta)$ multiplied by a ratio of factorials (Edmonds, eq. 4.1.16); the factorial sum above is the practical route for tabulating the low- j matrices below. It is derived by evaluating $e^{-i\beta J_y/\hbar}$ in the Schwinger boson realisation of §B.3; the full derivation is given immediately below.

Derivation of (B.14) (Schwinger boson realisation). In the two-boson realisation, $J_y = (J_+ - J_-)/2i = (\hbar/2i)(a_+^\dagger a_- - a_-^\dagger a_+)$ and the angular-momentum states are $|j, m\rangle = (a_+^\dagger)^{j+m} (a_-^\dagger)^{j-m} |0\rangle / \sqrt{(j+m)! (j-m)!}$. The rotation about y is

$$U(\beta) := e^{-i\beta J_y/\hbar} = e^{-(\beta/2)G}, \quad G := a_+^\dagger a_- - a_-^\dagger a_+, \quad (\text{B.15})$$

the beam-splitter unitary. From the boson commutators,

$$[G, a_+] = [a_+^\dagger a_-, a_+] - [a_-^\dagger a_+, a_+] = -a_-, \quad [G, a_-] = [a_+^\dagger a_-, a_-] - [a_-^\dagger a_+, a_-] = +a_+,$$

so $A(\theta) := e^{\theta G} a_+ e^{-\theta G}$ and $B(\theta) := e^{\theta G} a_- e^{-\theta G}$ obey $A' = -B$, $B' = +A$, solved by $A(\theta) = a_+ \cos \theta - a_- \sin \theta$, $B(\theta) = a_+ \sin \theta + a_- \cos \theta$. With $\theta = -\beta/2$ (since $U = e^{-(\beta/2)G}$) and taking the adjoint, U rotates the *creation* operators as

$$U a_+^\dagger U^\dagger = c a_+^\dagger + s a_-^\dagger, \quad U a_-^\dagger U^\dagger = -s a_+^\dagger + c a_-^\dagger, \quad c := \cos \frac{\beta}{2}, \quad s := \sin \frac{\beta}{2}, \quad (\text{B.16})$$

an $SU(2)$ rotation of the two-mode doublet by β about y — the boson avatar of the defining spin-1/2 rotation. The vacuum is invariant ($G|0\rangle = 0$, so $U|0\rangle = |0\rangle$), hence the rotated state is obtained by

replacing each $a_{\pm}^{\dagger}$ in $|j, m\rangle$ by its rotated image (B.16):

$$U(\beta) |j, m\rangle = \frac{(c a_+^{\dagger} + s a_-^{\dagger})^{j+m} (-s a_+^{\dagger} + c a_-^{\dagger})^{j-m}}{\sqrt{(j+m)!(j-m)!}} |0\rangle. \quad (\text{B.17})$$

Expand both binomials,

$$\begin{aligned} (c a_+^{\dagger} + s a_-^{\dagger})^{j+m} &= \sum_p \binom{j+m}{p} c^{j+m-p} s^p (a_+^{\dagger})^{j+m-p} (a_-^{\dagger})^p, \\ (-s a_+^{\dagger} + c a_-^{\dagger})^{j-m} &= \sum_q \binom{j-m}{q} (-s)^{j-m-q} c^q (a_+^{\dagger})^{j-m-q} (a_-^{\dagger})^q, \end{aligned}$$

and form $d_{m'm}^{(j)}(\beta) = \langle j, m' | U(\beta) | j, m \rangle$. The vacuum expectation

$$\langle 0 | (a_+)^{j+m'} (a_-)^{j-m'} (a_+^{\dagger})^r (a_-^{\dagger})^t | 0 \rangle = (j+m')! (j-m')! \delta_{r, j+m'} \delta_{t, j-m'}$$

forces the total creation powers $r = (j+m-p) + (j-m-q) = 2j-p-q$ and $t = p+q$ to equal $j+m'$ and $j-m'$ respectively — the same condition $p+q = j-m'$ from each — so the double sum collapses to a single sum over p (with $q = j-m'-p$). Collecting the prefactor and the surviving powers,

$$\begin{aligned} d_{m'm}^{(j)}(\beta) &= \frac{(j+m')! (j-m')!}{\sqrt{(j+m')! (j-m')! (j+m)! (j-m)!}} \\ &\times \sum_p \binom{j+m}{p} \binom{j-m}{j-m'-p} c^{2j+m-m'-2p} (-1)^{m'-m+p} s^{m'-m+2p}, \end{aligned}$$

where $c^{j+m-p} c^q = c^{2j+m-m'-2p}$ and $s^p (-s)^{j-m-q} = (-1)^{m'-m+p} s^{m'-m+2p}$ (using $j-m-q = m'-m+p$). Now write the binomials as factorials, $\binom{j+m}{p} \binom{j-m}{j-m'-p} = (j+m)! (j-m)! / [p! (j+m-p)! (j-m'-p)! (m'-m+p)!]$: the prefactor $(j+m')! (j-m')! / \sqrt{\cdots}$ combines with $(j+m)! (j-m)!$ to give $\sqrt{(j+m)! (j-m)! (j+m')! (j-m')!}$, and relabelling $p \rightarrow s$ yields exactly (B.14). The sign matches because $(-1)^{m'-m+s} = (-1)^{m-m'} (-1)^s$ (the two differ by $(-1)^{2(m-m')} = 1$). The single sum over s is the residue of the two binomial expansions under the vacuum's power-matching constraint — the same combinatorial mechanism that produces the Racah CG formula (6.78), of which the d -matrix is the rank- $2j$ instance. Two useful specializations follow directly from (B.14): the *top row*

$$d_{j,m}^{(j)}(\beta) = (-1)^{j-m} \sqrt{\binom{2j}{j-m}} \left(\cos \frac{\beta}{2} \right)^{j+m} \left(\sin \frac{\beta}{2} \right)^{j-m}, \quad (\text{B.18})$$

and the *bottom row* $d_{-j,m}^{(j)}(\beta) = \sqrt{\binom{2j}{j+m}} \left(\cos \frac{\beta}{2} \right)^{j-m} \left(\sin \frac{\beta}{2} \right)^{j+m}$ (all entries non-negative); the remaining rows follow from the ladder recurrences of §B.5.

Explicit low- j d -matrices. With $c = \cos(\beta/2)$, $s = \sin(\beta/2)$, and rows/columns ordered $m = j, j-1, \dots, -j$:

$$d^{(1/2)}(\beta) = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \quad (\text{B.19})$$

$$d^{(1)}(\beta) = \begin{pmatrix} \frac{1 + \cos \beta}{2} & -\frac{\sin \beta}{\sqrt{2}} & \frac{1 - \cos \beta}{2} \\ \frac{\sin \beta}{\sqrt{2}} & \cos \beta & -\frac{\sin \beta}{\sqrt{2}} \\ \frac{1 - \cos \beta}{2} & \frac{\sin \beta}{\sqrt{2}} & \frac{1 + \cos \beta}{2} \end{pmatrix}, \quad (\text{B.20})$$

$$d^{(3/2)}(\beta) = \begin{pmatrix} c^3 & -\sqrt{3} c^2 s & \sqrt{3} c s^2 & -s^3 \\ \sqrt{3} c^2 s & c^3 - 2 c s^2 & -(2 c^2 s - s^3) & \sqrt{3} c s^2 \\ \sqrt{3} c s^2 & 2 c^2 s - s^3 & c^3 - 2 c s^2 & -\sqrt{3} c^2 s \\ s^3 & \sqrt{3} c s^2 & \sqrt{3} c^2 s & c^3 \end{pmatrix}, \quad (\text{B.21})$$

$$d^{(2)}(\beta) = \begin{pmatrix} c^4 & -2 c^3 s & \sqrt{6} c^2 s^2 & -2 c s^3 & s^4 \\ 2 c^3 s & c^2(c^2 - 3 s^2) & -\sqrt{6} c s(c^2 - s^2) & s^2(3 c^2 - s^2) & -2 c s^3 \\ \sqrt{6} c^2 s^2 & \sqrt{6} c s(c^2 - s^2) & c^4 - 4 c^2 s^2 + s^4 & -\sqrt{6} c s(c^2 - s^2) & \sqrt{6} c^2 s^2 \\ 2 c s^3 & -s^2(3 c^2 - s^2) & \sqrt{6} c s(c^2 - s^2) & c^2(c^2 - 3 s^2) & -2 c^3 s \\ s^4 & 2 c s^3 & \sqrt{6} c^2 s^2 & 2 c^3 s & c^4 \end{pmatrix}. \quad (\text{B.22})$$

The $j = 1/2$ instance (B.19) underlies the spin- n basis and the half-angle law of Chapter 6 (6.35), (6.37); the $j = 1$ instance (B.20) is, in the spherical basis, exactly the rotation of a Cartesian vector (below). The half-angles $\beta/2$ in every entry are the signature of the double cover: $d^{(j)}(2\pi) = (-1)^{2j} I$.

Properties. The D-matrices satisfy

1. *Unitarity:* $\sum_{m'} D_{m'm}^{(j)*}(R) D_{m'm''}^{(j)}(R) = \delta_{mm''}$, since each $U(R)$ is unitary.
2. *Composition (representation property):* $\sum_{m''} D_{m'm''}^{(j)}(R_1) D_{m''m}^{(j)}(R_2) = D_{m'm}^{(j)}(R_1 R_2)$ — the matrix of the product is the product of the matrices.
3. *Complex conjugation:* $D_{m'm}^{(j)*}(\alpha, \beta, \gamma) = (-1)^{m-m'} D_{-m', -m}^{(j)}(\alpha, \beta, \gamma)$, from the Condon–Shortley reality and symmetry $d_{m'm}^{(j)}(\beta) = (-1)^{m-m'} d_{-m', -m}^{(j)}(\beta)$ of the small- d .
4. *Orthogonality (Peter–Weyl for SU(2)):* with the Haar measure (B.5),

$$\int_G dg D_{m'm}^{(j)*}(\alpha, \beta, \gamma) D_{\mu'\mu}^{(j')}(\alpha, \beta, \gamma) = \frac{1}{2j+1} \delta_{jj'} \delta_{m\mu} \delta_{m'\mu'}. \quad (\text{B.23})$$

This is the great orthogonality theorem (B.6) specialized to SU(2); the prefactor $1/(2j+1) = 1/d_\alpha$ is the Peter–Weyl norm (B.7).

Rotation of spherical harmonics. For integer $j = \ell$ the D-matrix column $m' = 0$ reduces to a spherical harmonic: with the Euler angles (α, β, γ) identified with the spherical coordinates $(\theta = \beta, \phi = \alpha)$ of the rotated z -axis,

$$D_{m0}^{(\ell)}(\alpha, \beta, \gamma) = \sqrt{\frac{4\pi}{2\ell+1}} Y_\ell^{m*}(\beta, \alpha), \quad (\text{B.24})$$

independent of γ (a rotation about the body z -axis leaves the $m' = 0$ column invariant). In particular $D_{00}^{(\ell)} = P_\ell(\cos \beta)$, the Legendre polynomial. The spherical harmonics are therefore the $m' = 0$ column of the D-matrix — the functions on the coset sphere $S^2 \cong \text{SO}(3)/\text{SO}(2)$. The general transformation

law under a finite rotation R follows:

$$Y_\ell^m(R^{-1}\hat{\mathbf{r}}) = \sum_{m'=-\ell}^{\ell} D_{m'm}^{(\ell)}(R) Y_\ell^{m'}(\hat{\mathbf{r}}), \quad (\text{B.25})$$

the active rotation of a scalar field on the sphere. This is the link between the group-theoretic D-matrices and the analytic spherical harmonics of Chapter 6 §6.4: rotating a spherical harmonic mixes the $(2\ell + 1)$ members of its multiplet by the D-matrix, exactly as rotating a state $|\ell, m\rangle$ mixes the basis kets.

Rotation of Cartesian vectors and tensors. The $j = 1$ representation is the vector representation in disguise. The spherical components of a vector $\mathbf{V}$,

$$V_{\pm 1}^{(1)} := \mp \frac{V_x \pm iV_y}{\sqrt{2}}, \quad V_0^{(1)} := V_z,$$

transform under a rotation R as $V_q^{(1)} \mapsto \sum_{q'} D_{q'q}^{(1)}(R) V_{q'}^{(1)}$, which is exactly the Cartesian rotation $\mathbf{V} \mapsto R\mathbf{V}$ rewritten in the spherical basis; the $d^{(1)}$ matrix (B.20) is the rotation about y in that basis.

A rank- k Cartesian tensor $T_{i_1 \dots i_k}$ decomposes under rotations into irreducible spherical tensors of ranks $k, k-2, \dots$ (the traceless symmetric part is rank k , the traces are lower rank); each irreducible piece transforms by $D^{(k)}$. This is the bridge from Cartesian tensor analysis to the irreducible-tensor operators of §B.9 and Chapter 6 §6.7.

The integral of three D-matrices. The product of three D-matrices, integrated over the group, collapses to a product of 3- j symbols:

$$\int_G dg D_{m_1 m'_1}^{(j_1)*}(g) D_{m_2 m'_2}^{(j_2)}(g) D_{m_3 m'_3}^{(j_3)}(g) = \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ m'_1 & m'_2 & m'_3 \end{pmatrix}, \quad (\text{B.26})$$

vanishing unless each triple (j_1, j_2, j_3) and (m_1, m_2, m_3) satisfies the triangle and m -sum conditions. This identity — a corollary of the CG decomposition $D^{(j_2)} \otimes D^{(j_3)} = \bigoplus D^{(J)}$ and the orthogonality (B.23) — is the group-theoretic origin of the 3- j symbol and the engine of the Wigner–Eckart theorem (§B.9): the geometric part of every matrix element is an integral of three D-matrices, hence a 3- j .

B.5 The Clebsch–Gordan Series

The Clebsch–Gordan problem is: given two irreps j_1, j_2 of $\text{SU}(2)$, decompose their tensor product into irreducibles. The answer is the Clebsch–Gordan series, proved here two ways — by characters (multiplicity counting) and by highest weights (explicit construction).

Theorem (Clebsch–Gordan series). *The tensor product decomposes as*

$$D^{(j_1)} \otimes D^{(j_2)} \cong \bigoplus_{J=|j_1-j_2|}^{j_1+j_2} D^{(J)}, \quad (\text{B.27})$$

each irrep J appearing once. Equivalently the coupled and uncoupled bases are related by

$$|j_1, j_2; J, M\rangle = \sum_{m_1, m_2} \langle j_1, m_1; j_2, m_2 | J, M \rangle |j_1, m_1\rangle \otimes |j_2, m_2\rangle, \quad (\text{B.28})$$

with real CG coefficients $\langle j_1, m_1; j_2, m_2 | J, M \rangle$ (Condon–Shortley), vanishing unless $m_1 + m_2 = M$

and $|j_1 - j_2| \leq J \leq j_1 + j_2$.

Proof 1 (characters). The character of a tensor product is the product of characters: $\chi^{(j_1 \otimes j_2)} = \chi^{(j_1)} \chi^{(j_2)}$. The multiplicity n_J of $D^{(J)}$ in the product is, by character orthogonality (B.11),

$$n_J = \frac{1}{\pi} \int_0^{2\pi} \chi^{(J)}(\theta)^* \chi^{(j_1)}(\theta) \chi^{(j_2)}(\theta) \sin^2 \frac{\theta}{2} d\theta.$$

Insert the geometric series (B.10); the integrand is $\sin^2(\theta/2) \sum_{m, m_1, m_2} e^{i(m-m_1-m_2)\theta}$, and using $\sin^2(\theta/2) = \frac{1}{2} - \frac{1}{4}(e^{i\theta} + e^{-i\theta})$ with $\frac{1}{\pi} \int_0^{2\pi} e^{in\theta} d\theta = 2\delta_{n0}$, the integral counts the number of triples (m, m_1, m_2) with $m = m_1 + m_2$ in the range $|m_i| \leq j_i$, $|m| \leq J$, minus the analogous count at $J \pm 1$. The counting gives $n_J = 1$ for $|j_1 - j_2| \leq J \leq j_1 + j_2$ and 0 otherwise (the boundary steps cancel), which is (B.27).

■

Proof 2 (highest weight). On $V^{(j_1)} \otimes V^{(j_2)}$ the total $J_z = J_{1z} + J_{2z}$ has eigenvalue $M = m_1 + m_2$, ranging $-(j_1 + j_2) \leq M \leq j_1 + j_2$. The top space $M = j_1 + j_2$ is one-dimensional, spanned by $|j_1, j_1\rangle \otimes |j_2, j_2\rangle$; it is a highest-weight vector (J_+ annihilates it), generating a copy of $D^{(j_1+j_2)}$. The next space $M = j_1 + j_2 - 1$ is two-dimensional; one combination is J_- of the top (belonging to $D^{(j_1+j_2)}$), the orthogonal combination is a new highest-weight vector generating $D^{(j_1+j_2-1)}$. Descending, the M -space dimension grows by one per step until $M = |j_1 - j_2|$, then stays constant and shrinks; each new highest-weight vector launches the next lower J . The process terminates at $J = |j_1 - j_2|$, having accounted for $\sum_J (2J+1) = (2j_1+1)(2j_2+1)$ states — the full product space. ■

Racah closed form. The CG coefficients are determined by (B.28), the recurrences of Chapter 6 (6.79), and the Condon–Shortley phase. Racah (1942) gave the single closed sum (6.78) (stated in Chapter 6), obtained by evaluating the overlap in the Schwinger boson realisation; the sum index runs over integers keeping all factorials nonnegative.

3- j symbols and symmetries. The CG coefficient treats the three angular momenta (j_1, j_2, J) asymmetrically (J is the resultant). Wigner’s **3- j symbol** restores the symmetry:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} := \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j_3+1}} \langle j_1, m_1; j_2, m_2 | j_3, -m_3 \rangle,$$

vanishing unless $m_1 + m_2 + m_3 = 0$ and the triangle $|j_i - j_k| \leq j_l \leq j_i + j_k$ holds. Its symmetries, derived from the Racah form, are: *even permutation of columns leaves it invariant; an odd permutation multiplies it by $(-1)^{j_1+j_2+j_3}$; simultaneous sign flip of all m_i multiplies it by $(-1)^{j_1+j_2+j_3}$* . The orthogonality (B.23) gives

$$\sum_{m_1, m_2} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j'_3 \\ m_1 & m_2 & m'_3 \end{pmatrix} = \frac{1}{2j_3+1} \delta_{j_3 j'_3} \delta_{m_3 m'_3},$$

and the special value $\begin{pmatrix} j & j & 0 \\ m & -m & 0 \end{pmatrix} = (-1)^{j-m} / \sqrt{2j+1}$. The 3- j is the preferred object whenever three or more angular momenta meet, as in the Wigner–Eckart theorem and the Racah algebra below.

12130

B.6 Racah Algebra: 6- j and 9- j Symbols

When three or four angular momenta are coupled, the order of coupling matters, and the unitary matrix converting one coupling scheme to another is built of **Racah coefficients** — the 6- j and 9- j symbols. They are the bookkeeping of recoupling.

12133

Definition (6- j symbol). The 6- j symbol

12134

$$\left\{ \begin{matrix} j_1 & j_2 & J_{12} \\ j_3 & J & J_{23} \end{matrix} \right\} \quad (\text{B.29})$$

is the recoupling coefficient connecting the two schemes for coupling three angular momenta to total J :

12137

$$\langle (j_1, (j_2, j_3) J_{23}) J, M \mid ((j_1, j_2) J_{12}, j_3) J, M \rangle = (-1)^{j_1+j_2+j_3+J} \sqrt{(2J_{12}+1)(2J_{23}+1)} \left\{ \begin{matrix} j_1 & j_2 & J_{12} \\ j_3 & J & J_{23} \end{matrix} \right\}. \quad (\text{B.30})$$

12138

It vanishes unless each of the four triads (j_1, j_2, J_{12}) , (j_2, j_3, J_{23}) , (j_1, J, J_{23}) , (J_{12}, j_3, J) satisfies the triangle condition — the four triangles are the four columns of the symbol read appropriately.

12139

Explicit Racah formula. With the triangle coefficient $\Delta(abc) := \sqrt{\frac{(a+b-c)!(a-b+c)!(-a+b+c)!}{(a+b+c+1)!}}$ (zero unless the triangle holds),

12142

$$\left\{ \begin{matrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{matrix} \right\} = \Delta(j_1 j_2 j_3) \Delta(j_4 j_5 j_3) \Delta(j_1 j_5 j_6) \Delta(j_4 j_2 j_6) \sum_t (-1)^t \frac{(t+1)!}{\prod_k (t-a_k)! \prod_l (b_l-t)!}, \quad (\text{B.31})$$

12143

where the four a_k are the triad sums $j_1 + j_2 + j_3$, $j_4 + j_5 + j_3$, $j_1 + j_5 + j_6$, $j_4 + j_2 + j_6$; the three b_l are $j_1 + j_2 + j_4 + j_5$, $j_2 + j_3 + j_5 + j_6$, $j_3 + j_1 + j_6 + j_4$; and t ranges over integers keeping all factorials nonnegative. The sum is finite and the 6- j is rational for integer/half-integer arguments.

12146

Symmetries (Regge). The 6- j is invariant under any permutation of its columns and under the interchange of the upper and lower elements in any two columns (72 symmetries in all). Regge's deeper symmetries pack the 6- j into a 3×3 array invariant under a larger group; these reduce tabulation and underlie the classical-limit asymptotics

12150

$$\left\{ \begin{matrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{matrix} \right\} \approx \frac{1}{\sqrt{12\pi V}} \cos(\Sigma + \frac{\pi}{4}), \quad j_i \gg 1, \quad (\text{B.32})$$

12151

where V is the volume of the tetrahedron whose six edges are the j_i and Σ the sum of its three pairs of opposite dihedral angles (Ponzano–Regge). The tetrahedral geometry of the asymptotic 6- j is the shadow of the recoupling of three large angular momenta.

12154

9- j symbol. The 9- j symbol recouples four angular momenta, converting between two schemes each of which couples two pairs:

12156

$$\left\{ \begin{matrix} j_{11} & j_{12} & J_1 \\ j_{21} & j_{22} & J_2 \\ J'_1 & J'_2 & J \end{matrix} \right\} \quad (\text{B.33})$$

12157

relates the scheme $((j_{11} j_{12}) J_1, (j_{21} j_{22}) J_2) J$ to $((j_{11} j_{21}) J'_1, (j_{12} j_{22}) J'_2) J$. It is a finite sum of products of *three* 6- j symbols (equivalently four 3- j 's), the contraction over the internal angular momentum x

12158

12159

carrying a factor $(-1)^{2x}(2x+1)$ in the standard form (Edmonds 6.4.1). It is invariant under even permutation of rows or columns and under transposition, and acquires $(-1)^{\sum j_{ij}}$ under odd permutations. The 9- j with one entry zero collapses to a single 6- j :

$$\begin{Bmatrix} a & b & c \\ d & e & f \\ g & h & 0 \end{Bmatrix} = \frac{(-1)^{b+d+c+g}}{\sqrt{(2c+1)(2g+1)}} \begin{Bmatrix} a & b & c \\ e & d & g \end{Bmatrix}, \quad (\text{B.34})$$

the case used to convert LS to jj coupling in atomic and nuclear shell models $\left(\begin{Bmatrix} l_1 & s_1 & j_1 \\ l_2 & s_2 & j_2 \\ L & S & 0 \end{Bmatrix} \right)$.

Worked example: three spin-1/2's. Coupling $j_1 = j_2 = j_3 = 1/2$, the total $J = 1/2$ can be reached by first coupling $(j_1 j_2)$ to $J_{12} = 0$ or 1, then adding j_3 . The two schemes $(j_1 j_2) J_{12} = 0$ and $J_{12} = 1$ give two orthogonal $J = 1/2$ doublets — the famous *mixed symmetry* of the three-spin system. The recoupling from the scheme $(j_2 j_3) J_{23}$ to $(j_1 j_2) J_{12}$ is

$$\begin{Bmatrix} 1/2 & 1/2 & J_{12} \\ 1/2 & 1/2 & J_{23} \end{Bmatrix} = \pm \frac{1}{2} \quad (J_{12}, J_{23} \in \{0, 1\}),$$

so the two coupling schemes differ by a $1/\sqrt{2}$ rotation — the algebraic root of the S_4 permutation symmetry of three identical spins. Four angular momenta recouple via 9- j symbols on the same pattern; the recoupling coefficients are the unitary bridges between any two binary coupling trees, and the 6- j /9- j are their atomic units.

B.7 Casimir Operators

Definition. A **Casimir operator** of a Lie algebra $\mathfrak{g}$ is an element of its universal enveloping algebra (a polynomial in the generators) that commutes with every generator. For a semisimple algebra the number of algebraically independent Casimirs equals the **rank** (the dimension of a maximal abelian/Cartan subalgebra); by Schur's lemma each Casimir is a scalar in an irrep, and (for compact groups) the joint eigenvalues of the Casimirs label the irrep.

For $\mathfrak{su}(2)$ the rank is one and the unique quadratic Casimir is $J^2 = J_x^2 + J_y^2 + J_z^2$, with eigenvalue $\hbar^2 j(j+1)$ in $D^{(j)}$ — the label of the irrep. For the Galilei algebra the central element M (mass) is a Casimir (it commutes with every generator), and the internal energy E_{int} of Chapter 4 is the Casimir of the subalgebra remaining after factoring out the central extension; the free-particle Hamiltonian $H = \mathbf{P}^2/2M + E_{\text{int}}$ is fixed by these two invariants. For the Lorentz group (named here, developed in relativistic texts) the two Casimirs are $P_\mu P^\mu$ (mass-squared) and $W_\mu W^\mu$ (the square of the Pauli-Lubanski vector, encoding spin); the Poincaré classification of particles by mass and spin is the physical harvest of its Casimirs.

B.8 Covering Groups

The $\text{SO}(3)/\text{SU}(2)$ double cover. The rotation group $\text{SO}(3)$ is not simply connected: $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$. Its universal covering group is $\text{SU}(2)$, simply connected ($\pi_1(\text{SU}(2)) = \{0\}$) and diffeomorphic to S^3 . The covering map $\Phi : \text{SU}(2) \rightarrow \text{SO}(3)$ is the 2:1 homomorphism of Chapter 6 (6.7): both U and $-U$ map to the same rotation, $\ker \Phi = \{\pm I\} \cong \mathbb{Z}_2$, so $\text{SO}(3) \cong \text{SU}(2)/\mathbb{Z}_2$. The algebras coincide ($\mathfrak{su}(2) \cong \mathfrak{so}(3)$), so the two groups share their Lie-algebra representation theory; they differ only in which algebra representations integrate to single-valued group representations.

Projective representations and the cover. By Wigner's theorem, the quantum-mechanical action of a symmetry group is a *projective* (ray) representation $U(g_1)U(g_2) = e^{i\xi(g_1, g_2)}U(g_1g_2)$. A theorem of Bargmann relates the two: for a connected Lie group G , every projective representation of G lifts to an ordinary representation of the universal cover $\tilde{G}$ (the phase is absorbed by going upstairs), and the distinct lifts are classified by the second cohomology $H^2(\mathfrak{g}, \mathbb{R})$. For $\text{SO}(3)$, $H^2 = 0$: every projective representation of $\text{SO}(3)$ is an ordinary representation of $\text{SU}(2)$. The integer-spin irreps ($j = 0, 1, 2, \dots$, $D^{(j)}(-I) = +I$) descend to true representations of $\text{SO}(3)$; the half-integer irreps ($D^{(j)}(-I) = -I$) are genuine projective representations of $\text{SO}(3)$, single-valued only on the cover. This is the group-theoretic origin of the spinor 4π periodicity: a 2π rotation in the laboratory ($\text{SO}(3)$, the identity) lifts to $-I$ in $\text{SU}(2)$.

The Lorentz group and $\text{SL}(2, \mathbb{C})$ (named). The proper orthochronous Lorentz group $\text{SO}^+(3, 1)$ is likewise doubly covered, by $\text{SL}(2, \mathbb{C})$; its finite-dimensional irreps are labelled by a pair (j_1, j_2) of $\text{SU}(2)$ spins, dimension $(2j_1 + 1)(2j_2 + 1)$. The $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ irreps are the left- and right-handed Weyl spinors; their direct sum $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ is the Dirac spinor. This nonrelativistic book names the structure without developing it; the rotation subgroup of the Lorentz group is the diagonal $\text{SU}(2)$, and nonrelativistic spin emerges on restricting the Lorentz classification to rotations.

Inönü–Wigner contraction. The Galilei group is the *contraction* of the Poincaré group as $c \rightarrow \infty$. Rescale the Lorentz boost generators $K_i^{(L)} = K_i/c$ (so that a finite boost is $e^{-iv \cdot \mathbf{K}^{(L)}/\hbar}$ with $\mathbf{v} = \beta c$). The Lorentz bracket $[K_i^{(L)}, K_j^{(L)}] = -(i\hbar/c^2)\epsilon_{ijk}J_k$ tends to 0 as $c \rightarrow \infty$, recovering the Galilean $[K_i, K_j] = 0$; the energy–momentum relation $E = \sqrt{\mathbf{p}^2 c^2 + m^2 c^4}$ expands to $mc^2 + \mathbf{p}^2/2m + \dots$, the rest energy (an additive constant, dropped into E_{int}) plus the nonrelativistic kinetic energy. The contraction is the structural reason the Galilean analysis of Chapter 4 is the exact $c \rightarrow \infty$ limit of the relativistic one — and the reason the Galilean mass superselection has no Poincaré analogue (mass is already a Casimir there, not a central extension).

B.9 Irreducible Tensor Operators and the Wigner–Eckart

Theorem

Chapter 6 §6.7 defines an irreducible tensor operator $T_q^{(k)}$ of rank k as a set of $2k + 1$ operators transforming as the state $|k, q\rangle$, proves the Wigner–Eckart theorem in Clebsch–Gordan form (6.93),

and extracts its selection rules. This section records the equivalent 3- j form and the projection theorem, the two refinements most used in spectroscopy.

Wigner–Eckart in 3- j form. Translating the CG coefficient in (6.93) via (6.80) gives

$$\langle \alpha', j', m' | T_q^{(k)} | \alpha, j, m \rangle = (-1)^{j'-m'} \begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix} \langle \alpha', j' || T^{(k)} || \alpha, j \rangle, \quad (\text{B.35})$$

where the double-bar (reduced) matrix element is independent of m, m', q . The 3- j form makes the m -selection rule $m' + (-q) + (-m) = 0$ (i.e. $m' = m + q$) and the triangle $|j - k| \leq j' \leq j + k$ immediate, and displays the symmetry under interchange of initial, final, and operator angular momenta — the form preferred in nuclear and atomic-structure codes.

The projection theorem. Within a fixed- j multiplet, every vector operator's matrix elements are proportional to those of $\mathbf{J}$ itself. Precisely, for a vector operator $\mathbf{V}$ ($k = 1$) diagonal in j ,

$$\langle \alpha, j, m | V_q | \alpha, j, m' \rangle = \frac{\langle \alpha, j || \mathbf{V} || \alpha, j \rangle}{\langle \alpha, j || \mathbf{J} || \alpha, j \rangle} \langle \alpha, j, m | J_q | \alpha, j, m' \rangle. \quad (\text{B.36})$$

Proof. Both sides are matrix elements of rank-1 tensors within the same j -multiplet, so each is a CG coefficient times its reduced matrix element; the CG coefficients are identical (same j, k, j'), so the ratio of the two sides is the ratio of reduced matrix elements, a constant independent of m, m', q . ■

The theorem underlies the Landé g -factor: the magnetic moment $\boldsymbol{\mu} = -\mu_B(\mathbf{L} + g_s\mathbf{S})/\hbar$ is a vector operator, so within a fixed- (L, S, J) multiplet $\langle \boldsymbol{\mu} \rangle = [\langle \boldsymbol{\mu} \cdot \mathbf{J} \rangle / \hbar^2 J(J+1)] \langle \mathbf{J} \rangle$, yielding the Landé formula $g_J = 1 + [J(J+1) + S(S+1) - L(L+1)]/2J(J+1)$ of Chapter 8 (8.10) with no radial integral. The same projection underlies the Zeeman splitting and the hyperfine-structure pattern: one reduced matrix element fixes an entire multiplet's worth of matrix elements.

Outlook. The machinery of this appendix — Lie algebras and their representations (B.1–B.3), the explicit finite-rotation D-matrices (B.4), the Clebsch–Gordan and Racah recoupling algebras (B.5–B.6), the Casimir classification (B.7), the covering-group topology (B.8), and the irreducible-tensor calculus (B.9) — is the complete group-theoretic apparatus on which Chapter 4's spacetime symmetries and Chapter 6's angular momentum rest. Every formula above is a consequence of the single commutator $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$ and the compactness of SU(2); the physics enters only in identifying which irrep a given system carries.

B.10 SO(4) and the Kepler Problem

The hydrogen spectrum of Chapter 6 §6.8 carries a degeneracy the rotation group cannot explain: every level E_n is n^2 -fold degenerate (without spin), but SO(3) forces only the $(2\ell + 1)$ -fold m -degeneracy within each ℓ . The extra degeneracy — all $\ell < n$ sharing one energy — is the signature of a *hidden* symmetry, larger than the rotation group. This section develops it: the Coulomb potential admits a second conserved vector, the Runge–Lenz vector, which together with $\mathbf{L}$ closes the algebra of SO(4). Pauli's algebraic solution of hydrogen (1926), which derives the Balmer formula and the n^2 degeneracy from symmetry alone, is the harvest. This section fulfills the promise made in §6.8.

The Runge–Lenz vector. For the Coulomb Hamiltonian $H = \mathbf{p}^2/2m - Ze^2/r$, the **Laplace–Runge–Lenz vector**

$$\mathbf{A} := \frac{1}{2m}(\mathbf{p} \times \mathbf{L} - \mathbf{L} \times \mathbf{p}) - Ze^2 \frac{\mathbf{r}}{r} \quad (\text{B.37})$$

is conserved, $[H, \mathbf{A}] = 0$. Classically $\mathbf{A}$ points along the major axis of the Kepler ellipse; its conservation is the reason closed Kepler orbits are ellipses with a fixed orientation, and it is special to the $1/r$ potential. Quantum-mechanically the cross product is symmetrized (since $\mathbf{p}$ and $\mathbf{L}$ do not commute), and $\mathbf{A}$ is a Hermitian vector operator. Direct computation from the canonical commutators gives the algebra

$$[L_i, A_j] = i\hbar \epsilon_{ijk} A_k, \quad [A_i, L_j] = i\hbar \epsilon_{ijk} A_k, \quad [A_i, A_j] = -2i\hbar \frac{H}{m} \epsilon_{ijk} L_k, \quad (\text{B.38})$$

together with the two identities

$$\mathbf{L} \cdot \mathbf{A} = 0, \quad \mathbf{A}^2 = \frac{2H}{m}(\mathbf{L}^2 + \hbar^2) + Z^2 e^4. \quad (\text{B.39})$$

The first says $\mathbf{A}$ lies in the plane perpendicular to $\mathbf{L}$ (the orbital plane); the second fixes its length. The bracket $[A_i, A_j]$ closes on $\mathbf{L}$ with a coefficient $-2H/m$ involving the Hamiltonian itself — the algebra is a *dynamical* symmetry, mixing kinematic ($\mathbf{L}$) and dynamic (H) objects.

The SO(4) algebra (bound states). On a bound-state eigenspace $H = E < 0$ the coefficient $-2H/m$ is a positive c-number. Rescale

$$\mathbf{M} := \frac{\mathbf{A}}{\sqrt{-2E/m}} \quad (\text{on the } E\text{-eigenspace}), \quad [M_i, M_j] = i\hbar \epsilon_{ijk} L_k, \quad (\text{B.40})$$

so $\mathbf{M}$ satisfies the same $\mathfrak{so}(3)$ bracket as $\mathbf{L}$, and $[L_i, M_j] = i\hbar \epsilon_{ijk} M_k$. The six generators $\{\mathbf{L}, \mathbf{M}\}$ close the Lie algebra of SO(4): defining the two linear combinations

$$\mathbf{J}^{(\pm)} := \frac{1}{2}(\mathbf{L} \pm \mathbf{M}), \quad (\text{B.41})$$

a direct substitution gives

$$[J_i^{(\pm)}, J_j^{(\pm)}] = i\hbar \epsilon_{ijk} J_k^{(\pm)}, \quad [J_i^{(+)}, J_j^{(-)}] = 0, \quad (\text{B.42})$$

two *commuting* $\mathfrak{su}(2)$ algebras. Hence $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, and (globally) $\text{SO}(4) \cong (\text{SU}(2) \times \text{SU}(2))/\mathbb{Z}_2$: the four-dimensional rotation group is two copies of the three-dimensional one, stitched together. The hydrogen atom's hidden symmetry is this doubled rotation group.

Pauli's algebraic solution (1926). The constraint $\mathbf{L} \cdot \mathbf{M} = 0$ (from (B.39)) forces the two SU(2)'s to carry the *same* spin: since $\mathbf{L} \cdot \mathbf{M} = \mathbf{J}^{(+)^2} - \mathbf{J}^{(-)^2}$ (the cross terms vanish by $[\mathbf{J}^{(+)}, \mathbf{J}^{(-)}] = 0$), $\mathbf{L} \cdot \mathbf{M} = 0$ gives $\mathbf{J}^{(+)^2} = \mathbf{J}^{(-)^2}$, hence $j_+ = j_- =: j$. The hydrogen bound states therefore transform in the SO(4) irrep (j, j) , of dimension $(2j+1)^2$.

Now use the length identity (B.39). From $\mathbf{M}^2 = (-m/2E)\mathbf{A}^2$ and (B.39),

$$\mathbf{L}^2 + \mathbf{M}^2 = -\hbar^2 - \frac{mZ^2 e^4}{2E}. \quad (\text{B.43})$$

But $\mathbf{L}^2 + \mathbf{M}^2 = 2(\mathbf{J}^{(+)^2} + \mathbf{J}^{(-)^2}) = 4j(j+1)\hbar^2$. Equating,

$$4j(j+1)\hbar^2 = -\hbar^2 - \frac{mZ^2 e^4}{2E} \implies E_n = -\frac{mZ^2 e^4}{2\hbar^2 n^2}, \quad n := 2j+1 = 1, 2, 3, \dots$$

(using $4j(j+1) + 1 = (2j+1)^2 = n^2$). This is the Balmer formula (6.108), derived without ever solving the Schrödinger equation — pure algebra from the SO(4) symmetry. The degeneracy is the

dimension of the (j, j) irrep,

$$\dim(j, j) = (2j + 1)^2 = n^2, \quad (\text{B.44})$$

the n^2 -fold degeneracy of §6.8 (doubled to $2n^2$ by spin). The orbital angular momentum $\mathbf{L} = \mathbf{J}^{(+)} + \mathbf{J}^{(-)}$ is the sum of two spin- j 's, so the $\text{SO}(3)$ content of the (j, j) irrep is $\ell = 0, 1, \dots, 2j = 0, \dots, n-1$ — exactly the allowed hydrogen orbital quantum numbers. The rotation group $\text{SO}(3)$ forces the $(2\ell+1)$ m -degeneracy; the Coulomb group $\text{SO}(4)$ forces the n^2 degeneracy; the distinction is why a non-Coulomb central potential lifts the ℓ -degeneracy but preserves the m -degeneracy.

Scattering states and $\text{SO}(3, 1)$. For positive-energy (scattering) states $E > 0$, the rescaling (B.40) uses $\sqrt{-2E/m} = i\sqrt{2E/m}$, and the bracket $[M_i, M_j]$ acquires the opposite sign: the algebra becomes $\mathfrak{so}(3, 1)$, the Lorentz algebra (§B.12). The Coulomb scattering states carry representations of the non-compact $\text{SO}(3, 1)$; the bound/scattering dichotomy is the signature $E < 0/E > 0$ of the same Runge–Lenz symmetry.

The spectrum-generating algebra $\text{SO}(4, 2)$. The $\text{SO}(4)$ symmetry acts *within* each n -shell; it does not connect different n . A larger algebra — $\text{SO}(4, 2)$, generated by $\mathbf{L}$, $\mathbf{A}$, the dilatation, and two further Runge–Lenz-like vectors — is a *spectrum-generating* (dynamical) algebra: its ladder operators raise and lower n , generating the entire hydrogen spectrum from a single ground state. The $\text{SO}(4, 2)$ construction is the algebraic underpinning of the hydrogen atom's exact solvability and the prototype of the dynamical-symmetry method, later deployed in nuclear and particle physics (the $\text{SU}(3)$ flavor of §B.11 among them). It is named here; the full development belongs to specialized treatises.

The Kepler problem is the paradigm of a hidden symmetry: a degeneracy unexplained by the obvious rotation group is the footprint of a larger one, found by exhibiting the extra conserved quantity ($\mathbf{A}$) and closing its algebra. The same pattern — a spectrum too degenerate for the manifest symmetry, resolved by a larger hidden one — recurs throughout physics, from the isotropic harmonic oscillator's $\text{SU}(3)$ to the quark model's flavor symmetry.

B.11 General Semisimple Lie Algebras

The representation theory of $\text{SU}(2)$ developed in §B.3 — highest weight, ladder termination, Casimir labelling — is the rank-one instance of a general construction that classifies *all* compact semisimple Lie algebras. This section sketches that classification (root systems and Dynkin diagrams) and works out the physically central example, $\text{SU}(3)$, the symmetry underlying the three-color and three-flavor quark models of Chapter 10.

Cartan subalgebra and roots. Let $\mathfrak{g}$ be a complex semisimple Lie algebra of dimension $\dim \mathfrak{g}$ and rank r . A **Cartan subalgebra** $\mathfrak{h} \subset \mathfrak{g}$ is a maximal abelian subalgebra (dimension $r = \text{rank } \mathfrak{g}$) whose elements are simultaneously diagonalizable in the *adjoint* representation. Choosing a basis $\{H_1, \dots, H_r\}$ for $\mathfrak{h}$, the remaining generators E_α can be labelled by simultaneous eigenvalues of ad_{H_i} : there are linear functionals $\alpha \in \mathfrak{h}^*$, the **roots**, with

$$[H_i, E_\alpha] = \alpha_i E_\alpha, \quad i = 1, \dots, r. \quad (\text{B.45})$$

The roots form a finite set $\Phi \subset \mathfrak{h}^*$ invariant under reflection in the hyperplanes perpendicular to each root (the **Weyl group**). If α is a root, so is $-\alpha$; the $E_{\pm\alpha}$ are the raising and lowering operators of the rank-one theory, generalized. For $\mathfrak{su}(2)$ there is one pair $\{\pm\alpha\}$ ($r = 1$); for $\mathfrak{su}(3)$, six roots forming a regular hexagon ($r = 2$).

Simple roots and Dynkin diagrams. Choose a hyperplane through the origin of $\mathfrak{h}^*$ avoiding all roots, splitting Φ into positive (Φ^+) and negative roots. The **simple roots** $\{\alpha_1, \dots, \alpha_r\}$ are the positive roots not expressible as a sum of other positive roots; every positive root is a non-negative integer combination of them. The geometry of the simple roots — their lengths and the angles between them — completely determines $\mathfrak{g}$ (the Cartan matrix $A_{ij} = 2(\alpha_i \cdot \alpha_j)/|\alpha_i|^2$). The **Dynkin diagram** encodes this geometry graphically: r nodes (one per simple root), with 0, 1, 2, 3 lines joining nodes i, j according as the roots meet at $90^\circ, 120^\circ, 135^\circ, 150^\circ$, and an arrow on multiply-joined edges pointing to the shorter root. The classification theorem (Cartan, Killing) states that the connected Dynkin diagrams are exactly the four infinite families and five exceptions:

$$\begin{array}{lll}
 A_r \ (r \geq 1) & \text{SU}(r+1) & \text{special unitary,} \\
 B_r \ (r \geq 2) & \text{SO}(2r+1) & \text{odd orthogonal,} \\
 C_r \ (r \geq 3) & \text{Sp}(2r) & \text{symplectic,} \\
 D_r \ (r \geq 4) & \text{SO}(2r) & \text{even orthogonal,} \\
 E_6, E_7, E_8, F_4, G_2 & \text{exceptional} & \text{(no classical matrix realization).}
 \end{array} \tag{B.46}$$

Every compact semisimple Lie algebra is a direct sum of these. The rank-one case A_1 is $\mathfrak{su}(2)$, the algebra of this appendix; the rotation algebra is the seed of the entire classification.

Highest-weight construction. An irrep of $\mathfrak{g}$ is labelled by its **highest weight** $\Lambda = \sum_i m_i \alpha_i$, $m_i \in \mathbb{Z}_{\geq 0}$, a non-negative integer combination of simple roots: the highest-weight state is annihilated by every E_α with $\alpha \in \Phi^+$, and the rest of the representation is generated by applying the lowering operators $E_{-\alpha}$, $\alpha \in \Phi^+$. This generalizes the $\text{SU}(2)$ ladder construction ($\Lambda = 2j\alpha$, $m_1 = 2j$). The dimension of the Λ -irrep is given by the **Weyl dimension formula**; the tensor-product decomposition generalizes the Clebsch–Gordan series and is efficiently computed by Young tableaux (each algebra has its own tableau rules; for $\text{SU}(n)$ the boxes are columns of $\leq n-1$ symmetrized indices).

SU(3): the eightfold way. The group $\text{SU}(3)$ is the set of 3×3 unitary matrices with determinant $+1$; its algebra $\mathfrak{su}(3)$ has rank 2 and dimension 8. A standard basis is the eight Gell-Mann matrices λ_a ($a = 1, \dots, 8$), the $\text{SU}(3)$ analogues of the Pauli matrices, with generators $T_a = \lambda_a/2$ satisfying

$$[T_a, T_b] = i f_{abc} T_c, \quad \{T_a, T_b\} = \frac{1}{3} \delta_{ab} I + d_{abc} T_c, \tag{B.47}$$

where f_{abc} (totally antisymmetric) and d_{abc} (totally symmetric) are the $\text{SU}(3)$ structure constants; the nonzero f_{abc} generalize ϵ_{ijk} (e.g. $f_{123} = 1$, $f_{147} = f_{246} = 1/2$, $f_{156} = f_{345} = -1/2$, etc.). The Cartan subalgebra is $\{T_3, T_8\}$; the six roots form the regular hexagon of the A_2 system, and the Weyl group is the dihedral group D_3 (reflections of the triangle).

$\text{SU}(3)$ has rank 2, hence two independent Casimirs: the quadratic $C_2 = \sum_a T_a^2$ and the cubic $C_3 = \sum_{abc} d_{abc} T_a T_b T_c$. The joint eigenvalues (C_2, C_3) label the irreps. The smallest irreps are the fundamental $\mathbf{3}$ (the defining representation on $\mathbb{C}^3$) and its conjugate $\bar{\mathbf{3}}$ (inequivalent to $\mathbf{3}$, unlike the $\text{SU}(2)$ case where $\mathbf{2} \cong \bar{\mathbf{2}}$); the adjoint $\mathbf{8}$ (dimension 8, the generators themselves); and the symmetric

products $\mathbf{6} = \text{Sym}^2 \mathbf{3}$ and $\mathbf{10} = \text{Sym}^3 \mathbf{3}$. The basic tensor decompositions, computed by Young tableaux or the Littlewood–Richardson rule, are

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}, \quad \mathbf{3} \otimes \mathbf{3} = \bar{\mathbf{3}} \oplus \mathbf{6}, \quad \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{10}, \quad (\text{B.48})$$

the last being the decomposition of three fundamentals into a singlet, two octets, and a fully symmetric decuplet — the group-theoretic backbone of the quark model.

Flavor SU(3) and the quark model. Gell-Mann and Ne’eman (1961) observed that the hadrons known at the time organize into SU(3) multiplets if SU(3) is treated as an (approximate) *flavor* symmetry acting on the three light quarks u, d, s ($\mathbf{3}$). The pseudoscalar mesons (π, K, η , eight states) fill the $\mathbf{8}$; the spin-1/2 baryons (N, Λ, Σ, Ξ , eight states) fill another $\mathbf{8}$; the spin-3/2 baryons (Δ, Σ^*, Ξ^* , nine states) fill all but one slot of the $\mathbf{10}$, the decuplet of (B.48). The missing tenth slot — the Ω^- , with predicted mass and quantum numbers — was discovered in 1964 with exactly the predicted properties, the triumphant confirmation of the SU(3) flavor symmetry and of the underlying quark composition (the $\mathbf{10}$ being the fully symmetric $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$ combination, three quarks in the decuplet). The same SU(3) reappears in Chapter 10 as the *color* symmetry of QCD (an exact, not approximate, symmetry), with the three colors furnishing the fundamental $\mathbf{3}$; the singlet-octet structure (B.48) is the group-theoretic reason QCD’s physical states are color singlets.

Outlook. The pattern is general: every compact semisimple symmetry in physics — flavor and color SU(3), isospin SU(2), the SO(4) of hydrogen (§B.10), the SO(3, 1) of scattering states — is one of the Dynkin algebras (B.46), its irreps labelled by highest weights, its tensor products decomposed by Young tableaux, its selection rules read off the Wigner–Eckart theorem generalized to the appropriate algebra. The rotation group of Chapter 6 is the seed from which the whole classification grows.

B.12 The Lorentz and Poincaré Groups

The nonrelativistic physics of this book lives under the Galilei group (Chapter 4 §4.7); its relativistic counterpart lives under the Poincaré group. This section develops the representation theory of the Lorentz and Poincaré groups far enough to state Wigner’s classification of relativistic particles by mass and spin — the relativistic analogue of the Galilean (m, s) labelling. It is the natural boundary of this nonrelativistic book, named throughout but built only here.

The Lorentz algebra. The proper orthochronous Lorentz group $\text{SO}^+(3, 1)$ is the group of real 4×4 matrices Λ preserving the Minkowski metric $\eta = \text{diag}(1, -1, -1, -1)$, $\Lambda^\top \eta \Lambda = \eta$, connected to the identity. Its Lie algebra is generated by three rotations J_i and three boosts K_i ,

$$[J_i, J_j] = i\hbar \epsilon_{ijk} J_k, \quad [J_i, K_j] = i\hbar \epsilon_{ijk} K_k, \quad [K_i, K_j] = -i\hbar \epsilon_{ijk} J_k, \quad (\text{B.49})$$

differing from the Galilean algebra only in the last bracket: boosts *do not commute* — their commutator is a rotation, the infinitesimal origin of the Thomas precession and of the Inönü–Wigner contraction (§B.8).

Complexification: $\mathfrak{so}(3, 1)_\mathbb{C} \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$. The sign flip in $[K_i, K_j]$ is removed by passing to the

complexified algebra and defining

$$\mathbf{A} := \frac{1}{2}(\mathbf{J} + i\mathbf{K}), \quad \mathbf{B} := \frac{1}{2}(\mathbf{J} - i\mathbf{K}), \quad (\text{B.50})$$

for which a direct substitution gives

$$[A_i, A_j] = i\hbar \epsilon_{ijk} A_k, \quad [B_i, B_j] = i\hbar \epsilon_{ijk} B_k, \quad [A_i, B_j] = 0. \quad (\text{B.51})$$

The complexified Lorentz algebra splits as two commuting copies of $\mathfrak{su}(2)$ — exactly the structure that made the $\text{SO}(4)$ of hydrogen (§B.10) a product of two rotations. The finite-dimensional irreducible representations are therefore labelled by a pair (j_1, j_2) of $\text{SU}(2)$ spins, of dimension $(2j_1 + 1)(2j_2 + 1)$, exactly as for $\text{SO}(4)$ but with no reality constraint (the Lorentz group is non-compact, so these finite-dimensional irreps are *not* unitary).

Spinors. The fundamental Lorentz irreps are:

- $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ — the two-component **Weyl spinors**, left- and right-handed, transforming independently under the two $\text{SU}(2)$ factors; each is two-dimensional.
- $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ — the four-component **Dirac spinor**, the representation on which the Dirac equation acts; parity interchanges the two Weyl components.
- $(1, 0)$ and $(0, 1)$ — the self-dual and anti-self-dual rank-2 antisymmetric tensors; $(1, 0) \oplus (0, 1)$ is the six-dimensional representation carried by the electromagnetic field strength $F_{\mu\nu}$.
- (j, j) — symmetric traceless rank- $2j$ tensors, the integer-spin representations; the rotation subgroup (the diagonal $\text{SU}(2)$, $\mathbf{J} = \mathbf{A} + \mathbf{B}$) sees total spin j . The half-integer spins are the (j_1, j_2) with $j_1 \neq j_2$.

The Lorentz group's universal cover is $\text{SL}(2, \mathbb{C})$ (the 2×2 complex matrices of determinant 1), a 2:1 cover constructed exactly as $\text{SU}(2)$ covers $\text{SO}(3)$ (§B.8): a Lorentz transformation acts on a Hermitian 2×2 matrix $x^\mu \sigma_\mu$ by $\Lambda \cdot X \mapsto MXM^\dagger$, $M \in \text{SL}(2, \mathbb{C})$. The Weyl spinors are the fundamental representation of this cover.

The Poincaré group and Wigner's classification. The Poincaré group adds four translations P_μ (energy and momentum) to the Lorentz transformations; its algebra extends (B.49) by $[J_i, P_j] = i\hbar \epsilon_{ijk} P_k$, $[K_i, P_0] = -i\hbar P_i$, $[K_i, P_j] = -i\hbar \delta_{ij} P_0/c^2$, and $[P_\mu, P_\nu] = 0$. The translations form an abelian normal subgroup, so Mackey's little-group method (§4.7.4) applies exactly as in the Galilean case: the irreps are induced from the orbits of the Lorentz group on energy–momentum space, classified by the Casimirs of the little group.

The Poincaré algebra has two Casimirs. The first is the mass-squared $P^2 := P_\mu P^\mu = P_0^2/c^2 - \mathbf{P}^2$, with eigenvalue $m^2 c^2$ on a representation of mass m . The second is the square of the **Pauli–Lubanski vector**

$$W_\mu := \frac{1}{2} \epsilon_{\mu\nu\alpha\beta} M^{\nu\alpha} P^\beta, \quad (\text{B.52})$$

($M^{\nu\alpha}$ the Lorentz generators); on a massive state $W^2 = W_\mu W^\mu = -m^2 c^2 s(s+1)\hbar^2$, with $s = 0, \frac{1}{2}, 1, \dots$ the spin. Wigner's theorem (1939) classifies the unitary irreps of the Poincaré group:

- **Massive particles** ($m > 0$): the little group of the rest momentum $p_0 = (mc, \mathbf{0})$ is $\text{SU}(2)$ (rotations); its irreps are the spin- s multiplets. The Poincaré irreps are labelled (m, s) , the relativistic counterpart of the Galilean (m, s) of Chapter 4 — but here m is a Casimir, not a

central extension, so there is no Galilean mass superselection.

- **Massless particles** ($m = 0$): the little group of a light-like momentum is $\text{ISO}(2)$ (rotations and translations of a plane); its unitary irreps are one-dimensional, labelled by the **helicity** $\lambda \in \mathbb{Z}$ (or $\mathbb{Z} + \frac{1}{2}$ for spinors). The photon ($\lambda = \pm 1$) and the graviton ($\lambda = \pm 2$) are the physical instances; the absence of a rest frame forbids a longitudinal polarization.
- **Tachyons** ($m^2 < 0$) and the trivial rep ($m = 0, \lambda = 0$, the vacuum): no known physical realization.

The classification is the relativistic harvest of the same Mackey machinery that produced the Galilean unirreps: a particle is an irrep of its spacetime symmetry group, labelled by the invariants (mass, spin/helicity) that the group forces. The Galilean analysis of Chapter 4 is the $c \rightarrow \infty$ contraction of this one (Inönü–Wigner, §B.8): the Poincaré Casimir $P^2 = m^2 c^2$ contracts to the Galilean mass M (a central charge), and the Poincaré spin s contracts to the Galilean spin, the two agreeing in the nonrelativistic limit. The Dirac equation, the field-strength tensor, and the photon helicity are all read off the Lorentz irreps above; their full development is the business of relativistic quantum mechanics and quantum field theory, the natural sequel to this nonrelativistic book.

B.13 Racah Algebra in Depth

The 6- j and 9- j symbols of §B.6 are the atomic units of recoupling, but the full *Racah algebra* is the calculus of identities they satisfy — the explicit closed form of the 3- j , the derivation of the 6- j formula, the sum rules (Biedenharn–Elliott) that reduce any recoupling to a product of a few 6- j ’s, and the graphical calculus (Yutsis) that makes the algebra visual. This section develops these, the machinery behind every advanced angular-momentum calculation.

The explicit 3- j factorial form. The 3- j symbol has the Racah closed form (complementary to the CG formula (6.78)):

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = (-1)^{j_1-j_2-m_3} \sqrt{\Delta(j_1 j_2 j_3)^2 \prod_i (j_i \pm m_i)!} \sum_k \frac{(-1)^k}{k! (j_1+j_2-j_3-k)! (j_1-m_1-k)! (j_2+m_2-k)!} \quad (\text{B.53})$$

with $\Delta(abc) = \sqrt{(a+b-c)!(a-b+c)!(-a+b+c)!/(a+b+c+1)!}$ the triangle coefficient (zero unless the triangle holds) and the product $\prod_i (j_i \pm m_i)! = (j_1+m_1)!(j_1-m_1)! \cdots (j_3-m_3)!$. The sum over k is finite (factorial poles terminate it). Every symmetry of the 3- j — the cyclic invariance, the $(-1)^{j_1+j_2+j_3}$ under odd permutation, the sign flip under $\mathbf{m} \rightarrow -\mathbf{m}$ — is read directly off this expression.

Derivation of the 6- j formula. The 6- j symbol is, by definition, a sum over magnetic quantum numbers of a product of four 3- j ’s (equivalently four CG coefficients):

$$\left\{ \begin{matrix} j_1 & j_2 & j_{12} \\ j_3 & J & j_{23} \end{matrix} \right\} = \sum_{\text{all } m} \begin{pmatrix} j_1 & j_2 & j_{12} \\ m_1 & m_2 & m_{12} \end{pmatrix} \begin{pmatrix} j_{12} & j_3 & J \\ -m_{12} & m_3 & M \end{pmatrix} \begin{pmatrix} j_2 & j_3 & j_{23} \\ -m_2 & m_3 & m_{23} \end{pmatrix} \begin{pmatrix} j_1 & j_{23} & J \\ m_1 & -m_{23} & -M \end{pmatrix}, \quad (\text{B.54})$$

which is the recoupling (B.30) expanded in the uncoupled basis. Insert the closed form (B.53) for

each 3- j : the four square-root prefactors combine to a single Δ -product, and the four sums over k become a multiple sum over k_1, k_2, k_3, k_4 . The m -sums are then combinatorial — each $\sum_m$ of a ratio of factorials is evaluated by the Vandermonde–Chu identity $\sum_m \binom{A}{m} \binom{B}{C-m} = \binom{A+B}{C}$, collapsing the multiple k -sum to the single sum over t of (B.31). The derivation is lengthy but mechanical: it is the Racah–CG derivation (6.78) run on four coupled 3- j ’s, with the m -sums done by Vandermonde. The result is the single-sum closed form (B.31), from which the 72 Regge symmetries and the tetrahedral asymptotics (B.32) are read off.

Orthogonality and the Biedenharn–Elliott identity. The 6- j satisfies the orthogonality

$$\sum_x (2x+1) \begin{Bmatrix} a & b & p \\ c & d & x \end{Bmatrix} \begin{Bmatrix} a & b & q \\ c & d & x \end{Bmatrix} = \frac{\delta_{pq}}{2p+1}, \quad (\text{B.55})$$

the statement that the recoupling matrices are orthogonal. The deeper identity is the **Biedenharn–Elliott sum rule** — the associativity of recoupling four angular momenta, which equates a sum over a triple product of 6- j ’s (one common column summed) to a product of two 6- j ’s:

$$\sum_x (-1)^{S+x} (2x+1) \begin{Bmatrix} a & b & e \\ c & d & x \end{Bmatrix} \begin{Bmatrix} c & d & f \\ a & b & x \end{Bmatrix} \begin{Bmatrix} f & a & g \\ d & c & x \end{Bmatrix} = \begin{Bmatrix} f & g & e \\ d & c & b \end{Bmatrix} \begin{Bmatrix} e & f & g \\ a & b & c \end{Bmatrix}, \quad (\text{B.56})$$

with $S = a + b + c + d + e + f + g$. Geometrically, the left side is the two ways of contracting the recouplings (the associativity $(ab)c \leftrightarrow a(bc)$); the identity is the 6- j avatar of the pentagon relation. The Biedenharn–Elliott identity is the *master identity* of the Racah algebra: the orthogonality (B.55), the 9- j expansion, and every recoupling-coefficient reduction are its consequences. Together with (B.55) it makes the 6- j symbols a closed algebraic system: any recoupling of any number of angular momenta reduces, by repeated application, to a finite product of 6- j ’s.

12- j and general $3nj$ symbols. The recoupling of four angular momenta already exceeds the 9- j (which couples two pairs of two); five and more introduce the **12- j symbols** (two inequivalent kinds, distinguished by which pair is recoupled) and, in general, the **$3nj$ symbols** for n angular momenta. Each $3nj$ symbol is a sum of products of 6- j ’s (by Biedenharn–Elliott), and the number of distinct recouplings grows factorially with n . In practice one does not tabulate them; one reduces to 6- j ’s algorithmically. The general theory (Yutsis, Levinson, Vanagas) shows that every $3nj$ symbol corresponds to a trivalent graph, and the Biedenharn–Elliott identity is the graph-reduction rule.

Yutsis graphical calculus. The 3- j symbol is represented as a *trivalent vertex* with three labelled edges (the j_i, m_i); contracting two vertices on a common edge (summing the shared m) composes them, and a closed graph (all edges contracted) is a scalar $3nj$ symbol. The 6- j is the complete graph K_4 (a tetrahedron, four vertices, six edges); the 9- j is a 3×3 grid; the 12- j ’s are larger cubic graphs. The calculus is a set of graph-reduction moves: edge elimination (3- j orthogonality), triangle contraction (a 6- j with a zero entry collapses to a δ), and the Biedenharn–Elliott “triangle move” that rewrites one trivalent subgraph as another — the graphical form of (B.56). Any $3nj$ graph reduces to a product of 6- j symbols by a finite sequence of these moves, giving a fully algorithmic (and visual) route to every recoupling coefficient. This is the engine of nuclear shell-model and atomic-structure codes, where recouplings of dozens of angular momenta are reduced to machine-computable 6- j products.

The Racah algebra is the mature form of the angular-momentum coupling theory of Chapter 6

§6.6: what there was the Clebsch–Gordan series and the 3- j notation is here, in the 6- j and its identities, a complete calculus of recoupling, sufficient to reduce any matrix element of any rotation-invariant operator in any coupling scheme to a product of tabulated 6- j ’s times a few reduced matrix elements.

B.14 Realizations of SU(2) and SU(n)

The Schwinger boson realization of §B.3 represents SU(2) with *two* oscillator modes. Two further realizations are standard in physics: the *Holstein–Primakoff* representation, which uses a *single* boson and underlies spin-wave (magnon) theory, and the *spin coherent states*, the SU(2) analogue of the harmonic-oscillator coherent states of Chapter 5 and the basis of the spin path integral of Chapter 9. Both generalize: the Jordan–Schwinger map represents any SU(n) with n boson modes. This section collects them.

Holstein–Primakoff. A spin- s representation can be realized on the Fock space of a *single* boson a , $[a, a^\dagger] = 1$, with number operator $N = a^\dagger a$. The **Holstein–Primakoff** map is

$$J_+ = a^\dagger \sqrt{2s - N}, \quad J_- = \sqrt{2s - N} a, \quad J_z = N - s. \quad (\text{B.57})$$

The Fock state $|n\rangle$ ($n = 0, 1, \dots, 2s$) represents the spin state $|s, m = s - n\rangle$, i.e. $|n\rangle$ is the state with n spin-flips (“magnons”) below the fully polarized $|s, s\rangle$. The square root truncates the Fock space at $n = 2s$: $\sqrt{2s - N} |2s\rangle = 0$, so $J_- |0\rangle \dots$ the space is exactly $(2s+1)$ -dimensional, and a direct computation using $[N, a^\dagger f(N)] = a^\dagger f(N)$ and $[f(N)a, a^\dagger g(N)] = f(N)g(N+1) \cdot (\dots)$ verifies $[J_z, J_\pm] = \pm \hbar J_\pm$, $[J_+, J_-] = 2J_z$ exactly — the rotation algebra, on a single-mode Fock space.

The **large- s expansion** is the seed of spin-wave theory. For $N \ll 2s$ (few magnons on a large spin), $\sqrt{2s - N} = \sqrt{2s} (1 - \frac{N}{4s} + \dots)$, so

$$J_+ \simeq \sqrt{2s} a^\dagger + \mathcal{O}(1/\sqrt{s}), \quad J_z \simeq -s + a^\dagger a : \quad (\text{B.58})$$

to leading order the spin is a classical vector of length s plus harmonic-oscillator fluctuations (the magnons). This linearization turns the Heisenberg ferromagnet into a gas of free bosons — the standard route to the low-temperature thermodynamics of magnets — and is the prototype of the $1/s$ (large-spin) expansion used throughout condensed-matter and nuclear physics.

Spin coherent states. The harmonic-oscillator coherent states of Chapter 5 §5.7 are the orbit of the ground state under the Heisenberg group; the SU(2) analogue is the orbit of $|s, s\rangle$ under the rotation group, the **spin coherent states** (Bloch coherent states). For a unit vector $\mathbf{n} \in S^2$ (equivalently a complex stereographic coordinate ζ), define

$$|\mathbf{n}\rangle := D(\mathbf{n}) |s, s\rangle, \quad D(\mathbf{n}) = e^{-i\theta \mathbf{e}_\phi \cdot \mathbf{J}/\hbar} \quad (\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)), \quad (\text{B.59})$$

the rotation carrying $\hat{z}$ to $\mathbf{n}$, applied to the highest-weight state. In the $|s, m\rangle$ basis this is

$$|\mathbf{n}\rangle = \sum_{m=-s}^s \sqrt{\binom{2s}{s+m}} \left(\cos \frac{\theta}{2}\right)^{s+m} \left(\sin \frac{\theta}{2} e^{i\phi}\right)^{s-m} |s, m\rangle,$$

a binomial superposition whose coefficients are the $d_{m,s}^{(s)}(\beta)$ column (B.18) times the z -rotation phase.

The states are *overcomplete* and resolve the identity:

$$\frac{2s+1}{4\pi} \int_{S^2} d\Omega |n\rangle \langle n| = I, \quad (\text{B.60})$$

the SU(2) analogue of $\int |\alpha\rangle \langle \alpha| d^2\alpha/\pi = I$. The spin coherent states saturate the angular-momentum uncertainty $\Delta J_i \Delta J_j \geq \frac{\hbar}{2} |\langle \mathbf{J} \rangle|$, are the closest quantum states to a classical angular-momentum vector, and make the Bloch sphere of Chapter 6 §6.3 into the coset SU(2)/U(1) (the highest-weight state's stabilizer is the U(1) of rotations about z). They are the basis of the spin path integral of Chapter 9 §9.7 and the large- s semiclassical limit of spin systems.

Jordan–Schwinger for SU(n). The Schwinger map of §B.3 — angular momentum as bilinears in two boson modes — generalizes to any SU(n). Introduce n boson modes a_α , $\alpha = 1, \dots, n$ ($[a_\alpha, a_\beta^\dagger] = \delta_{\alpha\beta}$), and the $n^2 - 1$ generators

$$T_A = \sum_{\alpha, \beta=1}^n a_\alpha^\dagger \frac{1}{2} (\lambda_A)_{\alpha\beta} a_\beta, \quad A = 1, \dots, n^2 - 1, \quad (\text{B.61})$$

where λ_A are the generalized Gell-Mann matrices (the Pauli matrices for $n = 2$, the Gell-Mann matrices of §B.11 for $n = 3$). The bilinears $a_\alpha^\dagger a_\beta$ close the $\mathfrak{u}(n)$ algebra; the traceless combinations (B.61) close $\mathfrak{su}(n)$. The n -mode Fock space decomposes into SU(n) irreps labelled by the total boson number $N = \sum_\alpha a_\alpha^\dagger a_\alpha$: the N -boson symmetric subspace (dimension $\binom{N+n-1}{n-1}$) is the SU(n) irrep with Young diagram a single row of N boxes — the fully symmetric tensor representation. For $n = 2$ this recovers the SU(2) Schwinger model ($|j, m\rangle = |n_+, n_-\rangle$, $j = N/2$); for $n = 3$ it gives the flavor/color multiplets of §B.11. The Jordan–Schwinger construction is the bridge between the oscillator methods of Chapter 5/Chapter 10 and the representation theory of this appendix: every SU(n) irrep is realized concretely as a bosonic Fock subspace, and every SU(n) tensor product is a re-occupation of the modes.

Coda. The three realizations — Schwinger (two modes, exact, the basis of the d -matrix derivation §B.4), Holstein–Primakoff (one mode, exact, the basis of magnon theory), and Jordan–Schwinger (n modes, the basis of SU(n) representation theory) — exhibit a single deep fact: a Lie algebra is an abstract commutation structure, and any set of operators satisfying it is a valid realization. The rotation group lives equally in the two-state spinor, the two-boson Fock space, the single-boson Holstein–Primakoff space, the orbital differential operators of Chapter 6 §6.4, and the n -mode Jordan–Schwinger space; the representation theory classifies what is common to all of them, and the choice of realization is a tactical matter of which physics is being computed. This freedom — the same algebra, many realizations — is the practical face of the representation theory developed in this appendix, and the reason a single body of group-theoretic machinery serves every chapter of the book.

Appendix C Gaussian Units and Constants



Note This book employs Gaussian (CGS) units throughout. The choice is deliberate: in Gaussian units the Coulomb force between two charges is e^2/r , the fine-structure constant is $\alpha = e^2/\hbar c$, and the electric and magnetic fields enter the Lorentz force symmetrically as $\mathbf{E}$ and $\mathbf{B}$. No factors of $4\pi\epsilon_0$ or $\mu_0/4\pi$ appear in the fundamental equations. This appendix collects the conversion rules between Gaussian and SI units and tabulates the fundamental constants used in the text.

C.1 The Gaussian System

In Gaussian units, the Coulomb force between two charges e_1 and e_2 separated by a distance r is

$$F = \frac{e_1 e_2}{r^2}, \quad (\text{C.1})$$

and the electrostatic potential of a point charge is $\phi = e/r$. The electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ have the same dimensions, and the Lorentz force on a charge e moving with velocity $\mathbf{v}$ is

$$\mathbf{F} = e \left(\mathbf{E} + \frac{\mathbf{v}}{c} \times \mathbf{B} \right). \quad (\text{C.2})$$

Maxwell's equations in vacuum take the form

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 4\pi\rho, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}, & \nabla \times \mathbf{B} &= \frac{4\pi}{c} \mathbf{J} + \frac{1}{c} \frac{\partial \mathbf{E}}{\partial t}. \end{aligned} \quad (\text{C.3})$$

The fine-structure constant — the dimensionless measure of the electromagnetic coupling — is

$$\alpha = \frac{e^2}{\hbar c} \approx \frac{1}{137.036}, \quad (\text{C.4})$$

a pure number in Gaussian units. In SI units the same constant is $\alpha = e^2/4\pi\epsilon_0\hbar c$, with the $4\pi\epsilon_0$ carrying the burden of the unit conversion.

The hydrogen atom Hamiltonian in Gaussian units is

$$H = \frac{\mathbf{p}^2}{2m} - \frac{e^2}{r}, \quad (\text{C.5})$$

and the Bohr radius is

$$a_0 = \frac{\hbar^2}{m_e e^2} \approx 0.529 \text{ \AA}. \quad (\text{C.6})$$

The Rydberg energy is

$$\text{Ry} = \frac{m_e e^4}{2\hbar^2} = \frac{e^2}{2a_0} \approx 13.606 \text{ eV}. \quad (\text{C.7})$$

C.2 Conversion Table: Gaussian to SI

The conversion of electromagnetic quantities between Gaussian and SI units is not a simple multiplicative factor for every quantity, because the two systems differ in their dimensional assignments. The table below gives the replacement rule: to convert a formula written in Gaussian units to its SI

equivalent, make the substitutions indicated. The reverse conversion (SI to Gaussian) follows from the same table read backwards.

Table C.1: Conversion of electromagnetic quantities from Gaussian to SI units. The symbol $\sqrt{4\pi\epsilon_0}$ is abbreviated as κ .

Quantity	Gaussian	SI replacement
Charge	e	$e/\sqrt{4\pi\epsilon_0}$
Electric field	$\mathbf{E}$	$\sqrt{4\pi\epsilon_0} \mathbf{E}$
Magnetic field	$\mathbf{B}$	$\mathbf{B}/\sqrt{4\pi\epsilon_0\mu_0} = c\sqrt{4\pi\epsilon_0} \mathbf{B}$
Scalar potential	ϕ	$\sqrt{4\pi\epsilon_0} \phi$
Vector potential	$\mathbf{A}$	$\mathbf{A}/\sqrt{4\pi\epsilon_0\mu_0}$
Coulomb force	e_1e_2/r^2	$e_1e_2/4\pi\epsilon_0r^2$
Fine-structure constant	$e^2/\hbar c$	$e^2/4\pi\epsilon_0\hbar c$
Bohr radius	$\hbar^2/m_e e^2$	$4\pi\epsilon_0\hbar^2/m_e e^2$
Rydberg energy	$m_e e^4/2\hbar^2$	$m_e e^4/2(4\pi\epsilon_0)^2\hbar^2$
Bohr magneton	$e\hbar/2m_e c$	$e\hbar/2m_e$

The conversion is systematic. Electric quantities (e , $\mathbf{E}$, ϕ) acquire a factor of $\sqrt{4\pi\epsilon_0}$ or its inverse; magnetic quantities ($\mathbf{B}$, $\mathbf{A}$) acquire an additional factor of c or $1/c$ because in Gaussian units $\mathbf{E}$ and $\mathbf{B}$ share the same dimensions. The Bohr magneton illustrates the pattern: $\mu_B = e\hbar/2m_e c$ in Gaussian units becomes $\mu_B = e\hbar/2m_e$ in SI, the c disappearing because the SI definition absorbs the $4\pi\epsilon_0$ and the speed of light into the unit conventions.

For macroscopic quantities not used in this book (current, resistance, capacitance), the conversion is more involved and the reader is referred to J. D. Jackson, *Classical Electrodynamics*, 3rd ed. (Wiley, 1999), Appendix A.

C.3 Fundamental Constants

The following table collects the fundamental constants used in the text, with their 2022 CODATA recommended values. The values are given in Gaussian units where applicable and in SI where the constant is system-independent. The number of significant figures shown exceeds what any calculation in this book needs; the extra digits are supplied so that the reader need not consult an external source for routine numerical work.

A note on the Gaussian magnetic constants. The Bohr magneton in Gaussian units is $\mu_B = e\hbar/2m_e c$, with the c in the denominator; its energy in a magnetic field B is $-\boldsymbol{\mu} \cdot \mathbf{B}$. Thus the Zeeman Hamiltonian of Chapter 6 (6.110) is $H_Z = (e/2m_e c)BL_z = \mu_B BL_z/\hbar$, giving the linear splitting $\Delta E = \mu_B Bm$.

The conversion factor $\hbar c \approx 197.327 \text{ MeV}\cdot\text{fm}$ (where $1 \text{ fm} = 10^{-13} \text{ cm}$) is useful for nuclear and particle-physics estimates, though those domains lie beyond the main text. For atomic-scale estimates,

Table C.2: Fundamental constants (2022 CODATA).

Symbol	Name	Value
$\hbar$	Reduced Planck constant	$1.054\,571\,817 \times 10^{-27} \text{ erg}\cdot\text{s}$ $= 6.582\,119\,569 \times 10^{-16} \text{ eV}\cdot\text{s}$
e	Elementary charge (esu)	$4.803\,204\,713 \times 10^{-10} \text{ esu}$
m_e	Electron mass	$9.109\,383\,713 \times 10^{-28} \text{ g}$ $= 0.510\,998\,950 \text{ MeV}/c^2$
m_p	Proton mass	$1.672\,621\,926 \times 10^{-24} \text{ g}$ $= 938.272\,089 \text{ MeV}/c^2$
k_B	Boltzmann constant	$1.380\,649 \times 10^{-16} \text{ erg/K}$
c	Speed of light	$2.997\,924\,58 \times 10^{10} \text{ cm/s (exact)}$
α	Fine-structure constant	$1/137.035\,999\,177$
a_0	Bohr radius	$5.291\,772\,105 \times 10^{-9} \text{ cm}$
R_∞	Rydberg constant	$109\,737.315\,681 \text{ cm}^{-1}$ $= 3.289\,841\,960 \times 10^{15} \text{ Hz}$
μ_B	Bohr magneton	$9.274\,009\,994 \times 10^{-21} \text{ erg/G}$
μ_N	Nuclear magneton	$5.050\,783\,739 \times 10^{-24} \text{ erg/G}$
$g_e/2$	Electron g -factor (half)	$1.001\,159\,652\,180$
N_A	Avogadro constant	$6.022\,140\,76 \times 10^{23} \text{ mol}^{-1}$

¹²⁶²² the combination $e^2 = \alpha \hbar c \approx 14.40 \text{ eV}\cdot\text{\AA}$ converts the Coulomb energy e^2/r into convenient units.

Appendix D Probability Theory Essentials

12623



Note *The quantum-mechanical probabilities of this book are interpreted in the frequentist (ensemble) sense: a probability is the limiting relative frequency of an outcome in an infinite sequence of identically prepared and independently measured trials. This appendix states the definitions and theorems of classical probability theory that the main text's probability arguments rest on, making explicit what Chapters 2 and 3 deploy without formal preamble. No claim is made that probability theory follows from quantum mechanics; the direction of dependence is the reverse: the Born rule is a bridge from the amplitude calculus to a pre-existing theory of frequencies.*

12624
12625
12626
12627
12628
12629
12630

D.1 The Ensemble Interpretation

12631

Quantum mechanics makes statistical predictions about ensembles, not about single events. When the main text states that $P(\psi \rightarrow \phi) = |\langle \phi | \psi \rangle|^2$, the statement is: in a large number N of identically prepared systems, each in state $|\psi\rangle$ and each subjected to the measurement that asks the question $\{|\phi\rangle, \text{all states orthogonal to } |\phi\rangle\}$, the fraction of trials that yield the outcome ϕ approaches $|\langle \phi | \psi \rangle|^2$ as $N \rightarrow \infty$.

12632
12633
12634
12635
12636

This is the **frequentist** (or **ensemble**) interpretation of probability, introduced at Chapter 2's Section 2.3 and held consistently throughout the book. It is not the only interpretation of probability in quantum mechanics — the Bayesian and propensity interpretations have their advocates — but it is the interpretation that underwrites every experimental test of the theory: a prediction of quantum mechanics is tested by counting frequencies, and a frequency that deviates from the Born-rule prediction by more than the statistical fluctuation either falsifies the theory or signals a failure of the preparation or measurement protocol.

12637
12638
12639
12640
12641
12642
12643

The ensemble interpretation has one operational consequence that bears directly on the book's architecture. A probability is a property of the *whole experimental arrangement* — preparation, waiting, and interrogation — not of the individual system. No single quantum event is assigned a probability; what is assigned a probability is the pattern of outcomes across the ensemble. This is why the theory's state vector $|\psi\rangle$ is a description of the preparation procedure (Chapter 2, §2.1), not of an individual system's hidden properties: the preparation *defines* the ensemble, and the state vector encodes the ensemble's statistical behavior.

12644
12645
12646
12647
12648
12649
12650

The same interpretation governs the measurement axioms of Chapter 3. The expectation value $\langle A \rangle_\psi$ is an ensemble average: it is not what a single measurement will yield (no such statement is made) but the arithmetic mean of the outcomes that would be obtained by measuring A on every member of the ensemble. The variance $(\Delta A)^2$ is the ensemble variance, and the uncertainty relations are constraints on the joint statistics of two ensembles (one for A , one for B , prepared identically but measured differently), not on the precision of a single joint measurement.

12651
12652
12653
12654
12655
12656

D.2 Conditional Probability

Let A and B be two events (subsets of the sample space of a probabilistic experiment). The **conditional probability** of A given B is

$$P(A|B) := \frac{P(A \cap B)}{P(B)}, \quad (\text{D.1})$$

provided $P(B) > 0$. It is the fraction of trials in which A occurs among those trials in which B has occurred.

The quantum-mechanical analogue is the state-update rule. After a measurement of an observable with outcome subspace P_ν , the conditional state (the Lüders update of Chapter 3 (3.47)) is $|\psi_\nu\rangle = P_\nu |\psi\rangle / \sqrt{p_\nu}$, and the conditional probability for a subsequent measurement of B is

$$P(B = b_\mu | \text{outcome } \nu) = \frac{\langle \psi | P_\nu P_\mu P_\nu | \psi \rangle}{\langle \psi | P_\nu | \psi \rangle} = \langle \psi_\nu | P_\mu | \psi_\nu \rangle. \quad (\text{D.2})$$

The analogy with (D.1) is exact: the numerator is the probability of the joint outcome (first ν , then μ), and the denominator is the probability of the conditioning outcome ν . The quantum formalism supplies the numerator through the composition principle for amplitudes (Chapter 2's (2.33)): the amplitude for the sequence $\psi \rightarrow \nu \rightarrow \mu$ is $\langle \mu | \nu \rangle \langle \nu | \psi \rangle$, and the squared modulus of that product, summed over the μ subspace and normalized by p_ν , is the conditional probability.

D.3 Expectation, Variance, and Moments

Let X be a real-valued random variable on a sample space, with probability distribution $p(x)$ (discrete) or density $\rho(x)$ (continuous). The **expectation** (mean) of X is

$$\langle X \rangle := \sum_x x p(x) \quad \text{or} \quad \langle X \rangle := \int x \rho(x) dx. \quad (\text{D.3})$$

The expectation is linear: $\langle aX + bY \rangle = a\langle X \rangle + b\langle Y \rangle$. The **variance** is the mean-square deviation from the mean:

$$\text{Var}(X) := \langle (X - \langle X \rangle)^2 \rangle = \langle X^2 \rangle - \langle X \rangle^2, \quad (\text{D.4})$$

and the standard deviation is $\Delta X := \sqrt{\text{Var}(X)}$.

In quantum mechanics, the observable A plays the role of the random variable, its spectral decomposition $A = \sum_n a_n P_n$ supplies the possible values a_n , and the state $|\psi\rangle$ supplies the distribution $p_n = \langle \psi | P_n | \psi \rangle$. The quantum expectation (3.24) and variance (3.26) are the direct transcriptions of (D.3) and (D.4) into the spectral language. The linearity of the quantum expectation, $\langle A + B \rangle_\psi = \langle A \rangle_\psi + \langle B \rangle_\psi$, follows from the linearity of the operator trace *in the state* (or, equivalently, from the additivity of the spectral sum), and it holds whether or not A and B commute. The failure of the *product* expectation to factor, $\langle AB \rangle_\psi \neq \langle A \rangle_\psi \langle B \rangle_\psi$ in general, is the statistical signature of noncommutation.

12688

D.4 The Law of Large Numbers

Theorem (Weak Law of Large Numbers; stated without proof). Let $X_1, X_2, \dots$ be independent, identically distributed random variables with finite expectation $\mu = \langle X \rangle$ and finite variance σ^2 . Define the sample mean $\bar{X}_N := (X_1 + \dots + X_N)/N$. Then for any $\varepsilon > 0$,

$$\lim_{N \rightarrow \infty} P(|\bar{X}_N - \mu| \geq \varepsilon) = 0. \quad (\text{D.5})$$

That is, the sample mean converges in probability to the expectation.

The proof is a direct application of Chebyshev's inequality: $\text{Var}(\bar{X}_N) = \sigma^2/N$, so $P(|\bar{X}_N - \mu| \geq \varepsilon) \leq \sigma^2/N\varepsilon^2 \rightarrow 0$. The theorem supplies the book's operational definition of probability as limiting frequency: when the main text writes $p_n = |\langle e_n | \psi \rangle|^2$, it asserts that in N independent trials the count of outcome n , divided by N , converges to p_n in the sense of (D.5).

The **Strong Law of Large Numbers** (Kolmogorov, 1933; stated without proof) strengthens the convergence to *almost sure* convergence: with probability one, $\lim_{N \rightarrow \infty} \bar{X}_N = \mu$. The distinction between the weak and strong laws is measure-theoretic and has no operational consequence for the finite ensembles of the laboratory, but it seals the logical consistency of the frequentist interpretation: the limiting frequency exists and equals the probability for almost every infinite sequence of trials.

D.5 Statistical Independence

12703

Two events A and B are **statistically independent** if $P(A \cap B) = P(A)P(B)$. In terms of conditional probabilities, $P(A|B) = P(A)$ — knowing that B occurred does not change the probability assigned to A . Two random variables X and Y are independent if the joint probability factorizes, $p(x, y) = p_X(x) p_Y(y)$, for all values.

In quantum mechanics, statistical independence has no direct counterpart at the level of pure states: if two observables A and B commute, they possess a joint probability distribution (the product of their spectral measures on the common eigenbasis), but the factorization $p(a, b) = p(a)p(b)$ obtains only for product states in a tensor-product Hilbert space (Chapter 7). For noncommuting observables, no joint distribution exists in the classical sense, and the failure is not a failure of independence but a failure of the very concept of a joint probability. This is the content of the compatibility theorem of Chapter 3 (3.49): $[A, B] = 0$ is the necessary and sufficient condition for the existence of a joint PVM, and the existence of a joint PVM is the quantum analogue of the existence of a joint probability. The classical notion of independence reappears in the composite-systems chapter as the definition of separable (unentangled) states.

D.6 Density Operators as Quantum Probability Measures

12718

A quantum state, on the ensemble interpretation, is a probability measure on the outcomes of every observable — an assignment, to each self-adjoint operator, of a probability distribution on its spectrum. Gleason's theorem (stated without proof in Chapter 3, §3.3) establishes that on a Hilbert

space of dimension ≥ 3 , every such assignment that is additive on orthogonal projectors is represented by a unique **density operator** ρ : a positive self-adjoint operator of unit trace, such that

$$p(P) = \text{tr}(\rho P) \quad (\text{D.6})$$

for every projector P . Pure states correspond to rank-one density operators $\rho = |\psi\rangle\langle\psi|$, for which $\text{tr}(\rho P) = \langle\psi|P|\psi\rangle$ recovers the Born rule. Mixed states correspond to higher-rank density operators, which cannot be represented by a single state vector and which describe ensembles that are not preparable as superpositions of a single procedure.

The density operator is the quantum analogue of the classical probability distribution. Its properties mirror the classical axioms: (i) $\rho^\dagger = \rho$ and $\rho \geq 0$ (positivity: every probability is nonnegative); (ii) $\text{tr}\rho = 1$ (normalization: the total probability is one); (iii) convexity: if ρ_1 and ρ_2 are density operators, so is $\lambda\rho_1 + (1 - \lambda)\rho_2$ for any $\lambda \in [0, 1]$, representing an ensemble formed by randomly choosing preparation procedure 1 with probability λ and procedure 2 with probability $1 - \lambda$. The set of all density operators on $\mathcal{H}$ is a convex set whose extreme points are exactly the pure states $|\psi\rangle\langle\psi|$. The dimension of the state space is $(\dim \mathcal{H})^2 - 1$ (real parameters of a Hermitian unit-trace matrix), which for a two-state system gives the three-parameter Bloch sphere of Chapter 6 (6.39).

The density operator is the proper language for the ensemble interpretation because it makes the ensemble explicit: the spectral decomposition $\rho = \sum_k w_k |k\rangle\langle k|$ (with $w_k \geq 0$, $\sum_k w_k = 1$) exhibits the state as a classical mixture of orthogonal pure states, prepared with classical probabilities w_k . The w_k are classical probabilities in the sense that they reflect ignorance of which pure state was actually prepared, not quantum amplitudes; they combine by ordinary addition, not by the composition principles of Chapter 2. The full development of the density operator as a tool — its equation of motion (the Liouville–von Neumann equation), the partial trace, entanglement measures, and the measurement problem — belongs to Chapter 7.